Code Archive

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[FILE] viirtuoz-vps-dev-rewrite-back-2/.dockerignore (406.0 B)

```
# Git and IDE
.git/
.gitignore
.cursor/
.trunk/
*.md

# Python
**/__pycache__/
*.py[cod]
.venv/
venv/
.pytest_cache/
.mypy_cache/

# Kubernetes (not needed in image)
kubernetes/
kubernetes_example/

# Docs and dev scripts
docs/
run-stub-mode.sh
docker-compose*
vm-template/

# Secrets and local config
.env
.env.*
juniper_*
config/*.local.*
*.example
*.iso
driver*/
auth*

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/.python-version (5.0 B)

```
3.11

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/Dockerfile (1.7 KB)

```
FROM quay.io/oauth2-proxy/oauth2-proxy:v7.14.2 AS oauth2-proxy

FROM python:3.11-slim-bookworm

COPY --from=ghcr.io/astral-sh/uv:latest /uv /uvx /usr/local/bin/

RUN apt-get update \
    && apt-get install -y --no-install-recommends curl openssl git nginx \
    && rm -rf /var/lib/apt/lists/* \
    && rm -f /etc/nginx/sites-enabled/default

COPY conf/default.conf /etc/nginx/conf.d/default.conf
COPY conf/gzip.conf /etc/nginx/conf.d/gzip.conf

RUN mkdir -p /etc/ssl/private \
    && openssl genrsa -out /etc/ssl/private/jwt_signing_key.pem 2048 \
    && chmod 644 /etc/ssl/private/jwt_signing_key.pem

RUN curl -LO "https://dl.k8s.io/release/$(curl -Ls https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl" \
    && chmod +x kubectl && mv kubectl /usr/local/bin/

RUN KUBEVIRT_VERSION=$(curl -s https://api.github.com/repos/kubevirt/kubevirt/releases/latest | grep tag_name | cut -...
    && curl -L -o virtctl "https://github.com/kubevirt/kubevirt/releases/download/${KUBEVIRT_VERSION}/virtctl-${KUBEV...
    && chmod +x virtctl && mv virtctl /usr/local/bin/

COPY --from=oauth2-proxy /bin/oauth2-proxy /bin/oauth2-proxy

WORKDIR /app

ENV UV_LINK_MODE=copy \
    PATH="/app/.venv/bin:$PATH"

COPY pyproject.toml uv.lock ./
RUN uv sync --frozen --no-dev --no-install-project

COPY app ./app
COPY entrypoint.sh ./

RUN git clone https://github.com/novnc/noVNC.git /app/app/static/novnc

RUN mkdir -p /docker-entrypoint.d
COPY entrypoint/90-oauth-proxy.sh entrypoint/91-startkubectl.sh /docker-entrypoint.d/
RUN chmod +x /docker-entrypoint.d/*.sh

COPY entrypoint.sh /entrypoint.sh
RUN chmod +x /entrypoint.sh

EXPOSE 8080

CMD ["/entrypoint.sh"]

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/README.md (19.1 KB)

```
# Viirtuoz VPS

[![License](https://img.shields.io/badge/license-MIT-blue.svg)](LICENSE)
[![Python](https://img.shields.io/badge/python-3.9+-blue.svg)](https://www.python.org/)
[![KubeVirt](https://img.shields.io/badge/KubeVirt-1.0+-green.svg)](https://kubevirt.io/)
[![Version](https://img.shields.io/badge/version-1.0--rc.0.0.1-blue.svg)](https://github.com/yourusername/kubevirt-ap...
[![Kubernetes](https://img.shields.io/badge/Kubernetes-1.24+-blue.svg)](https://kubernetes.io/)

> ? Modern web interface for managing KubeVirt virtual machines with Cloud-Init, GPU Passthrough, and Ceph storage su...

## ? Features

### Core Functionality

- **VM Lifecycle Management**: Create, start, stop, restart, pause, resume, and delete virtual machines
- **Cloud-Init Integration**: Automated user provisioning, SSH keys, package installation, and system configuration
- **GPU Passthrough**: Full NVIDIA GPU passthrough support with automatic audio device handling
- **Persistent Storage**: Automatic disk provisioning via Ceph RBD with CDI (Containerized Data Importer)
- **Real-time Monitoring**: Live VM status, resource utilization, and metrics visualization
- **VNC Console**: Browser-based VNC access to running virtual machines
- **LLM Chat Assistant**: Built-in AI assistant for VM management guidance

### Advanced Features

- **Wizard-based VM Creation**: Step-by-step guided workflow with validation
- **Resource Presets**: Quick deployment templates (Small, Medium, Large, GPU-enabled)
- **Dual Display (VNC + GPU)**: VNC remains accessible even with GPU passthrough
- **SSH Service Management**: Automatic NodePort creation with Juniper hardware NAT integration
- **Multi-user Support**: Namespace isolation per user with RBAC
- **Stub Mode**: Frontend development without Kubernetes cluster

---

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## ? Requirements

### Infrastructure

- **Kubernetes**: v1.24 or higher
- **KubeVirt**: v1.0 or higher (with HostDevices feature gate for GPU support)
- **CDI** (Containerized Data Importer): For VM disk image management
- **Ceph**: RBD for block storage, CephFS for application data (optional)

### Storage Classes

- `your-SC` - Block storage for VM disks
- `your-SC` - File storage for application database (optional)

_Note: Storage class names can be customized in configuration._

```

Optional Components

- **Juniper SRX Router**: For hardware NAT integration (external SSH access)
- **Multus CNI**: For advanced networking (optional)
- **Metrics Server**: For resource utilization monitoring

?? Architecture

Component Overview

```
??????????????????????????????????????????????????????????????????????
?                     Ingress / Load Balancer                  ?
?                    (kvm.example.com)                         ?
??????????????????????????????????????????????????????????????????????
                          ?
                          ?
??????????????????????????????????????????????????????????????????????
?                  KubeVirt API Manager Pod                     ?
?  ????????????????  ????????????????  ????????????????      ?
?  ?   Nginx      ?  ?    Flask     ?  ?   Juniper    ?      ?
?  ?  (Static)    ????  (Backend)   ????   Sidecar    ?      ?
?  ????????????????  ????????????????  ????????????????      ?
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                  ???????????????????????
                  ?                     ?
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        ??????????????????   ??????????????????
        ?   Kubernetes   ?   ?   KubeVirt     ?
        ?   API Server   ?   ?      CRs       ?
        ??????????????????   ??????????????????
                  ?                     ?
                  ???????????????????????
                            ?
                            ?
                ??????????????????????????
                ?  Virtual Machines     ?
                ?  (KubeVirt Pods)      ?
                ??????????????????????????
```

Key Components

Frontend

- **Vanilla JavaScript**: No framework dependencies, lightweight and fast
- **CSS Variables**: Theme support with dark/light mode
- **Modular Architecture**: Shared core modules and page-specific logic
- **Real-time Updates**: Auto-refresh and SSE (Server-Sent Events) for live data

Backend

- **Flask**: Python web framework
- **Kubernetes Python Client**: Direct API interaction
- **Jinja2 Templates**: Server-side rendering
- **Session Management**: User authentication and namespace isolation

Networking

- **Pod Network (Default)**: Masquerade mode for VM connectivity
- **NodePort Services**: SSH access via Kubernetes services
- **Juniper Integration**: Hardware NAT for external IP mapping

---

? Quick Start

Development Mode (No Kubernetes Required)

Perfect for frontend development and UI testing:

```bash
# Using Docker Compose
git clone https://github.com/yourusername/kubevirt-api-manager.git
cd kubevirt-api-manager
docker-compose up -d

# Access the application
open http://localhost:8083
# Login: example / your-secret-password (default, configurable via DEMO_USERNAME/DEMO_PASSWORD)
```

The application starts in **stub mode** with demo VMs:

- `demo-gpu-vm` - Running VM with NVIDIA A100 GPU
- `data-import-vm` - VM being provisioned
- `analytics-vm` - Stopped VM

See [DEVELOPMENT.md](docs/DEVELOPMENT.md) for more details.

? Installation

Prerequisites

1. **Label KubeVirt Nodes**

   ```bash
   kubectl label nodes <node-name> node-role.kubernetes.io/kubevirt=
   ```

2. **Configure GPU Passthrough** (if using GPUs)

   Edit KubeVirt CR:

   ```bash
   kubectl edit kubevirt kubevirt -n kubevirt
   ```

   Add:

   ```yaml
   spec:
     configuration:
       developerConfiguration:
         featureGates:
           - HostDevices
       permittedHostDevices:
         pciHostDevices:
           - pciVendorSelector: "10DE:1B06" # NVIDIA GTX 1080 Ti
             resourceName: "nvidia.com/1080ti"
           - pciVendorSelector: "10DE:10EF" # HDMI Audio
             resourceName: "nvidia.com/1080ti-audio"
   ```

3. **Verify CDI Installation**
   ```bash
   kubectl get pods -n cdi
   ```

Deploy to Kubernetes

```bash
# 1. Clone repository
git clone https://github.com/yourusername/kubevirt-api-manager.git
cd kubevirt-api-manager

# 2. Create namespace
```

```
kubectl create namespace kvm

# 3. Configure secrets (copy example manifests to kubernetes/ or use kubernetes_example/)
cp kubernetes_example/secrets-example.yaml kubernetes_example/secrets.yaml
# Edit secrets.yaml with your values (TODO: replace placeholders)
kubectl apply -f kubernetes_example/secrets.yaml

# 4. Deploy application (bundled manifest includes all resources)
kubectl apply -f kubernetes_example/bundled.yaml

# 5. Verify deployment
kubectl get pods -n kvm
kubectl get ingress -n kvm
```

For detailed deployment steps (individual manifests, prerequisites, troubleshooting), see [docs/KUBERNETES-DEPLOYMENT...

Access the Application

```bash
# Get ingress address
kubectl get ingress -n kvm

# Add to /etc/hosts if using local cluster
echo "192.168.1.100  kvm.example.com" | sudo tee -a /etc/hosts

# Open in browser
open https://kvm.example.com
```

?? Configuration

Environment Variables

| Variable | Default | Description ...
| --- | --- | --- ...
| `FRONTEND_STUB_MODE` | `false` | Enable stub mode for development ...
| `DEBUG` | `false` | Flask debug mode ...
| `SECRET_KEY` | (required) | Flask secret key for sessions ...
| `K8S_NAMESPACE_PREFIX` | `kv` | Prefix for user namespaces ...
| `DEFAULT_GPU_RESOURCE` | `nvidia.com/1080ti` | Default GPU resource name ...
| `STORAGE_CLASS_NAME` | `your-SC` | Storage class for VM disks ...
| `VM_STORAGE_NODE_SELECTOR` | (empty) | JSON nodeSelector for CDI importer pods. See [VM Scheduling (CSI)...
| `VM_RUN_STRATEGY` | `Manual` | VM shutdown/reboot behavior. See [VM Run Strategy](docs/VM_RUN_ST...
| `MAX_CPU_CORES` | `16` | Maximum CPU cores per VM ...
| `MAX_MEMORY_GB` | `64` | Maximum memory per VM (GB) ...
| `MAX_STORAGE_GB` | `500` | Maximum storage per VM (GB) ...
| `MAX_GPU_COUNT` | `2` | Maximum GPUs per VM ...
| `DEMO_USERNAME` | `example` | Demo login username ...
| `DEMO_PASSWORD` | (required) | Demo login password ...
| `LLM_API_KEY` | (optional) | For LLM chat features ...

See `app/config.py` for all options. Sensitive values should be set via Kubernetes Secrets.

Juniper Integration (Optional)

For external SSH access via hardware NAT:

```yaml
# In kubernetes_example/juniper-sidecar-cm.yaml
data:
  JUNIPER_HOST: "router.example.com"
  JUNIPER_USER: "automation"
  JUNIPER_PUBLIC_IP: "203.0.113.10"
```

SSH keys should be mounted as secrets:

```bash
```

```
kubectl create secret generic juniper-ssh-key \
  --from-file=id_rsa=./juniper_key \
  --from-file=id_rsa.pub=./juniper_key.pub \
  -n kvm
```

? Development

Local Development Setup

Option 1: Docker Compose (Recommended)

```bash
# Start services
docker-compose up -d

# View logs
docker-compose logs -f kubevirt-api

# Rebuild after code changes
docker-compose up -d --build

# Stop services
docker-compose down
```

Option 2: Native Python

```bash
# Create virtual environment
python3 -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Set environment variables
export FRONTEND_STUB_MODE=true
export DEBUG=true
export SECRET_KEY=dev-secret-key-change-in-production
export DEMO_USERNAME=admin
export DEMO_PASSWORD=your-secret-password

# Run Flask
python app/main.py
```

Access at `http://localhost:8000` (direct Flask) or `http://localhost:8083` (via Docker Compose with Nginx)

Code Structure

```
kubevirt-api-manager/
??? app/
?   ??? main.py                 # Flask application entry
?   ??? config.py               # All configuration (env vars, limits, LLM, Windows WINDOWS_*)
?   ??? routes/                 # API endpoints
?   ?   ??? __init__.py         # Dashboard, login, logout
?   ?   ??? vm_routes.py        # VM CRUD operations
?   ?   ??? storage_routes.py   # Storage management
?   ?   ??? vm_details_routes.py # VM details, LLM, services
?   ??? utils/                  # Business logic
?   ?   ??? k8s_utils.py        # Kubernetes interactions
?   ?   ??? vm_utils.py         # Linux VM manifest generation
?   ?   ??? vm_manifest_common.py # Shared VM manifest components
?   ?   ??? service_utils.py    # NodePort services & port allocation
?   ?   ??? network_policy_utils.py # Network policies
?   ?   ??? juniper_utils.py    # Juniper integration
?   ??? static/
```

```
?   ?   ??? js/
?   ?   ?   ??? core/           # Shared modules
?   ?   ?   ??? pages/          # Page-specific logic
?   ?   ??? css/
?   ??? templates/              # Jinja2 templates
??? kubernetes_example/         # K8s manifests (templates)
??? docs/                       # Documentation
??? Dockerfile
??? docker-compose.yml
??? requirements.txt
```

Making Changes

1. **Frontend**: Edit files in `app/static/` and `app/templates/`
2. **Backend**: Edit files in `app/routes/` and `app/utils/`
3. **Test in stub mode**: Use Docker Compose
4. **Test with K8s**: Deploy to dev cluster

See docs/DEVELOPMENT.md for development setup and contribution guidelines.

? API Documentation

VM Management Endpoints

Create VM

```http
POST /<username>/create-vm
Content-Type: application/x-www-form-urlencoded

name=my-ubuntu-vm&cpu=4&memory=16&storage=100&image=...
```

Start VM

```http
POST /<username>/vm/<vm_name>/start
```

Response:

```json
{
  "success": true,
  "message": "VM started successfully"
}
```

Stop VM

```http
POST /<username>/vm/<vm_name>/stop
```

Restart VM

```http
POST /<username>/vm/<vm_name>/restart
```

Delete VM

```http
POST /<username>/vm/<vm_name>/delete
```

Check VM Name Availability

```http
GET /<username>/check-vm-name/<vm_name>
```

Response:

```json
{
  "available": true,
  "exists": false
}
```

SSH Service Endpoints

Create SSH Service

```http
POST /<username>/vm/<vm_name>/create-ssh-service
```

Response:

```json
{
  "success": true,
  "port": 30123,
  "public_ip": "203.0.113.10",
  "ssh_command": "ssh user@203.0.113.10 -p 30123"
}
```

Delete SSH Service

```http
POST /<username>/vm/<vm_name>/delete-ssh-service
```

Metrics Endpoints

Get VM Metrics

```http
GET /<username>/vm/<vm_name>/metrics
```

Response:

```json
{
  "success": true,
  "metrics": {
    "cpu_usage": 125, // millicores
    "memory_usage": 2048, // MiB
    "timestamp": "2026-01-27T12:34:56Z"
  }
}
```

? Troubleshooting

Common Issues

VM Creation Fails

Symptom: VM creation returns error "Failed to create DataVolume"

Solution:

```bash
# Check CDI operator
kubectl get pods -n cdi

# Check storage class
kubectl get storageclass

# Check PVC events
kubectl describe pvc <vm-name>-rootdisk -n <namespace>

... [133 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/README_RU.md (23.0 KB)

```
# Viirtuoz VPS

[![License](https://img.shields.io/badge/license-MIT-blue.svg)](LICENSE)
[![Python](https://img.shields.io/badge/python-3.9+-blue.svg)](https://www.python.org/)
[![KubeVirt](https://img.shields.io/badge/KubeVirt-1.0+-green.svg)](https://kubevirt.io/)
[![Kubernetes](https://img.shields.io/badge/Kubernetes-1.24+-blue.svg)](https://kubernetes.io/)

> ? ??????????? ???-????????? ??? ?????????? ??????????? ???????? KubeVirt ? ?????????? Cloud-Init, ???????? GPU ? ?...

## ? ???????????

### ???????? ??????????

- **?????????? ????????? ?????? VM**: ????????, ??????, ?????????, ???????????, _____, ???????????? ? ???????? ????...
- **?????????? Cloud-Init**: ????????????????? ????????? ????????????????, SSH-??????, ????????? ??????? ? ??????????? ???...
- **??????? GPU**: ?????? ????????? ???????? NVIDIA GPU ? ????????????????? ?????????? ?????????????????
- **?????????? ??????????**: ????????????????? ????????? ?????? _____ Ceph RBD ? CDI (Containerized Data Importer)
- **?????????? ? ???????? ???????**: ?????????? ?????? VM, ???????????????? ???????? ? ??????????? ??????
- **VNC-????????**: ?????? ? VNC ??????? ?????????? ?????????? _____ _____ ???????
- **LLM-????????? ? ????**: ?????????? AI-????????? ??? ?????? ? ?????????? VM

### ????????????????? ??????????

- **?????? ???????? VM**: ????????? ?????????? ??????? ? ??????????
- **????????????????? ????????**: ??????? ???????????????? ???????? (Small, Medium, Large, ? GPU)
- **?????????? SSH-??????????**: ????????????????? ???????? NodePort ? ?????????? Juniper NAT
- **????????? ????**: ????????? ???????? _____ ??? ??????????
- **??????????????????????? ??????????**: ???????? namespace ?? ??????????????? ? RBAC
- **Stub-_____**: ?????????? frontend ??? ???????? Kubernetes

---

## ? ??????????

- [???????????](#-???????????)
- [??????????](#-??????????)
- [???????????](#-???????????)
- [??????? _____](#-???????-_____)
- [?????????](#-?????????)
- [????????????](#-????????????)
- [??????????](#-??????????)
- [???????????? API](#-????????????-api)
- [?????????? ?????????](#-??????????-?????????)
- [_____ ? ??????](#-_____-?-??????)
- [????????](#-????????)

---

## ? ??????????

### ?????????????????

- **Kubernetes**: v1.24 ??? ????
- **KubeVirt**: v1.0 ??? ???? (? feature gate HostDevices ??? ????????? GPU)
- **CDI** (Containerized Data Importer): ??? ?????????? ???????? ?????? VM
- **Ceph**: RBD ??? ???????? ?????????, CephFS ??? ?????? ?????????? (???????????)

### ?????? ?????????

- `your-SC` - ??????? ????????? ??? ?????? VM
- `your-SC` - ???????? ????????? ??? ???? ?????? ?????????? (???????????)

_??????????: _____ ??????? ????????? _____ ????????? ? ????????????._

### ???????????? ??????????
```

- **????????????? Juniper SRX**: ??? ????????? ??????????? NAT (??????? SSH-??????)
- **Multus CNI**: ??? ??????????? ????????? ???? (????????? ? v0.0.1)
- **Metrics Server**: ??? ??????????? ???????????? ????????

?? ??????????

????? ??????????

```
??????????????????????????????????????????????????????????????????????
?              Ingress / Load Balancer                        ?
?                (kvm.example.com)                            ?
??????????????????????????????????????????????????????????????????????
                          ?
                          ?
??????????????????????????????????????????????????????????????????????
?             Pod KubeVirt API Manager                        ?
?  ???????????????  ???????????????  ???????????????      ?
?  ?   Nginx      ?  ?    Flask     ?  ?   Juniper    ?      ?
?  ?  (Static)    ????  (Backend)   ????   Sidecar    ?      ?
?  ???????????????  ???????????????  ???????????????      ?
??????????????????????????????????????????????????????????????????????
                              ?
                  ??????????????????????
                  ?                    ?
                  ?                    ?
        ??????????????????   ??????????????????
        ?   Kubernetes   ?   ?   KubeVirt     ?
        ?   API Server   ?   ?      CRs       ?
        ??????????????????   ??????????????????
                  ?                    ?
                  ??????????????????????
                            ?
                            ?
                ??????????????????????????
                ?  ?????????? ??????   ?
                ?   (KubeVirt Pods)     ?
                ??????????????????????????
```

???????? ???????????

Frontend

- **Vanilla JavaScript**: ??? ??????????? ?? ??????????, ??????????? ? ???????
- **CSS Variables**: ????????? ??? ? ??????/??????? ???????
- **????????? ???????????**: ????? core-?????? ? ?????? ??? ????????? ???????
- **?????????? ? ???????? ???????**: ????????????? ? SSE (Server-Sent Events) ??? ?????????? ??????

Backend

- **Flask**: Python ???-?????????
- **Kubernetes Python Client**: ?????? ????????????? ? API
- **Jinja2 Templates**: ????????? ?? ??????? ???????
- **?????????? ????????**: ????????????? ???????????? ? ???????? namespace

????

- **Pod Network (?? ?????????)**: ????? masquerade ??? ??????????? VM
- **NodePort Services**: SSH-?????? ????? ??????? Kubernetes
- **?????????? Juniper**: ?????????? NAT ??? ???????????? ??????? IP

? ??????? ?????

????? ?????????? (??? Kubernetes)

```
???????? ??? ????????? frontend ? ??????????? UI:

```bash
# ????????? Docker Compose
git clone https://github.com/yourusername/kubevirt-api-manager.git
cd kubevirt-api-manager
docker-compose up -d

# ?????? ? ?????????
open http://localhost:8083
# ?????: example / your-secret-password (?? ?????????, ???????????? ????? DEMO_USERNAME/DEMO_PASSWORD)
```

????????? ?????????? ? **stub-??????** ? ???? VM:

- `demo-gpu-vm` - ????????? VM ? NVIDIA A100 GPU
- `data-import-vm` - VM ? ???????? ????????
- `analytics-vm` - ???????????? VM

????????? ??. docs/DEVELOPMENT.md.

? ?????????

??????????????? ?????????

1. **???????? ???? KubeVirt**

   ```bash
   kubectl label nodes <???-????> node-role.kubernetes.io/kubevirt=
   ```

2. **????????? ??????? GPU** (???? ??????????? GPU)

   ????????????? KubeVirt CR:

   ```bash
   kubectl edit kubevirt kubevirt -n kubevirt
   ```

   ????????:

   ```yaml
   spec:
     configuration:
       developerConfiguration:
         featureGates:
           - HostDevices
       permittedHostDevices:
         pciHostDevices:
           - pciVendorSelector: "10DE:1B06" # NVIDIA GTX 1080 Ti
             resourceName: "nvidia.com/1080ti"
           - pciVendorSelector: "10DE:10EF" # HDMI Audio
             resourceName: "nvidia.com/1080ti-audio"
   ```

3. **????????? ????????? CDI**
   ```bash
   kubectl get pods -n cdi
   ```

???????????? ? Kubernetes

```bash
# 1. ?????????? ??????????
git clone https://github.com/yourusername/kubevirt-api-manager.git
cd kubevirt-api-manager

# 2. ??????? namespace
kubectl create namespace kvm
```
```

```

# 3. ????????? secrets
cp kubernetes_example/secrets-example.yaml kubernetes_example/secrets.yaml
# ?????????????? secrets.yaml ?????? ?????????
kubectl apply -f kubernetes_example/secrets.yaml

# 4. ?????????? ?????????
kubectl apply -f kubernetes_example/bundled.yaml

# 5. ????????? ????????????
kubectl get pods -n kvm
kubectl get ingress -n kvm
```

?????? ? ?????????

```bash
# ???????? ????? ingress
kubectl get ingress -n kvm

# ???????? ? /etc/hosts ???? ??????????? ????????? ???????
echo "192.168.1.100  kvm.example.com" | sudo tee -a /etc/hosts

# ??????? ? ????????
open https://kvm.example.com
```

?? ???????????

?????????? ?????????

| ?????????? | ?? ????????? | ???????? ...
| --- | --- | --- ...
| `FRONTEND_STUB_MODE` | `false` | ???????? stub-????? ??? ?????????? ...
| `DEBUG` | `false` | ????? ??????? Flask ...
| `SECRET_KEY` | (??????????) | ????????? ???? Flask ??? ?????? ...
| `K8S_NAMESPACE_PREFIX` | `kv` | ??????? ??? ??????????????? namespace ...
| `DEFAULT_GPU_RESOURCE` | `nvidia.com/1080ti` | ??? ??????? GPU ?? ????????? ...
| `STORAGE_CLASS_NAME` | `your-SC` | ????? ????????? ??? ?????? VM ...
| `VM_STORAGE_NODE_SELECTOR` | (?????) | JSON nodeSelector ??? ????? CDI. ??. [VM-SCHEDULING-CSI](docs/V...
| `VM_RUN_STRATEGY` | `Manual` | ????????? VM ??? shutdown/reboot. ??. [VM_RUN_STRATEGY](docs/VM...
| `MAX_CPU_CORES` | `16` | ??????????? ????????? ???? CPU ?? VM ...
| `MAX_MEMORY_GB` | `64` | ??????????? ?????? ?? VM (??) ...
| `MAX_STORAGE_GB` | `500` | ??????????? ????????? ?? VM (??) ...
| `MAX_GPU_COUNT` | `2` | ??????????? ????????? GPU ?? VM ...
| `DEMO_USERNAME` | `example` | ??? ??????????? ??? ????? ...
| `DEMO_PASSWORD` | (??????????) | ?????? ??? ????? ...
| `LLM_API_KEY` | (??????????) | ??? ??????? LLM-???? ...

??. `app/config.py` ??? ???? ?????. ??????? ????????? ????? Kubernetes Secrets.

?????????? Juniper (??????????)

??? ???????? SSH-??????? ????? ?????????? NAT:

```yaml
# ? kubernetes_example/juniper-sidecar-cm.yaml (??? ? bundled.yaml ? ConfigMap juniper-sidecar-code)
data:
  JUNIPER_HOST: "router.example.com"
  JUNIPER_USER: "automation"
  JUNIPER_PUBLIC_IP: "203.0.113.10"
```

SSH-????? ?????? ???? ??????????? ??? secrets:

```bash
kubectl create secret generic juniper-ssh-key \
  --from-file=id_rsa=./juniper_key \
  --from-file=id_rsa.pub=./juniper_key.pub \
```

```
  -n kvm
```

? ??????????

????????? ????????? ??????????

??????? 1: Docker Compose (????????????)

```bash
# ????????? ???????
docker-compose up -d

# ???????? ?????
docker-compose logs -f kubevirt-api

# ??????????? ????? ????????? ????
docker-compose up -d --build

# ?????????? ???????
docker-compose down
```

??????? 2: ???????? Python

```bash
# ??????? ??????????? ?????????
python3 -m venv venv
source venv/bin/activate  # ?? Windows: venv\Scripts\activate

# ?????????? ??????????
pip install -r requirements.txt

# ?????????? ?????????? ?????????
export FRONTEND_STUB_MODE=true
export DEBUG=true
export SECRET_KEY=dev-secret-key-change-in-production
export DEMO_USERNAME=admin
export DEMO_PASSWORD=your-secret-password

# ????????? Flask
python app/main.py
```

?????? ?? ?????? `http://localhost:8000`

????????? ????

```
kubevirt-api-manager/
??? app/
?   ??? main.py                 # ????? ????? Flask
?   ??? config.py               # ??? ??????????? (env vars, ??????, LLM)
?   ??? routes/                 # API endpoints
?   ?   ??? __init__.py         # Dashboard, ?????, ?????
?   ?   ??? vm_routes.py        # CRUD ???????? ? VM
?   ?   ??? storage_routes.py   # ?????????? ??????????
?   ?   ??? vm_details_routes.py # ?????? VM, LLM, ???????
?   ??? utils/                  # ??????-??????
?   ?   ??? k8s_utils.py        # ????????????? ? Kubernetes
?   ?   ??? vm_utils.py         # ????????? ?????????? Linux VM
?   ?   ??? vm_manifest_common.py # ????? ?????????? ??????????
?   ?   ??? service_utils.py    # NodePort ??????? ? ?????
?   ?   ??? network_policy_utils.py # ??????? ????????
?   ?   ??? juniper_utils.py    # ?????????? Juniper
?   ??? static/
?   ?   ??? js/
?   ?   ?   ??? core/           # ????? ??????
?   ?   ?   ??? pages/          # ?????? ??? ????????? ???????
```

```
?   ?   ??? css/
?   ??? templates/              # ??????? Jinja2
??? kubernetes_example/        # ????????? K8s (???????)
??? docs/                       # ???????????
??? Dockerfile
??? docker-compose.yml
??? requirements.txt
```

???????? ????????

1. **Frontend**: ???????????? ????? ? `app/static/` ? `app/templates/`
2. **Backend**: ???????????? ????? ? `app/routes/` ? `app/utils/`
3. **???????????? ? stub-??????**: ??????????? Docker Compose
4. **???????????? ? K8s**: ????????? ? dev-????????

??. docs/DEVELOPMENT.md ??? ????????? ????????? ? [.github/copilot-instructions.md](.github/c...

? ??????????? API

Endpoints ????????? VM

??????? VM

```http
POST /<username>/create-vm
Content-Type: application/x-www-form-urlencoded

name=my-ubuntu-vm&cpu=4&memory=16&storage=100&image=...
```

???????? VM

```http
POST /<username>/vm/<vm_name>/start
```

?????:

```json
{
  "success": true,
  "message": "VM started successfully"
}
```

????????? VM

```http
POST /<username>/vm/<vm_name>/stop
```

???????????? VM

```http
POST /<username>/vm/<vm_name>/restart
```

??????? VM

```http
POST /<username>/vm/<vm_name>/delete
```

????????? ??????????? ????? VM

```http
GET /<username>/check-vm-name/<vm_name>
```

?????:

```json
{
  "available": true,
  "exists": false
}
```

Endpoints SSH-????????

??????? SSH-??????

```http
POST /<username>/vm/<vm_name>/create-ssh-service
```

?????:

```json
{
  "success": true,
  "port": 30123,
  "public_ip": "203.0.113.10",
  "ssh_command": "ssh user@203.0.113.10 -p 30123"
}
```

??????? SSH-??????

```http
POST /<username>/vm/<vm_name>/delete-ssh-service
```

Endpoints ??????

???????? ??????? VM

```http
GET /<username>/vm/<vm_name>/metrics
```

?????:

```json
{
  "success": true,
  "metrics": {
    "cpu_usage": 125, // millicores
    "memory_usage": 2048, // MiB
    "timestamp": "2026-01-27T12:34:56Z"
  }
}
```

? ?????????? ?????????

???????????????? ????????

?????? ???????? VM

????????: ???????? VM ?????????? ?????? "Failed to create DataVolume"

????????:

```bash
# ????????? ???????? CDI
kubectl get pods -n cdi
```

```

# ????????? ????? ?????????
kubectl get storageclass

# ????????? ??????? PVC
kubectl describe pvc <vm-name>-rootdisk -n <namespace>
```

GPU ?? ?????????

... [85 more lines truncated]

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/auth/__init__.py (51.0 B)

```
"""Auth module: dependencies, session helpers."""

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/auth/dependencies.py (3.0 KB)

```python
"""Auth dependencies for FastAPI: session user, namespace, role-based access."""

from __future__ import annotations

from dataclasses import dataclass
from typing import Annotated

from fastapi import Depends, HTTPException, Request
from fastapi.responses import RedirectResponse

from app.config import settings


def _get_db(request: Request):
    """Get database from app state. Returns None if DATABASE_URL not set."""
    return getattr(request.app.state, "db", None)


def get_user_namespace(username: str, user_id: int) -> str:
    """Build Kubernetes namespace for user: {prefix}-{username_safe}-{user_id}.
    Username underscores are replaced with hyphens (K8s namespace allows only a-z, 0-9, -).
    """
    prefix = settings.K8S_NAMESPACE_PREFIX or "kv"
    ns_safe = username.lower().replace("_", "-")
    namespace = f"{prefix}-{ns_safe}-{user_id}"
    if not namespace[0].isalpha():
        raise ValueError(
            "Namespace must start with a letter. Check K8S_NAMESPACE_PREFIX and username."
        )
    return namespace


@dataclass
class SessionUser:
    """Current user from session."""

    user_id: int
    username: str
    role: str
    namespace: str

    @property
    def is_admin(self) -> bool:
        return self.role == "admin"


async def get_current_user(request: Request) -> SessionUser:
    """Extract authenticated user from session."""
    session = request.session
    username = session.get("username")
    user_id = session.get("user_id")
    role = session.get("role", "user")

    if not username or user_id is None:
        if "/api/" in request.url.path or "/vms" in request.url.path:
            raise HTTPException(status_code=401, detail="Not authenticated")
        return RedirectResponse(url="/login", status_code=303)

    try:
        namespace = session.get("namespace") or get_user_namespace(username, user_id)
    except ValueError as e:
        raise HTTPException(status_code=400, detail=str(e))
    if namespace and not namespace[0].isalpha():
        raise HTTPException(
            status_code=400,
```

```
            detail="Namespace must start with a letter. Re-login or fix K8S_NAMESPACE_PREFIX.",
        )
    return SessionUser(user_id=user_id, username=username, role=role, namespace=namespace)


async def get_current_namespace(
    user: Annotated[SessionUser, Depends(get_current_user)],
) -> str:
    """Current user's Kubernetes namespace."""
    return user.namespace


async def require_owner(
    request: Request,
    username: str,
    user: Annotated[SessionUser, Depends(get_current_user)],
) -> SessionUser:
    """Require that path username matches session user."""
    if user.username == username:
        return user
    raise HTTPException(status_code=403, detail="Access denied")


async def require_admin(
    user: Annotated[SessionUser, Depends(get_current_user)],
) -> SessionUser:
    """Require admin role."""
    if not user.is_admin:
        raise HTTPException(status_code=403, detail="Admin access required")
    return user

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/bootstrap.py (1.4 KB)

```python
"""One-shot bootstrap: ensure DB schema and create the initial admin user."""

import asyncio
import logging
import sys

import bcrypt

from app.config import settings
from app.database import create_db

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(name)s: %(message)s",
    handlers=[logging.StreamHandler(sys.stdout)],
)
logger = logging.getLogger(__name__)


async def bootstrap() -> None:
    if not settings.DATABASE_URL:
        logger.info("DATABASE_URL not set ? skipping bootstrap")
        return

    async with create_db(settings.DATABASE_URL) as db:
        if not settings.ADMIN_USERNAME or not settings.ADMIN_PASSWORD:
            logger.info("ADMIN_USERNAME/ADMIN_PASSWORD not set ? skipping admin creation")
            return

        existing = await db.users.get_by_username(settings.ADMIN_USERNAME)
        if existing:
            logger.info("Admin user '%s' already exists (id=%s)", existing.username, existing.id)
            return

        pw_hash = bcrypt.hashpw(
            settings.ADMIN_PASSWORD.encode("utf-8"),
            bcrypt.gensalt(),
        ).decode("utf-8")

        admin = await db.users.create_user(
            settings.ADMIN_USERNAME,
            pw_hash,
            role="admin",
        )
        logger.info("Created bootstrap admin: %s (id=%s)", admin.username, admin.id)


if __name__ == "__main__":
    asyncio.run(bootstrap())

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/config.py (15.6 KB)

```python
"""Configuration file for kubevirt-api-manager.

All sensitive data and parameters should be set via environment variables.
"""

from __future__ import annotations

import json
import logging
from decimal import Decimal
from typing import Dict, List

from pydantic import BaseModel, ConfigDict, Field, field_validator, model_validator, ValidationInfo
from pydantic_settings import BaseSettings, SettingsConfigDict

logger = logging.getLogger(__name__)


# ---------------------------------------------------------------------------
# VM Pricing (used by the billing loop)
# ---------------------------------------------------------------------------

class VMPricing(BaseModel):
    model_config = ConfigDict(frozen=True)

    cpu_per_minute: Decimal = Decimal("0.001")
    memory_gb_per_minute: Decimal = Decimal("0.0005")
    gpu_default_per_minute: Decimal = Decimal("0.01")
    gpu_rates: dict[str, Decimal] = {}

    def cost_per_minute(
        self,
        cpu: int,
        memory_gb: float,
        gpu_count: int,
        gpu_model: str | None,
    ) -> Decimal:
        cost = Decimal(str(cpu)) * self.cpu_per_minute
        cost += Decimal(str(memory_gb)) * self.memory_gb_per_minute
        if gpu_count > 0 and gpu_model:
            rate = self.gpu_rates.get(gpu_model, self.gpu_default_per_minute)
            cost += Decimal(str(gpu_count)) * rate
        return cost


class Settings(BaseSettings):
    model_config = SettingsConfigDict(
        env_file=".env",
        env_file_encoding="utf-8",
        extra="ignore",
    )

    # ============================================
    # Flask Settings
    # ============================================
    SECRET_KEY: str = "qpjxiVAiF02qcNWpTf3RYOhOKpEt1h1SSYVCc7rPbK9pUXeLcR7q0xybMaEgxyD92rIp5sWNHNVZFU05pmSX5w"
    DEBUG: bool = False

    # ============================================
    # Development/Stub Mode Settings
    # ============================================
    FRONTEND_STUB_MODE: bool = False
    STUB_NODE_IP: str = "192.168.1.100"

    # ============================================
```

```
    # Authentication & Authorization
    # ============================================
    DEMO_USERNAME: str = "nlp"
    DEMO_PASSWORD: str = "kubevirt123"
    DEMO_USER_ID: str = "001"
    SESSION_TIMEOUT: int = 3600

    # Bootstrap admin (kept for compatibility with app/main.py bootstrap flow)
    ADMIN_USERNAME: str = "adm"
    ADMIN_PASSWORD: str = "psw"

    # ============================================
    # Database
    # ============================================
    DATABASE_URL: str = (
        "postgresql://user:pass@postgres.kvm-multus.svc.k8s.naint.tech:5432/db"
    )

    # ============================================
    # Kubernetes Settings
    # ============================================
    K8S_NAMESPACE_PREFIX: str = "kv"
    STORAGE_CLASS_NAME: str = "csi-rbd-k8s"

    # ============================================
    # LLM API Configuration
    # ============================================
    LLM_API_URL: str = ""
    # Deprecated: kept for backward compatibility only.
    LLM_API_KEY: str = ""
    LLM_MODEL: str = "qwen/qwen-coder-3b-instruct"

    # --- LLM per-token (llm-service): SK token from session ---
    LLM_SERVICE_URL: str = "llm-service.llm-backend-service.svc.k8s.naint.tech:8001"
    # Shared service token used by chat (no per-user billing in chat UI)
    LLM_CHAT_SHARED_TOKEN: str = ""
    # Max number of stored API keys per user
    LLM_USER_KEYS_MAX: int = 10
    # Symmetric key for encrypting stored API keys in DB
    LLM_KEYS_ENCRYPTION_KEY: str = ""

    # --- LLM Auth Service (balance bridge) ---
    LLM_AUTH_SERVICE_URL: str = "auth-service.llm-backend-service.svc.k8s.naint.tech:8000"
    LLM_AUTH_ADMIN_TOKEN: str = "cudo-BUKgZwMVpG3XfJT5zg7boQ"

    @model_validator(mode="after")
    def _llm_token_alias_fallback(self) -> "Settings":
        # Keep old env var working, but runtime should use one token variable.
        if not self.LLM_CHAT_SHARED_TOKEN and self.LLM_API_KEY:
            self.LLM_CHAT_SHARED_TOKEN = self.LLM_API_KEY
        if not self.LLM_API_KEY and self.LLM_CHAT_SHARED_TOKEN:
            self.LLM_API_KEY = self.LLM_CHAT_SHARED_TOKEN
        return self

    # ============================================
    # Billing
    # ============================================
    BILLING_ENABLED: bool = True
    BILLING_INTERVAL_SECONDS: int = 60
    BILLING_CPU_PER_MINUTE: str = "0.001"
    BILLING_MEMORY_GB_PER_MINUTE: str = "0.0005"
    BILLING_GPU_DEFAULT_PER_MINUTE: str = "0.01"
    BILLING_GPU_RATES: str = ""  # JSON: {"nvidia.com/A100": "0.05"}

    # ============================================
    # GPU Configuration
    # ============================================
    DEFAULT_GPU_RESOURCE: str = "nvidia.com/1080ti"
    GPU_RESOURCES_WITH_AUDIO: List[str] = []
    NVIDIA_DRIVER_VERSIONS: List[str] = []
    DEFAULT_NVIDIA_DRIVER: str = "535"
```

```
    NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER: List[str] = []
    NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY: str = (
        "vm.kubevirt.io/nvidia-custom-driver-token"
    )
    NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH: str = "nvidia-custom-driver-installed"
    APP_INTERNAL_BASE_URL: str = ""

    @field_validator(
        "GPU_RESOURCES_WITH_AUDIO",
        "NVIDIA_DRIVER_VERSIONS",
        "NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER",
        mode="before",
    )
    @classmethod
    def _split_csv(cls, v):
        if v is None or v == "":
            return []
        if isinstance(v, list):
            return [str(x).strip() for x in v if str(x).strip()]
        if isinstance(v, str):
            v = v.strip()
            if not v:
                return []
            if (v.startswith("[") and v.endswith("]")) or (v.startswith("(") and v.endswith(")")):
                try:
                    parsed = json.loads(v.replace("'", '"'))
                    return [str(x).strip() for x in parsed if str(x).strip()]
                except json.JSONDecodeError:
                    pass
            return [x.strip() for x in v.split(",") if x.strip()]
        return v

    # VM Resource Limits
    MAX_CPU_CORES: int = 16
    MAX_MEMORY_GB: int = 64
    MAX_STORAGE_GB: int = 500
    MAX_GPU_COUNT: int = 2

    # ==========================================
    # VM Defaults (cloud-init)
    # ==========================================
    VM_RUN_STRATEGY: str = "Manual"
    DEFAULT_VM_SETTINGS: Dict = Field(
        default_factory=lambda: {
            "username": "ubuntu",
            "package_update": False,
            "package_upgrade": False,
            "ssh_pwauth": False,
        }
    )
    DEFAULT_PACKAGES: List[str] = [
        "openssh-server",
        "qemu-guest-agent",
        "curl",
        "wget",
        "net-tools",
    ]

    @field_validator("DEFAULT_VM_SETTINGS", mode="before")
    @classmethod
    def _parse_vm_settings(cls, v):
        if isinstance(v, str):
            try:
                parsed = json.loads(v)
                return parsed if isinstance(parsed, dict) else v
            except json.JSONDecodeError:
                return v
        return v

    # Whitelist of allowed image domains for VM creation
    ALLOWED_IMAGE_DOMAINS: List[str] = [
```

```
        "cloud-images.ubuntu.com",
        "download.fedoraproject.org",
        "cloud.centos.org",
        "download.rockylinux.org",
        "cdimage.debian.org",
        "cloud.debian.org",
        "download.opensuse.org",
    ]

    @field_validator("ALLOWED_IMAGE_DOMAINS", mode="before")
    @classmethod
    def _split_domains(cls, v):
        if isinstance(v, str):
            return [d.strip() for d in v.split(",") if d.strip()]
        return v

    # ============================================
    # Network Policy Configuration
    # ============================================
    POD_NETWORK_CIDR: str = "192.168.192.0/18"
    SERVICE_NETWORK_CIDR: str = "10.96.0.0/18"
    SERVER_NETWORK_CIDRS: List[str] = ["192.168.16.0/24"]

    @field_validator("SERVER_NETWORK_CIDRS", mode="before")
    @classmethod
    def _split_cidrs(cls, v):
        if isinstance(v, str):
            return [c.strip() for c in v.split(",") if c.strip()]
        return v

    # ============================================
    # Juniper Router Configuration
    # ============================================
    JUNIPER_HOST: str = "192.168.16.1"
    JUNIPER_USER: str = "script"
    JUNIPER_SIDECAR_URL: str = "http://127.0.0.1:5001"
    JUNIPER_EXTERNAL_IP_POOL: List[str] = [
        "31.28.3.192",
        "31.28.3.193",
        "31.28.3.194",
        "31.28.3.195",
    ]

    @field_validator("JUNIPER_EXTERNAL_IP_POOL", mode="before")
    @classmethod
    def _split_ips(cls, v):
        if isinstance(v, str):
            return [ip.strip() for ip in v.split(",") if ip.strip()]
        return v

    @model_validator(mode="after")
    def _stub_defaults(self) -> "Settings":
        if self.FRONTEND_STUB_MODE and not self.JUNIPER_EXTERNAL_IP_POOL:
            self.JUNIPER_EXTERNAL_IP_POOL = ["192.168.100.1"]
        return self

    # ============================================
    # Windows VM Configuration
    # ============================================
    WINDOWS_ISO_DATAVOLUME: str = "windows-10-iso"
    WINDOWS_ISO_NAMESPACE: str = "kvm-multus"
    WINDOWS_GOLDEN_IMAGE_NAME: str = "windows-10-golden-image"
    WINDOWS_GOLDEN_IMAGE_NAMESPACE: str = "kvm-multus"
    WINDOWS_HYPERV_VENDORID: str = "1234567890ab"
    WINDOWS_ISO_STORAGE_GB: int = 7
    WINDOWS_DEFAULT_CPU: int = 4
    WINDOWS_DEFAULT_MEMORY: int = 8
    WINDOWS_DEFAULT_STORAGE: int = 60

    @field_validator(
        "WINDOWS_ISO_STORAGE_GB",
```

```
        "WINDOWS_DEFAULT_CPU",
        "WINDOWS_DEFAULT_MEMORY",
        "WINDOWS_DEFAULT_STORAGE",
        mode="before",
    )
    @classmethod
    def _empty_int_to_default(cls, v: object, info: ValidationInfo) -> object:
        if v == "" or v is None:
            defaults = {"WINDOWS_ISO_STORAGE_GB": 7, "WINDOWS_DEFAULT_CPU": 4, "WINDOWS_DEFAULT_MEMORY": 8, "WINDOWS_...
            return defaults.get(info.field_name, v)
        return v
    WINDOWS_MACHINE_TYPE: str = "q35"
    WINDOWS_DEFAULT_USERNAME: str = "Administrator"
    WINDOWS_HYPERV_FEATURES: Dict = Field(
        default_factory=lambda: {
            "relaxed": {"enabled": True},
            "vapic": {"enabled": True},
            "spinlocks": {"enabled": True, "spinlocks": 8191},
            "vpindex": {"enabled": True},
            "runtime": {"enabled": True},
            "synic": {"enabled": True},
            "stimer": {"enabled": True},
            "reset": {"enabled": True},
            "frequencies": {"enabled": True},
            "reenlightenment": {"enabled": True},
            "tlbflush": {"enabled": True},
            "ipi": {"enabled": True},
        }
    )
    VIRTIO_DRIVERS_IMAGE: str = "quay.io/kubevirt/virtio-container-disk:latest"


settings = Settings()

if settings.SECRET_KEY == "dev-secret-key-change-me":
    logger.warning(
        "Using default SECRET_KEY! Set SECRET_KEY environment variable in production!"
    )

if settings.DEMO_PASSWORD == "kubevirt123":
    logger.warning(
        "Using default weak password! Set DEMO_PASSWORD environment variable!"
    )

if not settings.LLM_CHAT_SHARED_TOKEN and not settings.FRONTEND_STUB_MODE:
    logger.warning(
        "LLM_CHAT_SHARED_TOKEN not set! Set LLM_CHAT_SHARED_TOKEN for LLM features."
    )


# Backward-compatible module-level exports for existing imports.
SECRET_KEY = settings.SECRET_KEY
DEBUG = settings.DEBUG
FRONTEND_STUB_MODE = settings.FRONTEND_STUB_MODE
STUB_NODE_IP = settings.STUB_NODE_IP
DEMO_USERNAME = settings.DEMO_USERNAME
DEMO_PASSWORD = settings.DEMO_PASSWORD
DEMO_USER_ID = settings.DEMO_USER_ID
SESSION_TIMEOUT = settings.SESSION_TIMEOUT
ADMIN_USERNAME = settings.ADMIN_USERNAME
ADMIN_PASSWORD = settings.ADMIN_PASSWORD
DATABASE_URL = settings.DATABASE_URL
K8S_NAMESPACE_PREFIX = settings.K8S_NAMESPACE_PREFIX
STORAGE_CLASS_NAME = settings.STORAGE_CLASS_NAME
LLM_API_URL = settings.LLM_API_URL
LLM_API_KEY = settings.LLM_CHAT_SHARED_TOKEN
LLM_MODEL = settings.LLM_MODEL
LLM_SERVICE_URL = settings.LLM_SERVICE_URL
LLM_CHAT_SHARED_TOKEN = settings.LLM_CHAT_SHARED_TOKEN
LLM_USER_KEYS_MAX = settings.LLM_USER_KEYS_MAX
LLM_KEYS_ENCRYPTION_KEY = settings.LLM_KEYS_ENCRYPTION_KEY
```

```
LLM_AUTH_SERVICE_URL = settings.LLM_AUTH_SERVICE_URL
LLM_AUTH_ADMIN_TOKEN = settings.LLM_AUTH_ADMIN_TOKEN
DEFAULT_GPU_RESOURCE = settings.DEFAULT_GPU_RESOURCE
GPU_RESOURCES_WITH_AUDIO = settings.GPU_RESOURCES_WITH_AUDIO
NVIDIA_DRIVER_VERSIONS = settings.NVIDIA_DRIVER_VERSIONS
DEFAULT_NVIDIA_DRIVER = settings.DEFAULT_NVIDIA_DRIVER
NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER = settings.NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER
NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY = (
    settings.NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY
)
NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH = settings.NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH
APP_INTERNAL_BASE_URL = settings.APP_INTERNAL_BASE_URL
MAX_CPU_CORES = settings.MAX_CPU_CORES
MAX_MEMORY_GB = settings.MAX_MEMORY_GB
MAX_STORAGE_GB = settings.MAX_STORAGE_GB
MAX_GPU_COUNT = settings.MAX_GPU_COUNT
VM_RUN_STRATEGY = settings.VM_RUN_STRATEGY
DEFAULT_PACKAGES = settings.DEFAULT_PACKAGES
DEFAULT_VM_SETTINGS = settings.DEFAULT_VM_SETTINGS
WINDOWS_ISO_DATAVOLUME = settings.WINDOWS_ISO_DATAVOLUME
WINDOWS_ISO_NAMESPACE = settings.WINDOWS_ISO_NAMESPACE
WINDOWS_GOLDEN_IMAGE_NAME = settings.WINDOWS_GOLDEN_IMAGE_NAME
WINDOWS_GOLDEN_IMAGE_NAMESPACE = settings.WINDOWS_GOLDEN_IMAGE_NAMESPACE
WINDOWS_HYPERV_VENDORID = settings.WINDOWS_HYPERV_VENDORID
WINDOWS_ISO_STORAGE_GB = settings.WINDOWS_ISO_STORAGE_GB
WINDOWS_DEFAULT_CPU = settings.WINDOWS_DEFAULT_CPU
WINDOWS_DEFAULT_MEMORY = settings.WINDOWS_DEFAULT_MEMORY
WINDOWS_DEFAULT_STORAGE = settings.WINDOWS_DEFAULT_STORAGE
WINDOWS_MACHINE_TYPE = settings.WINDOWS_MACHINE_TYPE
WINDOWS_DEFAULT_USERNAME = settings.WINDOWS_DEFAULT_USERNAME
WINDOWS_HYPERV_FEATURES = settings.WINDOWS_HYPERV_FEATURES
VIRTIO_DRIVERS_IMAGE = settings.VIRTIO_DRIVERS_IMAGE
ALLOWED_IMAGE_DOMAINS = settings.ALLOWED_IMAGE_DOMAINS
POD_NETWORK_CIDR = settings.POD_NETWORK_CIDR
SERVICE_NETWORK_CIDR = settings.SERVICE_NETWORK_CIDR
SERVER_NETWORK_CIDRS = settings.SERVER_NETWORK_CIDRS
JUNIPER_HOST = settings.JUNIPER_HOST
JUNIPER_USER = settings.JUNIPER_USER
JUNIPER_SIDECAR_URL = settings.JUNIPER_SIDECAR_URL
JUNIPER_EXTERNAL_IP_POOL = settings.JUNIPER_EXTERNAL_IP_POOL
JUNIPER_EXTERNAL_IP = (
    settings.JUNIPER_EXTERNAL_IP_POOL[0] if settings.JUNIPER_EXTERNAL_IP_POOL else ""
)


BILLING_ENABLED = settings.BILLING_ENABLED
BILLING_INTERVAL_SECONDS = settings.BILLING_INTERVAL_SECONDS


def build_vm_pricing(s: Settings | None = None) -> VMPricing:
    """Construct a VMPricing instance from Settings env-overridable fields."""
    s = s or settings
    gpu_rates: dict[str, Decimal] = {}
    if s.BILLING_GPU_RATES:
        try:
            raw = json.loads(s.BILLING_GPU_RATES)
            gpu_rates = {k: Decimal(str(v)) for k, v in raw.items()}
        except (json.JSONDecodeError, Exception) as exc:
            logger.warning("Failed to parse BILLING_GPU_RATES: %s", exc)
    return VMPricing(
        cpu_per_minute=Decimal(s.BILLING_CPU_PER_MINUTE),
        memory_gb_per_minute=Decimal(s.BILLING_MEMORY_GB_PER_MINUTE),
        gpu_default_per_minute=Decimal(s.BILLING_GPU_DEFAULT_PER_MINUTE),
        gpu_rates=gpu_rates,
    )


def is_valid_user_namespace(namespace: str | None, username: str) -> bool:
    """Check that namespace matches expected pattern for username.
    Username underscores are normalized to hyphens (K8s namespace: a-z, 0-9, - only).
    """
```

```
    if not namespace or not username:
        return False
    if not namespace[0].isalpha():
        return False
    if "_" in namespace:
        return False
    prefix = settings.K8S_NAMESPACE_PREFIX or "kv"
    ns_safe = username.lower().replace("_", "-")
    return namespace.startswith(f"{prefix}-{ns_safe}-")

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/config.py.example (9.6 KB)

```
"""
Production configuration for viirtuoz-vps.

Copy to config.py: cp app/config.py.example app/config.py
Sensitive values via env vars or .env (see env.example)
For stub mode: run-stub-mode.sh overrides with FRONTEND_STUB_MODE=true etc.
"""

from __future__ import annotations

import logging
from typing import Dict, List

from pydantic import Field, field_validator, model_validator
from pydantic_settings import BaseSettings, SettingsConfigDict

logger = logging.getLogger(__name__)


class Settings(BaseSettings):
    model_config = SettingsConfigDict(
        env_file=".env",
        env_file_encoding="utf-8",
        extra="ignore",
    )

    ADMIN_USERNAME: str = "admin"
    ADMIN_PASSWORD: str = ""

    # --- Core ---
    SECRET_KEY: str = ""
    DEBUG: bool = False

    # --- Stub / dev ---
    FRONTEND_STUB_MODE: bool = False
    STUB_NODE_IP: str = ""

    # --- Auth ---
    DEMO_USERNAME: str = ""
    DEMO_PASSWORD: str = ""
    DEMO_USER_ID: str = "1"
    SESSION_TIMEOUT: int = 3600

    # --- Database (production) ---
    DATABASE_URL: str = "postgresql://user:pass@postgres.kvm-multus.svc.k8s.naint.tech:5432/db"

    # --- Kubernetes ---
    K8S_NAMESPACE_PREFIX: str = "kv"
    STORAGE_CLASS_NAME: str = ""

    # --- LLM API ---
    LLM_API_URL: str = ""
    # Deprecated: falls back to LLM_CHAT_SHARED_TOKEN
    LLM_API_KEY: str = ""
    LLM_MODEL: str = ""

    # --- LLM per-token (llm-service): SK token from session ---
    LLM_SERVICE_URL: str = ""
    # Shared service token used by chat (no per-user billing in chat UI)
    LLM_CHAT_SHARED_TOKEN: str = ""
    # Max number of stored API keys per user
    LLM_USER_KEYS_MAX: int = 10
    # Symmetric key for encrypting stored API keys in DB
    LLM_KEYS_ENCRYPTION_KEY: str = ""

```

```
    # --- LLM Auth Service (balance bridge) ---
    LLM_AUTH_SERVICE_URL: str = ""
    LLM_AUTH_ADMIN_TOKEN: str = ""

    # --- Speech API ---
    SPEECH_API_URL: str = ""
    SPEECH_API_KEY: str = ""
    SPEECH_TTS_MODEL: str = ""
    SPEECH_TTS_OUTPUT_FORMAT: str = ""

    @model_validator(mode="after")
    def _speech_key_fallback(self) -> "Settings":
        if not self.SPEECH_API_KEY:
            self.SPEECH_API_KEY = self.LLM_API_KEY
        return self

    @model_validator(mode="after")
    def _stub_defaults(self) -> "Settings":
        if self.FRONTEND_STUB_MODE and not self.JUNIPER_EXTERNAL_IP_POOL:
            self.JUNIPER_EXTERNAL_IP_POOL = ["192.168.100.1"]
        return self

    # --- GPU ---
    DEFAULT_GPU_RESOURCE: str = "nvidia.com/1080ti"
    GPU_RESOURCES_WITH_AUDIO: List[str] = []
    NVIDIA_DRIVER_VERSIONS: List[str] = []
    DEFAULT_NVIDIA_DRIVER: str = "535"
    NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER: List[str] = []
    NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY: str = (
        "vm.kubevirt.io/nvidia-custom-driver-token"
    )
    NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH: str = "nvidia-custom-driver-installed"
    APP_INTERNAL_BASE_URL: str = ""

    @field_validator(
        "GPU_RESOURCES_WITH_AUDIO",
        "NVIDIA_DRIVER_VERSIONS",
        "NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER",
        mode="before",
    )
    @classmethod
    def _split_csv(cls, v):
        if isinstance(v, str):
            return [x.strip() for x in v.split(",") if x.strip()]
        return v

    # --- VM resource limits ---
    MAX_CPU_CORES: int = 16
    MAX_MEMORY_GB: int = 64
    MAX_STORAGE_GB: int = 500
    MAX_GPU_COUNT: int = 2

    # --- VM defaults ---
    VM_RUN_STRATEGY: str = "Manual"
    DEFAULT_PACKAGES: List[str] = [
        "openssh-server",
        "qemu-guest-agent",
        "curl",
        "wget",
        "net-tools",
    ]
    DEFAULT_VM_SETTINGS: Dict = Field(default_factory=lambda: {
        "username": "ubuntu",
        "package_update": False,
        "package_upgrade": False,
        "ssh_pwauth": False,
    })

    # --- Windows VM ---
    WINDOWS_ISO_DATAVOLUME: str = "windows-10-iso"
    WINDOWS_ISO_NAMESPACE: str = "kvm"
```

```
    WINDOWS_GOLDEN_IMAGE_NAME: str = "windows-10-golden-image"
    WINDOWS_GOLDEN_IMAGE_NAMESPACE: str = "kvm"
    WINDOWS_ISO_STORAGE_GB: int = 7
    WINDOWS_DEFAULT_CPU: int = 4
    WINDOWS_DEFAULT_MEMORY: int = 8
    WINDOWS_DEFAULT_STORAGE: int = 60
    WINDOWS_MACHINE_TYPE: str = "q35"
    WINDOWS_DEFAULT_USERNAME: str = "Administrator"
    WINDOWS_HYPERV_FEATURES: Dict = Field(default_factory=lambda: {
        "relaxed": {"enabled": True},
        "vapic": {"enabled": True},
        "spinlocks": {"enabled": True, "spinlocks": 8191},
        "vpindex": {"enabled": True},
        "runtime": {"enabled": True},
        "synic": {"enabled": True},
        "stimer": {"enabled": True},
        "reset": {"enabled": True},
        "frequencies": {"enabled": True},
        "reenlightenment": {"enabled": True},
        "tlbflush": {"enabled": True},
        "ipi": {"enabled": True},
    })
    VIRTIO_DRIVERS_IMAGE: str = (
        "quay.io/kubevirt/virtio-container-disk:latest"
    )

    # --- Image domains whitelist ---
    ALLOWED_IMAGE_DOMAINS: List[str] = []

    @field_validator("ALLOWED_IMAGE_DOMAINS", mode="before")
    @classmethod
    def _split_domains(cls, v):
        if isinstance(v, str):
            return [d.strip() for d in v.split(",") if d.strip()]
        return v

    # --- Network policy ---
    POD_NETWORK_CIDR: str = ""
    SERVICE_NETWORK_CIDR: str = ""
    SERVER_NETWORK_CIDRS: List[str] = []

    @field_validator("SERVER_NETWORK_CIDRS", mode="before")
    @classmethod
    def _split_cidrs(cls, v):
        if isinstance(v, str):
            return [c.strip() for c in v.split(",") if c.strip()]
        return v

    # --- Juniper router ---
    JUNIPER_HOST: str = ""
    JUNIPER_USER: str = ""
    JUNIPER_SIDECAR_URL: str = ""
    JUNIPER_EXTERNAL_IP_POOL: List[str] = []

    @field_validator("JUNIPER_EXTERNAL_IP_POOL", mode="before")
    @classmethod
    def _split_ips(cls, v):
        if isinstance(v, str):
            return [ip.strip() for ip in v.split(",") if ip.strip()]
        return v


settings = Settings()


# ---------------------------------------------------------------------------
# Backward-compatible module-level exports
# ---------------------------------------------------------------------------

SECRET_KEY = settings.SECRET_KEY
DEBUG = settings.DEBUG
```

```
FRONTEND_STUB_MODE = settings.FRONTEND_STUB_MODE
STUB_NODE_IP = settings.STUB_NODE_IP
DEMO_USERNAME = settings.DEMO_USERNAME
DEMO_PASSWORD = settings.DEMO_PASSWORD
DEMO_USER_ID = settings.DEMO_USER_ID
SESSION_TIMEOUT = settings.SESSION_TIMEOUT
DATABASE_URL = settings.DATABASE_URL
K8S_NAMESPACE_PREFIX = settings.K8S_NAMESPACE_PREFIX
STORAGE_CLASS_NAME = settings.STORAGE_CLASS_NAME
LLM_API_URL = settings.LLM_API_URL
LLM_API_KEY = settings.LLM_API_KEY
LLM_MODEL = settings.LLM_MODEL
LLM_SERVICE_URL = settings.LLM_SERVICE_URL
LLM_CHAT_SHARED_TOKEN = settings.LLM_CHAT_SHARED_TOKEN
LLM_USER_KEYS_MAX = settings.LLM_USER_KEYS_MAX
LLM_KEYS_ENCRYPTION_KEY = settings.LLM_KEYS_ENCRYPTION_KEY
LLM_AUTH_SERVICE_URL = settings.LLM_AUTH_SERVICE_URL
LLM_AUTH_ADMIN_TOKEN = settings.LLM_AUTH_ADMIN_TOKEN
SPEECH_API_URL = settings.SPEECH_API_URL
SPEECH_API_KEY = settings.SPEECH_API_KEY
SPEECH_TTS_MODEL = settings.SPEECH_TTS_MODEL
SPEECH_TTS_OUTPUT_FORMAT = settings.SPEECH_TTS_OUTPUT_FORMAT
DEFAULT_GPU_RESOURCE = settings.DEFAULT_GPU_RESOURCE
GPU_RESOURCES_WITH_AUDIO = settings.GPU_RESOURCES_WITH_AUDIO
NVIDIA_DRIVER_VERSIONS = settings.NVIDIA_DRIVER_VERSIONS
DEFAULT_NVIDIA_DRIVER = settings.DEFAULT_NVIDIA_DRIVER
NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER = settings.NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER
NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY = (
    settings.NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY
)
NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH = settings.NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH
APP_INTERNAL_BASE_URL = settings.APP_INTERNAL_BASE_URL
MAX_CPU_CORES = settings.MAX_CPU_CORES
MAX_MEMORY_GB = settings.MAX_MEMORY_GB
MAX_STORAGE_GB = settings.MAX_STORAGE_GB
MAX_GPU_COUNT = settings.MAX_GPU_COUNT
VM_RUN_STRATEGY = settings.VM_RUN_STRATEGY
DEFAULT_PACKAGES = settings.DEFAULT_PACKAGES
DEFAULT_VM_SETTINGS = settings.DEFAULT_VM_SETTINGS
WINDOWS_ISO_DATAVOLUME = settings.WINDOWS_ISO_DATAVOLUME
WINDOWS_ISO_NAMESPACE = settings.WINDOWS_ISO_NAMESPACE
WINDOWS_GOLDEN_IMAGE_NAME = settings.WINDOWS_GOLDEN_IMAGE_NAME
WINDOWS_GOLDEN_IMAGE_NAMESPACE = settings.WINDOWS_GOLDEN_IMAGE_NAMESPACE
WINDOWS_ISO_STORAGE_GB = settings.WINDOWS_ISO_STORAGE_GB
WINDOWS_DEFAULT_CPU = settings.WINDOWS_DEFAULT_CPU
WINDOWS_DEFAULT_MEMORY = settings.WINDOWS_DEFAULT_MEMORY
WINDOWS_DEFAULT_STORAGE = settings.WINDOWS_DEFAULT_STORAGE
WINDOWS_MACHINE_TYPE = settings.WINDOWS_MACHINE_TYPE
WINDOWS_DEFAULT_USERNAME = settings.WINDOWS_DEFAULT_USERNAME
WINDOWS_HYPERV_FEATURES = settings.WINDOWS_HYPERV_FEATURES
VIRTIO_DRIVERS_IMAGE = settings.VIRTIO_DRIVERS_IMAGE
ALLOWED_IMAGE_DOMAINS = settings.ALLOWED_IMAGE_DOMAINS
POD_NETWORK_CIDR = settings.POD_NETWORK_CIDR
SERVICE_NETWORK_CIDR = settings.SERVICE_NETWORK_CIDR
SERVER_NETWORK_CIDRS = settings.SERVER_NETWORK_CIDRS
JUNIPER_HOST = settings.JUNIPER_HOST
JUNIPER_USER = settings.JUNIPER_USER
JUNIPER_SIDECAR_URL = settings.JUNIPER_SIDECAR_URL
JUNIPER_EXTERNAL_IP_POOL = settings.JUNIPER_EXTERNAL_IP_POOL
JUNIPER_EXTERNAL_IP = (
    settings.JUNIPER_EXTERNAL_IP_POOL[0] if settings.JUNIPER_EXTERNAL_IP_POOL else ""
)


def is_valid_user_namespace(namespace: str | None, username: str) -> bool:
    """Check that *namespace* matches the expected pattern for *username*,
    starts with a letter, and contains no underscores.
    """
    if not namespace or not username:
        return False
    if not namespace[0].isalpha():
```

```
        return False
    if "_" in namespace:
        return False
    prefix = settings.K8S_NAMESPACE_PREFIX or "kv"
    return namespace.startswith(f"{prefix}-{username.lower()}-")

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/cursor_billing_feature_planning_for_vir.md (31.7 KB)

```
# Billing feature planning for virtual machines
_Exported on 2/26/2026 at 18:23:34 GMT+3 from Cursor (2.5.20)_

---

**User**

@viirtuoz-vps/app/routers @viirtuoz-vps/app/database.py

I want you to help me with planning for a feature
It does not involve k8s (almost) so lets do it here right now
1. I want to do billing for vms (lets base cost linearly depending on the resources used
like cpu, memory, gpu usage, gpu type etc) So we have like a data object with that info and we would have a method fo...

2. We would have some background process that would track usage and subtract money from the main user balance
3. When there are no money in the main balance we stop all vms
4. We track costs atomically so we do it once for all users and if there is not enough money we return smth like stop...
And we kill them in the main loop

5. I would also like to have a separate table in db where we would track all of the increases of the balance. I was a...

if you have questions lets discuss (but prioritize them) I cannot handle too many questions at once

---

**Cursor**

Let me explore the existing codebase first to understand the current data model and architecture before we plan.

Let me also look at the models and VM-related routers to understand the full picture.

Let me also check how the admin router handles balance and look at the schemas.

Good, now I have a solid picture of the codebase. Here's the plan, followed by a couple focused questions.

---

## Plan

### 1. VM Cost Calculator (`app/services/vm_billing.py`)

A pure data/logic module. No DB, no k8s -- just pricing math.

```python
from dataclasses import dataclass
from decimal import Decimal

@dataclass
class VMCostSpec:
    cpu: int
    memory_gb: float
    gpu_count: int
    gpu_model: str | None

    def cost_per_minute(self, pricing: "VMPricing") -> Decimal:
        cost = Decimal(str(self.cpu)) * pricing.cpu_per_minute
        cost += Decimal(str(self.memory_gb)) * pricing.memory_gb_per_minute
        if self.gpu_count > 0 and self.gpu_model:
            gpu_rate = pricing.gpu_rates.get(self.gpu_model, pricing.gpu_default_per_minute)
            cost += Decimal(str(self.gpu_count)) * gpu_rate
        return cost

@dataclass
class VMPricing:
    cpu_per_minute: Decimal
```
```

```
    memory_gb_per_minute: Decimal
    gpu_default_per_minute: Decimal
    gpu_rates: dict[str, Decimal]   # gpu_model -> rate per GPU per minute
```

`VMCostSpec` can be trivially constructed from an existing `VMRecord` or from live k8s VM spec data. The pricing conf...

2. Background billing loop (in lifespan)

A new `asyncio.Task` launched during the FastAPI lifespan, inside the `async with create_db(...)` block. Rough shape:

```
every BILLING_INTERVAL_SECONDS (e.g. 60):
  1. Fetch all active (non-deleted) VMRecords, grouped by user_id
  2. For each user_id:
     a. Sum cost_per_minute for all their *running* VMs
     b. Multiply by minutes_elapsed (time since last successful tick for this user)
     c. Try to deduct from balance via update_balance()
     d. If insufficient funds ? add to kill_list
  3. For each user in kill_list:
     ? stop all their VMs via k8s
     ? log event (VMEventType, maybe a new BILLING_STOPPED type)
  4. Snapshot balances
```

Key detail: the loop keeps a `last_billed_at: dict[int, datetime]` in memory to compute the actual elapsed interval p...

3. Insufficient balance ? stop VMs

Handled inside step 2d/3 above. The `update_balance` method already rolls back on negative balance and raises `ValueE...
- We add the user to the kill list
- After processing all users, we iterate the kill list and call `stop_virtual_machine` for each VM
- We log a new event type like `VMEventType.BILLING_STOPPED`

This matches your point 4 -- one pass over all users, collect the kill list, then act on it.

4. DB changes

4a. New table: `balance_topups`

Tracks every deposit/increase to either balance. One row per top-up event.

```sql
CREATE TABLE IF NOT EXISTS balance_topups (
    id BIGSERIAL PRIMARY KEY,
    user_id INTEGER REFERENCES users(id) NOT NULL,
    source TEXT NOT NULL CHECK (source IN ('main', 'llm')),
    amount NUMERIC(18,9) NOT NULL,
    admin_username TEXT,          -- who performed the top-up (nullable for system)
    created_at TIMESTAMPTZ DEFAULT NOW()
);
```

This gets a corresponding `BalanceTopupDB` class with `create()` and `get_cumulative(user_id, source)` methods. The `...

4b. New `VMEventType` entry

Add `BILLING_STOPPED = "billing_stopped"` to the enum so we can distinguish "admin stopped the VM" from "billing kill...

5. Where things live (file layout)

What	Where
`VMCostSpec`, `VMPricing`	`app/services/vm_billing.py`
`BalanceTopupDB`	`app/database.py` (new class, wired into `DB`)
Billing background task	`app/services/billing_loop.py`
Task startup	`app/main.py` lifespan (create task, cancel on shutdown)
Pricing config values	`app/config.py` / env vars

The routers don't change much -- the admin top-up endpoints just gain one extra `await db.balance_topups.create(...)`...

Questions (2, most critical first)

Q1 -- Billing only running VMs or also paused/stopped?
Right now `VMRecord` has `deleted_at` but no "running/stopped" status column. The live status comes from k8s. Should we:
- (a) Query k8s for actual running status each billing tick (accurate but adds k8s API calls every minute), or
- (b) Only bill VMs that are in `vm_records` and not deleted, i.e. bill for the existence of a VM regardless of wheth...

This is the single biggest design fork.

Q2 -- Pricing values: where should they come from?
Simplest option is env vars / `settings` (like `VM_BILLING_CPU_PER_MINUTE=0.001`). Alternative is a DB table so an ad...

User

q1: Only running (we can also track when was stopped so the user could not run vm for lets say 30s and get away with ...
q2: for now we will add an extra class in config.py with some defaults (this is an MVP mostly)

Cursor

Let me quickly check how your config is structured so I can make the pricing config part concrete.

Good. Here's the refined, complete plan with both answers baked in.

Refined Plan

A. Pricing config -- add to `config.py`

A simple dict in `Settings` that env vars can override, plus a helper dataclass for typed access:

```python
from decimal import Decimal
from dataclasses import dataclass, field

@dataclass
class VMPricing:
    cpu_per_minute: Decimal = Decimal("0.001")
    memory_gb_per_minute: Decimal = Decimal("0.0005")
    gpu_default_per_minute: Decimal = Decimal("0.01")
    gpu_rates: dict[str, Decimal] = field(default_factory=dict)

    def cost_per_minute(self, cpu: int, memory_gb: float, gpu_count: int, gpu_model: str | None) -> Decimal:
        cost = Decimal(str(cpu)) * self.cpu_per_minute
        cost += Decimal(str(memory_gb)) * self.memory_gb_per_minute
        if gpu_count > 0 and gpu_model:
            rate = self.gpu_rates.get(gpu_model, self.gpu_default_per_minute)
            cost += Decimal(str(gpu_count)) * rate
        return cost
```

In `Settings`, add env-overridable fields:

```python
# inside class Settings:
BILLING_ENABLED: bool = False
BILLING_INTERVAL_SECONDS: int = 60
BILLING_CPU_PER_MINUTE: str = "0.001"
BILLING_MEMORY_GB_PER_MINUTE: str = "0.0005"
BILLING_GPU_DEFAULT_PER_MINUTE: str = "0.01"
BILLING_GPU_RATES: str = ""  # JSON like {"nvidia.com/A100": "0.05"}
```

Then construct `VMPricing` from settings at startup. `BILLING_ENABLED = False` by default so existing deployments are...

B. DB schema changes

B1. New columns on `vm_records` -- tracked by the lifecycle endpoints, consumed by the billing loop:

```sql
ALTER TABLE vm_records
  ADD COLUMN is_running BOOLEAN DEFAULT FALSE NOT NULL,
  ADD COLUMN last_started_at TIMESTAMPTZ,
  ADD COLUMN last_stopped_at TIMESTAMPTZ,
  ADD COLUMN last_billed_at TIMESTAMPTZ;
```

Added via the usual `DO $$ ... IF NOT EXISTS ...` migration block in `VMRecordDB.ensure_schema()`.

B2. New methods on `VMRecordDB`:

- `set_running(namespace, vm_name, running: bool)` -- sets `is_running` and the corresponding timestamp
- `get_all_billable()` -- returns all non-deleted VMs where either `is_running = TRUE` or `last_stopped_at > last_bil...
- `update_last_billed(vm_id, billed_at)` -- stamps `last_billed_at` after deduction

B3. New table `balance_topups`:

```sql
CREATE TABLE IF NOT EXISTS balance_topups (
    id BIGSERIAL PRIMARY KEY,
    user_id INTEGER REFERENCES users(id) NOT NULL,
    source TEXT NOT NULL CHECK (source IN ('main', 'llm')),
    amount NUMERIC(18,9) NOT NULL,
    admin_username TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX IF NOT EXISTS idx_balance_topups_user
    ON balance_topups(user_id, source, created_at DESC);
```

New `BalanceTopupDB` class with:
- `create(user_id, source, amount, admin_username)`
- `get_cumulative(user_id, source)` -- `SELECT COALESCE(SUM(amount), 0) ...`
- `list_for_user(user_id, source, limit)` -- recent top-up history

B4. New event type: `BILLING_STOPPED = "billing_stopped"` in `VMEventType`.

C. Lifecycle endpoint changes (the anti-gaming part)

In `vms.py`, after each successful k8s call, also update the DB running state:

Endpoint	DB update
`create_vm` (already calls `_log_vm_created`)	`set_running(ns, name, True)`
`start_vm` / `restart_vm` / `unpause_vm`	`set_running(ns, name, True)`
`stop_vm` / `pause_vm`	`set_running(ns, name, False)`
`delete_vm`	`set_running(ns, name, False)`, then `mark_deleted` (already there)

Same for the `admin_stop_vm` / `admin_delete_vm` in `admin.py`.

This way the billing loop never needs to query k8s -- it relies entirely on DB state.

D. Billing loop (`app/services/billing_loop.py`)

```python
async def billing_tick(db: DB, pricing: VMPricing) -> list[int]:
    """One billing pass. Returns user_ids whose VMs should be killed."""
    now = datetime.now(timezone.utc)
    kill_user_ids: list[int] = []

    billable_vms = await db.vm_records.get_all_billable()

    # Group by user_id
    by_user: dict[int, list[VMRecord]] = {}
    for vm in billable_vms:
        by_user.setdefault(vm.user_id, []).append(vm)
```

```

    for user_id, vms in by_user.items():
        total_cost = Decimal("0")

        for vm in vms:
            if vm.is_running:
                start = max(filter(None, [vm.last_started_at, vm.last_billed_at])) or vm.created_at
                minutes = Decimal(str((now - start).total_seconds())) / 60
            elif vm.last_stopped_at and (not vm.last_billed_at or vm.last_stopped_at > vm.last_billed_at):
                start = max(filter(None, [vm.last_started_at, vm.last_billed_at])) or vm.created_at
                minutes = Decimal(str((vm.last_stopped_at - start).total_seconds())) / 60
            else:
                minutes = Decimal("0")

            if minutes > 0:
                total_cost += pricing.cost_per_minute(vm.cpu, vm.memory_gb, vm.gpu_count, vm.gpu_model) * minutes

        if total_cost <= 0:
            continue

        # Try to deduct
        bal = await db.users.get_balance(user_id)
        if bal is None:
            continue
        current_balance, version = bal

        try:
            new_balance, _ = await db.users.update_balance(user_id, -total_cost, version)
        except ValueError:
            kill_user_ids.append(user_id)
            continue

        # Stamp all VMs as billed
        for vm in vms:
            billed_at = now if vm.is_running else vm.last_stopped_at
            await db.vm_records.update_last_billed(vm.id, billed_at)

        # Snapshot
        await db.balance_snapshots.create(user_id, "main", new_balance)

    return kill_user_ids
```

Key properties:
- One pass over all users -- the "atomic" batch you described
- Returns a kill list; the caller (the loop in lifespan) handles stopping
- VMs stopped briefly between ticks still get billed because `last_stopped_at > last_billed_at`
- No k8s API calls inside billing

E. Background task in lifespan (`app/main.py`)

Started inside the `async with create_db(...)` block:

```python
async def _billing_worker(app: FastAPI):
    from app.services.billing_loop import billing_tick
    pricing = build_vm_pricing(settings)  # construct VMPricing from settings

    while True:
        await asyncio.sleep(settings.BILLING_INTERVAL_SECONDS)
        try:
            db = app.state.db
            kill_list = await billing_tick(db, pricing)
            for user_id in kill_list:
                await _stop_all_user_vms(db, user_id)  # stop via k8s + log BILLING_STOPPED events
        except Exception:
            logger.exception("Billing tick failed")
```

`_stop_all_user_vms` fetches active VM records for that user, calls `stop_virtual_machine` for each, logs `BILLING_ST...

```
```

The task is created with `asyncio.create_task()` and cancelled in the lifespan teardown, gated behind `if settings.BI...

F. Admin top-up changes

The existing `topup_main_balance` and `topup_llm_balance` endpoints get one extra line each:

```python
await db.balance_topups.create(user_id, "main", amount, admin.username)
```

Same for transfers that add to a balance.

G. Summary of all files touched

File	Change
`app/config.py`	`VMPricing` dataclass, `BILLING_*` settings fields, `build_vm_pricing()` helper
`app/models.py`	Add `is_running`, `last_started_at`, `last_stopped_at`, `last_billed_at` to `VMRecord`; add `BILL...
`app/database.py`	Migration columns on `vm_records`; new methods `set_running`, `get_all_billable`, `update_last_...
`app/services/billing_loop.py`	**New file** -- `billing_tick()` + `_stop_all_user_vms()`
`app/main.py`	Launch `_billing_worker` task in lifespan (guarded by `BILLING_ENABLED`)
`app/routers/vms.py`	Call `db.vm_records.set_running(...)` in lifecycle endpoints
`app/routers/admin.py`	Same for admin stop/delete; add `balance_topups.create()` in top-up endpoints
`app/schemas.py`	(optional) response schema for billing status if you want an API for it

That's the full scope. Does this look right, or do you want to adjust anything before we start implementing?

User

maybe you have some other questions
I would also ask you to use pydantic only and not python dataclasses

---

Cursor

Good call on Pydantic -- consistent with everything else in the codebase. `VMPricing` becomes a `BaseModel` (or a fro...

A few more questions, grouped by priority:

1. What should happen to already-running VMs when billing is first enabled?

Every existing VM will have `last_started_at = NULL` and `last_billed_at = NULL`. When the billing loop first sees th...

2. Should the billing kill stop VMs or delete them?

The plan currently says "stop all VMs" when a user runs out of funds. Stopped VMs still occupy storage. Do you want:
- (a) Stop only (user can top up and start again) -- this is what I assumed
- (b) Stop, then delete after some grace period

3. Should admins be exempt from billing?

Right now the billing loop would process all users equally. Should users with `role = "admin"` skip billing entirely?

User

1. Lets use last_billed_at = NOW() sounds reasonable
2. I think we should stop everything for some time. Deleting everything would be too cruel
3. Well we can exempt them, that would be nice
Any other questions?

Cursor

Two more, then I think we're complete:

1. Should we prevent users from starting a VM if their balance is already zero/negative?

Right now `start_vm` doesn't check balance at all. It would be odd to let someone start a VM only for the billing loo...

2. For the `balance_topups` tracking -- should LLM balance transfers also be recorded?

Currently there are three ways money enters a balance:
- Admin tops up main balance (`topup_main_balance`)
- Admin tops up LLM balance (`topup_llm_balance`)
- Admin transfers between main and LLM (`transfer_balance`)

For the "track cumulative deposits" requirement, should we log all three? A transfer from main to LLM would be a nega...

User

1. We should deny that method
2. We care about all net positive inflows (i guess it is the right term)
So when we add stuff we log it, when we transfer we should also track it (even though it does not change the total am...
the goal is to have a valid amount of money spent in the end

Cursor

Clear. That makes the accounting straightforward: `total_spent = cumulative_topups - current_balance` for each source.

I think we've covered everything. Here's the final consolidated plan with all decisions locked in:

Final Plan -- VM Billing MVP

Decisions summary

Decision	Answer
Bill running VMs only	Yes, track start/stop times to prevent gaming
Pricing config	Pydantic model in `config.py`, env-overridable
Existing VMs on first deploy	`last_billed_at = NOW()` migration, bill forward only
Insufficient funds	Stop all VMs (not delete)
Admins	Exempt from billing
VM start/create with zero balance	Denied with error
Topup tracking	Log all net-positive inflows per balance (top-ups + receiving side of transfers)

Files and changes

1. `app/config.py` -- new `VMPricing` Pydantic model + `BILLING_*` settings

```python
class VMPricing(BaseModel):
    model_config = ConfigDict(frozen=True)
    cpu_per_minute: Decimal = Decimal("0.001")
    memory_gb_per_minute: Decimal = Decimal("0.0005")
    gpu_default_per_minute: Decimal = Decimal("0.01")
    gpu_rates: dict[str, Decimal] = {}

    def cost_per_minute(self, cpu, memory_gb, gpu_count, gpu_model) -> Decimal: ...
```

Settings fields: `BILLING_ENABLED`, `BILLING_INTERVAL_SECONDS`, `BILLING_CPU_PER_MINUTE`, `BILLING_MEMORY_GB_PER_MINU...

2. `app/models.py`
- `VMRecord`: add `is_running`, `last_started_at`, `last_stopped_at`, `last_billed_at`
- `VMEventType`: add `BILLING_STOPPED = "billing_stopped"`

3. `app/database.py`
- `VMRecordDB`: migration for new columns (with `last_billed_at = NOW()` for existing active VMs); new methods `set_r...
- New `BalanceTopupDB` class with `create()`, `get_cumulative()`, `list_for_user()`
- Wire `BalanceTopupDB` into `DB` aggregate + `create_db()`

```

**4. `app/services/billing_loop.py`** -- **new file**
- `billing_tick(db, pricing)` -- one pass over all users: compute costs, deduct, return kill list
- `stop_all_user_vms(db, user_id)` -- stop via k8s, log `BILLING_STOPPED` events, set `is_running = False`
- Admin users (`role = 'admin'`) skipped entirely

**5. `app/main.py`**
- Launch `_billing_worker` asyncio task inside lifespan, guarded by `settings.BILLING_ENABLED`
- Cancel on shutdown

... [295 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/database.py (40.4 KB)

```
"""
PostgreSQL-backed persistence via raw asyncpg (no ORM).

Pattern follows the ``db_ops.py`` approach: a shared ``asyncpg.Pool`` is created
once at app startup (inside the FastAPI *lifespan*) and closed on shutdown.
Each DB class receives the pool and acquires connections per-operation.

Usage (inside FastAPI lifespan)::

    async with create_db(settings.DATABASE_URL) as db:
        app.state.db = db
        yield
"""

from __future__ import annotations

import json
import logging
import uuid
from contextlib import asynccontextmanager
from datetime import datetime
from decimal import Decimal

import asyncpg

from app.models import User, UserRole, VMEvent, VMEventType, VMRecord
from app.models import LLMApiKey, LLMApiKeyStatus, LLMUsageLog

logger = logging.getLogger(__name__)


# ---------------------------------------------------------------------------
# UserDB
# ---------------------------------------------------------------------------

class UserDB:
    def __init__(self, pool: asyncpg.Pool):
        self.pool = pool

    async def ensure_schema(self) -> None:
        async with self.pool.acquire() as conn:
            await conn.execute("""
                CREATE TABLE IF NOT EXISTS users (
                    id SERIAL PRIMARY KEY,
                    username TEXT UNIQUE NOT NULL,
                    password_hash TEXT NOT NULL,
                    role TEXT DEFAULT 'user' NOT NULL,
                    is_active BOOLEAN DEFAULT TRUE,
                    balance NUMERIC(18,9) DEFAULT 0 NOT NULL,
                    balance_version INTEGER DEFAULT 0 NOT NULL,
                    created_at TIMESTAMPTZ DEFAULT NOW()
                )
            """)
            await conn.execute("""
                CREATE INDEX IF NOT EXISTS idx_users_username
                ON users(username)
            """)
            # Migration: add columns if missing (existing DBs)
            await conn.execute("""
                DO $$
                BEGIN
                  IF NOT EXISTS (
                    SELECT 1 FROM information_schema.columns
                    WHERE table_name='users' AND column_name='role'
                  ) THEN
```

```
                    ALTER TABLE users ADD COLUMN role TEXT DEFAULT 'user' NOT NULL;
                  END IF;
                  IF NOT EXISTS (
                    SELECT 1 FROM information_schema.columns
                    WHERE table_name='users' AND column_name='balance'
                  ) THEN
                    ALTER TABLE users ADD COLUMN balance NUMERIC(18,9) DEFAULT 0 NOT NULL;
                  END IF;
                  IF NOT EXISTS (
                    SELECT 1 FROM information_schema.columns
                    WHERE table_name='users' AND column_name='balance_version'
                  ) THEN
                    ALTER TABLE users ADD COLUMN balance_version INTEGER DEFAULT 0 NOT NULL;
                  END IF;
                END $$;
            """)

    async def create_user(
        self,
        username: str,
        password_hash: str,
        role: str = UserRole.USER.value,
    ) -> User:
        async with self.pool.acquire() as conn:
            row = await conn.fetchrow("""
                INSERT INTO users (username, password_hash, role)
                VALUES ($1, $2, $3)
                RETURNING *
            """, username, password_hash, role)
            return User.from_row(row)

    async def get_by_username(self, username: str) -> User | None:
        async with self.pool.acquire() as conn:
            row = await conn.fetchrow(
                "SELECT * FROM users WHERE username = $1", username,
            )
            return User.from_row(row) if row else None

    async def get_by_id(self, user_id: int) -> User | None:
        async with self.pool.acquire() as conn:
            row = await conn.fetchrow(
                "SELECT * FROM users WHERE id = $1", user_id,
            )
            return User.from_row(row) if row else None

    async def deactivate(self, user_id: int) -> bool:
        async with self.pool.acquire() as conn:
            result = await conn.execute(
                "UPDATE users SET is_active = FALSE WHERE id = $1", user_id,
            )
            return result == "UPDATE 1"

    async def list_all(self, include_inactive: bool = False) -> list[User]:
        async with self.pool.acquire() as conn:
            if include_inactive:
                rows = await conn.fetch("SELECT * FROM users ORDER BY username")
            else:
                rows = await conn.fetch(
                    "SELECT * FROM users WHERE is_active = TRUE ORDER BY username"
                )
            return [User.from_row(r) for r in rows]

    async def set_role(self, user_id: int, role: str) -> bool:
        async with self.pool.acquire() as conn:
            result = await conn.execute(
                "UPDATE users SET role = $1 WHERE id = $2", role, user_id,
            )
            return result == "UPDATE 1"

    async def get_balance(self, user_id: int) -> tuple[Decimal, int] | None:
        """Return (balance, balance_version) or None if user not found."""
```

```
        async with self.pool.acquire() as conn:
            row = await conn.fetchrow(
                "SELECT balance, balance_version FROM users WHERE id = $1", user_id,
            )
            if not row:
                return None
            return Decimal(str(row["balance"])), row["balance_version"]

    async def update_balance(
        self,
        user_id: int,
        amount: Decimal,
        expected_version: int,
    ) -> tuple[Decimal, int]:
        """Atomic balance update with optimistic locking.

        Returns (new_balance, new_version).
        Raises ValueError on version mismatch or insufficient funds.
        """
        async with self.pool.acquire() as conn:
            row = await conn.fetchrow("""
                UPDATE users
                SET balance = balance + $1,
                    balance_version = balance_version + 1
                WHERE id = $2 AND balance_version = $3
                RETURNING balance, balance_version
            """, amount, user_id, expected_version)
            if not row:
                raise ValueError("Balance update failed: version mismatch or user not found")
            new_balance = Decimal(str(row["balance"]))
            if new_balance < 0:
                # Rollback: the UPDATE already committed the negative balance,
                # so revert it immediately.
                await conn.fetchrow("""
                    UPDATE users
                    SET balance = balance - $1,
                        balance_version = balance_version + 1
                    WHERE id = $2
                    RETURNING balance, balance_version
                """, amount, user_id)
                raise ValueError("Insufficient funds")
            return new_balance, row["balance_version"]


# ---------------------------------------------------------------------------
# BalanceSnapshotDB
# ---------------------------------------------------------------------------

class BalanceSnapshotDB:
    def __init__(self, pool: asyncpg.Pool):
        self.pool = pool

    async def ensure_schema(self) -> None:
        async with self.pool.acquire() as conn:
            await conn.execute("""
                CREATE TABLE IF NOT EXISTS balance_snapshots (
                    id BIGSERIAL PRIMARY KEY,
                    user_id INTEGER REFERENCES users(id) NOT NULL,
                    source TEXT NOT NULL CHECK (source IN ('main', 'llm')),
                    balance NUMERIC(18,9) NOT NULL,
                    created_at TIMESTAMPTZ DEFAULT NOW()
                )
            """)
            await conn.execute("""
                CREATE INDEX IF NOT EXISTS idx_balance_snapshots_user_source_created
                ON balance_snapshots(user_id, source, created_at DESC)
            """)

    async def create(
        self,
        user_id: int,
```

```
        source: str,
        balance: Decimal | float | int,
    ) -> None:
        if source not in {"main", "llm"}:
            raise ValueError("Invalid balance snapshot source")
        async with self.pool.acquire() as conn:
            await conn.execute("""
                INSERT INTO balance_snapshots (user_id, source, balance)
                VALUES ($1, $2, $3)
            """, user_id, source, Decimal(str(balance)))

    async def summary_llm_spend(
        self,
        days: int = 30,
        user_id: int | None = None,
    ) -> dict:
        """Aggregate LLM spend from balance snapshots.

        Spend is calculated as sum of negative balance deltas only:
        max(prev_balance - current_balance, 0).
        Positive deltas (top-ups/transfers-in) are ignored.
        """
        async with self.pool.acquire() as conn:
            per_user_rows = await conn.fetch("""
                WITH window_start AS (
                    SELECT NOW() - ($1::int || ' days')::interval AS ts
                ),
                relevant AS (
                    SELECT bs.user_id, u.username, bs.created_at, bs.balance
                    FROM balance_snapshots bs
                    JOIN users u ON u.id = bs.user_id
                    WHERE bs.source = 'llm'
                      AND ($2::int IS NULL OR bs.user_id = $2)
                      AND (
                        bs.created_at >= (SELECT ts FROM window_start)
                        OR bs.created_at = (
                            SELECT MAX(bs2.created_at)
                            FROM balance_snapshots bs2
                            WHERE bs2.user_id = bs.user_id
                              AND bs2.source = 'llm'
                              AND bs2.created_at < (SELECT ts FROM window_start)
                        )
                      )
                ),
                deltas AS (
                    SELECT
                        user_id,
                        username,
                        created_at,
                        balance,
                        LAG(balance) OVER (PARTITION BY user_id ORDER BY created_at) AS prev_balance
                    FROM relevant
                ),
                spend_by_user AS (
                    SELECT
                        user_id,
                        SUM(
                            GREATEST(
                                COALESCE(prev_balance, balance) - balance,
                                0
                            )
                        ) AS total_spent
                    FROM deltas
                    WHERE created_at >= (SELECT ts FROM window_start)
                    GROUP BY user_id
                )
                SELECT
                    u.username,
                    COALESCE(s.total_spent, 0) AS total_cost
                FROM users u
                LEFT JOIN spend_by_user s ON s.user_id = u.id
```

```
                WHERE ($2::int IS NULL OR u.id = $2)
                ORDER BY total_cost DESC, u.username ASC
            """, days, user_id)

            per_day_rows = await conn.fetch("""
                WITH window_start AS (
                    SELECT NOW() - ($1::int || ' days')::interval AS ts
                ),
                relevant AS (
                    SELECT bs.user_id, bs.created_at, bs.balance
                    FROM balance_snapshots bs
                    WHERE bs.source = 'llm'
                      AND ($2::int IS NULL OR bs.user_id = $2)
                      AND (
                        bs.created_at >= (SELECT ts FROM window_start)
                        OR bs.created_at = (
                            SELECT MAX(bs2.created_at)
                            FROM balance_snapshots bs2
                            WHERE bs2.user_id = bs.user_id
                              AND bs2.source = 'llm'
                              AND bs2.created_at < (SELECT ts FROM window_start)
                        )
                      )
                ),
                deltas AS (
                    SELECT
                        user_id,
                        created_at,
                        balance,
                        LAG(balance) OVER (PARTITION BY user_id ORDER BY created_at) AS prev_balance
                    FROM relevant
                )
                SELECT
                    to_char(date_trunc('day', created_at), 'YYYY-MM-DD') AS day,
                    SUM(
                        GREATEST(
                            COALESCE(prev_balance, balance) - balance,
                            0
                        )
                    ) AS total_cost
                FROM deltas
                WHERE created_at >= (SELECT ts FROM window_start)
                GROUP BY day
                ORDER BY day ASC
            """, days, user_id)

            per_user_vm_rows = await conn.fetch("""
                WITH window_start AS (
                    SELECT NOW() - ($1::int || ' days')::interval AS ts
                ),
                relevant AS (
                    SELECT bs.user_id, u.username, bs.created_at, bs.balance
                    FROM balance_snapshots bs
                    JOIN users u ON u.id = bs.user_id
                    WHERE bs.source = 'main'
                      AND ($2::int IS NULL OR bs.user_id = $2)
                      AND (
                        bs.created_at >= (SELECT ts FROM window_start)
                        OR bs.created_at = (
                            SELECT MAX(bs2.created_at)
                            FROM balance_snapshots bs2
                            WHERE bs2.user_id = bs.user_id
                              AND bs2.source = 'main'
                              AND bs2.created_at < (SELECT ts FROM window_start)
                        )
                      )
                ),
                deltas AS (
                    SELECT
                        user_id,
                        username,
```

```
                        created_at,
                        balance,
                        LAG(balance) OVER (PARTITION BY user_id ORDER BY created_at) AS prev_balance
                    FROM relevant
                ),
                spend_by_user AS (
                    SELECT
                        user_id,
                        SUM(
                            GREATEST(
                                COALESCE(prev_balance, balance) - balance,
                                0
                            )
                        ) AS total_spent
                    FROM deltas
                    WHERE created_at >= (SELECT ts FROM window_start)
                    GROUP BY user_id
                )
                SELECT
                    u.username,
                    COALESCE(s.total_spent, 0) AS total_cost
                FROM users u
                LEFT JOIN spend_by_user s ON s.user_id = u.id
                WHERE ($2::int IS NULL OR u.id = $2)
                ORDER BY total_cost DESC, u.username ASC
            """, days, user_id)

            per_day_vm_rows = await conn.fetch("""
                WITH window_start AS (
                    SELECT NOW() - ($1::int || ' days')::interval AS ts
                ),
                relevant AS (
                    SELECT bs.user_id, bs.created_at, bs.balance
                    FROM balance_snapshots bs
                    WHERE bs.source = 'main'
                      AND ($2::int IS NULL OR bs.user_id = $2)
                      AND (
                        bs.created_at >= (SELECT ts FROM window_start)
                        OR bs.created_at = (
                            SELECT MAX(bs2.created_at)
                            FROM balance_snapshots bs2
                            WHERE bs2.user_id = bs.user_id
                              AND bs2.source = 'main'
                              AND bs2.created_at < (SELECT ts FROM window_start)
                        )
                      )
                ),
                deltas AS (
                    SELECT
                        user_id,
                        created_at,
                        balance,
                        LAG(balance) OVER (PARTITION BY user_id ORDER BY created_at) AS prev_balance
                    FROM relevant
                )
                SELECT
                    to_char(date_trunc('day', created_at), 'YYYY-MM-DD') AS day,
                    SUM(
                        GREATEST(
                            COALESCE(prev_balance, balance) - balance,
                            0
                        )
                    ) AS total_cost
                FROM deltas
                WHERE created_at >= (SELECT ts FROM window_start)
                GROUP BY day
                ORDER BY day ASC
            """, days, user_id)

        total_cost = sum(Decimal(str(r["total_cost"] or 0)) for r in per_user_rows)
        total_vm_cost = sum(Decimal(str(r["total_cost"] or 0)) for r in per_user_vm_rows)
```

```
        vm_by_user = {str(r["username"]): float(r["total_cost"] or 0) for r in per_user_vm_rows}
        vm_by_day = {str(r["day"]): float(r["total_cost"] or 0) for r in per_day_vm_rows}

        return {
            "days": days,
            "total_cost": float(total_cost),
            "total_vm_cost": float(total_vm_cost),
            "total_tokens": 0,
            "per_user": [
                {
                    "username": r["username"],
                    "total_cost": float(r["total_cost"] or 0),
                    "vm_total_cost": vm_by_user.get(str(r["username"]), 0.0),
                }
                for r in per_user_rows
            ],
            "per_model": [],
            "per_day": [
                {
                    "day": r["day"],
                    "total_cost": float(r["total_cost"] or 0),
                    "vm_total_cost": vm_by_day.get(str(r["day"]), 0.0),
                    "total_tokens": 0,
                }
                for r in per_day_rows
            ],
        }


# ---------------------------------------------------------------------------
# VMRecordDB
# ---------------------------------------------------------------------------

class VMRecordDB:
    def __init__(self, pool: asyncpg.Pool):
        self.pool = pool

    async def ensure_schema(self) -> None:
        async with self.pool.acquire() as conn:
            await conn.execute("""
                CREATE TABLE IF NOT EXISTS vm_records (
                    id SERIAL PRIMARY KEY,
                    user_id INTEGER REFERENCES users(id),
                    vm_name TEXT NOT NULL,
                    namespace TEXT NOT NULL,
                    cpu INTEGER NOT NULL,
                    memory_gb DOUBLE PRECISION NOT NULL,
                    storage_gb DOUBLE PRECISION NOT NULL,
                    gpu_count INTEGER DEFAULT 0,
                    gpu_model TEXT,
                    image_url TEXT NOT NULL,
                    os_type TEXT DEFAULT 'linux',
                    is_running BOOLEAN DEFAULT FALSE NOT NULL,
                    last_started_at TIMESTAMPTZ,
                    last_stopped_at TIMESTAMPTZ,
                    last_billed_at TIMESTAMPTZ,
                    created_at TIMESTAMPTZ DEFAULT NOW(),
                    deleted_at TIMESTAMPTZ
                )
            """)
            await conn.execute("""
                CREATE INDEX IF NOT EXISTS idx_vm_records_user_id
                ON vm_records(user_id)
            """)
            await conn.execute("""
                CREATE INDEX IF NOT EXISTS idx_vm_records_namespace
                ON vm_records(namespace, vm_name)
            """)
            # Migration: add billing columns if missing (existing DBs)
            await conn.execute("""
                DO $$
```

```
                BEGIN
                  IF NOT EXISTS (
                    SELECT 1 FROM information_schema.columns
                    WHERE table_name='vm_records' AND column_name='is_running'
                  ) THEN
                    ALTER TABLE vm_records ADD COLUMN is_running BOOLEAN DEFAULT FALSE NOT NULL;
                    ALTER TABLE vm_records ADD COLUMN last_started_at TIMESTAMPTZ;
                    ALTER TABLE vm_records ADD COLUMN last_stopped_at TIMESTAMPTZ;
                    ALTER TABLE vm_records ADD COLUMN last_billed_at TIMESTAMPTZ;

... [543 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/debug_log.py (888.0 B)

```python
"""Tiny NDJSON logger for agent debug sessions."""

from __future__ import annotations

import json
import time
from pathlib import Path
from typing import Any

DEBUG_LOG_PATH = Path("/home/ubuntu/.cursor/debug-54c5fc.log")


def write_debug_log(
    *,
    run_id: str,
    hypothesis_id: str,
    location: str,
    message: str,
    data: dict[str, Any],
) -> None:
    payload = {
        "sessionId": "54c5fc",
        "runId": run_id,
        "hypothesisId": hypothesis_id,
        "location": location,
        "message": message,
        "data": data,
        "timestamp": int(time.time() * 1000),
    }
    try:
        with DEBUG_LOG_PATH.open("a", encoding="utf-8") as fh:
            fh.write(json.dumps(payload, ensure_ascii=False) + "\n")
    except Exception:
        # Never break app flow because of debug instrumentation.
        pass

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/main.py (11.7 KB)

```python
"""
FastAPI application entrypoint.

Initialises the database pool, K8s client, session middleware,
and registers all routers.
"""

from __future__ import annotations

import asyncio
import logging
import os
import sys
from contextlib import asynccontextmanager

from fastapi import FastAPI, HTTPException
from fastapi.responses import JSONResponse
from fastapi.staticfiles import StaticFiles
from starlette.middleware.sessions import SessionMiddleware

from app.config import settings
from app.debug_log import write_debug_log
from app.utils.k8s_utils import init_k8s_client

# ---------------------------------------------------------------------------
# Logging
# ---------------------------------------------------------------------------

log_level = logging.DEBUG if settings.DEBUG else logging.INFO
logging.basicConfig(
    level=log_level,
    format="%(asctime)s [%(levelname)s] %(name)s: %(message)s",
    handlers=[logging.StreamHandler(sys.stdout)],
)
logger = logging.getLogger(__name__)


# ---------------------------------------------------------------------------
# Billing background worker
# ---------------------------------------------------------------------------

async def _billing_worker(app: FastAPI) -> None:
    from app.config import build_vm_pricing
    from app.services.billing_loop import billing_tick, stop_all_user_vms

    pricing = build_vm_pricing(settings)

    while True:
        await asyncio.sleep(settings.BILLING_INTERVAL_SECONDS)
        try:
            db = app.state.db
            if db is None:
                continue
            kill_list = await billing_tick(db, pricing)
            for user_id in kill_list:
                n = await stop_all_user_vms(db, user_id)
                logger.info(
                    "Billing: stopped %d VMs for user %d (insufficient funds)",
                    n, user_id,
                )
        except asyncio.CancelledError:
            raise
        except Exception:
            logger.exception("Billing tick failed")

```

```python

# ---------------------------------------------------------------------------
# Lifespan: DB pool + K8s client + global network policy
# ---------------------------------------------------------------------------

@asynccontextmanager
async def lifespan(app: FastAPI):
    # --- K8s client ---
    if not init_k8s_client():
        logger.critical("Failed to initialise Kubernetes client")
        raise SystemExit("Unable to load Kubernetes configuration")

    # --- Global network policy ---
    if not settings.FRONTEND_STUB_MODE:
        try:
            from app.utils.network_policy_utils import create_vm_global_network_policy

            cidrs = [c for c in settings.SERVER_NETWORK_CIDRS if c]
            create_vm_global_network_policy(
                settings.POD_NETWORK_CIDR, settings.SERVICE_NETWORK_CIDR, cidrs,
            )
            logger.info("GlobalNetworkPolicy initialised")
        except Exception as e:
            logger.warning("GlobalNetworkPolicy setup failed: %s (continuing)", e)

    # --- LLM service (chat + available-models via /v1/models) ---
    if settings.LLM_SERVICE_URL:
        logger.info(
            "LLM service: %s",
            settings.LLM_SERVICE_URL,
        )
        if not settings.LLM_CHAT_SHARED_TOKEN:
            logger.warning("LLM chat shared token is not configured (LLM_CHAT_SHARED_TOKEN)")
    else:
        logger.warning("LLM service: LLM_SERVICE_URL empty ? LLM features unavailable")

    # --- Database ---
    db = None
    if settings.DATABASE_URL:
        import bcrypt

        from app.database import create_db

        async with create_db(settings.DATABASE_URL) as db:
            app.state.db = db
            logger.info("Database pool ready")

            # Bootstrap: create first admin if no users exist
            if settings.ADMIN_USERNAME and settings.ADMIN_PASSWORD:
                users = await db.users.list_all(include_inactive=True)
                if not users:
                    pw_hash = bcrypt.hashpw(
                        settings.ADMIN_PASSWORD.encode("utf-8"),
                        bcrypt.gensalt(),
                    ).decode("utf-8")
                    admin = await db.users.create_user(
                        settings.ADMIN_USERNAME,
                        pw_hash,
                        role="admin",
                    )
                    logger.info("Created bootstrap admin user: %s (id=%s)", admin.username, admin.id)

            # Billing background task
            billing_task: asyncio.Task | None = None
            if settings.BILLING_ENABLED:
                billing_task = asyncio.create_task(
                    _billing_worker(app), name="billing-worker",
                )
                logger.info(
                    "Billing worker started (interval=%ds)",
                    settings.BILLING_INTERVAL_SECONDS,
```

```
                )

            yield

            if billing_task is not None:
                billing_task.cancel()
                try:
                    await billing_task
                except asyncio.CancelledError:
                    pass
                logger.info("Billing worker stopped")
    else:
        logger.warning("DATABASE_URL not set ? running without persistence")
        app.state.db = None
        yield


# ---------------------------------------------------------------------------
# App factory
# ---------------------------------------------------------------------------

app = FastAPI(
    title="Viirtuoz VPS",
    version="2.0.0",
    lifespan=lifespan,
)


@app.middleware("http")
async def agent_llm_route_trace(request: Request, call_next):
    path = request.url.path
    is_llm_flow = path.startswith("/adm/llm") or path.endswith("/llm/chat")
    if is_llm_flow:
        # region agent log
        write_debug_log(
            run_id="run1",
            hypothesis_id="H2",
            location="app/main.py:agent_llm_route_trace:entry",
            message="llm request entered app",
            data={"method": request.method, "path": path},
        )
        # endregion

    response = await call_next(request)

    if is_llm_flow:
        # region agent log
        write_debug_log(
            run_id="run1",
            hypothesis_id="H2",
            location="app/main.py:agent_llm_route_trace:exit",
            message="llm request completed",
            data={"method": request.method, "path": path, "status_code": response.status_code},
        )
        # endregion
    return response

# Disable cache for HTML in debug mode (avoid stale UI)
if settings.DEBUG:
    from starlette.middleware.base import BaseHTTPMiddleware
    from starlette.requests import Request
    from starlette.responses import Response

    @app.middleware("http")
    async def no_cache_html(request: Request, call_next):
        response = await call_next(request)
        if "text/html" in response.headers.get("content-type", ""):
            response.headers["Cache-Control"] = "no-store, no-cache, must-revalidate"
            response.headers["Pragma"] = "no-cache"
            response.headers["Expires"] = "0"
        return response
```

```python

# Signed-cookie sessions (replaces Flask server-side sessions)
app.add_middleware(
    SessionMiddleware,
    secret_key=settings.SECRET_KEY,
    max_age=settings.SESSION_TIMEOUT,
    same_site="lax",
    session_cookie="session",
)

# Static files
app.mount("/static", StaticFiles(directory="app/static"), name="static")


# ---------------------------------------------------------------------------
# Health / readiness
# ---------------------------------------------------------------------------

@app.get("/health", tags=["ops"])
async def health():
    """Health check. Includes app version to verify deployed code."""
    return {"status": "healthy", "version": app.version}


@app.get("/ready", tags=["ops"])
async def ready():
    if settings.FRONTEND_STUB_MODE:
        return {"status": "ready", "mode": "stub"}
    try:
        from app.utils.k8s_utils import get_core_api
        from app.utils.async_helpers import run_sync

        api = get_core_api()
        await run_sync(api.list_namespace, limit=1)
        return {"status": "ready"}
    except Exception as e:
        logger.error("Readiness check failed: %s", e)
        return {"status": "not ready", "error": str(e)}, 503


# ---------------------------------------------------------------------------
# Login page (Jinja2 template, for backward compat)
# ---------------------------------------------------------------------------

from fastapi import Request
from fastapi.responses import HTMLResponse
from app.templating import Jinja2Templates

_templates = Jinja2Templates(directory="app/templates")


def _is_api_request_path(path: str) -> bool:
    return (
        path.startswith("/api/")
        or path.startswith("/admin/api/")
        or path.startswith("/auth/")
        or path.startswith("/health")
        or path.startswith("/ready")
        or "/api/" in path
    )


def _prefers_html(request: Request) -> bool:
    accept = (request.headers.get("accept") or "").lower()
    return "text/html" in accept


@app.exception_handler(HTTPException)
async def http_exception_handler(request: Request, exc: HTTPException):
    path = request.url.path
    if _is_api_request_path(path) or not _prefers_html(request):
```

```
        return JSONResponse(
            status_code=exc.status_code,
            content={"detail": exc.detail},
        )

    return _templates.TemplateResponse(
        "error.html",
        {
            "request": request,
            "status_code": exc.status_code,
            "error_title": "Something went wrong",
            "error_message": str(exc.detail) if exc.detail else "Unexpected request error",
            "error_path": path,
        },
        status_code=exc.status_code,
    )


@app.exception_handler(Exception)
async def unhandled_exception_handler(request: Request, exc: Exception):
    logger.exception("Unhandled error for path %s: %s", request.url.path, exc)
    path = request.url.path
    if _is_api_request_path(path) or not _prefers_html(request):
        return JSONResponse(
            status_code=500,
            content={"detail": "Internal server error"},
        )

    return _templates.TemplateResponse(
        "error.html",
        {
            "request": request,
            "status_code": 500,
            "error_title": "Server error",
            "error_message": "Please try again or contact support if the issue persists.",
            "error_path": path,
        },
        status_code=500,
    )


@app.get("/login", include_in_schema=False)
async def login_page(request: Request):
    return _templates.TemplateResponse("login.html", {"request": request})


@app.get("/logout", include_in_schema=False)
async def logout_redirect(request: Request):
    """Logout: clear session and redirect to login."""
    from fastapi.responses import RedirectResponse
    request.session.clear()
    return RedirectResponse(url="/login", status_code=303)


# ---------------------------------------------------------------------------
# Register routers
# ---------------------------------------------------------------------------

from app.routers import admin, auth, dashboard, llm, storage, vm_details, vms

app.include_router(auth.router, prefix="/auth")
app.include_router(admin.router)
app.include_router(dashboard.router)
app.include_router(vms.router)
app.include_router(vm_details.router)
app.include_router(storage.router)
app.include_router(llm.router)

if __name__ == "__main__":
    import uvicorn

```

```
    host = "0.0.0.0"
    port = int(os.environ.get("PORT", "8080"))
    workers = 1
    log_level = os.environ.get("LOG_LEVEL", "info").lower()

    uvicorn.run(
        "app.main:app",
        host=host,
        port=port,
        workers=workers,
        log_level=log_level,
        proxy_headers=True,
        forwarded_allow_ips="*",
    )

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/models.py (7.3 KB)

```python
"""
Pydantic domain models for persistence (users, VM records, VM events).

These are *not* request/response schemas ? see ``schemas.py`` for those.
They mirror the DB rows and are returned by the DB layer.
"""

from __future__ import annotations

from datetime import datetime
from decimal import Decimal
from enum import Enum
from typing import Any

import asyncpg
from pydantic import BaseModel, Field, field_validator


# ---------------------------------------------------------------------------
# Enums
# ---------------------------------------------------------------------------

class UserRole(str, Enum):
    ADMIN = "admin"
    USER = "user"


class VMEventType(str, Enum):
    CREATED = "created"
    STARTED = "started"
    STOPPED = "stopped"
    RESTARTED = "restarted"
    PAUSED = "paused"
    UNPAUSED = "unpaused"
    DELETED = "deleted"
    ERROR = "error"
    BILLING_STOPPED = "billing_stopped"


class LLMApiKeyStatus(str, Enum):
    ACTIVE = "active"
    REVOKED = "revoked"


# ---------------------------------------------------------------------------
# User
# ---------------------------------------------------------------------------

class User(BaseModel):
    id: int
    username: str
    password_hash: str
    role: str = "user"
    is_active: bool = True
    balance: Decimal = Decimal("0")
    balance_version: int = 0
    created_at: datetime

    @property
    def is_admin(self) -> bool:
        return self.role == UserRole.ADMIN.value

    @classmethod
    def from_row(cls, row: asyncpg.Record) -> User:
        return cls(
```

```
            id=row["id"],
            username=row["username"],
            password_hash=row["password_hash"],
            role=row.get("role", "user"),
            is_active=row["is_active"],
            balance=Decimal(str(row["balance"])) if row.get("balance") is not None else Decimal("0"),
            balance_version=row.get("balance_version", 0),
            created_at=row["created_at"],
        )


# ---------------------------------------------------------------------------
# VM record (audit / metadata persisted alongside K8s objects)
# ---------------------------------------------------------------------------

class VMRecord(BaseModel):
    id: int
    user_id: int
    vm_name: str
    namespace: str
    cpu: int
    memory_gb: float
    storage_gb: float
    gpu_count: int = 0
    gpu_model: str | None = None
    image_url: str
    os_type: str = "linux"
    is_running: bool = False
    last_started_at: datetime | None = None
    last_stopped_at: datetime | None = None
    last_billed_at: datetime | None = None
    created_at: datetime
    deleted_at: datetime | None = None

    @classmethod
    def from_row(cls, row: asyncpg.Record) -> VMRecord:
        return cls(
            id=row["id"],
            user_id=row["user_id"],
            vm_name=row["vm_name"],
            namespace=row["namespace"],
            cpu=row["cpu"],
            memory_gb=row["memory_gb"],
            storage_gb=row["storage_gb"],
            gpu_count=row["gpu_count"],
            gpu_model=row["gpu_model"],
            image_url=row["image_url"],
            os_type=row["os_type"],
            is_running=row.get("is_running", False),
            last_started_at=row.get("last_started_at"),
            last_stopped_at=row.get("last_stopped_at"),
            last_billed_at=row.get("last_billed_at"),
            created_at=row["created_at"],
            deleted_at=row["deleted_at"],
        )


# ---------------------------------------------------------------------------
# VM event log (append-only)
# ---------------------------------------------------------------------------

class VMEvent(BaseModel):
    event_id: str
    vm_name: str
    namespace: str
    user_id: int
    event_type: VMEventType
    metadata: dict[str, Any] = Field(default_factory=dict)
    created_at: datetime

    @field_validator("metadata", mode="before")
```

```
    @classmethod
    def _parse_metadata(cls, v: Any) -> Any:
        if isinstance(v, str):
            import json
            return json.loads(v)
        return v

    @classmethod
    def from_row(cls, row: asyncpg.Record) -> VMEvent:
        return cls(
            event_id=row["event_id"],
            vm_name=row["vm_name"],
            namespace=row["namespace"],
            user_id=row["user_id"],
            event_type=VMEventType(row["event_type"]),
            metadata=row["metadata"] if row["metadata"] else {},
            created_at=row["created_at"],
        )


# ---------------------------------------------------------------------------
# LLM API keys (stored masked + encrypted)
# ---------------------------------------------------------------------------

class LLMApiKey(BaseModel):
    id: int
    user_id: int
    external_ref: str
    token_masked: str
    token_ciphertext: str
    status: LLMApiKeyStatus = LLMApiKeyStatus.ACTIVE
    created_at: datetime
    revoked_at: datetime | None = None
    last_used_at: datetime | None = None

    @property
    def is_active(self) -> bool:
        return self.status == LLMApiKeyStatus.ACTIVE

    @classmethod
    def from_row(cls, row: asyncpg.Record) -> "LLMApiKey":
        return cls(
            id=row["id"],
            user_id=row["user_id"],
            external_ref=row["external_ref"],
            token_masked=row["token_masked"],
            token_ciphertext=row["token_ciphertext"],
            status=LLMApiKeyStatus(row["status"]),
            created_at=row["created_at"],
            revoked_at=row["revoked_at"],
            last_used_at=row["last_used_at"],
        )


# ---------------------------------------------------------------------------
# LLM usage log (spend analytics source)
# ---------------------------------------------------------------------------

class LLMUsageLog(BaseModel):
    id: int
    user_id: int
    model: str
    provider: str | None = None
    prompt_tokens: int = 0
    completion_tokens: int = 0
    total_tokens: int = 0
    estimated_cost: Decimal = Decimal("0")
    currency: str = "USD"
    request_id: str | None = None
    created_at: datetime

```

```
    @classmethod
    def from_row(cls, row: asyncpg.Record) -> "LLMUsageLog":
        return cls(
            id=row["id"],
            user_id=row["user_id"],
            model=row["model"],
            provider=row["provider"],
            prompt_tokens=row["prompt_tokens"],
            completion_tokens=row["completion_tokens"],
            total_tokens=row["total_tokens"],
            estimated_cost=Decimal(str(row["estimated_cost"])),
            currency=row["currency"],
            request_id=row["request_id"],
            created_at=row["created_at"],
        )


# ---------------------------------------------------------------------------
# Balance operation ledger (admin-initiated mutations)
# ---------------------------------------------------------------------------

class BalanceOperation(BaseModel):
    id: int
    user_id: int
    source: str
    amount: Decimal
    op_type: str
    admin_username: str | None = None
    created_at: datetime

    @classmethod
    def from_row(cls, row: asyncpg.Record) -> "BalanceOperation":
        return cls(
            id=row["id"],
            user_id=row["user_id"],
            source=row["source"],
            amount=Decimal(str(row["amount"])),
            op_type=row["op_type"],
            admin_username=row["admin_username"],
            created_at=row["created_at"],
        )

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/routers/__init__.py (0.0 B)

```
1
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/routers/admin.py (23.0 KB)

```
"""Admin router: user management, VM control, and balance management (admin only)."""

from __future__ import annotations

import logging
from decimal import Decimal

from fastapi import APIRouter, Depends, HTTPException, Query, Request
from fastapi.responses import HTMLResponse
from app.templating import Jinja2Templates

from app.auth.dependencies import SessionUser, get_user_namespace, require_admin
from app.config import settings
from app.schemas import (
    BalanceTransferRequest,
    BalanceUpdateRequest,
    CreateUserRequest,
    SetRoleRequest,
    UserPublic,
)
from app.services.llm_auth_client import LLMAuthClient
from app.utils.async_helpers import run_sync
from app.utils.k8s_utils import (
    delete_virtual_machine,
    list_vms_in_namespace,
    stop_virtual_machine,
)
from app.utils.service_utils import get_vm_allocated_ip

logger = logging.getLogger(__name__)

router = APIRouter(tags=["admin"], prefix="/admin")
templates = Jinja2Templates(directory="app/templates")


@router.get("", include_in_schema=False)
async def admin_page(
    request: Request,
    user: SessionUser = Depends(require_admin),
):
    """Admin dashboard - redirect to users list."""
    return HTMLResponse("<script>location.href='/admin/users';</script>")


def _stub_users(current_username: str) -> list:
    """Stub user list when DB unavailable (e.g. stub mode)."""
    return [
        {"id": 1, "username": current_username, "role": "admin", "is_active": True, "main_balance": None, "llm_balanc...
    ]


@router.get("/users", include_in_schema=False)
async def admin_users_page(
    request: Request,
    user: SessionUser = Depends(require_admin),
):
    """User management page."""
    db = getattr(request.app.state, "db", None)
    if not db:
        if settings.FRONTEND_STUB_MODE:
            user_list = _stub_users(user.username)
        else:
            raise HTTPException(503, "Database not available")
    else:
        users = await db.users.list_all(include_inactive=True)
```

```
        llm_client = LLMAuthClient()
        llm_available = llm_client.available()

        user_list = []
        for u in users:
            entry = {
                "id": u.id,
                "username": u.username,
                "role": u.role,
                "is_active": u.is_active,
                "main_balance": float(u.balance),
                "llm_balance": None,
            }
            if llm_available:
                llm_uid = await llm_client.get_user_id(u.username)
                if llm_uid:
                    entry["llm_balance"] = await llm_client.get_balance(llm_uid)
            user_list.append(entry)

    return templates.TemplateResponse("admin_users.html", {
        "request": request,
        "username": user.username,
        "namespace": user.namespace,
        "is_admin": user.is_admin,
        "users": user_list,
        "has_db": db is not None,
    })


@router.get("/llm-spend", include_in_schema=False)
async def admin_llm_spend_page(
    request: Request,
    user: SessionUser = Depends(require_admin),
):
    """Admin spend dashboard page."""
    db = getattr(request.app.state, "db", None)
    summary = {
        "days": 30,
        "total_cost": 0.0,
        "total_vm_cost": 0.0,
        "total_tokens": 0,
        "per_user": [],
        "per_model": [],
        "per_day": [],
    }
    if db:
        if hasattr(db, "balance_snapshots"):
            summary = await db.balance_snapshots.summary_llm_spend(days=30)
        elif hasattr(db, "llm_usage_logs"):
            summary = await db.llm_usage_logs.summary(days=30)
    return templates.TemplateResponse("admin_llm_spend.html", {
        "request": request,
        "username": user.username,
        "namespace": user.namespace,
        "is_admin": user.is_admin,
        "page": "spend",
        "summary": summary,
        "spend_title": "???????",
        "spend_api_url": "/admin/api/llm-spend/summary",
        "spend_is_user_view": False,
    })


@router.get("/api/users")
async def list_users(
    request: Request,
    user: SessionUser = Depends(require_admin),
):
    """List all users with main and LLM balances."""
    db = getattr(request.app.state, "db", None)
    if not db:
```

```
        if settings.FRONTEND_STUB_MODE:
            return {"users": _stub_users(user.username)}
        raise HTTPException(503, "Database not available")

    users = await db.users.list_all(include_inactive=True)
    llm_client = LLMAuthClient()
    llm_available = llm_client.available()

    result = []
    for u in users:
        entry: dict = {
            "id": u.id,
            "username": u.username,
            "role": u.role,
            "is_active": u.is_active,
            "main_balance": float(u.balance),
            "llm_balance": None,
            "llm_user_id": None,
        }
        if llm_available:
            llm_uid = await llm_client.get_user_id(u.username)
            if llm_uid:
                entry["llm_user_id"] = llm_uid
                entry["llm_balance"] = await llm_client.get_balance(llm_uid)
        result.append(entry)

    return {"users": result}


@router.get("/api/llm-spend/summary")
async def admin_llm_spend_summary(
    request: Request,
    days: int = Query(default=30, ge=1, le=365),
    user: SessionUser = Depends(require_admin),
):
    """Return aggregated LLM spend stats for admin dashboard."""
    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")
    if hasattr(db, "balance_snapshots"):
        return await db.balance_snapshots.summary_llm_spend(days=days)
    if hasattr(db, "llm_usage_logs"):
        return await db.llm_usage_logs.summary(days=days)
    return {
        "days": days,
        "total_cost": 0.0,
        "total_vm_cost": 0.0,
        "total_tokens": 0,
        "per_user": [],
        "per_model": [],
        "per_day": [],
    }


@router.post("/api/users")
async def create_user(
    request: Request,
    body: CreateUserRequest,
    user: SessionUser = Depends(require_admin),
):
    """Create new user. Rejects if username already exists in viirtuoz or in LLM auth-service."""
    import bcrypt

    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")

    existing = await db.users.get_by_username(body.username)
    if existing:
        raise HTTPException(400, "Username already exists")

```

```
    llm_client = LLMAuthClient()
    if llm_client.available():
        existing_llm_uid = await llm_client.get_user_id(body.username)
        if existing_llm_uid:
            raise HTTPException(
                400,
                "User with this username already exists in the system (LLM auth)",
            )

    pw_hash = bcrypt.hashpw(body.password.encode("utf-8"), bcrypt.gensalt()).decode("utf-8")
    new_user = await db.users.create_user(body.username, pw_hash, role=body.role)
    logger.info("Admin %s created user %s (role=%s)", user.username, new_user.username, new_user.role)
    return {"id": new_user.id, "username": new_user.username, "role": new_user.role}


@router.patch("/api/users/{user_id}/role")
async def set_user_role(
    request: Request,
    user_id: int,
    body: SetRoleRequest,
    user: SessionUser = Depends(require_admin),
):
    """Change user role."""
    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")

    target = await db.users.get_by_id(user_id)
    if not target:
        raise HTTPException(404, "User not found")

    if target.id == user.user_id and body.role != "admin":
        raise HTTPException(400, "Cannot demote yourself")

    ok = await db.users.set_role(user_id, body.role)
    if not ok:
        raise HTTPException(500, "Failed to update role")
    return {"username": target.username, "role": body.role}


@router.post("/api/users/{user_id}/deactivate")
async def deactivate_user(
    request: Request,
    user_id: int,
    user: SessionUser = Depends(require_admin),
):
    """Deactivate user."""
    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")

    if user_id == user.user_id:
        raise HTTPException(400, "Cannot deactivate yourself")

    target = await db.users.get_by_id(user_id)
    if not target:
        raise HTTPException(404, "User not found")

    ok = await db.users.deactivate(user_id)
    if not ok:
        raise HTTPException(500, "Failed to deactivate")
    return {"username": target.username, "deactivated": True}


# ---------------------------------------------------------------------------
# Admin VM management
# ---------------------------------------------------------------------------

async def _get_target_user_or_stub(db, user_id: int, current_user: SessionUser):
    """Get target user from DB, or use current user in stub mode."""
    if db:
```

```
        target = await db.users.get_by_id(user_id)
        if target:
            return target.username, target.id
    if settings.FRONTEND_STUB_MODE and user_id == current_user.user_id:
        return current_user.username, current_user.user_id
    return None, None


@router.get("/api/users/{user_id}/vms")
async def list_user_vms(
    request: Request,
    user_id: int,
    user: SessionUser = Depends(require_admin),
):
    """List VMs for a user (admin only)."""
    db = getattr(request.app.state, "db", None)
    target_username, target_id = await _get_target_user_or_stub(db, user_id, user)
    if not target_username:
        raise HTTPException(404, "User not found")

    try:
        namespace = get_user_namespace(target_username, target_id)
    except ValueError as e:
        raise HTTPException(400, str(e))
    vms_raw = await run_sync(list_vms_in_namespace, namespace)

    vm_list = []
    for vm in vms_raw:
        vm_name = vm.get("metadata", {}).get("name", "Unknown")
        if vm.get("datavolume_status"):
            vm_status = vm["datavolume_status"]
            running = False
        elif vm.get("status", {}).get("printableStatus"):
            vm_status = vm["status"]["printableStatus"]
            running = vm_status == "Running"
        else:
            vm_status = "Unknown"
            running = False

        spec = vm.get("spec", {}).get("template", {}).get("spec", {})
        domain = spec.get("domain", {})
        cpu = domain.get("cpu", {}).get("cores")
        memory = domain.get("resources", {}).get("requests", {}).get("memory")

        gpu_devices = domain.get("devices", {}).get("gpus", [])
        gpu_count = len(gpu_devices)
        gpu_model = gpu_devices[0].get("deviceName", "") if gpu_devices else None

        allocated_ip = await run_sync(get_vm_allocated_ip, namespace, vm_name)

        vm_list.append({
            "name": vm_name,
            "status": vm_status,
            "running": running,
            "cpu": cpu,
            "memory": memory,
            "gpu_count": gpu_count,
            "gpu_model": gpu_model,
            "allocated_ip": allocated_ip,
        })

    return {
        "success": True,
        "vms": vm_list,
        "username": target_username,
        "namespace": namespace,
        "stub_mode": settings.FRONTEND_STUB_MODE,
    }


@router.post("/api/users/{user_id}/vm/{vm_name}/stop")
```

```
350 async def admin_stop_vm(
351     request: Request,
352     user_id: int,
353     vm_name: str,
354     user: SessionUser = Depends(require_admin),
355 ):
356     """Stop user's VM (admin only)."""
357     db = getattr(request.app.state, "db", None)
358     target_username, target_id = await _get_target_user_or_stub(db, user_id, user)
359     if not target_username:
360         raise HTTPException(404, "User not found")
361 
362     try:
363         namespace = get_user_namespace(target_username, target_id)
364     except ValueError as e:
365         raise HTTPException(400, str(e))
366     ok = await run_sync(stop_virtual_machine, namespace, vm_name)
367     if not ok:
368         raise HTTPException(500, "Failed to stop VM")
369 
370     if db:
371         try:
372             await db.vm_records.set_running(namespace, vm_name, False)
373         except Exception:
374             pass
375 
376     logger.info("Admin %s stopped VM %s for user %s", user.username, vm_name, target_username)
377     return {"success": True, "message": f"VM {vm_name} stopping"}
378 
379 
380 @router.post("/api/users/{user_id}/vm/{vm_name}/delete")
381 async def admin_delete_vm(
382     request: Request,
383     user_id: int,
384     vm_name: str,
385     user: SessionUser = Depends(require_admin),
386 ):
387     """Delete user's VM (admin only)."""
388     db = getattr(request.app.state, "db", None)
389     target_username, target_id = await _get_target_user_or_stub(db, user_id, user)
390     if not target_username:
391         raise HTTPException(404, "User not found")
392 
393     try:
394         namespace = get_user_namespace(target_username, target_id)
395     except ValueError as e:
396         raise HTTPException(400, str(e))
397     ok = await run_sync(delete_virtual_machine, namespace, vm_name)
398     if not ok:
399         raise HTTPException(500, "Failed to delete VM")
400 
401     try:
402         from app.utils.service_utils import delete_vm_service
403         await run_sync(delete_vm_service, namespace, vm_name)
404     except Exception as e:
405         logger.warning("Service cleanup failed for %s: %s", vm_name, e)
406 
407     if db:
408         try:
409             await db.vm_records.set_running(namespace, vm_name, False)
410             await db.vm_records.mark_deleted(namespace, vm_name)
411         except Exception:
412             pass
413 
414     logger.info("Admin %s deleted VM %s for user %s", user.username, vm_name, target_username)
415     return {"success": True, "message": f"VM {vm_name} deleted"}
416 
417 
418 # ----------------------------------------------------------------------------
419 # Balance management
420 # ----------------------------------------------------------------------------
```

```python

def _require_db(request: Request):
    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")
    return db


async def _snapshot_balance(db, user_id: int, source: str, balance: Decimal | float | int) -> None:
    """Best-effort balance snapshot for dashboard analytics."""
    snapshots = getattr(db, "balance_snapshots", None)
    if not snapshots:
        return
    try:
        await snapshots.create(user_id=user_id, source=source, balance=balance)
    except Exception as exc:
        logger.warning(
            "Failed to write %s balance snapshot for user %s: %s",
            source,
            user_id,
            exc,
        )


async def _log_balance_op(
    db, user_id: int, source: str, amount: Decimal, op_type: str, admin_username: str,
) -> None:
    """Best-effort ledger entry for an admin balance mutation."""
    ops = getattr(db, "balance_operations", None)
    if not ops:
        return
    try:
        await ops.create(
            user_id=user_id, source=source, amount=amount,
            op_type=op_type, admin_username=admin_username,
        )
    except Exception as exc:
        logger.warning(
            "Failed to log %s balance operation for user %s: %s",
            source, user_id, exc,
        )


@router.get("/api/users/{user_id}/balance")
async def get_user_balance(
    request: Request,
    user_id: int,
    admin: SessionUser = Depends(require_admin),
):
    """Get both main and LLM balances for a user, plus cumulative admin-administered totals."""
    db = _require_db(request)
    target = await db.users.get_by_id(user_id)
    if not target:
        raise HTTPException(404, "User not found")

    result: dict = {
        "username": target.username,
        "main_balance": float(target.balance),
        "llm_balance": None,
        "llm_user_id": None,
        "main_total_administered": 0.0,
        "llm_total_administered": 0.0,
    }

    ops = getattr(db, "balance_operations", None)
    if ops:
        result["main_total_administered"] = float(
            await ops.get_cumulative(user_id, "main")
        )
        result["llm_total_administered"] = float(
            await ops.get_cumulative(user_id, "llm")
```

```
        )

    llm_client = LLMAuthClient()
    if llm_client.available():
        llm_uid = await llm_client.get_user_id(target.username)
        if llm_uid:
            result["llm_user_id"] = llm_uid
            result["llm_balance"] = await llm_client.get_balance(llm_uid)


... [187 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/routers/dashboard.py (14.9 KB)

```python
"""
Dashboard page and VM list API.
"""

from __future__ import annotations

import logging
from decimal import Decimal

from fastapi import APIRouter, Depends, HTTPException, Query, Request
from fastapi.responses import HTMLResponse, RedirectResponse
from app.templating import Jinja2Templates

from app.auth.dependencies import SessionUser, get_current_namespace, get_current_user, require_owner
from app.config import (
    NVIDIA_DRIVER_VERSIONS,
    NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER,
    settings,
)
from app.schemas import BalanceTransferRequest
from app.services.llm_auth_client import LLMAuthClient
from app.utils.async_helpers import run_sync
from app.utils.k8s_utils import discover_gpu_resources, list_vms_in_namespace
from app.utils.vm_utils import format_gpu_display_name, resolve_gpu_resource_name

logger = logging.getLogger(__name__)

router = APIRouter(tags=["dashboard"])

templates = Jinja2Templates(directory="app/templates")


def _require_db(request: Request):
    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")
    return db


async def _snapshot_balance(db, user_id: int, source: str, balance: Decimal | float | int) -> None:
    """Best-effort balance snapshot for analytics."""
    snapshots = getattr(db, "balance_snapshots", None)
    if not snapshots:
        return
    try:
        await snapshots.create(user_id=user_id, source=source, balance=balance)
    except Exception as exc:
        logger.warning(
            "Failed to write %s balance snapshot for user %s: %s",
            source,
            user_id,
            exc,
        )


async def _log_balance_op(
    db, user_id: int, source: str, amount: Decimal, op_type: str, actor_username: str,
) -> None:
    """Best-effort ledger entry for user/admin balance mutation."""
    ops = getattr(db, "balance_operations", None)
    if not ops:
        return
    try:
        await ops.create(
            user_id=user_id,
```

```
            source=source,
            amount=amount,
            op_type=op_type,
            admin_username=actor_username,
        )
    except Exception as exc:
        logger.warning(
            "Failed to log %s balance operation for user %s: %s",
            source,
            user_id,
            exc,
        )


@router.get("/", include_in_schema=False)
async def index():
    return RedirectResponse(url="/login")


@router.get("/dashboard", include_in_schema=False)
async def dashboard_redirect(request: Request):
    username = request.session.get("username")
    if username:
        return RedirectResponse(url=f"/{username}/dashboard")
    return RedirectResponse(url="/login")


@router.get("/{username}/api/balance", include_in_schema=False)
async def get_my_balance(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
):
    """Return current user's main and LLM balance (for sidebar display)."""
    db = getattr(request.app.state, "db", None)
    main_balance = None
    if db:
        target = await db.users.get_by_username(username)
        if target:
            bal = await db.users.get_balance(target.id)
            if bal:
                main_balance = float(bal[0])
    llm_balance = None
    llm_client = LLMAuthClient()
    if llm_client.available():
        llm_uid = await llm_client.get_user_id(username)
        if llm_uid:
            llm_balance = await llm_client.get_balance(llm_uid)
            if llm_balance is not None:
                llm_balance = float(llm_balance)
    return {"main_balance": main_balance, "llm_balance": llm_balance}


@router.post("/{username}/api/balance/transfer", include_in_schema=False)
async def transfer_my_balance(
    request: Request,
    username: str,
    body: BalanceTransferRequest,
    user: SessionUser = Depends(require_owner),
):
    """Transfer funds between own main and LLM balances."""
    db = _require_db(request)
    target = await db.users.get_by_id(user.user_id)
    if not target:
        raise HTTPException(404, "User not found")

    llm_client = LLMAuthClient()
    if not llm_client.available():
        raise HTTPException(503, "LLM auth-service not configured")

    llm_uid = await llm_client.ensure_user(username)
```

```
    if not llm_uid:
        raise HTTPException(502, "Could not resolve user in LLM auth-service")

    if body.source == "main" and body.target == "llm":
        bal = await db.users.get_balance(user.user_id)
        if bal is None:
            raise HTTPException(404, "User not found")
        current_balance, version = bal

        amount = current_balance if body.transfer_all else Decimal(str(body.amount))
        if amount <= 0:
            raise HTTPException(400, "Nothing to transfer")
        if current_balance < amount:
            raise HTTPException(400, "Insufficient main balance")

        try:
            new_main, new_version = await db.users.update_balance(user.user_id, -amount, version)
        except ValueError as exc:
            raise HTTPException(400, str(exc))

        new_llm = await llm_client.adjust_balance(llm_uid, amount)
        if new_llm is None:
            await db.users.update_balance(user.user_id, amount, new_version)
            raise HTTPException(502, "LLM auth-service unreachable ? transfer rolled back")

        await _snapshot_balance(db, user.user_id, "main", new_main)
        await _snapshot_balance(db, user.user_id, "llm", new_llm)
        await _log_balance_op(db, user.user_id, "main", -amount, "transfer_out", user.username)
        await _log_balance_op(db, user.user_id, "llm", amount, "transfer_in", user.username)
        return {"main_balance": float(new_main), "llm_balance": new_llm}

    if body.source == "llm" and body.target == "main":
        current_llm = await llm_client.get_balance(llm_uid)
        if current_llm is None:
            raise HTTPException(502, "Could not read LLM balance")

        amount = Decimal(str(current_llm)) if body.transfer_all else Decimal(str(body.amount))
        if amount <= 0:
            raise HTTPException(400, "Nothing to transfer")
        if current_llm < float(amount):
            raise HTTPException(400, "Insufficient LLM balance")

        new_llm = await llm_client.adjust_balance(llm_uid, -amount)
        if new_llm is None:
            raise HTTPException(502, "LLM auth-service deduction failed")

        bal = await db.users.get_balance(user.user_id)
        if bal is None:
            await llm_client.adjust_balance(llm_uid, amount)
            raise HTTPException(404, "User not found")
        _, version = bal

        try:
            new_main, _ = await db.users.update_balance(user.user_id, amount, version)
        except ValueError:
            await llm_client.adjust_balance(llm_uid, amount)
            raise HTTPException(500, "Local balance update failed ? transfer rolled back")

        await _snapshot_balance(db, user.user_id, "llm", new_llm)
        await _snapshot_balance(db, user.user_id, "main", new_main)
        await _log_balance_op(db, user.user_id, "llm", -amount, "transfer_out", user.username)
        await _log_balance_op(db, user.user_id, "main", amount, "transfer_in", user.username)
        return {"main_balance": float(new_main), "llm_balance": new_llm}

    raise HTTPException(400, "Invalid transfer direction")


@router.get("/{username}/dashboard", include_in_schema=False)
async def dashboard(
    request: Request,
    username: str,
```

```
    user: SessionUser = Depends(get_current_user),
    user_ns: str = Depends(get_current_namespace),
):
    if user.username != username:
        return RedirectResponse(url="/login")

    return templates.TemplateResponse("dashboard_home.html", {
        "request": request,
        "username": user.username,
        "namespace": user_ns,
        "is_admin": user.is_admin,
        "page": "dashboard",
    })


@router.get("/{username}/vms", include_in_schema=False)
async def vms_page(
    request: Request,
    username: str,
    user: SessionUser = Depends(get_current_user),
    user_ns: str = Depends(get_current_namespace),
):
    if user.username != username:
        return RedirectResponse(url="/login")

    vms = await run_sync(list_vms_in_namespace, user_ns)

    discovered = await run_sync(discover_gpu_resources)
    default_gpu = resolve_gpu_resource_name()
    gpu_resources = sorted(
        set(discovered or [default_gpu])
        | set(NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER)
    ) or [default_gpu]
    gpu_labels = {res: format_gpu_display_name(res) for res in gpu_resources}
    if default_gpu not in gpu_labels:
        gpu_labels[default_gpu] = format_gpu_display_name(default_gpu)

    return templates.TemplateResponse("dashboard.html", {
        "request": request,
        "username": user.username,
        "namespace": user_ns,
        "is_admin": user.is_admin,
        "page": "vms",
        "vms": vms,
        "gpu_resources": gpu_resources,
        "gpu_resources_custom_driver": NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER,
        "default_gpu_resource": default_gpu,
        "gpu_labels": gpu_labels,
        "nvidia_driver_versions": NVIDIA_DRIVER_VERSIONS,
        "default_nvidia_driver": settings.DEFAULT_NVIDIA_DRIVER,
        "max_cpu_cores": settings.MAX_CPU_CORES,
        "max_memory_gb": settings.MAX_MEMORY_GB,
        "max_storage_gb": settings.MAX_STORAGE_GB,
        "max_gpu_count": settings.MAX_GPU_COUNT,
    })


@router.get("/{username}/dashboard/vms")
async def get_dashboard_vms(
    username: str,
    user: SessionUser = Depends(get_current_user),
    user_ns: str = Depends(get_current_namespace),
):
    if user.username != username:
        return {"error": "Unauthorized"}, 401

    vms_raw = await run_sync(list_vms_in_namespace, user_ns)

    from app.utils.service_utils import get_vm_allocated_ip

    vm_list = []
```

```
    for vm in vms_raw:
        vm_name = vm.get("metadata", {}).get("name", "Unknown")

        dv_st = vm.get("datavolume_status")
        printable = vm.get("status", {}).get("printableStatus") or ""
        if dv_st:
            vm_status = "Provisioning"
            running = False
        elif printable in ("WaitingForVolumeBinding", "WaitingForDataVolume"):
            vm_status = "Provisioning"
            running = False
        elif printable:
            vm_status = printable
            running = vm_status == "Running"
        else:
            vm_status = "Unknown"
            running = False

        spec = vm.get("spec", {}).get("template", {}).get("spec", {})
        domain = spec.get("domain", {})
        resources = domain.get("resources", {}).get("requests", {})
        cpu = domain.get("cpu", {}).get("cores")
        memory = resources.get("memory")

        gpu_devices = domain.get("devices", {}).get("gpus", [])
        gpu_count = len(gpu_devices)
        gpu_model = gpu_devices[0].get("deviceName", "") if gpu_devices else None

        vm_data: dict = {
            "name": vm_name,
            "status": vm_status,
            "running": running,
            "cpu": cpu,
            "memory": memory,
        }
        if gpu_count > 0:
            vm_data["gpu_count"] = gpu_count
            vm_data["gpu_model"] = gpu_model

        vm_os_type = (
            vm.get("metadata", {}).get("labels", {}).get("vm.kubevirt.io/os", "linux")
        )
        vm_data["os"] = vm_os_type

        allocated_ip = await run_sync(get_vm_allocated_ip, user_ns, vm_name)
        if allocated_ip:
            vm_data["allocated_ip"] = allocated_ip

        vm_list.append(vm_data)

    return {"success": True, "vms": vm_list, "stub_mode": settings.FRONTEND_STUB_MODE}


@router.get("/{username}/spend", include_in_schema=False)
async def my_spend_page(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
):
    """User spend dashboard (own data only)."""
    db = getattr(request.app.state, "db", None)
    summary = {
        "days": 30,
        "total_cost": 0.0,
        "total_vm_cost": 0.0,
        "total_tokens": 0,
        "per_user": [{"username": user.username, "total_cost": 0.0, "vm_total_cost": 0.0}],
        "per_model": [],
        "per_day": [],
    }
    if db:
```

```
        if hasattr(db, "balance_snapshots"):
            summary = await db.balance_snapshots.summary_llm_spend(days=30, user_id=user.user_id)
        elif hasattr(db, "llm_usage_logs"):
            legacy = await db.llm_usage_logs.summary(days=30)
            own = next((r for r in legacy.get("per_user", []) if r.get("username") == user.username), None)
            summary = {
                "days": 30,
                "total_cost": float((own or {}).get("total_cost", 0.0)),
                "total_vm_cost": 0.0,
                "total_tokens": 0,
                "per_user": [{
                    "username": user.username,
                    "total_cost": float((own or {}).get("total_cost", 0.0)),
                    "vm_total_cost": 0.0,
                }],
                "per_model": [],
                "per_day": [],
            }
    return templates.TemplateResponse("admin_llm_spend.html", {
        "request": request,
        "username": user.username,
        "namespace": user.namespace,
        "is_admin": user.is_admin,
        "page": "spend",
        "summary": summary,
        "spend_title": "???????",
        "spend_api_url": f"/{user.username}/api/spend/summary",
        "spend_is_user_view": True,
    })


@router.get("/{username}/api/spend/summary", include_in_schema=False)
async def my_spend_summary(
    request: Request,
    username: str,
    days: int = Query(default=30, ge=1, le=365),
    user: SessionUser = Depends(require_owner),
):
    """Return user spend summary (own data only)."""
    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")
    if hasattr(db, "balance_snapshots"):
        return await db.balance_snapshots.summary_llm_spend(days=days, user_id=user.user_id)
    if hasattr(db, "llm_usage_logs"):
        legacy = await db.llm_usage_logs.summary(days=days)
        own = next((r for r in legacy.get("per_user", []) if r.get("username") == user.username), None)
        return {
            "days": days,
            "total_cost": float((own or {}).get("total_cost", 0.0)),
            "total_vm_cost": 0.0,
            "total_tokens": 0,
            "per_user": [{
                "username": user.username,
                "total_cost": float((own or {}).get("total_cost", 0.0)),
                "vm_total_cost": 0.0,
            }],
            "per_model": [],
            "per_day": [],
        }
    return {
        "days": days,
        "total_cost": 0.0,
        "total_vm_cost": 0.0,
        "total_tokens": 0,
        "per_user": [],
        "per_model": [],
        "per_day": [],
    }

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/routers/llm.py (19.9 KB)

```
"""LLM chat, keys management and models routes."""

from __future__ import annotations

import json
import logging
import time
import uuid
from decimal import Decimal
from typing import Any

import httpx
from fastapi import APIRouter, Depends, HTTPException, Request
from fastapi.responses import RedirectResponse, StreamingResponse
from app.templating import Jinja2Templates

from app.auth.dependencies import SessionUser, get_current_namespace, require_owner
from app.config import settings
from app.debug_log import write_debug_log
from app.schemas import LLMChatRequest
from app.services.llm_auth_client import LLMAuthClient
from app.utils.llm_key_crypto import decrypt_token, encrypt_token, mask_token

logger = logging.getLogger(__name__)

router = APIRouter(tags=["llm"])
templates = Jinja2Templates(directory="app/templates")

# ??? ????????? LLM-??????? (??????????? ?????? 5 ???)
_llm_models_cache: tuple[list[dict[str, Any]], float] | None = None
_LLM_MODELS_CACHE_TTL = 300  # 5 ?????


async def _fetch_available_llm_models() -> list[dict[str, Any]]:
    """????????? ?????? LLM ?? llm-service /v1/models."""
    base = (settings.LLM_SERVICE_URL or "").rstrip("/")
    if not base:
        logger.warning("available-models: LLM_SERVICE_URL is empty, returning []")
        return []
    shared_token = (settings.LLM_CHAT_SHARED_TOKEN or "").strip()
    headers = {"Authorization": f"Bearer {shared_token}"} if shared_token else None

    async with httpx.AsyncClient(timeout=15) as client:
        try:
            r = await client.get(f"{base}/health", headers=headers)
            logger.info("LLM service %s/health: HTTP %d", base, r.status_code)
        except Exception as e:
            logger.warning("LLM service %s unreachable: %s", base, e)

        try:
            r = await client.get(f"{base}/v1/models", headers=headers)
            if r.status_code != 200:
                logger.warning("LLM /v1/models: HTTP %d %s", r.status_code, r.text[:200])
                return []
            data = r.json()
        except Exception as e:
            logger.warning("LLM /v1/models failed: %s", e)
            return []

    items: list[Any]
    if isinstance(data, list):
        items = data
    elif isinstance(data, dict):
        items = data.get("data") or data.get("models") or data.get("items") or data.get("results") or []
    else:
```

```
        items = []

    result: list[dict[str, Any]] = []
    for m in items:
        if isinstance(m, dict):
            row = dict(m)
            row.setdefault("_source", "llm")
            result.append(row)
    return result


def _llm_service_base() -> str:
    """LLM service base URL without trailing slash."""
    url = (settings.LLM_SERVICE_URL or "").rstrip("/")
    return url


def _require_db(request: Request):
    db = getattr(request.app.state, "db", None)
    if not db:
        raise HTTPException(503, "Database not available")
    return db


def _shared_chat_token() -> str:
    return (settings.LLM_CHAT_SHARED_TOKEN or "").strip()


async def _ensure_first_user_api_key(
    request: Request,
    username: str,
    user: SessionUser,
) -> str | None:
    """Return an active decrypted user key; create the first one if absent."""
    db = getattr(request.app.state, "db", None)
    if not db or not hasattr(db, "llm_api_keys"):
        return None

    secret = (settings.LLM_KEYS_ENCRYPTION_KEY or "").strip()
    if not secret:
        logger.warning("llm-models: LLM_KEYS_ENCRYPTION_KEY is not configured")
        return None

    try:
        keys = await db.llm_api_keys.list_for_user(user.user_id)
        for key in keys:
            if not key.is_active:
                continue
            try:
                return decrypt_token(key.token_ciphertext, secret)
            except Exception as e:
                logger.warning("llm-models: failed to decrypt key %s: %s", key.id, e)

        if settings.LLM_USER_KEYS_MAX > 0:
            active_count = await db.llm_api_keys.count_active_for_user(user.user_id)
            if active_count >= settings.LLM_USER_KEYS_MAX:
                logger.warning(
                    "llm-models: cannot auto-create key, limit reached (%s) for user_id=%s",
                    settings.LLM_USER_KEYS_MAX,
                    user.user_id,
                )
                return None

        llm_client = LLMAuthClient()
        if not llm_client.available():
            logger.warning("llm-models: LLM auth-service not configured")
            return None
        token_data = await llm_client.create_token_details(username)
        if not token_data:
            logger.warning("llm-models: auth-service create-token failed")
            return None
```

```python

        token_value = token_data["token"]
        external_ref = token_data["external_ref"]
        await db.llm_api_keys.create(
            user_id=user.user_id,
            external_ref=external_ref,
            token_masked=mask_token(token_value),
            token_ciphertext=encrypt_token(token_value, secret),
        )
        return token_value
    except Exception as e:
        logger.warning("llm-models: ensure first key failed: %s", e)
        return None


def _user_message_from_status(status_code: int, body: dict | None) -> str:
    """User-facing message from auth/llm-service status code and response body."""
    if body and isinstance(body.get("error"), dict):
        msg = body["error"].get("message") or body["error"].get("detail")
        code = body["error"].get("code", "")
        if code == "insufficient_balance":
            return "Insufficient balance. Please top up your account."
        if "quota" in (msg or "").lower() or "billing" in (msg or "").lower():
            return "Insufficient balance. Check billing."
    if status_code == 401:
        return "Invalid or expired key. Check your token."
    if status_code == 403:
        return "Access denied. Check account status."
    if status_code == 402:
        return "Insufficient balance. Please top up your account."
    if status_code == 429:
        return "Rate limit exceeded or insufficient funds."
    if status_code >= 500:
        return "LLM server error. Try again later."
    return "LLM access error."


@router.get("/{username}/llm", include_in_schema=False)
async def llm_chat_page(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    # region agent log
    write_debug_log(
        run_id="run1",
        hypothesis_id="H1",
        location="app/routers/llm.py:llm_chat_page",
        message="llm page route hit",
        data={"method": request.method, "path": request.url.path, "username": username},
    )
    # endregion
    llm_display = settings.LLM_MODEL.split("/")[-1] if "/" in settings.LLM_MODEL else settings.LLM_MODEL
    return templates.TemplateResponse("llm_chat.html", {
        "request": request,
        "username": username,
        "namespace": user_ns,
        "is_admin": user.is_admin,
        "page": "llm",
        "llm_model": llm_display,
        "llm_default_model": settings.LLM_MODEL,
    })


@router.get("/{username}/llm/chat", include_in_schema=False)
async def llm_chat_page_alias(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
```

```
):
    # region agent log
    write_debug_log(
        run_id="post-fix",
        hypothesis_id="H1",
        location="app/routers/llm.py:llm_chat_page_alias",
        message="llm chat alias route hit, redirecting",
        data={"method": request.method, "path": request.url.path, "username": username},
    )
    # endregion
    return RedirectResponse(url=f"/{username}/llm", status_code=307)


@router.get("/{username}/api/llm/available-models", include_in_schema=False)
async def get_available_llm_models(
    user: SessionUser = Depends(require_owner),
):
    """?????? ????????? LLM ?? llm-service /v1/models (??? 5 ???)."""
    global _llm_models_cache
    now = time.monotonic()
    if _llm_models_cache is not None:
        cached, cached_at = _llm_models_cache
        if now - cached_at < _LLM_MODELS_CACHE_TTL:
            return {"models": cached, "cached": True}
    models = await _fetch_available_llm_models()
    logger.info("available-models: fetched %d models (LLM_SERVICE_URL=%s)", len(models), settings.LLM_SERVICE_URL or ...
    _llm_models_cache = (models, now)
    return {"models": models, "cached": False}


@router.get("/{username}/llm/models", include_in_schema=False)
async def llm_models_page(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    """???????? «????????? ??????»."""
    llm_example_api_key = await _ensure_first_user_api_key(request, username, user)
    return templates.TemplateResponse("llm_models.html", {
        "request": request,
        "username": username,
        "namespace": user_ns,
        "is_admin": user.is_admin,
        "page": "llm",
        "llm_example_api_key": llm_example_api_key or "",
    })


@router.get("/{username}/llm/keys", include_in_schema=False)
async def llm_keys_page(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    """???????? «????? API»."""
    return templates.TemplateResponse("llm_keys.html", {
        "request": request,
        "username": username,
        "namespace": user_ns,
        "is_admin": user.is_admin,
        "page": "llm",
        "llm_user_keys_max": settings.LLM_USER_KEYS_MAX,
    })


@router.get("/{username}/llm/status", include_in_schema=False)
async def llm_status(user: SessionUser = Depends(require_owner)):
    """Compatibility endpoint: chat always uses shared token."""
    return {"sk_mode": False, "connected": True, "create_token_available": True}
```

```python


@router.get("/{username}/api/llm/keys", include_in_schema=False)
async def llm_keys_list(
    request: Request,
    user: SessionUser = Depends(require_owner),
):
    db = _require_db(request)
    keys = await db.llm_api_keys.list_for_user(user.user_id)
    return {
        "keys": [
            {
                "id": k.id,
                "external_ref": k.external_ref,
                "token_masked": k.token_masked,
                "status": k.status.value,
                "created_at": k.created_at.isoformat(),
                "revoked_at": k.revoked_at.isoformat() if k.revoked_at else None,
                "last_used_at": k.last_used_at.isoformat() if k.last_used_at else None,
            }
            for k in keys
        ],
        "max_keys": settings.LLM_USER_KEYS_MAX,
    }


@router.post("/{username}/api/llm/keys", include_in_schema=False)
async def llm_keys_create(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
):
    db = _require_db(request)
    if settings.LLM_USER_KEYS_MAX > 0:
        active_count = await db.llm_api_keys.count_active_for_user(user.user_id)
        if active_count >= settings.LLM_USER_KEYS_MAX:
            raise HTTPException(400, f"API keys limit reached ({settings.LLM_USER_KEYS_MAX})")

    llm_client = LLMAuthClient()
    if not llm_client.available():
        raise HTTPException(503, "LLM auth-service not configured")
    token_data = await llm_client.create_token_details(username)
    if not token_data:
        raise HTTPException(502, "Auth-service create-token failed")
    token_value = token_data["token"]
    external_ref = token_data["external_ref"]

    secret = (settings.LLM_KEYS_ENCRYPTION_KEY or "").strip()
    if not secret:
        raise HTTPException(503, "LLM_KEYS_ENCRYPTION_KEY is not configured")

    row = await db.llm_api_keys.create(
        user_id=user.user_id,
        external_ref=external_ref,
        token_masked=mask_token(token_value),
        token_ciphertext=encrypt_token(token_value, secret),
    )
    return {
        "id": row.id,
        "token": token_value,  # show once to user
        "token_masked": row.token_masked,
        "status": row.status.value,
        "created_at": row.created_at.isoformat(),
    }


@router.get("/{username}/api/llm/keys/{key_id}/reveal", include_in_schema=False)
async def llm_keys_reveal(
    request: Request,
    key_id: int,
    user: SessionUser = Depends(require_owner),
```

```
350 ):
351     """???????? ?????? ???? (??????????? ?? ??). ?????? ????????."""
352     db = _require_db(request)
353     key = await db.llm_api_keys.get_by_id_for_user(user.user_id, key_id)
354     if not key:
355         raise HTTPException(404, "Key not found")
356     if not key.is_active:
357         raise HTTPException(400, "Cannot reveal revoked key")
358     secret = (settings.LLM_KEYS_ENCRYPTION_KEY or "").strip()
359     if not secret:
360         raise HTTPException(503, "LLM_KEYS_ENCRYPTION_KEY is not configured")
361     try:
362         token = decrypt_token(key.token_ciphertext, secret)
363     except Exception as e:
364         logger.warning("llm_keys_reveal decrypt failed: %s", e)
365         raise HTTPException(500, "Failed to decrypt key")
366     return {"token": token}
367
368
369 @router.post("/{username}/llm/chat")
370 async def llm_chat_api(
371     request: Request,
372     username: str,
373     body: LLMChatRequest,
374     user: SessionUser = Depends(require_owner),
375 ):
376     # region agent log
377     write_debug_log(
378         run_id="run1",
379         hypothesis_id="H3",
380         location="app/routers/llm.py:llm_chat_api",
381         message="llm chat api route hit",
382         data={
383             "method": request.method,
384             "path": request.url.path,
385             "username": username,
386             "messages_count": len(body.messages),
387             "model": body.model or "",
388         },
389     )
390     # endregion
391     system_message = {
392         "role": "system",
393         "content": (
394             "?? AI-????????? ????????? ???????????? ????????.\n"
395             "??????? ?? ??????? ?????, ???? ??????????? ?? ???????? ?????.\n"
396             "?????? ?????? ? ??????????????, Linux, DevOps, Kubernetes.\n"
397             "?????????? ?????? ? ????????????? Markdown."
398         ),
399     }
400     full_messages = [system_message] + body.messages
401
402     base_url = _llm_service_base()
403     request_id = str(uuid.uuid4())
404
405     async def generate():
406         api_url: str
407         auth_header: str
408         provider = "llm-service"
409         if base_url:
410             token = _shared_chat_token()
411             if not token:
412                 yield f"data: {json.dumps({'error': 'LLM_CHAT_SHARED_TOKEN is not configured'})}\n\n"
413                 return
414             api_url = f"{base_url}/v1/chat/completions"
415             auth_header = f"Bearer {token}"
416         else:
417             token = _shared_chat_token()
418             if not token:
419                 if settings.FRONTEND_STUB_MODE:
420                     yield f"data: {json.dumps({'content': '?? **Stub Mode**: LLM chat token is missing.'})}\n\n"
```

```
                    yield "data: [DONE]\n\n"
                    return
                yield f"data: {json.dumps({'error': 'LLM_CHAT_SHARED_TOKEN is not configured'})}\n\n"
                return
            api_url = settings.LLM_API_URL
            auth_header = f"Bearer {token}"
            provider = "legacy-llm-api"

        model_id = (body.model or "").strip() or settings.LLM_MODEL
        if not model_id:
            model_id = "gpt-4o-mini"  # fallback if no default

        prompt_tokens = 0
        completion_tokens = 0
        total_tokens = 0
        estimated_cost = Decimal("0")

        try:
            async with httpx.AsyncClient(timeout=60) as client:
                payload = {
                    "model": model_id,
                    "messages": full_messages,
                    "stream": True,
                    "stream_options": {"include_usage": True},
                }
                if body.temperature is not None:
                    payload["temperature"] = body.temperature
                if body.max_tokens is not None:
                    payload["max_tokens"] = body.max_tokens

                async with client.stream(
                    "POST",
                    api_url,
                    headers={
                        "Content-Type": "application/json",
                        "Authorization": auth_header,
                    },
                    json=payload,
                ) as resp:
                    if resp.status_code != 200:
                        try:
                            err_body = json.loads(await resp.aread())
                        except Exception:
                            err_body = None
                        msg = _user_message_from_status(resp.status_code, err_body)
                        yield f"data: {json.dumps({'error': msg})}\n\n"
                        return

                    async for line in resp.aiter_lines():
                        if not line:
                            continue
                        if line.startswith("data: "):
                            data_part = line[6:]
                            if data_part.strip() == "[DONE]":
                                yield "data: [DONE]\n\n"
                                break
                            try:
                                chunk = json.loads(data_part)
                                usage = chunk.get("usage") or {}
                                if usage:
                                    prompt_tokens = int(usage.get("prompt_tokens") or prompt_tokens or 0)
                                    completion_tokens = int(usage.get("completion_tokens") or completion_tokens or 0)
                                    total_tokens = int(usage.get("total_tokens") or total_tokens or 0)
                                    raw_cost = usage.get("estimated_cost")
                                    if raw_cost is None:
                                        raw_cost = usage.get("cost")
                                    if raw_cost is not None:
                                        estimated_cost = Decimal(str(raw_cost))
                                choices = chunk.get("choices", [])
                                if choices:
                                    content = choices[0].get("delta", {}).get("content", "")
```

```
                                    if content:
                                        yield f"data: {json.dumps({'content': content})}\n\n"
                            except json.JSONDecodeError:
                                continue
            db = getattr(request.app.state, "db", None)
            if db and hasattr(db, "llm_usage_logs"):
                await db.llm_usage_logs.create(
                    user_id=user.user_id,
                    model=model_id,

... [29 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/routers/storage.py (2.3 KB)

```python
"""
DataVolume storage management routes.
"""

from __future__ import annotations

import logging

from fastapi import APIRouter, Depends, HTTPException

from app.auth.dependencies import SessionUser, get_current_namespace, require_owner
from app.utils.async_helpers import run_sync
from app.utils.k8s_utils import delete_data_volume, get_custom_api

logger = logging.getLogger(__name__)

router = APIRouter(tags=["storage"])


@router.get("/{username}/api/datavolumes")
async def get_user_datavolumes(
    username: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    api = get_custom_api()

    dv_response = await run_sync(
        api.list_namespaced_custom_object,
        "cdi.kubevirt.io", "v1beta1", user_ns, "datavolumes",
    )
    dvs = dv_response.get("items", [])

    vm_response = await run_sync(
        api.list_namespaced_custom_object,
        "kubevirt.io", "v1", user_ns, "virtualmachines",
    )
    vms = vm_response.get("items", [])

    vm_disks_map: dict[str, str] = {}
    for vm in vms:
        for vol in (
            vm.get("spec", {}).get("template", {}).get("spec", {}).get("volumes", [])
        ):
            if "persistentVolumeClaim" in vol:
                claim = vol["persistentVolumeClaim"]["claimName"]
                vm_disks_map[claim] = vm["metadata"]["name"]

    result = []
    for dv in dvs:
        phase = dv.get("status", {}).get("phase", "Unknown")
        progress = dv.get("status", {}).get("progress", "N/A")
        pvc_name = dv["metadata"]["name"]
        result.append({
            "name": dv["metadata"]["name"],
            "phase": phase,
            "progress": progress,
            "size": dv["spec"]["pvc"]["resources"]["requests"]["storage"],
            "pvc_name": pvc_name,
            "vm_name": vm_disks_map.get(pvc_name),
        })
    return result


@router.delete("/{username}/api/datavolumes/{name}")
```

```
async def delete_user_datavolume(
    username: str,
    name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    ok, err = await run_sync(delete_data_volume, user_ns, name)
    if not ok:
        raise HTTPException(500, err)
    return {"success": True}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/routers/vm_details.py (10.7 KB)

```
"""
VM details page, VM info API, and service (port) management.

Replaces ``routes/vm_details_routes.py``.  The old subprocess-based
cloud-init status check has been replaced with ``console_utils``.
"""

from __future__ import annotations

import logging

from fastapi import APIRouter, Depends, HTTPException, Request
from fastapi.responses import HTMLResponse
from app.templating import Jinja2Templates

from app.auth.dependencies import SessionUser, get_current_namespace, require_owner
from app.config import JUNIPER_EXTERNAL_IP, settings
from app.schemas import CreateServiceRequest, ServicePortInfo, VMInfoResponse
from app.utils.async_helpers import run_sync
from app.utils.console_utils import get_cloudinit_status
from app.utils.k8s_utils import (
    get_custom_api,
    get_datavolume_status_for_vm,
    get_vm_details,
    get_vm_username,
)

logger = logging.getLogger(__name__)

router = APIRouter(tags=["vm-details"])

templates = Jinja2Templates(directory="app/templates")


# ---------------------------------------------------------------------------
# VM Details page (SSR)
# ---------------------------------------------------------------------------

@router.get("/{username}/vm/{vm_name}", include_in_schema=False)
async def vm_details_page(
    request: Request,
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    vm = await run_sync(get_vm_details, user_ns, vm_name)
    vm_os = (
        vm.get("metadata", {}).get("labels", {}).get("vm.kubevirt.io/os", "linux")
        if vm else "linux"
    )
    llm_model_display = (
        settings.LLM_MODEL.split("/")[-1] if "/" in settings.LLM_MODEL else settings.LLM_MODEL
    )
    return templates.TemplateResponse("vm_details.html", {
        "request": request,
        "username": username,
        "namespace": user_ns,
        "is_admin": user.is_admin,
        "vm_name": vm_name,
        "vm_os": vm_os,
        "llm_model": llm_model_display,
    })


```

```python
# ----------------------------------------------------------------------------
# VM Info JSON API
# ----------------------------------------------------------------------------

@router.get("/{username}/vm/{vm_name}/info")
async def vm_info(
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    vm = await run_sync(get_vm_details, user_ns, vm_name)
    if not vm:
        raise HTTPException(404, "VM not found")

    from app.utils.service_utils import get_vm_ssh_service

    ssh_exists, ssh_node_port = await run_sync(get_vm_ssh_service, user_ns, vm_name)

    vm_status = vm.get("status", {}).get("printableStatus", "Unknown")

    dv_status = await run_sync(get_datavolume_status_for_vm, user_ns, vm)
    if dv_status:
        vm_status = "Provisioning"
    elif vm_status in ("WaitingForVolumeBinding", "WaitingForDataVolume"):
        vm_status = "Provisioning"

    running = vm_status.lower() == "running"

    spec = vm.get("spec", {}).get("template", {}).get("spec", {})
    domain = spec.get("domain", {})
    resources = domain.get("resources", {}).get("requests", {})
    cpu = domain.get("cpu", {}).get("cores", "N/A")
    memory = resources.get("memory", "N/A")

    gpu_devices = domain.get("devices", {}).get("gpus", [])
    gpu_count = len(gpu_devices)
    gpu_model = gpu_devices[0].get("deviceName", "Unknown") if gpu_devices else None

    volumes = spec.get("volumes", [])
    disks_info = []
    for vol in volumes:
        if "dataVolume" in vol:
            disks_info.append(vol["dataVolume"]["name"])
        elif "persistentVolumeClaim" in vol:
            disks_info.append(vol["persistentVolumeClaim"]["claimName"])
    storage = ", ".join(disks_info) if disks_info else "N/A"

    # Cloud-init status (replaces the old subprocess.run kubectl calls)
    cloudinit_status = None
    cloudinit_message = ""

    if running:
        try:
            api = get_custom_api()
            vmi = await run_sync(
                api.get_namespaced_custom_object,
                "kubevirt.io", "v1", user_ns, "virtualmachineinstances", vm_name,
            )
            conditions = vmi.get("status", {}).get("conditions", [])
            agent_connected = any(
                c.get("type") == "AgentConnected" and c.get("status") == "True"
                for c in conditions
            )
            vm_ready = any(
                c.get("type") == "Ready" and c.get("status") == "True"
                for c in conditions
            )

            vm_os_type = (
                vm.get("metadata", {}).get("labels", {}).get("vm.kubevirt.io/os", "linux")
```

```
            )
            is_windows = "windows" in vm_os_type.lower()

            finished, _ = await get_cloudinit_status(user_ns, vm_name, is_windows)

            if finished:
                cloudinit_status = "completed"
                cloudinit_message = "?????? ? ??????"
            elif agent_connected:
                cloudinit_status = "running"
                cloudinit_message = "?????????????..."
            elif vm_ready:
                cloudinit_status = "running"
                cloudinit_message = "????????..."
        except Exception as e:
            logger.debug("Could not get VMI conditions: %s", e)

    vm_user = "ubuntu"
    if ssh_exists:
        vm_user = await run_sync(get_vm_username, user_ns, vm_name)

    from app.utils.service_utils import get_vm_allocated_ip

    allocated_ip = await run_sync(get_vm_allocated_ip, user_ns, vm_name)

    result: dict = {
        "success": True,
        "status": vm_status,
        "cpu": str(cpu),
        "memory": memory,
        "storage": storage,
        "running": running,
        "cloudinit_status": cloudinit_status,
        "cloudinit_message": cloudinit_message,
        "datavolume_status": dv_status,
        "allocated_ip": allocated_ip,
        "ssh_service": {
            "exists": ssh_exists,
            "node_port": ssh_node_port,
            "public_ip": allocated_ip if ssh_exists else JUNIPER_EXTERNAL_IP,
            "command": (
                f"ssh {vm_user}@{allocated_ip} -p {ssh_node_port}"
                if ssh_exists and allocated_ip else None
            ),
        },
    }
    if gpu_count > 0:
        result["gpu_count"] = gpu_count
        result["gpu_model"] = gpu_model

    return result


# ---------------------------------------------------------------------------
# SSH service shortcut
# ---------------------------------------------------------------------------

@router.post("/{username}/vm/{vm_name}/ssh-service")
async def create_ssh_service(
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    from app.utils.k8s_utils import get_vm_username
    from app.utils.service_utils import add_port_to_vm_service, get_vm_allocated_ip

    vm_user = await run_sync(get_vm_username, user_ns, vm_name)
    ok, message, assigned = await run_sync(
        add_port_to_vm_service, user_ns, vm_name, 22, "ssh", None,
    )
```

```
    if not ok:
        raise HTTPException(400, message)

    allocated_ip = await run_sync(get_vm_allocated_ip, user_ns, vm_name)
    public_ip = allocated_ip or JUNIPER_EXTERNAL_IP

    return {
        "message": message,
        "nodePort": assigned,
        "publicIp": public_ip,
        "sshCommand": f"ssh {vm_user}@{public_ip} -p {assigned}",
    }


@router.delete("/{username}/vm/{vm_name}/ssh-service")
async def delete_ssh_service(
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    from app.utils.service_utils import remove_port_from_vm_service

    ok, msg = await run_sync(remove_port_from_vm_service, user_ns, vm_name, 22)
    if not ok:
        raise HTTPException(400, msg)
    return {"message": msg}


# ---------------------------------------------------------------------------
# Generic service / port management
# ---------------------------------------------------------------------------

@router.get("/{username}/vm/{vm_name}/services")
async def list_services(
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    from app.utils.k8s_utils import get_vm_username
    from app.utils.service_utils import get_vm_allocated_ip, list_vm_services

    services = await run_sync(list_vm_services, user_ns, vm_name)
    allocated_ip = await run_sync(get_vm_allocated_ip, user_ns, vm_name)
    public_ip = allocated_ip or JUNIPER_EXTERNAL_IP

    for svc in services:
        if svc.get("name") == "ssh" and svc.get("nodePort"):
            vm_user = await run_sync(get_vm_username, user_ns, vm_name)
            svc["ssh_command"] = f"ssh {vm_user}@{public_ip} -p {svc['nodePort']}"

    return {"services": services}


@router.post("/{username}/vm/{vm_name}/services")
async def create_service(
    username: str,
    vm_name: str,
    body: CreateServiceRequest,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    from app.utils.service_utils import add_port_to_vm_service, get_vm_allocated_ip

    ok, msg, assigned = await run_sync(
        add_port_to_vm_service, user_ns, vm_name, body.port, body.type,
    )
    if not ok:
        raise HTTPException(400, msg)

```

```
    allocated_ip = await run_sync(get_vm_allocated_ip, user_ns, vm_name)
    public_ip = allocated_ip or JUNIPER_EXTERNAL_IP

    return {
        "message": msg,
        "port": assigned,
        "nodePort": assigned,
        "target_port": body.port,
        "type": body.type,
        "public_ip": public_ip,
    }


@router.delete("/{username}/vm/{vm_name}/services/{service_type}")
async def delete_service(
    username: str,
    vm_name: str,
    service_type: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    from app.utils.service_utils import list_vm_services, remove_port_from_vm_service

    ports = await run_sync(list_vm_services, user_ns, vm_name)
    target_port = None
    for p in ports:
        if p["name"] == service_type:
            target_port = p["port"]
            break
    if target_port is None:
        raise HTTPException(404, f"Port type '{service_type}' not found")

    ok, msg = await run_sync(remove_port_from_vm_service, user_ns, vm_name, target_port)
    if not ok:
        raise HTTPException(400, msg)
    return {"message": msg}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/routers/vms.py (21.3 KB)

```
"""
VM CRUD and lifecycle routes: create, start/stop/restart, delete, pause/unpause.

Replaces the old ``routes/vm_routes.py``.
"""

from __future__ import annotations

import logging
import secrets

from fastapi import APIRouter, Depends, HTTPException, Request, status
from fastapi.responses import RedirectResponse

from app.auth.dependencies import (
    SessionUser,
    _get_db,
    get_current_namespace,
    get_current_user,
    require_owner,
)
from app.config import (
    NVIDIA_DRIVER_VERSIONS,
    NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER,
    settings,
)
from app.database import DB
from app.models import VMEventType
from app.schemas import CreateVMRequest, VMActionResponse, VMNameAvailability
from app.utils.async_helpers import run_sync
from app.utils.k8s_utils import (
    create_cloud_init_secret,
    create_data_volume,
    create_virtual_machine,
    delete_data_volume,
    delete_secret,
    delete_virtual_machine,
    ensure_namespace,
    get_vm_details,
    pause_virtual_machine,
    restart_virtual_machine,
    start_virtual_machine,
    stop_virtual_machine,
    unpause_virtual_machine,
    update_secret_owner,
    vm_exists_in_namespace,
)
from app.utils.vm_utils import (
    generate_cloud_init_userdata,
    generate_vm_manifest,
    get_default_username_for_image,
    sanitize_cloud_init_field,
    validate_ssh_key,
)

logger = logging.getLogger(__name__)

router = APIRouter(tags=["vms"])


async def _require_positive_balance(
    request: Request, user: SessionUser,
) -> None:
    """Reject the request if billing is enabled and the user has no funds.

```

```
    Admins are exempt.
    """
    if not settings.BILLING_ENABLED or user.is_admin:
        return
    db = _get_db(request)
    if not db:
        return
    bal = await db.users.get_balance(user.user_id)
    if bal is None:
        return
    current_balance, _ = bal
    if current_balance <= 0:
        raise HTTPException(
            status_code=status.HTTP_402_PAYMENT_REQUIRED,
            detail="Insufficient balance to perform this action",
        )


async def _set_running_safe(
    request: Request, namespace: str, vm_name: str, running: bool,
) -> None:
    """Best-effort update of the DB running state."""
    try:
        db = _get_db(request)
        if db:
            await db.vm_records.set_running(namespace, vm_name, running)
    except Exception as exc:
        logger.warning("set_running(%s/%s, %s) failed: %s", namespace, vm_name, running, exc)


# ---------------------------------------------------------------------------
# Create VM
# ---------------------------------------------------------------------------

@router.post("/{username}/create-vm")
async def create_vm(
    request: Request,
    username: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
):
    await _require_positive_balance(request, user)

    form = await request.form()
    raw = dict(form)

    if "sshkey" in raw:
        raw["ssh_key"] = raw.pop("sshkey")
    raw.pop("nvidia_custom_driver_installer_type", None)

    for key in ("package_update", "ssh_pwauth"):
        if key in raw:
            raw[key] = raw[key] in ("true", "on", "1", True)

    for key in ("gpu_model", "hostname", "username", "ssh_key", "password",
                "additional_packages", "nvidia_driver_version", "custom_cloudinit"):
        if key in raw and raw[key] == "":
            raw[key] = None

    body = CreateVMRequest(**raw)
    db = _get_db(request)

    is_windows_installer = body.image == "windows-installer"
    is_windows_golden = body.image == "windows-golden-image"

    # --- Cloud-init / Windows parameters ---
    if not is_windows_installer and not is_windows_golden:
        hostname = sanitize_cloud_init_field(body.hostname or body.name, "hostname")
        vm_username = body.username
        if not vm_username:
            vm_username = get_default_username_for_image(body.image)
```

```
        vm_username = sanitize_cloud_init_field(vm_username, "username")

        ssh_key = body.ssh_key or ""
        password = body.password or ""
        ssh_pwauth = "true" if body.ssh_pwauth else "false"
        package_update = "true" if body.package_update else "false"
        custom_cloudinit = body.custom_cloudinit or ""

        if not custom_cloudinit:
            has_ssh = bool(ssh_key.strip())
            has_password = ssh_pwauth == "true" and bool(password)
            if not has_ssh and not has_password:
                raise HTTPException(400, "SSH key or password (with ssh_pwauth) required")

        if ssh_key:
            ok, err = validate_ssh_key(ssh_key)
            if not ok:
                raise HTTPException(400, f"SSH key validation: {err}")

        nvidia_driver_version = body.nvidia_driver_version or settings.DEFAULT_NVIDIA_DRIVER
        if (
            body.gpu > 0
            and nvidia_driver_version
            and nvidia_driver_version not in NVIDIA_DRIVER_VERSIONS
        ):
            nvidia_driver_version = settings.DEFAULT_NVIDIA_DRIVER

        import re
        packages_list = None
        if body.additional_packages:
            packages_list = []
            for pkg in body.additional_packages.split():
                pkg = pkg.strip()
                if pkg and re.match(r"^[a-zA-Z0-9._-]+$", pkg):
                    packages_list.append(pkg)
                elif pkg:
                    raise HTTPException(400, f"Invalid package name: {pkg}")

    elif is_windows_installer:
        vm_username = "Administrator"
        hostname = body.name
        ssh_key = None
        password = None
        ssh_pwauth = "false"
        package_update = "false"
        packages_list = None
        custom_cloudinit = None
        nvidia_driver_version = None
    else:
        hostname = sanitize_cloud_init_field(body.hostname or body.name, "hostname")
        vm_username = sanitize_cloud_init_field(body.username or "Administrator", "username")
        password = body.password
        if not password:
            raise HTTPException(400, "Password required for Windows VM")
        ssh_key = None
        ssh_pwauth = "false"
        package_update = "false"
        packages_list = None
        custom_cloudinit = None
        nvidia_driver_version = None

    # --- Ensure namespace ---
    ns_ok = await run_sync(ensure_namespace, user_ns)
    if not ns_ok:
        raise HTTPException(500, f"Failed to ensure namespace: {user_ns}")

    try:
        from app.utils.network_policy_utils import ensure_namespace_network_policy
        await run_sync(ensure_namespace_network_policy, user_ns)
    except Exception as e:
        logger.warning("NetworkPolicy setup failed for %s: %s", user_ns, e)
```

```
    gpu_node_selector = (
        {"node-role.kubernetes.io/kubevirt": ""} if body.gpu > 0 else None
    )
    gpu_model = body.gpu_model if body.gpu > 0 else None

    # --- Windows installer flow ---
    if is_windows_installer:
        from app.utils.windows_utils import generate_windows_installer_vm_manifest

        manifest = generate_windows_installer_vm_manifest(
            vm_name=body.name, namespace=user_ns,
            cpu=body.cpu, memory=body.memory, storage=body.storage,
            gpu_model=gpu_model, gpu_count=body.gpu,
            gpu_node_selector=gpu_node_selector,
        )
        ok, err, vm_obj = await run_sync(create_virtual_machine, user_ns, manifest)
        if not ok:
            raise HTTPException(500, f"Windows VM creation failed: {err}")

        await _log_vm_created(db, user, body, user_ns, "windows-installer")
        return RedirectResponse(
            url=f"/{user.username}/vm/{body.name}", status_code=303,
        )

    # --- Windows golden image flow ---
    if is_windows_golden:
        from app.utils.windows_utils import (
            generate_cloudbase_init_userdata,
            generate_windows_vm_from_golden_image,
        )

        cloudbase_userdata = generate_cloudbase_init_userdata(
            hostname=hostname, username=vm_username, password=password,
        )
        secret_name = f"{body.name}-cloudinit"

        ok, err = await run_sync(
            create_cloud_init_secret, user_ns, secret_name, cloudbase_userdata, None,
        )
        if not ok and "??? ??????????" in err:
            await run_sync(delete_secret, user_ns, secret_name)
            ok, err = await run_sync(
                create_cloud_init_secret, user_ns, secret_name, cloudbase_userdata, None,
            )
        if not ok:
            raise HTTPException(500, f"Cloudbase-Init secret failed: {err}")

        manifest = generate_windows_vm_from_golden_image(
            vm_name=body.name, namespace=user_ns,
            cpu=body.cpu, memory=body.memory, storage=body.storage,
            secret_name=secret_name,
            golden_image_name=settings.WINDOWS_GOLDEN_IMAGE_NAME,
            golden_image_namespace=settings.WINDOWS_GOLDEN_IMAGE_NAMESPACE,
            gpu_model=gpu_model, gpu_count=body.gpu,
            gpu_node_selector=gpu_node_selector, vm_username=vm_username,
        )
        ok, err, vm_obj = await run_sync(create_virtual_machine, user_ns, manifest)
        if not ok:
            await run_sync(delete_secret, user_ns, secret_name)
            raise HTTPException(500, f"Windows VM creation failed: {err}")

        if vm_obj:
            await run_sync(update_secret_owner, user_ns, secret_name, vm_obj)

        await _log_vm_created(db, user, body, user_ns, "windows-golden")
        return RedirectResponse(
            url=f"/{user.username}/vm/{body.name}", status_code=303,
        )

    # --- Regular Linux VM flow ---
```

```
    dv_name = f"{body.name}-rootdisk"
    use_custom_nvidia = (
        body.gpu > 0
        and body.gpu_model in NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER
        and settings.APP_INTERNAL_BASE_URL
    )

    ok, err = await run_sync(
        create_data_volume, user_ns, dv_name, body.image,
        f"{body.storage}Gi" if not str(body.storage).endswith("Gi") else str(body.storage),
        settings.STORAGE_CLASS_NAME,
    )
    if not ok:
        raise HTTPException(500, f"DataVolume creation failed: {err}")

    callback_url = None
    callback_token = None
    if use_custom_nvidia and settings.APP_INTERNAL_BASE_URL:
        callback_token = secrets.token_urlsafe(32)
        base = settings.APP_INTERNAL_BASE_URL.strip().rstrip("/")
        if not base.startswith(("http://", "https://")):
            base = f"http://{base}"
        callback_url = (
            f"{base}/{user.username}/vm/{body.name}/"
            f"{settings.NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH}"
            f"?token={callback_token}&namespace={user_ns}"
        )

    try:
        cloud_init_data = generate_cloud_init_userdata(
            hostname, vm_username, ssh_key, package_update, package_update,
            ssh_pwauth, password, custom_cloudinit,
            packages=packages_list, image_url=body.image,
            gpu_count=body.gpu, nvidia_driver_version=nvidia_driver_version,
            gpu_model=gpu_model,
            nvidia_custom_driver_callback_url=callback_url,
            nvidia_custom_driver_installer_type="run" if use_custom_nvidia else "auto",
        )
    except Exception as e:
        await run_sync(delete_data_volume, user_ns, dv_name)
        raise HTTPException(500, f"Cloud-init generation failed: {e}")

    secret_name = f"{body.name}-cloudinit"
    try:
        manifest = generate_vm_manifest(
            body.name, user_ns, body.cpu, body.memory, dv_name, secret_name,
            gpu_model=gpu_model, gpu_count=body.gpu,
            gpu_node_selector=None, vm_username=vm_username,
            nvidia_custom_driver_callback_token=callback_token,
        )
    except Exception as e:
        await run_sync(delete_data_volume, user_ns, dv_name)
        raise HTTPException(500, f"Manifest generation failed: {e}")

    ok, err, vm_obj = await run_sync(create_virtual_machine, user_ns, manifest)
    if not ok:
        await run_sync(delete_data_volume, user_ns, dv_name)
        raise HTTPException(500, f"VM creation failed: {err}")

    ok, err = await run_sync(
        create_cloud_init_secret, user_ns, secret_name, cloud_init_data, vm_obj,
    )
    if not ok and "??? ??????????" in err:
        await run_sync(delete_secret, user_ns, secret_name)
        ok, err = await run_sync(
            create_cloud_init_secret, user_ns, secret_name, cloud_init_data, vm_obj,
        )
    if not ok:
        logger.warning("VM created but cloud-init secret failed: %s", err)

    await _log_vm_created(db, user, body, user_ns, "linux")
```

```
    return RedirectResponse(
        url=f"/{user.username}/vms", status_code=303,
    )


async def _log_vm_created(
    db: DB | None, user: SessionUser, body: CreateVMRequest, namespace: str, os_type: str,
) -> None:
    if not db:
        return
    try:
        await db.vm_records.create(
            user_id=user.user_id, vm_name=body.name, namespace=namespace,
            cpu=body.cpu, memory_gb=body.memory, storage_gb=body.storage,
            image_url=body.image, gpu_count=body.gpu, gpu_model=body.gpu_model,
            os_type=os_type,
        )
        await db.vm_events.log_event(
            vm_name=body.name, namespace=namespace, user_id=user.user_id,
            event_type=VMEventType.CREATED,
            metadata={"cpu": body.cpu, "memory_gb": body.memory, "gpu": body.gpu},
        )
    except Exception as e:
        logger.warning("DB logging failed for VM create: %s", e)


# ---------------------------------------------------------------------------
# VM name availability check
# ---------------------------------------------------------------------------

@router.get("/{username}/check-vm-name/{vm_name}")
async def check_vm_name(
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
) -> VMNameAvailability:
    from app.utils.vm_utils import validate_vm_name

    ok, err = validate_vm_name(vm_name)
    if not ok:
        raise HTTPException(400, err or "Invalid VM name")

    exists = await run_sync(vm_exists_in_namespace, user_ns, vm_name)
    return VMNameAvailability(available=not exists, exists=exists)


# ---------------------------------------------------------------------------
# NVIDIA custom driver callback
# ---------------------------------------------------------------------------

# @router.api_route(
#     "/{username}/vm/{vm_name}/{callback_path}",
#     methods=["POST", "GET"],
# )
# async def nvidia_custom_driver_installed(
#     username: str,
#     vm_name: str,
#     callback_path: str,
#     token: str | None = None,
#     namespace: str | None = None,
# ):
#     if callback_path != settings.NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH:
#         raise HTTPException(404)
#     if not token or not namespace:
#         raise HTTPException(400, "Missing token or namespace")
#     if not namespace[0].isalpha():
#         raise HTTPException(400, "Namespace must start with a letter")
#     if "_" in namespace:
#         raise HTTPException(400, "Namespace must not contain underscores")
#     if not namespace.startswith(f"{settings.K8S_NAMESPACE_PREFIX or 'kv'}-"):
```

```
#         raise HTTPException(403, "Invalid namespace")

#     vm = await run_sync(get_vm_details, namespace, vm_name)
#     if not vm:
#         raise HTTPException(404, "VM not found")

#     ann = (
#         vm.get("spec", {})
#         .get("template", {})
#         .get("metadata", {})
#         .get("annotations", {})
#     )
#     expected = ann.get(settings.NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY)
#     if not expected or token != expected:
#         raise HTTPException(403, "Invalid token")

#     logger.info("Custom driver callback OK for %s/%s", namespace, vm_name)
#     return {"success": True}


# ---------------------------------------------------------------------------
# Lifecycle: start / stop / restart / delete / pause / unpause / metrics
# ---------------------------------------------------------------------------

@router.post("/{username}/vm/{vm_name}/start")
async def start_vm(
    request: Request,
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
) -> VMActionResponse:
    await _require_positive_balance(request, user)

    ok = await run_sync(start_virtual_machine, user_ns, vm_name)
    if not ok:
        raise HTTPException(500, "Failed to start VM")
    await _set_running_safe(request, user_ns, vm_name, True)
    await _log_lifecycle(request, user, user_ns, vm_name, VMEventType.STARTED)
    return VMActionResponse(success=True, message=f"VM {vm_name} starting")


@router.post("/{username}/vm/{vm_name}/stop")
async def stop_vm(
    request: Request,
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
) -> VMActionResponse:
    ok = await run_sync(stop_virtual_machine, user_ns, vm_name)
    if not ok:
        raise HTTPException(500, "Failed to stop VM")
    await _set_running_safe(request, user_ns, vm_name, False)
    await _log_lifecycle(request, user, user_ns, vm_name, VMEventType.STOPPED)
    return VMActionResponse(success=True, message=f"VM {vm_name} stopping")


@router.post("/{username}/vm/{vm_name}/restart")
async def restart_vm(
    request: Request,
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),
    user_ns: str = Depends(get_current_namespace),
) -> VMActionResponse:
    ok = await run_sync(restart_virtual_machine, user_ns, vm_name)
    if not ok:
        raise HTTPException(500, "Failed to restart VM (must be running)")
    await _set_running_safe(request, user_ns, vm_name, True)
    await _log_lifecycle(request, user, user_ns, vm_name, VMEventType.RESTARTED)
```

```
    return VMActionResponse(success=True, message=f"VM {vm_name} restarting")


@router.post("/{username}/vm/{vm_name}/delete")
async def delete_vm_route(
    request: Request,
    username: str,
    vm_name: str,
    user: SessionUser = Depends(require_owner),

... [92 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/scheduling_config.py (3.9 KB)

```
"""
????????? ???????????: VM (affinity, tolerations) ? ????? CDI importer (nodePlacement).

VM ? ?????? ?? ????? ? KubeVirt (affinity/tolerations). ???? ????????? CDI (importer-prime-*)
????? ???? ?????????? ?? nodeSelector ?/??? affinity ????? env.
??? ????????? ?????; ?????????? ????? env (??. env.example).
"""

import json
import logging
import os
from typing import Any, Dict

logger = logging.getLogger(__name__)

# --- Affinity: ????????? VM ?????? ?? ????? ? KubeVirt ---
VM_NODE_AFFINITY: Dict[str, Any] = {
    "nodeAffinity": {
        "requiredDuringSchedulingIgnoredDuringExecution": {
            "nodeSelectorTerms": [
                {
                    "matchExpressions": [
                        {
                            "key": "node-role.kubernetes.io/kubevirt",
                            "operator": "Exists",
                        }
                    ]
                }
            ]
        }
    }
}

# --- Tolerations: ?????? ? ????? ? taints (cephfs, data, dedicated) ---
VM_TOLERATIONS: list = [
    {
        "key": "dedicated",
        "operator": "Equal",
        "value": "kubevirt",
        "effect": "NoSchedule",
    },
    {"key": "cephfs", "operator": "Exists", "effect": "NoSchedule"},
    {"key": "data", "operator": "Exists", "effect": "NoSchedule"},
]


def get_storage_node_selector() -> Dict[str, str]:
    """
    NodeSelector ??? ??? ? CSI-????????? (???????? Ceph RBD).

    ??????????? ?????? ??? ????? CDI importer/transfer (importer-prime-*), ???????
    ??????? ? ???????? ????? DataVolume. VM ???? ?????????? ?????? ?? affinity
    (???? ? KubeVirt). ???????? ????? env VM_STORAGE_NODE_SELECTOR (JSON),
    ???????? {"csi-rbd":"ceph"}. ???? ????? ? ?????????? ?? ????????????.
    """
    raw = (os.getenv("VM_STORAGE_NODE_SELECTOR") or "your-SC").strip()
    if not raw:
        return {}
    try:
        sel = json.loads(raw)
        if not isinstance(sel, dict):
            logger.warning("VM_STORAGE_NODE_SELECTOR must be a JSON object, ignoring")
            return {}
        return {str(k): str(v) for k, v in sel.items()}
    except json.JSONDecodeError as e:
```

```
        logger.warning("Invalid VM_STORAGE_NODE_SELECTOR JSON: %s", e)
        return {}

def get_storage_node_affinity() -> Dict[str, Any]:
    """
    Affinity ??? ????? CDI importer/transfer (importer-prime-*).

    ???????? ????? env VM_STORAGE_NODE_AFFINITY:
      - "kubevirt" -> ??????????? VM_NODE_AFFINITY
      - JSON ?????? affinity -> ??????????? ??? ????
      - ????? -> affinity ?? ????????
    """
    raw = (os.getenv("VM_STORAGE_NODE_AFFINITY") or "").strip()
    if not raw:
        return {}

    if raw.lower() == "kubevirt":
        return VM_NODE_AFFINITY

    try:
        affinity = json.loads(raw)
        if not isinstance(affinity, dict):
            logger.warning("VM_STORAGE_NODE_AFFINITY must be a JSON object, ignoring")
            return {}
        return affinity
    except json.JSONDecodeError as e:
        logger.warning("Invalid VM_STORAGE_NODE_AFFINITY JSON: %s", e)
        return {}


def get_storage_node_placement() -> Dict[str, Any]:
    """
    ??????? nodePlacement ??? CDI DataVolume importer/transfer pod-??.

    ?????????? VM_STORAGE_NODE_SELECTOR ? VM_STORAGE_NODE_AFFINITY.
    """
    node_placement: Dict[str, Any] = {}

    selector = get_storage_node_selector()
    if selector:
        node_placement["nodeSelector"] = selector

    affinity = get_storage_node_affinity()
    if affinity:
        node_placement["affinity"] = affinity

    return node_placement
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/schemas.py (7.3 KB)

```python
"""
Pydantic request / response schemas for the FastAPI routes.
"""

from __future__ import annotations

from typing import Any
from urllib.parse import urlparse

from pydantic import BaseModel, Field, field_validator, model_validator

from app.config import ALLOWED_IMAGE_DOMAINS, settings


# ---------------------------------------------------------------------------
# Auth
# ---------------------------------------------------------------------------

class LoginRequest(BaseModel):
    username: str = Field(..., min_length=1, max_length=64)
    password: str = Field(..., min_length=1)


class RegisterRequest(BaseModel):
    username: str = Field(..., min_length=3, max_length=64, pattern=r"^[a-z0-9-]+$")
    password: str = Field(..., min_length=6, max_length=128)


class UserPublic(BaseModel):
    id: int
    username: str
    role: str = "user"
    is_active: bool


class CreateUserRequest(BaseModel):
    username: str = Field(
        ...,
        min_length=3,
        max_length=64,
        pattern=r"^[a-z0-9-]+$",
        description="?????? ???????? ?????, ????? ? ????? (a-z, 0-9, -)",
    )
    password: str = Field(..., min_length=6, max_length=128)
    role: str = Field(default="user", pattern=r"^(admin|user)$")


class SetRoleRequest(BaseModel):
    role: str = Field(..., pattern=r"^(admin|user)$")


class BalanceUpdateRequest(BaseModel):
    amount: float = Field(..., gt=0, description="Amount to add")


class BalanceTransferRequest(BaseModel):
    source: str = Field(..., pattern=r"^(main|llm)$")
    target: str = Field(..., pattern=r"^(main|llm)$")
    amount: float | None = Field(default=None, gt=0)
    transfer_all: bool = Field(default=False, description="Transfer entire source balance (ignores amount)")

    @model_validator(mode="after")
    def validate_transfer(self) -> "BalanceTransferRequest":
        if self.source == self.target:
            raise ValueError("source and target must be different")
```

```
        if not self.transfer_all and self.amount is None:
            raise ValueError("amount is required when transfer_all is false")
        return self


# ---------------------------------------------------------------------------
# VM Creation
# ---------------------------------------------------------------------------

class CreateVMRequest(BaseModel):
    name: str = Field(..., min_length=1, max_length=253)
    cpu: int = Field(..., ge=1)
    memory: float = Field(..., ge=1)
    storage: float = Field(..., ge=10)
    image: str
    gpu: int = Field(default=0, ge=0)
    gpu_model: str | None = None

    hostname: str | None = None
    username: str | None = None
    ssh_key: str | None = None
    password: str | None = None
    ssh_pwauth: bool = False
    package_update: bool = False
    nvidia_driver_version: str | None = None
    additional_packages: str | None = None
    custom_cloudinit: str | None = None

    @field_validator("name")
    @classmethod
    def validate_dns_name(cls, v: str) -> str:
        import re
        if not re.match(r"^[a-z0-9]([-a-z0-9]*[a-z0-9])?$", v):
            raise ValueError(
                "Must be lowercase alphanumeric / '-', start and end with alphanumeric"
            )
        return v

    @field_validator("cpu")
    @classmethod
    def cap_cpu(cls, v: int) -> int:
        if v > settings.MAX_CPU_CORES:
            raise ValueError(f"CPU must be <= {settings.MAX_CPU_CORES}")
        return v

    @field_validator("memory")
    @classmethod
    def cap_memory(cls, v: float) -> float:
        if v > settings.MAX_MEMORY_GB:
            raise ValueError(f"Memory must be <= {settings.MAX_MEMORY_GB} GB")
        return v

    @field_validator("storage")
    @classmethod
    def cap_storage(cls, v: float) -> float:
        if v > settings.MAX_STORAGE_GB:
            raise ValueError(f"Storage must be <= {settings.MAX_STORAGE_GB} GB")
        return v

    @field_validator("gpu")
    @classmethod
    def cap_gpu(cls, v: int) -> int:
        if v > settings.MAX_GPU_COUNT:
            raise ValueError(f"GPU count must be <= {settings.MAX_GPU_COUNT}")
        return v

    @model_validator(mode="after")
    def gpu_model_required(self) -> "CreateVMRequest":
        if self.gpu > 0 and not self.gpu_model:
            raise ValueError("gpu_model is required when gpu > 0")
        return self
```

```python

    @field_validator("image")
    @classmethod
    def validate_image(cls, v: str) -> str:
        if v in ("windows-installer", "windows-golden-image"):
            return v
        if not v.startswith(("http://", "https://")):
            raise ValueError("Image URL must use http:// or https://")
        parsed = urlparse(v)
        if ALLOWED_IMAGE_DOMAINS and parsed.netloc not in ALLOWED_IMAGE_DOMAINS:
            raise ValueError(
                f"Domain not allowed: {parsed.netloc}. "
                f"Allowed: {', '.join(ALLOWED_IMAGE_DOMAINS)}"
            )
        return v


# ---------------------------------------------------------------------------
# VM Lifecycle (responses)
# ---------------------------------------------------------------------------

class VMActionResponse(BaseModel):
    success: bool
    message: str


class VMNameAvailability(BaseModel):
    available: bool
    exists: bool


# ---------------------------------------------------------------------------
# VM Details (responses)
# ---------------------------------------------------------------------------

class SSHServiceInfo(BaseModel):
    exists: bool
    node_port: int | None = None
    public_ip: str | None = None
    command: str | None = None


class VMInfoResponse(BaseModel):
    success: bool = True
    status: str
    cpu: str
    memory: str
    storage: str
    running: bool
    cloudinit_status: str | None = None
    cloudinit_message: str = ""
    datavolume_status: str | None = None
    allocated_ip: str | None = None
    ssh_service: SSHServiceInfo
    gpu_count: int | None = None
    gpu_model: str | None = None


# ---------------------------------------------------------------------------
# Services
# ---------------------------------------------------------------------------

class CreateServiceRequest(BaseModel):
    port: int = Field(..., ge=1, le=65535)
    type: str = "custom"


class ServicePortInfo(BaseModel):
    name: str
    port: int
    nodePort: int
```

```
    protocol: str = "TCP"
    ssh_command: str | None = None


# ---------------------------------------------------------------------------
# Storage
# ---------------------------------------------------------------------------

class DataVolumeInfo(BaseModel):
    name: str
    phase: str
    progress: str
    size: str
    pvc_name: str
    vm_name: str | None = None


# ---------------------------------------------------------------------------
# LLM
# ---------------------------------------------------------------------------

class LLMChatRequest(BaseModel):
    messages: list[dict[str, str]] = Field(..., min_length=1)
    model: str | None = None  # Model ID from available-models (e.g. google/gemma-3-27b-it)
    temperature: float | None = Field(default=None, ge=0.0, le=2.0)
    max_tokens: int | None = Field(default=None, ge=1, le=32768)
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/services/__init__.py (0.0 B)

```
1
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/services/billing_loop.py (4.7 KB)

```python
"""
Background billing loop: compute VM costs, deduct from user balances,
stop VMs for users with insufficient funds.
"""

from __future__ import annotations

import logging
from datetime import datetime, timezone
from decimal import Decimal

from app.config import VMPricing
from app.database import DB
from app.models import VMEventType, VMRecord
from app.utils.async_helpers import run_sync

logger = logging.getLogger(__name__)


async def billing_tick(db: DB, pricing: VMPricing) -> list[int]:
    """One billing pass over all users.

    Returns user_ids whose VMs must be force-stopped (insufficient funds).
    """
    now = datetime.now(timezone.utc)
    kill_user_ids: list[int] = []

    billable_vms = await db.vm_records.get_all_billable()
    if not billable_vms:
        return kill_user_ids

    by_user: dict[int, list[VMRecord]] = {}
    for vm in billable_vms:
        by_user.setdefault(vm.user_id, []).append(vm)

    admin_ids = await _get_admin_ids(db)

    for user_id, vms in by_user.items():
        if user_id in admin_ids:
            for vm in vms:
                await db.vm_records.update_last_billed(vm.id, now)
            continue

        total_cost = Decimal("0")

        for vm in vms:
            minutes = _compute_billable_minutes(vm, now)
            if minutes > 0:
                vm_cost = pricing.cost_per_minute(
                    vm.cpu, vm.memory_gb, vm.gpu_count, vm.gpu_model,
                )
                total_cost += vm_cost * minutes

        if total_cost <= 0:
            continue

        bal = await db.users.get_balance(user_id)
        if bal is None:
            continue
        current_balance, version = bal

        try:
            new_balance, _ = await db.users.update_balance(
                user_id, -total_cost, version,
            )
```

```
        except ValueError:
            kill_user_ids.append(user_id)
            continue

        for vm in vms:
            billed_at = now if vm.is_running else (vm.last_stopped_at or now)
            await db.vm_records.update_last_billed(vm.id, billed_at)

        try:
            await db.balance_snapshots.create(user_id, "main", new_balance)
        except Exception as exc:
            logger.warning("Balance snapshot failed for user %d: %s", user_id, exc)

    return kill_user_ids


def _compute_billable_minutes(vm: VMRecord, now: datetime) -> Decimal:
    """Compute billable minutes for a VM since it was last billed."""
    if vm.is_running:
        candidates = [t for t in (vm.last_started_at, vm.last_billed_at) if t is not None]
        start = max(candidates) if candidates else vm.created_at
        elapsed = (now - start).total_seconds()
    elif vm.last_stopped_at and (
        not vm.last_billed_at or vm.last_stopped_at > vm.last_billed_at
    ):
        candidates = [t for t in (vm.last_started_at, vm.last_billed_at) if t is not None]
        start = max(candidates) if candidates else vm.created_at
        elapsed = (vm.last_stopped_at - start).total_seconds()
    else:
        return Decimal("0")

    if elapsed <= 0:
        return Decimal("0")
    return Decimal(str(elapsed)) / 60


async def _get_admin_ids(db: DB) -> set[int]:
    users = await db.users.list_all(include_inactive=False)
    return {u.id for u in users if u.is_admin}


async def stop_all_user_vms(db: DB, user_id: int) -> int:
    """Stop all running VMs for a user (billing enforcement).

    Returns number of VMs stopped.
    """
    from app.utils.k8s_utils import stop_virtual_machine

    active_vms = await db.vm_records.get_active_for_user(user_id)
    stopped = 0

    for vm in active_vms:
        if not vm.is_running:
            continue
        try:
            ok = await run_sync(stop_virtual_machine, vm.namespace, vm.vm_name)
            if ok:
                await db.vm_records.set_running(vm.namespace, vm.vm_name, False)
                stopped += 1
            else:
                logger.warning(
                    "Failed to stop VM %s/%s for billing enforcement",
                    vm.namespace, vm.vm_name,
                )
        except Exception as exc:
            logger.error(
                "Error stopping VM %s/%s for billing: %s",
                vm.namespace, vm.vm_name, exc,
            )

        try:
```

```
            await db.vm_events.log_event(
                vm_name=vm.vm_name,
                namespace=vm.namespace,
                user_id=user_id,
                event_type=VMEventType.BILLING_STOPPED,
                metadata={"reason": "insufficient_funds"},
            )
        except Exception as exc:
            logger.warning("Failed to log BILLING_STOPPED event: %s", exc)

    return stopped

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/services/llm_auth_client.py (7.0 KB)

```python
"""
HTTP client for the LLM auth-service admin API.

Bridges viirtuoz-vps user accounts to the LLM billing system.
The auth-service uses UUIDs for user IDs; viirtuoz-vps uses integers.
The shared key between the two systems is ``username``.
"""
from __future__ import annotations

import logging
from datetime import datetime, timedelta, timezone
from decimal import Decimal
from typing import Any

import httpx

from app.config import settings

logger = logging.getLogger(__name__)

_TIMEOUT = 10


class LLMAuthClient:
    """Thin wrapper around the LLM auth-service admin endpoints."""

    def _headers(self) -> dict[str, str]:
        return {"X-Cudo-Admin": f"Bearer {settings.LLM_AUTH_ADMIN_TOKEN}"}

    @property
    def _base(self) -> str:
        return settings.LLM_AUTH_SERVICE_URL.rstrip("/")

    def available(self) -> bool:
        return bool(settings.LLM_AUTH_SERVICE_URL and settings.LLM_AUTH_ADMIN_TOKEN)

    # ------------------------------------------------------------------
    # User lookup / registration
    # ------------------------------------------------------------------

    async def get_user_id(self, username: str) -> str | None:
        """Resolve a VPS username to an auth-service UUID."""
        try:
            async with httpx.AsyncClient(timeout=_TIMEOUT) as c:
                r = await c.post(
                    f"{self._base}/get-user-id",
                    json={"username": username},
                    headers=self._headers(),
                )
                if r.status_code == 200:
                    return r.json()["user_id"]
        except httpx.HTTPError as exc:
            logger.warning("LLM auth-service get_user_id failed: %s", exc)
        return None

    async def ensure_user(self, username: str) -> str | None:
        """Register the user in auth-service if absent, return their UUID."""
        uid = await self.get_user_id(username)
        if uid:
            return uid
        try:
            async with httpx.AsyncClient(timeout=_TIMEOUT) as c:
                r = await c.post(
                    f"{self._base}/register",
                    json={"username": username},
```

```
                    headers=self._headers(),
                )
                if r.status_code == 200:
                    return str(r.json().get("id"))
        except httpx.HTTPError as exc:
            logger.warning("LLM auth-service register failed: %s", exc)
        return await self.get_user_id(username)

    async def verify_token_owner(self, token: str) -> str | None:
        """Verify SK token via auth-service and return the auth-service user_id it belongs to.
        Returns None if token is invalid or auth-service is unavailable.
        """
        if not self.available():
            return None
        try:
            async with httpx.AsyncClient(timeout=_TIMEOUT) as c:
                r = await c.get(
                    f"{self._base}/verify-token",
                    params={"token": token},
                    headers=self._headers(),
                )
                if r.status_code == 200:
                    return str(r.json().get("user_id", "")) or None
        except httpx.HTTPError as exc:
            logger.warning("LLM auth-service verify_token failed: %s", exc)
        return None

    # ------------------------------------------------------------------
    # Balance operations
    # ------------------------------------------------------------------

    async def get_balance(self, llm_user_id: str) -> float | None:
        """Get the current LLM balance (uses the amount=0 trick)."""
        try:
            async with httpx.AsyncClient(timeout=_TIMEOUT) as c:
                r = await c.post(
                    f"{self._base}/balance/increase-by-id",
                    json={"user_id": llm_user_id, "amount": "0"},
                    headers=self._headers(),
                )
                if r.status_code == 200:
                    raw = r.json().get("balance")
                    if raw is None:
                        return None
                    return float(raw)
        except httpx.HTTPError as exc:
            logger.warning("LLM auth-service get_balance failed: %s", exc)
        return None

    def create_token_available(self) -> bool:
        """Whether create-token is available (auth-service configured)."""
        return self.available()

    async def create_token(self, username: str) -> str | None:
        """Backward-compatible shortcut that returns only token value."""
        data = await self.create_token_details(username)
        return data.get("token") if data else None

    async def create_token_details(self, username: str) -> dict[str, str] | None:
        """Create SK token and return token + external reference for storage."""
        if not self.available():
            return None
        llm_uid = await self.ensure_user(username)
        if not llm_uid:
            return None
        url = f"{self._base}/generate-token-by-id"
        try:
            async with httpx.AsyncClient(timeout=_TIMEOUT) as c:
                r = await c.post(url, params={"user_id": llm_uid}, headers=self._headers())
                if r.status_code != 200:
                    logger.warning(
```

```
                        "auth-service generate-token-by-id %s: HTTP %d ? %s",
                        username, r.status_code, r.text,
                    )
                    return None
                data = r.json()
                token = data.get("access_token")
                if not token:
                    return None
                external_ref = (
                    data.get("token_id")
                    or data.get("id")
                    or data.get("token_ref")
                    or str(token)
                )
                return {"token": str(token), "external_ref": str(external_ref)}
        except httpx.HTTPError as exc:
            logger.warning("auth-service create_token failed: %s", exc)
            return None

    async def adjust_balance(self, llm_user_id: str, amount: Decimal) -> float | None:
        """Increase (positive) or decrease (negative) the LLM balance.

        Returns the new balance, or None on failure.
        """
        try:
            async with httpx.AsyncClient(timeout=_TIMEOUT) as c:
                r = await c.post(
                    f"{self._base}/balance/increase-by-id",
                    json={"user_id": llm_user_id, "amount": str(amount)},
                    headers=self._headers(),
                )
                if r.status_code == 200:
                    return r.json()["balance"]
                logger.warning(
                    "LLM auth-service adjust_balance %s: HTTP %d ? %s",
                    llm_user_id, r.status_code, r.text,
                )
        except httpx.HTTPError as exc:
            logger.warning("LLM auth-service adjust_balance failed: %s", exc)
        return None

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/services/vm_billing.py (2.1 KB)

```python
"""
VM billing helpers: resource spec model and k8s-to-spec parser.

The billing loop uses these to compute per-minute costs from live k8s state.
"""

from __future__ import annotations

import re
from pydantic import BaseModel


class VMBillingSpec(BaseModel):
    """Billable resource snapshot for a single VM, parsed from k8s."""
    vm_name: str
    namespace: str
    user_id: int | None = None
    cpu: int
    memory_gb: float
    gpu_count: int = 0
    gpu_model: str | None = None


_MEM_PATTERN = re.compile(r"^(\d+(?:\.\d+)?)\s*(Gi|Mi|Ki|G|M|K|Ti)?$", re.IGNORECASE)


def parse_memory_to_gb(raw: str | int | float) -> float:
    """Convert a k8s memory string to GB.  '8Gi' -> 8.0, '512Mi' -> 0.5, etc."""
    if isinstance(raw, (int, float)):
        return float(raw)
    raw = str(raw).strip()
    m = _MEM_PATTERN.match(raw)
    if not m:
        return 0.0
    value = float(m.group(1))
    unit = (m.group(2) or "").upper()
    multipliers = {
        "GI": 1.0, "G": 1.0,
        "MI": 1.0 / 1024, "M": 1.0 / 1024,
        "KI": 1.0 / (1024 * 1024), "K": 1.0 / (1024 * 1024),
        "TI": 1024.0,
    }
    return value * multipliers.get(unit, 1.0)


def vm_billing_spec_from_k8s(
    vm: dict,
    user_id: int | None = None,
) -> VMBillingSpec:
    """Build a VMBillingSpec from a raw k8s VirtualMachine dict."""
    meta = vm.get("metadata", {})
    spec = vm.get("spec", {}).get("template", {}).get("spec", {})
    domain = spec.get("domain", {})

    cpu = domain.get("cpu", {}).get("cores", 1)
    memory_raw = domain.get("resources", {}).get("requests", {}).get("memory", "0Gi")
    memory_gb = parse_memory_to_gb(memory_raw)

    gpu_devices = domain.get("devices", {}).get("gpus", [])

    return VMBillingSpec(
        vm_name=meta.get("name", ""),
        namespace=meta.get("namespace", ""),
        user_id=user_id,
        cpu=int(cpu) if cpu != "N/A" else 0,
```

```
        memory_gb=memory_gb,
        gpu_count=len(gpu_devices),
        gpu_model=gpu_devices[0].get("deviceName") if gpu_devices else None,
    )

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/cloud-inits/cloud-init-debian.yaml (910.0 B)

```
#cloud-config
hostname: {hostname}
manage_etc_hosts: true
preserve_hostname: false
ssh_pwauth: {ssh_pwauth}

{write_files_section}

# ??????????? ?? ????????? ???????
bootcmd:
  - mkdir -p /etc/systemd/resolved.conf.d
  - |
    cat >/etc/systemd/resolved.conf.d/custom-dns.conf << 'EOF'
    [Resolve]
    DNS=8.8.8.8 1.1.1.1
    FallbackDNS=1.0.0.1
    Domains=~.
    DNSSEC=no
    EOF
  - systemctl restart systemd-resolved || true
  - sleep 3
  - |
    for i in 1 2 3 4 5; do
      if getent hosts deb.debian.org >/dev/null 2>&1; then
        echo "DNS OK"
        break
      fi
      echo "DNS not ready yet, retry..."
      sleep 2
    done

users:
  - name: {username}
    groups: sudo
    shell: /bin/bash
    sudo: 'ALL=(ALL) NOPASSWD:ALL'
{ssh_keys}
{password_config}

{package_update}

packages:
{packages}

runcmd:
{runcmd}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/cloud-inits/cloud-init-fedora.yaml (923.0 B)

```
#cloud-config
hostname: {hostname}
manage_etc_hosts: true
preserve_hostname: false
ssh_pwauth: {ssh_pwauth}

{write_files_section}

# ??????????? ?? ????????? ???????
bootcmd:
  - mkdir -p /etc/systemd/resolved.conf.d
  - |
    cat >/etc/systemd/resolved.conf.d/custom-dns.conf << 'EOF'
    [Resolve]
    DNS=8.8.8.8 1.1.1.1
    FallbackDNS=1.0.0.1
    Domains=~.
    DNSSEC=no
    EOF
  - systemctl restart systemd-resolved || true
  - sleep 3
  - |
    for i in 1 2 3 4 5; do
      if getent hosts download.fedoraproject.org >/dev/null 2>&1; then
        echo "DNS OK"
        break
      fi
      echo "DNS not ready yet, retry..."
      sleep 2
    done

users:
  - name: {username}
    groups: wheel
    shell: /bin/bash
    sudo: 'ALL=(ALL) NOPASSWD:ALL'
{ssh_keys}
{password_config}

{package_update}

packages:
{packages}

runcmd:
{runcmd}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/cloud-inits/cloud-init-opensuse.yaml (918.0 B)

```
#cloud-config
hostname: {hostname}
manage_etc_hosts: true
preserve_hostname: false
ssh_pwauth: {ssh_pwauth}

{write_files_section}

# ??????????? ?? ????????? ???????
bootcmd:
  - mkdir -p /etc/systemd/resolved.conf.d
  - |
    cat >/etc/systemd/resolved.conf.d/custom-dns.conf << 'EOF'
    [Resolve]
    DNS=8.8.8.8 1.1.1.1
    FallbackDNS=1.0.0.1
    Domains=~.
    DNSSEC=no
    EOF
  - systemctl restart systemd-resolved || true
  - sleep 3
  - |
    for i in 1 2 3 4 5; do
      if getent hosts download.opensuse.org >/dev/null 2>&1; then
        echo "DNS OK"
        break
      fi
      echo "DNS not ready yet, retry..."
      sleep 2
    done

users:
  - name: {username}
    groups: wheel
    shell: /bin/bash
    sudo: 'ALL=(ALL) NOPASSWD:ALL'
{ssh_keys}
{password_config}

{package_update}

packages:
{packages}

runcmd:
{runcmd}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/cloud-inits/cloud-init-ubuntu.yaml (914.0 B)

```
#cloud-config
hostname: {hostname}
manage_etc_hosts: true
preserve_hostname: false
ssh_pwauth: {ssh_pwauth}

{write_files_section}

# ??????????? ?? ????????? ???????
bootcmd:
  - mkdir -p /etc/systemd/resolved.conf.d
  - |
    cat >/etc/systemd/resolved.conf.d/custom-dns.conf << 'EOF'
    [Resolve]
    DNS=8.8.8.8 1.1.1.1
    FallbackDNS=1.0.0.1
    Domains=~.
    DNSSEC=no
    EOF
  - systemctl restart systemd-resolved || true
  - sleep 3
  - |
    for i in 1 2 3 4 5; do
      if getent hosts archive.ubuntu.com >/dev/null 2>&1; then
        echo "DNS OK"
        break
      fi
      echo "DNS not ready yet, retry..."
      sleep 2
    done

users:
  - name: {username}
    groups: sudo
    shell: /bin/bash
    sudo: 'ALL=(ALL) NOPASSWD:ALL'
{ssh_keys}
{password_config}

{package_update}

packages:
{packages}

runcmd:
{runcmd}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/admin-llm.css (5.5 KB)

```
/* LLM Spend & Admin LLM pages - readable layout */

.admin-llm-page {
  max-width: 1200px;
  margin: 0 auto;
}

.admin-llm-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 16px;
  margin-bottom: 24px;
}

.admin-llm-header h1 {
  font-size: 22px;
  font-weight: 700;
  color: var(--text-primary);
  margin: 0;
}

.admin-llm-controls {
  display: flex;
  align-items: center;
  gap: 12px;
}

.admin-llm-controls label {
  font-size: 14px;
  color: var(--text-secondary);
}

.admin-llm-controls select {
  padding: 8px 12px;
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius);
  background: var(--surface);
  color: var(--text-primary);
  font-size: 14px;
}

/* Summary cards */
.admin-llm-summary {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 16px;
  margin-bottom: 32px;
}

.admin-llm-summary-card {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  padding: 20px;
  box-shadow: var(--shadow-sm);
}

.admin-llm-summary-card .label {
  font-size: 13px;
  color: var(--text-muted);
  margin-bottom: 4px;
}

```

```
.admin-llm-summary-card .value {
  font-size: 20px;
  font-weight: 700;
  color: var(--text-primary);
  font-variant-numeric: tabular-nums;
}

.admin-llm-summary-card .value.mono {
  font-family: 'SF Mono', Monaco, Consolas, monospace;
}

/* Section headers */
.admin-llm-section {
  margin-bottom: 28px;
}

.admin-llm-section h3 {
  font-size: 16px;
  font-weight: 600;
  color: var(--text-primary);
  margin: 0 0 12px 0;
  padding-bottom: 8px;
  border-bottom: 1px solid var(--border-subtle);
}

/* Tables */
.admin-llm-table-wrap {
  overflow-x: auto;
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius);
  box-shadow: var(--shadow-sm);
}

.admin-llm-table {
  width: 100%;
  border-collapse: collapse;
  font-size: 14px;
}

.admin-llm-table th,
.admin-llm-table td {
  padding: 12px 16px;
  text-align: left;
  border-bottom: 1px solid var(--border-subtle);
}

.admin-llm-table th {
  background: var(--surface-muted);
  font-weight: 600;
  color: var(--text-secondary);
  font-size: 12px;
  text-transform: uppercase;
  letter-spacing: 0.04em;
}

.admin-llm-table td {
  color: var(--text-primary);
}

.admin-llm-table tbody tr:last-child td {
  border-bottom: none;
}

.admin-llm-table tbody tr:hover {
  background: var(--surface-muted);
}

.admin-llm-table .num {
  font-variant-numeric: tabular-nums;
  font-family: 'SF Mono', Monaco, Consolas, monospace;
```

```
}

.admin-llm-table .empty-row td {
  color: var(--text-muted);
  font-style: italic;
  padding: 24px;
  text-align: center;
}

/* Keys page */
.llm-keys-page .admin-llm-page {
  max-width: 900px;
}

.llm-keys-hint {
  font-size: 14px;
  color: var(--text-muted);
  margin: 0 0 20px 0;
  line-height: 1.5;
}

.keys-count-meta {
  font-size: 14px;
  color: var(--text-secondary);
}

.key-status-badge {
  display: inline-block;
  padding: 4px 10px;
  border-radius: var(--border-radius);
  font-size: 12px;
  font-weight: 500;
}

.key-status-badge.active {
  background: var(--status-success-bg);
  color: var(--status-success);
}

.key-status-badge.revoked {
  background: var(--surface-muted);
  color: var(--text-muted);
}

.keys-table .key-cell {
  font-family: 'SF Mono', Monaco, Consolas, monospace;
  font-size: 13px;
  color: var(--text-secondary);
}

.keys-table .key-cell .key-full {
  color: var(--text-primary);
}

.keys-table .key-actions {
  display: flex;
  gap: 8px;
}

.spend-charts-grid {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
  gap: 16px;
  margin: 0 0 28px;
}

.spend-chart-card {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  padding: 16px;
```

```
  box-shadow: var(--shadow-sm);
}

.spend-chart-card h3 {
  margin: 0 0 12px;
  font-size: 15px;
  font-weight: 600;
}

.spend-donut-wrap {
  display: flex;
  align-items: center;
  gap: 12px;
}

.spend-donut {
  width: 120px;
  height: 120px;
  border-radius: 999px;
  position: relative;
  flex: 0 0 auto;
}

.spend-donut::after {
  content: "";
  position: absolute;
  inset: 22px;
  border-radius: 999px;
  background: var(--surface);
}

.spend-legend {
  list-style: none;
  margin: 0;
  padding: 0;
  width: 100%;
  display: flex;
  flex-direction: column;
  gap: 7px;
}

.spend-legend-item {
  display: flex;
  justify-content: space-between;
  gap: 8px;
  font-size: 13px;
}

.spend-legend-label {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  color: var(--text-secondary);
}

.spend-legend-dot {
  width: 10px;
  height: 10px;
  border-radius: 999px;
  display: inline-block;
}

.spend-legend-empty,
.spend-bars-empty {
  color: var(--text-muted);
  font-size: 13px;
}

.spend-bars {
  display: flex;
  flex-direction: column;
```

```
  gap: 10px;
}

.spend-bar-row {
  display: grid;
  grid-template-columns: 84px 1fr auto;
  gap: 8px;
  align-items: center;
}

.spend-bar-label {
  font-size: 12px;
  color: var(--text-secondary);
}

.spend-bar-track {
  height: 8px;
  border-radius: 999px;
  background: var(--surface-muted);
  overflow: hidden;
}

.spend-bar-fill {
  height: 100%;
  border-radius: 999px;
  background: linear-gradient(90deg, #2563eb, #7c3aed);
}

.spend-bar-value {
  font-size: 12px;
  color: var(--text-muted);
  font-variant-numeric: tabular-nums;
  min-width: 72px;
  text-align: right;
}

@media (max-width: 1100px) {
  .spend-charts-grid {
    grid-template-columns: 1fr;
  }
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/chat-common.css (3.3 KB)

```
/* ============================================
   Common Chat Styles
   Used by llm-chat.css and vm-details-new.css
   ============================================ */

/* Chat Message Text Styling */
.chat-message-text {
  padding: 14px 18px;
  border-radius: 18px;
  font-size: 15px;
  line-height: 1.6;
}

.chat-message-user .chat-message-text {
  background: var(--surface-muted);
  color: var(--text-primary);
  border: 1px solid var(--border-subtle);
  border-bottom-right-radius: 6px;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.06);
}

.chat-message-assistant .chat-message-text {
  background: var(--surface-muted);
  color: var(--text-primary);
  border-bottom-left-radius: 6px;
  border: 1px solid var(--border-subtle);
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.06);
}

/* Typing Indicator */
.typing-indicator {
  display: inline-flex;
  gap: 5px;
  align-items: center;
  padding: 6px 0;
}

.typing-indicator span {
  width: 8px;
  height: 8px;
  background: var(--text-muted);
  border-radius: 50%;
  animation: typingBounce 1.4s ease-in-out infinite;
}

.typing-indicator span:nth-child(1) { animation-delay: 0s; }
.typing-indicator span:nth-child(2) { animation-delay: 0.2s; }
.typing-indicator span:nth-child(3) { animation-delay: 0.4s; }

@keyframes typingBounce {
  0%, 60%, 100% {
    transform: translateY(0);
    opacity: 0.4;
  }
  30% {
    transform: translateY(-8px);
    opacity: 1;
  }
}

/* Markdown Styles in Chat */
.chat-message-text p {
  margin: 0 0 14px;
}

```

```
.chat-message-text p:last-child {
  margin-bottom: 0;
}

.chat-message-text h1,
.chat-message-text h2,
.chat-message-text h3,
.chat-message-text h4 {
  margin: 18px 0 10px;
  font-weight: 600;
  color: var(--text-primary);
}

.chat-message-text h1:first-child,
.chat-message-text h2:first-child,
.chat-message-text h3:first-child,
.chat-message-text h4:first-child {
  margin-top: 0;
}

.chat-message-text ul,
.chat-message-text ol {
  margin: 10px 0;
  padding-left: 24px;
}

.chat-message-text li {
  margin: 6px 0;
}

.chat-message-text code {
  font-family: 'SF Mono', Monaco, Consolas, monospace;
  font-size: 13px;
  background: var(--surface);
  padding: 3px 8px;
  border-radius: 6px;
  border: 1px solid var(--border-subtle);
}

.chat-message-text pre {
  margin: 14px 0;
  background: #1e1e2e;
  border-radius: 10px;
  overflow: hidden;
}

.chat-message-text pre code {
  display: block;
  padding: 16px 18px;
  background: transparent;
  border: none;
  color: #cdd6f4;
  overflow-x: auto;
  font-size: 13px;
  line-height: 1.5;
}

.chat-message-text a {
  color: var(--brand-blue-mid);
  text-decoration: none;
}

.chat-message-text a:hover {
  text-decoration: underline;
}

.chat-message-text blockquote {
  margin: 14px 0;
  padding: 10px 18px;
  border-left: 3px solid var(--brand-blue-mid);
  background: var(--surface);
```

```
  border-radius: 0 10px 10px 0;
}

.chat-message-text table {
  width: 100%;
  border-collapse: collapse;
  margin: 14px 0;
  font-size: 13px;
}

.chat-message-text th,
.chat-message-text td {
  padding: 10px 14px;
  border: 1px solid var(--border-subtle);
  text-align: left;
}

.chat-message-text th {
  background: var(--surface);
  font-weight: 600;
}

.chat-error {
  color: var(--status-error);
  font-style: italic;
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/create-vm.css (13.8 KB)

```
/* ============================================
   Create VM Modal - Viiruoz Theme
   Consistent with main design system
   ============================================ */

.modal-overlay {
  position: fixed;
  inset: 0;
  background: rgba(0, 0, 0, 0.5);
  backdrop-filter: blur(4px);
  z-index: 999;
}

.modal {
  position: fixed;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%);
  background: var(--surface, #FFFFFF);
  color: var(--text-primary, #111827);
  border: 1px solid var(--border-subtle, #E5E7EB);
  border-radius: 16px;
  width: min(720px, calc(100vw - 32px));
  max-height: calc(100vh - 60px);
  overflow-y: auto;
  padding: 28px;
  box-shadow: 0 20px 40px rgba(0, 0, 0, 0.15);
  z-index: 1000;
}

[data-theme="dark"] .modal {
  box-shadow: 0 20px 40px rgba(0, 0, 0, 0.4);
}

.modal-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 24px;
  padding-bottom: 16px;
  border-bottom: 1px solid var(--border-subtle, #E5E7EB);
}

.modal-header h2 {
  margin: 0;
  font-size: 18px;
  font-weight: 700;
  color: var(--text-primary, #111827);
}

.modal-header button {
  border: 1px solid var(--border-subtle, #E5E7EB);
  background: var(--surface-muted, #F3F4F6);
  color: var(--text-secondary, #6B7280);
  width: 32px;
  height: 32px;
  border-radius: 8px;
  font-size: 18px;
  cursor: pointer;
  transition: all 0.15s ease;
  display: flex;
  align-items: center;
  justify-content: center;
}

```

```
.modal-header button:hover {
  background: var(--status-error-bg, #FEE2E2);
  border-color: var(--status-error, #DC2626);
  color: var(--status-error, #DC2626);
}

.create-form {
  display: flex;
  flex-direction: column;
  gap: 16px;
}

/* Form spacing utilities (replaces inline styles) */
.form-section-spacing {
  margin-bottom: 24px;
}

.preset-block-spacing {
  margin-bottom: 16px;
}

/* Form Tabs */
.form-tabs {
  display: inline-flex;
  background: var(--surface-muted, #F3F4F6);
  border-radius: 8px;
  padding: 4px;
  gap: 4px;
  align-self: flex-start;
}

.form-tab {
  border: none;
  background: transparent;
  padding: 8px 16px;
  border-radius: 6px;
  font-weight: 600;
  font-size: 13px;
  color: var(--text-muted, #9CA3AF);
  cursor: pointer;
  transition: all 0.15s ease;
}

.form-tab:hover {
  color: var(--text-secondary, #6B7280);
}

.form-tab.active {
  background: var(--surface, #FFFFFF);
  color: var(--brand-blue-mid, #3B82A5);
  box-shadow: 0 1px 3px rgba(0, 0, 0, 0.1);
}

[data-theme="dark"] .form-tab.active {
  background: var(--surface-elevated, #111827);
}

.tab-panel {
  display: none;
}

.tab-panel.active {
  display: block;
}

/* Preset Block */
.preset-block {
  background: var(--surface-muted, #F3F4F6);
  border: 1px solid var(--border-subtle, #E5E7EB);
  border-radius: 12px;
  padding: 18px;
```

```
  display: flex;
  flex-direction: column;
  gap: 14px;
}

.manual-grid {
  margin-top: 8px;
}

.preset-title {
  margin: 0;
  font-weight: 700;
  font-size: 14px;
  color: var(--text-primary, #111827);
}

.preset-hint {
  margin: 4px 0 0;
  color: var(--text-secondary, #6B7280);
  font-size: 13px;
}

.preset-list {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(160px, 1fr));
  gap: 10px;
}

.preset-card {
  border: 2px solid var(--border-subtle, #E5E7EB);
  border-radius: 10px;
  padding: 12px 14px;
  background: var(--surface, #FFFFFF);
  cursor: pointer;
  text-align: left;
  display: flex;
  flex-direction: column;
  gap: 4px;
  transition: all 0.15s ease;
}

.preset-card:hover {
  border-color: var(--brand-blue-mid, #3B82A5);
}

.preset-card.active,
.preset-card.selected {
  border-color: var(--brand-blue-mid, #3B82A5);
  background: var(--nav-active-bg, #E5F3F7);
}

[data-theme="dark"] .preset-card.selected {
  background: var(--nav-active-bg, #0F172A);
}

.preset-name {
  font-weight: 600;
  font-size: 13px;
  color: var(--text-primary, #111827);
}

.preset-meta {
  font-size: 11px;
  color: var(--text-secondary, #6B7280);
}

/* Form Grid */
.form-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 14px;
```

```
}

label {
  font-size: 13px;
  font-weight: 600;
  display: flex;
  flex-direction: column;
  gap: 6px;
  color: var(--text-secondary, #6B7280);
}

input,
select,
textarea {
  border-radius: 8px;
  border: 1px solid var(--border-subtle, #E5E7EB);
  background: var(--surface-muted, #F3F4F6);
  color: var(--text-primary, #111827);
  padding: 10px 12px;
  font-size: 13px;
  font-family: inherit;
  transition: all 0.15s ease;
}

input::placeholder,
textarea::placeholder {
  color: var(--text-muted, #9CA3AF);
}

input:focus,
select:focus,
textarea:focus {
  outline: none;
  border-color: var(--brand-blue-mid, #3B82A5);
  background: var(--surface, #FFFFFF);
}

select {
  cursor: pointer;
  appearance: none;
  background-image: url("data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' width='12' height='12' fill='%2...
  background-repeat: no-repeat;
  background-position: right 10px center;
  padding-right: 32px;
}

select option {
  background: var(--surface, #FFFFFF);
  color: var(--text-primary, #111827);
}

textarea {
  resize: vertical;
  min-height: 72px;
}

.full-width {
  width: 100%;
}

/* Checkbox Group */
.checkbox-group {
  display: flex;
  flex-direction: column;
  gap: 10px;
  margin: 14px 0;
}

.checkbox-group label {
  display: flex;
  align-items: center;
```

```
  gap: 8px;
  font-weight: 500;
  cursor: pointer;
  flex-direction: row;
  color: var(--text-primary, #111827);
}

.checkbox-group input[type="checkbox"],
.checkbox-group input[type="radio"] {
  width: 16px;
  height: 16px;
  margin: 0;
  cursor: pointer;
  accent-color: var(--brand-blue-mid, #3B82A5);
}

/* Wizard Steps */
.wizard-steps {
  display: flex;
  justify-content: center;
  gap: 40px;
  margin: 16px 0 28px;
}

.wizard-step {
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 8px;
  opacity: 0.4;
  transition: opacity 0.2s ease;
}

.wizard-step.active {
  opacity: 1;
}

.wizard-step.completed {
  opacity: 0.7;
}

.step-number {
  display: flex;
  align-items: center;
  justify-content: center;
  width: 40px;
  height: 40px;
  border-radius: 50%;
  background: var(--surface-muted, #F3F4F6);
  color: var(--text-muted, #9CA3AF);
  font-weight: 700;
  font-size: 14px;
  border: 2px solid var(--border-subtle, #E5E7EB);
  transition: all 0.2s ease;
}

.wizard-step.active .step-number {
  background: var(--brand-blue-deep, #0F3D5E);
  color: #FFFFFF;
  border-color: var(--brand-blue-deep, #0F3D5E);
}

[data-theme="dark"] .wizard-step.active .step-number {
  background: var(--brand-blue-mid, #3B82A5);
  border-color: var(--brand-blue-mid, #3B82A5);
}

.wizard-step.completed .step-number {
  background: var(--status-success, #059669);
  color: #FFFFFF;
  border-color: var(--status-success, #059669);
```

```
}

.step-label {
  font-size: 12px;
  font-weight: 600;
  color: var(--text-muted, #9CA3AF);
}

.wizard-step.active .step-label {
  color: var(--text-primary, #111827);
}

/* Wizard Actions */
.wizard-actions {
  display: flex;
  gap: 10px;
  justify-content: flex-end;
  margin-top: 24px;
  padding-top: 20px;
  border-top: 1px solid var(--border-subtle, #E5E7EB);
}

.wizard-actions button,
.form-actions button {
  padding: 10px 18px;
  border: none;
  border-radius: 8px;
  font-size: 13px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.15s ease;
}

.wizard-actions .primary-btn,
.form-actions .primary-btn {
  background: var(--primary-bg, var(--brand-blue-deep, #0F3D5E));
  color: var(--primary-text, #FFFFFF);
}

.wizard-actions .primary-btn:hover,
.form-actions .primary-btn:hover {
  background: var(--primary-hover, var(--brand-blue-mid, #3B82A5));
}

.wizard-actions .ghost-btn,
.form-actions .ghost-btn {
  background: transparent;
  color: var(--text-secondary, #6B7280);
  border: 1px solid var(--border-subtle, #E5E7EB);
}

.wizard-actions .ghost-btn:hover,
.form-actions .ghost-btn:hover {
  background: var(--surface-muted, #F3F4F6);
  color: var(--text-primary, #111827);
}

/* Info Block */
.info-block {
  background: var(--nav-active-bg, #E5F3F7);
  border: 1px solid var(--brand-blue-mid, #3B82A5);
  border-radius: 8px;
  padding: 12px 14px;
  color: var(--text-secondary, #6B7280);
  font-size: 12px;
  line-height: 1.5;
  margin-top: 10px;
}

[data-theme="dark"] .info-block {
  background: var(--nav-active-bg, #0F172A);
```

```
}

/* GPU Section */
.gpu-section {
  margin-top: 20px;
  padding-top: 16px;
  border-top: 1px solid var(--border-subtle, #E5E7EB);
}

.gpu-section h3 {
  font-size: 14px;
  font-weight: 700;
  margin-bottom: 12px;
  color: var(--text-primary, #111827);
}

#gpu-count-label {
  display: none;
}

#gpu-count-label.gpu-count-visible {
  display: flex;
}

#gpu-count-label.visible {
  display: flex;
}

.gpu-count-hidden {
  display: none;
}

/* Password Field */
.additional-packages-field {
  margin-top: 14px;
}

#password-field.hidden {
  display: none;
}

/* Wizard Navigation Buttons */
#wizard-back,
#wizard-submit {
  /* Default display style - visibility controlled by .hidden utility class */
  display: inline-flex;
  align-items: center;
  justify-content: center;
}

#wizard-back.visible,
#wizard-submit.visible {
  /* Legacy support if .visible is used anywhere else */
  display: inline-flex;
}

.hidden-permanently {
  display: none;
}

/* Responsive */
@media (max-width: 768px) {
  .modal {
    width: calc(100vw - 24px);
    padding: 20px;
    top: 16px;
    transform: translate(-50%, 0);
  }

  .wizard-steps {
    gap: 24px;
```

```
492   }
493
494   .step-number {
495     width: 36px;
496     height: 36px;
497     font-size: 13px;
498   }
499
500   .preset-list {
501
502 ... [169 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/dashboard-home.css (3.0 KB)

```
.dh-wrap {
  display: grid;
  gap: 18px;
}

.dh-welcome h1 {
  font-size: 28px;
  font-weight: 700;
  color: var(--text-primary);
}

.dh-welcome p {
  margin-top: 4px;
  color: var(--text-secondary);
}

.dh-card {
  display: block;
  text-decoration: none;
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  padding: 18px;
  transition: border-color var(--transition-fast), box-shadow var(--transition-fast);
}

.dh-card:hover {
  border-color: var(--brand-blue-mid);
  box-shadow: var(--shadow-sm);
}

.dh-card-head {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 10px;
  margin-bottom: 14px;
}

.dh-card-head h2 {
  font-size: 20px;
  color: var(--text-primary);
}

.dh-card-link {
  font-size: 13px;
  color: var(--text-secondary);
}

.dh-balance-grid {
  display: grid;
  grid-template-columns: repeat(2, minmax(0, 1fr));
  gap: 16px;
}

.dh-stat-label {
  font-size: 12px;
  text-transform: uppercase;
  letter-spacing: 0.03em;
  color: var(--text-muted);
  margin-bottom: 6px;
}

.dh-stat-value {
  font-size: 30px;
```

```
  font-weight: 700;
  color: var(--text-primary);
}

.dh-transfer {
  margin-top: 16px;
  padding-top: 14px;
  border-top: 1px solid var(--border-subtle);
}

.dh-transfer-head {
  font-size: 13px;
  color: var(--text-secondary);
  margin-bottom: 8px;
}

.dh-transfer-row {
  display: grid;
  grid-template-columns: minmax(160px, 1fr) minmax(120px, 1fr) auto auto;
  gap: 8px;
}

.dh-transfer-row select,
.dh-transfer-row input {
  width: 100%;
  padding: 8px 10px;
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-sm);
  background: var(--surface-muted);
  color: var(--text-primary);
}

.dh-transfer-error {
  margin-top: 8px;
  font-size: 12px;
  color: var(--status-error);
}

.dh-vm-stats {
  margin-top: 2px;
}

.dh-quickstart {
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius);
  background: var(--surface-muted);
  overflow: hidden;
}

.dh-quickstart-head {
  padding: 10px 12px;
  border-bottom: 1px solid var(--border-subtle);
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 12px;
}

.dh-quickstart-actions {
  display: flex;
  align-items: center;
  gap: 8px;
}

.dh-tabs {
  display: inline-flex;
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-sm);
  overflow: hidden;
}

```

```
.dh-tab {
  border: none;
  background: transparent;
  color: var(--text-secondary);
  padding: 6px 9px;
  cursor: pointer;
  font-size: 12px;
}

.dh-tab.active {
  background: var(--surface);
  color: var(--text-primary);
}

.dh-code {
  margin: 0;
  padding: 12px;
  font-size: 12px;
  line-height: 1.5;
  overflow-x: auto;
  color: var(--text-primary);
  background: transparent;
}

@media (max-width: 900px) {
  .dh-transfer-row {
    grid-template-columns: 1fr 1fr;
  }
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/dashboard-layout.css (37.0 KB)

```
/* ============================================
   ???????? - Modern Dashboard Layout
   Design System matching logo colors
   ============================================ */

/* Brand Palette (from logo) */
:root {
  --brand-bg: #111827;
  --brand-blue-deep: #0f3d5e;
  --brand-blue-mid: #3b82a5;
  --brand-aqua-light: #c4f0f2;
}

/* Light Theme (default) */
:root,
[data-theme="light"] {
  /* Backgrounds */
  --app-bg: #f9fafb;
  --surface: #ffffff;
  --surface-muted: #f3f4f6;
  --surface-elevated: #ffffff;

  /* Text */
  --text-primary: #111827;
  --text-secondary: #6b7280;
  --text-muted: #9ca3af;
  --text-disabled: #d1d5db;

  /* Borders */
  --border-subtle: #e5e7eb;
  --border-accent: var(--brand-blue-mid);

  /* Primary Actions */
  --primary-bg: var(--brand-blue-deep);
  --primary-hover: var(--brand-blue-mid);
  --primary-text: #ffffff;

  /* Gradient Buttons */
  --button-gradient-text: #020617;

  /* Navigation */
  --nav-active-bg: #e5f3f7;
  --nav-active-border: var(--brand-blue-mid);
  --nav-hover-bg: #f3f4f6;

  /* Badges & Tags */
  --badge-bg: var(--brand-aqua-light);
  --badge-text: #0f172a;

  /* Status Colors */
  --status-success: #059669;
  --status-success-bg: #d1fae5;
  --status-warning: #d97706;
  --status-warning-bg: #fef3c7;
  --status-error: #dc2626;
  --status-error-bg: #fee2e2;
  --status-info: var(--brand-blue-mid);
  --status-info-bg: #e5f3f7;

  /* Shadows */
  --shadow-sm: 0 1px 2px rgba(0, 0, 0, 0.05);
  --shadow-md:
    0 4px 6px -1px rgba(0, 0, 0, 0.1), 0 2px 4px -1px rgba(0, 0, 0, 0.06);
  --shadow-lg:
    0 10px 15px -3px rgba(0, 0, 0, 0.1), 0 4px 6px -2px rgba(0, 0, 0, 0.05);
```

```
  /* Sidebar */
  --sidebar-bg: var(--surface-muted);
  --sidebar-border: var(--border-subtle);

  /* Topbar */
  --topbar-bg: var(--surface);
  --topbar-border: var(--border-subtle);
}

/* Dark Theme */
[data-theme="dark"] {
  /* Backgrounds */
  --app-bg: #020617;
  --surface: #0b1120;
  --surface-muted: #020617;
  --surface-elevated: #111827;

  /* Text */
  --text-primary: #f9fafb;
  --text-secondary: #9ca3af;
  --text-muted: #6b7280;
  --text-disabled: #4b5563;

  /* Borders */
  --border-subtle: #1f2937;
  --border-accent: var(--brand-aqua-light);

  /* Primary Actions */
  --primary-bg: var(--brand-blue-mid);
  --primary-hover: var(--brand-aqua-light);
  --primary-text: #020617;

  /* Gradient Buttons */
  --button-gradient-text: #ffffff;

  /* Navigation */
  --nav-active-bg: #0f172a;
  --nav-active-border: var(--brand-aqua-light);
  --nav-hover-bg: #111827;

  /* Badges & Tags */
  --badge-bg: #1e293b;
  --badge-text: #e5f3f7;

  /* Status Colors */
  --status-success: #10b981;
  --status-success-bg: rgba(16, 185, 129, 0.15);
  --status-warning: #f59e0b;
  --status-warning-bg: rgba(245, 158, 11, 0.15);
  --status-error: #ef4444;
  --status-error-bg: rgba(239, 68, 68, 0.15);
  --status-info: var(--brand-aqua-light);
  --status-info-bg: rgba(196, 240, 242, 0.1);

  /* Shadows */
  --shadow-sm: 0 1px 2px rgba(0, 0, 0, 0.3);
  --shadow-md: 0 4px 6px -1px rgba(0, 0, 0, 0.4);
  --shadow-lg: 0 10px 15px -3px rgba(0, 0, 0, 0.5);

  /* Sidebar */
  --sidebar-bg: var(--surface);
  --sidebar-border: var(--border-subtle);

  /* Topbar */
  --topbar-bg: var(--surface);
  --topbar-border: var(--border-subtle);
}

/* Layout Variables */
:root {
```

```
  --sidebar-width: 260px;
  --sidebar-collapsed-width: 72px;
  --topbar-height: 60px;
  --content-max-width: 1400px;
  --border-radius: 8px;
  --border-radius-sm: 6px;
  --border-radius-lg: 12px;
  --font-family:
    "Inter", -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
  --transition-fast: 0.15s ease;
  --transition-normal: 0.2s ease;
}

/* Reset & Base */
*,
*::before,
*::after {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

html {
  font-size: 14px;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
}

body {
  font-family: var(--font-family);
  background: var(--app-bg);
  color: var(--text-primary);
  line-height: 1.5;
  min-height: 100vh;
}

/* Dashboard Layout */
.dashboard-layout {
  display: flex;
  min-height: 100vh;
  background: var(--app-bg);
}

/* ============================================
   SIDEBAR
   ============================================ */
.sidebar {
  position: fixed;
  left: 0;
  top: 0;
  bottom: 0;
  width: var(--sidebar-width);
  background: var(--sidebar-bg);
  border-right: 1px solid var(--sidebar-border);
  display: flex;
  flex-direction: column;
  z-index: 100;
  transition:
    transform var(--transition-normal),
    width var(--transition-normal);
}

.sidebar-header {
  padding: 20px;
  border-bottom: 1px solid var(--border-subtle);
  background: var(--brand-bg);
}

.logo-link {
  display: block;
  text-decoration: none;
```

```
}

.logo-img {
  width: 100%;
  max-width: 160px;
  height: auto;
  display: block;
  margin: 0 auto;
}

/* Sidebar Navigation */
.sidebar-nav {
  flex: 1;
  padding: 16px 12px;
  overflow-y: auto;
}

.nav-list {
  list-style: none;
  display: flex;
  flex-direction: column;
  gap: 4px;
}

.nav-link {
  display: flex;
  align-items: center;
  gap: 12px;
  width: 100%;
  padding: 12px 14px;
  border-radius: var(--border-radius);
  text-decoration: none;
  color: var(--text-secondary);
  font-weight: 500;
  font-size: 14px;
  transition: all var(--transition-fast);
  border-left: 3px solid transparent;
}

.nav-link-accordion {
  justify-content: flex-start;
  border: none;
  font-family: inherit;
  text-align: left;
  background: transparent;
  cursor: pointer;
}

.nav-link:hover:not(.disabled) {
  background: var(--nav-hover-bg);
  color: var(--text-primary);
}

.nav-link.active {
  background: var(--nav-active-bg);
  color: var(--brand-blue-deep);
  border-left-color: var(--nav-active-border);
}

[data-theme="dark"] .nav-link.active {
  color: var(--brand-aqua-light);
}

.nav-link.disabled {
  color: var(--text-disabled);
  cursor: not-allowed;
}

.nav-link.disabled:hover {
  background: transparent;
}
```

```
.nav-icon {
  width: 20px;
  height: 20px;
  flex-shrink: 0;
  opacity: 0.7;
}

.nav-chevron {
  width: 16px;
  height: 16px;
  margin-left: auto;
  opacity: 0.7;
  transition: transform var(--transition-fast);
}

.nav-item-llm.is-open .nav-chevron {
  transform: rotate(90deg);
}

.nav-submenu {
  list-style: none;
  margin: 4px 0 6px 18px;
  padding-left: 14px;
  border-left: 1px solid var(--border-subtle);
  display: none;
}

.nav-submenu.submenu-open {
  display: block;
}

.nav-sublink {
  display: block;
  padding: 8px 10px;
  color: var(--text-secondary);
  text-decoration: none;
  border-radius: var(--border-radius-sm);
  font-size: 13px;
  font-weight: 500;
  transition: all var(--transition-fast);
}

.nav-sublink:hover {
  background: var(--nav-hover-bg);
  color: var(--text-primary);
}

.nav-sublink.active {
  background: var(--nav-active-bg);
  color: var(--brand-blue-deep);
}

[data-theme="dark"] .nav-sublink.active {
  color: var(--brand-aqua-light);
}

.nav-link.active .nav-icon {
  opacity: 1;
}

.badge {
  font-size: 10px;
  font-weight: 600;
  padding: 2px 8px;
  border-radius: 10px;
  text-transform: uppercase;
  letter-spacing: 0.3px;
  margin-left: auto;
  pointer-events: none;
}
```

```

.badge-soon {
  background: var(--badge-bg);
  color: var(--badge-text);
}

.badge-new {
  background: linear-gradient(
    135deg,
    var(--brand-aqua-mid),
    var(--brand-blue-mid)
  );
  color: var(--button-gradient-text) !important;
  animation: badgePulse 2s ease-in-out infinite;
}

@keyframes badgePulse {
  0%,
  100% {
    opacity: 1;
  }
  50% {
    opacity: 0.7;
  }
}

/* Sidebar Footer */
.sidebar-footer {
  padding: 16px;
  border-top: 1px solid var(--border-subtle);
}

.user-info-sidebar {
  display: flex;
  align-items: center;
  gap: 12px;
}

.user-avatar {
  width: 36px;
  height: 36px;
  border-radius: var(--border-radius);
  background: var(--brand-blue-deep);
  display: flex;
  align-items: center;
  justify-content: center;
  font-weight: 600;
  font-size: 13px;
  color: var(--primary-text);
  flex-shrink: 0;
}

[data-theme="dark"] .user-avatar {
  background: var(--brand-blue-mid);
  color: var(--brand-bg);
}

.user-details {
  display: flex;
  flex-direction: column;
  gap: 2px;
  min-width: 0;
}

.user-name-sidebar {
  font-size: 13px;
  font-weight: 600;
  color: var(--text-primary);
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
```

```
}

.user-namespace {
  font-size: 11px;
  color: var(--text-muted);
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}

.sidebar-collapse-toggle {
  width: 100%;
  margin-top: 12px;
  height: 34px;
  border: 1px solid var(--border-subtle);
  background: var(--surface);
  border-radius: var(--border-radius-sm);
  color: var(--text-secondary);
  cursor: pointer;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  transition: all var(--transition-fast);
}

.sidebar-collapse-toggle:hover {
  color: var(--text-primary);
  background: var(--surface-muted);
}

.sidebar-collapse-toggle svg {
  width: 18px;
  height: 18px;
}

.sidebar-collapse-toggle .icon-collapse-right {
  display: none;
}

.user-balance-sidebar .balance-label {
  display: block;
  font-size: 11px;
  font-weight: 600;
  color: var(--text-muted);
  margin-bottom: 4px;
  text-transform: uppercase;
  letter-spacing: 0.02em;
}

.user-balance-sidebar {
  margin-top: 10px;
  padding-top: 10px;
  border-top: 1px solid var(--border-subtle);
  font-size: 12px;
  color: var(--text-muted);
  line-height: 1.4;
}

.user-balance-sidebar .balance-values {
  word-break: break-word;
}

.dropdown-balance {
  padding: 8px 12px;
  font-size: 12px;
  color: var(--text-muted);
  border-bottom: 1px solid var(--border-subtle);
}
.dropdown-balance-label {
  font-weight: 600;
  margin-right: 6px;
```

```
492 }
493
494 /* ==========================================
495    MAIN WRAPPER
496    ========================================== */
497 .main-wrapper {
498   flex: 1;
499   margin-left: var(--sidebar-width);
500   display: flex;
501
502 ... [1402 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/llm-chat.css (14.1 KB)

```
/* ============================================
   LLM Chat Page Styles
   ============================================ */

.llm-page {
  height: calc(100vh - 120px);
  display: flex;
  flex-direction: column;
}

.llm-playground {
  display: block;
  flex: 1;
  min-height: 0;
}

.chat-layout {
  display: grid;
  grid-template-columns: minmax(0, 1fr) 320px;
  gap: 12px;
  flex: 1;
  min-height: 0;
  height: 100%;
}

.chat-container {
  display: flex;
  flex-direction: column;
  min-width: 0;
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  overflow: hidden;
}

.playground-settings {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  padding: 12px;
  min-height: 0;
  display: flex;
  flex-direction: column;
  gap: 10px;
}

.chat-settings-drawer {
  position: relative;
  display: flex;
  flex-direction: column;
  align-items: stretch;
  gap: 0;
  padding: 8px;
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  min-height: 0;
}

.chat-settings-toggle {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  border: 1px solid var(--border-subtle);
  background: var(--surface);
```

```
  color: var(--text-secondary);
  border-radius: 999px;
  padding: 7px 10px;
  font-size: 12px;
  font-weight: 600;
  cursor: pointer;
  transition: all var(--transition-fast);
}

.chat-settings-toggle svg {
  width: 14px;
  height: 14px;
}

.chat-settings-toggle:hover {
  border-color: var(--brand-blue-mid);
  color: var(--brand-blue-mid);
}

.chat-settings-toggle::after {
  content: "";
}

.chat-settings-toggle[aria-expanded="true"]::after {
  content: "";
}

.chat-settings-drawer .playground-settings {
  width: 100%;
  max-height: none;
  height: 100%;
  overflow-y: auto;
  margin-top: 0;
  box-shadow: none;
  z-index: 1;
}

.settings-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 8px;
}

.playground-settings h4 {
  margin: 0;
  font-size: 14px;
  color: var(--text-primary);
}

.settings-caption {
  margin: 0;
  color: var(--text-muted);
  font-size: 11px;
  line-height: 1.35;
}

.settings-reset-btn {
  border: 1px solid var(--border-subtle);
  background: var(--surface-muted);
  color: var(--text-secondary);
  font-size: 11px;
  border-radius: 999px;
  padding: 3px 9px;
  cursor: pointer;
  transition: all var(--transition-fast);
}

.settings-reset-btn:hover {
  border-color: var(--brand-blue-mid);
  color: var(--brand-blue-mid);
```

```
}

.settings-section {
  border: 1px solid var(--border-subtle);
  border-radius: 12px;
  background: var(--surface-muted);
  padding: 10px;
}

.settings-section h5 {
  margin: 0 0 8px;
  font-size: 11px;
  letter-spacing: 0.03em;
  text-transform: uppercase;
  color: var(--text-muted);
}

.playground-field {
  margin-bottom: 8px;
}

.settings-section .playground-field:last-child {
  margin-bottom: 0;
}

.playground-field label {
  display: block;
  margin-bottom: 4px;
  color: var(--text-secondary);
  font-size: 11px;
  font-weight: 600;
}

.playground-field-head {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 4px;
}

.playground-field-head span {
  font-size: 11px;
  color: var(--text-primary);
}

.value-chip {
  min-width: 52px;
  text-align: center;
  border: 1px solid var(--border-subtle);
  background: var(--surface);
  border-radius: 999px;
  padding: 2px 7px;
  font-family: "SF Mono", Monaco, Consolas, monospace;
}

.playground-field input[type="range"] {
  width: 100%;
  accent-color: var(--brand-blue-mid);
}

.playground-field-meta {
  display: flex;
  justify-content: space-between;
  margin-top: 4px;
  font-size: 10px;
  color: var(--text-muted);
}

.settings-presets {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
```

```
  gap: 6px;
}

.settings-preset-btn {
  border: 1px solid var(--border-subtle);
  background: var(--surface);
  color: var(--text-secondary);
  border-radius: 8px;
  padding: 6px 5px;
  font-size: 11px;
  font-weight: 600;
  cursor: pointer;
  transition: all var(--transition-fast);
}

.settings-actions {
  display: flex;
  align-items: center;
  gap: 6px;
}

.settings-close-btn {
  border: 1px solid var(--border-subtle);
  background: var(--surface-muted);
  color: var(--text-secondary);
  border-radius: 8px;
  width: 24px;
  height: 24px;
  line-height: 1;
  padding: 0;
  cursor: pointer;
  transition: all var(--transition-fast);
}

.settings-close-btn:hover {
  border-color: var(--brand-blue-mid);
  color: var(--brand-blue-mid);
}

.settings-preset-btn:hover {
  border-color: var(--brand-blue-mid);
  color: var(--brand-blue-mid);
}

.settings-preset-btn.active {
  background: rgba(0, 156, 224, 0.12);
  border-color: var(--brand-blue-mid);
  color: var(--brand-blue-mid);
}

.chat-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 16px 20px;
  background: var(--surface-muted);
  border-bottom: 1px solid var(--border-subtle);
}

.chat-title {
  display: flex;
  align-items: center;
  gap: 14px;
}

.chat-title-icon {
  width: 44px;
  height: 44px;
  display: flex;
  align-items: center;
  justify-content: center;
```

```
  background: linear-gradient(135deg, var(--brand-blue-mid), var(--brand-aqua-mid));
  border-radius: 12px;
}

.chat-title-icon svg {
  width: 24px;
  height: 24px;
  color: var(--button-gradient-text) !important;
}

.chat-title h3 {
  font-size: 16px;
  font-weight: 600;
  color: var(--text-primary);
  margin: 0;
}

.chat-model {
  font-size: 12px;
  color: var(--text-muted);
  font-family: 'SF Mono', Monaco, Consolas, monospace;
}

.chat-model-select {
  font-size: 13px;
  color: var(--text-primary);
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius);
  padding: 4px 8px;
  max-width: 240px;
  cursor: pointer;
  width: 100%;
  max-width: none;
}

.chat-llm-balance {
  font-size: 12px;
  color: var(--text-muted);
}

.chat-clear-btn {
  display: flex;
  align-items: center;
  justify-content: center;
  width: 40px;
  height: 40px;
  padding: 0;
  background: transparent;
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius);
  color: var(--text-secondary);
  cursor: pointer;
  transition: all var(--transition-fast);
}

.chat-clear-btn:hover {
  background: var(--status-error-bg);
  border-color: var(--status-error);
  color: var(--status-error);
}

.chat-clear-btn svg {
  width: 18px;
  height: 18px;
}

/* LLM Connect (SK token) */
.llm-connect-section {
  display: flex;
  align-items: center;
```

```
  gap: 12px;
  padding: 12px 20px;
  background: var(--surface);
  border-bottom: 1px solid var(--border-subtle);
}

.llm-connect-section.llm-token-only {
  flex-wrap: wrap;
  gap: 8px;
}

.llm-token-hint {
  font-size: 12px;
  color: var(--text-muted);
}

.llm-connect-status {
  flex-shrink: 0;
}

.llm-status-badge {
  font-size: 12px;
  font-weight: 500;
  padding: 4px 10px;
  border-radius: 20px;
}

.llm-status-connected {
  background: var(--status-success-bg, rgba(34, 197, 94, 0.15));
  color: var(--status-success, #22c55e);
}

.llm-status-disconnected {
  background: var(--status-warning-bg, rgba(234, 179, 8, 0.15));
  color: var(--status-warning, #eab308);
}

.llm-connect-form {
  display: flex;
  gap: 8px;
  flex: 1;
  max-width: 400px;
  align-items: center;
}

.llm-create-token-btn {
  margin-left: auto;
  flex-shrink: 0;
}

.llm-sk-input {
  flex: 1;
  padding: 8px 14px;
  font-size: 13px;
  font-family: 'SF Mono', Monaco, Consolas, monospace;
  border: 1px solid var(--border-subtle);
  border-radius: 8px;
  background: var(--surface-muted);
  color: var(--text-primary);
}

.llm-sk-input:focus {
  outline: none;
  border-color: var(--brand-blue-mid);
}

.llm-sk-input::placeholder {
  color: var(--text-muted);
}

.llm-connect-btn {
```

```
  padding: 8px 16px;
  font-size: 13px;
  font-weight: 500;
  background: var(--brand-blue-mid);
  color: white;
  border: none;
  border-radius: 8px;
  cursor: pointer;
  transition: opacity var(--transition-fast);
}

.llm-connect-btn:hover {
  opacity: 0.9;
}

.llm-disconnect-btn {
  padding: 8px 16px;
  font-size: 13px;
  font-weight: 500;
  background: transparent;
  color: var(--text-secondary);
  border: 1px solid var(--border-subtle);
  border-radius: 8px;
  cursor: pointer;
  transition: opacity var(--transition-fast);
}

.llm-disconnect-btn:hover {
  opacity: 0.9;
  border-color: var(--status-error, #ef4444);
  color: var(--status-error, #ef4444);
}

.llm-create-token-btn {
  padding: 8px 16px;
  font-size: 13px;
  font-weight: 500;
  background: transparent;
  color: var(--text-secondary);
  border: 1px solid var(--border-subtle);
  border-radius: 8px;
  cursor: pointer;
  transition: opacity var(--transition-fast);
}

.llm-create-token-btn:hover {
  opacity: 0.9;
  border-color: var(--status-success, #22c55e);
  color: var(--status-success, #22c55e);
}

/* Messages Container */
.chat-messages {
  flex: 1;
  overflow-y: auto;
  padding: 24px;
  display: flex;
  flex-direction: column;
  gap: 20px;
  scroll-behavior: smooth;
}

/* Welcome Screen */
.chat-welcome {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  text-align: center;
  padding: 60px 20px;
  height: 100%;
```

```
492 }
493
494 .chat-welcome-icon {
495   width: 80px;
496   height: 80px;
497   display: flex;
498   align-items: center;
499   justify-content: center;
500   background: linear-gradient(135deg, var(--brand-blue-mid), var(--brand-aqua-mid));
501
502 ... [230 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/llm-models.css (6.5 KB)

```
/* ????????? ?????? */

.llm-models-page {
  height: auto;
  min-height: calc(100vh - 120px);
}

.llm-subnav {
  display: flex;
  gap: 4px;
  margin-bottom: 16px;
  padding: 4px;
  background: var(--surface-muted);
  border-radius: var(--border-radius);
  width: fit-content;
}

.llm-subnav-link {
  padding: 8px 16px;
  border-radius: var(--border-radius);
  font-size: 14px;
  font-weight: 500;
  color: var(--text-secondary);
  text-decoration: none;
  transition: color 0.15s, background 0.15s;
}

.llm-subnav-link:hover {
  color: var(--text-primary);
  background: var(--surface);
}

.llm-subnav-link.active {
  background: var(--surface);
  color: var(--brand-blue-mid);
  box-shadow: 0 1px 2px rgba(0, 0, 0, 0.05);
}

.llm-models-container {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  padding: 24px;
}

.llm-models-header h2 {
  margin: 0 0 8px 0;
  font-size: 18px;
  font-weight: 600;
  color: var(--text-primary);
}

.llm-models-hint {
  margin: 0 0 12px 0;
  font-size: 13px;
  color: var(--text-muted);
}

.llm-models-meta {
  font-size: 12px;
  color: var(--text-muted);
}

.llm-models-refresh {
  margin-left: 12px;
```

```
}

.llm-models-loading,
.llm-models-error {
  padding: 24px;
  text-align: center;
  color: var(--text-muted);
}

.llm-models-error {
  color: var(--status-error);
}

.llm-models-list {
  display: grid;
  grid-template-columns: 1fr;
  gap: 16px;
  margin-top: 20px;
}

.llm-model-card {
  padding: 16px;
  background: var(--surface-muted);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius);
  transition: border-color 0.15s, background 0.15s;
}

.llm-model-card:hover {
  border-color: var(--brand-blue-mid);
}

.llm-model-card-head {
  display: flex;
  align-items: flex-start;
  justify-content: space-between;
  gap: 12px;
}

.llm-model-main {
  min-width: 0;
}

.llm-model-actions {
  display: flex;
  align-items: center;
  gap: 8px;
}

.model-expand-btn {
  border: 1px solid var(--border-subtle);
  background: var(--surface);
  color: var(--text-secondary);
  border-radius: 999px;
  width: 30px;
  height: 30px;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  padding: 0;
  font-size: 14px;
  line-height: 1;
  cursor: pointer;
  transition: color 0.15s, border-color 0.15s, transform 0.15s;
}

.model-expand-btn:hover {
  color: var(--text-primary);
  border-color: var(--brand-blue-mid);
}

```

```
.model-expand-btn span {
  display: inline-block;
  transform: rotate(0deg);
  transition: transform 0.15s ease;
}

.model-expand-btn[aria-expanded="true"] span {
  transform: rotate(180deg);
}

.llm-model-card h4 {
  margin: 0 0 6px 0;
  font-size: 14px;
  font-weight: 600;
  color: var(--text-primary);
  font-family: 'SF Mono', Monaco, Consolas, monospace;
}

.llm-model-card .model-source {
  display: inline-block;
  font-size: 10px;
  padding: 2px 6px;
  border-radius: 4px;
  background: var(--brand-blue-mid);
  color: white;
  margin-bottom: 8px;
}

.llm-model-card .model-source.cudo {
  background: var(--brand-aqua-mid);
}

.llm-model-card .model-desc {
  font-size: 12px;
  color: var(--text-muted);
  line-height: 1.4;
  margin: 0 0 8px 0;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
}

.llm-model-card .model-meta {
  font-size: 11px;
  color: var(--text-muted);
}

.model-prices {
  margin-top: 8px;
  display: flex;
  gap: 18px;
  font-size: 13px;
  color: var(--text-secondary);
}

.model-price-value {
  color: var(--text-primary);
  font-weight: 600;
}

.model-example {
  margin-top: 12px;
  border-top: 1px solid var(--border-subtle);
  padding-top: 12px;
  display: none;
}

.model-example.open {
  display: block;
}
```

```
.model-example-head {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 8px;
}

.model-example-head strong {
  font-size: 12px;
  color: var(--text-secondary);
}

.model-copy-btn {
  border: 1px solid var(--border-subtle);
  background: var(--surface);
  color: var(--text-secondary);
  border-radius: var(--border-radius-sm);
  padding: 5px 9px;
  font-size: 12px;
  cursor: pointer;
}

.model-code {
  margin: 0;
  padding: 10px;
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-sm);
  overflow-x: auto;
  color: var(--text-primary);
  font-size: 12px;
  line-height: 1.45;
}

/* ============================================
   Keys page - specific overrides
   ============================================ */
.llm-keys-page .llm-models-container {
  max-width: 720px;
}

.llm-keys-page .llm-models-header {
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  gap: 12px;
  margin-bottom: 16px;
}

.llm-keys-page .llm-models-header h2 {
  margin: 0;
}

.llm-keys-page .llm-models-refresh {
  margin: 0 0 16px 0;
  font-size: 13px;
  color: var(--text-muted);
  line-height: 1.5;
}

.llm-keys-page .llm-models-list {
  grid-template-columns: 1fr;
  gap: 12px;
}

.llm-keys-page .llm-model-card {
  padding: 16px 20px;
  display: flex;
  flex-direction: column;
  gap: 8px;
```

```
}

.llm-keys-page .llm-model-card h4 {
  font-size: 13px;
  font-weight: 500;
  word-break: break-all;
  line-height: 1.5;
}

.llm-keys-page .llm-model-card .key-display {
  font-family: 'SF Mono', Monaco, Consolas, monospace;
  font-size: 12px;
  color: var(--text-secondary);
}

.llm-keys-page .llm-model-card .key-full {
  color: var(--text-primary);
}

.llm-keys-page .llm-model-card .model-meta {
  font-size: 12px;
}

.llm-keys-page .key-reveal-btn,
.llm-keys-page .key-delete-btn {
  font-size: 13px;
  padding: 6px 12px;
  border-radius: var(--border-radius);
  cursor: pointer;
  border: 1px solid var(--border-subtle);
  background: var(--surface);
  color: var(--text-secondary);
  transition: all 0.15s;
}

.llm-keys-page .key-reveal-btn:hover,
.llm-keys-page .key-delete-btn:hover {
  background: var(--surface-muted);
  color: var(--text-primary);
}

.llm-keys-page .key-delete-btn {
  color: var(--status-error);
  border-color: var(--status-error);
}

.llm-keys-page .key-delete-btn:hover {
  background: var(--status-error-bg);
}

.llm-keys-page .model-source.cudo {
  background: var(--text-muted);
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/services-ui.css (5.9 KB)

```
/* app/static/css/services-ui.css - Services UI Styles */

.services-container {
    display: flex;
    flex-direction: column;
    gap: 2rem;
    width: 100%;
    margin-top: 1rem;
}

.quick-ports {
    display: flex;
    gap: 1rem;
    flex-wrap: nowrap;
    align-items: center;
    overflow-x: auto;
}

/* ?????? ???????????????? ?????? - ?? ?????????? .action-btn ????? ???????? ?????????? */
.services-container .btn-service {
    background: linear-gradient(135deg, var(--brand-blue-deep) 0%, var(--brand-blue-mid) 100%);
    border: 2px solid var(--brand-blue-mid);
    color: white;
    padding: 0.5rem 1rem;
    border-radius: 0.5rem;
    cursor: pointer;
    font-size: 0.875rem;
    font-weight: 600;
    transition: all 0.2s;
    min-width: 100px;
    text-align: center;
    display: inline-flex;
    align-items: center;
    justify-content: center;
    box-shadow: 0 2px 4px rgba(15, 61, 94, 0.15);
    white-space: nowrap;
}

.services-container .btn-service:hover {
    background: linear-gradient(135deg, #0a2a3f 0%, var(--brand-blue-deep) 100%);
    border-color: var(--brand-blue-deep);
    color: white;
    transform: translateY(-2px);
    box-shadow: 0 4px 8px rgba(15, 61, 94, 0.3);
}

.services-container .btn-service:active {
    transform: translateY(0);
    box-shadow: 0 1px 2px rgba(15, 61, 94, 0.2);
}

/* RDP button styling (Windows green theme) */
.services-container .btn-service-rdp {
    background: linear-gradient(135deg, #059669 0%, #10b981 100%);
    border-color: #059669;
}

.services-container .btn-service-rdp:hover {
    background: linear-gradient(135deg, #047857 0%, #059669 100%);
    border-color: #047857;
}

.custom-port-form {
    display: flex;
    gap: 1rem;
```

```
    align-items: center;
    padding-top: 0.75rem;
    padding-bottom: 0.5rem;
}

.services-container .port-input {
    padding: 0.625rem 1rem;
    border: 2px solid rgba(15, 61, 94, 0.3);
    border-radius: 0.5rem;
    background-color: var(--surface);
    color: var(--text-primary);
    flex: 1;
    max-width: 350px;
    min-width: 200px;
    font-size: 0.9375rem;
    transition: all 0.2s;
}

[data-theme="dark"] .services-container .port-input {
    border-color: rgba(255, 255, 255, 0.2);
}

.services-container .port-input:focus {
    outline: none;
    border-color: var(--brand-blue-deep);
    box-shadow: 0 0 0 3px rgba(15, 61, 94, 0.1);
}

[data-theme="dark"] .services-container .port-input:focus {
    border-color: var(--brand-aqua);
    box-shadow: 0 0 0 3px rgba(0, 255, 255, 0.1);
}

/* ?????? ?????????? ?????????? ????? */
.services-container .btn-add-custom-port {
    background-color: var(--brand-blue-deep);
    color: white;
    border: none;
    padding: 0.625rem 1.25rem;
    border-radius: 0.5rem;
    cursor: pointer;
    font-size: 0.9375rem;
    font-weight: 600;
    transition: all 0.2s;
    white-space: nowrap;
}

.services-container .btn-add-custom-port:hover {
    background-color: var(--brand-blue-hover);
    transform: translateY(-1px);
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.15);
}

.services-list {
    display: flex;
    flex-direction: column;
    gap: 0.75rem;
    border-top: 1px solid var(--border);
    padding-top: 1.5rem;
    margin-top: 1rem;
}

.service-item {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 1rem;
    background-color: var(--background);
    border-radius: 0.5rem;
    border: 1px solid var(--border);
    transition: all 0.2s;
```

```
}

.service-item:hover {
    border-color: var(--brand-blue-deep);
    box-shadow: 0 2px 6px rgba(0, 0, 0, 0.08);
}

.service-info {
    display: flex;
    flex-direction: column;
    gap: 0.75rem;
    flex: 1;
}

.service-header {
    display: flex;
    align-items: center;
    gap: 1rem;
}

.service-name {
    font-weight: 600;
    text-transform: uppercase;
    font-size: 0.75rem;
    background: var(--brand-blue-deep);
    color: white;
    padding: 0.125rem 0.375rem;
    border-radius: 0.25rem;
}

.btn-delete-service {
    background: none;
    border: none;
    color: var(--status-error);
    cursor: pointer;
    padding: 0.25rem;
    opacity: 0.7;
    transition: opacity 0.2s;
}

.btn-delete-service:hover {
    opacity: 1;
}

.loading-services {
    color: var(--text-secondary);
    font-size: 0.875rem;
    text-align: center;
    padding: 1.5rem;
    font-style: italic;
}

.no-services-message {
    color: var(--text-secondary);
    font-size: 0.875rem;
    text-align: center;
    padding: 1.5rem;
    font-style: italic;
}

/* SSH Command Display */
.ssh-command-display {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    padding: 0.5rem 0.75rem;
    background: var(--surface-raised);
    border: 1px solid var(--border-subtle);
    border-radius: 0.375rem;
    font-family: 'Monaco', 'Menlo', 'Ubuntu Mono', monospace;
}
```

```

.ssh-command-display code {
    color: var(--status-success);
    font-size: 0.8125rem;
    font-weight: 500;
    flex: 1;
}

.btn-copy-ssh {
    background: none;
    border: none;
    color: var(--text-secondary);
    cursor: pointer;
    padding: 0.25rem;
    border-radius: 0.25rem;
    display: flex;
    align-items: center;
    justify-content: center;
    transition: all 0.2s;
    opacity: 0.7;
}

.btn-copy-ssh:hover {
    opacity: 1;
    background: rgba(0, 0, 0, 0.05);
    color: var(--brand-blue-deep);
}

[data-theme="dark"] .btn-copy-ssh:hover {
    background: rgba(255, 255, 255, 0.1);
    color: var(--brand-aqua);
}

[data-theme="dark"] .ssh-command-display {
    background: rgba(255, 255, 255, 0.05);
    border-color: rgba(255, 255, 255, 0.1);
}

[data-theme="dark"] .ssh-command-display code {
    color: #4ade80;
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/storage.css (1.6 KB)

```
/* Storage Tab Styles */
.progress-container {
  width: 100%;
  background-color: #eee;
  height: 6px;
  border-radius: 3px;
  overflow: hidden;
  margin-top: 4px;
}

.progress-bar {
  height: 100%;
  background-color: #27ae60;
  transition: width 0.3s ease;
}

.font-medium {
  font-weight: 500;
}

.text-small {
  font-size: 0.85rem;
}

.text-tiny {
  font-size: 0.75rem;
}

.text-muted {
  color: #666;
}

.vm-link {
  display: inline-flex;
  align-items: center;
  color: #2c3e50;
  text-decoration: none;
  font-weight: 500;
}

.vm-link:hover {
  color: #3498db;
}

.vm-icon {
  margin-right: 6px;
  font-size: 1.1em;
}

/* Action button icon only */
.action-btn-icon {
  display: flex;
  align-items: center;
  justify-content: center;
  width: 32px;
  height: 32px;
  border-radius: 6px;
  border: none;
  background-color: #f1f1f1;
  color: #555;
  cursor: pointer;
  transition: all 0.2s;
}

.action-btn-icon:hover {
```

```
  background-color: #e0e0e0;
}

.action-btn-icon.danger:hover {
  background-color: #ffeaea;
  color: #e74c3c;
}

.action-btn-icon:disabled {
  opacity: 0.5;
  cursor: not-allowed;
  background-color: #fafafa;
}

.loading-spinner-small {
  display: inline-block;
  width: 20px;
  height: 20px;
  border: 2px solid #f3f3f3;
  border-top: 2px solid #3498db;
  border-radius: 50%;
  animation: spin 1s linear infinite;
  margin-right: 10px;
  vertical-align: middle;
}

.empty-state-small {
  color: #999;
  padding: 2rem;
  font-size: 0.95rem;
}

.text-error {
  color: #e74c3c;
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/utilities.css (3.0 KB)

```
/**
 * Utility Classes
 * Common utility classes used across the application
 * All colors use CSS variables for theme compatibility
 */

/* ========================================
   Display Utilities
   ======================================== */

.hidden {
  display: none !important;
}

.visible {
  display: block !important;
}

.flex-visible {
  display: flex !important;
}

.invisible {
  visibility: hidden !important;
}

/* ========================================
   Text State Utilities
   ======================================== */

.text-error {
  color: var(--status-error) !important;
}

.text-warning {
  color: var(--status-warning) !important;
}

.text-success {
  color: var(--status-success) !important;
}

.text-secondary {
  color: var(--text-secondary) !important;
}

.text-primary {
  color: var(--text-primary) !important;
}

/* ========================================
   Progress Bar State Utilities
   ======================================== */

.progress-bar--error {
  background: var(--status-error) !important;
}

.progress-bar--warning {
  background: var(--status-warning) !important;
}

.progress-bar--success {
  background: var(--status-success) !important;
}
```

```
/* ========================================
   Width Control Utilities
   ======================================== */

.width-auto {
  width: auto !important;
}

.width-30 {
  width: 30% !important;
}

.width-60 {
  width: 60% !important;
}

.width-90 {
  width: 90% !important;
}

.width-100 {
  width: 100% !important;
}

/* ========================================
   Cursor Utilities
   ======================================== */

.cursor-default {
  cursor: default !important;
}

.cursor-pointer {
  cursor: pointer !important;
}

.cursor-not-allowed {
  cursor: not-allowed !important;
}

/* ========================================
   Component-Specific Utilities
   ======================================== */

/* Cloud-init icon */
.cloudinit-icon {
  margin-left: 8px;
}

/* SSH Service components */
.ssh-service-btn {
  padding: 4px 8px;
  font-size: 0.8rem;
  width: auto;
  height: auto;
}

.ssh-service-info {
  color: var(--text-secondary);
}

.ssh-nodeport {
  color: var(--text-primary);
  font-weight: 600;
}

/* Auto-resize textarea */
.auto-resize {
  overflow: hidden;
  resize: none;
```

```
}

/* Page balance highlight box (VM main balance, LLM balance) */
.page-balance-box {
  display: inline-block;
  padding: 0.35rem 0.75rem;
  border: 1px solid var(--border-subtle, #e2e8f0);
  border-radius: 6px;
  background: var(--surface-muted, #f8fafc);
  font-size: 0.9rem;
  color: var(--text-primary);
  vertical-align: middle;
}

/* Push balance to the right half of the header */
.page-balance-right {
  margin-left: auto;
}

[data-theme="dark"] .page-balance-box {
  border-color: var(--border-subtle);
  background: var(--bg-muted, #1e293b);
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/css/vm-details-new.css (10.5 KB)

```
/* ============================================
   VM Details Page - Extended Styles
   Consistent with main design system
   ============================================ */

/* Breadcrumb */
.breadcrumb {
  display: flex;
  align-items: center;
  gap: 8px;
  margin-bottom: 20px;
  font-size: 13px;
}

.breadcrumb-link {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  color: var(--text-secondary);
  text-decoration: none;
  transition: color var(--transition-fast);
}

.breadcrumb-link:hover {
  color: var(--brand-blue-mid);
}

.breadcrumb-link svg {
  width: 14px;
  height: 14px;
}

.breadcrumb-separator {
  color: var(--text-muted);
}

.breadcrumb-current {
  color: var(--text-primary);
  font-weight: 500;
}

/* Header with Status */
.vm-status-header {
  display: flex;
  align-items: center;
  gap: 10px;
  margin-top: 6px;
}

.header-actions {
  display: flex;
  gap: 8px;
  flex-wrap: wrap;
}

.header-actions .action-btn {
  padding: 8px 14px;
}

.header-actions .btn-delete {
  background: var(--status-error-bg);
  color: var(--status-error);
}

.header-actions .btn-delete:hover {
```

```
  background: var(--status-error);
  color: #ffffff;
}

/* Tabs */
.vm-tabs {
  display: flex;
  gap: 4px;
  padding: 4px;
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius);
  margin-bottom: 20px;
}

.tab-btn {
  flex: 1;
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 8px;
  padding: 10px 16px;
  background: transparent;
  border: none;
  border-radius: var(--border-radius-sm);
  font-size: 13px;
  font-weight: 600;
  color: var(--text-secondary);
  cursor: pointer;
  transition: all var(--transition-fast);
}

.tab-btn svg {
  width: 16px;
  height: 16px;
}

.tab-btn:hover {
  color: var(--text-primary);
  background: var(--surface-muted);
}

.tab-btn.active {
  background: var(--nav-active-bg);
  color: var(--brand-blue-mid);
}

[data-theme="dark"] .tab-btn.active {
  color: var(--brand-aqua-light);
}

.tab-content {
  display: none;
  animation: fadeIn 0.2s ease;
}

.tab-content.active {
  display: block;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(8px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

```

```
/* Cloud-init Status Banner */
.cloud-init-status {
  display: flex;
  align-items: flex-start;
  gap: 12px;
  padding: 16px;
  margin-bottom: 20px;
  border-radius: var(--border-radius-lg);
  border: 1px solid;
  animation: fadeIn 0.3s ease;
}

.cloud-init-status.running {
  background: linear-gradient(
    135deg,
    rgba(255, 193, 7, 0.1),
    rgba(255, 152, 0, 0.05)
  );
  border-color: rgba(255, 193, 7, 0.3);
}

.cloud-init-status.completed {
  background: linear-gradient(
    135deg,
    rgba(118, 255, 122, 0.1),
    rgba(0, 200, 83, 0.05)
  );
  border-color: rgba(118, 255, 122, 0.3);
}

[data-theme="dark"] .cloud-init-status.running {
  background: linear-gradient(
    135deg,
    rgba(255, 193, 7, 0.08),
    rgba(255, 152, 0, 0.03)
  );
}

[data-theme="dark"] .cloud-init-status.completed {
  background: linear-gradient(
    135deg,
    rgba(118, 255, 122, 0.08),
    rgba(0, 200, 83, 0.03)
  );
}

.cloud-init-icon {
  flex-shrink: 0;
  width: 40px;
  height: 40px;
  display: flex;
  align-items: center;
  justify-content: center;
  border-radius: 50%;
}

.cloud-init-status.running .cloud-init-icon {
  background: rgba(255, 193, 7, 0.2);
  color: var(--warning);
}

.cloud-init-status.completed .cloud-init-icon {
  background: rgba(118, 255, 122, 0.2);
  color: var(--brand-green-light);
}

.cloud-init-status.running .cloud-init-icon svg {
  animation: rotate 2s linear infinite;
}

@keyframes rotate {
```

```
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}

.cloud-init-icon svg {
  width: 20px;
  height: 20px;
}

.cloud-init-content {
  flex: 1;
}

.cloud-init-content strong {
  display: block;
  font-size: 14px;
  font-weight: 600;
  margin-bottom: 4px;
}

.cloud-init-status.running .cloud-init-content strong {
  color: var(--warning);
}

.cloud-init-status.completed .cloud-init-content strong {
  color: var(--brand-green-light);
}

.cloud-init-content p {
  font-size: 13px;
  color: var(--text-secondary);
  margin: 0;
}

/* Info Grid */
.info-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
  gap: 20px;
}

.info-card {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  padding: 20px;
}

.info-card-header {
  display: flex;
  align-items: center;
  gap: 10px;
  margin-bottom: 16px;
  padding-bottom: 12px;
  border-bottom: 1px solid var(--border-subtle);
}

.info-card-header svg {
  width: 20px;
  height: 20px;
  color: var(--brand-blue-mid);
}

.info-card-header h3 {
  font-size: 14px;
  font-weight: 600;
  color: var(--text-primary);
```

```
  margin: 0;
}

.info-row {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 10px 0;
  border-bottom: 1px solid var(--border-subtle);
}

.info-row:last-child {
  border-bottom: none;
  padding-bottom: 0;
}

.info-label {
  font-size: 13px;
  color: var(--text-secondary);
}

.info-value {
  font-size: 13px;
  font-weight: 600;
  color: var(--text-primary);
}

/* Console Container */
.console-container {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  overflow: hidden;
  min-height: 480px;
  display: flex;
  flex-direction: column;
}

.console-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 12px 16px;
  background: var(--surface-muted);
  border-bottom: 1px solid var(--border-subtle);
}

.console-title {
  display: flex;
  align-items: center;
  gap: 8px;
}

.console-title svg {
  width: 18px;
  height: 18px;
  color: var(--brand-blue-mid);
}

.console-title h3 {
  font-size: 14px;
  font-weight: 600;
  color: var(--text-primary);
  margin: 0;
}

.console-status {
  display: flex;
  align-items: center;
  gap: 6px;
  font-size: 12px;
```

```
  color: var(--text-secondary);
}

.status-dot {
  width: 6px;
  height: 6px;
  border-radius: 50%;
  background: var(--status-warning);
  animation: pulse 2s infinite;
}

.console-status.connected .status-dot {
  background: var(--status-success);
}

.vnc-iframe {
  flex: 1;
  width: 100%;
  border: none;
  background: var(--app-bg);
}

/* Metrics Banner */
.info-banner {
  display: flex;
  gap: 14px;
  padding: 16px;
  margin-bottom: 20px;
  background: var(--nav-active-bg);
  border: 1px solid var(--brand-blue-mid);
  border-radius: var(--border-radius);
}

.info-banner svg {
  width: 20px;
  height: 20px;
  color: var(--brand-blue-mid);
  flex-shrink: 0;
}

.info-banner h3 {
  font-size: 14px;
  font-weight: 600;
  color: var(--text-primary);
  margin: 0 0 4px;
}

.info-banner p {
  font-size: 12px;
  color: var(--text-secondary);
  margin: 0;
}

/* Metrics Grid */
.metrics-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
  gap: 20px;
}

.metric-card {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--border-radius-lg);
  padding: 20px;
}

.metric-header {
  display: flex;
  align-items: center;
  gap: 8px;
```

```
  margin-bottom: 16px;
}

.metric-header svg {
  width: 18px;
  height: 18px;
  color: var(--brand-blue-mid);
}

.metric-header h3 {
  font-size: 14px;
  font-weight: 600;
  color: var(--text-primary);
  margin: 0;
}

.metric-value-wrapper {
  text-align: center;
  margin-bottom: 10px;
}

.metric-value {
  font-size: 40px;
  font-weight: 700;
  color: var(--brand-blue-mid);
}

.metric-unit {
  font-size: 16px;
  color: var(--text-muted);
  margin-left: 4px;
}

.metric-allocated {
  font-size: 12px;
  color: var(--text-secondary);
  text-align: center;
  margin-bottom: 12px;
}

.memory-note {
  font-size: 11px;
  color: var(--text-muted);
  font-style: italic;
  text-align: center;
  margin: 6px 0 12px;
}

.metric-chart {
  max-height: 160px;
}

/* Status Badges in Details */
#vm-status-badge {
  font-size: 12px;
}

/* Responsive */
@media (max-width: 768px) {
  .breadcrumb {
    font-size: 12px;
  }

  .header-actions {
    width: 100%;
    margin-top: 12px;
  }

  .header-actions .action-btn {
    flex: 1;
    justify-content: center;
```

```
492   }
493
494   .vm-tabs {
495     flex-direction: column;
496   }
497
498   .tab-btn {
499     justify-content: flex-start;
500   }
501
502 ... [90 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/core/app-state.js (3.3 KB)

```
"use strict";

/**
 * Lightweight shared app store for cross-page UI synchronization.
 * Keeps balances/profile in one place and notifies subscribers on changes.
 */
(function initAppState(global) {
  if (global.AppState) return;

  var state = {
    balances: {
      mainBalance: null,
      llmBalance: null,
      updatedAt: null,
    },
    profile: {
      username: document.body ? document.body.dataset.username || "" : "",
      namespace: "",
    },
    vmSummary: {
      total: null,
      running: null,
      updatedAt: null,
    },
  };

  var listeners = [];

  function notify(source) {
    for (var i = 0; i < listeners.length; i += 1) {
      try {
        listeners[i](state, source || "unknown");
      } catch (_) {}
    }
    try {
      global.dispatchEvent(
        new CustomEvent("appstate:change", {
          detail: {
            source: source || "unknown",
            state: state,
          },
        }),
      );
    } catch (_) {}
  }

  function subscribe(listener) {
    if (typeof listener !== "function") return function noop() {};
    listeners.push(listener);
    return function unsubscribe() {
      listeners = listeners.filter(function (cb) {
        return cb !== listener;
      });
    };
  }

  function setState(nextPartial, source) {
    if (!nextPartial || typeof nextPartial !== "object") return;
    if (nextPartial.balances) {
      state.balances = Object.assign({}, state.balances, nextPartial.balances);
    }
    if (nextPartial.profile) {
      state.profile = Object.assign({}, state.profile, nextPartial.profile);
    }
    if (nextPartial.vmSummary) {
```

```
      state.vmSummary = Object.assign({}, state.vmSummary, nextPartial.vmSummary);
    }
    notify(source || "setState");
  }

  function setBalances(rawBalances, source) {
    if (!rawBalances || typeof rawBalances !== "object") return;
    var mainVal = Number(rawBalances.main_balance);
    var llmVal = Number(rawBalances.llm_balance);
    setState(
      {
        balances: {
          mainBalance: Number.isFinite(mainVal) ? mainVal : null,
          llmBalance: Number.isFinite(llmVal) ? llmVal : null,
          updatedAt: Date.now(),
        },
      },
      source || "setBalances",
    );
  }

  function formatMoney(value) {
    var num = Number(value);
    return Number.isFinite(num) ? num.toFixed(2) : "?";
  }

  function loadBalances(username) {
    var user = username || state.profile.username;
    if (!user) return Promise.resolve(null);
    return fetch("/" + encodeURIComponent(user) + "/api/balance", {
      credentials: "same-origin",
    })
      .then(function (r) {
        if (r.status === 401) {
          global.location.href = "/login";
          return null;
        }
        if (!r.ok) return Promise.reject(new Error("Failed to load balances"));
        return r.json();
      })
      .then(function (data) {
        if (!data) return null;
        setBalances(data, "loadBalances");
        return data;
      });
  }

  global.AppState = {
    getState: function getState() {
      return JSON.parse(JSON.stringify(state));
    },
    subscribe: subscribe,
    setState: setState,
    setBalances: setBalances,
    loadBalances: loadBalances,
    formatMoney: formatMoney,
  };
})(window);

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/core/chat-manager.js (6.6 KB)

```
/**
 * Chat Manager Module
 *
 * Shared chat functionality for LLM chat and VM details chat.
 * Handles message sending, streaming, markdown rendering, and UI updates.
 *
 * @module chat-manager
 */

/**
 * Escapes HTML special characters to prevent XSS attacks
 *
 * @param {string} text - Text to escape
 * @returns {string} Escaped HTML-safe text
 *
 * @example
 * escapeHtml('<script>alert("xss")</script>');
 * // Returns: '&lt;script&gt;alert("xss")&lt;/script&gt;'
 */
function escapeHtml(text) {
  const div = document.createElement("div");
  div.textContent = text;
  return div.innerHTML;
}

/**
 * Renders markdown text to HTML using marked.js library
 *
 * @param {string} text - Markdown text to render
 * @returns {string} Rendered HTML
 *
 * @example
 * renderMarkdown('# Hello\n\n```python\nprint("world")\n```');
 * // Returns HTML with syntax highlighting
 */
function renderMarkdown(text) {
  if (typeof marked !== "undefined") {
    const html = marked.parse(text);

    // Apply syntax highlighting if highlight.js is available
    if (typeof hljs !== "undefined") {
      setTimeout(() => {
        document
          .querySelectorAll("pre code:not(.hljs)")
          .forEach((block) => hljs.highlightElement(block));
      }, 50);
    }

    return html;
  }
  return escapeHtml(text);
}

/**
 * Adds a message to the chat container
 *
 * @param {string} role - Message role ("user" or "assistant")
 * @param {string} content - Message content (HTML for assistant, plain text for user)
 * @param {string|null} [id=null] - Optional message ID for updates
 * @param {string} [chatContainerId='chat-messages'] - ID of the chat container element
 * @returns {HTMLElement} The created message element
 *
 * @example
 * addMessage('user', 'Hello, how are you?');
 * addMessage('assistant', '<p>I am well, thank you!</p>', 'msg-123');
```

```
 */
function addMessage(role, content, id = null, chatContainerId = "chat-messages") {
  const chatMessages = document.getElementById(chatContainerId);
  if (!chatMessages) {
    console.error(`Chat container #${chatContainerId} not found`);
    return null;
  }

  const messageDiv = document.createElement("div");
  messageDiv.className = `chat-message chat-message--${role}`;
  if (id) messageDiv.id = id;

  messageDiv.innerHTML = `
    <div class="chat-message-text">${
      role === "user" ? escapeHtml(content) : content || '<span class="typing-indicator"><span></span><span></span><s...
    }</div>
  `;

  chatMessages.appendChild(messageDiv);
  chatMessages.scrollTop = chatMessages.scrollHeight;

  return messageDiv;
}

/**
 * Clears all messages from the chat
 *
 * @param {string} [chatContainerId='chat-messages'] - ID of the chat container element
 * @returns {void}
 *
 * @example
 * clearChat();
 * clearChat('vm-chat-messages');
 */
function clearChat(chatContainerId = "chat-messages") {
  const chatMessages = document.getElementById(chatContainerId);
  if (!chatMessages) {
    console.error(`Chat container #${chatContainerId} not found`);
    return;
  }

  chatMessages.innerHTML = "";
  if (window.showToast) {
    showToast("??????? ???? ???????", "info");
  }
}

/**
 * Streams LLM response from server and updates message in real-time
 *
 * @async
 * @param {string} messageId - ID of the message element to update
 * @param {string} apiUrl - URL of the streaming API endpoint
 * @param {Array<Object>} messages - Chat history to send to API
 * @param {string} [chatContainerId='chat-messages'] - ID of the chat container element
 * @returns {Promise<void>}
 *
 * @throws {Error} If network error or API error occurs
 *
 * @sideEffects
 * - Updates message element content in real-time
 * - Scrolls chat container to bottom
 * - Shows error toast on failure
 *
 * @example
 * const messages = [
 *   { role: 'user', content: 'What is KubeVirt?' }
 * ];
 * await streamResponse('msg-123', '/admin/llm/chat', messages);
 */
async function streamResponse(messageId, apiUrl, messages, chatContainerId = "chat-messages") {
```

```
const chatMessages = document.getElementById(chatContainerId);
const messageDiv = document.getElementById(messageId);

if (!messageDiv) {
  console.error(`Message element #${messageId} not found`);
  return;
}

const textEl = messageDiv.querySelector(".chat-message-text");
let fullContent = "";

try {
  const response = await fetch(apiUrl, {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    credentials: "same-origin",
    body: JSON.stringify({ messages }),
  });

  if (response.status === 401) {
    window.location.href = "/login";
    return;
  }

  if (!response.ok) {
    textEl.innerHTML = `<span class="chat-error">??????: HTTP ${response.status}</span>`;
    if (window.showToast) {
      showToast(`?????? API: ${response.status}`, "error");
    }
    return;
  }

  const reader = response.body.getReader();
  const decoder = new TextDecoder();

  while (true) {
    const { done, value } = await reader.read();
    if (done) break;

    const chunk = decoder.decode(value);
    const lines = chunk.split("\n");

    for (const line of lines) {
      if (line.startsWith("data: ")) {
        const data = line.slice(6);
        if (data.trim() === "[DONE]") {
          textEl.innerHTML = renderMarkdown(fullContent);
          chatMessages.scrollTop = chatMessages.scrollHeight;
          return;
        }

        try {
          const parsed = JSON.parse(data);
          if (parsed.error) {
            textEl.innerHTML = `<span class="chat-error">??????: ${escapeHtml(parsed.error)}</span>`;
            if (window.showToast) {
              showToast(parsed.error, "error");
            }
            return;
          }

          if (parsed.content) {
            fullContent += parsed.content;
            textEl.innerHTML = renderMarkdown(fullContent);
            chatMessages.scrollTop = chatMessages.scrollHeight;
          }
        } catch (e) {
          console.error("JSON parse error:", e, "Data:", data);
        }
      }
    }
```

```
    }
  } catch (error) {
    console.error("Stream error:", error);
    textEl.innerHTML = `<span class="chat-error">?????? ??????????: ${escapeHtml(error.message)}</span>`;
    if (window.showToast) {
      showToast("?????? ???? ??? ??????? ??????????", "error");
    }
  }
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/core/theme.js (1.2 KB)

```
/**
 * Theme Manager
 *
 * Handles light/dark theme switching and persistence.
 */

/**
 * Applies saved theme from localStorage
 * Called immediately on page load to prevent flash.
 *
 * @returns {void}
 */
(function applyTheme() {
  const savedTheme = localStorage.getItem("viiruoz-theme") || "light";
  document.documentElement.setAttribute("data-theme", savedTheme);
})();

/**
 * Initialize theme toggle button
 * Attaches event listener to toggle button when DOM is ready
 *
 * @returns {void}
 */
function initThemeToggle() {
  const themeToggle = document.getElementById('theme-toggle');

  if (!themeToggle) {
    return; // No theme toggle button on this page
  }

  themeToggle.addEventListener('click', () => {
    const currentTheme = document.documentElement.getAttribute('data-theme');
    const newTheme = currentTheme === 'dark' ? 'light' : 'dark';
    document.documentElement.setAttribute('data-theme', newTheme);
    localStorage.setItem('viiruoz-theme', newTheme);
  });
}

// Initialize theme toggle when DOM is ready
if (document.readyState === 'loading') {
  document.addEventListener('DOMContentLoaded', initThemeToggle);
} else {
  initThemeToggle();
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/core/ui-components.js (7.0 KB)

```
/**
 * UI Components Module
 *
 * Provides global UI components like toast notifications, sidebar toggle, theme switcher.
 */

/**
 * Shows a toast notification
 *
 * @param {string} message - Message to display
 * @param {('success'|'error'|'warning'|'info')} type - Notification type
 * @param {number} [duration=4000] - Display duration in milliseconds
 * @returns {void}
 *
 * @example
 * showToast('VM created successfully', 'success');
 * showToast('Invalid input', 'error', 5000);
 */
function showToast(message, type = "info", duration = 4000) {
  const container = document.getElementById("toast-container");
  if (!container) return;

  const toast = document.createElement("div");
  toast.className = `toast toast-${type}`;

  let icon = "";
  switch (type) {
    case "success":
      icon =
        '<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M22 11.08V12a10 10 0 1 ...
      break;
    case "error":
      icon =
        '<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle cx="12" cy="12" r="10"/>...
      break;
    case "warning":
      icon =
        '<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M10.29 3.86L1.82 18a2 2...
      break;
    default:
      icon =
        '<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle cx="12" cy="12" r="10"/>...
  }

  toast.innerHTML = `
    <span class="toast-icon">${icon}</span>
    <span class="toast-message">${message}</span>
    <button class="toast-close" onclick="this.parentElement.remove()">×</button>
  `;

  container.appendChild(toast);

  // Trigger animation
  requestAnimationFrame(() => {
    toast.classList.add("toast-show");
  });

  // Auto remove
  setTimeout(() => {
    toast.classList.remove("toast-show");
    setTimeout(() => toast.remove(), 200);
  }, duration);
}

/**
```

```
 * Sets theme and saves to localStorage
 * @param {string} theme - Theme name ('light' or 'dark')
 * @returns {void}
 */
function setTheme(theme) {
  document.documentElement.setAttribute("data-theme", theme);
  try {
    localStorage.setItem("viiruoz-theme", theme);
  } catch (_) {}
}

// Initialize UI components on page load
document.addEventListener("DOMContentLoaded", () => {
  const safeStorage = {
    get(key) {
      try {
        return localStorage.getItem(key);
      } catch (_) {
        return null;
      }
    },
    set(key, value) {
      try {
        localStorage.setItem(key, value);
      } catch (_) {}
    },
  };

  const rootLayout = document.querySelector(".dashboard-layout");

  // Theme toggle
  const themeToggle = document.getElementById("theme-toggle");
  if (themeToggle) {
    themeToggle.addEventListener("click", () => {
      const currentTheme = document.documentElement.getAttribute("data-theme");
      const newTheme = currentTheme === "dark" ? "light" : "dark";
      setTheme(newTheme);
    });
  }

  // Mobile menu toggle
  const menuToggle = document.getElementById("menu-toggle");
  const sidebar = document.getElementById("sidebar");

  if (menuToggle && sidebar) {
    menuToggle.addEventListener("click", () => {
      sidebar.classList.toggle("sidebar-open");
    });

    // Close sidebar when clicking outside
    document.addEventListener("click", (e) => {
      if (
        sidebar.classList.contains("sidebar-open") &&
        !sidebar.contains(e.target) &&
        !menuToggle.contains(e.target)
      ) {
        sidebar.classList.remove("sidebar-open");
      }
    });
  }

  // Desktop sidebar collapse/expand
  const collapseBtn = document.getElementById("sidebar-collapse-toggle");
  const sidebarCollapsedKey = "viiruoz-sidebar-collapsed";
  function setSidebarCollapsed(isCollapsed) {
    if (!rootLayout) return;
    rootLayout.classList.toggle("sidebar-collapsed", isCollapsed);
    if (collapseBtn) {
      collapseBtn.setAttribute(
        "aria-label",
        isCollapsed ? "??????????? ??????" : "???????? ??????",
```

```
      );
    }
  }
  if (rootLayout) {
    const isInitiallyCollapsed =
      safeStorage.get(sidebarCollapsedKey) === "1";
    setSidebarCollapsed(isInitiallyCollapsed);
  }
  if (collapseBtn) {
    collapseBtn.addEventListener("click", () => {
      if (!rootLayout) return;
      const nextCollapsed = !rootLayout.classList.contains("sidebar-collapsed");
      setSidebarCollapsed(nextCollapsed);
      safeStorage.set(sidebarCollapsedKey, nextCollapsed ? "1" : "0");
    });
  }

  // LLM accordion in sidebar
  const llmTrigger = document.getElementById("llm-nav-trigger");
  const llmSubmenu = document.getElementById("llm-submenu");
  const currentPath = window.location.pathname;
  if (llmSubmenu) {
    llmSubmenu.querySelectorAll(".nav-sublink").forEach((link) => {
      if (link.getAttribute("href") === currentPath) {
        link.classList.add("active");
      }
    });
  }
  if (llmTrigger && llmSubmenu) {
    const llmItem = llmTrigger.closest(".nav-item-llm");
    const setOpen = (isOpen) => {
      llmSubmenu.classList.toggle("submenu-open", isOpen);
      if (llmItem) llmItem.classList.toggle("is-open", isOpen);
      llmTrigger.setAttribute("aria-expanded", isOpen ? "true" : "false");
    };

    // Open accordion by default on LLM pages
    const llmPageActive =
      llmItem && llmItem.classList.contains("is-active") ? true : false;
    setOpen(llmSubmenu.classList.contains("submenu-open") || llmPageActive);

    llmTrigger.addEventListener("click", () => {
      if (rootLayout && rootLayout.classList.contains("sidebar-collapsed")) {
        rootLayout.classList.remove("sidebar-collapsed");
        safeStorage.set(sidebarCollapsedKey, "0");
      }
      setOpen(!llmSubmenu.classList.contains("submenu-open"));
    });
  }

  // User dropdown toggle
  const userChip = document.getElementById("user-chip");
  const userDropdown = document.getElementById("user-dropdown");

  if (userChip && userDropdown) {
    userChip.addEventListener("click", (e) => {
      e.stopPropagation();
      userDropdown.classList.toggle("dropdown-open");
    });

    document.addEventListener("click", () => {
      userDropdown.classList.remove("dropdown-open");
    });
  }

  // Logout handler
  const logoutBtn = document.getElementById("logout-btn");
  if (logoutBtn) {
    logoutBtn.addEventListener("click", () => {
      if (confirm("?? ???????, ??? ?????? ??????")) {
        window.location.href = "/logout";
```

```
      }
    });
  }
});

// Expose globally
window.showToast = showToast;

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/core/utils.js (2.9 KB)

```
/**
 * Utility Functions Module
 *
 * Common utility functions used across multiple pages.
 *
 * @module utils
 */

/**
 * Opens VNC console for a virtual machine in a new tab
 *
 * @param {string} vmName - Name of the VM to open VNC for
 *
 * @example
 * openVNC('my-ubuntu-vm');
 */
function openVNC(vmName) {
// Get username from data attribute, window global, or URL fallback
  const username = document.body?.dataset?.username ||
                   window.username ||
                   window.location.pathname.split("/")[1];
  window.open(`/${username}/vm/${vmName}#console`, "_blank");
}

/**
 * Toggles the "more actions" dropdown menu
 *
 * @param {HTMLElement} btn - The button element that was clicked
 * @param {Event} event - The click event
 *
 * @example
 * <button onclick="toggleMoreMenu(this, event)">...</button>
 */
function toggleMoreMenu(btn, event) {
  event.stopPropagation();
  const dropdown = btn.nextElementSibling;
  const isOpen = dropdown.classList.contains("dropdown-open");

  // Close all other dropdowns first
  document.querySelectorAll(".more-dropdown.dropdown-open").forEach((d) => {
    d.classList.remove("dropdown-open");
  });

  // Toggle the clicked dropdown
  if (!isOpen) {
    dropdown.classList.add("dropdown-open");
  }
}

/**
 * Closes all open dropdown menus (global click handler)
 * Should be attached to document click event
 */
function closeAllDropdowns() {
  document.querySelectorAll(".more-dropdown.dropdown-open").forEach((d) => {
    d.classList.remove("dropdown-open");
  });
}

/**
 * Robust copy helper with Clipboard API + fallback.
 *
 * @param {string} text - Text to copy
 * @returns {Promise<boolean>} true when copied, false otherwise
 */
```

```
async function copyTextToClipboard(text) {
  const value = String(text ?? "");
  if (!value) return false;

  // Primary path for modern browsers and secure contexts.
  if (navigator.clipboard && typeof navigator.clipboard.writeText === "function") {
    try {
      await navigator.clipboard.writeText(value);
      return true;
    } catch (_) {}
  }

  // Fallback for restricted contexts (e.g. iframe / blocked clipboard API).
  try {
    const ta = document.createElement("textarea");
    ta.value = value;
    ta.setAttribute("readonly", "");
    ta.style.position = "fixed";
    ta.style.top = "0";
    ta.style.left = "0";
    ta.style.opacity = "0";
    document.body.appendChild(ta);
    ta.focus();
    ta.select();
    ta.setSelectionRange(0, ta.value.length);
    const ok = document.execCommand("copy");
    document.body.removeChild(ta);
    return !!ok;
  } catch (_) {
    return false;
  }
}

// Auto-attach global click handler for dropdowns
document.addEventListener("click", closeAllDropdowns);

// Expose helpers globally for page-specific scripts.
window.copyTextToClipboard = copyTextToClipboard;

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/core/vm-actions.js (9.8 KB)

```
/**
 * VM Actions Module
 *
 * Centralized VM management functions used across dashboard and VM details pages.
 * Provides consistent API for starting, stopping, restarting, deleting, pausing, and resuming VMs.
 *
 * @module vm-actions
 * @requires globals.js (for username variable)
 */

/**
 * Starts a virtual machine
 *
 * @param {string} vmName - Name of the VM to start
 * @returns {Promise<void>}
 *
 * @example
 * startVM('my-ubuntu-vm');
 */
function startVM(vmName) {
  if (!confirm(`????????? ??????????? ?????? «${vmName}»?`)) return;

  const username = document.getElementById("user-name")?.textContent?.trim();
  if (!username) {
    showToast("??????: ?? ??????? ????????? ???????????", "error");
    return;
  }

  fetch(`/${username}/vm/${vmName}/start`, {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
    },
    credentials: "same-origin",
  })
    .then((response) => {
      if (response.status === 401) {
        window.location.href = "/login";
        return null;
      }
      if (response.status === 402) {
        showToast(
          "Insufficient balance. Please contact your admin to top up your balance.",
          "error",
        );
        return null;
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return;
      if (data.success) {
        showToast(`??????????? ?????? «${vmName}» ??????? ????????`, "success");
        setTimeout(() => {
          // Trigger reload based on current page
          if (typeof updateVMList === "function") {
            updateVMList();
          } else {
            location.reload();
          }
        }, 1000);
      } else {
        showToast(
          "??????: " + (data.error || "?? ??????? ????????? VM"),
          "error",
```

```
        );
      }
    })
    .catch((error) => {
      console.error("Error starting VM:", error);
      showToast("?????? ??? ??????? VM", "error");
    });
}

/**
 * Stops a virtual machine
 *
 * @param {string} vmName - Name of the VM to stop
 * @returns {Promise<void>}
 *
 * @example
 * stopVM('my-ubuntu-vm');
 */
function stopVM(vmName) {
  if (!confirm(`????????? ?????????? ?????? «${vmName}»?`)) return;

  const username = document.getElementById("user-name")?.textContent?.trim();
  if (!username) {
    showToast("??????: ?? ??????? ????????? ???????????", "error");
    return;
  }

  fetch(`/${username}/vm/${vmName}/stop`, {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
    },
    credentials: "same-origin",
  })
    .then((response) => {
      if (response.status === 401) {
        window.location.href = "/login";
        return null;
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return;
      if (data.success) {
        showToast(`?????????? ?????? «${vmName}» ??????????`, "success");
        setTimeout(() => {
          if (typeof updateVMList === "function") {
            updateVMList();
          } else {
            location.reload();
          }
        }, 1000);
      } else {
        showToast(
          "??????: " + (data.error || "?? ??????? ????????? VM"),
          "error",
        );
      }
    })
    .catch((error) => {
      console.error("Error stopping VM:", error);
      showToast("?????? ??? ???????? VM", "error");
    });
}

/**
 * Restarts a virtual machine
 *
 * @param {string} vmName - Name of the VM to restart
 * @returns {Promise<void>}
 *
```

```
 * @example
 * restartVM('my-ubuntu-vm');
 */
function restartVM(vmName) {
  if (!confirm(`???????????? ?????????? ?????? «${vmName}»?`)) return;

  const username = document.getElementById("user-name")?.textContent?.trim();
  if (!username) {
    showToast("??????: ?? ??????? ??????????? ???????????", "error");
    return;
  }

  fetch(`/${username}/vm/${vmName}/restart`, {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
    },
    credentials: "same-origin",
  })
    .then((response) => {
      if (response.status === 401) {
        window.location.href = "/login";
        return null;
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return;
      if (data.success) {
        showToast(`?????????? ?????? «${vmName}» ??????????????`, "success");
        setTimeout(() => {
          if (typeof updateVMList === "function") {
            updateVMList();
          } else {
            location.reload();
          }
        }, 1000);
      } else {
        showToast(
          "??????: " + (data.error || "?? ??????? ???????????? VM"),
          "error",
        );
      }
    })
    .catch((error) => {
      console.error("Error restarting VM:", error);
      showToast("?????? ??? ??????????? VM", "error");
    });
}

/**
 * Deletes a virtual machine (with confirmation)
 *
 * @param {string} vmName - Name of the VM to delete
 * @returns {Promise<void>}
 *
 * @example
 * deleteVM('my-ubuntu-vm');
 */
function deleteVM(vmName) {
  if (
    !confirm(
      `????????! ??????? ?????????? ?????? «${vmName}»?\n\n??? ???????? ???????????!`,
    )
  )
    return;

  const username = document.getElementById("user-name")?.textContent?.trim();
  if (!username) {
    showToast("??????: ?? ??????? ??????????? ???????????", "error");
    return;
```

```
  }

  fetch(`/${username}/vm/${vmName}/delete`, {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
    },
    credentials: "same-origin",
  })
    .then((response) => {
      if (response.status === 401) {
        window.location.href = "/login";
        return null;
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return;
      if (data.success) {
        showToast(`??????????? ?????? «${vmName}» ???????`, "success");
        // Redirect to VMs list after successful deletion
        setTimeout(() => {
          window.location.href = `/${username}/vms`;
        }, 1500);
      } else {
        showToast(
          "??????: " + (data.error || "?? ??????? ??????? VM"),
          "error",
        );
      }
    })
    .catch((error) => {
      console.error("Error deleting VM:", error);
      showToast("?????? ??? ???????? VM", "error");
    });
}

/**
 * Pauses a virtual machine
 *
 * @param {string} vmName - Name of the VM to pause
 * @returns {Promise<void>}
 *
 * @example
 * pauseVM('my-ubuntu-vm');
 */
function pauseVM(vmName) {
  if (!confirm(`???????????? ??????????? ?????? «${vmName}»?`)) return;

  const username = document.getElementById("user-name")?.textContent?.trim();
  if (!username) {
    showToast("??????: ?? ??????? ?????????? ???????????", "error");
    return;
  }

  fetch(`/${username}/vm/${vmName}/pause`, {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
    },
    credentials: "same-origin",
  })
    .then((response) => {
      if (response.status === 401) {
        window.location.href = "/login";
        return null;
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return;
```

```
      if (data.success) {
        showToast(`VM «${vmName}» ??????????????`, "success");
        setTimeout(() => location.reload(), 1500);
      } else {
        showToast(
          "??????: " + (data.error || "?? ??????? ???????????????? VM"),
          "error",
        );
      }
    })
    .catch((error) => {
      console.error("Error pausing VM:", error);
      showToast("?????? ??? ??????????????? VM", "error");
    });
}

/**
 * Resumes a paused virtual machine
 *
 * @param {string} vmName - Name of the VM to resume
 * @returns {Promise<void>}
 *
 * @example
 * unpauseVM('my-ubuntu-vm');
 */
function unpauseVM(vmName) {
  const username = document.getElementById("user-name")?.textContent?.trim();
  if (!username) {
    showToast("??????: ?? ??????? ??????????? ????????????", "error");
    return;
  }

  fetch(`/${username}/vm/${vmName}/unpause`, {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
    },
    credentials: "same-origin",
  })
    .then((response) => {
      if (response.status === 401) {
        window.location.href = "/login";
        return null;
      }
      if (response.status === 402) {
        showToast(
          "Insufficient balance. Please contact your admin to top up your balance.",
          "error",
        );
        return null;
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return;
      if (data.success) {
        showToast(`VM «${vmName}» ????????????`, "success");
        setTimeout(() => location.reload(), 1500);
      } else {
        showToast(
          "??????: " + (data.error || "?? ??????? ??????????? VM"),
          "error",
        );
      }
    })
    .catch((error) => {
      console.error("Error resuming VM:", error);
      showToast("?????? ??? ???????????????? VM", "error");
    });
}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/core/windows-logic.js (6.0 KB)

```
/**
 * Windows VM Management Logic
 * Encapsulates all Windows-specific behavior for the frontend
 */

const WindowsManager = {
    /**
     * Constants for Windows VM configuration
     */
    CONSTANTS: {
        INSTALLER_IMAGE_ID: 'windows-installer',
        DEFAULT_USERNAME: 'Administrator',
        NAMESPACE_ISO: 'kvm',
        DEFAULTS: {
            CPU: '4',
            MEMORY: '8', // GB
            STORAGE: '60' // GB
        }
    },

    /**
     * Checks if the selected image is a Windows installer
     * @param {string} imageUrl - The image identifier/URL
     * @returns {boolean} True if Windows installer
     */
    isWindowsInstaller(imageUrl) {
        return imageUrl === this.CONSTANTS.INSTALLER_IMAGE_ID;
    },

    /**
     * Sets recommended placeholders for Windows (does NOT override user values)
     * @param {Object} inputElements - Object containing DOM elements inputs
     */
    setRecommendedPlaceholders(inputElements) {
        const { cpuInput, memoryInput, storageInput } = inputElements;

        if (cpuInput) {
            cpuInput.placeholder = `????????????: ${this.CONSTANTS.DEFAULTS.CPU}`;
        }
        if (memoryInput) {
            memoryInput.placeholder = `????????????: ${this.CONSTANTS.DEFAULTS.MEMORY} GB`;
        }
        if (storageInput) {
            storageInput.placeholder = `????????????: ${this.CONSTANTS.DEFAULTS.STORAGE} GB`;
        }
    },

    /**
     * Gets recommended defaults for Windows
     * @returns {Object} Object with CPU, MEMORY, STORAGE defaults
     */
    getRecommendedDefaults() {
        return {
            cpu: this.CONSTANTS.DEFAULTS.CPU,
            memory: this.CONSTANTS.DEFAULTS.MEMORY,
            storage: this.CONSTANTS.DEFAULTS.STORAGE
        };
    },

    /**
     * Validates Windows username format (allows Uppercase)
     * @param {string} username - The username to validate
     * @returns {boolean} True if valid
     */
    isValidUsername(username) {
```

```
        // Windows: allow alphanumeric, underscore, hyphen (case-insensitive)
        const windowsUsernamePattern = /^[a-zA-Z_][a-zA-Z0-9_-]*$/;
        return windowsUsernamePattern.test(username);
    },

    /**
     * Configures the UI for Windows selection
     * Marks step 3 panel as Windows mode (actual field visibility controlled by updateStep3Visibility)
     * @param {HTMLElement} cloudInitSection - The cloud-init wizard panel
     * @param {HTMLInputElement} usernameInput - Username input field (deprecated, kept for compatibility)
     */
    configureUI(cloudInitSection, usernameInput) {
        // Mark cloud-init panel as Windows mode
        // Actual field configuration happens in updateStep3Visibility() when step 3 is shown
        if (cloudInitSection) {
            cloudInitSection.dataset.osType = 'windows';
        }

        // Note: Username is now set by updateStep3Visibility() to avoid premature field manipulation
    },

    /**
     * Restores UI for Linux selection
     * @param {HTMLElement} cloudInitSection - The cloud-init wizard panel
     */
    restoreLinuxUI(cloudInitSection) {
        // Mark as Linux mode - do NOT change display style
        // Wizard navigation controls panel visibility via CSS classes
        if (cloudInitSection) {
            cloudInitSection.dataset.osType = 'linux';
        }
    },

    /**
     * Configures GPU section for Windows installation
     * Shows info about GPU driver installation
     * @param {HTMLElement} gpuSection - The GPU configuration section
     */
    configureGPUSection(gpuSection) {
        if (!gpuSection) return;

        // Find or create info block
        let infoBlock = gpuSection.querySelector('.info-block.windows-gpu-info');

        if (!infoBlock) {
            // Create new info block for Windows GPU
            infoBlock = document.createElement('div');
            infoBlock.className = 'info-block windows-gpu-info';
            infoBlock.style.marginTop = '12px';
            infoBlock.style.backgroundColor = 'var(--warning-bg, #fef3c7)';
            infoBlock.style.borderLeft = '3px solid var(--warning, #f59e0b)';
            infoBlock.style.padding = '12px';
            infoBlock.style.borderRadius = '4px';
            infoBlock.innerHTML = `
                <strong>?? Windows + GPU Passthrough:</strong><br>
                <div style="margin-top: 8px;">
                    <strong>?? ????? ????????? Windows:</strong><br>
                    ? GPU ????? ???????? ????? ????????? ?????????<br>
                    ? VirtIO ???????? ????????? ?? ?????????? ????? (?????? E:)<br><br>

                    <strong>????? ????????? Windows:</strong><br>
                    1. ???????? ???? <code>E:\\</code> (VirtIO Drivers)<br>
                    2. ????????? ????????? ????????? VirtIO ??? Windows 10<br>
                    3. ????????? ???????? NVIDIA ?? ??????????? ?????<br>
                    4. ???????????? VM<br>
                    5. GPU ????? ? ????????????! ?
                </div>
            `;

            // Insert after GPU controls
            const existingInfo = gpuSection.querySelector('.info-block');
```

```
            if (existingInfo) {
                existingInfo.insertAdjacentElement('afterend', infoBlock);
            } else {
                gpuSection.appendChild(infoBlock);
            }
        }

        infoBlock.style.display = 'block';
    },

    /**
     * Removes Windows-specific GPU info
     * @param {HTMLElement} gpuSection - The GPU configuration section
     */
    restoreGPUSection(gpuSection) {
        if (!gpuSection) return;

        const infoBlock = gpuSection.querySelector('.info-block.windows-gpu-info');
        if (infoBlock) {
            infoBlock.style.display = 'none';
        }
    }
};

// Expose to window for usage in other scripts
window.WindowsManager = WindowsManager;

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/create-vm.js (48.1 KB)

```
document.addEventListener("DOMContentLoaded", () => {

  /**
   * Updates GPU configuration fields based on selected GPU mode
   *
   * Toggles visibility of GPU model and count fields.
   * Resets GPU values when switching between modes.
   *
   * @returns {void}
   */
  /** Update driver UI: show version/installer only when a GPU model is selected; for custom-driver GPUs show install...
  function updateDriverLabelVisibility() {
    const section = document.getElementById("gpu-config-section");
    const customDriverInfo = document.getElementById("gpu-custom-driver-info");
    const nvidiaDriverLabel = document.getElementById("nvidia-driver-label");
    const customInstallerLabel = document.getElementById("nvidia-custom-installer-label");
    const gpuModelSelect = document.getElementById("gpu-model-select");
    if (!section || !customDriverInfo || !nvidiaDriverLabel || !gpuModelSelect) return;
    const customList = (section.getAttribute("data-custom-driver-resources") || "").split(",").map((s) => s.trim()).f...
    const selected = (gpuModelSelect.value || "").trim();
    if (!selected) {
      nvidiaDriverLabel.classList.add("gpu-count-hidden");
      nvidiaDriverLabel.classList.remove("gpu-count-visible");
      if (customInstallerLabel) customInstallerLabel.classList.add("hidden");
      customDriverInfo.classList.add("hidden");
      return;
    }
    const isCustomDriver = customList.indexOf(selected) !== -1;
    if (isCustomDriver) {
      nvidiaDriverLabel.classList.add("gpu-count-hidden");
      nvidiaDriverLabel.classList.remove("gpu-count-visible");
      if (customInstallerLabel) customInstallerLabel.classList.remove("hidden");
      customDriverInfo.classList.remove("hidden");
    } else {
      nvidiaDriverLabel.classList.remove("gpu-count-hidden");
      nvidiaDriverLabel.classList.add("gpu-count-visible");
      if (customInstallerLabel) customInstallerLabel.classList.add("hidden");
      customDriverInfo.classList.add("hidden");
    }
  }

  function updateGpuConfig() {
      const gpuModeSelect = document.getElementById("gpu-mode-select");
      const gpuModelSelect = document.getElementById("gpu-model-select");
      const gpuModelLabel = document.getElementById("gpu-model-label");
      const gpuCountLabel = document.getElementById("gpu-count-label");
      const nvidiaDriverLabel = document.getElementById("nvidia-driver-label");
      const gpuInput = document.getElementById("gpu-input");
      const hiddenGpuModel = document.getElementById("hidden-gpu-model");
      const hiddenGpu = document.getElementById("hidden-gpu");

    if (!gpuModeSelect || !gpuModelSelect || !gpuModelLabel) return;

    if (gpuModeSelect.value === "gpu") {
      gpuModelLabel.classList.remove("hidden");
      gpuCountLabel.classList.remove("gpu-count-hidden");
      gpuCountLabel.classList.add("gpu-count-visible");
      gpuModelSelect.value = "";
      gpuInput.value = "";
      if (hiddenGpuModel) hiddenGpuModel.value = "";
      if (hiddenGpu) hiddenGpu.value = "";
      updateDriverLabelVisibility();
    } else {
      gpuModelLabel.classList.add("hidden");
      gpuCountLabel.classList.add("gpu-count-hidden");
```

```
      gpuCountLabel.classList.remove("gpu-count-visible");
      if (nvidiaDriverLabel) {
        nvidiaDriverLabel.classList.add("gpu-count-hidden");
        nvidiaDriverLabel.classList.remove("gpu-count-visible");
      }
      gpuModelSelect.value = "";
      gpuInput.value = "";
      if (hiddenGpu) hiddenGpu.value = "0";
      if (hiddenGpuModel) hiddenGpuModel.value = "";
      const customDriverInfo = document.getElementById("gpu-custom-driver-info");
      if (customDriverInfo) customDriverInfo.classList.add("hidden");
    }
  }

  // Tab switching logic
  const tabs = document.querySelectorAll(".form-tab");

  tabs.forEach((tab) => {
    tab.addEventListener("click", () => {
      const tabContainer = tab.closest(".form-tabs");
      if (!tabContainer) return;

      tabContainer
        .querySelectorAll(".form-tab")
        .forEach((t) => t.classList.remove("active"));
      tab.classList.add("active");

      const targetPanel = tab.dataset.tab;
      const parentSection = tabContainer.parentElement;
      const panels = parentSection.querySelectorAll(":scope > .tab-panel");

      panels.forEach((panel) => {
        if (panel.dataset.panel === targetPanel) {
          panel.classList.add("active");
        } else {
          panel.classList.remove("active");
        }
      });
    });
  });

  // GPU selection logic
  const gpuModeSelect = document.getElementById("gpu-mode-select");
  const gpuModelSelect = document.getElementById("gpu-model-select");
  const gpuInput = document.getElementById("gpu-input");
  const hiddenGpuModel = document.getElementById("hidden-gpu-model");
  const hiddenGpu = document.getElementById("hidden-gpu");

  if (gpuModeSelect) {
    gpuModeSelect.addEventListener("change", updateGpuConfig);
  }

  if (gpuModelSelect) {
    gpuModelSelect.addEventListener("change", function () {
      if (hiddenGpuModel) hiddenGpuModel.value = this.value;
      updateDriverLabelVisibility();
    });
  }

  if (gpuInput) {
    gpuInput.addEventListener("change", function () {
      if (hiddenGpu) hiddenGpu.value = this.value;
    });
    // Sync initial hidden value
    if (hiddenGpu) hiddenGpu.value = "0";
  }

  // Initialize state
  updateGpuConfig();

  // OS Selection Logic
```

```
const osCards = document.querySelectorAll('.os-card:not(.disabled)');
const hiddenImage = document.getElementById('hidden-image');

// Pre-select first OS (Ubuntu)
if (osCards.length > 0) {
    const firstCard = osCards[0];
    selectOSCard(firstCard);
}

osCards.forEach(card => {
    card.addEventListener('click', () => {
        selectOSCard(card);
    });
});

/**
 * Gets default username for OS image
 *
 * @param {string} imageUrl - Cloud image URL
 * @returns {string} Default username for the OS
 */
function getDefaultUsernameForImage(imageUrl) {
    if (!imageUrl) return 'ubuntu';

    const url = imageUrl.toLowerCase();

    if (url.includes('ubuntu')) return 'ubuntu';
    if (url.includes('debian')) return 'debian';
    if (url.includes('fedora')) return 'fedora';
    if (url.includes('opensuse') || url.includes('suse')) return 'opensuse';
    if (url.includes('centos')) return 'centos';
    if (url.includes('rocky')) return 'rocky';

    return 'ubuntu'; // Default fallback
}

function selectOSCard(card) {
    if (card.classList.contains('disabled')) return;

    // Update UI
    osCards.forEach(c => c.classList.remove('selected'));
    card.classList.add('selected');

    // Update hidden input
    const imageUrl = card.dataset.image;
    if (hiddenImage) {
        hiddenImage.value = imageUrl;
    }

    // Check if Windows installer or golden image
    const isWindowsInstaller = window.WindowsManager ? window.WindowsManager.isWindowsInstaller(imageUrl) : imageUr...
    const isWindowsGolden = imageUrl === 'windows-golden-image';
    const isWindows = isWindowsInstaller || isWindowsGolden;

    const cloudInitSection = document.querySelector('.wizard-panel[data-wizard-step="3"]');

    if (isWindows && window.WindowsManager) {
        window.WindowsManager.configureUI(cloudInitSection, null);

        const cpuInput = document.getElementById('cpu-input');
        const memoryInput = document.getElementById('memory-input');
        const storageInput = document.getElementById('storage-input');
        window.WindowsManager.setRecommendedPlaceholders({ cpuInput, memoryInput, storageInput });

        const gpuSection = document.querySelector('.gpu-section');
        window.WindowsManager.configureGPUSection(gpuSection);
    } else {
        if (window.WindowsManager) {
            window.WindowsManager.restoreLinuxUI(cloudInitSection);
            const gpuSection = document.querySelector('.gpu-section');
            window.WindowsManager.restoreGPUSection(gpuSection);
```

```
        } else if (cloudInitSection) {
            cloudInitSection.dataset.osType = 'linux';
        }
    }
}

// Wizard Navigation Logic
const wizardSteps = document.querySelectorAll(".wizard-step");
const wizardPanels = document.querySelectorAll(".wizard-panel");
const wizardNext = document.getElementById("wizard-next");
const wizardBack = document.getElementById("wizard-back");
const wizardSubmit = document.getElementById("wizard-submit");
const createVmForm = document.getElementById("create-vm-form");

let currentStep = 1;

// ????????? ?????? ????? ?? ?????????? ????? - ??????? ?????? ????? ??????
wizardSteps.forEach((step) => {
  step.addEventListener("click", (e) => {
    e.preventDefault();
    e.stopPropagation();
    // console.log('Step indicator clicked - ignoring, use Next/Back buttons instead');
    return false;
  });
  // ??????? cursor:pointer ???? ?? ????
  step.classList.add("cursor-default");
});

// ????????? ???????? ????? ?? Enter - ?????? ????? ?????? wizard
if (createVmForm) {
  createVmForm.addEventListener("keydown", (e) => {
    if (e.key === "Enter" && e.target.tagName !== "TEXTAREA") {
      e.preventDefault();
      // console.log('Enter pressed - handling wizard navigation instead of form submit');

      // ???? ?? ???? 1 ??? 2, ????????? ??????? "?????"
      if ((currentStep === 1 || currentStep === 2) && wizardNext) {
        wizardNext.click();
      }
      // ???? ?? ???? 3, ?????????? - ??????????? ?????? ???? ?????? "???????"

      return false;
    }
  });
}

/**
 * Updates visibility of fields on Step 3 based on selected OS type
 * For Windows: Show only hostname, username, password
 * For Linux: Show all cloud-init fields
 */
function updateStep3Visibility() {
  const hiddenImage = document.getElementById('hidden-image');
  const imageUrl = hiddenImage ? hiddenImage.value : '';

  const isWindowsGolden = imageUrl === 'windows-golden-image';
  const isWindowsInstaller = window.WindowsManager
      ? window.WindowsManager.isWindowsInstaller(imageUrl)
      : imageUrl === 'windows-installer';
  const isWindows = isWindowsGolden || isWindowsInstaller;

  const sshKeyField = document.querySelector('textarea[name="sshkey"]')?.closest('label');
  const infoBlock = document.querySelector('.wizard-panel[data-wizard-step="3"] .info-block');
  const additionalPackagesField = document.querySelector('.additional-packages-field');
  const checkboxGroup = document.querySelector('.wizard-panel[data-wizard-step="3"] .checkbox-group');
  const passwordField = document.getElementById('password-field');
  const passwordInput = document.getElementById('user-password');
  const usernameInput = document.getElementById('username-input');
  const sshKeyTextarea = document.querySelector('textarea[name="sshkey"]');
  const pwAuthCheck = document.getElementById('ssh-pwauth-check');

```

```
  if (isWindows) {
      // ===== WINDOWS MODE =====
      if (sshKeyField) sshKeyField.classList.add('hidden');
      if (infoBlock) infoBlock.classList.add('hidden');
      if (additionalPackagesField) additionalPackagesField.classList.add('hidden');
      if (checkboxGroup) checkboxGroup.classList.add('hidden');

      if (passwordField) {
          passwordField.classList.remove('hidden');
          passwordField.classList.add('flex-visible');
      }
      if (passwordInput) {
          passwordInput.setAttribute('required', 'required');
          passwordInput.placeholder = '?????? ??? Windows (??????????, ???. 8 ????????)';
      }

      if (pwAuthCheck) {
          pwAuthCheck.checked = true;
          pwAuthCheck.disabled = true;
      }

      if (usernameInput) {
          usernameInput.value = 'Administrator';
          usernameInput.readOnly = false;
      }

      if (sshKeyTextarea) {
          sshKeyTextarea.removeAttribute('required');
      }
  } else {
      // ===== LINUX MODE =====
      if (sshKeyField) sshKeyField.classList.remove('hidden');
      if (infoBlock) infoBlock.classList.remove('hidden');
      if (additionalPackagesField) additionalPackagesField.classList.remove('hidden');
      if (checkboxGroup) checkboxGroup.classList.remove('hidden');
      if (pwAuthCheck) pwAuthCheck.disabled = false;

      if (pwAuthCheck && pwAuthCheck.checked) {
          if (passwordField) { passwordField.classList.remove('hidden'); passwordField.classList.add('flex-visible'...
      } else {
          if (passwordField) { passwordField.classList.remove('flex-visible'); passwordField.classList.add('hidden'...
      }
      if (passwordInput) { passwordInput.removeAttribute('required'); passwordInput.placeholder = '?????? ????????...
      if (usernameInput) {
          if (usernameInput.value === 'Administrator' || usernameInput.value === '') {
              usernameInput.value = getDefaultUsernameForImage(imageUrl);
          }
          usernameInput.readOnly = false;
      }
      if (sshKeyTextarea) sshKeyTextarea.setAttribute('required', 'required');
  }
}

/**
 * Updates wizard UI based on current step
 * Supports 3 steps: 1 (OS), 2 (Resources), 3 (Config)
 */
function updateWizardUI() {
  // Update step indicators
  wizardSteps.forEach((step) => {
    const stepNum = parseInt(step.dataset.step);
    if (stepNum === currentStep) {
      step.classList.add("active");
      step.classList.remove("completed");
    } else if (stepNum < currentStep) {
      step.classList.add("completed");
      step.classList.remove("active");
    } else {
      step.classList.remove("active", "completed");
    }
  });
```

```

    // Update panels
    wizardPanels.forEach((panel) => {
      const panelStep = parseInt(panel.dataset.wizardStep);
      if (panelStep === currentStep) {
        panel.classList.add("active");

        // If this is step 3 (Configuration), adjust visibility based on OS type
        if (panelStep === 3) {
          updateStep3Visibility();
        }
      } else {
        panel.classList.remove("active");
      }
    });


    // Update button visibility
    // Always show Cancel button (Reset)
    if (wizardBack) {
        // Back availability
        if (currentStep === 1) {
             // On Step 1 (System), Back acts like Cancel/Reset? No, user requested Back specifically on Resources/Sy...
             // If we really want "Back" on Step 1, it must go to Dashboard or be hidden.
             // Per instruction "On Resources/System tabs there should be Back buttons".
             // Step 1 = System. Step 2 = Resources.
             // If we enable Back here, we must handle it.
             // For now, let's keep Back hidden on Step 1 as it's the start.
             // But enable it on Step 2 (Resources) and Step 3 (Config).
             wizardBack.classList.add("hidden", "invisible");
             wizardBack.disabled = true;
        } else {
             wizardBack.classList.remove("hidden", "invisible");
             wizardBack.disabled = false;
        }
    }

    if (currentStep === 1) {
      if (wizardNext) {
        wizardNext.classList.remove("hidden", "invisible");
      }
      if (wizardSubmit) {
        wizardSubmit.classList.add("hidden", "invisible");
      }
    } else if (currentStep === 2) {
      if (wizardNext) {
        wizardNext.classList.remove("hidden", "invisible");
      }
      if (wizardSubmit) {
        wizardSubmit.classList.add("hidden", "invisible");
      }
    } else if (currentStep === 3) {
      if (wizardNext) {
        wizardNext.classList.add("hidden", "invisible");
      }
      if (wizardSubmit) {
        wizardSubmit.classList.remove("hidden", "invisible");
        // Force display to ensure visibility
        wizardSubmit.style.display = 'inline-flex';
      }
    }
  }

  // Bind Cancel button to Reset
  const cancelButton = document.getElementById("close-create-vm-bottom");
  if (cancelButton) {
      // Remove old listeners to avoid duplicates if any (though this script runs once)
      // Actually we just add a new one, handle logic inside.
      cancelButton.addEventListener('click', (e) => {
          e.preventDefault();
          resetWizard();
```

```
        // Also close the modal
        const modal = document.getElementById("create-vm-modal");
        const overlay = document.getElementById("create-vm-overlay");
        if (modal) modal.hidden = true;
        if (overlay) overlay.hidden = true;
    });
}

// Listen for reset event from modal close (X button or overlay click)
document.addEventListener('resetCreateVMWizard', () => {
    resetWizard();
});

function resetWizard() {
    // Reset form
    if (createVmForm) createVmForm.reset();

    // Reset step
    currentStep = 1;

    // Reset hidden fields
    document.querySelectorAll('input[type="hidden"]').forEach(input => {
        // Keep default gpu model
        if (input.name === 'gpu_model' && input.dataset.defaultModel) {
            input.value = input.dataset.defaultModel;
        } else if (input.name === 'gpu') {
            input.value = "0";
        } else {
            input.value = "";
        }
    });

    // Reset OS selection
    const osCards = document.querySelectorAll('.os-card');
    osCards.forEach(c => c.classList.remove('selected'));
    // Select first supported OS
    const firstCard = document.querySelector('.os-card:not(.disabled)');
    if (firstCard) {
        selectOSCard(firstCard);
    }

    // Reset UI
    updateWizardUI();

    // Reset Presets/Manual tabs to default (Presets)
    const presetsTab = document.querySelector('.form-tab[data-tab="presets"]');
    if (presetsTab) presetsTab.click();

    // Reset GPU UI
    const gpuModeSelect = document.getElementById("gpu-mode-select");
    if (gpuModeSelect) {
        gpuModeSelect.value = "no-gpu";
        // Trigger change event manually or call updateGpuConfig
        updateGpuConfig();
    }
}


/**
 * Syncs input fields from manual tab to hidden fields
 */
function syncManualFieldsToHidden() {
    // Check which tab is active (presets or manual)
    const manualTab = document.querySelector('.tab-panel[data-panel="manual"]');
    if (manualTab && manualTab.classList.contains("active")) {
        const hiddenCpu = document.getElementById("hidden-cpu");
        const hiddenMemory = document.getElementById("hidden-memory");
        const hiddenStorage = document.getElementById("hidden-storage");

        const cpuInput = document.getElementById("cpu-input");
        const memoryInput = document.getElementById("memory-input");
```

```
492           const storageInput = document.getElementById("storage-input");
493
494           if (cpuInput && hiddenCpu) hiddenCpu.value = cpuInput.value;
495           if (memoryInput && hiddenMemory) hiddenMemory.value = memoryInput.value;
496           if (storageInput && hiddenStorage) hiddenStorage.value = storageInput.value;
497
498           // GPU sync
499           const gpuModeSelect = document.getElementById("gpu-mode-select");
500           const gpuInput = document.getElementById("gpu-input");
501
502 ... [740 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/dashboard-filters.js (2.6 KB)

```
/**
 * Dashboard filtering and VM actions
 */

// Client-side filtering
const searchInput = document.getElementById("search-input");
const statusFilter = document.getElementById("status-filter");
const regionFilter = document.getElementById("region-filter");
const tableBody = document.getElementById("vm-table-body");
const filteredCount = document.getElementById("filtered-count");

/**
 * Filters the VM table based on search term and status filter
 *
 * @returns {void}
 *
 * @sideEffects
 * - Shows/hides table rows based on filters
 * - Updates filtered count display
 * - Shows/hides empty state row
 *
 * @example
 * searchInput.addEventListener('input', filterTable);
 * statusFilter.addEventListener('change', filterTable);
 */
function filterTable() {
  const searchTerm = searchInput.value.toLowerCase();
  const statusValue = statusFilter.value.toLowerCase();
  const rows = tableBody.querySelectorAll("tr:not(.empty-row)");
  let visibleCount = 0;

  rows.forEach((row) => {
    const vmName = row.dataset.vmName ? row.dataset.vmName.toLowerCase() : "";
    const vmStatus = row.dataset.status || "";
    const ipCell = row.querySelector(".ip-cell");
    const ipAddress = ipCell ? ipCell.textContent.toLowerCase().trim() : "";

    // Check search
    const matchesSearch =
      !searchTerm ||
      vmName.includes(searchTerm) ||
      ipAddress.includes(searchTerm);

    // Check status
    let matchesStatus = !statusValue;
    if (statusValue === "running") {
      matchesStatus = vmStatus === "running";
    } else if (statusValue === "stopped") {
      matchesStatus = vmStatus === "stopped" || vmStatus === "halted";
    }

    if (matchesSearch && matchesStatus) {
      row.style.display = "";
      visibleCount++;
    } else {
      row.style.display = "none";
    }
  });

  // Update count
  const suffix =
    visibleCount === 1
      ? "??????"
      : visibleCount >= 2 && visibleCount <= 4
        ? "??????"
```

```
        : "?????";
  filteredCount.textContent = `${visibleCount} ${suffix}`;

  // Show empty state if no results
  const emptyRow = tableBody.querySelector(".empty-row");
  if (emptyRow) {
    emptyRow.style.display = visibleCount === 0 ? "" : "none";
  }
}

if (searchInput) searchInput.addEventListener("input", filterTable);
if (statusFilter) statusFilter.addEventListener("change", filterTable);
if (regionFilter) regionFilter.addEventListener("change", filterTable);

// NOTE: toggleMoreMenu() and openVNC() are now imported from core/utils.js
// Removed duplicate implementations to avoid conflicts

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/dashboard-home.js (6.8 KB)

```
"use strict";

document.addEventListener("DOMContentLoaded", () => {
  const appState = window.AppState;
  const username = document.body?.dataset?.username;
  if (!appState || !username) return;

  const elMainBalance = document.getElementById("dh-main-balance");
  const elLlmBalance = document.getElementById("dh-llm-balance");
  const elVmTotal = document.getElementById("dh-vm-total");
  const elVmRunning = document.getElementById("dh-vm-running");

  const elDirection = document.getElementById("dh-transfer-direction");
  const elAmount = document.getElementById("dh-transfer-amount");
  const elAll = document.getElementById("dh-transfer-all");
  const elSubmit = document.getElementById("dh-transfer-submit");
  const elError = document.getElementById("dh-transfer-error");

  const copyBtn = document.getElementById("dh-copy-snippet");
  const snippetEl = document.getElementById("dh-snippet");
  const copyHelper = window.copyTextToClipboard;

  const snippets = {
    python: `import openai

client = openai.OpenAI(
    base_url="https://api.virtuoz.com/v1",
    api_key="???_API_????",
)

response = client.chat.completions.create(
    model="qwen/qwen-coder-3b-instruct",
    messages=[{"role": "user", "content": "Hello!"}],
)
print(response.choices[0].message.content)`,
    typescript: `import OpenAI from "openai";

const client = new OpenAI({
  baseURL: "https://api.virtuoz.com/v1",
  apiKey: "???_API_????",
});

const response = await client.chat.completions.create({
  model: "qwen/qwen-coder-3b-instruct",
  messages: [{ role: "user", content: "Hello!" }],
});

console.log(response.choices[0].message.content);`,
    curl: `curl https://api.virtuoz.com/v1/chat/completions \\
  -H "Content-Type: application/json" \\
  -H "Authorization: Bearer ???_API_????" \\
  -d '{
    "model": "qwen/qwen-coder-3b-instruct",
    "messages": [{"role": "user", "content": "Hello!"}]
  }'`,
  };
  let currentLang = "python";

  function setTransferError(message) {
    if (!elError) return;
    elError.hidden = !message;
    elError.textContent = message || "";
  }

  function render(state) {
```

```
  const balances = state?.balances || {};
  const vm = state?.vmSummary || {};
  if (elMainBalance) elMainBalance.textContent = appState.formatMoney(balances.mainBalance);
  if (elLlmBalance) elLlmBalance.textContent = appState.formatMoney(balances.llmBalance);
  if (elVmTotal) elVmTotal.textContent = Number.isFinite(vm.total) ? String(vm.total) : "?";
  if (elVmRunning) elVmRunning.textContent = Number.isFinite(vm.running) ? String(vm.running) : "?";
}

function sourceBalanceByDirection(state) {
  const d = elDirection?.value || "main-llm";
  const balances = state?.balances || {};
  return d === "main-llm" ? balances.mainBalance : balances.llmBalance;
}

async function loadVmSummary() {
  try {
    const res = await fetch(`/${encodeURIComponent(username)}/dashboard/vms`, {
      credentials: "same-origin",
    });
    if (!res.ok) return;
    const data = await res.json();
    const vms = Array.isArray(data.vms) ? data.vms : [];
    const running = vms.filter((vm) => vm.running).length;
    appState.setState(
      {
        vmSummary: {
          total: vms.length,
          running,
          updatedAt: Date.now(),
        },
      },
      "dashboardHomeVmSummary",
    );
  } catch (_) {}
}

if (elSubmit && elDirection && elAmount) {
  elSubmit.addEventListener("click", async () => {
    setTransferError("");
    const state = appState.getState();
    const rawAmount = Number.parseFloat(elAmount.value || "");
    const amount = Number.isFinite(rawAmount) ? rawAmount : NaN;
    const maxAvailable = Number(sourceBalanceByDirection(state));

    if (!Number.isFinite(amount) || amount <= 0) {
      setTransferError("??????? ?????????? ?????.");
      return;
    }
    if (!Number.isFinite(maxAvailable)) {
      setTransferError("?????? ??????????, ?????????? ?????.");
      return;
    }
    if (amount > maxAvailable) {
      setTransferError("????? ???????? ????????? ????????? ??????.");
      return;
    }

    const source = elDirection.value === "main-llm" ? "main" : "llm";
    const target = source === "main" ? "llm" : "main";

    elSubmit.disabled = true;
    try {
      const res = await fetch(`/${encodeURIComponent(username)}/api/balance/transfer`, {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        credentials: "same-origin",
        body: JSON.stringify({ source, target, amount }),
      });
      const data = await res.json().catch(() => ({}));
      if (!res.ok) {
        const msg = typeof data.detail === "string" ? data.detail : "?????? ????????";
```

```
          throw new Error(msg);
        }
        elAmount.value = "";
        appState.setBalances(data, "dashboardHomeTransfer");
        if (window.showToast) window.showToast("??????? ????????", "success");
      } catch (err) {
        setTransferError(err.message || "?????? ????????");
      } finally {
        elSubmit.disabled = false;
      }
    });
  }

  if (elAll && elAmount && elDirection) {
    elAll.addEventListener("click", () => {
      const state = appState.getState();
      const sourceValue = Number(sourceBalanceByDirection(state));
      if (!Number.isFinite(sourceValue) || sourceValue <= 0) {
        setTransferError("??????????? ??????? ??? ????????.");
        return;
      }
      setTransferError("");
      elAmount.value = Number(sourceValue).toFixed(2);
    });
  }

  if (copyBtn && snippetEl && typeof copyHelper === "function") {
    copyBtn.addEventListener("click", async () => {
      const ok = await copyHelper(snippets[currentLang] || "");
      copyBtn.textContent = ok ? "???????????" : "??????";
      setTimeout(() => {
        copyBtn.textContent = "??????????";
      }, 1200);
    });
  }

  document.querySelectorAll(".dh-tab").forEach((tab) => {
    tab.addEventListener("click", () => {
      currentLang = tab.dataset.lang || "python";
      document.querySelectorAll(".dh-tab").forEach((t) => t.classList.remove("active"));
      tab.classList.add("active");
      if (snippetEl) snippetEl.textContent = snippets[currentLang] || "";
    });
  });

  if (snippetEl) snippetEl.textContent = snippets[currentLang];
  render(appState.getState());
  appState.subscribe((state) => render(state));
  appState.loadBalances(username).catch(() => {});
  loadVmSummary();
});

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/globals.js (223.0 B)

```
/**
 * Global variables shared across dashboard scripts
 */

// Get username from URL path safe check
const pathParts = window.location.pathname.split("/");
const username = pathParts.length > 1 ? pathParts[1] : "";

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/llm-chat.js (15.8 KB)

```
// LLM Chat Page JavaScript

// Extract username from data attribute
const username = document.body.dataset.username;

const llmPage = document.querySelector(".llm-page");
const llmDefaultModel = (llmPage?.dataset.llmDefaultModel || "").trim();

let chatHistory = [];
let isGenerating = false;
let temperature = 0.7;
let maxTokens = 2048;
const DEFAULT_TEMPERATURE = 0.7;
const DEFAULT_MAX_TOKENS = 2048;

// Initialize marked.js with highlight.js
document.addEventListener("DOMContentLoaded", () => {
  if (typeof marked !== "undefined") {
    marked.setOptions({
      highlight: function (code, lang) {
        if (typeof hljs !== "undefined" && lang && hljs.getLanguage(lang)) {
          try {
            return hljs.highlight(code, { language: lang }).value;
          } catch (e) {}
        }
        return code;
      },
      breaks: true,
      gfm: true,
    });
  }

  initChat();
  initModelSelector();
  initPlaygroundSettings();
  updateSendButtonState();
});

function initPlaygroundSettings() {
  const temperatureEl = document.getElementById("chat-temperature");
  const temperatureValueEl = document.getElementById("chat-temperature-value");
  const maxTokensEl = document.getElementById("chat-max-tokens");
  const maxTokensValueEl = document.getElementById("chat-max-tokens-value");
  const resetBtn = document.getElementById("chat-settings-reset");
  const presetButtons = Array.from(document.querySelectorAll(".settings-preset-btn"));

  const updatePresetState = () => {
    presetButtons.forEach((btn) => {
      const temp = Number.parseFloat(btn.dataset.temp || "");
      const tokens = Number.parseInt(btn.dataset.tokens || "", 10);
      const isActive = Math.abs(temp - temperature) < 0.0001 && tokens === maxTokens;
      btn.classList.toggle("active", isActive);
      btn.setAttribute("aria-pressed", isActive ? "true" : "false");
    });
  };

  const applyValues = (nextTemp, nextTokens) => {
    temperature = Number.isFinite(nextTemp) ? nextTemp : DEFAULT_TEMPERATURE;
    maxTokens = Number.isFinite(nextTokens) ? nextTokens : DEFAULT_MAX_TOKENS;
    if (temperatureEl) temperatureEl.value = String(temperature);
    if (temperatureValueEl) temperatureValueEl.textContent = String(temperature.toFixed(1));
    if (maxTokensEl) maxTokensEl.value = String(maxTokens);
    if (maxTokensValueEl) maxTokensValueEl.textContent = String(maxTokens);
    updatePresetState();
  };
```

```

  if (temperatureEl && temperatureValueEl) {
    temperature = Number.parseFloat(temperatureEl.value || String(DEFAULT_TEMPERATURE));
    temperatureValueEl.textContent = String(temperature.toFixed(1));
    temperatureEl.addEventListener("input", () => {
      const parsed = Number.parseFloat(temperatureEl.value || String(DEFAULT_TEMPERATURE));
      temperature = Number.isFinite(parsed) ? parsed : DEFAULT_TEMPERATURE;
      temperatureValueEl.textContent = String(temperature.toFixed(1));
      updatePresetState();
    });
  }

  if (maxTokensEl && maxTokensValueEl) {
    maxTokens = Number.parseInt(maxTokensEl.value || String(DEFAULT_MAX_TOKENS), 10);
    maxTokensValueEl.textContent = String(maxTokens);
    maxTokensEl.addEventListener("input", () => {
      const parsed = Number.parseInt(maxTokensEl.value || String(DEFAULT_MAX_TOKENS), 10);
      maxTokens = Number.isFinite(parsed) ? parsed : DEFAULT_MAX_TOKENS;
      maxTokensValueEl.textContent = String(maxTokens);
      updatePresetState();
    });
  }

  presetButtons.forEach((btn) => {
    btn.addEventListener("click", () => {
      const temp = Number.parseFloat(btn.dataset.temp || "");
      const tokens = Number.parseInt(btn.dataset.tokens || "", 10);
      applyValues(temp, tokens);
    });
  });

  if (resetBtn) {
    resetBtn.addEventListener("click", () => {
      applyValues(DEFAULT_TEMPERATURE, DEFAULT_MAX_TOKENS);
    });
  }

  applyValues(temperature, maxTokens);
}

/**
 * Initializes the chat interface
 * Sets up event listeners for input, send, and clear buttons.
 * @returns {void}
 */
function initChat() {
  const chatInput = document.getElementById("chat-input");
  const sendBtn = document.getElementById("chat-send-btn");
  const clearBtn = document.getElementById("chat-clear-btn");

  if (!chatInput || !sendBtn) return;

  // Auto-resize textarea
  chatInput.classList.add("auto-resize");
  chatInput.addEventListener("input", updateSendButtonState);

  // Send on Enter (Shift+Enter for newline)
  chatInput.addEventListener("keydown", (e) => {
    if (e.key === "Enter" && !e.shiftKey) {
      e.preventDefault();
      if (chatInput.value.trim() && !isGenerating) {
        sendMessage();
      }
    }
  });

  // Send button click
  sendBtn.addEventListener("click", () => {
    if (chatInput.value.trim() && !isGenerating) {
      sendMessage();
    }
```

```
  });

  // Clear chat
  if (clearBtn) {
    clearBtn.addEventListener("click", clearChat);
  }

  // Suggestion buttons
  initSuggestionButtons();
}

/**
 * ????????? ?????? ??????? ? ????????? ????????????? ??????.
 */
async function initModelSelector() {
  const selectEl = document.getElementById("chat-model-select");
  if (!selectEl) return;

  try {
    const r = await fetch(`/${username}/api/llm/available-models`, { credentials: "same-origin" });
    const data = await r.json();
    const models = data.models || [];

    selectEl.innerHTML = "";
    if (models.length === 0) {
      selectEl.innerHTML = '<option value="">??? ????????? ???????</option>';
      return;
    }

    for (const m of models) {
      const id = m.id || m.name || "";
      const name = m.name || m.id || id;
      const source = m._source === "cudo" ? " (Cudo)" : m._source === "openrouter" ? " (OpenRouter)" : "";
      const opt = document.createElement("option");
      opt.value = id;
      opt.textContent = `${name}${source}`;
      selectEl.appendChild(opt);
    }
    const norm = (v) => (v || "").trim().toLowerCase();
    const defaultFull = norm(llmDefaultModel);
    const defaultShort = norm(llmDefaultModel.split("/").pop());
    let preferredIndex = -1;
    if (defaultFull || defaultShort) {
      for (let i = 0; i < selectEl.options.length; i++) {
        const value = norm(selectEl.options[i].value);
        const valueShort = norm(selectEl.options[i].value.split("/").pop());
        if (value === defaultFull || value === defaultShort || valueShort === defaultShort) {
          preferredIndex = i;
          break;
        }
      }
    }
    selectEl.selectedIndex = preferredIndex >= 0 ? preferredIndex : 0;
  } catch (e) {
    selectEl.innerHTML = '<option value="">?????? ????????</option>';
  }
}

function initSuggestionButtons() {
  document.querySelectorAll(".suggestion-btn").forEach((btn) => {
    btn.addEventListener("click", () => {
      const suggestion = btn.dataset.suggestion;
      const chatInput = document.getElementById("chat-input");
      if (suggestion && !isGenerating && chatInput) {
        chatInput.value = suggestion;
        chatInput.dispatchEvent(new Event("input"));
        sendMessage();
      }
    });
  });
}
```

```

function updateSendButtonState() {
  const chatInput = document.getElementById("chat-input");
  const sendBtn = document.getElementById("chat-send-btn");
  if (!sendBtn || !chatInput) return;
  const hasText = !!chatInput.value.trim();
  const canSend = hasText && !isGenerating;
  sendBtn.disabled = !canSend;
}

/**
 * Sends a user message to the LLM API
 * Adds user message to chat, initiates streaming response.
 * @returns {void}
 */
function sendMessage() {
  const chatInput = document.getElementById("chat-input");
  const sendBtn = document.getElementById("chat-send-btn");
  const messagesContainer = document.getElementById("chat-messages");

  const message = chatInput.value.trim();
  if (!message) return;

  clearWelcomeState();

  // Add user message
  addMessage("user", message);
  chatHistory.push({ role: "user", content: message });

  // Clear input
  chatInput.value = "";
  sendBtn.disabled = true;

  // Start generating
  isGenerating = true;
  sendBtn.classList.add("loading");

  // Add assistant message placeholder
  const assistantMsgId = "msg-" + Date.now();
  addMessage("assistant", "", assistantMsgId);

  // Stream the response
  streamResponse(assistantMsgId);
}

/**
 * Adds a message to the chat UI
 * @param {string} role - "user" or "assistant"
 * @param {string} content - Message text (can be empty for streaming)
 * @param {string|null} id - Optional message ID for later updates
 * @returns {void}
 */
function addMessage(role, content, id = null) {
  const messagesContainer = document.getElementById("chat-messages");

  const msgDiv = document.createElement("div");
  msgDiv.className = `chat-message chat-message-${role}`;
  if (id) msgDiv.id = id;

  const avatarHtml =
    role === "user"
      ? `<div class="chat-avatar chat-avatar-user">${typeof username !== "undefined" ? username.slice(0, 2).toUpperCa...
      : `<div class="chat-avatar chat-avatar-assistant"><svg viewBox="0 0 24 24" fill="none" stroke="currentColor" st...

  msgDiv.innerHTML = `
    ${avatarHtml}
    <div class="chat-message-content">
      <div class="chat-message-text">${role === "user" ? escapeHtml(content) : content || '<span class="typing-indica...
    </div>
  `;

```

```
  messagesContainer.appendChild(msgDiv);
  messagesContainer.scrollTop = messagesContainer.scrollHeight;
}

function escapeHtml(text) {
  const div = document.createElement("div");
  div.textContent = text;
  return div.innerHTML;
}

async function streamResponse(messageId) {
  const messageDiv = document.getElementById(messageId);
  const textEl = messageDiv?.querySelector(".chat-message-text");

  if (!textEl) return;

  let fullContent = "";

  try {
    const response = await fetch(`/${username}/llm/chat`, {
      method: "POST",
      headers: {
        "Content-Type": "application/json",
      },
      credentials: "same-origin",
      body: JSON.stringify({
        messages: chatHistory,
        model: document.getElementById("chat-model-select")?.value || undefined,
        temperature,
        max_tokens: maxTokens,
      }),
    });

    if (!response.ok) {
      throw new Error(`HTTP ${response.status}`);
    }

    const reader = response.body.getReader();
    const decoder = new TextDecoder();

    const MAX_ITERATIONS = 10000;
    const TIMEOUT = 300000; // 5 ?????
    let iterations = 0;
    const startTime = Date.now();

    while (iterations < MAX_ITERATIONS) {
      if (Date.now() - startTime > TIMEOUT) {
        throw new Error("Stream timeout exceeded");
      }

      const { done, value } = await reader.read();

      if (done) break;
      iterations++;

      const chunk = decoder.decode(value, { stream: true });
      const lines = chunk.split("\n");

      for (const line of lines) {
        if (line.startsWith("data: ")) {
          const data = line.slice(6);

          if (data === "[DONE]") {
            break;
          }

          try {
            const parsed = JSON.parse(data);

            if (parsed.error) {
              textEl.innerHTML = `<span class="chat-error">??????: ${escapeHtml(parsed.error)}</span>`;
```

```
              break;
            }

            if (parsed.content) {
              fullContent += parsed.content;
              // Render markdown as we receive content
              textEl.innerHTML = renderMarkdown(fullContent);

              // Apply syntax highlighting to code blocks
              textEl.querySelectorAll("pre code").forEach((block) => {
                if (
                  typeof hljs !== "undefined" &&
                  !block.classList.contains("hljs")
                ) {
                  hljs.highlightElement(block);
                }
              });

              // Scroll to bottom
              const messagesContainer =
                document.getElementById("chat-messages");
              messagesContainer.scrollTop = messagesContainer.scrollHeight;
            }
          } catch (e) {
            // Skip malformed JSON
          }
        }
      }
    }

    // Save assistant response to history
    if (fullContent) {
      chatHistory.push({ role: "assistant", content: fullContent });
    }
  } catch (error) {
    console.error("Chat error:", error);
    textEl.innerHTML = `<span class="chat-error">?????? ??????????: ${escapeHtml(error.message)}</span>`;
  } finally {
    isGenerating = false;
    const sendBtn = document.getElementById("chat-send-btn");
    if (sendBtn) {
      sendBtn.classList.remove("loading");
      updateSendButtonState();
    }
  }
}

function renderMarkdown(text) {
  if (typeof marked !== "undefined") {
    return marked.parse(text);
  }
  // Fallback: basic formatting
  return text
    .replace(/```(\w+)?\n([\s\S]*?)```/g, "<pre><code>$2</code></pre>")
    .replace(/`([^`]+)`/g, "<code>$1</code>")
    .replace(/\*\*([^*]+)\*\*/g, "<strong>$1</strong>")
    .replace(/\*([^*]+)\*/g, "<em>$1</em>")
    .replace(/\n/g, "<br>");
}

/**
 * Clears the chat history
 * Resets conversation state and shows welcome message.
 * @returns {void}
 */
function clearChat() {
  const messagesContainer = document.getElementById("chat-messages");

  // Clear messages
  showWelcomeState(messagesContainer);

```

```
  // Clear history
  chatHistory = [];

  if (window.showToast) {
    showToast("??? ??????", "success");
  }
}

function clearWelcomeState() {
  const messagesContainer = document.getElementById("chat-messages");
  if (!messagesContainer) return;
  const welcome = messagesContainer.querySelector(".chat-welcome");
  if (welcome) {
    welcome.remove();
  }
}

function showWelcomeState(messagesContainer) {
  if (!messagesContainer) return;
  messagesContainer.innerHTML = `
    <div class="chat-welcome">
      <div class="chat-welcome-icon">
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
          <path d="M21 16V8a2 2 0 0 0-1-1.73l-7-4a2 2 0 0 0-2 0l-7 4A2 2 0 0 0 3 8v8a2 2 0 0 0 1 1.73l7 4a2 2 0 0 0 2...
          <polyline points="7.5 4.21 12 6.81 16.5 4.21"></polyline>
          <polyline points="7.5 19.79 7.5 14.6 3 12"></polyline>
          <polyline points="21 12 16.5 14.6 16.5 19.79"></polyline>
          <polyline points="3.27 6.96 12 12.01 20.73 6.96"></polyline>
          <line x1="12" y1="22.08" x2="12" y2="12"></line>
        </svg>
      </div>
      <h3>????? ????????? ? LLM-???</h3>
      <p>??????? ?????? ?? ????????????, Linux, DevOps ??? Kubernetes.</p>
      <div class="chat-suggestions">
        <button class="suggestion-btn" data-suggestion="??? ????????? ???????? CPU ? RAM ?? Linux?">????????? CPU/RA...
        <button class="suggestion-btn" data-suggestion="??? ?????? ????????? SSH-????? ?? ????????">????????? SSH-???...
        <button class="suggestion-btn" data-suggestion="??? ????????? Docker Compose ?????? ?? VPS?">Docker Compose ...
      </div>
    </div>
  `;
  initSuggestionButtons();
}
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/llm-keys.js (7.0 KB)

```
(function () {
  const pageEl = document.querySelector(".llm-models-page");
  if (!pageEl) return;

  const username = pageEl.dataset.username;
  const maxKeys = Number(pageEl.dataset.maxKeys || "10");
  const loadingEl = document.getElementById("keys-loading");
  const errorEl = document.getElementById("keys-error");
  const listEl = document.getElementById("keys-list");
  const countEl = document.getElementById("keys-count");
  const createBtn = document.getElementById("keys-create-btn");
  const copyHelper = window.copyTextToClipboard;

  function fmtDate(iso) {
    if (!iso) return "?";
    const d = new Date(iso);
    return d.toLocaleString("ru-RU");
  }

  function escapeHtml(s) {
    const div = document.createElement("div");
    div.textContent = s;
    return div.innerHTML;
  }

  function renderKeys(keys) {
    const sectionEl = document.getElementById("keys-section");
    countEl.textContent = String(keys.length);
    if (!keys.length) {
      listEl.innerHTML = '<tr class="empty-row"><td colspan="5">?????? ???? ???</td></tr>';
      sectionEl.style.display = "block";
      return;
    }

    listEl.innerHTML = keys.map((k) => {
      const isActive = k.status === "active";
      return `
        <tr data-key-id="${k.id}">
          <td><span class="key-status-badge ${isActive ? "active" : "revoked"}">${isActive ? "???????" : "???????"}</...
          <td class="key-cell">
            <span class="key-masked">${escapeHtml(k.token_masked || "?")}</span>
            <span class="key-full" style="display:none;word-break:break-all;font-family:monospace;"></span>
          </td>
          <td>${escapeHtml(fmtDate(k.created_at))}</td>
          <td>${escapeHtml(fmtDate(k.last_used_at))}</td>
          <td>${
            isActive
              ? `<div class="key-actions">
                  <button class="btn-secondary key-reveal-btn" data-key-id="${k.id}">????????</button>
                  <button class="btn-secondary key-copy-btn" data-key-id="${k.id}">??????????</button>
                </div>`
              : "?"
          }</td>
        </tr>
      `;
    }).join("");

    async function revealKey(row, keyId, revealBtn) {
      const maskedEl = row.querySelector(".key-masked");
      const fullEl = row.querySelector(".key-full");
      if (!row || !keyId || !maskedEl || !fullEl) return null;
      if (fullEl.style.display !== "none" && fullEl.textContent) {
        return fullEl.textContent;
      }
      const btn = revealBtn || row.querySelector(".key-reveal-btn");
```

```
      if (btn) {
        btn.disabled = true;
        btn.textContent = "?";
      }
      try {
        const r = await fetch(`/${username}/api/llm/keys/${keyId}/reveal`, { credentials: "same-origin" });
        const data = await r.json().catch(() => ({}));
        if (!r.ok) throw new Error(data.detail || "?????? ????????");
        fullEl.textContent = data.token || "";
        fullEl.style.display = "inline";
        maskedEl.style.display = "none";
        if (btn) btn.textContent = "??????";
        return data.token || "";
      } catch (e) {
        if (btn) btn.textContent = "????????";
        throw e;
      } finally {
        if (btn) btn.disabled = false;
      }
    }

    listEl.querySelectorAll(".key-reveal-btn").forEach((btn) => {
      btn.addEventListener("click", async () => {
        const keyId = btn.dataset.keyId;
        if (!keyId) return;
        const row = btn.closest("tr");
        const maskedEl = row.querySelector(".key-masked");
        const fullEl = row.querySelector(".key-full");
        if (fullEl.style.display === "none") {
          try {
            await revealKey(row, keyId, btn);
          } catch (e) {
            if (window.showToast) window.showToast(e.message || "??????", "error");
          }
        } else {
          fullEl.style.display = "none";
          fullEl.textContent = "";
          maskedEl.style.display = "inline";
          btn.textContent = "????????";
        }
      });
    });

    listEl.querySelectorAll(".key-copy-btn").forEach((btn) => {
      btn.addEventListener("click", async () => {
        const keyId = btn.dataset.keyId;
        if (!keyId) return;
        const row = btn.closest("tr");
        const fullEl = row.querySelector(".key-full");
        btn.disabled = true;
        const prevText = btn.textContent;
        btn.textContent = "?";
        try {
          let token = (fullEl && fullEl.style.display !== "none" && fullEl.textContent) ? fullEl.textContent : "";
          if (!token) {
            token = await revealKey(row, keyId, row.querySelector(".key-reveal-btn"));
          }
          if (!token) throw new Error("???? ??????");
          const ok = typeof copyHelper === "function" ? await copyHelper(token) : false;
          if (!ok) throw new Error("?? ??????? ??????????");
          if (window.showToast) window.showToast("??????????? ? ?????", "success");
          btn.textContent = "???????????";
          setTimeout(() => {
            btn.textContent = prevText;
          }, 1200);
        } catch (e) {
          btn.textContent = prevText;
          if (window.showToast) window.showToast(e.message || "??????", "error");
        } finally {
          btn.disabled = false;
        }
```

```
      });
    });
    sectionEl.style.display = "block";
  }

  async function loadKeys() {
    loadingEl.style.display = "block";
    errorEl.style.display = "none";
    document.getElementById("keys-section").style.display = "none";
    try {
      const r = await fetch(`/${username}/api/llm/keys`, { credentials: "same-origin" });
      const data = await r.json().catch(() => ({}));
      if (!r.ok) throw new Error(data.detail || `HTTP ${r.status}`);
      renderKeys(data.keys || []);
      const activeCount = (data.keys || []).filter((k) => k.status === "active").length;
      createBtn.disabled = activeCount >= maxKeys;
    } catch (e) {
      errorEl.textContent = "?????? ????????: " + (e.message || "??????????");
      errorEl.style.display = "block";
      document.getElementById("keys-section").style.display = "none";
    } finally {
      loadingEl.style.display = "none";
    }
  }

  createBtn.addEventListener("click", async () => {
    createBtn.disabled = true;
    try {
      const r = await fetch(`/${username}/api/llm/keys`, {
        method: "POST",
        credentials: "same-origin",
      });
      const data = await r.json().catch(() => ({}));
      if (!r.ok) throw new Error(data.detail || "?????? ????????");
      if (window.showToast) window.showToast("???? ??????", "success");
      await loadKeys();
    } catch (e) {
      if (window.showToast) window.showToast(e.message || "?????? ????????", "error");
    } finally {
      createBtn.disabled = false;
    }
  });

  loadKeys();
})();

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/llm-models.js (9.0 KB)

```
/**
 * ????????? LLM-?????? ? ????????, ???? ? ?????? ????.
 */
(function () {
  const REFRESH_INTERVAL_MS = 5 * 60 * 1000;
  const pageEl = document.querySelector(".llm-models-page");
  let exampleApiKey =
    (pageEl && pageEl.dataset.exampleApiKey && pageEl.dataset.exampleApiKey.trim()) ||
    "???_API_????";
  const username =
    (pageEl && pageEl.dataset.username) ||
    (window.location.pathname.match(/^\/([^/]+)\//) || [])[1];
  if (!username) return;

  const apiUrl = `/${username}/api/llm/available-models`;
  const loadingEl = document.getElementById("models-loading");
  const errorEl = document.getElementById("models-error");
  const listEl = document.getElementById("models-list");
  const countEl = document.getElementById("models-count");
  const refreshEl = document.getElementById("models-last-refresh");
  const copyHelper = window.copyTextToClipboard;
  const FALLBACK_MODEL_PRICING = {
    "qwen/qwen-coder-3b-instruct": { input: "0.15", output: "0.30" },
    "asi1-mini": { input: "0.05", output: "0.10" },
  };

  function escapeHtml(s) {
    const div = document.createElement("div");
    div.textContent = String(s ?? "");
    return div.innerHTML;
  }

  function formatTime() {
    return new Date().toLocaleTimeString("ru-RU", {
      hour: "2-digit",
      minute: "2-digit",
      second: "2-digit",
    });
  }

  function formatPriceValue(val) {
    const num = Number(val);
    if (!Number.isFinite(num)) return String(val);
    if (num === 0) return "0";
    if (Math.abs(num) >= 1) return num.toFixed(4).replace(/\.?0+$/, "");
    if (Math.abs(num) >= 0.0001) return num.toFixed(8).replace(/\.?0+$/, "");
    return num.toExponential(4);
  }

  function pickPrice(model, keys) {
    const pricing = model.pricing || {};
    for (const key of keys) {
      const val = pricing[key] ?? model[key];
      if (val == null || val === "") continue;
      return formatPriceValue(val);
    }
    return "?";
  }

  function getFallbackPrice(modelId, direction) {
    const key = String(modelId || "").trim().toLowerCase();
    const row = FALLBACK_MODEL_PRICING[key];
    return row ? row[direction] || null : null;
  }

```

```
  function buildCodeExample(modelId) {
    return `import openai

client = openai.OpenAI(
    base_url="https://api.virtuoz.com/v1",
    api_key="${exampleApiKey}",
)

response = client.chat.completions.create(
    model="${modelId}",
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": "Hello!"}
    ]
)

print(response.choices[0].message.content)`;
  }

  async function ensureExampleApiKey() {
    if (exampleApiKey && exampleApiKey !== "???_API_????") return;
    try {
      const response = await fetch(`/${username}/api/llm/keys`, {
        method: "POST",
        credentials: "same-origin",
      });
      if (!response.ok) return;
      const data = await response.json();
      if (data && typeof data.token === "string" && data.token.trim()) {
        exampleApiKey = data.token.trim();
      }
    } catch (error) {
      // Keep placeholder if auto-create is unavailable.
    }
  }

  function renderModels(models) {
    if (!models || models.length === 0) {
      listEl.innerHTML = '<p class="llm-models-empty">??? ????????? ???????</p>';
      countEl.textContent = "0";
      refreshEl.textContent = `?????????: ${formatTime()}`;
      return;
    }

    listEl.innerHTML = models
      .map((m, idx) => {
        const modelId = m.id || m.name || `model-${idx}`;
        const modelName = m.name || m.id || "?";
        const desc = m.description || "";
        const ctx = m.context_length
          ? `${Math.round(Number(m.context_length) / 1000)}k`
          : "";
        const source = m._source || "unknown";
        const sourceLabel =
          source === "openrouter" ? "OpenRouter" : source === "cudo" ? "Cudo" : "LLM";
        const sourceClass = source === "cudo" ? "cudo" : "";
        const inputPrice = pickPrice(m, [
          "prompt",
          "input",
          "in",
          "prompt_per_million",
          "input_per_million",
          "input_price",
          "prompt_price",
          "input_cost",
          "input_cost_per_token",
          "prompt_cost_per_token",
        ]);
        const outputPrice = pickPrice(m, [
          "completion",
          "output",
```

```
          "out",
          "completion_per_million",
          "output_per_million",
          "output_price",
          "completion_price",
          "output_cost",
          "output_cost_per_token",
          "completion_cost_per_token",
        ]);
        const fallbackInput = getFallbackPrice(modelId, "input");
        const fallbackOutput = getFallbackPrice(modelId, "output");
        const resolvedInputPrice = inputPrice === "?" && fallbackInput ? fallbackInput : inputPrice;
        const resolvedOutputPrice = outputPrice === "?" && fallbackOutput ? fallbackOutput : outputPrice;
        const codeExample = buildCodeExample(modelId);
        return `
          <article class="llm-model-card" data-model-card>
            <div class="llm-model-card-head">
              <div class="llm-model-main">
                <span class="model-source ${sourceClass}">${sourceLabel}</span>
                <h4>${escapeHtml(modelName)}</h4>
                ${desc ? `<p class="model-desc">${escapeHtml(desc)}</p>` : ""}
                <div class="model-meta">${ctx ? `????????: ${escapeHtml(ctx)}` : ""}</div>
                <div class="model-prices">
                  <span>????: <span class="model-price-value">${escapeHtml(resolvedInputPrice)}</span></span>
                  <span>?????: <span class="model-price-value">${escapeHtml(resolvedOutputPrice)}</span></span>
                </div>
              </div>
              <div class="llm-model-actions">
                <button
                  type="button"
                  class="model-expand-btn"
                  data-expand-btn
                  aria-expanded="false"
                  aria-label="???????? ?????? ????"
                  title="???????? ?????? ????"
                >
                  <span aria-hidden="true">?</span>
                </button>
              </div>
            </div>
            <div class="model-example" data-example-block>
              <div class="model-example-head">
                <strong>?????? ???????????? (OpenAI SDK)</strong>
                <button type="button" class="model-copy-btn" data-copy-btn>??????????</button>
              </div>
              <pre class="model-code" data-code-block>${escapeHtml(codeExample)}</pre>
            </div>
          </article>
        `;
      })
      .join("");

    listEl.querySelectorAll("[data-expand-btn]").forEach((btn) => {
      btn.addEventListener("click", () => {
        const card = btn.closest("[data-model-card]");
        const block = card ? card.querySelector("[data-example-block]") : null;
        if (!block) return;
        const open = block.classList.toggle("open");
        btn.setAttribute("aria-expanded", open ? "true" : "false");
        btn.setAttribute("aria-label", open ? "?????? ?????? ????" : "???????? ?????? ????");
        btn.title = open ? "?????? ?????? ????" : "???????? ?????? ????";
      });
    });

    listEl.querySelectorAll("[data-copy-btn]").forEach((btn) => {
      btn.addEventListener("click", async () => {
        const card = btn.closest("[data-model-card]");
        const codeEl = card ? card.querySelector("[data-code-block]") : null;
        const text = codeEl ? codeEl.textContent || "" : "";
        const ok =
          typeof copyHelper === "function" ? await copyHelper(text) : false;
```

```
        btn.textContent = ok ? "??????????" : "??????";
        setTimeout(() => {
          btn.textContent = "??????????";
        }, 1200);
      });
    });

    countEl.textContent = String(models.length);
    refreshEl.textContent = `?????????: ${formatTime()}`;
  }

  async function loadModels() {
    loadingEl.style.display = "block";
    errorEl.style.display = "none";
    listEl.style.display = "none";
    try {
      await ensureExampleApiKey();
      const r = await fetch(apiUrl, { credentials: "same-origin" });
      if (!r.ok) throw new Error(`HTTP ${r.status}`);
      const data = await r.json();
      renderModels(data.models || []);
      listEl.style.display = "grid";
    } catch (e) {
      errorEl.textContent = `?????? ????????: ${e.message || "?????????? ??????"}`;
      errorEl.style.display = "block";
    } finally {
      loadingEl.style.display = "none";
    }
  }

  loadModels();
  setInterval(loadModels, REFRESH_INTERVAL_MS);
})();

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/pages/vm-details-init.js (1.2 KB)

```
/**
 * VM Details Page Initialization
 *
 * Handles page-specific initialization tasks:
 * - Reads VM details from data attributes (no inline script in HTML)
 * - VNC iframe setup
 * - Global variable setup for vm-details.js
 */

/**
 * Reads VM details from data container and sets window globals
 * @returns {void}
 */
function initVMDetailsData() {
  const el = document.getElementById("vm-details-data");
  if (!el) return;

  window.vmName = el.dataset.vmName || "";
  window.username = el.dataset.username || "";
  window.namespace = el.dataset.namespace || "";
  window.vmOS = el.dataset.vmOs || "";
  window.vncPath = el.dataset.vncPath || "";
}

/**
 * Initializes VNC iframe with proper path
 * Sets up autoconnect and scaling parameters.
 *
 * @returns {void}
 */
function initVNC() {
  const iframe = document.getElementById("vnc-frame");
  if (!iframe) return;

  const vncPath = window.vncPath;
  if (vncPath) {
    iframe.src = `/static/novnc/vnc.html?autoconnect=1&resize=scale&path=${encodeURIComponent(vncPath)}`;
  }
}

// Initialize on page load
window.addEventListener("DOMContentLoaded", () => {
  initVMDetailsData();
  initVNC();
});

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/storage.js (5.6 KB)

```
document.addEventListener("DOMContentLoaded", () => {
  const refreshBtn = document.getElementById("refresh-storage-btn");
  const storageTableBody = document.getElementById("storage-table-body");
  const username = window.location.pathname.split("/")[1];

  if (refreshBtn) {
    refreshBtn.addEventListener("click", loadStorageData);
  }

  // Load data when tab is activated
  const observer = new MutationObserver((mutations) => {
    mutations.forEach((mutation) => {
      if (
        mutation.target.id === "storage-content" &&
        mutation.target.classList.contains("active") &&
        mutation.attributeName === "class"
      ) {
        loadStorageData();
      }
    });
  });

  const storageContent = document.getElementById("storage-content");
  if (storageContent) {
    observer.observe(storageContent, { attributes: true });
  }

  async function loadStorageData() {
    if (!storageTableBody) return;

    storageTableBody.innerHTML = `
            <tr class="loading-row">
                <td colspan="6" class="text-center">
                    <div class="loading-spinner-small"></div>
                    ???????? ?????? ? ??????...
                </td>
            </tr>
        `;

    try {
      const response = await fetch(`/${username}/api/datavolumes`);
      if (!response.ok) throw new Error("Failed to load storage data");

      const data = await response.json();
      renderTable(data);
    } catch (error) {
      console.error("Error loading storage:", error);
      storageTableBody.innerHTML = `
                <tr class="error-row">
                    <td colspan="6" class="text-center text-error">
                        ?????? ???????? ??????: ${error.message}
                    </td>
                </tr>
            `;
    }
  }

  function renderTable(data) {
    if (!data || data.length === 0) {
      storageTableBody.innerHTML = `
                <tr class="empty-row">
                    <td colspan="6" class="text-center">
                         <div class="empty-state-small">
                            ??? ????????? ??????
                        </div>
```

```
                    </td>
                </tr>
            `;
      return;
    }

    storageTableBody.innerHTML = data
      .map((dv) => {
        const statusClass = getStatusClass(dv.phase);
        const isAttached = dv.vm_name ? true : false;

        return `
                <tr>
                    <td>
                        <div class="font-medium">${dv.name}</div>
                        <div class="text-muted text-small">${dv.pvc_name || "-"}</div>
                    </td>
                    <td>
                        <span class="status-badge ${statusClass}">${dv.phase}</span>
                    </td>
                    <td>${dv.size}</td>
                    <td>
                        ${
                          dv.vm_name
                            ? `<a href="/${username}/vm/${dv.vm_name}" class="vm-link">
                                <span class="vm-icon">??</span> ${dv.vm_name}
                             </a>`
                            : '<span class="text-muted">?? ?????????????</span>'
                        }
                    </td>
                    <td>
                        <div class="progress-container" title="${dv.progress}">
                             <div class="progress-bar" style="width: ${dv.progress}"></div>
                        </div>
                        <div class="text-tiny text-muted mt-1">${dv.progress}</div>
                    </td>
                    <td>
                        <div class="actions-cell">
                            <button class="action-btn-icon danger"
                                    onclick="deleteDataVolume('${dv.name}')"
                                    title="??????? ????"
                                    ${isAttached ? "disabled" : ""}>
                                <svg viewBox="0 0 24 24" width="18" height="18" fill="none" stroke="currentColor" str...
                                    <polyline points="3 6 5 6 21 6"></polyline>
                                    <path d="M19 6v14a2 2 0 0 1-2 2H7a2 2 0 0 1-2-2V6m3 0V4a2 2 0 0 1 2-2h4a2 2 0 0 1...
                                </svg>
                            </button>
                        
                    </td>
                </tr>
            `;
      })
      .join("");
  }

  function getStatusClass(phase) {
    switch (phase?.toLowerCase()) {
      case "succeeded":
        return "status-running";
      case "importinprogress":
      case "cloneinprogress":
        return "status-provisioning";
      case "failed":
        return "status-stopped";
      default:
        return "status-unknown";
    }
  }

  window.deleteDataVolume = async function (dvName) {
    if (
```

```
      !confirm(
        `?? ???????, ??? ?????? ??????? ???? ${dvName}? ??? ???????? ?????? ????????.`,
      )
    ) {
      return;
    }

    try {
      const response = await fetch(`/${username}/api/datavolumes/${dvName}`, {
        method: "DELETE",
      });

      if (response.ok) {
        showToast(`???? ${dvName} ??????`, "success");
        loadStorageData();
      } else {
        const err = await response.json();
        showToast(`?????? ????????: ${err.error}`, "error");
      }
    } catch (error) {
      showToast(`??????: ${error.message}`, "error");
    }
  };
});

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/user-dashboard.js (26.0 KB)

```
// Modal controls
const openModalBtn = document.getElementById("open-create-vm");
const closeModalBtn = document.getElementById("close-create-vm");
const closeModalBottomBtn = document.getElementById("close-create-vm-bottom");
const modal = document.getElementById("create-vm-modal");
const overlay = document.getElementById("create-vm-overlay");

// Store interval ID for cleanup
let __agentLastVmRenderTs = 0;

// #region agent log
function __agentLog(hypothesisId, location, message, data) {
  fetch("http://localhost:7242/ingest/5d6e8a84-0852-4c51-9dde-f3a26bfdf00d", {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
      "X-Debug-Session-Id": "54c5fc",
    },
    body: JSON.stringify({
      sessionId: "54c5fc",
      runId: "run2",
      hypothesisId,
      location,
      message,
      data,
      timestamp: Date.now(),
    }),
  }).catch(() => {});
}
// #endregion

const PINNED_STORAGE_KEY = "viirtuoz-pinned-vms";
const appState = window.AppState || null;

function syncVmSummaryState(total, running) {
  if (!appState) return;
  appState.setState(
    {
      vmSummary: {
        total: Number.isFinite(total) ? total : null,
        running: Number.isFinite(running) ? running : null,
        updatedAt: Date.now(),
      },
    },
    "vmSummarySync",
  );
}

function getPinnedVMs(username) {
  try {
    const raw = localStorage.getItem(`${PINNED_STORAGE_KEY}-${username}`);
    return raw ? JSON.parse(raw) : [];
  } catch {
    return [];
  }
}

function setPinnedVMs(username, vms) {
  try {
    localStorage.setItem(`${PINNED_STORAGE_KEY}-${username}`, JSON.stringify(vms));
  } catch (_) {}
}

window.togglePinVM = function (vmName) {
  const username = document.getElementById("user-name")?.textContent?.trim();
```

```
  if (!username) return;
  const pinned = getPinnedVMs(username);
  const idx = pinned.indexOf(vmName);
  if (idx >= 0) {
    pinned.splice(idx, 1);
  } else {
    pinned.push(vmName);
  }
  setPinnedVMs(username, pinned);
  updateVMList();
};

/**
 * Resets the Create VM form to its initial state
 *
 * Clears all form fields, hidden inputs, manual input fields, sliders,
 * and restores default wizard step to step 1.
 *
 * @returns {void}
 *
 * @example
 * resetForm(); // Called when closing modal or after successful VM creation
 */
function resetForm() {
  // Reset all form fields
  const form = document.getElementById("create-vm-form");
  if (form) form.reset();

  // Clear hidden fields
  const hiddenCpu = document.getElementById("hidden-cpu");
  const hiddenMemory = document.getElementById("hidden-memory");
  const hiddenStorage = document.getElementById("hidden-storage");
  const hiddenGpu = document.getElementById("hidden-gpu");
  const hiddenGpuModel = document.getElementById("hidden-gpu-model");

  if (hiddenCpu) hiddenCpu.value = "";
  if (hiddenMemory) hiddenMemory.value = "";
  if (hiddenStorage) hiddenStorage.value = "";
  if (hiddenGpu) hiddenGpu.value = "0";
  if (hiddenGpuModel) hiddenGpuModel.value = "";

  // Clear manual input fields explicitly
  const cpuInput = document.getElementById("cpu-input");
  const memInput = document.getElementById("memory-input");
  const storageInput = document.getElementById("storage-input");
  const gpuInput = document.getElementById("gpu-input");
  const gpuModelSelect = document.getElementById("gpu-model-input");
  const gpuCountLabel = document.getElementById("gpu-count-label");
  const nvidiaDriverLabel = document.getElementById("nvidia-driver-label");

  if (cpuInput) cpuInput.value = "";
  if (memInput) memInput.value = "";
  if (storageInput) storageInput.value = "";
  if (gpuInput) gpuInput.value = "0";
  if (gpuModelSelect) gpuModelSelect.value = ""; // Reset to "??? GPU"
  if (gpuCountLabel) {
    gpuCountLabel.classList.add("gpu-count-hidden");
    gpuCountLabel.classList.remove("gpu-count-visible");
  }
  if (nvidiaDriverLabel) {
    nvidiaDriverLabel.classList.add("gpu-count-hidden");
    nvidiaDriverLabel.classList.remove("gpu-count-visible");
  }

  // Clear VM name
  const vmNameInput = document.getElementById("vm-name-input");
  if (vmNameInput) vmNameInput.value = "";

  // Remove selection from preset cards
  const presetCards = document.querySelectorAll(".preset-card");
  presetCards.forEach((c) => c.classList.remove("selected"));
```

```

  // Clear password field
  const passwordInput = document.getElementById("user-password");
  if (passwordInput) passwordInput.value = "";

  // Uncheck checkboxes
  const upgradeCheck = document.getElementById("full-upgrade-check");
  const pwAuthCheck = document.getElementById("ssh-pwauth-check");
  if (upgradeCheck) upgradeCheck.checked = false;
  if (pwAuthCheck) pwAuthCheck.checked = false;

  // Hide password field
  const passwordField = document.getElementById("password-field");
  if (passwordField) passwordField.classList.add("hidden");

  // Reset to first tab (presets)
  const presetsTab = document.querySelector('.form-tab[data-tab="presets"]');
  const manualTab = document.querySelector('.form-tab[data-tab="manual"]');
  const presetsPanel = document.querySelector(
    '.tab-panel[data-panel="presets"]',
  );
  const manualPanel = document.querySelector('.tab-panel[data-panel="manual"]');

  if (presetsTab) presetsTab.classList.add("active");
  if (manualTab) manualTab.classList.remove("active");
  if (presetsPanel) presetsPanel.classList.add("active");
  if (manualPanel) manualPanel.classList.remove("active");

  // Note: Wizard state is now managed by create-vm.js
}

if (openModalBtn && modal && overlay) {
  openModalBtn.addEventListener("click", () => {
    resetForm(); // Reset form BEFORE opening modal

    // Reset wizard state to step 1 and update UI
    const resetEvent = new CustomEvent('resetCreateVMWizard');
    document.dispatchEvent(resetEvent);

    modal.removeAttribute("hidden");
    overlay.removeAttribute("hidden");
  });

  const closeModal = () => {
    modal.setAttribute("hidden", "hidden");
    overlay.setAttribute("hidden", "hidden");
    resetForm(); // Reset form when closing modal

    // Also reset wizard state (same as Cancel button behavior)
    // Check if resetWizard function exists in create-vm.js scope
    const resetEvent = new CustomEvent('resetCreateVMWizard');
    document.dispatchEvent(resetEvent);
  };

  if (closeModalBtn) closeModalBtn.addEventListener("click", closeModal);
  if (closeModalBottomBtn)
    closeModalBottomBtn.addEventListener("click", closeModal);
  if (overlay) overlay.addEventListener("click", closeModal);
}

// Wizard navigation is now handled by create-vm.js

// VM Management Functions are now in core/vm-actions.js
// Functions available globally: startVM(), stopVM(), restartVM(), deleteVM(), pauseVM(), unpauseVM()

/**
 * Fetches and updates the VM list in the dashboard
 *
 * Makes an AJAX call to refresh the VM table without full page reload.
 * Redirects to login if session expired (401).
 *
```

```
 * @returns {Promise<void>}
 *
 * @sideEffects
 * - Updates #vm-table-body HTML content
 * - Redirects to /login on 401
 * - Shows toast notification on error
 *
 * @example
 * updateVMList(); // Called after VM operations (start/stop/delete)
 * setInterval(updateVMList, 5000); // Auto-refresh every 5 seconds
 */
function updateVMList() {
  const username = document.getElementById("user-name")?.textContent?.trim();
  if (!username) {
    return;
  }
  // #region agent log
  __agentLog("H1", "user-dashboard.js:updateVMList:start", "vm refresh start", {
    username,
  });
  // #endregion

  fetch(`/${username}/dashboard/vms`, {
    credentials: "same-origin",
  })
    .then((response) => {
      if (response.status === 401) {
        // Session expired, redirect to login
        window.location.href = "/login";
        return null;
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return; // Handle null from 401 redirect
      if (data.success && data.vms) {
        const tableBody = document.getElementById("vm-table-body");
        if (!tableBody) return;
        // #region agent log
        __agentLog(
          "H1",
          "user-dashboard.js:updateVMList:beforeReplace",
          "replacing vm table body",
          {
            existingRows: tableBody.querySelectorAll("tr").length,
            incomingRows: Array.isArray(data.vms) ? data.vms.length : -1,
          },
        );
        // #endregion

        // Clear existing rows
        tableBody.innerHTML = "";

        // Check if there are VMs
        if (data.vms.length === 0) {
          // Show "No VMs" message
          const emptyRow = document.createElement("tr");
          emptyRow.className = "empty-row";
          emptyRow.innerHTML = `
            <td colspan="7">
              <div class="empty-state">
                <svg class="empty-state-icon" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.5">
                  <rect x="2" y="3" width="20" height="14" rx="2"/>
                  <path d="M8 21h8"/>
                  <path d="M12 17v4"/>
                  <path d="M9 10h6"/>
                </svg>
                <h3>??? ??????????? ?????</h3>
                <p>???????? ???? ?????? ??????????? ??????, ????? ?????? ????</p>
              </div>
            </td>
```

```
          `;
          tableBody.appendChild(emptyRow);
        } else {
          const stubMode = data.stub_mode === true;
          const pinnedVMs = getPinnedVMs(username);
          // Add updated VM rows
          data.vms.forEach((vm) => {
            const row = document.createElement("tr");
            row.dataset.vmName = vm.name;
            row.dataset.status = vm.status
              ? vm.status.toLowerCase()
              : "unknown";
            const isPinned = pinnedVMs.includes(vm.name);
            if (isPinned) row.classList.add("vm-row-pinned");

            // Determine status class and text
            const statusLower = vm.status ? vm.status.toLowerCase() : "unknown";
            let statusClass = "status-unknown";
            let statusText = vm.status || "??????????";

            if (statusLower === "running") {
              statusClass = "status-running";
              statusText = "????????";
            } else if (statusLower === "paused") {
              statusClass = "status-paused";
              statusText = "??????????????";
            } else if (statusLower === "stopped" || statusLower === "halted") {
              statusClass = "status-stopped";
              statusText = "??????????";
            } else if (statusLower === "starting") {
              statusClass = "status-starting";
              statusText = "??????????";
            } else if (statusLower === "stopping") {
              statusClass = "status-stopping";
              statusText = "??????????????";
            } else if (
              statusLower.includes("creating") ||
              statusLower.includes("provisioning") ||
              statusLower === "??????????" ||
              statusLower === "waitingforvolumebinding" ||
              statusLower === "waitingfordatavolume" ||
              statusLower.startsWith("datavolume ")
            ) {
              statusClass = "status-creating";
              const progressMatch = statusLower.match(/(\d+)%/);
              if (progressMatch) {
                statusText = `???????? ?????... ${progressMatch[1]}%`;
              } else if (statusLower === "??????????") {
                statusText = "??????????";
              } else {
                statusText = "???????? ?????...";
              }
            }

            const stubBadge = stubMode ? ' <span class="badge badge-demo">????</span>' : "";
            const starClass = isPinned ? "vm-pin-btn pinned" : "vm-pin-btn";
            const starTitle = isPinned ? "?????? ??????????" : "???????? VM";
            row.innerHTML = `
              <td>
                <div class="vm-name-cell">
                  <button type="button" class="${starClass}" onclick="togglePinVM('${vm.name}')" title="${starTitle}"...
                    <svg viewBox="0 0 24 24" fill="${isPinned ? "currentColor" : "none"}" stroke="currentColor" strok...
                  </button>
                  <a href="/${username}/vm/${vm.name}" class="vm-name-link">${vm.name}</a>${stubBadge}
                </div>
              </td>
              <td>
                <span class="status-badge ${statusClass}">${statusText}</span>
              </td>
              <td>
                <div class="resources-cell">
```

```
                <span class="resource-tag">${vm.cpu || "-"} vCPU</span>
                <span class="resource-tag">${vm.memory || "-"} RAM</span>
                ${vm.gpu_count ? `<span class="resource-tag resource-gpu">${vm.gpu_count}x ${vm.gpu_model.split("/"...
              </div>
            </td>
            <td>
              <span class="text-muted">??????</span>
            </td>
            <td class="ip-cell">
              ${vm.allocated_ip ? `<span class="ip-badge">${vm.allocated_ip}</span>` : '<span class="text-muted">?<...
            </td>
            <td>
              <div class="console-buttons">
                <button class="vnc-btn" onclick="openVNC('${vm.name}')" ${!vm.running ? "disabled" : ""}>
                  <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                    <rect x="2" y="3" width="20" height="14" rx="2"/>
                    <path d="M8 21h8"/>
                    <path d="M12 17v4"/>
                  </svg>
                  ${vm.running ? "VNC" : "VNC ??????????"}
                </button>
                ${
                  vm.os && vm.os.toLowerCase().includes("windows")
                    ? `
                <button class="rdp-btn" onclick="toggleRDP('${vm.name}', ${vm.running})" ${!vm.running ? "disabled"...
                  <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                    <rect x="3" y="3" width="18" height="18" rx="2"/>
                    <rect x="7" y="7" width="3" height="9"/>
                    <rect x="14" y="7" width="3" height="5"/>
                  </svg>
                  ${vm.running ? "RDP" : "RDP ??????????"}
                </button>
                `
                    : ""
                }
              </div>
            </td>
            <td>
              <div class="actions-cell">
                ${
                  vm.running
                    ? `
                  <button class="action-btn btn-stop" onclick="stopVM('${vm.name}')">
                    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                      <rect x="6" y="6" width="12" height="12" rx="1"/>
                    </svg>
                    <span>??????????</span>
                  </button>
                `
                    : statusLower === "paused"
                      ? `
                  <button class="action-btn btn-resume" onclick="unpauseVM('${vm.name}')">
                    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                      <polygon points="5 3 19 12 5 21 5 3"/>
                    </svg>
                    <span>???????????</span>
                  </button>
                `
                      : `
                  <button class="action-btn btn-start" onclick="startVM('${vm.name}')">
                    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                      <polygon points="5 3 19 12 5 21 5 3"/>
                    </svg>
                    <span>?????????</span>
                  </button>
                `
                }
                <div class="more-actions">
                  <button class="more-btn" onclick="toggleMoreMenu(this, event)">
                    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                      <circle cx="12" cy="12" r="1"/>
```

```
                      <circle cx="12" cy="5" r="1"/>
                      <circle cx="12" cy="19" r="1"/>
                    </svg>
                  </button>
                  <div class="more-dropdown">
                    ${
                      vm.running
                        ? `
                    <button class="more-dropdown-item" onclick="restartVM('${vm.name}')">
                      <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                        <path d="M23 4v6h-6"/>
                        <path d="M20.49 15a9 9 0 0 1 1-2.12-9.36L23 10"/>
                      </svg>
                      ????????????
                    </button>
                    <button class="more-dropdown-item" onclick="pauseVM('${vm.name}')">
                      <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                        <rect x="6" y="4" width="4" height="16"/>
                        <rect x="14" y="4" width="4" height="16"/>
                      </svg>
                      ????????????
                    </button>
                    `
                        : statusLower === "paused"
                          ? `
                    <button class="more-dropdown-item" onclick="stopVM('${vm.name}')">
                      <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                        <rect x="6" y="6" width="12" height="12" rx="1"/>
                      </svg>
                      ???????????
                    </button>
                    `
                          : ""
                    }
                    <button class="more-dropdown-item danger" onclick="deleteVM('${vm.name}')">
                      <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                        <polyline points="3 6 5 6 21 6"/>
                        <path d="M19 6v14a2 2 0 0 1-2 2H7a2 2 0 0 1-2-2V6m3 0V4a2 2 0 0 1 2-2h4a2 2 0 0 1 2 2v2"/>
                      </svg>
                      ???????
                    </button>
                  </div>
                </div>
              </div>
            </td>
          `;
          tableBody.appendChild(row);
        });
      }

      // Update VM count in stats
      const vmCountTotal = document.getElementById("vm-count-total");
      const vmCountRunning = document.getElementById("vm-count-running");
      const filteredCount = document.getElementById("filtered-count");

      if (vmCountTotal) {
        vmCountTotal.textContent = data.vms.length;
      }

      if (vmCountRunning) {
        const runningCount = data.vms.filter((vm) => vm.running).length;
        vmCountRunning.textContent = runningCount;
      }

      if (filteredCount) {
        const suffix =
          data.vms.length === 1
            ? "??????"
            : data.vms.length >= 2 && data.vms.length <= 4
              ? "??????"
              : "?????";
```

```
492           filteredCount.textContent = `${data.vms.length} ${suffix}`;
493         }
494
495         const runningCount = data.vms.filter((vm) => vm.running).length;
496         syncVmSummaryState(data.vms.length, runningCount);
497         __agentLastVmRenderTs = Date.now();
498         // #region agent log
499         __agentLog("H1", "user-dashboard.js:updateVMList:end", "vm refresh end", {
500           renderedRows: data.vms.length,
501
502 ... [201 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/static/js/vm-details.js (36.0 KB)

```
// VM Details JavaScript

// Variables vmName, username, namespace are set in vm-details-init.js (window.*)
// Fallback: parse from URL path /<username>/vm/<vm_name>
function getVMDetailsContext() {
  const pathParts = window.location.pathname.split("/");
  return {
    user: window.username || (pathParts.length > 3 ? pathParts[1] : ""),
    vm: window.vmName || (pathParts.length > 3 ? pathParts[3] : ""),
  };
}

// Store interval ID for cleanup
let refreshInterval = null;
let metricsInterval = null;
let isUpdatingMetrics = false;

// ---- Service Management Functions ----

/**
 * Loads and displays active services for the VM
 */
function loadServices() {
    const listContainer = document.getElementById("active-services-list");
    const { user, vm } = getVMDetailsContext();
    if (!user || !vm) return;

    fetch(`/${user}/vm/${vm}/services`, {
        credentials: "same-origin"
    })
    .then(res => {
        if (res.status === 401) {
            window.location.href = "/login";
            return null;
        }
        return res.json();
    })
    .then(data => {
        if (!data || !data.services) return;
        if (data.services.length === 0) {
            listContainer.innerHTML = '<div class="loading-services">??? ???????? ?????? ????????</div>';
            return;
        }

        listContainer.innerHTML = ''; // Clear list

        data.services.forEach(svc => {
            const item = document.createElement("div");
            item.className = "service-item";
            // Determine display name
            let typeLabel = svc.name.toUpperCase();
            if(typeLabel.startsWith("CUSTOM-")) {
                const parts = svc.name.split("-");
                typeLabel = `CUSTOM ${parts[1] || ""}`.trim();
            }

            // Build service info HTML
            let serviceInfoHTML = `
                <div class="service-info">
                    <div class="service-header">
                        <span class="service-name">${typeLabel}</span>
                        <span class="service-ports">
                            <span style="color:var(--text-secondary)">VM Port:</span> <strong>${svc.port}</strong>
                            <span style="margin:0 0.5rem">?</span>
                            <span style="color:var(--text-secondary)">Ext Port:</span> <strong>${svc.nodePort}</strong>
```

```
                        </span>
                    </div>
            `;

            // Add SSH command if available
            if (svc.ssh_command) {
                serviceInfoHTML += `
                    <div class="ssh-command-display">
                        <code>${svc.ssh_command}</code>
                        <button class="btn-copy-ssh" onclick="copySSHCommand('${svc.ssh_command.replace(/'/g, "\\'")}...
                            <svg viewBox="0 0 24 24" width="16" height="16" fill="none" stroke="currentColor" stroke-...
                                <rect x="9" y="9" width="13" height="13" rx="2" ry="2"></rect>
                                <path d="M5 15H4a2 2 0 0 1-2-2V4a2 2 0 0 1 2-2h9a2 2 0 0 1 2 2v1"></path>
                            </svg>
                        </button>
                    </div>
                `;
            }

            serviceInfoHTML += `</div>`;

            item.innerHTML = serviceInfoHTML + `
                <button class="btn-delete-service" onclick="deleteService('${svc.name}')" title="??????? ???????">
                    <svg viewBox="0 0 24 24" width="18" height="18" fill="none" stroke="currentColor" stroke-width="2">
                        <path d="M18 6L6 18M6 6l12 12"></path>
                    </svg>
                </button>
            `;

            listContainer.appendChild(item);
        });
    })
    .catch(err => {
        console.error("Failed to load services:", err);
        listContainer.innerHTML = '<div class="loading-services text-error">?????? ????????</div>';
    });
}

/**
 * Adds a new predefined service (HTTP, HTTPS, SSH)
 * @param {string} type - Service type (http, https, ssh)
 * @param {number} port - Target VM port
 */
function addServiceMode(type, port) {
    createService(type, port);
}

/**
 * Adds a custom service from input
 */
function addCustomService() {
    const input = document.getElementById("custom-port-input");
    const port = parseInt(input.value);

    if (!port || port < 1 || port > 65535) {
        showToast("??????? ????????? ????? ????? (1-65535)", "error");
        return;
    }

    // Find next available custom slot index (1-5) logic could be here or backend.
    // Backend handles name collision, but we need to generate a unique "type" name if we want multiple customs.
    // The requirement says "custom-x (1 to 5)".
    // Let's implement a simple check or try random?
    // Better: Try to find which custom slots are free by checking UI list or API.
    // Simplification for now: Use random ID or count existing.

    // Actually, backend needs "type" to be passed.
    // Let's check currently loaded services to find next slot.
    const { user, vm } = getVMDetailsContext();
    fetch(`/${user}/vm/${vm}/services`)
        .then(res => res.json())
```

```
        .then(data => {
            const existingCustoms = data.services
                .map(s => s.name)
                .filter(n => n.startsWith("custom-"))
                .map(n => parseInt(n.split("-")[1] || 0));

            if (existingCustoms.length >= 5) {
                showToast("???????? 5 ????????? ??????", "error");
                return;
            }

            // Find first free slot
            let slot = 1;
            while(existingCustoms.includes(slot)) slot++;
            createService(`custom-${slot}`, port);
            input.value = ""; // clear input
        });
}

/**
 * API call to create service
 */
function createService(type, port) {
    if(window.showToast) showToast(`???????? ??????? ${type}...`, "info");

    const { user, vm } = getVMDetailsContext();
    fetch(`/${user}/vm/${vm}/services`, {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        credentials: "same-origin",
        body: JSON.stringify({
            port: port,
            type: type
        })
    })
    .then(res => res.json())
    .then(data => {
        if(data.error) {
             showToast(data.error, "error");
        } else {
             showToast(`???? ${port} ??????? ??????????? ?? ${data.port}`, "success");
             loadServices(); // Refresh list
        }
    })
    .catch(err => {
        console.error(err);
        showToast("?????? ????", "error");
    });
}

/**
 * Copy SSH command to clipboard
 */
function copySSHCommand(command) {
    const helper = window.copyTextToClipboard;
    if (typeof helper !== "function") {
        showToast('?????? ??? ???????????', 'error');
        return;
    }
    helper(command).then((ok) => {
        if (ok) {
            showToast('SSH ??????? ??????????? ? ????? ??????', 'success');
        } else {
            showToast('?????? ??? ???????????', 'error');
        }
    });
}

/**
 * API call to delete service
 */
```

```
function deleteService(serviceName) {
    if(!confirm("??????? ??? ??????? ?????????")) return;

    // serviceName might be "custom-1", or "ssh", "http"
    // Our backend route expects "type" used in construction.
    // The serviceName we get from list_vm_services matches the type used (vm_utils.py:195)
    // because list returns "ssh", "http", "custom-1" as names.

    // Wait... list_vm_services returns 'name' as 'ssh' or 'custom-1'.
    // The DELETE route expects <service_type>.
    // So if serviceName is "ssh", type is "ssh".
    // If serviceName is "custom-1", type is "custom-1".

    const { user, vm } = getVMDetailsContext();
    fetch(`/${user}/vm/${vm}/services/${serviceName}`, {
        method: "DELETE",
        credentials: "same-origin"
    })
    .then(res => res.json())
    .then(data => {
        if(data.error) {
             showToast(data.error, "error");
        } else {
             showToast("??????? ???????", "success");
             loadServices();
        }
    })
    .catch(err => {
        showToast("?????? ????????", "error");
    });
}


// ---- End Service Management Functions ----

/**
 * Loads VM information from API
 * Updates VM status, cloud-init status, and SSH service info.
 * @returns {void}
 */
function loadVMInfo() {
  const { user, vm } = getVMDetailsContext();

  if (!user || !vm) {
      console.error("loadVMInfo missing requirements:", {user, vm});
      return;
  }

  fetch(`/${user}/vm/${vm}/info`, {
    credentials: "same-origin",
  })
    .then((response) => {
      // Handle session expiration - redirect to login (reload can fail with Ingress Basic Auth)
      if (response.status === 401 || response.type === 'opaqueredirect') {
        console.warn("Session expired or unauthorized. Redirecting to login.");
        window.location.href = "/login";
        return null;
      }
      if (!response.ok) {
          throw new Error(`HTTP error! status: ${response.status}`);
      }
      return response.json();
    })
    .then((data) => {
      if (!data) return;
      if (data.success) {

        // Refresh services list periodically (or just on load)
        // Let's refresh only on explicit action or load for now to save requests,
        // OR add it here if we want real-time updates.
        // loadServices(); // Uncommenting this might be too heavy every 5s if list is long
```

```

        const statusEl = document.getElementById("vm-status");
        const statusBadge = document.getElementById("vm-status-badge");

        // Check for DataVolume status first (VM being created)
        let statusText, statusLower;

        if (data.datavolume_status) {
          statusLower = "creating";
          const dvStatus = data.datavolume_status.toLowerCase();
          const progressMatch = dvStatus.match(/(\d+)%/);
          if (progressMatch) {
            statusText = `???????? ?????... ${progressMatch[1]}%`;
          } else if (dvStatus === "??????????") {
            statusText = "??????????";
          } else {
            statusText = "???????? ?????...";
          }
        } else {
          // Normal VM status
          statusLower = data.status.toLowerCase().replace(/\s+/g, "-");

          // Map status to Russian text with comprehensive status coverage
          statusText = data.status;
          if (statusLower === "running") statusText = "????????";
          else if (statusLower === "stopped" || statusLower === "halted")
            statusText = "???????????";
          else if (statusLower === "starting") statusText = "???????????";
          else if (statusLower === "stopping") statusText = "????????????????";
          else if (statusLower === "provisioning" || statusLower === "waitingfordatavolume" || statusLower === "waiti...
            statusText = "???????? ??????";
          else if (statusLower === "paused") statusText = "??????????????";
          else if (statusLower === "migrating") statusText = "????????";
          else if (statusLower === "terminating") statusText = "????????";
          else if (statusLower === "crashloopbackoff") statusText = "?????? ???????";
        }

        if (statusEl) {
          statusEl.textContent = statusText;
          statusEl.className = "status-badge status-" + statusLower;
        }

        if (statusBadge) {
          statusBadge.textContent = statusText;
          statusBadge.className = "status-badge status-" + statusLower;
        }

        // Update action buttons based on status
        const startBtn = document.getElementById("start-btn");
        const stopBtn = document.getElementById("stop-btn");
        const rebootBtn = document.getElementById("reboot-btn");

        if (data.running) {
          if (startBtn) startBtn.classList.add("hidden");
          if (stopBtn) stopBtn.classList.remove("hidden");
          if (rebootBtn) rebootBtn.disabled = false;
        } else {
          if (startBtn) startBtn.classList.remove("hidden");
          if (stopBtn) stopBtn.classList.add("hidden");
          if (rebootBtn) rebootBtn.disabled = true;
        }

        // Update console status
        const consoleStatus = document.getElementById("console-status");
        if (consoleStatus) {
          if (data.running) {
            consoleStatus.classList.add("connected");
            consoleStatus.innerHTML =
              '<span class="status-dot"></span>??????????';
          } else {
            consoleStatus.classList.remove("connected");
```

```
            consoleStatus.innerHTML =
              '<span class="status-dot"></span>VM ??????????';
          }
        }

        document.getElementById("vm-cpu").textContent = data.cpu + " vCPU";
        document.getElementById("vm-memory").textContent = data.memory;
        document.getElementById("vm-storage").textContent = data.storage;

        // Update allocated IP
        const allocatedIpEl = document.getElementById("vm-allocated-ip");
        if (allocatedIpEl) {
          if (data.allocated_ip) {
            allocatedIpEl.textContent = data.allocated_ip;
          } else {
            allocatedIpEl.textContent = "-";
          }
        }

        // Show GPU info if available
        if (data.gpu_count && data.gpu_count > 0) {
          const gpuRow = document.getElementById("gpu-row");
          const gpuValue = document.getElementById("vm-gpu");
          const gpuModelName = data.gpu_model
            ? data.gpu_model.split("/").pop().toUpperCase()
            : "GPU";
          if (gpuValue)
            gpuValue.textContent = `${data.gpu_count}x ${gpuModelName}`;
          if (gpuRow) gpuRow.classList.remove("hidden");
        }

        // Update cloud-init status icon (????????? ?????? ??? Running VM)
        const cloudInitIcon = document.getElementById("cloudinit-icon");
        if (cloudInitIcon) {
          const status = data.cloudinit_status;

          if (status === "running") {
            // ???????? ???? - ???????????? ????
            cloudInitIcon.innerHTML = "?";
            cloudInitIcon.title = data.cloudinit_message || "????????????...";
            cloudInitIcon.classList.remove("hidden");
          } else if (status === "completed") {
            // ??????? ??????? - ??????
            cloudInitIcon.innerHTML = "?";
            cloudInitIcon.title = data.cloudinit_message || "?????? ? ??????";
            cloudInitIcon.classList.remove("hidden");
          } else {
            // ?? ????????? ?????? ???? ?????? unknown ??? null
            cloudInitIcon.classList.add("hidden");
          }
        }
      } else {
        console.error("Failed to load VM info:", data.error);
      }
    })
    .catch((error) => {
      console.error("Error loading VM info:", error);
    });
}

// Load VM info on page load and setup auto-refresh
document.addEventListener("DOMContentLoaded", () => {
  // Tab Switching
  document.querySelectorAll(".tab-btn").forEach((btn) => {
    btn.addEventListener("click", () => {
      const tabName = btn.dataset.tab;

      // Update buttons
      document
        .querySelectorAll(".tab-btn")
        .forEach((b) => b.classList.remove("active"));
```

```
      btn.classList.add("active");

      // Update content
      document
        .querySelectorAll(".tab-content")
        .forEach((c) => c.classList.remove("active"));
      document
        .querySelector(`[data-content="${tabName}"]`)
        ?.classList.add("active");
    });
  });

  // Hash-based navigation (e.g., #console from VNC button)
  const hash = window.location.hash.substring(1);
  if (hash && ["overview", "console", "metrics", "assistant"].includes(hash)) {
    const targetBtn = document.querySelector(`[data-tab="${hash}"]`);
    if (targetBtn) targetBtn.click();
  }

  loadVMInfo();
  loadServices(); // Initial load of services

  // Auto-refresh VM status every 5 seconds
  refreshInterval = setInterval(loadVMInfo, 5000);
});

// Cleanup interval when leaving page
window.addEventListener("beforeunload", () => {
  if (refreshInterval) {
    clearInterval(refreshInterval);
  }
  if (metricsInterval) {
    clearInterval(metricsInterval);
  }
});

// VM Control Functions
function createSSHService() {
  const btn = document.getElementById("ssh-service-btn");
  btn.disabled = true;
  btn.textContent = "????????...";

  const { user, vm } = getVMDetailsContext();
  fetch(`/${user}/vm/${vm}/ssh-service`, {
    method: "POST",
    credentials: "same-origin",
  })
    .then((response) => response.json())
    .then((data) => {
      if (data.nodePort) {
        if (window.showToast) {
          showToast("SSH ?????? ??????? ??????", "success");
        } else {
          alert("SSH ?????? ??????? ??????");
        }
        document.getElementById("ssh-service-btn").classList.add("hidden");
        const infoSpan = document.getElementById("ssh-service-info");
        infoSpan.classList.remove("hidden");

        if (data.sshCommand) {
          // Show full command if available (with public IP)
          document.getElementById("ssh-nodeport").innerHTML = `
            ${data.nodePort} <br>
            <span class="ssh-command-text">
              ${data.sshCommand}
            </span>
          `;
        } else {
          document.getElementById("ssh-nodeport").textContent = data.nodePort;
        }
      } else {
```

```
492         const errorMsg = data.error || "Unknown error";
493         if (window.showToast) {
494           showToast(`??????: ${errorMsg}`, "error");
495         } else {
496           alert(`??????: ${errorMsg}`);
497         }
498         btn.textContent = "??????? ???? SSH";
499         btn.disabled = false;
500       }
501
502 ... [603 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/admin_llm_spend.html (8.4 KB)

```
{% extends "base.html" %}
{% block title %}{{ spend_title or '???????' }}{% endblock %}
{% block nav_spend %}active{% endblock %}
{% block topbar_title %}{{ spend_title or '???????' }}{% endblock %}
{% block extra_css %}
<link rel="stylesheet" href="{{ url_for('static', path='css/admin-llm.css') }}" />
{% endblock %}

{% block content %}
<div class="admin-llm-page">
  <div class="admin-llm-header">
    <h1>{{ spend_title or '???????' }}</h1>
    <div class="admin-llm-controls">
      <label for="spend-days">??????:</label>
      <select id="spend-days">
        <option value="7">7 ????</option>
        <option value="30" selected>30 ????</option>
        <option value="90">90 ????</option>
        <option value="365">365 ????</option>
      </select>
      <button id="spend-refresh-btn" class="btn-secondary">????????</button>
    </div>
  </div>

  <div class="admin-llm-summary">
    <div class="admin-llm-summary-card">
      <div class="label">????? ???????</div>
      <div class="value mono" id="total-cost">{{ "{:.4f}".format(summary.total_cost or 0) }}</div>
    </div>
    <div class="admin-llm-summary-card">
      <div class="label">VM ???????</div>
      <div class="value mono" id="total-vm-cost">{{ "{:.4f}".format(summary.total_vm_cost or 0) }}</div>
    </div>
    <div class="admin-llm-summary-card">
      <div class="label">????? ??????</div>
      <div class="value" id="total-tokens">{{ "{:,}".format(summary.total_tokens or 0) }}</div>
    </div>
  </div>

  <div class="spend-charts-grid">
    <section class="spend-chart-card">
      <h3>????????? ???????? (pie)</h3>
      <div class="spend-donut-wrap">
        <div id="spend-source-donut" class="spend-donut"></div>
        <ul id="spend-source-legend" class="spend-legend"></ul>
      </div>
    </section>
  </div>

  <div class="admin-llm-section">
    <h3>?? ?????????????</h3>
    <div class="admin-llm-table-wrap">
      <table class="admin-llm-table">
        <thead>
          <tr><th>???????????</th><th class="num">LLM</th><th class="num">VM</th></tr>
        </thead>
        <tbody id="spend-per-user-body">
          {% for row in summary.per_user %}
          <tr>
            <td>{{ row.username }}</td>
            <td class="num">{{ "{:.4f}".format(row.total_cost) }}</td>
            <td class="num">{{ "{:.4f}".format(row.vm_total_cost or 0) }}</td>
          </tr>
          {% endfor %}
        </tbody>
```

```
      </table>
    </div>
  </div>

  <div class="admin-llm-section">
    <h3>?? ???????</h3>
    <div class="admin-llm-table-wrap">
      <table class="admin-llm-table">
        <thead>
          <tr><th>??????</th><th class="num">??????</th><th class="num">???????</th></tr>
        </thead>
        <tbody id="spend-per-model-body">
          {% for row in summary.per_model %}
          <tr><td>{{ row.model }}</td><td class="num">{{ "{:,}".format(row.total_tokens) }}</td><td class="num">{{ "{...
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>

</div>

<script>
(function() {
  const daysEl = document.getElementById("spend-days");
  const refreshBtn = document.getElementById("spend-refresh-btn");
  const totalCostEl = document.getElementById("total-cost");
  const totalVmCostEl = document.getElementById("total-vm-cost");
  const totalTokensEl = document.getElementById("total-tokens");
  const perUserBody = document.getElementById("spend-per-user-body");
  const perModelBody = document.getElementById("spend-per-model-body");
  const sourceDonut = document.getElementById("spend-source-donut");
  const sourceLegend = document.getElementById("spend-source-legend");

  function fmtNum(n) { return Number(n || 0).toLocaleString("ru-RU", { minimumFractionDigits: 0, maximumFractionDigit...
  function fmtCost(n) { return Number(n || 0).toFixed(4); }
  function escapeHtml(v) {
    return String(v == null ? "" : v)
      .replace(/&/g, "&amp;")
      .replace(/</g, "&lt;")
      .replace(/>/g, "&gt;")
      .replace(/"/g, "&quot;")
      .replace(/'/g, "&#39;");
  }

  function renderRows(container, rows, renderRow, colCount) {
    container.innerHTML = "";
    if (!rows.length) {
      container.innerHTML = `<tr class="empty-row"><td colspan="${colCount || 3}">??? ??????</td></tr>`;
      return;
    }
    container.innerHTML = rows.map(renderRow).join("");
  }

  function renderDonutAndLegend(items) {
    if (!sourceDonut || !sourceLegend) return;
    var total = items.reduce(function(sum, item) { return sum + Number(item.value || 0); }, 0);
    if (total <= 0) {
      sourceDonut.style.background = "conic-gradient(#e2e8f0 0deg 360deg)";
      sourceLegend.innerHTML = '<li class="spend-legend-empty">??? ??????</li>';
      return;
    }
    var angle = 0;
    var gradient = items.map(function(item) {
      var val = Number(item.value || 0);
      if (val <= 0) return null;
      var part = (val / total) * 360;
      var start = angle;
      angle += part;
      return item.color + " " + start.toFixed(1) + "deg " + angle.toFixed(1) + "deg";
    }).filter(Boolean);
```

```
  sourceDonut.style.background = "conic-gradient(" + gradient.join(", ") + ")";
  sourceLegend.innerHTML = items.map(function(item) {
    var val = Number(item.value || 0);
    var pct = total > 0 ? Math.round((val / total) * 100) : 0;
    return (
      '<li class="spend-legend-item">' +
        '<span class="spend-legend-label"><span class="spend-legend-dot" style="background:' + item.color + '"></sp...
        '<strong>' + fmtCost(val) + ' (' + pct + '%)</strong>' +
      '</li>'
    );
  }).join("");
}

function renderBars(container, rows, getLabel, getValue, valueFormatter) {
  if (!container) return;
  if (!rows.length) {
    container.innerHTML = '<div class="spend-bars-empty">??? ??????</div>';
    return;
  }
  var max = rows.reduce(function(m, row) { return Math.max(m, Number(getValue(row) || 0)); }, 0) || 1;
  container.innerHTML = rows.map(function(row) {
    var label = getLabel(row);
    var value = Number(getValue(row) || 0);
    var pct = Math.max(0, Math.min(100, (value / max) * 100));
    return (
      '<div class="spend-bar-row">' +
        '<span class="spend-bar-label">' + escapeHtml(label) + '</span>' +
        '<div class="spend-bar-track"><div class="spend-bar-fill" style="width:' + pct.toFixed(1) + '%"></div></div...
        '<span class="spend-bar-value">' + valueFormatter(value) + '</span>' +
      '</div>'
    );
  }).join("");
}

function renderCharts(data) {
  var llm = Number(data.total_cost || 0);
  var vm = Number(data.total_vm_cost || 0);
  renderDonutAndLegend([
    { label: "LLM", value: llm, color: "#3b82f6" },
    { label: "VM", value: vm, color: "#8b5cf6" }
  ]);
}

async function refresh() {
  refreshBtn.disabled = true;
  try {
    const days = Number(daysEl.value || "30");
    const apiUrl = `{{ spend_api_url or '/admin/api/llm-spend/summary' }}`;
    const r = await fetch(`${apiUrl}?days=${days}`, { credentials: "same-origin" });
    const data = await r.json().catch(() => ({}));
    if (!r.ok) throw new Error(data.detail || "?????? ????????");

    totalCostEl.textContent = fmtCost(data.total_cost);
    totalVmCostEl.textContent = data.total_vm_cost == null ? "?" : fmtCost(data.total_vm_cost);
    totalTokensEl.textContent = fmtNum(data.total_tokens);

    renderRows(
      perUserBody,
      data.per_user || [],
      (row) => `<tr><td>${row.username}</td><td class="num">${fmtCost(row.total_cost)}</td><td class="num">${row.vm...
      3
    );
    renderRows(perModelBody, data.per_model || [], (row) => `<tr><td>${row.model}</td><td class="num">${fmtNum(row....
    renderCharts(data);
  } catch (e) {
    if (window.showToast) window.showToast(e.message || "??????", "error");
  } finally {
    refreshBtn.disabled = false;
  }
}

```

```
  refreshBtn.addEventListener("click", refresh);
  renderCharts({{ summary | tojson }});
})();
</script>
{% endblock %}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/admin_users.html (29.8 KB)

```
{% extends "base.html" %}
{% block title %}??????????? ? ?????{% endblock %}
{% block nav_admin %}active{% endblock %}
{% block topbar_title %}?????????? ??????????????{% endblock %}

{% block content %}
<div class="page-header">
  <h1>???????????</h1>
  {% if is_admin and has_db %}
  <button class="btn-secondary" id="refresh-balances-btn" title="???????? ??????? ? ???????">???????? ???????</button>
  {% endif %}
  <button class="btn-primary" id="create-user-btn">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" width="18" height="18">
      <line x1="12" y1="5" x2="12" y2="19" /><line x1="5" y1="12" x2="19" y2="12" />
    </svg>
    ??????? ???????????
  </button>
</div>

<div class="admin-users-table-wrap">
  <table class="admin-users-table">
    <thead>
      <tr>
        <th>ID</th>
        <th>?????</th>
        <th>????</th>
        {% if is_admin %}
        <th>????????</th>
        <th>LLM</th>
        {% endif %}
        <th>??????</th>
        <th>????????</th>
      </tr>
    </thead>
    <tbody>
      {% for u in users %}
      <tr data-user-id="{{ u.id }}">
        <td>{{ u.id }}</td>
        <td>{{ u.username }}</td>
        <td>
          <select class="role-select" data-user-id="{{ u.id }}" data-username="{{ u.username }}">
            <option value="user" {% if u.role == 'user' %}selected{% endif %}>???????????</option>
            <option value="admin" {% if u.role == 'admin' %}selected{% endif %}>????????????</option>
          </select>
        </td>
        {% if is_admin %}
        <td class="balance-cell">{{ '{:.2f}'.format(u.main_balance) if u.main_balance is not none else '?' }}</td>
        <td class="balance-cell">{{ '{:.4f}'.format(u.llm_balance) if u.llm_balance is not none else '?' }}</td>
        {% endif %}
        <td>
          {% if u.is_active %}
          <span class="badge badge-success">???????</span>
          {% else %}
          <span class="badge badge-muted">????????</span>
          {% endif %}
        </td>
        <td>
          {% if is_admin and has_db %}
          <button class="btn-small btn-balance" data-user-id="{{ u.id }}" data-username="{{ u.username }}" title="???...
          {% endif %}
          <button class="btn-small btn-vms" data-user-id="{{ u.id }}" data-username="{{ u.username }}" title="VM ????...
          {% if u.is_active and u.role != 'admin' %}
          <button class="btn-small btn-danger deactivate-btn" data-user-id="{{ u.id }}" data-username="{{ u.username ...
          {% endif %}
        </td>
```

```
      </tr>
      {% endfor %}
    </tbody>
  </table>
</div>

<!-- Create User Modal -->
<div id="create-user-modal" class="modal-overlay" style="display:none">
  <div class="modal-card">
    <h2>??????? ????????????</h2>
    <form id="create-user-form">
      <div class="form-group">
        <label for="new-username">?????</label>
        <input type="text" id="new-username" name="username" pattern="[a-z0-9-]+" required minlength="3" placeholder=...
      </div>
      <div class="form-group">
        <label for="new-password">??????</label>
        <input type="password" id="new-password" name="password" required minlength="6" />
      </div>
      <div class="form-group">
        <label for="new-role">????</label>
        <select id="new-role" name="role">
          <option value="user">????????????</option>
          <option value="admin">?????????????</option>
        </select>
      </div>
      <div class="modal-actions">
        <button type="button" class="btn-secondary" id="cancel-create">??????</button>
        <button type="submit" class="btn-primary">???????</button>
      </div>
    </form>
    <div id="create-error" class="error-message" style="display:none; margin-top:8px"></div>
  </div>
</div>

{% if is_admin %}
<!-- Balance Modal -->
<div id="balance-modal" class="modal-overlay" style="display:none">
  <div class="modal-card" style="min-width: 400px;">
    <h2 id="balance-modal-title">??????</h2>
    <div id="balance-loading" class="vms-loading">????????...</div>
    <div id="balance-content" style="display:none">
      <div class="balance-summary" style="margin-bottom: 1rem;">
        <p><strong>????????:</strong> <span id="balance-main">?</span></p>
        <p><strong>LLM:</strong> <span id="balance-llm">?</span></p>
        <p><strong>????????? ?? VM:</strong> <span id="balance-spent-main">?</span></p>
        <p><strong>????????? ?? LLM:</strong> <span id="balance-spent-llm">?</span></p>
      </div>
      <div class="form-group">
        <label>????????? ????????</label>
        <div style="display: flex; gap: 8px;">
          <input type="number" id="topup-main-amount" step="0.01" min="0.01" placeholder="?????" style="flex:1" />
          <button type="button" class="btn-small btn-primary" id="topup-main-btn">?????????</button>
        </div>
      </div>
      <div class="form-group">
        <label>????????? LLM</label>
        <div style="display: flex; gap: 8px;">
          <input type="number" id="topup-llm-amount" step="0.01" min="0.01" placeholder="?????" style="flex:1" />
          <button type="button" class="btn-small btn-primary" id="topup-llm-btn">?????????</button>
        </div>
      </div>
      <hr style="margin: 1rem 0; border: none; border-top: 1px solid var(--border-color, #e2e8f0);" />
      <div class="form-group">
        <label>???????</label>
        <div style="display: flex; gap: 8px; flex-wrap: wrap; align-items: center;">
          <select id="transfer-direction">
            <option value="main-llm">???????? ? LLM</option>
            <option value="llm-main">LLM ? ????????</option>
          </select>
          <input type="number" id="transfer-amount" step="0.01" min="0.01" placeholder="?????" style="width: 100px;" />
```

```
          <label style="display: flex; align-items: center; gap: 4px; margin: 0;"><input type="checkbox" id="transfer...
          <button type="button" class="btn-small btn-primary" id="transfer-btn">??????????</button>
        </div>
      </div>
      <div id="balance-error" class="error-message" style="display:none; margin-top: 8px;"></div>
    </div>
    <div class="modal-actions" style="margin-top: 1rem;">
      <button type="button" class="btn-secondary" id="close-balance-modal">???????</button>
    </div>
  </div>
</div>
{% endif %}

<!-- User VMs Modal -->
<div id="user-vms-modal" class="modal-overlay" style="display:none">
  <div class="modal-card modal-vms" style="min-width: 520px; max-width: 90vw;">
    <h2 id="user-vms-title">VM ???????????</h2>
    <div id="user-vms-loading" class="vms-loading">????????...</div>
    <div id="user-vms-list" class="vms-list" style="display:none"></div>
    <div id="user-vms-empty" class="vms-empty" style="display:none">??? ?????????? ?????</div>
    <div class="modal-actions" style="margin-top: 1rem;">
      <button type="button" class="btn-secondary" id="close-vms-modal">???????</button>
    </div>
  </div>
</div>

<style>
.admin-users-table-wrap { overflow-x: auto; margin-top: 1rem; }
.admin-users-table { width: 100%; border-collapse: collapse; }
.admin-users-table th, .admin-users-table td { padding: 10px 12px; text-align: left; border-bottom: 1px solid var(--b...
.admin-users-table th { background: var(--bg-muted, #f8fafc); font-weight: 600; }
.role-select { padding: 4px 8px; border-radius: 4px; border: 1px solid var(--border-color); }
.btn-small { padding: 4px 10px; font-size: 0.85rem; margin-right: 4px; }
.btn-balance { background: #059669; color: white; border: none; }
.btn-balance:hover { background: #047857; }
.btn-vms { background: #3b82f6; color: white; border: none; }
.btn-vms:hover { background: #2563eb; }
.btn-danger { background: #dc2626; color: white; border: none; }
.btn-danger:hover { background: #b91c1c; }
.btn-stop { background: #f59e0b; color: white; border: none; }
.btn-stop:hover { background: #d97706; }
.vms-loading, .vms-empty { padding: 1rem; color: var(--text-muted, #64748b); }
.vms-list { max-height: 360px; overflow-y: auto; }
.vms-list table { width: 100%; border-collapse: collapse; }
.vms-list th, .vms-list td { padding: 8px 10px; text-align: left; border-bottom: 1px solid var(--border-color, #e2e8f...
.vms-list th { font-size: 0.85rem; color: var(--text-muted, #64748b); }
.vm-row-actions { display: flex; gap: 6px; }
.vm-name-link { color: #3b82f6; text-decoration: none; }
.vm-name-link:hover { text-decoration: underline; }
.badge-success { background: #22c55e; color: white; padding: 2px 8px; border-radius: 4px; font-size: 0.8rem; }
.badge-muted { background: #94a3b8; color: white; padding: 2px 8px; border-radius: 4px; font-size: 0.8rem; }
.badge-demo { background: #8b5cf6; color: white; padding: 2px 8px; border-radius: 4px; font-size: 0.75rem; margin-lef...
.vm-row-stub { background: rgba(139, 92, 246, 0.08); }
.modal-overlay { position: fixed; inset: 0; background: rgba(0,0,0,0.5); display: flex; align-items: center; justify-...
.modal-card { background: var(--bg-primary, #fff); padding: 1.5rem; border-radius: 8px; min-width: 320px; }
.form-group { margin-bottom: 1rem; }
.form-group label { display: block; margin-bottom: 4px; font-weight: 500; }
.form-group input, .form-group select { width: 100%; padding: 8px; border-radius: 4px; border: 1px solid var(--border...
.modal-actions { display: flex; gap: 8px; justify-content: flex-end; margin-top: 1rem; }
</style>

<script>
(function() {
  // #region agent log
  function __agentLog(hypothesisId, location, message, data) {
    fetch("http://localhost:7242/ingest/5d6e8a84-0852-4c51-9dde-f3a26bfdf00d", {
      method: "POST",
      headers: {
        "Content-Type": "application/json",
        "X-Debug-Session-Id": "54c5fc",
      },
```

```
      body: JSON.stringify({
        sessionId: "54c5fc",
        runId: "run2",
        hypothesisId: hypothesisId,
        location: location,
        message: message,
        data: data,
        timestamp: Date.now(),
      }),
    }).catch(function() {});
  }
  // #endregion

  function fmtBalance(v, decimals) {
    if (v == null) return '?';
    var d = decimals != null ? decimals : 2;
    return parseFloat(v).toFixed(d);
  }
  function fmtBalanceLlm(v) { return fmtBalance(v, 4); }

  const refreshBalancesBtn = document.getElementById('refresh-balances-btn');
  if (refreshBalancesBtn) {
    refreshBalancesBtn.addEventListener('click', async function() {
      this.disabled = true;
      try {
        const r = await fetch('/admin/api/users', { credentials: 'same-origin', cache: 'no-store' });
        const data = await r.json().catch(function() { return {}; });
        if (!r.ok) throw new Error(formatApiError(data.detail) || '?????? ????????');
        const users = data.users || [];
        users.forEach(function(u) {
          const row = document.querySelector('tr[data-user-id="' + String(u.id) + '"]');
          if (!row) return;
          const cells = row.querySelectorAll('.balance-cell');
          if (cells.length >= 2) {
            cells[0].textContent = fmtBalance(u.main_balance);
            cells[1].textContent = fmtBalanceLlm(u.llm_balance);
          }
        });
        if (window.showToast) window.showToast('??????? ????????? ? ???????', 'success');
      } catch (err) {
        if (window.showToast) window.showToast(err.message || '??????', 'error');
      } finally {
        this.disabled = false;
      }
    });
  }

  const createBtn = document.getElementById('create-user-btn');
  const modal = document.getElementById('create-user-modal');
  const form = document.getElementById('create-user-form');
  const cancelBtn = document.getElementById('cancel-create');
  const errorEl = document.getElementById('create-error');

  function showModal() { modal.style.display = 'flex'; }
  function hideModal() { modal.style.display = 'none'; errorEl.style.display = 'none'; }

  function formatApiError(detail) {
    if (!detail) return null;
    if (typeof detail === 'string') return detail;
    if (Array.isArray(detail) && detail.length > 0) {
      var msgs = {
        'string_too_short': '??????? ???????? ????????',
        'string_too_long': '??????? ??????? ????????',
        'string_pattern_mismatch': '?????: ?????? ???????? ????? (a-z), ????? (0-9) ? ????? (-)',
        'value_error': '??????????? ????????'
      };
      return detail.map(function(e) {
        var loc = (e.loc || []).slice(1).join('.');
        var msg = e.msg || e.message || '';
        if (e.type && msgs[e.type]) msg = msgs[e.type];
        else if (typeof msg === 'string' && msg.indexOf('pattern') !== -1) msg = msgs.string_pattern_mismatch || msg;
```

```
      return (loc ? loc + ': ' : '') + (typeof msg === 'string' && msg ? msg : '?????? ?????????');
    }).join('; ');
  }
  if (detail && typeof detail === 'object' && (detail.msg || detail.message)) return detail.msg || detail.message;
  return String(detail);
}

createBtn?.addEventListener('click', showModal);
cancelBtn?.addEventListener('click', hideModal);
modal?.addEventListener('click', function(e) { if (e.target === modal) hideModal(); });

form?.addEventListener('submit', async function(e) {
  e.preventDefault();
  const username = document.getElementById('new-username').value.trim();
  const password = document.getElementById('new-password').value;
  const role = document.getElementById('new-role').value;
  errorEl.style.display = 'none';
  try {
    const r = await fetch('/admin/api/users', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ username, password, role }),
    });
    const data = await r.json().catch(() => ({}));
    if (!r.ok) {
      errorEl.textContent = formatApiError(data.detail) || data.message || '?????? ????????';
      errorEl.style.display = 'block';
      return;
    }
    location.reload();
  } catch (err) {
    errorEl.textContent = '?????? ????';
    errorEl.style.display = 'block';
  }
});

document.querySelectorAll('.role-select').forEach(function(sel) {
  sel.addEventListener('change', async function() {
    const userId = this.dataset.userId;
    const role = this.value;
    try {
      const r = await fetch('/admin/api/users/' + userId + '/role', {
        method: 'PATCH',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ role }),
      });
      const roleData = await r.json();
      if (!r.ok) throw new Error(formatApiError(roleData.detail) || '??????');
      if (window.showToast) window.showToast('???? ?????????', 'success');
    } catch (err) {
      if (window.showToast) window.showToast(err.message || '??????', 'error');
      location.reload();
    }
  });
});

document.querySelectorAll('.deactivate-btn').forEach(function(btn) {
  btn.addEventListener('click', async function() {
    if (!confirm('????????? ??????????? ' + this.dataset.username + '?')) return;
    const userId = this.dataset.userId;
    try {
      const r = await fetch('/admin/api/users/' + userId + '/deactivate', { method: 'POST' });
      const deactData = await r.json();
      if (!r.ok) throw new Error(formatApiError(deactData.detail) || '??????');
      location.reload();
    } catch (err) {
      if (window.showToast) window.showToast(err.message || '??????', 'error');
    }
  });
});

```

```
// User VMs modal
const vmsModal = document.getElementById('user-vms-modal');
const vmsTitle = document.getElementById('user-vms-title');
const vmsLoading = document.getElementById('user-vms-loading');
const vmsList = document.getElementById('user-vms-list');
const vmsEmpty = document.getElementById('user-vms-empty');
const closeVmsBtn = document.getElementById('close-vms-modal');

function closeVmsModal() { vmsModal.style.display = 'none'; }
closeVmsBtn?.addEventListener('click', closeVmsModal);
vmsModal?.addEventListener('click', function(e) { if (e.target === vmsModal) closeVmsModal(); });

document.querySelectorAll('.btn-vms').forEach(function(btn) {
  btn.addEventListener('click', async function() {
    const userId = this.dataset.userId;
    const username = this.dataset.username;
    vmsTitle.textContent = 'VM ??????????? ' + username;
    vmsModal.style.display = 'flex';
    vmsLoading.style.display = 'block';
    vmsList.style.display = 'none';
    vmsEmpty.style.display = 'none';

    try {
      const r = await fetch('/admin/api/users/' + userId + '/vms');
      let data;
      try {
        data = await r.json();
      } catch (parseErr) {
        vmsLoading.style.display = 'none';
        throw new Error(r.status >= 500 ? '?????? ???????' : '?????? ????????');
      }
      vmsLoading.style.display = 'none';

      if (!r.ok) throw new Error(formatApiError(data.detail) || '?????? ????????');

      const vms = data.vms || [];
      const stubMode = data.stub_mode === true;
      if (vms.length === 0) {
        vmsEmpty.style.display = 'block';
        return;
      }

      let html = '<table><thead><tr><th>VM</th><th>??????</th><th>CPU</th><th>??????</th><th></th></tr></thead><tbo...
      vms.forEach(function(vm) {
        const statusCls = vm.running ? 'badge-success' : 'badge-muted';
        const statusTxt = vm.running ? '????????' : (vm.status || '?');
        const vmName = vm.name || '?';
        const vmLink = vm.name
          ? '<a href="/' + encodeURIComponent(username) + '/vm/' + encodeURIComponent(vm.name) + '" class="vm-name-...
          : vmName;
        const rowClass = stubMode ? ' class="vm-row-stub"' : '';
        const stubBadge = stubMode ? ' <span class="badge badge-demo">????</span>' : '';
        html += '<tr data-vm="' + (vm.name || '') + '"' + rowClass + '>';
        html += '<td><strong>' + vmLink + stubBadge + '</strong></td>';
        html += '<td><span class="badge ' + statusCls + '">' + statusTxt + '</span></td>';
        html += '<td>' + (vm.cpu || '?') + '</td>';
        html += '<td>' + (vm.memory || '?') + '</td>';
        html += '<td class="vm-row-actions">';
        if (vm.running) {
          html += '<button class="btn-small btn-stop vm-stop-btn" data-user-id="' + userId + '" data-vm="' + (vm.na...
        }
        html += '<button class="btn-small btn-danger vm-delete-btn" data-user-id="' + userId + '" data-vm="' + (vm....
        html += '</td></tr>';
      });
      html += '</tbody></table>';
      vmsList.innerHTML = html;
      vmsList.style.display = 'block';

      vmsList.querySelectorAll('.vm-stop-btn').forEach(function(b) {
        b.addEventListener('click', async function() {
          const uid = this.dataset.userId, vn = this.dataset.vm;
```

```
            if (!confirm('????????? VM ' + vn + '?')) return;
            try {
              const rr = await fetch('/admin/api/users/' + uid + '/vm/' + vn + '/stop', { method: 'POST' });
              const dd = await rr.json();
              if (!rr.ok) throw new Error(formatApiError(dd.detail) || '??????');
              if (window.showToast) window.showToast('VM ??????????????', 'success');
              this.closest('tr').querySelector('.badge')?.classList.replace('badge-success', 'badge-muted');
              this.closest('tr').querySelector('.badge').textContent = '???????????????';
              this.remove();
            } catch (e) {
              if (window.showToast) window.showToast(e.message || '??????', 'error');
            }
          });
        });

        vmsList.querySelectorAll('.vm-delete-btn').forEach(function(b) {
          b.addEventListener('click', async function() {
            const uid = this.dataset.userId, vn = this.dataset.vm;
            if (!confirm('??????? VM ' + vn + '? ???? ????? ?????? ???????????.')) return;
            try {
              const rr = await fetch('/admin/api/users/' + uid + '/vm/' + vn + '/delete', { method: 'POST' });
              const dd = await rr.json();
              if (!rr.ok) throw new Error(formatApiError(dd.detail) || '??????');
              if (window.showToast) window.showToast('VM ???????', 'success');
              this.closest('tr').remove();
              if (vmsList.querySelectorAll('tr').length === 0) {
                vmsList.style.display = 'none';
                vmsEmpty.style.display = 'block';
              }
            } catch (e) {
              if (window.showToast) window.showToast(e.message || '??????', 'error');
            }
          });
        });
      } catch (err) {
        vmsLoading.style.display = 'none';
        vmsEmpty.textContent = err.message || '?????? ????????';
        vmsEmpty.style.display = 'block';
      }
    });
  });

  // Balance modal
  {% if has_db %}
  const balanceModal = document.getElementById('balance-modal');
  const balanceTitle = document.getElementById('balance-modal-title');
  const balanceLoading = document.getElementById('balance-loading');
  const balanceContent = document.getElementById('balance-content');
  const balanceMainEl = document.getElementById('balance-main');
  const balanceLlmEl = document.getElementById('balance-llm');
  const balanceSpentMainEl = document.getElementById('balance-spent-main');
  const balanceSpentLlmEl = document.getElementById('balance-spent-llm');
  const balanceError = document.getElementById('balance-error');
  const closeBalanceBtn = document.getElementById('close-balance-modal');

  let currentBalanceUserId = null;

  function showBalanceError(msg) {
    balanceError.textContent = msg || '';
    balanceError.style.display = msg ? 'block' : 'none';
  }

  function updateBalanceDisplay(data) {
    balanceMainEl.textContent = fmtBalance(data.main_balance);
    balanceLlmEl.textContent = fmtBalanceLlm(data.llm_balance);

    var hasMainTotals = data.main_total_administered != null && data.main_balance != null;
    var hasLlmTotals = data.llm_total_administered != null && data.llm_balance != null;

    if (balanceSpentMainEl) {
      balanceSpentMainEl.textContent = hasMainTotals
```

```
        ? fmtBalance(Number(data.main_total_administered) - Number(data.main_balance))
        : '?';
    }
    if (balanceSpentLlmEl) {
      balanceSpentLlmEl.textContent = hasLlmTotals
        ? fmtBalanceLlm(Number(data.llm_total_administered) - Number(data.llm_balance))
        : '?';
    }
  }

... [147 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/base.html (20.3 KB)

```
<!doctype html>
<html lang="ru" data-theme="light">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>{% block title %}????????{% endblock %}</title>
    <link
      rel="icon"
      type="image/png"
      href="{{ url_for('static', path='images/favicon.png') }}"
    />
    <link rel="preconnect" href="https://fonts.googleapis.com" />
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
    <link
      href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap"
      rel="stylesheet"
    />
    <link
      rel="stylesheet"
      href="{{ url_for('static', path='css/dashboard-layout.css') }}"
    />
    <link
      rel="stylesheet"
      href="{{ url_for('static', path='css/utilities.css') }}"
    />
    {% block extra_css %}{% endblock %}
    <script src="{{ url_for('static', path='js/core/theme.js') }}"></script>
  </head>
  <body data-username="{{ username }}" data-page="{{ page|default('') }}">
    <div class="dashboard-layout">
      <!-- Sidebar -->
      <aside class="sidebar" id="sidebar">
        <div class="sidebar-header">
          <a href="/{{ username }}/dashboard" class="logo-link">
            <img
              src="{{ url_for('static', path='images/viiruoz.png') }}"
              alt="????????"
              class="logo-img"
            />
          </a>
        </div>

        <nav class="sidebar-nav">
          <ul class="nav-list">
            <li class="nav-item">
              <a
                href="/{{ username }}/dashboard"
                class="nav-link {% block nav_dashboard %}{% endblock %}"
              >
                <svg
                  class="nav-icon"
                  viewBox="0 0 24 24"
                  fill="none"
                  stroke="currentColor"
                  stroke-width="2"
                >
                  <rect x="3" y="3" width="7" height="7" />
                  <rect x="14" y="3" width="7" height="7" />
                  <rect x="14" y="14" width="7" height="7" />
                  <rect x="3" y="14" width="7" height="7" />
                </svg>
                <span>?????? ??????????</span>
              </a>
            </li>
            <li class="nav-item">
```

```
            <a
              href="/{{ username }}/vms"
              class="nav-link {% block nav_vms %}{% endblock %}"
            >
              <svg
                class="nav-icon"
                viewBox="0 0 24 24"
                fill="none"
                stroke="currentColor"
                stroke-width="2"
              >
                <rect x="2" y="3" width="20" height="14" rx="2" />
                <path d="M8 21h8" />
                <path d="M12 17v4" />
              </svg>
              <span>??????????? ??????</span>
            </a>
          </li>
          <li class="nav-item nav-item-llm {% if page == 'llm' %}is-active{% endif %}">
            <button
              type="button"
              class="nav-link nav-link-accordion {% block nav_llm %}{% endblock %}"
              id="llm-nav-trigger"
              aria-expanded="{% if page == 'llm' %}true{% else %}false{% endif %}"
              aria-controls="llm-submenu"
            >
              <svg
                class="nav-icon"
                viewBox="0 0 24 24"
                fill="none"
                stroke="currentColor"
                stroke-width="2"
              >
                <path
                  d="M21 16V8a2 2 0 0 0-1-1.73l-7-4a2 2 0 0 0-2 0l-7 4A2 2 0 0 0 3 8v8a2 2 0 0 0 1 1.73l7 4a2 2 0 0...
                />
                <polyline points="7.5 4.21 12 6.81 16.5 4.21" />
                <polyline points="7.5 19.79 7.5 14.6 3 12" />
                <polyline points="21 12 16.5 14.6 16.5 19.79" />
                <polyline points="3.27 6.96 12 12.01 20.73 6.96" />
                <line x1="12" y1="22.08" x2="12" y2="12" />
              </svg>
              <span>LLM-????????</span>
              <svg class="nav-chevron" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                <path d="M9 6l6 6-6 6" />
              </svg>
            </button>
            <ul class="nav-submenu {% if page == 'llm' %}submenu-open{% endif %}" id="llm-submenu">
              <li>
                <a href="/{{ username }}/llm" class="nav-sublink">???</a>
              </li>
              <li>
                <a href="/{{ username }}/llm/models" class="nav-sublink">??????</a>
              </li>
              <li>
                <a href="/{{ username }}/llm/keys" class="nav-sublink">API ?????</a>
              </li>
            </ul>
          </li>
          {% if is_admin|default(false) %}
          <li class="nav-item">
            <a href="/admin/users" class="nav-link {% block nav_admin %}{% endblock %}">
              <svg class="nav-icon" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                <path d="M16 21v-2a4 4 0 0 0-4-4H6a4 4 0 0 0-4 4v2" />
                <circle cx="9" cy="7" r="4" />
                <path d="M22 21v-2a4 4 0 0 0-3-3.87" />
                <path d="M16 3.13a4 4 0 0 1 0 7.75" />
              </svg>
              <span>????????????</span>
            </a>
          </li>
```

```
            {% endif %}
            <li class="nav-item">
              <a
                href="{{ '/admin/llm-spend' if is_admin|default(false) else '/' + username + '/spend' }}"
                class="nav-link {% block nav_spend %}{% endblock %}"
              >
                <svg class="nav-icon" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                  <path d="M3 3v18h18" />
                  <path d="M7 14l4-4 3 3 5-6" />
                </svg>
                <span>???????</span>
              </a>
            </li>
          </ul>
        </nav>

        <div class="sidebar-footer">
          <div class="user-info-sidebar">
            <div class="user-avatar">
              {{ username[:2]|upper if username else 'U' }}
            </div>
            <div class="user-details">
              <span class="user-name-sidebar">{{ username }}</span>
              <span class="user-namespace">{{ namespace }}</span>
            </div>
          </div>
          <button class="sidebar-collapse-toggle" id="sidebar-collapse-toggle" type="button" aria-label="???????? ???...
            <svg class="icon-collapse-left" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
              <path d="M15 18l-6-6 6-6" />
            </svg>
            <svg class="icon-collapse-right" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
              <path d="M9 18l6-6-6-6" />
            </svg>
          </button>
        </div>
      </aside>

      <!-- Main Content Area -->
      <div class="main-wrapper">
        <!-- Top Header -->
        <header class="topbar">
          <button
            class="menu-toggle"
            id="menu-toggle"
            aria-label="??????? ????"
          >
            <svg
              viewBox="0 0 24 24"
              fill="none"
              stroke="currentColor"
              stroke-width="2"
            >
              <line x1="3" y1="6" x2="21" y2="6" />
              <line x1="3" y1="12" x2="21" y2="12" />
              <line x1="3" y1="18" x2="21" y2="18" />
            </svg>
          </button>

          <div class="topbar-title">
            {% block topbar_title %}?????? ??????????{% endblock %}
          </div>

          <div class="topbar-actions">
            <div class="topbar-balance topbar-balance-pill">
              <span class="topbar-balance-label">???????? ??????:</span>
              <span class="topbar-balance-value" id="topbar-main-balance-value">?</span>
            </div>
            <div class="topbar-balance topbar-balance-pill">
              <span class="topbar-balance-label">???????? ??????:</span>
              <span class="topbar-balance-value" id="topbar-llm-balance-value">?</span>
            </div>
```

```
            {% if page == 'spend' %}
            <button class="btn-secondary btn-balance-transfer-trigger" id="open-balance-transfer" type="button">
              ???????
            </button>
            <div id="balance-transfer-panel" class="balance-transfer-panel" hidden>
              <div class="balance-transfer-header">
                <h3 id="balance-transfer-title">??????? ???????</h3>
                <button id="balance-transfer-close" class="balance-transfer-close-btn" type="button" aria-label="????...
              </div>
              <div class="balance-transfer-summary">
                <p><strong>????????:</strong> <span id="balance-transfer-main">?</span></p>
                <p><strong>LLM:</strong> <span id="balance-transfer-llm">?</span></p>
              </div>
              <div class="balance-transfer-form">
                <select id="balance-transfer-direction">
                  <option value="main-llm">???????? ? LLM</option>
                  <option value="llm-main">LLM ? ????????</option>
                </select>
                <input type="number" id="balance-transfer-amount" step="0.01" min="0.01" placeholder="?????" />
                <label class="balance-transfer-all-label"><input type="checkbox" id="balance-transfer-all" /> ???</la...
                <button id="balance-transfer-submit" class="btn-primary" type="button">?????????</button>
              </div>
              <div id="balance-transfer-error" class="balance-transfer-error" hidden></div>
            </div>
            {% endif %}
            <div class="user-chip" id="user-chip">
              <div class="user-avatar-small" id="user-avatar">
                {{ username[:2]|upper if username else 'U' }}
              </div>
              <span class="user-email">{{ username }}</span>
              <svg
                class="dropdown-icon"
                viewBox="0 0 24 24"
                fill="none"
                stroke="currentColor"
                stroke-width="2"
              >
                <path d="M6 9l6 6 6-6" />
              </svg>
            </div>
            <div class="user-dropdown" id="user-dropdown">
              <div class="dropdown-header">
                <strong id="user-name" data-username="{{ username }}"
                  >{{ username }}</strong
                >
                <span id="project-name">{{ namespace }}</span>
              </div>
              <div class="dropdown-divider"></div>
              <button id="logout-btn" class="dropdown-item logout-item">
                <svg
                  viewBox="0 0 24 24"
                  fill="none"
                  stroke="currentColor"
                  stroke-width="2"
                >
                  <path d="M9 21H5a2 2 0 0 1-2-2V5a2 2 0 0 1 2-2h4" />
                  <polyline points="16 17 21 12 16 7" />
                  <line x1="21" y1="12" x2="9" y2="12" />
                </svg>
                ?????
              </button>
            </div>
          </div>
        </header>

        <!-- Page Content -->
        <main class="main-content">{% block content %}{% endblock %}</main>
      </div>
    </div>

    <!-- Toast Notification Container -->
```

```
    <div id="toast-container" class="toast-container"></div>
    <!-- Core Modules (loaded globally) -->
    <script src="{{ url_for('static', path='js/core/app-state.js') }}"></script>
    <script src="{{ url_for('static', path='js/core/windows-logic.js') }}"></script>
    <script src="{{ url_for('static', path='js/core/ui-components.js') }}?v=20260303"></script>
    <script src="{{ url_for('static', path='js/core/vm-actions.js') }}"></script>
    <script src="{{ url_for('static', path='js/core/utils.js') }}"></script>
    <script>
    (function() {
      var appState = window.AppState;
      if (!appState) return;
      var fallbackUsername = document.body ? (document.body.dataset.username || '') : '';
      var fallbackPage = document.body ? (document.body.dataset.page || '') : '';
      var namespaceEl = document.getElementById('project-name');
      var mainBalanceValueEl = document.getElementById('topbar-main-balance-value');
      var llmBalanceValueEl = document.getElementById('topbar-llm-balance-value');
      var transferPanel = document.getElementById('balance-transfer-panel');
      var transferOpenBtn = document.getElementById('open-balance-transfer');
      var transferCloseBtn = document.getElementById('balance-transfer-close');
      var transferMainEl = document.getElementById('balance-transfer-main');
      var transferLlmEl = document.getElementById('balance-transfer-llm');
      var transferDirectionEl = document.getElementById('balance-transfer-direction');
      var transferAmountEl = document.getElementById('balance-transfer-amount');
      var transferAllEl = document.getElementById('balance-transfer-all');
      var transferSubmitEl = document.getElementById('balance-transfer-submit');
      var transferErrorEl = document.getElementById('balance-transfer-error');

      appState.setState({
        profile: {
          username: fallbackUsername,
          namespace: namespaceEl ? (namespaceEl.textContent || '').trim() : '',
        },
      }, 'baseProfileInit');

      if (!fallbackUsername || !mainBalanceValueEl || !llmBalanceValueEl) return;

      var isLoading = false;
      var balanceRefreshInterval = null;

      function renderBalances(stateSnapshot) {
        if (!stateSnapshot || !stateSnapshot.balances) return;
        mainBalanceValueEl.textContent = appState.formatMoney(stateSnapshot.balances.mainBalance);
        llmBalanceValueEl.textContent = appState.formatMoney(stateSnapshot.balances.llmBalance);
      }

      function parseApiError(detail) {
        if (!detail) return null;
        if (typeof detail === 'string') return detail;
        if (Array.isArray(detail) && detail.length > 0) {
          return detail.map(function(e) {
            if (e && (e.msg || e.message)) return e.msg || e.message;
            return '?????? ?????????';
          }).join('; ');
        }
        if (typeof detail === 'object' && (detail.msg || detail.message)) return detail.msg || detail.message;
        return String(detail);
      }

      function setTransferError(msg) {
        if (!transferErrorEl) return;
        if (!msg) {
          transferErrorEl.hidden = true;
          transferErrorEl.textContent = '';
          return;
        }
        transferErrorEl.hidden = false;
        transferErrorEl.textContent = msg;
      }

      function renderTransferBalances(stateSnapshot) {
        if (!transferMainEl || !transferLlmEl) return;
```

```
    if (!stateSnapshot || !stateSnapshot.balances) return;
    transferMainEl.textContent = appState.formatMoney(stateSnapshot.balances.mainBalance);
    transferLlmEl.textContent = appState.formatMoney(stateSnapshot.balances.llmBalance);
  }

  function refreshTransferBalances() {
    return appState.loadBalances(username)
      .then(function(data) {
        if (!data) return null;
        var snapshot = appState.getState();
        renderTransferBalances(snapshot);
        renderBalances(snapshot);
        return snapshot;
      })
      .catch(function() {
        setTransferError('?? ??????? ????????? ??????');
        return null;
      });
  }

  function openTransferModal() {
    if (!transferPanel) return;
    transferPanel.hidden = false;
    setTransferError(null);
    refreshTransferBalances();
  }

  function closeTransferModal() {
    if (!transferPanel) return;
    transferPanel.hidden = true;
    setTransferError(null);
  }

  function refreshBalance() {
    if (isLoading) return;
    isLoading = true;
    appState.loadBalances(fallbackUsername)
      .then(function() {
        renderBalances(appState.getState());
      })
      .catch(function() {
        mainBalanceValueEl.textContent = '?';
        llmBalanceValueEl.textContent = '?';
      })
      .finally(function() { isLoading = false; });
  }

  if (transferOpenBtn && transferPanel) {
    transferOpenBtn.addEventListener('click', function() {
      openTransferModal();
    });
  }

  if (transferCloseBtn) {
    transferCloseBtn.addEventListener('click', function() {
      closeTransferModal();
    });
  }

  document.addEventListener('click', function(e) {
    if (!transferPanel || transferPanel.hidden) return;
    var clickedInsidePanel = transferPanel.contains(e.target);
    var clickedBalance = box && box.contains(e.target);
    var clickedTrigger = transferOpenBtn && transferOpenBtn.contains(e.target);
    if (!clickedInsidePanel && !clickedBalance && !clickedTrigger) {
      closeTransferModal();
    }
  });

  document.addEventListener('keydown', function(e) {
    if (e.key === 'Escape') closeTransferModal();
```

```
421       });
422
423       if (transferAllEl && transferAmountEl) {
424         transferAllEl.addEventListener('change', function() {
425           transferAmountEl.disabled = transferAllEl.checked;
426           if (transferAllEl.checked) transferAmountEl.value = '';
427         });
428       }
429
430       if (transferSubmitEl && transferDirectionEl && transferAmountEl && transferAllEl) {
431         transferSubmitEl.addEventListener('click', function() {
432           var transferAll = transferAllEl.checked;
433           var amount = transferAll ? null : parseFloat(transferAmountEl.value);
434           if (!transferAll && (!amount || amount <= 0)) {
435             setTransferError('??????? ????? ??? ???????? «???»');
436             return;
437           }
438
439           var direction = transferDirectionEl.value;
440           var source = direction === 'main-llm' ? 'main' : 'llm';
441           var target = direction === 'main-llm' ? 'llm' : 'main';
442           var body = { source: source, target: target };
443           if (transferAll) body.transfer_all = true; else body.amount = amount;
444
445           setTransferError(null);
446           transferSubmitEl.disabled = true;
447           fetch('/' + encodeURIComponent(fallbackUsername) + '/api/balance/transfer', {
448             method: 'POST',
449             headers: { 'Content-Type': 'application/json' },
450             credentials: 'same-origin',
451             body: JSON.stringify(body),
452           })
453             .then(function(r) {
454               return r.json().catch(function() { return {}; }).then(function(data) {
455                 if (!r.ok) throw new Error(parseApiError(data.detail) || '?????? ????????');
456                 return data;
457               });
458             })
459             .then(function(data) {
460               transferAmountEl.value = '';
461               transferAllEl.checked = false;
462               transferAmountEl.disabled = false;
463               appState.setBalances(data, 'transferSuccess');
464               var snapshot = appState.getState();
465               renderTransferBalances(snapshot);
466               renderBalances(snapshot);
467               if (window.showToast) window.showToast('??????? ????????', 'success');
468             })
469             .catch(function(err) {
470               setTransferError(err.message || '?????? ????????');
471             })
472             .finally(function() {
473               transferSubmitEl.disabled = false;
474             });
475         });
476       }
477
478       renderBalances(appState.getState());
479       refreshBalance();
480       appState.subscribe(function(nextState, source) {
481         if (!source) return;
482         renderBalances(nextState);
483         renderTransferBalances(nextState);
484       });
485       balanceRefreshInterval = setInterval(refreshBalance, 5000);
486
487       document.addEventListener('visibilitychange', function() {
488         if (!document.hidden && (fallbackPage === 'dashboard' || fallbackPage === 'llm' || fallbackPage === 'spend')) {
489           refreshBalance();
490         }
491       });
```

```

      window.addEventListener('beforeunload', function() {
        if (balanceRefreshInterval) clearInterval(balanceRefreshInterval);
      });
    })();
    </script>
    {% block extra_js %}{% endblock %}
  </body>
</html>

... [1 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/dashboard.html (28.5 KB)

```
{% extends "base.html" %} {% block title %}??????????? ?????? ? ????????{%
endblock %} {% block nav_vms %}active{% endblock %} {% block topbar_title
%}??????????? ??????{% endblock %} {% block extra_css %}
<link
  rel="stylesheet"
  href="{{ url_for('static', path='css/create-vm.css') }}"
/>
<link
  rel="stylesheet"
  href="{{ url_for('static', path='css/storage.css') }}"
/>
{% endblock %} {% block content %}
<!-- Page Header -->
<div class="page-header" id="dashboard-page-header">
  <div class="page-header-text">
    <h1>??????????? ??????</h1>
    <p>????????????? ??????????????? ???????? ? ?????? ????????</p>
  </div>
  <button class="btn-primary" id="open-create-vm">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <line x1="12" y1="5" x2="12" y2="19" />
      <line x1="5" y1="12" x2="19" y2="12" />
    </svg>
    ??????? ??????????? ??????
  </button>
</div>

<!-- Sub-navigation Tabs -->
<div class="sub-nav-tabs">
  <button class="sub-nav-tab active" data-tab="dashboard">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <rect x="3" y="3" width="7" height="7" />
      <rect x="14" y="3" width="7" height="7" />
      <rect x="14" y="14" width="7" height="7" />
      <rect x="3" y="14" width="7" height="7" />
    </svg>
    ???????
  </button>
  <button class="sub-nav-tab" data-tab="storage">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <path d="M6 9l6 6 6-6" />
      <path d="M21 12v7a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-7" />
      <path d="M3 5a2 2 0 0 1 2-2h14a2 2 0 0 1 2 2v7H3z" />
    </svg>
    ?????
  </button>
</div>

<!-- Dashboard Tab Content -->
<div id="dashboard-content" class="tab-content active">
  <!-- Stats Grid -->
  <div class="stats-grid">
    <div class="stat-card">
      <div class="stat-card-label">????? ?????</div>
      <div class="stat-card-value accent" id="vm-count-total">
        {{ vms|length }}
      </div>
    </div>
    <div class="stat-card">
      <div class="stat-card-label">????????</div>
      <div class="stat-card-value" id="vm-count-running">
        {{ vms|selectattr('status')|selectattr('status.printableStatus',
        'equalto', 'Running')|list|length }}
      </div>
    </div>
```

```
    <div class="stat-card">
      <div class="stat-card-label">Namespace</div>
      <div class="stat-card-value">{{ namespace }}</div>
    </div>
  </div>

  <!-- Filters Bar -->
  <div class="filters-bar">
    <div class="filter-search">
      <svg
        class="filter-search-icon"
        viewBox="0 0 24 24"
        fill="none"
        stroke="currentColor"
        stroke-width="2"
      >
        <circle cx="11" cy="11" r="8" />
        <path d="M21 21l-4.35-4.35" />
      </svg>
      <input
        type="text"
        id="search-input"
        placeholder="????? ?? ????? ??? IP-??????"
        autocomplete="off"
      />
    </div>
    <div class="filter-select">
      <select id="status-filter">
        <option value="">??? ???????</option>
        <option value="running">????????</option>
        <option value="stopped">??????????</option>
      </select>
    </div>
    <div class="filter-select">
      <select id="region-filter">
        <option value="">??? ???????</option>
        <option value="moscow">??????</option>
        <option value="frankfurt">??????????</option>
        <option value="amsterdam">??????????</option>
      </select>
    </div>
    <button class="btn-secondary" id="refresh-btn" onclick="updateVMList()">
      <svg
        viewBox="0 0 24 24"
        fill="none"
        stroke="currentColor"
        stroke-width="2"
      >
        <path d="M23 4v6h-6" />
        <path d="M1 20v-6h6" />
        <path
          d="M3.51 9a9 9 0 0 1 14.85-3.36L23 10M1 14l4.64 4.36A9 9 0 0 0 20.49 15"
        />
      </svg>
      ???????? ??????
    </button>
  </div>

  <!-- VM Table Card -->
  <div class="card">
    <div class="card-header">
      <h2 class="card-title">?????? ?????????? ?????</h2>
      <span class="vm-count-badge" id="filtered-count"
        >{{ vms|length }} ?????</span
      >
    </div>
    <div class="card-body">
      <table class="vm-table">
        <thead>
          <tr>
            <th>???</th>
```

```
            <th>??????</th>
            <th>???????</th>
            <th>??????</th>
            <th>IP-?????</th>
            <th>VNC</th>
            <th>????????</th>
          </tr>
        </thead>
        <tbody id="vm-table-body">
          {% for vm in vms %}
          <tr
            data-vm-name="{{ vm.metadata.name }}"
            data-status="{{ vm.status.printableStatus|lower if vm.status and vm.status.printableStatus else 'unknown'...
            data-region=""
          >
            <td>
              <div class="vm-name-cell">
                <a
                  href="{{ url_for('vm_details_page', username=username, vm_name=vm.metadata.name) }}"
                  class="vm-name-link"
                >
                  {{ vm.metadata.name }}
                </a>
              </div>
            </td>
            <td>
              {% if vm.datavolume_status %}
                <span class="status-badge status-creating">???????? ?????...</span>
              {% elif vm.status and vm.status.printableStatus %}
                {% set ps = vm.status.printableStatus %}
                {% set status_class = ps|lower|replace(' ', '-') %}
                {% if status_class == 'running' %}
                  <span class="status-badge status-running">????????</span>
                {% elif status_class == 'stopped' or status_class == 'halted' %}
                  <span class="status-badge status-stopped">??????????</span>
                {% elif status_class == 'starting' %}
                  <span class="status-badge status-starting">??????????</span>
                {% elif status_class == 'stopping' %}
                  <span class="status-badge status-stopping">???????????????</span>
                {% elif status_class in ('provisioning', 'waitingforvolumebinding', 'waitingfordatavolume') %}
                  <span class="status-badge status-creating">???????? ??????</span>
                {% else %}
                  <span class="status-badge status-{{ status_class }}">{{ ps }}</span>
                {% endif %}
              {% else %}
                <span class="status-badge status-unknown">??????????</span>
              {% endif %}
            </td>
            <td>
              <div class="resources-cell">
                {% set spec = vm.spec.template.spec if vm.spec and
                vm.spec.template else {} %} {% set domain = spec.domain if spec
                else {} %} {% set cpu = domain.cpu.cores if domain and
                domain.cpu else '-' %} {% set memory =
                domain.resources.requests.memory if domain and domain.resources
                and domain.resources.requests else '-' %} {% set gpu_devices =
                domain.devices.gpus if domain and domain.devices and
                domain.devices.gpus else [] %} {% set gpu_count =
                gpu_devices|length %} {% set gpu_model =
                gpu_devices[0].deviceName if gpu_devices else '' %}
                <span class="resource-tag">{{ cpu }} vCPU</span>
                <span class="resource-tag">{{ memory }} RAM</span>
                {% if gpu_count > 0 %}
                <span class="resource-tag resource-gpu"
                  >{{ gpu_count }}x {{ gpu_model.split('/')[-1]|upper }}</span
                >
                {% endif %}
              </div>
            </td>
            <td>
              <span class="text-muted">??????</span>
```

```
            </td>
            <td class="ip-cell">
              {% if vm.status and vm.status.interfaces %} {{
              vm.status.interfaces[0].ipAddress|default('-') }} {% else %}
              <span class="text-muted">?</span>
              {% endif %}
            </td>
            <td>
              {% set is_running = vm.status and vm.status.printableStatus and
              vm.status.printableStatus|lower == 'running' %}
              <button
                class="vnc-btn"
                onclick="openVNC('{{ vm.metadata.name }}')"
                {%
                if
                not
                is_running
                %}disabled{%
                endif
                %}
              >
                <svg
                  viewBox="0 0 24 24"
                  fill="none"
                  stroke="currentColor"
                  stroke-width="2"
                >
                  <rect x="2" y="3" width="20" height="14" rx="2" />
                  <path d="M8 21h8" />
                  <path d="M12 17v4" />
                </svg>
                {% if is_running %}??????? VNC{% else %}VNC ??????????{% endif
                %}
              </button>
            </td>
            <td>
              <div class="actions-cell">
                {% set is_running = vm.status and vm.status.printableStatus and
                vm.status.printableStatus|lower == 'running' %} {% if is_running
                %}
                <button
                  class="action-btn btn-stop"
                  onclick="stopVM('{{ vm.metadata.name }}')"
                >
                  <svg
                    viewBox="0 0 24 24"
                    fill="none"
                    stroke="currentColor"
                    stroke-width="2"
                  >
                    <rect x="6" y="6" width="12" height="12" rx="1" />
                  </svg>
                  <span>??????????</span>
                </button>
                <button
                  class="action-btn btn-reboot"
                  onclick="restartVM('{{ vm.metadata.name }}')"
                >
                  <svg
                    viewBox="0 0 24 24"
                    fill="none"
                    stroke="currentColor"
                    stroke-width="2"
                  >
                    <path d="M23 4v6h-6" />
                    <path d="M20.49 15a9 9 0 0 1 1-2.12-9.36L23 10" />
                  </svg>
                  <span>?????????????</span>
                </button>
                {% else %}
                <button
```

```
                  class="action-btn btn-start"
                  onclick="startVM('{{ vm.metadata.name }}')"
                >
                  <svg
                    viewBox="0 0 24 24"
                    fill="none"
                    stroke="currentColor"
                    stroke-width="2"
                  >
                    <polygon points="5 3 19 12 5 21 5 3" />
                  </svg>
                  <span>?????????</span>
                </button>
                <button class="action-btn btn-reboot" disabled>
                  <svg
                    viewBox="0 0 24 24"
                    fill="none"
                    stroke="currentColor"
                    stroke-width="2"
                  >
                    <path d="M23 4v6h-6" />
                    <path d="M20.49 15a9 9 0 0 1 1-2.12-9.36L23 10" />
                  </svg>
                  <span>????????????</span>
                </button>
                {% endif %}

                <!-- More Actions Dropdown -->
                <div class="more-actions">
                  <button
                    class="more-btn"
                    onclick="toggleMoreMenu(this, event)"
                  >
                    <svg
                      viewBox="0 0 24 24"
                      fill="none"
                      stroke="currentColor"
                      stroke-width="2"
                    >
                      <circle cx="12" cy="12" r="1" />
                      <circle cx="12" cy="5" r="1" />
                      <circle cx="12" cy="19" r="1" />
                    </svg>
                  </button>
                  <div class="more-dropdown">
                    <button
                      class="more-dropdown-item danger"
                      onclick="deleteVM('{{ vm.metadata.name }}')"
                    >
                      <svg
                        viewBox="0 0 24 24"
                        fill="none"
                        stroke="currentColor"
                        stroke-width="2"
                      >
                        <polyline points="3 6 5 6 21 6" />
                        <path
                          d="M19 6v14a2 2 0 0 1-2 2H7a2 2 0 0 1-2-2V6m3 0V4a2 2 0 0 1 2-2h4a2 2 0 0 1 2 2v2"
                        />
                        <line x1="10" y1="11" x2="10" y2="17" />
                        <line x1="14" y1="11" x2="14" y2="17" />
                      </svg>
                      ???????
                    </button>
                  </div>
                </div>
              </div>
            </td>
          </tr>
          {% else %}
          <tr class="empty-row">
```

```
            <td colspan="7">
              <div class="empty-state">
                <svg
                  class="empty-state-icon"
                  viewBox="0 0 24 24"
                  fill="none"
                  stroke="currentColor"
                  stroke-width="1.5"
                >
                  <rect x="2" y="3" width="20" height="14" rx="2" />
                  <path d="M8 21h8" />
                  <path d="M12 17v4" />
                  <path d="M9 10h6" />
                </svg>
                <h3>??? ??????????? ?????</h3>
                <p>
                  ???????? ???? ?????? ??????????? ??????, ????? ?????? ????
                </p>
              </div>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>

  <!-- Create VM Modal -->
  <section id="create-vm-overlay" class="modal-overlay" hidden></section>
  <section id="create-vm-modal" class="modal" hidden>
    <header class="modal-header">
      <h2>???????? ??????????? ??????</h2>
      <button type="button" id="close-create-vm" aria-label="???????">×</button>
    </header>

    <!-- Wizard Steps Indicator -->
    <div class="wizard-steps">
      <div class="wizard-step active" data-step="1">
        <span class="step-number">1</span>
        <span class="step-label">???????</span>
      </div>
      <div class="wizard-step" data-step="2">
        <span class="step-number">2</span>
        <span class="step-label">???????</span>
      </div>
      <div class="wizard-step" data-step="3">
        <span class="step-number">3</span>
        <span class="step-label">????????????</span>
      </div>
    </div>

    <form
      id="create-vm-form"
      class="create-form"
      method="POST"
      action="{{ url_for('create_vm', username=username) }}"
      novalidate
      data-max-cpu="{{ max_cpu_cores }}"
      data-max-memory="{{ max_memory_gb }}"
      data-max-storage="{{ max_storage_gb }}"
      data-max-gpu="{{ max_gpu_count }}"
    >
      <!-- ??????? ???? -->
      <input type="hidden" name="image" id="hidden-image" />
      <input type="hidden" name="cpu" id="hidden-cpu" />
      <input type="hidden" name="memory" id="hidden-memory" />
      <input type="hidden" name="storage" id="hidden-storage" />
      <input type="hidden" name="gpu" id="hidden-gpu" value="0" />
      <input
        type="hidden"
        name="gpu_model"
```

```
        id="hidden-gpu-model"
        value="{{ default_gpu_resource }}"
        data-default-model="{{ default_gpu_resource }}"
      />

      <!-- ??? 1: ????? ?? -->
      <section class="wizard-panel active" data-wizard-step="1">
        <label class="full-width form-section-spacing">
          ??? ??????????? ??????
          <input
            type="text"
            name="name"
            id="vm-name-input"
            placeholder="my-server-01"
            pattern="[a-z0-9]([a-z0-9\-]*[a-z0-9])?"
            title="?????? ???????? ?????, ????? ? ??????"
            minlength="3"
            maxlength="63"
            required
          />
        </label>

        {% include 'partials/os_cards.html' %}
      </section>

      <!-- ??? 2: ??????? -->
      <section class="wizard-panel" data-wizard-step="2">
        <div class="form-tabs sub-tabs form-section-spacing">
          <button type="button" class="form-tab active" data-tab="presets">
            ???????
          </button>
          <button type="button" class="form-tab" data-tab="manual">
            ????????? ???????????
          </button>
        </div>

        <section class="tab-panel active" data-panel="presets">
          <div class="preset-block">
            <div class="preset-block-spacing">
              <p class="preset-title">??????? ???????</p>
              <p class="preset-hint">
                ???????? ??????????? ???????? ??? ????? ??????.
              </p>
            </div>
            <div class="preset-list">
              <button
                type="button"
                class="preset-card"
                data-preset="shared-small"
                data-cpu="2"
                data-memory="4"
                data-storage="40"
                data-gpu="0"
                data-gpu-model=""
              >
                <span class="preset-name">S ? ?????????</span>
                <span class="preset-meta">2 vCPU ? 4 ?? RAM ? 40 ?? SSD</span>
              </button>
              <button
                type="button"
                class="preset-card"
                data-preset="shared-medium"
                data-cpu="4"
                data-memory="8"
                data-storage="80"
                data-gpu="0"
                data-gpu-model=""
              >
                <span class="preset-name">M ? ??????????</span>
                <span class="preset-meta">4 vCPU ? 8 ?? RAM ? 80 ?? SSD</span>
              </button>
```

```
492                <button
493                  type="button"
494                  class="preset-card"
495                  data-preset="shared-large"
496                  data-cpu="8"
497                  data-memory="16"
498                  data-storage="160"
499                  data-gpu="0"
500                  data-gpu-model=""
501
502 ... [283 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/dashboard_home.html (3.5 KB)

```
{% extends "base.html" %}
{% block title %}?????? ?????????? ? ????????{% endblock %}
{% block nav_dashboard %}active{% endblock %}
{% block topbar_title %}?????? ??????????{% endblock %}
{% block extra_css %}
<link rel="stylesheet" href="{{ url_for('static', path='css/dashboard-home.css') }}" />
{% endblock %}
{% block content %}
<div class="dh-wrap" data-username="{{ username }}">
  <section class="dh-welcome">
    <h1>????? ??????????, {{ username }} ?</h1>
    <p>???????? ?????? ??? ????????? ????????.</p>
  </section>

  <a href="/{{ username }}/spend" class="dh-card dh-card-balance">
    <div class="dh-card-head">
      <h2>??????</h2>
      <span class="dh-card-link">????????? ? ???????? ?</span>
    </div>
    <div class="dh-balance-grid">
      <div>
        <div class="dh-stat-label">???????? ??????</div>
        <div class="dh-stat-value" id="dh-main-balance">?</div>
      </div>
      <div>
        <div class="dh-stat-label">???????? ??????</div>
        <div class="dh-stat-value" id="dh-llm-balance">?</div>
      </div>
    </div>
    <div class="dh-transfer" onclick="event.preventDefault();event.stopPropagation();">
      <div class="dh-transfer-head">??????? ???????</div>
      <div class="dh-transfer-row">
        <select id="dh-transfer-direction">
          <option value="main-llm">???. ? ????????</option>
          <option value="llm-main">???????? ? ???.</option>
        </select>
        <input id="dh-transfer-amount" type="number" min="0.01" step="0.01" placeholder="?????" />
        <button id="dh-transfer-all" type="button" class="btn-secondary">???</button>
        <button id="dh-transfer-submit" type="button" class="btn-primary">?????????</button>
      </div>
      <div id="dh-transfer-error" class="dh-transfer-error" hidden></div>
    </div>
  </a>

  <a href="/{{ username }}/vms" class="dh-card">
    <div class="dh-card-head">
      <h2>?????????? ??????</h2>
      <span class="dh-card-link">????????? ?? ?</span>
    </div>
    <div class="dh-vm-stats">
      <div class="dh-stat-label">?????? (?????/???)</div>
      <div class="dh-stat-value"><span id="dh-vm-total">?</span> / <span id="dh-vm-running">?</span></div>
    </div>
  </a>

  <a href="/{{ username }}/llm" class="dh-card">
    <div class="dh-card-head">
      <h2>????????</h2>
      <span class="dh-card-link">??????? ? ??? ?</span>
    </div>
    <div class="dh-quickstart" onclick="event.preventDefault();event.stopPropagation();">
      <div class="dh-quickstart-head">
        <strong>??????? ????? (API)</strong>
        <div class="dh-quickstart-actions">
          <button id="dh-copy-snippet" type="button" class="btn-secondary">??????????</button>
```

```
          <div class="dh-tabs" role="tablist">
            <button type="button" class="dh-tab active" data-lang="python">Python</button>
            <button type="button" class="dh-tab" data-lang="typescript">TypeScript</button>
            <button type="button" class="dh-tab" data-lang="curl">cURL</button>
          </div>
        </div>
      </div>
      <pre id="dh-snippet" class="dh-code"></pre>
    </div>
  </a>
</div>
{% endblock %}
{% block extra_js %}
<script src="{{ url_for('static', path='js/dashboard-home.js') }}"></script>
{% endblock %}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/error.html (2.1 KB)

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Error</title>
    <style>
      :root {
        color-scheme: light dark;
      }
      body {
        margin: 0;
        font-family: Inter, system-ui, -apple-system, Segoe UI, Roboto, sans-serif;
        background: #0f172a;
        color: #e2e8f0;
        display: grid;
        place-items: center;
        min-height: 100vh;
        padding: 20px;
      }
      .card {
        width: 100%;
        max-width: 640px;
        background: rgba(15, 23, 42, 0.88);
        border: 1px solid rgba(148, 163, 184, 0.35);
        border-radius: 14px;
        padding: 20px;
        box-sizing: border-box;
      }
      h1 {
        margin: 0 0 8px;
        font-size: 22px;
      }
      p {
        margin: 0 0 10px;
        color: #cbd5e1;
      }
      .meta {
        margin-top: 14px;
        padding: 10px;
        border-radius: 8px;
        background: rgba(30, 41, 59, 0.7);
        font-size: 13px;
      }
      .actions {
        display: flex;
        gap: 10px;
        margin-top: 16px;
      }
      .btn {
        display: inline-block;
        border: 1px solid rgba(148, 163, 184, 0.4);
        background: #1d4ed8;
        color: #fff;
        text-decoration: none;
        padding: 8px 12px;
        border-radius: 8px;
        font-size: 14px;
      }
      .btn.secondary {
        background: #334155;
      }
    </style>
  </head>
  <body>
```

```
    <main class="card">
      <h1>{{ error_title or "Something went wrong" }}</h1>
      <p>{{ error_message or "Please try again later." }}</p>
      <div class="meta">
        <div><strong>Status:</strong> {{ status_code or 500 }}</div>
        <div><strong>Path:</strong> {{ error_path or request.url.path }}</div>
      </div>
      <div class="actions">
        <a class="btn" href="/dashboard">Go to dashboard</a>
        <a class="btn secondary" href="#" onclick="history.back(); return false;">Go back</a>
      </div>
    </main>
  </body>
</html>

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/highlight_js_block.html (941.0 B)

```
{# Common highlight.js includes for code syntax highlighting #}
<script src="https://cdn.jsdelivr.net/npm/marked/marked.min.js"></script>
<link rel="stylesheet" href="https://cdn.jsdelivr.net/gh/highlightjs/cdn-release@11.9.0/build/styles/github-dark.min....
<script src="https://cdn.jsdelivr.net/gh/highlightjs/cdn-release@11.9.0/build/highlight.min.js"></script>
<script src="https://cdn.jsdelivr.net/gh/highlightjs/cdn-release@11.9.0/build/languages/bash.min.js"></script>
<script src="https://cdn.jsdelivr.net/gh/highlightjs/cdn-release@11.9.0/build/languages/python.min.js"></script>
<script src="https://cdn.jsdelivr.net/gh/highlightjs/cdn-release@11.9.0/build/languages/javascript.min.js"></script>
<script src="https://cdn.jsdelivr.net/gh/highlightjs/cdn-release@11.9.0/build/languages/yaml.min.js"></script>
<script src="https://cdn.jsdelivr.net/gh/highlightjs/cdn-release@11.9.0/build/languages/json.min.js"></script>

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/llm_chat.html (7.4 KB)

```
{% extends "base.html" %}
{% block title %}LLM-???????? ? ????????{% endblock %}
{% block nav_llm %}active{% endblock %}
{% block topbar_title %}LLM-????????{% endblock %}
{% block extra_css %}
<link rel="stylesheet" href="{{ url_for('static', path='css/chat-common.css') }}" />
<link rel="stylesheet" href="{{ url_for('static', path='css/llm-chat.css') }}" />
<link rel="stylesheet" href="{{ url_for('static', path='css/llm-models.css') }}" />
{% include 'highlight_js_block.html' %}
{% endblock %}
{% block content %}
<div class="llm-page" data-username="{{ username }}" data-llm-default-model="{{ llm_default_model|default('') }}">
  <div class="llm-subnav">
    <a href="/{{ username }}/llm" class="llm-subnav-link active">???</a>
    <a href="/{{ username }}/llm/models" class="llm-subnav-link">??????</a>
    <a href="/{{ username }}/llm/keys" class="llm-subnav-link">????? API</a>
  </div>

  <div class="llm-playground">
    <div class="chat-layout">
    <div class="chat-container">
      <div class="chat-header">
        <div class="chat-title">
          <div class="chat-title-icon">
            <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
              <path d="M21 16V8a2 2 0 0 0-1-1.73l-7-4a2 2 0 0 0-2 0l-7 4A2 2 0 0 0 3 8v8a2 2 0 0 0 1 1.73l7 4a2 2 0 0...
              <polyline points="7.5 4.21 12 6.81 16.5 4.21" />
              <polyline points="7.5 19.79 7.5 14.6 3 12" />
              <polyline points="21 12 16.5 14.6 16.5 19.79" />
              <polyline points="3.27 6.96 12 12.01 20.73 6.96" />
              <line x1="12" y1="22.08" x2="12" y2="12" />
            </svg>
          </div>
          <div><h3>???</h3></div>
        </div>
        <button class="chat-clear-btn" id="chat-clear-btn" title="???????? ???">
          <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
            <polyline points="3 6 5 6 21 6" />
            <path d="M19 6v14a2 2 0 0 0 1-2 2H7a2 2 0 0 0 1-2-2V6m3 0V4a2 2 0 0 0 1 2-2h4a2 2 0 0 0 1 2 2v2" />
          </svg>
        </button>
      </div>

      <div class="chat-messages" id="chat-messages">
        <div class="chat-welcome">
          <div class="chat-welcome-icon">
            <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
              <path d="M21 16V8a2 2 0 0 0-1-1.73l-7-4a2 2 0 0 0-2 0l-7 4A2 2 0 0 0 3 8v8a2 2 0 0 0 1 1.73l7 4a2 2 0 0...
              <polyline points="7.5 4.21 12 6.81 16.5 4.21" />
              <polyline points="7.5 19.79 7.5 14.6 3 12" />
              <polyline points="21 12 16.5 14.6 16.5 19.79" />
              <polyline points="3.27 6.96 12 12.01 20.73 6.96" />
              <line x1="12" y1="22.08" x2="12" y2="12" />
            </svg>
          </div>
          <h3>????? ??????????? ? LLM-???</h3>
          <p>??????? ?????? ?? ????????????, Linux, DevOps ??? Kubernetes.</p>
          <div class="chat-suggestions">
            <button class="suggestion-btn" data-suggestion="??? ????????? ???????? CPU ? RAM ?? Linux?">?????????? CP...
            <button class="suggestion-btn" data-suggestion="??? ?????? ????????? SSH-????? ?? ????????">????????? SSH...
            <button class="suggestion-btn" data-suggestion="??? ?????????? Docker Compose ?????? ?? VPS?">Docker Comp...
          </div>
        </div>
      </div>

```

```
      <div class="chat-input-container">
        <textarea id="chat-input" class="chat-input" placeholder="???????? ???-??????..." rows="1"></textarea>
        <button class="chat-send-btn" id="chat-send-btn" disabled>
          <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
            <line x1="22" y1="2" x2="11" y2="13" />
            <polygon points="22 2 15 22 11 13 2 9 22 2" />
          </svg>
        </button>
      </div>
    </div>
    <div class="chat-settings-drawer" id="chat-settings-drawer">
        <aside
          id="chat-settings-panel"
          class="playground-settings"
          aria-label="????????? ?????????"
        >
          <div class="settings-header">
            <h4>??????????</h4>
            <div class="settings-actions">
              <button type="button" id="chat-settings-reset" class="settings-reset-btn">????????</button>
            </div>
          </div>
          <p class="settings-caption">????????? ?????? ? ????????? ??????. ????????? ??????????? ? ?????????? ???????...

          <section class="settings-section" aria-labelledby="settings-model-title">
            <h5 id="settings-model-title">Model</h5>
            <div class="playground-field">
              <label for="chat-model-select">??????</label>
              <select id="chat-model-select" class="chat-model-select" title="???????? ??????">
                <option value="">????????</option>
              </select>
            </div>
          </section>

          <section class="settings-section" aria-labelledby="settings-generation-title">
            <h5 id="settings-generation-title">Generation</h5>
            <div class="playground-field">
              <div class="playground-field-head">
                <label for="chat-temperature">Temperature</label>
                <span class="value-chip" id="chat-temperature-value">0.7</span>
              </div>
              <input id="chat-temperature" type="range" min="0" max="2" step="0.1" value="0.7" />
              <div class="playground-field-meta">
                <span>??????</span>
                <span>??????????</span>
              </div>
            </div>
            <div class="playground-field">
              <div class="playground-field-head">
                <label for="chat-max-tokens">Output Length</label>
                <span class="value-chip" id="chat-max-tokens-value">2048</span>
              </div>
              <input id="chat-max-tokens" type="range" min="256" max="8192" step="256" value="2048" />
              <div class="playground-field-meta">
                <span>???????</span>
                <span>????????</span>
              </div>
            </div>
          </section>

          <section class="settings-section" aria-labelledby="settings-presets-title">
            <h5 id="settings-presets-title">???????</h5>
            <div class="settings-presets" role="group" aria-label="??????? ??????? ??????????">
              <button type="button" class="settings-preset-btn" data-temp="0.2" data-tokens="1024">??????</button>
              <button type="button" class="settings-preset-btn" data-temp="0.7" data-tokens="2048">???????.</button>
              <button type="button" class="settings-preset-btn" data-temp="1.2" data-tokens="4096">???????</button>
            </div>
          </section>
        </aside>
    </div>
    </div>
```

```
  </div>
</div>
{% endblock %}
{% block extra_js %}
<script src="{{ url_for('static', path='js/llm-chat.js') }}"></script>
{% endblock %}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/llm_keys.html (2.2 KB)

```
{% extends "base.html" %} {% block title %}????? API ? ????????{% endblock %}
{% block nav_llm %}active{% endblock %} {% block topbar_title %}????? API{% endblock %}
{% block extra_css %}
<link rel="stylesheet" href="{{ url_for('static', path='css/llm-chat.css') }}" />
<link rel="stylesheet" href="{{ url_for('static', path='css/llm-models.css') }}" />
<link rel="stylesheet" href="{{ url_for('static', path='css/admin-llm.css') }}" />
{% endblock %} {% block content %}
<div class="llm-page llm-models-page llm-keys-page" data-username="{{ username }}" data-max-keys="{{ llm_user_keys_ma...
  <div class="llm-subnav">
    <a href="/{{ username }}/llm" class="llm-subnav-link">???</a>
    <a href="/{{ username }}/llm/models" class="llm-subnav-link">??????</a>
    <a href="/{{ username }}/llm/keys" class="llm-subnav-link active">????? API</a>
  </div>

  <div class="admin-llm-page">
    <div class="admin-llm-header">
      <h1>????? API</h1>
      <div class="admin-llm-controls">
        <span class="keys-count-meta"><span id="keys-count">?</span> / {{ llm_user_keys_max }}</span>
        <button id="keys-create-btn" class="btn-primary">??????? ????</button>
      </div>
    </div>

    <p class="llm-keys-hint">????? ???????? ? ??????? ??? ??????. ??????? «????????», ????? ??????? ??????????? ????....

    <div id="keys-loading" class="llm-models-loading">??????????</div>
    <div id="keys-error" class="llm-models-error" style="display:none"></div>

    <div class="admin-llm-section" id="keys-section" style="display:none">
      <div class="admin-llm-table-wrap">
        <table class="admin-llm-table keys-table">
          <thead>
            <tr>
              <th>??????</th>
              <th>????</th>
              <th>??????</th>
              <th>????????? ????????????</th>
              <th>????????</th>
            </tr>
          </thead>
          <tbody id="keys-list"></tbody>
        </table>
      </div>
    </div>
  </div>
</div>
{% endblock %} {% block extra_js %}
<script src="{{ url_for('static', path='js/llm-keys.js') }}"></script>
{% endblock %}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/llm_models.html (1.5 KB)

```
{% extends "base.html" %} {% block title %}?????? ? ????????{% endblock %}
{% block nav_llm %}active{% endblock %} {% block topbar_title %}LLM-????????{% endblock %}
{% block extra_css %}
<link rel="stylesheet" href="{{ url_for('static', path='css/llm-chat.css') }}" />
<link rel="stylesheet" href="{{ url_for('static', path='css/llm-models.css') }}" />
{% endblock %} {% block content %}
<div class="llm-page llm-models-page" data-username="{{ username }}" data-example-api-key="{{ llm_example_api_key|def...
  <div class="llm-subnav">
    <a href="/{{ username }}/llm" class="llm-subnav-link">???</a>
    <a href="/{{ username }}/llm/models" class="llm-subnav-link active">??????</a>
    <a href="/{{ username }}/llm/keys" class="llm-subnav-link">????? API</a>
  </div>

  <div class="llm-models-container">
    <div class="llm-models-header">
      <h2>????????? ??????</h2>
      <div class="llm-models-meta">
        <span id="models-count">?</span> ???????
        <span class="llm-models-refresh" id="models-last-refresh"></span>
      </div>
    </div>

    <div id="models-loading" class="llm-models-loading">??????????</div>
    <div id="models-error" class="llm-models-error" style="display:none"></div>
    <div id="models-list" class="llm-models-list" style="display:none"></div>
  </div>
</div>
{% endblock %} {% block extra_js %}
<script src="{{ url_for('static', path='js/llm-models.js') }}"></script>
{% endblock %}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/login.html (8.2 KB)

```
<!DOCTYPE html>
<html lang="ru" data-theme="light">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>???? ? ????????</title>
  <link rel="icon" type="image/png" href="{{ url_for('static', path='images/favicon.png') }}" />
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="stylesheet">
  <script src="{{ url_for('static', path='js/core/theme.js') }}"></script>
  <style>
    /* Brand Colors */
    :root {
      --brand-bg: #111827;
      --brand-blue-deep: #0F3D5E;
      --brand-blue-mid: #3B82A5;
      --brand-aqua-light: #C4F0F2;
    }

    /* Light Theme */
    :root, [data-theme="light"] {
      --app-bg: #F9FAFB;
      --surface: #FFFFFF;
      --surface-muted: #F3F4F6;
      --text-primary: #111827;
      --text-secondary: #6B7280;
      --text-muted: #9CA3AF;
      --border-subtle: #E5E7EB;
      --primary-bg: var(--brand-blue-deep);
      --primary-hover: var(--brand-blue-mid);
      --primary-text: #FFFFFF;
      --error-bg: #FEE2E2;
      --error-text: #DC2626;
    }

    /* Dark Theme */
    [data-theme="dark"] {
      --app-bg: #020617;
      --surface: #0B1120;
      --surface-muted: #111827;
      --text-primary: #F9FAFB;
      --text-secondary: #9CA3AF;
      --text-muted: #6B7280;
      --border-subtle: #1F2937;
      --primary-bg: var(--brand-blue-mid);
      --primary-hover: var(--brand-aqua-light);
      --primary-text: #020617;
      --error-bg: rgba(239, 68, 68, 0.15);
      --error-text: #EF4444;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    html {
      font-size: 14px;
      -webkit-font-smoothing: antialiased;
    }

    body {
      font-family: 'Inter', -apple-system, sans-serif;
```

```
  min-height: 100vh;
  display: flex;
  align-items: center;
  justify-content: center;
  background: var(--app-bg);
  color: var(--text-primary);
  padding: 20px;
}

.login-container {
  width: 100%;
  max-width: 400px;
}

.login-header {
  text-align: center;
  margin-bottom: 32px;
}

.logo-wrapper {
  background: var(--brand-bg);
  padding: 24px;
  border-radius: 12px;
  margin-bottom: 24px;
  display: inline-block;
}

.logo-img {
  width: 180px;
  height: auto;
}

.login-title {
  font-size: 20px;
  font-weight: 700;
  color: var(--text-primary);
  margin-bottom: 8px;
}

.login-subtitle {
  font-size: 14px;
  color: var(--text-secondary);
  line-height: 1.5;
}

.login-card {
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: 12px;
  padding: 32px 28px;
}

.form-group {
  margin-bottom: 20px;
}

.form-label {
  display: block;
  font-size: 13px;
  font-weight: 600;
  color: var(--text-secondary);
  margin-bottom: 8px;
}

.form-input {
  width: 100%;
  padding: 12px 14px;
  background: var(--surface-muted);
  border: 1px solid var(--border-subtle);
  border-radius: 8px;
  color: var(--text-primary);
```

```
  font-size: 14px;
  font-family: inherit;
  transition: all 0.15s ease;
}

.form-input::placeholder {
  color: var(--text-muted);
}

.form-input:focus {
  outline: none;
  border-color: var(--brand-blue-mid);
  background: var(--surface);
}

.btn-login {
  width: 100%;
  padding: 12px 20px;
  background: var(--primary-bg);
  color: var(--primary-text);
  font-size: 14px;
  font-weight: 600;
  border: none;
  border-radius: 8px;
  cursor: pointer;
  transition: all 0.15s ease;
}

.btn-login:hover {
  background: var(--primary-hover);
}

.btn-login:active {
  transform: scale(0.98);
}

.error-message {
  background: var(--error-bg);
  color: var(--error-text);
  padding: 12px 14px;
  border-radius: 8px;
  font-size: 13px;
  font-weight: 500;
  margin-top: 16px;
  text-align: center;
}

.login-footer {
  text-align: center;
  margin-top: 24px;
  font-size: 12px;
  color: var(--text-muted);
}

.theme-toggle {
  position: fixed;
  top: 20px;
  right: 20px;
  width: 40px;
  height: 40px;
  display: flex;
  align-items: center;
  justify-content: center;
  background: var(--surface);
  border: 1px solid var(--border-subtle);
  border-radius: 8px;
  cursor: pointer;
  color: var(--text-secondary);
  transition: all 0.15s ease;
}
```

```
    .theme-toggle:hover {
      color: var(--text-primary);
      border-color: var(--brand-blue-mid);
    }

    .theme-toggle svg {
      width: 20px;
      height: 20px;
    }

    .icon-moon { display: block; }
    .icon-sun { display: none; }

    [data-theme="dark"] .icon-moon { display: none; }
    [data-theme="dark"] .icon-sun { display: block; }

    @media (max-width: 480px) {
      .login-card {
        padding: 24px 20px;
      }

      .logo-img {
        width: 150px;
      }

      .login-title {
        font-size: 18px;
      }
    }
  </style>
</head>
<body>
  <div class="login-container">
    <div class="login-header">
      <div class="logo-wrapper">
        <img src="{{ url_for('static', path='images/viiruoz.png') }}" alt="????????" class="logo-img" />
      </div>
      <h1 class="login-title">???? ? ?????? ??????????</h1>
      <p class="login-subtitle">???????? ????????? ??? ???????</p>
    </div>

    <div class="login-card">
      <form id="login-form">
        <div class="form-group">
          <label class="form-label" for="username">?????</label>
          <input type="text" id="username" name="username" class="form-input"
                 placeholder="??????? ??? ???????????"
                 autocomplete="username" required />
        </div>

        <div class="form-group">
          <label class="form-label" for="password">??????</label>
          <input type="password" id="password" name="password" class="form-input"
                 placeholder="??????? ??????"
                 autocomplete="current-password" required />
        </div>

        <button type="submit" class="btn-login" id="login-btn">?????</button>

        <div id="login-error" class="error-message" style="display:none"></div>
      </form>

    </div>

    <div class="login-footer">
      © 2025 ????????
    </div>
  </div>

  <script>
  document.getElementById('login-form').addEventListener('submit', async function(e) {
```

```
    e.preventDefault();
    var errorEl = document.getElementById('login-error');
    var btn = document.getElementById('login-btn');
    errorEl.style.display = 'none';
    btn.disabled = true;
    btn.textContent = '????...';

    try {
      var resp = await fetch('/auth/login', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          username: document.getElementById('username').value,
          password: document.getElementById('password').value
        })
      });
      if (resp.ok) {
        var data = await resp.json();
        window.location.href = '/' + data.username + '/dashboard';
      } else {
        var err = await resp.json().catch(function() { return {}; });
        errorEl.textContent = err.detail || '???????? ????? ??? ??????';
        errorEl.style.display = 'block';
        btn.disabled = false;
        btn.textContent = '?????';
      }
    } catch (_) {
      errorEl.textContent = '?????? ??????????? ? ???????';
      errorEl.style.display = 'block';
      btn.disabled = false;
      btn.textContent = '?????';
    }
  });
  </script>
</body>
</html>

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/partials/os_cards.html (3.5 KB)

```
        <div class="os-selection-grid">
          <!-- Ubuntu Card -->
          <div
            class="os-card"
            data-os="ubuntu"
            data-image="https://cloud-images.ubuntu.com/jammy/current/jammy-server-cloudimg-amd64.img"
          >
            <div class="os-icon ubuntu-icon"></div>
            <div class="os-info">
              <h3>Ubuntu</h3>
              <p>22.04 LTS</p>
            </div>
            <div class="os-select-indicator"></div>
          </div>
          <div
            class="os-card"
            data-os="ubuntu24"
            data-image="https://cloud-images.ubuntu.com/noble/current/noble-server-cloudimg-amd64.img"
          >
            <div class="os-icon ubuntu-icon"></div>
            <div class="os-info">
              <h3>Ubuntu</h3>
              <p>24.04 LTS</p>
            </div>
            <div class="os-select-indicator"></div>
          </div>

          <!-- Fedora Card -->
          <div
            class="os-card"
            data-os="fedora"
            data-image="https://download.fedoraproject.org/pub/fedora/linux/releases/39/Cloud/x86_64/images/Fedora-Cl...
          >
            <div class="os-icon fedora-icon"></div>
            <div class="os-info">
              <h3>Fedora</h3>
              <p>39 Cloud</p>
            </div>
            <div class="os-select-indicator"></div>
          </div>

          <!-- Debian Card -->
          <div
            class="os-card"
            data-os="debian"
            data-image="https://cloud.debian.org/images/cloud/bookworm/latest/debian-12-generic-amd64.qcow2"
          >
            <div class="os-icon debian-icon"></div>
            <div class="os-info">
              <h3>Debian</h3>
              <p>12 Bookworm</p>
            </div>
            <div class="os-select-indicator"></div>
          </div>

          <!-- OpenSUSE Card -->
          <div
            class="os-card"
            data-os="opensuse"
            data-image="https://download.opensuse.org/distribution/leap/16.0/appliances/Leap-16.0-Minimal-VM.x86_64-C...
          >
            <div class="os-icon suse-icon"></div>
            <div class="os-info">
              <h3>OpenSUSE</h3>
              <p>Leap 16</p>
```

```
            </div>
            <div class="os-select-indicator"></div>
          </div>

          <!-- Windows 10 Installer - Clean Installation -->
          <div
            class="os-card"
            data-os="windows-installer"
            data-image="windows-installer"
            title="?????? ????????? Windows 10 ? ISO (??? ???????? golden image)"
          >
            <div class="os-icon windows-icon"></div>
            <div class="os-info">
              <h3>Windows 10</h3>
              <p>?????????</p>
            </div>
            <div class="os-select-indicator"></div>
          </div>

          <!-- Windows 10 Clone - From Golden Image -->
          <div
            class="os-card"
            data-os="windows-golden"
            data-image="windows-golden-image"
            title="Windows 10 ?? ???????? ?????? (??????? ???????????? ? Cloudbase-Init)"
          >
            <div class="os-icon windows-icon"></div>
            <div class="os-info">
              <h3>Windows 10</h3>
              <p>Golden Image</p>
            </div>
            <div class="os-select-indicator"></div>
          </div>
        </div>

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templates/vm_details.html (15.8 KB)

```
{% extends "base.html" %} {% block title %}{{ vm_name }} ? ????????{% endblock
%} {% block nav_vms %}active{% endblock %} {% block topbar_title %}{{ vm_name
}}{% endblock %} {% block extra_css %}
<link
  rel="stylesheet"
  href="{{ url_for('static', path='css/chat-common.css') }}"
/>
<link
  rel="stylesheet"
  href="{{ url_for('static', path='css/llm-chat.css') }}"
/>
<link
  rel="stylesheet"
  href="{{ url_for('static', path='css/vm-details-new.css') }}"
/>
<link
  rel="stylesheet"
  href="{{ url_for('static', path='css/services-ui.css') }}"
/>
<script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
{% include 'highlight_js_block.html' %} {% endblock %} {% block content %}
<!-- Breadcrumb -->
<nav class="breadcrumb">
  <a href="/{{ username }}/vms" class="breadcrumb-link">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <path d="M15 18l-6-6 6-6" />
    </svg>
    ?????????? ??????
  </a>
  <span class="breadcrumb-separator">/</span>
  <span class="breadcrumb-current">{{ vm_name }}</span>
</nav>

<!-- Page Header -->
<div class="page-header">
  <div class="page-header-text">
    <h1>{{ vm_name }}</h1>
    <div class="vm-status-header">
      <span class="status-badge" id="vm-status-badge">????????...</span>
    </div>
  </div>
  <div class="header-actions">
    <button
      type="button"
      class="action-btn btn-start"
      id="start-btn"
      onclick="startVM('{{ vm_name }}')"
    >
      <svg
        viewBox="0 0 24 24"
        fill="none"
        stroke="currentColor"
        stroke-width="2"
      >
        <polygon points="5 3 19 12 5 21 5 3" />
      </svg>
      <span>?????????</span>
    </button>
    <button
      type="button"
      class="action-btn btn-stop"
      id="stop-btn"
      onclick="stopVM('{{ vm_name }}')"
    >
      <svg
```

```
        viewBox="0 0 24 24"
        fill="none"
        stroke="currentColor"
        stroke-width="2"
      >
        <rect x="6" y="6" width="12" height="12" rx="1" />
      </svg>
      <span>??????????</span>
    </button>
    <button
      type="button"
      class="action-btn btn-reboot"
      id="reboot-btn"
      onclick="restartVM('{{ vm_name }}')"
    >
      <svg
        viewBox="0 0 24 24"
        fill="none"
        stroke="currentColor"
        stroke-width="2"
      >
        <path d="M23 4v6h-6" />
        <path d="M20.49 15a9 9 0 0 1 1-2.12-9.36L23 10" />
      </svg>
      <span>?????????????</span>
    </button>
    <button
      type="button"
      class="action-btn btn-delete"
      onclick="deleteVM('{{ vm_name }}')"
    >
      <svg
        viewBox="0 0 24 24"
        fill="none"
        stroke="currentColor"
        stroke-width="2"
      >
        <polyline points="3 6 5 6 21 6" />
        <path
          d="M19 6v14a2 2 0 0 1-2 2H7a2 2 0 0 1-2-2V6m3 0V4a2 2 0 0 1 2-2h4a2 2 0 0 1 2 2v2"
        />
      </svg>
      <span>???????</span>
    </button>
  </div>
</div>

<!-- Tabs -->
<div class="vm-tabs">
  <button class="tab-btn active" data-tab="overview">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <circle cx="12" cy="12" r="10" />
      <line x1="12" y1="16" x2="12" y2="12" />
      <line x1="12" y1="8" x2="12.01" y2="8" />
    </svg>
    ?????
  </button>
  <button class="tab-btn" data-tab="console">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <rect x="2" y="3" width="20" height="14" rx="2" />
      <path d="M8 21h8" />
      <path d="M12 17v4" />
    </svg>
    ???????
  </button>
  <button class="tab-btn" data-tab="metrics">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <line x1="12" y1="20" x2="12" y2="10" />
      <line x1="18" y1="20" x2="18" y2="4" />
      <line x1="6" y1="20" x2="6" y2="16" />
    </svg>
```

```
    ???????
  </button>
  <button class="tab-btn" data-tab="assistant">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <path d="M12 2a3 3 0 0 0-3 3v4a3 3 0 0 0 6 0V5a3 3 0 0 0-3-3Z" />
      <path d="M19 10v2a7 7 0 0 1-14 0v-2" />
      <line x1="12" x2="12" y1="19" y2="22" />
    </svg>
    AI ?????????
  </button>
</div>

<!-- Overview Tab -->
<section class="tab-content active" data-content="overview">
  <div class="info-grid">
    <div class="info-card">
      <div class="info-card-header">
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <circle cx="12" cy="12" r="10" />
          <line x1="12" y1="16" x2="12" y2="12" />
          <line x1="12" y1="8" x2="12.01" y2="8" />
        </svg>
        <h3>???????? ??????????</h3>
      </div>
      <div class="info-row">
        <span class="info-label">???</span>
        <span class="info-value">{{ vm_name }}</span>
      </div>
      <div class="info-row">
        <span class="info-label">Namespace</span>
        <span class="info-value">{{ namespace }}</span>
      </div>
      <div class="info-row">
        <span class="info-label">??????</span>
        <span class="info-value">
          <span class="status-badge" id="vm-status"
            >????????...</span
          >
          <span
            id="cloudinit-icon"
            class="cloudinit-icon hidden"
            title="Cloud-init ??????"
          ></span>
        </span>
      </div>
      <div class="info-row">
        <span class="info-label">??????</span>
        <span class="info-value">??????</span>
      </div>
      <div class="info-row" id="ip-row">
        <span class="info-label">IP ?????</span>
        <span class="info-value" id="vm-allocated-ip">-</span>
      </div>
    </div>

    <div class="info-card">
      <div class="info-card-header">
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <rect x="4" y="4" width="16" height="16" rx="2" />
          <rect x="9" y="9" width="6" height="6" />
          <path d="M9 1v3" />
```

```
          <path d="M15 1v3" />
          <path d="M9 20v3" />
          <path d="M15 20v3" />
          <path d="M20 9h3" />
          <path d="M20 14h3" />
          <path d="M1 9h3" />
          <path d="M1 14h3" />
        </svg>
        <h3>???????</h3>
      </div>
      <div class="info-row">
        <span class="info-label">CPU</span>
        <span class="info-value" id="vm-cpu">-</span>
      </div>
      <div class="info-row">
        <span class="info-label">??????</span>
        <span class="info-value" id="vm-memory">-</span>
      </div>
      <div class="info-row">
        <span class="info-label">????</span>
        <span class="info-value" id="vm-storage">-</span>
      </div>
      <div class="info-row hidden" id="gpu-row">
        <span class="info-label">GPU</span>
        <span class="info-value" id="vm-gpu">-</span>
      </div>
    </div>

    <!-- Network Access Card -->
    <div class="info-card">
      <div class="info-card-header">
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <path d="M5 12.55a11 11 0 0 1 14.08 0" />
          <path d="M1.42 9a16 16 0 0 1 21.16 0" />
          <path d="M8.53 16.11a6 6 0 0 1 6.95 0" />
          <line x1="12" y1="20" x2="12" y2="20" />
        </svg>
        <h3>??????? ??????</h3>
      </div>

      <div class="services-container">
        <!-- Predefined Buttons -->
        <div class="quick-ports">
           <button class="btn-service" onclick="addServiceMode('http', 80)">HTTP (80)</button>
           <button class="btn-service" onclick="addServiceMode('https', 443)">HTTPS (443)</button>
           <button class="btn-service" onclick="addServiceMode('ssh', 22)">SSH (22)</button>
           {% if vm_os and 'windows' in vm_os.lower() %}
           <button class="btn-service btn-service-rdp" onclick="addServiceMode('rdp', 3389)">RDP (3389)</button>
           {% endif %}
        </div>

        <!-- Custom Port Form -->
        <div class="custom-port-form">
          <input type="number" id="custom-port-input" placeholder="???? (????. 8080)" min="1" max="65535" class="port...
          <button class="btn-add-custom-port" onclick="addCustomService()">+ ???????? ????</button>
        </div>

        <!-- Active Services List -->
        <div id="active-services-list" class="services-list">
             <!-- Populated by JS -->
             <div class="loading-services">???????? ??????...</div>
        </div>
      </div>
    </div>
  </div>
</section>
```

```
<!-- Console Tab -->
<section class="tab-content" data-content="console">
  <div class="console-container">
    <div class="console-header">
      <div class="console-title">
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <rect x="2" y="3" width="20" height="14" rx="2" />
          <path d="M8 21h8" />
          <path d="M12 17v4" />
        </svg>
        <h3>VNC-???????</h3>
      </div>
      <div class="console-status" id="console-status">
        <span class="status-dot"></span>
        ??????????...
      </div>
    </div>
    <iframe id="vnc-frame" class="vnc-iframe" title="VNC ???????"></iframe>
  </div>
</section>

<!-- Metrics Tab -->
<section class="tab-content" data-content="metrics">
  <div class="info-banner metrics-banner">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
      <line x1="12" y1="20" x2="12" y2="10" />
      <line x1="18" y1="20" x2="18" y2="4" />
      <line x1="6" y1="20" x2="6" y2="16" />
    </svg>
    <div>
      <h3>?????????? ????????</h3>
      <p>
        ?????????? ?????? 10 ??????. ????????? ???????????? metrics-server ?
        ????????.
      </p>
    </div>
  </div>

  <div class="metrics-grid">
    <div class="metric-card">
      <div class="metric-header">
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <rect x="4" y="4" width="16" height="16" rx="2" />
          <rect x="9" y="9" width="6" height="6" />
        </svg>
        <h3>???????? CPU</h3>
      </div>
      <div class="metric-value-wrapper">
        <span id="cpu-value" class="metric-value">-</span>
        <span class="metric-unit" id="cpu-unit">%</span>
      </div>
      <div class="metric-allocated" id="cpu-allocated"></div>
      <canvas id="cpu-chart" class="metric-chart"></canvas>
    </div>

    <div class="metric-card">
      <div class="metric-header">
        <svg
          viewBox="0 0 24 24"
          fill="none"
```

```
          stroke="currentColor"
          stroke-width="2"
        >
          <path d="M6 19v-3" />
          <path d="M10 19v-6" />
          <path d="M14 19V9" />
          <path d="M18 19V5" />
        </svg>
        <h3>????????????? ??????</h3>
      </div>
      <div class="metric-value-wrapper">
        <span id="memory-value" class="metric-value">-</span>
        <span class="metric-unit" id="memory-unit">GB</span>
      </div>
      <div class="metric-allocated" id="memory-allocated"></div>
      <p class="memory-note">???????? ??? ? ??????</p>
      <canvas id="memory-chart" class="metric-chart"></canvas>
    </div>
  </div>
</section>

<!-- AI Assistant Tab -->
<section class="tab-content" data-content="assistant">
  <div class="chat-container">
    <div class="chat-header">
      <div class="chat-title">
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <path d="M12 2a3 3 0 0 0-3 3v4a3 3 0 0 0 6 0V5a3 3 0 0 0-3-3Z" />
          <path d="M19 10v2a7 7 0 0 1-14 0v-2" />
          <line x1="12" x2="12" y1="19" y2="22" />
        </svg>
        <div>
          <h3>AI ?????????</h3>
          <span class="chat-model">{{ llm_model|default('qwen-coder-3b-instruct') }}</span>
        </div>
      </div>
      <button class="chat-clear-btn" id="chat-clear-btn" title="???????? ???">
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <polyline points="3 6 5 6 21 6" />
          <path
            d="M19 6v14a2 2 0 0 1-2 2H7a2 2 0 0 1-2-2V6m3 0V4a2 2 0 0 1 2-2h4a2 2 0 0 1 2 2v2"
          />
        </svg>
      </button>
    </div>

    <div class="chat-messages" id="chat-messages">
      <div class="chat-welcome">
        <div class="chat-welcome-icon">
          <svg
            viewBox="0 0 24 24"
            fill="none"
            stroke="currentColor"
            stroke-width="2"
          >
            <path d="M12 2a3 3 0 0 0-3 3v4a3 3 0 0 0 6 0V5a3 3 0 0 0-3-3Z" />
            <path d="M19 10v2a7 7 0 0 1-14 0v-2" />
            <line x1="12" x2="12" y1="19" y2="22" />
          </svg>
        </div>
        <h3>??????! ? AI-?????????</h3>
```

```
        <p>
          ?????? ? ?????????? ????? ?????????? ??????
          <strong>{{ vm_name }}</strong>. ??????? ????? ?????? ? Linux,
          ????????? ?????????????????? ??? DevOps.
        </p>
        <div class="chat-suggestions">
          <button
            class="suggestion-btn"
            data-suggestion="??? ????????? ????????? ????? ?? ??????"
          >
            ? ????????? ????? ?? ?????
          </button>
          <button
            class="suggestion-btn"
            data-suggestion="??? ????????? SSH-??????"
          >
            ? ????????? SSH
          </button>
          <button
            class="suggestion-btn"
            data-suggestion="??? ????????? Docker ?? Ubuntu?"
          >
            ? ????????? Docker
          </button>
          <button
            class="suggestion-btn"
            data-suggestion="??? ????????? firewall ?? Linux?"
          >
            ?? ????????? firewall
          </button>
        </div>
      </div>
    </div>

    <div class="chat-input-container">
      <textarea
        id="chat-input"
        class="chat-input"
        placeholder="???????? ???-??????..."
        rows="1"
      ></textarea>
      <button class="chat-send-btn" id="chat-send-btn" disabled>
        <svg
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
        >
          <line x1="22" y1="2" x2="11" y2="13" />
          <polygon points="22 2 15 22 11 13 2 9 22 2" />
        </svg>
      </button>
    </div>
  </div>
</section>
<!-- VM details data for JS (no inline script) -->
<div id="vm-details-data" class="hidden" data-vm-name="{{ vm_name }}" data-username="{{ username }}" data-namespace="...
{% endblock %} {% block extra_js %}
<script src="{{ url_for('static', path='js/pages/vm-details-init.js') }}"></script>
<script src="{{ url_for('static', path='js/vm-details.js') }}"></script>
{% endblock %}

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/templating.py (981.0 B)

```
"""Jinja2Templates subclass that emits path-only URLs.

Starlette's default ``url_for`` generates absolute URLs
(``https://host/static/css/?``).  Behind a Kubernetes ingress the
scheme/host derived from proxy headers is often wrong, causing the
browser to request CSS/JS from an unreachable internal address.

This subclass overrides ``url_for`` to return only the path component
(``/static/css/?``), which always resolves relative to the current
origin regardless of reverse-proxy configuration.
"""

from jinja2 import pass_context
from starlette.templating import Jinja2Templates as _Base


class Jinja2Templates(_Base):
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)

        @pass_context
        def url_for(context: dict, name: str, /, **path_params) -> str:
            request = context["request"]
            return request.url_for(name, **path_params).path

        self.env.globals["url_for"] = url_for

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/__init__.py (52.0 B)

```
"""
Utility package for kubevirt-api-manager
"""

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/async_helpers.py (802.0 B)

```
"""
Thin async wrappers for running synchronous Kubernetes SDK calls
off the event loop via ``asyncio.to_thread``.
"""

from __future__ import annotations

import asyncio
from functools import partial
from typing import Any, Callable, TypeVar

T = TypeVar("T")


async def run_sync(func: Callable[..., T], *args: Any, **kwargs: Any) -> T:
    """Run a blocking function in a thread-pool executor.

    This keeps the event loop free while the synchronous ``kubernetes``
    client makes HTTP calls to the API server.

    Usage::

        from app.utils.k8s_utils import start_virtual_machine
        ok = await run_sync(start_virtual_machine, namespace, vm_name)
    """
    if kwargs:
        func = partial(func, **kwargs)
    return await asyncio.to_thread(func, *args)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/console_utils.py (3.2 KB)

```python
"""
Replace ``subprocess.run(["kubectl", ...])`` calls with native K8s Python API.

The old ``vm_details_routes.py`` shelled out to ``kubectl`` to:
  1. Find the virt-launcher pod for a VM
  2. Read ``guest-console-log`` container logs to detect cloud-init completion

This module provides the same functionality via the Kubernetes Python client,
wrapped with ``run_sync`` so callers can ``await`` from async context.
"""

from __future__ import annotations

import logging
from typing import Tuple

from app.config import FRONTEND_STUB_MODE
from app.utils.async_helpers import run_sync
from app.utils.k8s_utils import get_core_api

logger = logging.getLogger(__name__)


def _find_launcher_pod_name(namespace: str, vm_name: str) -> str | None:
    """Return the virt-launcher pod name for *vm_name*, or ``None``."""
    if FRONTEND_STUB_MODE:
        return None
    core = get_core_api()
    try:
        pods = core.list_namespaced_pod(
            namespace=namespace,
            label_selector=f"kubevirt.io/domain={vm_name}",
            limit=1,
        )
        if pods.items:
            return pods.items[0].metadata.name
    except Exception as exc:
        logger.debug("Could not find launcher pod for %s: %s", vm_name, exc)
    return None


def _read_guest_console_log(
    namespace: str,
    pod_name: str,
    tail_lines: int = 100,
) -> str:
    """Read the last *tail_lines* lines from the ``guest-console-log`` container."""
    core = get_core_api()
    try:
        return core.read_namespaced_pod_log(
            name=pod_name,
            namespace=namespace,
            container="guest-console-log",
            tail_lines=tail_lines,
        ) or ""
    except Exception as exc:
        logger.debug("Could not read guest-console-log for %s: %s", pod_name, exc)
        return ""


def _detect_cloudinit_finished(console_log: str, is_windows: bool) -> bool:
    """Parse console log output and return whether cloud-init / Cloudbase-Init completed."""
    if not console_log:
        return False

```

```
    if is_windows:
        return any(marker in console_log for marker in (
            "Cloudbase-Init complete",
            "Execution of Cloudbase-Init is done",
            "Cloudbase-Init finished",
        ))

    return "Cloud-init v." in console_log and "finished at" in console_log


def get_cloudinit_status_sync(
    namespace: str,
    vm_name: str,
    is_windows: bool = False,
) -> Tuple[bool, str | None]:
    """Synchronous helper that checks cloud-init/cloudbase-init completion.

    Returns ``(finished: bool, console_log: str | None)``.
    """
    pod_name = _find_launcher_pod_name(namespace, vm_name)
    if not pod_name:
        return False, None
    log_text = _read_guest_console_log(namespace, pod_name)
    finished = _detect_cloudinit_finished(log_text, is_windows)
    return finished, log_text


async def get_cloudinit_status(
    namespace: str,
    vm_name: str,
    is_windows: bool = False,
) -> Tuple[bool, str | None]:
    """Async wrapper ? calls the sync helper off the event loop."""
    return await run_sync(get_cloudinit_status_sync, namespace, vm_name, is_windows)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/datavolume_utils.py (2.9 KB)

```
"""
????? ????????? ? ??????? ??? DataVolume (CDI): ????, ????? ?? VM spec, ??????? ?????????.
??????????? ? k8s_utils ??? ??????? ???????? ??????.
"""

# ???? CDI DataVolume, ??? ??????? ??????/???? ??? ???? (?? ?????????)
CDI_IN_PROGRESS_PHASES = (
    "Pending",
    "PVCBound",
    "WaitForFirstConsumer",
    "WaitingForVolumeBinding",
    "ImportScheduled",
    "ImportInProgress",
    "CloneScheduled",
    "CloneInProgress",
    "UploadScheduled",
    "UploadInProgress",
    "Unknown",
    "",
)


def rootdisk_dv_name(vm_name: str) -> str:
    """??? DataVolume rootdisk ??? VM (??????????: <vm_name>-rootdisk)."""
    return f"{vm_name}-rootdisk"


def get_vm_datavolume_names(vm: dict) -> list:
    """
    ?????? ???? DataVolumes, ??????????? ? VM (?? spec.template.spec.volumes).
    ?????????? ?????? ?????? ??? ?????? ??? ?????????? dataVolume.
    """
    if not vm:
        return []
    try:
        volumes = (
            vm.get("spec", {})
            .get("template", {})
            .get("spec", {})
            .get("volumes", [])
        )
        names = []
        for vol in volumes:
            if "dataVolume" in vol and vol["dataVolume"].get("name"):
                names.append(vol["dataVolume"]["name"])
        return names
    except Exception:
        return []


def parse_dv_progress_pct(progress_str: str) -> int | None:
    """
    ?????? ?????? ????????? DV (???????? "27.10%" ??? "27.10") ? ????? ??????? 0..99.
    ?????????? None ??? ??????.
    """
    if progress_str is None:
        return None
    try:
        pct = float(str(progress_str).strip().rstrip("%"))
        return min(99, max(0, int(pct)))
    except (ValueError, TypeError):
        return None


def get_dv_fallback_status(dv_names: list, datavolumes: dict) -> str | None:
```

```
    """
    ????????? ????????? DataVolumes ??? VM.
    - ???? ?????-?? DV ? ?????? (?? Succeeded ? ?? in_progress) ? "DataVolume <phase>".
    - ???? ?????-?? DV ??? ? ???????? ???????? ? "?????????".
    - None ?????? ????? ??? DV ? Succeeded (??? ?????? ????).
    """
    has_in_progress = False
    for name in dv_names:
        if name not in datavolumes:
            has_in_progress = True
            continue
        phase = datavolumes[name].get("status", {}).get("phase", "")
        if phase == "Succeeded":
            continue
        if phase in CDI_IN_PROGRESS_PHASES:
            has_in_progress = True
            continue
        return f"DataVolume {phase}"
    if has_in_progress:
        return "?????????"
    return None

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/juniper_utils.py (3.5 KB)

```python
import logging

import requests

from app.config import JUNIPER_SIDECAR_URL

logger = logging.getLogger(__name__)


def add_port_forward(
    vm_name, dest_ip, port, service_name=None, external_ip=None, namespace=None
):
    """
    Add port forwarding rule to Juniper via Sidecar.

    Args:
        vm_name: Name of the VM
        dest_ip: Internal node IP
        port: NodePort
        service_name: e.g. 'ssh', 'http', 'rdp' for NAT naming
        external_ip: External IP to use for NAT (from pool)
        namespace: Kubernetes namespace (for DNAT naming: k8s-kvirt_{vm}_{ns}_{svc}_{port})
    """
    url = f"{JUNIPER_SIDECAR_URL}/add_rule"
    payload = {"vm_name": vm_name, "dest_ip": dest_ip, "port": port}
    if service_name:
        payload["service_name"] = service_name
    if external_ip:
        payload["external_ip"] = external_ip
    if namespace:
        payload["namespace"] = namespace

    logger.info(f"Requesting Juniper Sidecar add_rule: {payload}")

    try:
        resp = requests.post(url, json=payload, timeout=60)
        resp.raise_for_status()
        data = resp.json()

        # ??????????? ????? ??? ?????? ????????? ? ?????
        output = data.get("output", "")
        status = data.get("status", "unknown")

        if status == "success":
            logger.info(
                f"? NAT rule added successfully for {vm_name} -> {dest_ip}:{port}"
            )
            # ???????? ??????? ????????? ?????? ??????? ??????
            if "commit complete" in output:
                logger.info("  Juniper commit completed")
        else:
            logger.warning(f"? NAT rule addition failed for {vm_name}: {status}")

        return True, output
    except Exception as e:
        logger.exception("Failed to call Juniper Sidecar (Add Rule)")
        return False, str(e)


def delete_port_forward(
    vm_name, dest_ip, port, service_name=None, external_ip=None, namespace=None
):
    """
    Delete port forwarding rule from Juniper via Sidecar.

```

```
    Args:
        vm_name: Name of the VM
        dest_ip: Internal node IP
        port: NodePort
        service_name: e.g. 'ssh', 'http', 'rdp' for NAT naming
        external_ip: External IP used for NAT (from pool)
        namespace: Kubernetes namespace (for DNAT naming: k8s-kvirt_{vm}_{ns}_{svc}_{port})
    """
    url = f"{JUNIPER_SIDECAR_URL}/delete_rule"
    payload = {"vm_name": vm_name, "dest_ip": dest_ip, "port": port}
    if service_name:
        payload["service_name"] = service_name
    if external_ip:
        payload["external_ip"] = external_ip
    if namespace:
        payload["namespace"] = namespace

    logger.info(f"Requesting Juniper Sidecar delete_rule: {payload}")

    try:
        resp = requests.post(url, json=payload, timeout=60)
        resp.raise_for_status()
        data = resp.json()

        # ??????????? ????? ??? ?????? ?????????? ? ?????
        output = data.get("output", "")
        status = data.get("status", "unknown")

        if status == "success":
            logger.info(
                f"? NAT rule deleted successfully for {vm_name} -> {dest_ip}:{port}"
            )
            if "commit complete" in output:
                logger.info("  Juniper commit completed")
        else:
            logger.warning(f"? NAT rule deletion failed for {vm_name}: {status}")

        return True, output
    except Exception as e:
        logger.exception("Failed to call Juniper Sidecar (Delete Rule)")
        return False, str(e)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/k8s_utils.py (50.3 KB)

```
"""
??????? ??? ?????? ? Kubernetes ? KubeVirt: VM, DataVolume, namespace, start/stop subresource.
"""

import copy
import json
import logging
import os
from pathlib import Path

try:
    from kubernetes.utils.quantity import parse_quantity
except Exception:
    # Fallback when kubernetes.utils.quantity is missing or broken (e.g. some installs)
    def _parse_quantity(qty):
        from decimal import Decimal

        s = str(qty).strip()
        if not s:
            raise ValueError("Empty quantity")
        if s[-1] in "iI" and len(s) >= 2 and s[-2] in "KMGTPE":
            return int(
                Decimal(s[:-2])
                * (
                    1024
                    ** {"K": 1, "M": 2, "G": 3, "T": 4, "P": 5, "E": 6}[s[-2].upper()]
                )
            )
        if s[-1] in "KMGTPEkmgtpe":
            return int(
                Decimal(s[:-1])
                * (
                    1000
                    ** {"k": 1, "m": 2, "g": 3, "t": 4, "p": 5, "e": 6}[s[-1].lower()]
                )
            )
        return int(Decimal(s))

    parse_quantity = _parse_quantity

from app.config import (
    FRONTEND_STUB_MODE,
    STORAGE_CLASS_NAME,
)
from app.scheduling_config import get_storage_node_placement
from app.utils.datavolume_utils import (
    get_dv_fallback_status,
    get_vm_datavolume_names,
    parse_dv_progress_pct,
    rootdisk_dv_name,
)
from kubernetes import client, config

logger = logging.getLogger(__name__)


STUB_GPU_RESOURCES = [
    "nvidia.com/a100-80gb",
    "nvidia.com/l4",
    "amd.com/mi300",
    "intel.com/gpu-max",
]


def _build_stub_vm(
```

```
    name,
    printable_status,
    cpu_cores,
    memory_gi,
    storage_gi,
    gpu_resource=None,
    gpu_count=0,
    datavolume_status=None,
):
    """???????? ?????? VM ??? stub-?????? (?????? ?????? ? ?????? API)."""
    vm = {
        "metadata": {"name": name, "namespace": "", "uid": f"stub-{name}"},
        "status": {"printableStatus": printable_status},
        "spec": {
            "template": {
                "spec": {
                    "domain": {
                        "cpu": {"cores": cpu_cores},
                        "resources": {"requests": {"memory": f"{memory_gi}Gi"}},
                        "devices": {
                            "disks": [{"name": "rootdisk", "disk": {"bus": "virtio"}}]
                        },
                    },
                    "volumes": [
                        {
                            "name": "rootdisk",
                            "persistentVolumeClaim": {"claimName": f"{name}-rootdisk"},
                        }
                    ],
                }
            }
        },
        "datavolume_status": datavolume_status,
    }

    if gpu_resource and gpu_count > 0:
        vm["spec"]["template"]["spec"]["domain"]["devices"]["gpus"] = [
            {"name": f"gpu{i}", "deviceName": gpu_resource} for i in range(gpu_count)
        ]

    return vm


_STUB_VM_FILE = Path(__file__).resolve().parent.parent / ".stub_vms.json"


DEFAULT_STUB_VMS = [
    _build_stub_vm(
        name="demo-gpu-vm",
        printable_status="Running",
        cpu_cores=8,
        memory_gi=32,
        storage_gi=200,
        gpu_resource=STUB_GPU_RESOURCES[0],
        gpu_count=1,
    ),
    _build_stub_vm(
        name="data-import-vm",
        printable_status="Provisioning",
        cpu_cores=4,
        memory_gi=16,
        storage_gi=100,
        datavolume_status="Creating... DataVolume ImportInProgress 72%",
    ),
    _build_stub_vm(
        name="analytics-vm",
        printable_status="Stopped",
        cpu_cores=6,
        memory_gi=24,
        storage_gi=150,
    ),
```

```
]


def _load_stub_vms() -> list[dict]:
    try:
        if _STUB_VM_FILE.exists():
            data = json.loads(_STUB_VM_FILE.read_text())
            if isinstance(data, list) and data:
                return data
        return copy.deepcopy(DEFAULT_STUB_VMS)
    except Exception:
        return copy.deepcopy(DEFAULT_STUB_VMS)


def _save_stub_vms() -> None:
    try:
        _STUB_VM_FILE.write_text(json.dumps(STUB_VM_TEMPLATES, indent=2))
    except Exception as e:
        logger.warning("Failed to persist stub VMs: %s", e)


STUB_VM_TEMPLATES = _load_stub_vms()


def init_k8s_client():
    """???????????? ??????? K8s: in-cluster (??? ? ????????) ??? kubeconfig (????????)."""
    if FRONTEND_STUB_MODE:
        logger.warning(
            "FRONTEND_STUB_MODE enabled - skipping Kubernetes client initialization"
        )
        return True

    try:
        if os.getenv("KUBERNETES_SERVICE_HOST"):
            config.load_incluster_config()
        else:
            config.load_kube_config()
        return True
    except Exception as e:
        logger.error(f"Error loading k8s config: {e}")
        return False


def get_core_api():
    """?????? CoreV1Api (Pods, Services, Namespaces ? ?.?.)."""
    return client.CoreV1Api()


def get_custom_api():
    """?????? CustomObjectsApi (VirtualMachines, DataVolumes ? ??. CRD)."""
    return client.CustomObjectsApi()


def discover_gpu_resources():
    """?????? GPU-???????? ?? allocatable/capacity ??? (nvidia/amd/intel). ????????? kvm/tun/vhost-net."""
    if FRONTEND_STUB_MODE:
        return STUB_GPU_RESOURCES

    api = get_core_api()
    try:
        nodes = api.list_node().items
    except Exception as exc:  # pragma: no cover - defensive logging
        logger.error(f"Error discovering GPU resources from nodes: {exc}")
        return []

    gpu_resources = {}

    # List of KubeVirt device resources that are NOT GPUs
    kubevirt_non_gpu_devices = {
        "devices.kubevirt.io/kvm",
        "devices.kubevirt.io/tun",
```

```
        "devices.kubevirt.io/vhost-net",
    }

    def record_resources(resource_map):
        for name, qty in resource_map.items():
            # Filter only real GPU resources
            is_nvidia = name.startswith("nvidia.com/")
            is_amd = name.startswith("amd.com/")
            is_intel = name.startswith("intel.com/gpu")

            # Skip if not a GPU vendor resource
            if not (is_nvidia or is_amd or is_intel):
                continue

            # Skip KubeVirt device plugins (kvm, tun, vhost-net)
            if name in kubevirt_non_gpu_devices:
                continue

            # Skip audio devices (mounted automatically with GPU, not shown in UI)
            if "audio" in name.lower():
                continue

            # Skip generic nvidia.com/gpu resource if present (prefer specific models)
            if name == "nvidia.com/gpu":
                continue

            try:
                numeric_qty = parse_quantity(qty)
            except Exception:
                try:
                    numeric_qty = int(qty)
                except Exception:
                    continue

            if numeric_qty <= 0:
                continue

            gpu_resources[name] = max(gpu_resources.get(name, 0), numeric_qty)

    for node in nodes:
        status = node.status or {}
        for source in (
            getattr(status, "allocatable", {}) or {},
            getattr(status, "capacity", {}) or {},
        ):
            record_resources(source)

    return sorted(gpu_resources.keys())


def ensure_namespace(namespace_name):
    """??????? namespace ??? ?????????? ? ??????? NetworkPolicy ??? ???????? VM."""
    if FRONTEND_STUB_MODE:
        logger.info(f"[stub] Pretending namespace {namespace_name} exists")
        return True

    api = get_core_api()

    try:
        api.read_namespace(name=namespace_name)
        logger.info(f"Namespace {namespace_name} already exists.")
    except client.exceptions.ApiException as e:
        if e.status == 404:
            logger.info(f"Namespace {namespace_name} not found. Creating...")
            ns_body = client.V1Namespace(
                metadata=client.V1ObjectMeta(name=namespace_name)
            )
            try:
                api.create_namespace(body=ns_body)
                logger.info(f"Namespace {namespace_name} created.")
            except Exception as create_error:
```

```
                logger.error(f"Failed to create namespace: {create_error}")
                return False
        else:
            logger.error(f"Error checking namespace: {e}")
            return False

    # Ensure NetworkPolicy exists for the namespace
    try:
        from app.utils.network_policy_utils import ensure_namespace_network_policy

        ensure_namespace_network_policy(namespace_name)
    except Exception as np_error:
        logger.warning(
            f"Failed to create NetworkPolicy for namespace {namespace_name}: {np_error}"
        )
        # Don't fail namespace creation if NetworkPolicy fails

    return True


def vm_exists_in_namespace(namespace, vm_name):
    """Check if a VM with given name exists in namespace"""
    if FRONTEND_STUB_MODE:
        existing_names = [vm["metadata"]["name"] for vm in STUB_VM_TEMPLATES]
        return vm_name in existing_names

    api = get_custom_api()
    try:
        api.get_namespaced_custom_object(
            group="kubevirt.io",
            version="v1",
            namespace=namespace,
            plural="virtualmachines",
            name=vm_name,
        )
        return True
    except client.exceptions.ApiException as e:
        if e.status == 404:
            return False
        logger.error(f"Error checking VM existence: {e}")
        return False
    except Exception as e:
        logger.error(f"Unexpected error checking VM existence: {e}")
        return False


def list_vms_in_namespace(namespace):
    """List all VirtualMachines in a namespace with DataVolume status"""
    if FRONTEND_STUB_MODE:
        vms = []
        for template in STUB_VM_TEMPLATES:
            vm = copy.deepcopy(template)
            vm["metadata"]["namespace"] = namespace
            vms.append(vm)
        return vms

    api = get_custom_api()
    try:
        # ???????? VM
        vm_resource = api.list_namespaced_custom_object(
            group="kubevirt.io",
            version="v1",
            namespace=namespace,
            plural="virtualmachines",
        )
        vms = vm_resource.get("items", [])
        datavolumes = _fetch_datavolumes_for_namespace(namespace)
        if not datavolumes:
            logger.warning("No DataVolumes found in namespace %s", namespace)

        # ????????? VM ??????????? ? DataVolume (?????? ?????? ? get_datavolume_status_for_vm)
```

```
        for vm in vms:
            if "status" not in vm:
                vm["status"] = {}
            if "printableStatus" not in vm.get("status", {}):
                vm["status"]["printableStatus"] = None
            vm["datavolume_status"] = get_datavolume_status_for_vm(
                namespace, vm, datavolumes=datavolumes
            )
        return vms
    except Exception as e:
        logger.error(f"Error fetching VMs: {e}")
        return []


def _fetch_datavolumes_for_namespace(namespace, ensure_names=None):
    """
    ?????? DataVolumes ? namespace (cdi.kubevirt.io v1beta1/v1).
    ???? ??????? ensure_names, ??????????? ???????????????? get ?? ?????.
    ?????????? dict: ??? DV -> ?????? DV.
    """
    if FRONTEND_STUB_MODE:
        return {}
    api = get_custom_api()
    datavolumes = {}
    for cdi_version in ("v1beta1", "v1"):
        try:
            dvs = api.list_namespaced_custom_object(
                group="cdi.kubevirt.io",
                version=cdi_version,
                namespace=namespace,
                plural="datavolumes",
            )
            datavolumes = {dv["metadata"]["name"]: dv for dv in dvs.get("items", [])}
            if datavolumes:
                break
        except Exception as e:
            logger.debug("DataVolumes list cdi.kubevirt.io/%s: %s", cdi_version, e)
    if ensure_names:
        for name in ensure_names:
            if name in datavolumes:
                continue
            for cdi_version in ("v1beta1", "v1"):
                try:
                    dv = api.get_namespaced_custom_object(
                        group="cdi.kubevirt.io",
                        version=cdi_version,
                        namespace=namespace,
                        plural="datavolumes",
                        name=name,
                    )
                    datavolumes[name] = dv
                    break
                except Exception:
                    continue
    return datavolumes


def _importer_pod_exists(namespace, dv_name):
    """?????????, ???? ?? ??? ??????? CDI ??? ??????? DataVolume (importer-<dv_name> ??? importer-<dv_name>-<suffix>)...
    if FRONTEND_STUB_MODE:
        return False
    try:
        core_api = get_core_api()
        pods = core_api.list_namespaced_pod(namespace=namespace, label_selector=None)
        prefix_exact = f"importer-{dv_name}"
        prefix_with_dash = f"importer-{dv_name}-"
        for p in pods.items:
            name = p.metadata.name
            if name == prefix_exact or name.startswith(prefix_with_dash):
                return True
        return False
```

```
    except Exception:
        return False


def _get_importer_log_progress(namespace, dv_name):
    """
    ???? ??? ??????? CDI ??? DataVolume (importer-<dv_name> ??? importer-<dv_name>-<suffix>),
    ?????? ???? ?????????? ? ?????????? ???????? ?? ?????:
    - (qemu_pct, True) ???? ????????? ?????? ????????? ? qemu.go:283] XX.XX (???????)
    - (None, True) ???? ????????? ?????? ? prometheus.go:78] (???? ??? ????????, ?????????? «?????????»)
    - (None, False) ???? ??? ?? ?????? ??? ????? ???.
    """
    import re

    if FRONTEND_STUB_MODE:
        return None, False
    core_api = get_core_api()
    try:
        pods = core_api.list_namespaced_pod(
            namespace=namespace,
            label_selector=None,
        )
        # CDI: ??? ????? ?????????? importer-<dv_name> ??? importer-<dv_name>-<suffix>
        prefix_exact = f"importer-{dv_name}"
        prefix_with_dash = f"importer-{dv_name}-"
        importer_pod = None
        for p in pods.items:
            name = p.metadata.name
            if name == prefix_exact or name.startswith(prefix_with_dash):
                importer_pod = p
                break
        if not importer_pod:
            return None, False
        pod_name = importer_pod.metadata.name
        container_name = "importer"
        try:
            log_stream = core_api.read_namespaced_pod_log(
                name=pod_name,
                namespace=namespace,
                container=container_name,
                tail_lines=500,
            )
        except Exception:
            try:
                log_stream = core_api.read_namespaced_pod_log(
                    name=pod_name,
                    namespace=namespace,
                    container="import",
                    tail_lines=500,
                )
            except Exception:
                return None, False
        if not log_stream:
            return None, False
        lines = (log_stream or "").strip().split("\n")
        qemu_re = re.compile(r"qemu\.go:\d+\]\s*([\d.]+)")
        prometheus_re = re.compile(r"prometheus\.go:\d+\]\s*([\d.]+)")
        last_qemu_pct = None
        last_qemu_line_no = -1
        last_prometheus_line_no = -1
        for i, line in enumerate(lines):
            m = qemu_re.search(line)
            if m:
                try:
                    last_qemu_pct = float(m.group(1))
                    last_qemu_line_no = i
                except (ValueError, TypeError):
                    pass
            if prometheus_re.search(line):
                last_prometheus_line_no = i
        if last_qemu_line_no >= 0:
```

```
            pct = int(round(last_qemu_pct)) if last_qemu_pct is not None else None
            if pct is not None:
                return min(99, max(0, pct)), True
        if last_prometheus_line_no >= 0:
            return None, True
        return None, False
    except Exception as e:
        logger.debug("Importer log progress for DV %s: %s", dv_name, e)
        return None, False

... [977 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/llm_key_crypto.py (1.2 KB)

```python
from __future__ import annotations

import base64
import hashlib


def _derive_keystream(secret: str, size: int) -> bytes:
    if not secret:
        raise ValueError("LLM_KEYS_ENCRYPTION_KEY is not configured")
    seed = secret.encode("utf-8")
    out = bytearray()
    counter = 0
    while len(out) < size:
        digest = hashlib.sha256(seed + counter.to_bytes(4, "big")).digest()
        out.extend(digest)
        counter += 1
    return bytes(out[:size])


def encrypt_token(plain: str, secret: str) -> str:
    raw = plain.encode("utf-8")
    key = _derive_keystream(secret, len(raw))
    encrypted = bytes(a ^ b for a, b in zip(raw, key))
    return base64.urlsafe_b64encode(encrypted).decode("ascii")


def decrypt_token(ciphertext: str, secret: str) -> str:
    enc = base64.urlsafe_b64decode(ciphertext.encode("ascii"))
    key = _derive_keystream(secret, len(enc))
    plain = bytes(a ^ b for a, b in zip(enc, key))
    return plain.decode("utf-8")


def mask_token(token: str) -> str:
    token = token.strip()
    if len(token) <= 10:
        return "*" * len(token)
    return f"{token[:5]}{'*' * (len(token) - 9)}{token[-4:]}"

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/network_policy_utils.py (12.1 KB)

```python
"""
??????? ????????: NetworkPolicy ?? namespace ? GlobalNetworkPolicy ??? ???????? VM ?? ?????????? ?????.
"""

import logging
from typing import List

from kubernetes.client import NetworkingV1Api
from kubernetes.client.rest import ApiException

from app.config import FRONTEND_STUB_MODE
from app.utils.k8s_utils import get_custom_api
from kubernetes import client

logger = logging.getLogger(__name__)


def create_vm_namespace_network_policy(namespace: str) -> bool:
    """??????? NetworkPolicy ? namespace: ingress 22/80/443 ? egress ? ??? ?? namespace + ??????."""
    if FRONTEND_STUB_MODE:
        logger.info(f"[stub] Would create NetworkPolicy for namespace {namespace}")
        return True

    try:
        # Use NetworkingV1Api for NetworkPolicy operations
        api = NetworkingV1Api()

        # Check if NetworkPolicy already exists and validate it
        existing_policy = None
        try:
            existing_policy = api.read_namespaced_network_policy(
                namespace=namespace, name="vm-namespace-policy"
            )
            logger.info(
                f"NetworkPolicy vm-namespace-policy already exists in {namespace}, validating..."
            )

            # Validate existing policy - check if it has correct ingress rules (ports 22, 80, 443)
            needs_update = False

            # Check ingress rules
            if (
                not existing_policy.spec.ingress
                or len(existing_policy.spec.ingress) == 0
            ):
                logger.warning(
                    f"Existing NetworkPolicy in {namespace} has no ingress rules, will update"
                )
                needs_update = True
            else:
                # Check if we have rules with ports 22, 80, 443
                has_correct_ports = False
                for rule in existing_policy.spec.ingress:
                    if rule.ports:
                        port_numbers = [
                            p.port for p in rule.ports if p.port in [22, 80, 443]
                        ]
                        if len(port_numbers) == 3:
                            has_correct_ports = True
                            break

                if not has_correct_ports:
                    logger.warning(
                        f"Existing NetworkPolicy in {namespace} doesn't have correct port rules (22, 80, 443), will u...
                    )
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                    needs_update = True

            # Check egress rules - should allow same namespace pods and external
            if not existing_policy.spec.egress or len(existing_policy.spec.egress) == 0:
                logger.warning(
                    f"Existing NetworkPolicy in {namespace} has no egress rules, will update"
                )
                needs_update = True
            else:
                has_namespace_egress = False
                has_external_egress = False
                for rule in existing_policy.spec.egress:
                    if rule.to and len(rule.to) > 0:
                        # Check if rule allows same namespace pods
                        for peer in rule.to:
                            if peer.pod_selector:
                                has_namespace_egress = True
                                break
                    elif not rule.to or len(rule.to) == 0:
                        # Empty rule means allow all external
                        has_external_egress = True

                if not (has_namespace_egress and has_external_egress):
                    logger.warning(
                        f"Existing NetworkPolicy in {namespace} doesn't have correct egress rules (needs namespace po...
                    )
                    needs_update = True

            if not needs_update:
                logger.info(
                    f"NetworkPolicy vm-namespace-policy in {namespace} is correct, no update needed"
                )
                return True
            else:
                logger.info(
                    f"Updating NetworkPolicy vm-namespace-policy in {namespace} (existing policy is incorrect)"
                )
                # Delete existing policy to recreate with correct rules
                try:
                    api.delete_namespaced_network_policy(
                        namespace=namespace, name="vm-namespace-policy"
                    )
                    logger.info(
                        f"Deleted existing NetworkPolicy vm-namespace-policy in {namespace}"
                    )
                    # Wait a moment for deletion to complete
                    import time

                    time.sleep(1)
                except ApiException as e:
                    logger.warning(
                        f"Error deleting existing NetworkPolicy: {e.reason}, will try to create anyway"
                    )
                    # Continue to create - if it exists, we'll get an error and can handle it
        except ApiException as e:
            if e.status != 404:
                raise

        # Create NetworkPolicy
        # Allow all ingress, but restrict egress to pods in the same namespace
        network_policy = client.V1NetworkPolicy(
            api_version="networking.k8s.io/v1",
            kind="NetworkPolicy",
            metadata=client.V1ObjectMeta(
                name="vm-namespace-policy",
                namespace=namespace,
                labels={
                    "app": "kubevirt-api-manager",
                    "policy-type": "vm-namespace-isolation",
                },
            ),
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            spec=client.V1NetworkPolicySpec(
                # Select VM pods (pods with kubevirt.io/domain label)
                pod_selector=client.V1LabelSelector(
                    match_expressions=[
                        client.V1LabelSelectorRequirement(
                            key="kubevirt.io/domain", operator="Exists"
                        )
                    ]
                ),
                policy_types=["Ingress", "Egress"],
                ingress=[
                    # Allow SSH (22), HTTP (80), and HTTPS (443) from any source
                    client.V1NetworkPolicyIngressRule(
                        ports=[
                            client.V1NetworkPolicyPort(port=22, protocol="TCP"),
                            client.V1NetworkPolicyPort(port=80, protocol="TCP"),
                            client.V1NetworkPolicyPort(port=443, protocol="TCP"),
                        ]
                    ),
                    # Allow all other incoming traffic from any source
                    client.V1NetworkPolicyIngressRule(),
                ],
                egress=[
                    # Allow egress to pods in the same namespace (for VM-to-VM communication)
                    client.V1NetworkPolicyEgressRule(
                        to=[
                            client.V1NetworkPolicyPeer(
                                pod_selector=client.V1LabelSelector(
                                    match_expressions=[
                                        client.V1LabelSelectorRequirement(
                                            key="kubevirt.io/domain", operator="Exists"
                                        )
                                    ]
                                )
                            )
                        ]
                    ),
                    # Allow egress to external networks (internet)
                    # GlobalNetworkPolicy will block Pod/Service/server networks
                    client.V1NetworkPolicyEgressRule(),
                ],
            ),
        )

        api.create_namespaced_network_policy(namespace=namespace, body=network_policy)
        logger.info(
            f"Created NetworkPolicy vm-namespace-policy in namespace {namespace}"
        )
        return True

    except ApiException as e:
        error_details = ""
        if e.body:
            import json

            try:
                error_body = json.loads(e.body) if isinstance(e.body, str) else e.body
                error_details = (
                    f" - Details: {error_body.get('message', str(error_body))}"
                )
            except Exception:
                error_details = f" - Body: {str(e.body)[:500]}"
        logger.error(
            f"Error creating NetworkPolicy: {e.reason} (Status: {e.status}){error_details}"
        )
        return False
    except Exception as e:
        logger.error(f"Unexpected error creating NetworkPolicy: {e}")
        return False
```

```
def create_vm_global_network_policy(
    pod_network_cidr: str, service_network_cidr: str, server_network_cidrs: List[str]
) -> bool:
    """????????? ???????? Calico: ?????? egress ?? VM-????? ? Pod/Service/????????? ???? (selector: kubevirt.io/doma...
    if FRONTEND_STUB_MODE:
        logger.info("[stub] Would create GlobalNetworkPolicy for VM pods")
        return True

    try:
        api = get_custom_api()

        # Check if GlobalNetworkPolicy already exists
        try:
            api.get_cluster_custom_object(
                group="crd.projectcalico.org",
                version="v1",
                plural="globalnetworkpolicies",
                name="vm-deny-internal-networks",
            )
            logger.info("GlobalNetworkPolicy vm-deny-internal-networks already exists")
            return True
        except ApiException as e:
            if e.status != 404:
                raise

        # Create GlobalNetworkPolicy
        global_network_policy = {
            "apiVersion": "crd.projectcalico.org/v1",
            "kind": "GlobalNetworkPolicy",
            "metadata": {
                "name": "vm-deny-internal-networks",
                "labels": {
                    "app": "kubevirt-api-manager",
                    "policy-type": "vm-network-restriction",
                },
            },
            "spec": {
                # Select VM pods only
                "selector": "has(kubevirt.io/domain)",
                "order": 1000,  # Lower order = higher priority
                "types": ["Ingress", "Egress"],
                "egress": [
                    # Deny egress to Pod network
                    {"action": "Deny", "destination": {"nets": [pod_network_cidr]}},
                    # Deny egress to Service network
                    {"action": "Deny", "destination": {"nets": [service_network_cidr]}},
                    # Deny egress to server networks
                    *[
                        {"action": "Deny", "destination": {"nets": [cidr]}}
                        for cidr in server_network_cidrs
                    ],
                    # Allow all other egress
                    {"action": "Allow"},
                ],
                "ingress": [
                    # Allow all ingress
                    {"action": "Allow"}
                ],
            },
        }

        api.create_cluster_custom_object(
            group="crd.projectcalico.org",
            version="v1",
            plural="globalnetworkpolicies",
            body=global_network_policy,
        )
        logger.info("Created GlobalNetworkPolicy vm-deny-internal-networks")
        return True

    except ApiException as e:
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```
        logger.error(f"Error creating GlobalNetworkPolicy: {e.reason}")
        if e.body:
            import json

            try:
                error_details = json.loads(e.body)
                logger.error(f"Error details: {error_details}")
            except Exception:
                logger.error(f"Error body: {e.body[:500]}")
        return False
    except Exception as e:
        logger.error(f"Unexpected error creating GlobalNetworkPolicy: {e}")
        return False


def ensure_namespace_network_policy(namespace: str) -> bool:
    """????????? ??????? NetworkPolicy ? namespace ? ??????? ??? ??????????."""
    return create_vm_namespace_network_policy(namespace)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/service_utils.py (21.8 KB)

```python
"""Utilities for managing a single NodePort Service per VM with multiple ports.
Each VM has at most one Kubernetes Service named `<vm>-service` that may contain
multiple port entries. Juniper NAT is configured per NodePort. When adding/removing
ports we only touch the specific NodePort for that operation.
"""

import json
import logging
import random
from typing import List, Optional, Tuple

from app.config import FRONTEND_STUB_MODE, JUNIPER_EXTERNAL_IP_POOL, STUB_NODE_IP
from app.utils.juniper_utils import add_port_forward, delete_port_forward
from app.utils.k8s_utils import get_core_api
from kubernetes import client

logger = logging.getLogger(__name__)


def _get_vm_service_name(vm_name: str) -> str:
    return f"{vm_name}-service"


def get_ip_allocation_stats() -> dict:
    """Get statistics of IP usage across all VMs.
    Returns dict: {ip: count_of_services_using_it}
    """
    if FRONTEND_STUB_MODE:
        stub_ip = JUNIPER_EXTERNAL_IP_POOL[0] if JUNIPER_EXTERNAL_IP_POOL else "10.0.0.1"
        return {stub_ip: 1}

    api = get_core_api()
    ip_usage = {ip: 0 for ip in JUNIPER_EXTERNAL_IP_POOL}

    try:
        svcs = api.list_service_for_all_namespaces().items
        for svc in svcs:
            if svc.spec.type == "NodePort" and svc.metadata.annotations:
                allocated_ip = svc.metadata.annotations.get("kubevirt.kv/allocated-ip")
                if allocated_ip and allocated_ip in ip_usage:
                    ip_usage[allocated_ip] += 1
    except Exception as e:
        logger.error(f"Failed to get IP allocation stats: {e}")

    return ip_usage


def get_least_used_ip() -> str:
    """Get IP address with least number of services.
    Returns IP from pool with minimal usage.
    """
    if FRONTEND_STUB_MODE:
        return JUNIPER_EXTERNAL_IP_POOL[0] if JUNIPER_EXTERNAL_IP_POOL else "10.0.0.1"

    ip_usage = get_ip_allocation_stats()
    # Return IP with minimum usage
    return min(ip_usage.items(), key=lambda x: x[1])[0]


def get_vm_allocated_ip(namespace: str, vm_name: str) -> Optional[str]:
    """Get IP address allocated to VM (from service annotation).
    Returns None if no service exists or no IP allocated.
    """
    if FRONTEND_STUB_MODE:
        return JUNIPER_EXTERNAL_IP_POOL[0] if JUNIPER_EXTERNAL_IP_POOL else "10.0.0.1"
```

```python

    service_name = _get_vm_service_name(vm_name)
    api = get_core_api()

    try:
        svc = api.read_namespaced_service(name=service_name, namespace=namespace)
        if svc.metadata.annotations:
            return svc.metadata.annotations.get("kubevirt.kv/allocated-ip")
    except client.exceptions.ApiException as e:
        if e.status == 404:
            return None
        logger.error(f"Failed to get VM allocated IP: {e}")
    except Exception as e:
        logger.error(f"Error getting VM allocated IP: {e}")

    return None


def get_used_node_ports() -> set:
    """Return set of NodePorts currently allocated in cluster."""
    if FRONTEND_STUB_MODE:
        return {31999}

    api = get_core_api()
    used = set()
    try:
        svcs = api.list_service_for_all_namespaces().items
        for svc in svcs:
            if svc.spec.type == "NodePort" and svc.spec.ports:
                for p in svc.spec.ports:
                    if getattr(p, "node_port", None):
                        used.add(p.node_port)
    except Exception as e:
        logger.error(f"Failed to list services for used ports: {e}")
    return used


def find_free_node_port(start: int = 30000, end: int = 31000) -> int:
    used = get_used_node_ports()
    for _ in range(100):
        c = random.randint(start, end)
        if c not in used:
            return c
    for p in range(start, end + 1):
        if p not in used:
            return p
    raise Exception(f"No free NodePorts in range {start}-{end}")


def get_ingress_node_ip() -> Optional[str]:
    if FRONTEND_STUB_MODE:
        return STUB_NODE_IP
    try:
        api = get_core_api()
        nodes = api.list_node().items
        nodes.sort(key=lambda x: x.metadata.name)
        for node in nodes:
            for addr in node.status.addresses:
                if addr.type == "InternalIP":
                    return addr.address
        if nodes:
            return nodes[0].status.addresses[0].address
    except Exception as e:
        logger.error(f"Failed to get ingress node IP: {e}")
    return None


def _create_vm_service(
    namespace: str,
    vm_name: str,
    target_port: int,
```

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    service_type: str,
    node_port: int,
    node_ip: str,
    external_ip: Optional[str] = None,
) -> Tuple[bool, str, Optional[int]]:
    """Create VM service with IP allocation from pool.

    Args:
        namespace: Kubernetes namespace
        vm_name: VM name
        target_port: Target port on VM
        service_type: Service type (ssh, http, etc)
        node_port: NodePort to allocate
        node_ip: Internal node IP
        external_ip: External IP to use (if None, allocates from pool)

    Returns:
        (success, message, assigned_node_port)
    """
    service_name = _get_vm_service_name(vm_name)
    api = get_core_api()

    # Allocate external IP from pool if not provided
    if external_ip is None:
        external_ip = get_least_used_ip()
        logger.info(f"Allocated IP {external_ip} from pool for VM {vm_name}")

    port_spec = {
        "name": service_type,
        "protocol": "TCP",
        "port": int(target_port),
        "targetPort": int(target_port),
        "nodePort": int(node_port),
    }

    manifest = {
        "apiVersion": "v1",
        "kind": "Service",
        "metadata": {
            "name": service_name,
            "namespace": namespace,
            "labels": {
                "kubevirt-manager.io/vm-name": vm_name,
                "kubevirt-manager.io/managed": "true",
            },
            "annotations": {
                "juniper.kubevirt.io/dest-ip": node_ip,
                "kubevirt.kv/allocated-ip": external_ip,  # Store allocated IP
                "kubevirt.kv/ports": json.dumps(
                    [{"name": service_type, "port": target_port, "nodePort": node_port}]
                ),
            },
        },
        "spec": {
            "type": "NodePort",
            "selector": {"vmi.kubevirt.io/id": vm_name},
            "ports": [port_spec],
        },
    }

    try:
        svc = api.create_namespaced_service(namespace=namespace, body=manifest)
        assigned = svc.spec.ports[0].node_port
        if not FRONTEND_STUB_MODE:
            logger.info(
                f"Configuring Juniper NAT for {service_name} on {external_ip}:{assigned} -> {node_ip}:{assigned}"
            )
            ok, msg = add_port_forward(
                vm_name, node_ip, assigned, service_type, external_ip, namespace
            )
            if not ok:
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                logger.error(f"Juniper add failed: {msg}")
                return (
                    True,
                    f"Service created, but Juniper config failed: {msg}",
                    assigned,
                )
        return True, "Service created and port added", assigned
    except client.exceptions.ApiException as e:
        logger.error(f"K8s API error creating service: {e}")
        if e.status == 409:
            return False, "Service already exists (race)", None
        return False, str(e), None
    except Exception as e:
        logger.error(f"Error creating service: {e}")
        return False, str(e), None


def add_port_to_vm_service(
    namespace: str,
    vm_name: str,
    target_port: int,
    service_type: str = "custom",
    node_port: Optional[int] = None,
) -> Tuple[bool, str, Optional[int]]:
    """Add a port to the single-per-VM Service. Creates the Service if missing.

    Returns (success, message, assigned_node_port)
    """
    service_name = _get_vm_service_name(vm_name)
    if FRONTEND_STUB_MODE:
        assigned = node_port or (31000 + (target_port % 1000))
        logger.info(f"STUB add port {target_port} -> {assigned} for {service_name}")
        return True, "Port added (stub)", assigned

    api = get_core_api()
    node_ip = get_ingress_node_ip() or "127.0.0.1"

    try:
        svc = api.read_namespaced_service(name=service_name, namespace=namespace)
        # service exists - ensure port not present
        for p in svc.spec.ports:
            if int(p.port) == int(target_port):
                return False, f"Port {target_port} already exposed", None

        # Get already allocated IP from service annotation
        external_ip = None
        if svc.metadata.annotations:
            external_ip = svc.metadata.annotations.get("kubevirt.kv/allocated-ip")

        if not external_ip:
            # This should not happen, but allocate if missing
            external_ip = get_least_used_ip()
            logger.warning(
                f"Service {service_name} missing allocated-ip annotation, allocating {external_ip}"
            )

        if node_port is None:
            try:
                node_port = find_free_node_port(30000, 31000)
            except Exception as e:
                return False, f"Port allocation failed: {e}", None

        # Create clean port dicts manually to avoid serialization issues
        ports_dicts = []
        for p in (svc.spec.ports if svc.spec.ports else []):
            ports_dicts.append(
                {
                    "name": p.name,
                    "protocol": p.protocol,
                    "port": int(p.port),
                    "targetPort": (
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                        int(p.target_port)
                        if hasattr(p, "target_port") and p.target_port
                        else int(p.port)
                    ),
                    "nodePort": int(p.node_port),
                }
            )

        # Add new port
        ports_dicts.append(
            {
                "name": service_type,
                "protocol": "TCP",
                "port": int(target_port),
                "targetPort": int(target_port),
                "nodePort": int(node_port),
            }
        )

        # Update annotation ports mapping
        annotations = svc.metadata.annotations or {}
        ports_meta = []
        if "kubevirt.kv/ports" in annotations:
            try:
                ports_meta = json.loads(annotations["kubevirt.kv/ports"])
            except Exception:
                ports_meta = []
        ports_meta.append(
            {"name": service_type, "port": int(target_port), "nodePort": int(node_port)}
        )
        annotations["kubevirt.kv/ports"] = json.dumps(ports_meta)

        body = {
            "spec": {"ports": ports_dicts},
            "metadata": {"annotations": annotations},
        }
        logger.info(
            f"Patching service {service_name} to add port {target_port}, total ports: {len(ports_dicts)}"
        )
        updated = api.patch_namespaced_service(
            name=service_name, namespace=namespace, body=body
        )

        assigned = None
        for p in updated.spec.ports:
            if int(p.port) == int(target_port):
                assigned = p.node_port
                break

        # Configure Juniper
        if not FRONTEND_STUB_MODE and assigned:
            ok, msg = add_port_forward(
                vm_name, node_ip, assigned, service_type, external_ip, namespace
            )
            if not ok:
                logger.error(f"Juniper add failed for {assigned}: {msg}")
                return True, f"Port added, but Juniper config failed: {msg}", assigned

        return True, f"Port {target_port} added", assigned

    except client.exceptions.ApiException as e:
        if e.status == 404:
            # create new service
            if node_port is None:
                try:
                    node_port = find_free_node_port(30000, 31000)
                except Exception as ex:
                    return False, f"Port allocation failed: {ex}", None
            return _create_vm_service(
                namespace, vm_name, target_port, service_type, node_port, node_ip
            )
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        logger.error(f"Error reading service: {e}")
        return False, str(e), None
    except Exception as e:
        logger.error(f"Error adding port to service: {e}")
        return False, str(e), None


def remove_port_from_vm_service(
    namespace: str, vm_name: str, target_port: int
) -> Tuple[bool, str]:
    """Remove a port from VM's service; delete service if no ports left."""
    service_name = _get_vm_service_name(vm_name)
    if FRONTEND_STUB_MODE:
        logger.info(f"STUB remove port {target_port} for {service_name}")
        return True, "Port removed (stub)"

    api = get_core_api()
    try:
        svc = api.read_namespaced_service(name=service_name, namespace=namespace)
        ports = list(svc.spec.ports) if svc.spec.ports else []
        remaining = []
        node_port_to_remove = None
        for p in ports:
            if int(p.port) == int(target_port):
                node_port_to_remove = p.node_port
            else:
                remaining.append(p)

        if node_port_to_remove is None:
            return False, f"Port {target_port} not found"

        dest_ip = (
            svc.metadata.annotations.get("juniper.kubevirt.io/dest-ip")
            if svc.metadata.annotations
            else None
        )
        if not dest_ip:
            dest_ip = get_ingress_node_ip()

        # Get allocated external IP
        external_ip = None
        if svc.metadata.annotations:
            external_ip = svc.metadata.annotations.get("kubevirt.kv/allocated-ip")

        # Find service_name for this port (from port name)
        service_name_to_delete = None
        for p in ports:
            if int(p.port) == int(target_port):
                service_name_to_delete = p.name
                break

        # Remove Juniper NAT for this node_port
        if not FRONTEND_STUB_MODE and node_port_to_remove and dest_ip:
            ok, msg = delete_port_forward(
                vm_name,
                dest_ip,
                node_port_to_remove,
                service_name_to_delete,
                external_ip,
                namespace,
            )
            if not ok:
                logger.error(f"Juniper delete failed for {node_port_to_remove}: {msg}")
                # continue anyway

        if len(remaining) == 0:
            # delete service entirely
            try:
                api.delete_namespaced_service(name=service_name, namespace=namespace)
            except client.exceptions.ApiException as e:
                if e.status != 404:
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                logger.error(f"Failed to delete service: {e}")
                return False, str(e)
        return True, f"Port {target_port} removed, service deleted"

    # Patch service with remaining ports - create clean dicts manually
    remaining_ports_dicts = []
    for p in remaining:
        port_dict = {
            "name": p.name,
            "protocol": p.protocol,
            "port": int(p.port),
            "targetPort": (
                int(p.target_port)
                if hasattr(p, "target_port") and p.target_port
                else int(p.port)
            ),
            "nodePort": int(p.node_port),
        }
        remaining_ports_dicts.append(port_dict)

    logger.info(
        f"BEFORE PATCH: Service {service_name} has {len(ports)} ports: {[p.name + ':' + str(p.port) for p in port...
    )
    logger.info(
        f"AFTER FILTER: Will keep {len(remaining_ports_dicts)} ports: {[p['name'] + ':' + str(p['port']) for p in...
    )

    # Update annotation mapping
    annotations = svc.metadata.annotations or {}
    ports_meta = []
    if "kubevirt.kv/ports" in annotations:
        try:
            ports_meta = json.loads(annotations["kubevirt.kv/ports"])
        except Exception:
            ports_meta = []
    ports_meta = [
        m for m in ports_meta if int(m.get("port", -1)) != int(target_port)
    ]
    annotations["kubevirt.kv/ports"] = json.dumps(ports_meta)

    # CRITICAL: Strategic merge patch doesn't replace arrays, it merges them!
    # We need to use replace_namespaced_service instead
    logger.info(
        f"Replacing service {service_name} to remove port {target_port}, remaining ports: {len(remaining_ports_di...
    )
    logger.debug(
        f"Replacement ports: {json.dumps(remaining_ports_dicts, indent=2)}"
    )

    # Update the service object with new ports and annotations
    svc.spec.ports = remaining_ports_dicts
    svc.metadata.annotations = annotations

    # Use replace instead of patch to ensure ports array is replaced, not merged
    result = api.replace_namespaced_service(
        name=service_name, namespace=namespace, body=svc
    )

    logger.info(
        f"REPLACE response: Service now has {len(result.spec.ports)} ports: {[p.name + ':' + str(p.port) for p in...
    )

    # Verify replacement was applied
    if len(result.spec.ports) != len(remaining_ports_dicts):
        logger.error(
            f"REPLACE FAILED! Expected {len(remaining_ports_dicts)} ports but got {len(result.spec.ports)} ports"
        )
        # Re-read service to check actual state
        svc_verify = api.read_namespaced_service(
            name=service_name, namespace=namespace
        )
```

```
            logger.error(
                f"Re-read service: {len(svc_verify.spec.ports)} ports: {[p.name + ':' + str(p.port) for p in svc_veri...
            )

        return True, f"Port {target_port} removed"
    except client.exceptions.ApiException as e:
        if e.status == 404:
            return True, "Service not found"
        logger.error(f"K8s API error removing port: {e}")

... [116 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/vm_manifest_common.py (2.1 KB)

```python
"""????? ????? ????????? VM: affinity, tolerations, ????????? GPU (nodeSelector ?????? ??? GPU)."""

from typing import Any, Dict, Optional

from app.config import GPU_RESOURCES_WITH_AUDIO
from app.scheduling_config import VM_NODE_AFFINITY, VM_TOLERATIONS

# ???????? ????????????
KUBEVIRT_NODE_AFFINITY = VM_NODE_AFFINITY
KUBEVIRT_TOLERATIONS = VM_TOLERATIONS


def add_gpu_to_manifest(
    manifest: Dict[str, Any],
    gpu_model: str,
    gpu_count: int,
    gpu_node_selector: Optional[Dict[str, str]] = None,
) -> None:
    """????????? ? ???????? GPU, UEFI/IOMMU ? nodeSelector ?????? ??? GPU (VM ?????? ?? ????? kubevirt).
    ????? hostDevices ?????????? ?????? ??? ???????? ?? GPU_RESOURCES_WITH_AUDIO (consumer 1080 Ti ? ?.?.).
    Enterprise/mdev (RTX Pro 6000) ?? ????? ?????????? ?????-?????????? ? ??? ??? hostDevices ?? ??????????.
    """
    if gpu_count <= 0 or not gpu_model:
        return

    domain = manifest["spec"]["template"]["spec"]["domain"]
    devices = domain.setdefault("devices", {})

    devices["gpus"] = [
        {"name": f"gpu{i}", "deviceName": gpu_model} for i in range(gpu_count)
    ]
    # Only add HDMI audio device for GPUs that have it in permittedHostDevices (e.g. 1080 Ti).
    # RTX Pro 6000 (mdev/vGPU) and similar enterprise cards have no separate audio ? skip.
    if gpu_model in GPU_RESOURCES_WITH_AUDIO:
        audio_resource = f"{gpu_model}-audio"
        devices["hostDevices"] = [
            {"name": f"audio{i}", "deviceName": audio_resource} for i in range(gpu_count)
        ]
    else:
        devices["hostDevices"] = []

    domain["machine"] = {"type": "q35"}
    domain["firmware"] = {"bootloader": {"efi": {"secureBoot": False}}}
    if "cpu" not in domain:
        domain["cpu"] = {}
    domain["cpu"]["model"] = "host-passthrough"
    domain["ioThreadsPolicy"] = "auto"

    if gpu_node_selector:
        manifest["spec"]["template"]["spec"]["nodeSelector"] = gpu_node_selector

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/vm_utils.py (25.1 KB)

```python
"""VM manifest generation utilities."""

import logging
import os
import re
from pathlib import Path

from app.config import (
    DEFAULT_GPU_RESOURCE,
    DEFAULT_NVIDIA_DRIVER,
    DEFAULT_PACKAGES,
    NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY,
    NVIDIA_DRIVER_VERSIONS,
    NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER,
    VM_RUN_STRATEGY,
)
from app.utils.k8s_utils import discover_gpu_resources
from app.utils.vm_manifest_common import (
    KUBEVIRT_NODE_AFFINITY,
    KUBEVIRT_TOLERATIONS,
    add_gpu_to_manifest,
)

logger = logging.getLogger(__name__)


def sanitize_cloud_init_field(
    value: str, field_name: str, allow_newlines: bool = False
) -> str:
    """Sanitize cloud-init field to prevent injection attacks.

    Args:
        value: Input value to sanitize
        field_name: Name of the field (for error messages)
        allow_newlines: Whether to allow newline characters

    Returns:
        Sanitized value

    Raises:
        ValueError: If dangerous patterns detected
    """
    if not value:
        return value

    # ????????? ??????? ???????
    dangerous_chars = ["|", ">", "$", "`", "\\", ";"]
    if not allow_newlines:
        dangerous_chars.extend(["\n", "\r"])

    for char in dangerous_chars:
        if char in value:
            raise ValueError(f"{field_name} contains dangerous character: {char}")

    # ????????? ?? injection ????????
    injection_patterns = [
        r"runcmd\s*:",
        r"bootcmd\s*:",
        r"write_files\s*:",
        r"curl.*\|.*bash",
        r"wget.*\|.*sh",
        r"\$\(.*\)",
        r"`.*`",
    ]

```

```
    for pattern in injection_patterns:
        if re.search(pattern, value, re.IGNORECASE):
            raise ValueError(f"{field_name} contains suspicious pattern: {pattern}")

    return value


def resolve_gpu_resource_name(requested_resource: str | None = None) -> str:
    """Resolve the GPU resource name to use for device passthrough.

    Preference order:
    1. Explicit value passed by caller (e.g., from the UI form)
    2. Environment variable ``GPU_RESOURCE_NAME``
    3. First discovered GPU resource from cluster nodes
    4. Static default (GeForce 1080Ti)
    """

    if requested_resource and requested_resource.strip():
        return requested_resource.strip()

    env_resource = (os.getenv("GPU_RESOURCE_NAME") or "").strip()
    if env_resource:
        return env_resource

    discovered_gpu_resources = discover_gpu_resources()
    if discovered_gpu_resources:
        return discovered_gpu_resources[0]

    return DEFAULT_GPU_RESOURCE


def format_gpu_display_name(resource_name: str) -> str:
    """Return a human-friendly GPU model label for UI hints."""

    suffix = resource_name.split("/")[-1]
    normalized = suffix.replace("_", " ").replace("-", " ")
    if suffix.lower() == "1080ti":
        return "GEFORCE 1080Ti"

    if suffix.lower().startswith("nvidia.com-"):
        normalized = suffix.split("-", 1)[-1].replace("-", " ")

    return normalized.upper()


def get_default_username_for_image(image_url: str) -> str:
    """
    ??????????? ??? ??????????????? ?? ????????? ?? ?????? URL ??????

    Args:
        image_url: URL cloud-??????

    Returns:
        ??? ??????????????? ?? ????????? ??? ????? ??????
    """
    if not image_url:
        return "ubuntu"

    image_lower = image_url.lower()

    if "ubuntu" in image_lower or "cloud-images.ubuntu.com" in image_lower:
        return "ubuntu"
    elif "debian" in image_lower or "cloud.debian.org" in image_lower:
        return "debian"
    elif "fedora" in image_lower or "download.fedoraproject.org" in image_lower:
        return "fedora"
    elif "opensuse" in image_lower or "download.opensuse.org" in image_lower:
        return "opensuse"
    elif "centos" in image_lower or "cloud.centos.org" in image_lower:
        return "centos"
    elif "rocky" in image_lower or "download.rockylinux.org" in image_lower:
```

```
        return "rocky"

    # ?? ????????? ????????? ubuntu (???????? ??????????????????)
    return "ubuntu"


def detect_distro_from_image(image_url: str) -> str:
    """Detect distribution type from image URL.

    Returns:
        'fedora' for Fedora images
        'ubuntu' for Ubuntu images (default)
    """
    if not image_url:
        return "ubuntu"

    image_lower = image_url.lower()

    # ????????? Fedora ?? ????????? ?????????
    fedora_patterns = ["fedora", "fedora-cloud", "fedora-server", "fedoraproject.org"]

    for pattern in fedora_patterns:
        if pattern in image_lower:
            return "fedora"

    # Check for Debian
    debian_patterns = ["debian", "debian-cloud"]
    for pattern in debian_patterns:
        if pattern in image_lower:
            return "debian"

    # Check for OpenSUSE
    opensuse_patterns = ["opensuse", "suse", "leap", "tumbleweed"]
    for pattern in opensuse_patterns:
        if pattern in image_lower:
            return "opensuse"

    # ?? ????????? Ubuntu
    return "ubuntu"


def load_cloud_init_template(distro: str = "ubuntu"):
    """Load cloud-init template from file based on distribution type.

    Args:
        distro: Distribution type ('ubuntu', 'fedora', 'debian', 'opensuse')

    Returns: template string or None if file not found
    """
    # ???????? ??? ????? ??????? ? ??????????? ?? ???????????
    if distro == "fedora":
        template_filename = "cloud-init-fedora.yaml"
    elif distro == "debian":
        template_filename = "cloud-init-debian.yaml"
    elif distro == "opensuse":
        template_filename = "cloud-init-opensuse.yaml"
    else:
        template_filename = "cloud-init-ubuntu.yaml"

    # ????????? ????????? ????????? ????
    possible_paths = [
        Path(__file__).parent.parent / "static" / "cloud-inits" / template_filename,
        Path("/app/app/static/cloud-inits")
        / template_filename,  # ?????????? ???? ? ?????????
        Path("app/static/cloud-inits")
        / template_filename,  # ???????????? ?? ??????? ?????????
    ]

    for template_path in possible_paths:
        try:
            if template_path.exists():
```

```
                with open(template_path, "r", encoding="utf-8") as f:
                    content = f.read()
                    logger.info(f"Cloud-init template loaded from: {template_path}")
                    return content
        except Exception as e:
            logger.warning(f"Failed to load template from {template_path}: {e}")
            continue

    logger.warning(
        f"Cloud-init template not found. Tried paths: {[str(p) for p in possible_paths]}"
    )
    return None


def validate_vm_name(vm_name):
    """Validate VM name according to Kubernetes DNS-1123 subdomain standard
    Returns: (is_valid: bool, error_message: str)
    """
    if not vm_name:
        return False, "VM name cannot be empty"

    if len(vm_name) > 253:
        return False, "VM name must be at most 253 characters"

    # DNS-1123 subdomain must consist of lower case alphanumeric characters, '-' or '.'
    # and must start and end with an alphanumeric character
    pattern = r"^[a-z0-9]([-a-z0-9]*[a-z0-9])?$"
    if not re.match(pattern, vm_name):
        return (
            False,
            "VM name must consist of lowercase alphanumeric characters or '-', start and end with alphanumeric",
        )

    return True, ""


def validate_ssh_key(ssh_key):
    """Validate SSH public key format
    Returns: (is_valid: bool, error_message: str)
    """
    if not ssh_key or not ssh_key.strip():
        return True, ""  # Empty is OK

    lines = ssh_key.strip().split("\n")
    for line in lines:
        line = line.strip()
        if not line:
            continue

        # SSH key should start with key type
        valid_types = [
            "ssh-rsa",
            "ssh-dss",
            "ssh-ed25519",
            "ecdsa-sha2-nistp256",
            "ecdsa-sha2-nistp384",
            "ecdsa-sha2-nistp521",
        ]

        parts = line.split()
        if len(parts) < 2:
            return False, "SSH key format is invalid (too few parts)"

        if parts[0] not in valid_types:
            return False, f"SSH key type '{parts[0]}' is not supported"

    return True, ""


def generate_cloud_init_userdata(
    hostname,
```

```
    username,
    ssh_key,
    package_update,
    package_upgrade,
    ssh_pwauth,
    password=None,
    custom_data=None,
    packages=None,
    image_url=None,
    gpu_count=0,
    nvidia_driver_version=None,
    gpu_model=None,
    nvidia_custom_driver_callback_url=None,
    nvidia_custom_driver_installer_type="auto",
):
    """Generate cloud-init user data from form parameters or template.

    Args:
        image_url: VM image URL (used to detect distribution type)
        nvidia_custom_driver_installer_type: "auto" | "run" | "deb" | "rpm" ? ????? ????
            ??????????? ? ?????? (auto = ?? ???????????).
    """

    # DNS bootcmd - ???????? ??? ?????? apt/yum ? KubeVirt
    dns_bootcmd = """bootcmd:
  - mkdir -p /etc/systemd/resolved.conf.d
  - |
    cat >/etc/systemd/resolved.conf.d/custom-dns.conf << 'EOF'
    [Resolve]
    DNS=8.8.8.8 1.1.1.1
    FallbackDNS=1.0.0.1
    Domains=~.
    DNSSEC=no
    EOF
  - systemctl restart systemd-resolved || true
  - sleep 3
  - |
    for i in 1 2 3 4 5; do
      if getent hosts archive.ubuntu.com >/dev/null 2>&1; then
        echo "DNS OK"
        break
      fi
      echo "DNS not ready yet, retry..."
      sleep 2
    done
  # Set VNC resolution to 1920x1080 via kernel parameters
  - |
    if [ -f /etc/default/grub ]; then
      # Backup original grub config
      cp /etc/default/grub /etc/default/grub.bak
      # Add video parameter for 1920x1080 resolution
      if ! grep -q "video=" /etc/default/grub; then
        sed -i 's/GRUB_CMDLINE_LINUX_DEFAULT="/GRUB_CMDLINE_LINUX_DEFAULT="video=1920x1080 /' /etc/default/grub || \
        sed -i 's/GRUB_CMDLINE_LINUX="/GRUB_CMDLINE_LINUX="video=1920x1080 /' /etc/default/grub || true
      fi
      # Update grub
      update-grub 2>/dev/null || grub2-mkconfig -o /boot/grub2/grub.cfg 2>/dev/null || true
    fi
"""

    # ???? ???? custom cloud-init, ????????? DNS bootcmd ???? ??? ??? ???
    if custom_data and custom_data.strip():
        custom = custom_data.strip()
        # ?????????, ???? ?? ??? bootcmd ? ????????? ???????
        if "bootcmd:" not in custom.lower():
            # ????????? DNS bootcmd ????? #cloud-config
            if custom.startswith("#cloud-config"):
                lines = custom.split("\n", 1)
                if len(lines) == 2:
                    return f"{lines[0]}\n\n{dns_bootcmd}\n{lines[1]}"
                else:
```

```
                    return f"{custom}\n\n{dns_bootcmd}"
        return custom

    # ??????????? ??????????? ? hostname ? username
    hostname = hostname.replace(":", "-").replace("/", "-")
    username = re.sub(r"[^a-z0-9_-]", "", username.lower())

    # ??????????? ??????????? ?? image_url
    distro = detect_distro_from_image(image_url)
    logger.info(f"Detected distro: {distro} from image: {image_url}")

    # ????????? ?????? ??? ???????????????? ???????????
    template = load_cloud_init_template(distro)

    # ???? ?????? ?? ?????? - ??? ??????
    if not template:
        raise FileNotFoundError(
            f"Cloud-init template not found for {distro}. Check that app/templates/ contains the template."
        )

    # ??????????? SSH ???????????
    ssh_config = ""
    runcmd_list = []
    write_files_section = ""
    write_files_list = []

    if ssh_pwauth == "true":
        ssh_config = """  - path: /etc/ssh/sshd_config.d/99-cloud-init.conf
    content: |
      PasswordAuthentication yes
      PermitRootLogin yes
      ChallengeResponseAuthentication no
    permissions: '0644'"""
        write_files_list.append(ssh_config)
        runcmd_list.append("systemctl restart sshd || systemctl restart ssh")

    # ??????????? SSH ??????
    ssh_keys = ""
    if ssh_key and ssh_key.strip():
        keys_list = []
        for key in ssh_key.strip().split("\n"):
            if key.strip():
                keys_list.append(f"      - {key.strip()}")
        if keys_list:
            ssh_keys = "    ssh_authorized_keys:\n" + "\n".join(keys_list)

    # ??????????? ??????????? ??????
    password_config = ""
    if ssh_pwauth == "true" and password:
        password_config = f"    plain_text_passwd: {password}\n    lock_passwd: false"
    else:
        password_config = "    lock_passwd: true"

    # ??????????? package_update
    package_update_str = ""
    if package_update == "true":
        package_update_str = "package_update: true\npackage_upgrade: true"

    # ??????????? ?????? ???????
    base_packages = DEFAULT_PACKAGES

    if packages:
        if isinstance(packages, list):
            all_packages = base_packages + [
                pkg for pkg in packages if pkg not in base_packages
            ]
            pkg_list = list(dict.fromkeys(all_packages))
        else:
            pkg_list = [pkg.strip() for pkg in str(packages).split(",") if pkg.strip()]
            for base_pkg in base_packages:
                if base_pkg not in pkg_list:
```

```
                    pkg_list.insert(0, base_pkg)
            pkg_list = list(dict.fromkeys(pkg_list))
    else:
        pkg_list = base_packages

    packages_str = "\n".join([f"  - {pkg}" for pkg in pkg_list])

    # ????????????? runcmd (??? DNS ?????? - ??? ? bootcmd)
    runcmd_list.extend(
        ["systemctl enable qemu-guest-agent", "systemctl start qemu-guest-agent"]
    )

    # ????????? ?????????? VNC ?? Full HD (1920x1080)
    # ? KubeVirt VNC ?????????????????? QEMU, ? ?? VNC ???????? ?????? VM
    # ?????????? ????????????? ????? kernel ????????? ??? ?????????? ??????
    # ???? ?????????? X ??????, ????? ???????????? ????????? ????? xrandr
    # ?? ???????? ????????? ??? ??????? ? bootcmd ????? kernel ?????????

    # ????????? ??????????? ????????? ??? X ??????? (???? ??????????)
    x11_resolution_config = """  - path: /etc/X11/xorg.conf.d/99-vnc-resolution.conf
    content: |
      Section "Screen"
        Identifier "Screen0"
        SubSection "Display"
          Modes "1920x1080"
          Virtual 1920 1080
        EndSubSection
      EndSection
    permissions: '0644'
  - path: /usr/local/bin/set-vnc-resolution.sh
    content: |
      #!/bin/bash
      # Set VNC resolution to 1920x1080 (optional, for X server if installed)
      if command -v xrandr >/dev/null 2>&1; then
        export DISPLAY=:0
        # Create new mode if it doesn't exist
        xrandr --newmode "1920x1080_60.00" 173.00 1920 2048 2248 2576 1080 1083 1088 1120 -hsync +vsync 2>/dev/null |...
        # Try to add mode to any available output
        for output in $(xrandr 2>/dev/null | grep -E "connected|disconnected" | awk '{print $1}'); do
          xrandr --addmode "$output" 1920x1080_60.00 2>/dev/null || true
          xrandr --output "$output" --mode 1920x1080_60.00 2>/dev/null || true
        done
      fi
    permissions: '0755'"""
    write_files_list.append(x11_resolution_config)

    # ????????? ??????? ??? ??????????? ????????? X ??????? (???? ??????????)
    runcmd_list.extend(
        [
            # ??????? ?????????? ??? xorg.conf.d ???? ?? ???
            "mkdir -p /etc/X11/xorg.conf.d 2>/dev/null || true",
            # ??????? ?????????? ?????????? ????? xrandr (???? X ?????? ??????????)
            "/usr/local/bin/set-vnc-resolution.sh 2>/dev/null || true",
        ]
    )

    # ???? VM ? GPU, ????????? ????????? ????????? NVIDIA (??????????? ??? ????????? ? ????? ??? ????? ??????, ??????...
    use_custom_driver = (
        gpu_count
        and int(gpu_count) > 0
        and gpu_model
        and (gpu_model in NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER)
        and nvidia_custom_driver_callback_url
    )
    if gpu_count and int(gpu_count) > 0:
        if use_custom_driver:
            # ????????? ???????: ?????? ????????? ?? ???????? (????? ?????????? ????, ???????? ????? Docker).
            # ?? VM ???????? ????????? ? ??????? ? callback URL ??? ??????????? ????? ?????????.
            pass
        else:
            # ??????????? ????????? ?? ??????????? (1080 Ti, 3080 Ti, 3090 ? ?.?.)
```

```
            driver_ver = (nvidia_driver_version or "").strip() or DEFAULT_NVIDIA_DRIVER
            if driver_ver not in NVIDIA_DRIVER_VERSIONS:
                driver_ver = DEFAULT_NVIDIA_DRIVER
            if driver_ver == "auto":
                install_cmd = (
                    "ubuntu-drivers autoinstall || apt-get install -y nvidia-driver-580"
                )
            else:
                install_cmd = f"apt-get install -y nvidia-driver-{driver_ver}"

... [196 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/app/utils/windows_utils.py (16.2 KB)

```python
import logging
from typing import Any, Dict, Optional

from app.config import (
    STORAGE_CLASS_NAME,
    VIRTIO_DRIVERS_IMAGE,
    VM_RUN_STRATEGY,
    WINDOWS_DEFAULT_USERNAME,
    WINDOWS_HYPERV_FEATURES,
    WINDOWS_ISO_DATAVOLUME,
    WINDOWS_ISO_NAMESPACE,
    WINDOWS_ISO_STORAGE_GB,
    WINDOWS_MACHINE_TYPE,
)
from app.scheduling_config import get_storage_node_selector
from app.utils.vm_manifest_common import (
    KUBEVIRT_NODE_AFFINITY,
    KUBEVIRT_TOLERATIONS,
    add_gpu_to_manifest,
)

logger = logging.getLogger(__name__)


def _dv_spec_with_node_placement(source: Dict, pvc: Dict) -> Dict:
    """???????? spec DataVolume ? nodePlacement ??? ????????? (???? ? CSI)."""
    spec = {"source": source, "pvc": pvc}
    sel = get_storage_node_selector()
    if sel:
        spec["nodePlacement"] = {"nodeSelector": sel}
    return spec


def _windows_installer_data_volume_templates(
    dv_name: str, iso_dv_name: str, storage: int
) -> list:
    """DataVolumeTemplates ??? Windows installer: blank rootdisk + clone ISO."""
    root_spec = _dv_spec_with_node_placement(
        {"blank": {}},
        {
            "accessModes": ["ReadWriteOnce"],
            "resources": {"requests": {"storage": f"{storage}Gi"}},
            "storageClassName": STORAGE_CLASS_NAME,
        },
    )
    iso_spec = _dv_spec_with_node_placement(
        {
            "pvc": {
                "name": WINDOWS_ISO_DATAVOLUME,
                "namespace": WINDOWS_ISO_NAMESPACE,
            }
        },
        {
            "accessModes": ["ReadWriteOnce"],
            "resources": {"requests": {"storage": f"{WINDOWS_ISO_STORAGE_GB}Gi"}},
            "storageClassName": STORAGE_CLASS_NAME,
        },
    )
    return [
        {
            "apiVersion": "cdi.kubevirt.io/v1beta1",
            "kind": "DataVolume",
            "metadata": {"name": dv_name},
            "spec": root_spec,
        },
```

```
        {
            "apiVersion": "cdi.kubevirt.io/v1beta1",
            "kind": "DataVolume",
            "metadata": {
                "name": iso_dv_name,
                "annotations": {
                    "cdi.kubevirt.io/storage.bind.immediate.requested": "true"
                },
            },
            "spec": iso_spec,
        },
    ]


def generate_windows_installer_vm_manifest(
    vm_name: str,
    namespace: str,
    cpu: int,
    memory: int,
    storage: int,
    gpu_model: Optional[str] = None,
    gpu_count: int = 0,
    gpu_node_selector: Optional[Dict[str, str]] = None,
) -> Dict[str, Any]:
    """
    Generates VM manifest for Windows 10 installation

    Creates:
    1. Blank DataVolume for root disk (virtio)
    2. Cloned DataVolume with Windows ISO (sata cdrom, boot order 1) - cloned from kvm
    3. ContainerDisk with VirtIO drivers (sata cdrom)

    Args:
        vm_name: Name of the VM
        namespace: User namespace
        cpu: Number of CPU cores
        memory: Memory in GB
        storage: Disk size in GB
        gpu_model: GPU resource name (optional)
        gpu_count: Number of GPUs (optional)
        gpu_node_selector: Node selector for GPU placement (optional)

    Returns:
        dict: VirtualMachine manifest
    """
    logger.info(
        f"Generating Windows installer VM: {vm_name}, CPU={cpu}, RAM={memory}GB, Storage={storage}GB"
    )

    # DataVolume name for root disk
    dv_name = f"{vm_name}-rootdisk"
    # ISO DataVolume name (will be cloned to user namespace)
    iso_dv_name = f"{vm_name}-windows-iso"

    manifest = {
        "apiVersion": "kubevirt.io/v1",
        "kind": "VirtualMachine",
        "metadata": {
            "name": vm_name,
            "namespace": namespace,
            "labels": {
                "kubevirt.io/vm": vm_name,
                "vm.kubevirt.io/os": "windows-10",
                "vm.kubevirt.io/type": "installer",
                "vm.kubevirt.io/username": WINDOWS_DEFAULT_USERNAME,
            },
        },
        "spec": {
            "runStrategy": "Manual",  # Allow manual control, won't auto-start
            "dataVolumeTemplates": _windows_installer_data_volume_templates(
                dv_name, iso_dv_name, storage
```

```
            ),
            "template": {
                "metadata": {
                    "labels": {
                        "kubevirt.io/vm": vm_name,
                    }
                },
                "spec": {
                    "domain": {
                        "ioThreadsPolicy": "auto",
                        "cpu": {
                            "cores": cpu,
                            "sockets": 1,
                            "threads": 1,
                        },
                        "devices": {
                            "blockMultiQueue": True,
                            "disks": [
                                # Root disk for Windows installation
                                {
                                    "name": "rootdisk",
                                    "disk": {
                                        "bus": "virtio",
                                        "cache": "none",
                                        "io": "native",
                                    },
                                    "bootOrder": 2,
                                },
                                # Windows ISO
                                {
                                    "name": "windows-iso",
                                    "cdrom": {"bus": "sata"},
                                    "bootOrder": 1,
                                },
                                # VirtIO drivers
                                {
                                    "name": "virtio-drivers",
                                    "cdrom": {"bus": "sata"},
                                },
                            ],
                            "interfaces": [{"name": "default", "masquerade": {}}],
                            # Serial console for Cloudbase-Init logs
                            "serials": [{"type": "serial", "name": "serial0"}],
                        },
                        "machine": {"type": WINDOWS_MACHINE_TYPE},
                        "resources": {
                            "requests": {
                                "memory": f"{memory}Gi",
                                "cpu": str(cpu),
                            }
                        },
                        "features": {
                            "acpi": {"enabled": True},
                            "apic": {},
                            "hyperv": {
                                **WINDOWS_HYPERV_FEATURES,
                            },
                            "kvm": {"hidden": True},
                        },
                    },
                    "networks": [{"name": "default", "pod": {}}],
                    "affinity": KUBEVIRT_NODE_AFFINITY,
                    "tolerations": KUBEVIRT_TOLERATIONS,
                    "volumes": [
                        # Root disk volume
                        {
                            "name": "rootdisk",
                            "dataVolume": {"name": dv_name},
                        },
                        # Windows ISO volume (cloned from kvm)
                        {
```

```
                            "name": "windows-iso",
                            "dataVolume": {"name": iso_dv_name},
                        },
                        # VirtIO drivers from containerDisk
                        {
                            "name": "virtio-drivers",
                            "containerDisk": {"image": VIRTIO_DRIVERS_IMAGE},
                        },
                    ],
                },
            },
        },
    }

    if gpu_count > 0 and gpu_model:
        logger.info(f"Adding {gpu_count}x {gpu_model} to Windows installer VM")
        add_gpu_to_manifest(manifest, gpu_model, gpu_count, gpu_node_selector)
    else:
        # No GPU: use BIOS (SeaBIOS) for faster boot than UEFI/TianoCore
        manifest["spec"]["template"]["spec"]["domain"]["firmware"] = {
            "bootloader": {"bios": {}}
        }

    return manifest


def generate_cloudbase_init_userdata(
    hostname: str,
    username: str,
    password: str,
) -> str:
    """
    Generates Cloudbase-Init userdata for Windows VM

    Cloudbase-Init is Windows equivalent of cloud-init.
    Uses YAML format similar to cloud-init.

    Args:
        hostname: Computer name for Windows
        username: Username to create/configure
        password: Password for the user

    Returns:
        str: Cloudbase-Init userdata in YAML format
    """
    logger.info(
        f"Generating Cloudbase-Init userdata: hostname={hostname}, user={username}"
    )

    userdata = f"""#cloud-config
# Cloudbase-Init configuration for Windows
# This file is processed by Cloudbase-Init on first boot

# Set computer name
set_hostname: {hostname}
hostname: {hostname}

# User configuration
users:
  - name: {username}
    gecos: {username}
    primary_group: Administrators
    groups: Users,Administrators
    passwd: {password}
    inactive: false

# Additional configuration
preserve_hostname: false
manage_etc_hosts: true

# Cloudbase-Init plugins to run
```

```
plugins:
  - cloudbaseinit.plugins.common.mtu.MTUPlugin
  - cloudbaseinit.plugins.common.sethostname.SetHostNamePlugin
  - cloudbaseinit.plugins.windows.createuser.CreateUserPlugin
  - cloudbaseinit.plugins.windows.extendvolumes.ExtendVolumesPlugin
  - cloudbaseinit.plugins.common.userdata.UserDataPlugin
  - cloudbaseinit.plugins.common.localscripts.LocalScriptsPlugin
"""

    return userdata


def generate_windows_vm_from_golden_image(
    vm_name: str,
    namespace: str,
    cpu: int,
    memory: int,
    storage: int,
    secret_name: str,
    golden_image_name: str = "windows-10-golden-image",
    golden_image_namespace: str = "kvm",
    gpu_model: Optional[str] = None,
    gpu_count: int = 0,
    gpu_node_selector: Optional[Dict[str, str]] = None,
    vm_username: str = "Administrator",
) -> Dict[str, Any]:
    """
    Generates VM manifest for Windows VM cloned from golden image

    Creates:
    1. DataVolume template that clones from golden image PVC
    2. CloudInit volume with Cloudbase-Init userdata
    3. Windows-optimized domain configuration

    Args:
        vm_name: Name of the VM
        namespace: User namespace
        cpu: Number of CPU cores
        memory: Memory in GB
        storage: Disk size in GB (must be >= golden image size)
        secret_name: Name of secret containing Cloudbase-Init userdata
        golden_image_name: Name of golden image PVC
        golden_image_namespace: Namespace of golden image PVC
        gpu_model: GPU resource name (optional)
        gpu_count: Number of GPUs (optional)
        gpu_node_selector: Node selector for GPU placement (optional)
        vm_username: Windows username for SSH label

    Returns:
        dict: VirtualMachine manifest
    """
    logger.info(
        f"Generating Windows VM from golden image: {vm_name}, "
        f"source={golden_image_namespace}/{golden_image_name}, "
        f"CPU={cpu}, RAM={memory}GB, Storage={storage}GB"
    )

    manifest = {
        "apiVersion": "kubevirt.io/v1",
        "kind": "VirtualMachine",
        "metadata": {
            "name": vm_name,
            "namespace": namespace,
            "labels": {
                "app": "kubevirt-vm",
                "vm.kubevirt.io/name": vm_name,
                "vm.kubevirt.io/os": "windows10",
                "vm.kubevirt.io/username": vm_username,
            },
        },
        "spec": {
```

```
            "runStrategy": VM_RUN_STRATEGY,  # Use configured strategy (default: Manual)
            "dataVolumeTemplates": [
                {
                    "metadata": {"name": f"{vm_name}-rootdisk"},
                    "spec": _dv_spec_with_node_placement(
                        {
                            "pvc": {
                                "name": golden_image_name,
                                "namespace": golden_image_namespace,
                            }
                        },
                        {
                            "accessModes": ["ReadWriteOnce"],
                            "resources": {"requests": {"storage": f"{storage}Gi"}},
                            "storageClassName": STORAGE_CLASS_NAME,
                        },
                    ),
                }
            ],
            "template": {
                "metadata": {
                    "labels": {
                        "vm.kubevirt.io/name": vm_name,
                        "vm.kubevirt.io/os": "windows10",
                    }
                },
                "spec": {
                    "domain": {
                        "ioThreadsPolicy": "auto",
                        "firmware": {
                            "bootloader": {
                                "efi": {
                                    "secureBoot": False,
                                }
                            }
                        },
                        "cpu": {
                            "cores": cpu,
                        },
                        "devices": {
                            "blockMultiQueue": True,
                            "disks": [
                                {
                                    "name": "rootdisk",
                                    "bootOrder": 1,
                                    "disk": {
                                        "bus": "virtio",
                                        "cache": "none",
                                        "io": "native",
                                    },
                                },
                                {
                                    "name": "cloudinit",
                                    "disk": {
                                        "bus": "sata",
                                    },
                                },
                            ],
                            "interfaces": [
                                {
                                    "name": "default",
                                    "masquerade": {},
                                }
                            ],
                            # Serial console for Cloudbase-Init logs
                            "serials": [
                                {
                                    "type": "serial",
                                    "name": "serial0",
                                }
                            ],
```

```
                        },
                        "machine": {
                            "type": WINDOWS_MACHINE_TYPE,
                        },
                        "resources": {
                            "requests": {
                                "memory": f"{memory}Gi",
                            }
                        },
                        "features": {
                            "acpi": {"enabled": True},
                            "apic": {},
                            "hyperv": {
                                **WINDOWS_HYPERV_FEATURES,
                            },
                            "kvm": {"hidden": True},
                        },
                    },
                    "networks": [
                        {
                            "name": "default",
                            "pod": {},
                        }
                    ],
                    "affinity": KUBEVIRT_NODE_AFFINITY,
                    "tolerations": KUBEVIRT_TOLERATIONS,
                    "volumes": [
                        {
                            "name": "rootdisk",
                            "dataVolume": {
                                "name": f"{vm_name}-rootdisk",
                            },
                        },
                        {
                            "name": "cloudinit",
                            "cloudInitNoCloud": {
                                "secretRef": {
                                    "name": secret_name,
                                }
                            },
                        },
                    ],
                },
            },
        },
    }

    if gpu_count > 0 and gpu_model:
        logger.info(f"Adding {gpu_count}x {gpu_model} to Windows VM")
        add_gpu_to_manifest(manifest, gpu_model, gpu_count, gpu_node_selector)

    logger.info(f"Windows VM manifest generated successfully for {vm_name}")
    return manifest

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/conf/default.conf (2.5 KB)

```
map $http_upgrade $connection_upgrade {
    default upgrade;
    '' close;
}

map $http_x_forwarded_proto $forwarded_proto {
    default $http_x_forwarded_proto;
    ''      $scheme;
}

server {
    listen 8080 default_server;
    listen [::]:8080 default_server;
    server_name _;

    access_log /dev/stdout combined;
    error_log /dev/stderr warn;

    # Normalize /k8s -> /k8s/
    location = /k8s {
        return 301 /k8s/;
    }

    # Kube API proxy path used by noVNC
    location /k8s/ {
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $forwarded_proto;
        proxy_set_header Authorization "";
        proxy_pass_request_body on;
        proxy_pass_request_headers on;
        client_max_body_size 5g;

        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
        proxy_http_version 1.1;
        proxy_read_timeout 86400s;
        proxy_send_timeout 86400s;
        proxy_connect_timeout 86400s;
        proxy_buffering off;
        proxy_request_buffering off;

        proxy_pass http://127.0.0.1:8001/k8s/;
    }

    location /internal-auth/ {
        internal;
        proxy_pass_request_body off;
        proxy_set_header Content-Length "";
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-Uri $request_uri;
        proxy_pass http://127.0.0.1:4180/oauth2/auth;
    }

    location /oauth2/ {
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-Uri $request_uri;
        proxy_redirect off;
        proxy_pass http://127.0.0.1:4180/oauth2/;
    }

    # Let FastAPI serve /static to avoid alias/mime mismatches
    location / {
```

```
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $forwarded_proto;

        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
        proxy_http_version 1.1;

        proxy_connect_timeout 60s;
        proxy_send_timeout 60s;
        proxy_read_timeout 60s;
        proxy_buffering off;
    }

    include /etc/nginx/location.d/*.conf;
}
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/conf/gzip.conf (510.0 B)

```
# GZIP Module
#
gzip_disable "msie6";

gzip_vary on;
gzip_proxied any;
gzip_comp_level 6;
gzip_buffers 16 8k;
gzip_http_version 1.1;
gzip_min_length 256;
gzip_types
  application/atom+xml
  application/geo+json
  application/javascript
  application/x-javascript
  application/json
  application/ld+json
  application/manifest+json
  application/rdf+xml
  application/rss+xml
  application/xhtml+xml
  application/xml
  font/eot
  font/otf
  font/ttf
  image/svg+xml
  text/css
  text/javascript
  text/plain;

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docker-compose.stub.yml (733.0 B)

```
# Stub mode with real PostgreSQL
# Use: docker compose -f docker-compose.stub.yml up -d postgres
# Then: ./run-stub-mode.sh --with-db
# Or run both: docker compose -f docker-compose.stub.yml up
#
version: "3.9"

services:
  postgres:
    image: postgres:16-alpine
    container_name: viirtuoz-stub-postgres
    restart: unless-stopped
    environment:
      POSTGRES_USER: viirtuoz
      POSTGRES_PASSWORD: stub
      POSTGRES_DB: viirtuoz
    ports:
      - "127.0.0.1:5434:5432"
    volumes:
      - viirtuoz_stub_pgdata:/var/lib/postgresql/data
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U viirtuoz"]
      interval: 5s
      timeout: 3s
      retries: 5

volumes:
  viirtuoz_stub_pgdata:

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docker-compose.yml.example (1.4 KB)

```
# Production docker-compose for kubevirt-api-manager
# Copy to docker-compose.yml and customize. Use env.example for variables.
# Use for: docker-compose build && docker push <image>
#
version: "3.9"

services:
  kubevirt-api:
    build:
      context: .
      dockerfile: Dockerfile
    image: harbor.k8s.naint.tech/kubevirt/kv_api_manager:latest
    container_name: kubevirt-api-manager
    restart: unless-stopped
    ports:
      - "8083:8080"   # Nginx + UI
      - "8004:8001"   # kubectl proxy (/k8s API)
    environment:
      SECRET_KEY: ${SECRET_KEY}
      DEBUG: ${DEBUG:-false}
      FRONTEND_STUB_MODE: ${FRONTEND_STUB_MODE:-false}
      DATABASE_URL: ${DATABASE_URL}
      STORAGE_CLASS_NAME: ${STORAGE_CLASS_NAME}
      LOG_LEVEL: ${LOG_LEVEL:-info}
      LLM_SERVICE_URL: ${LLM_SERVICE_URL}
      LLM_CHAT_SHARED_TOKEN: ${LLM_CHAT_SHARED_TOKEN}
      LLM_USER_KEYS_MAX: ${LLM_USER_KEYS_MAX:-10}
      LLM_KEYS_ENCRYPTION_KEY: ${LLM_KEYS_ENCRYPTION_KEY}
      LLM_AUTH_SERVICE_URL: ${LLM_AUTH_SERVICE_URL}
      LLM_AUTH_ADMIN_TOKEN: ${LLM_AUTH_ADMIN_TOKEN}
      LLM_API_KEY: ${LLM_API_KEY}
      LLM_API_URL: ${LLM_API_URL}
      LLM_MODEL: ${LLM_MODEL}
    volumes:
      - ${HOME}/.kube:/root/.kube:ro
      - ./conf/oauth.d:/etc/nginx/oauth.d:ro
    healthcheck:
      test: ["CMD", "curl", "-f", "http://127.0.0.1:8080/ready"]
      interval: 30s
      timeout: 5s
      retries: 5
      start_period: 20s

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/CDI-SETUP.md (2.3 KB)

```
# ????????? Containerized Data Importer (CDI)

CDI ????????? ??? ??????? ??????? ?????? (PVC) ??? ??????????? ?????.

## ?????????

??????????? ???????????: [https://kubevirt.io/user-guide/storage/containerized_data_importer/](https://kubevirt.io/u...

?? ???????????? ?????? v1.55+ ? ??????? ?????????.

### ??????????? ??????????? ??? ?????? ????????

??? ?????????? ?????? ? ????? ?????, ??? ???? ????????? ?? ????? (GPU vs CPU/Storage), ????????? ??????? ????????? C...

1. **????????? ?????????**:
   - `cdi-operator.yaml`
   - `cdi-cr.yaml`

2. **??????? (Tolerations & Affinity)**:
   ????????? ? ??? ??????????? Taints `cephfs`, `data` ? `dedicated`, ????????? CDI ?????? ????? ?? "???????" (tole...

   ? `cdi-cr.yaml` ????????? ?????? `infra` ? `workloads`:

   ```yaml
   spec:
     infra:
       nodePlacement:
         affinity:
           nodeAffinity:
             requiredDuringSchedulingIgnoredDuringExecution:
               nodeSelectorTerms:
                 - matchExpressions:
                     - key: node-role.kubernetes.io/control-plane
                       operator: Exists
                     - key: node-role.kubernetes.io/gpu-worker
                       operator: DoesNotExist
         tolerations:
           - key: cephfs
             operator: Exists
           - key: data
             operator: Exists
     workloads:
       # ??????????? ??????????? ??? worker-????? (importer/uploadproxy)
   ```

3. **??????????**:

   ```bash
   kubectl apply -f kubevirt-proj/cdi-operator.yaml
   kubectl apply -f kubevirt-proj/cdi-cr.yaml
   ```

   ??? ??????????, ??? ?????????? CDI ?? ????????? ?????????????? ?? GPU ???? (??? ??? ??????? ? CEPH) ? ?????? ??...

```

## [FILE] viirtuoz-vps-dev-rewrite-back-2/docs/CLOUD_INIT_FILES_EXAMPLES.md (4.3 KB)

```
??????? ??????????? ?????? ? cloud-init

? ??????? ????????? write_files ???????

??? ?????? ??????????? ????????? ????????? YAML ??? ?????? `write_files`.

??? ???????? ????????????? ?????

?????? 1: ???????? custom ???????????????? ????

```yaml
write_files:
  # DNS Configuration (???????????)
  - path: /etc/systemd/resolved.conf.d/custom-dns.conf
    content: |
      [Resolve]
      DNS=8.8.8.8 1.1.1.1
      FallbackDNS=1.0.0.1
      Domains=~.
      DNSSEC=no
    permissions: '0644'

  # ??? custom ????
  - path: /etc/myapp/config.conf
    content: |
      SERVER_URL=https://example.com
      API_KEY=your-api-key
      TIMEOUT=30
    permissions: '0600'
    owner: root:root
```

?????? 2: ??????? systemd service

```yaml
write_files:
  # ... ??????????? ????? ...

  - path: /etc/systemd/system/myapp.service
    content: |
      [Unit]
      Description=My Custom Application
      After=network.target

      [Service]
      Type=simple
      User=ubuntu
      ExecStart=/usr/local/bin/myapp
      Restart=always

      [Install]
      WantedBy=multi-user.target
    permissions: '0644'
```

?????? 3: ??????? ??????

```yaml
write_files:
  # ... ??????????? ????? ...

  - path: /usr/local/bin/init-script.sh
    content: |
      #!/bin/bash
      echo "Running custom initialization"
```
```

```
      apt-get update
      apt-get install -y custom-package
    permissions: '0755'
    owner: root:root
```

?????? 4: ???????? cron ??????

```yaml
write_files:
  # ... ??????????? ????? ...

  - path: /etc/cron.d/custom-job
    content: |
      # Backup every day at 2 AM
      0 2 * * * root /usr/local/bin/backup.sh
    permissions: '0644'
```

?????? ?????? ??????????? ???????

???? ?????? ????????? ??????? ??????, ???????? ????? ????? `{ssh_config}`:

```yaml
#cloud-config
hostname: {hostname}
manage_etc_hosts: true
preserve_hostname: false

write_files:
  # DNS Configuration
  - path: /etc/systemd/resolved.conf.d/custom-dns.conf
    content: |
      [Resolve]
      DNS=8.8.8.8 1.1.1.1
      FallbackDNS=1.0.0.1
      Domains=~.
      DNSSEC=no
    permissions: '0644'

  # Custom application config
  - path: /etc/myapp/config.yml
    content: |
      app:
        name: MyApp
        port: 8080
        debug: false
    permissions: '0644'

  # Init script
  - path: /usr/local/bin/custom-init.sh
    content: |
      #!/bin/bash
      echo "Custom initialization started"
      # Your commands here
    permissions: '0755'

{ssh_config}

users:
  - name: {username}
    groups: sudo
    shell: /bin/bash
    sudo: 'ALL=(ALL) NOPASSWD:ALL'
{ssh_keys}
{password_config}

{package_update}

packages:
{packages}
```

```

runcmd:
{runcmd}
  # Run custom init script
  - /usr/local/bin/custom-init.sh
```

?????? ???????

1. **Permissions**:
   - `'0644'` - ???????? ?????, ???????????? ?????????
   - `'0600'` - ?????? ????????
   - `'0755'` - ?????????? ????

2. **Owner**: ?? ????????? `root:root`, ????? ??????? `ubuntu:ubuntu`

3. **Content**: ?????????? `|` ??? ????????????? ??????????

4. **Encoding**: ????? ??????????? `encoding: b64` ??? base64 ?????????????? ????????

???????? ?????????

????? ????????? ??????? ????????? ???:

```bash
python3 << 'EOF'
from app.utils.vm_utils import generate_cloud_init_userdata
import yaml

result = generate_cloud_init_userdata(
    hostname="test", username="ubuntu", ssh_key="",
    package_update="false", package_upgrade="false",
    ssh_pwauth="false"
)

try:
    yaml.safe_load(result)
    print("? ????????? ?????????")
except yaml.YAMLError as e:
    print(f"? ??????: {e}")
EOF
```

```
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/CUSTOM-DRIVER-INTEGRATION.md (12.8 KB)

```
# ?????????? ?????? ?????????? ????????

???????? ??? ???????????: ??? ??? ??????? ? ??????????, ???? ???????? ???? ?????? ????????? ?????????? ???????? (???...

---

## 1. ??? ??? ????

### 1.1 UI ? ????? «????????? ???????»

- ? ????? ???????? VM ??? ?????? GPU ?? ?????? **??????????** ??????? (`NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER`) ?????????...
  - ???? «??? ??????????? ? ??????» ? ?????? `nvidia_custom_driver_installer_type` (run/deb/rpm/auto);
  - ???????? «??? ????????? ?????? ????? ??????????? ????????? ????????».
- ????? ????? ?????, ??????? ??????? ?? ???????:
  - `nvidia_custom_driver_installer_type` ? ???????? ??????? (run, deb, rpm, auto).

?????:

- ??????: `app/templates/dashboard.html` ? ???? GPU, ?????? `#nvidia-custom-installer-select`, ???? `#gpu-custom-driv...
- ?????? GPU ??? «??????????» ???????? ?????????? ? ?????? ??? `gpu_resources_custom_driver` ?? `app/routes/__init__....

### 1.2 ????? ??????????? ???????? «????????? ???????»

? `app/routes/vm_routes.py` (???????? VM):

- ???????:
  `use_custom_nvidia_driver = (gpu_count > 0 and gpu_resource_name in NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER and APP_INTE...
- ???? ??? ???????:
  - ??????????? ?????????? **?????** ? **callback URL**.
  - ????? ??????????? ? **????????** VM (???? ?? `NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY`).
  - ? cloud-init ? ? ???????? VM ??????????:
    - `nvidia_custom_driver_callback_url`,
    - `nvidia_custom_driver_installer_type` (?? ????? ??? `"run"` ?? ????????? ??? ??????????).

?? ???? ??? ???????? VM ? ????? GPU ??????? ?????????? ??? «????????» VM ??? ?????????? ???????? ? ??????? URL ??? ??...

### 1.3 ??? ???????? ? cloud-init ? ???????? VM

- **???????? VM** (`app/utils/vm_utils.py`, `generate_vm_manifest`):
  - ????????? ?? `spec.template.metadata.annotations`:
    - ????: `NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY` (?? ????????? `vm.kubevirt.io/nvidia-custom-driver-token`),
    - ????????: ?????????? ?????.
- **Cloud-init** (`app/utils/vm_utils.py`, `generate_cloud_init_userdata`):
  - ??? `use_custom_driver == True` ?????? **?????? ?? ???????????** ? ?????? ?????????? ? ????, ??? ???? ????? ????...
  - ? ??????? ??? ??????????:
    - `nvidia_custom_driver_callback_url` ? ?????? URL ??? POST ????? ?????????;
    - `nvidia_custom_driver_installer_type` ? ????????? ??? ??????????.

????? ????? ???? ???????? ? cloud-init ???????/??????, ??????? ????????? ???? ????????? ? ? ????? ???????? callback, ...

---

## 2. ???? ???????? ?????? ?????????

### 2.1 ??????? A: ?????? ?????? cloud-init (?????? / ??????? ? ???????? ??)

????: **`app/utils/vm_utils.py`**.

???????: **`generate_cloud_init_userdata`**.

?????: ???? `if gpu_count and int(gpu_count) > 0:` ? ????? `if use_custom_driver:` (?????? ??? ?????? `pass`).

????????? ????? ? ???? (?????? ~486?489):

```python
```
```

```
if use_custom_driver:
    # ????????? ???????: ?????? ????????? ?? ???????? (????? ?????????? ????, ???????? ????? Docker).
    # ?? VM ???????? ????????? ? ??????? ? callback URL ??? ?????????? ????? ?????????.
    pass
```

??? ????? ???????:

- ???????? ? `write_files_list` ?????? ??????? (????????, `/tmp/install-nvidia-custom.sh`) ??? ???????, ??????? ?????...
- ???????? ? `runcmd_list` ???????, ???????:
  - ????????? ???? ????????? (Docker, ??????, ????? ?????? ? ?.?.);
  - ? ????? ????????? ????? callback (???????? `curl -sf -X POST "<callback_url>"`).

?????: ? cloud-init YAML ?????? ? `callback_url` ????? ??????????? (???????, ????????? ? `http://`). ?????? ????????...

2.2 ??????? B: ?????? ??????? (Kubernetes Job / Pod ? Docker ? ?.?.)

???? ????????? ??????????? ?? ?? cloud-init, ? ????????? ?????/Job???:

- Callback URL ? ????? ????? ????? ? ??????? VM:
  - **?????:** ????????? ?? VM `spec.template.metadata.annotations[NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY]`.
  - **URL:** ??????? ?? ???? ?? ???????, ??? ? ? ?????????? (??. ????), ??? ??????? ? ?????????/ConfigMap ??? ???????...

? ???? ?????? ? `vm_utils.py` ? ????? `if use_custom_driver:` ????? ?????? ?? ????????? ? cloud-init, ? ?????? ??? ??...

---

3. Callback ????? ????????? ????????

3.1 ??????????

????? callback ???????? ??????????, ??? ????????? ?????????? ???????? ????????? (??????? ??? ? ??????? ? ?? ????? ???...

3.2 URL

?????? (???????? ? `app/routes/vm_routes.py` ??? ???????? VM):

```
{APP_INTERNAL_BASE_URL}/{username}/vm/{vm_name}/{NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH}?token={token}&namespace={namespace}
```

??????:

```
http://kubevirt-api-manager.kvm.svc.cluster.local/alice/vm/myvm/nvidia-custom-driver-installed?token=...&namespace=kv...
```

- `username` ? ????? ??????????? ? ??????????.
- `vm_name` ? ??? VM.
- `token` ? ?????????? ????? ?? ????????? VM.
- `namespace` ? namespace VM (???????? `kv-alice-001`).

????????? ???????? ????? query; ?????: **GET ??? POST** (??? ?????????????).

3.3 ??? ???????? callback

- ???? ????????? ? cloud-init: ? ????? ?????? ???????/??????? ? `runcmd`, ????????:
  `curl -sf -X POST '...'` ? ???????????? URL (??? ?????????? ?? write_files).
- ???? ????????? ? ????????? Pod/Job: ????? ???????? ????????? ????????? HTTP-?????? ?? ???? URL ?? ???? (????? ?????...

3.4 ?????????? callback ? ??????????

????: **`app/routes/vm_routes.py`**.

???????: **`nvidia_custom_driver_installed`** (??????? `/<username>/vm/<vm_name>/{NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH}`).

??????? ????????:

1. ?????? ?? query: `token`, `namespace`.
2. ????????, ??? `namespace` ? ????????? `K8S_NAMESPACE_PREFIX`.
3. ????????? VM ?? `namespace` ? `vm_name`.

```
4. ?????????? `token` ? ?????????? VM ?? ????? `NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY`.
5. ??? ?????????? ???????? ? ?????????? `{"success": true}`.

???? ????? ???????? ???? ?????? (?????????? ?????????, ????????, ??????? ????? ? ?.?.).

---

## 4. ??????????? (config ? env)

????: **`app/config.py`**. ??????? ?????????? ?????????: **`env.example`**.

| ?????????? | ?????????? |
|------------|------------|
| `NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER` | ?????? GPU-???????? (????? ???????), ??? ??????? ??????????? ???????? «????...
| `NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH` | ??????? ???? callback (?? ????????? `nvidia-custom-driver-installed`). |
| `NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY` | ???? ????????? ?? VM ? ??????? (?? ????????? `vm.kubevirt.io/nvidia-c...
| `APP_INTERNAL_BASE_URL` | Base URL ?????????? ??? ?????? callback (?????? ???? ???????? ?? ????/VM). |

????????? ??????????, ????????? ?? ?????? ????????? PVC/ISO, ???????.

---

## 5. ??????? ????? ?? ??????

| ?????? | ???? | ????? |
|--------|------|--------|
| ???????? ???????/?????? ????????? ? cloud-init ??? ????????? ???????? | `app/utils/vm_utils.py` | `generate_cloud_i...
| ???????? callback URL ? ???????? ?? (write_files) | `app/utils/vm_utils.py` | ??? ??, ????? `write_files_list` ? ??...
| ?????????? ????????? ????? ????????? (?????????, ???? ? ?.?.) | `app/routes/vm_routes.py` | ??????? `nvidia_custom_...
| ???????? ?????? GPU ??? ?????????? ???????? | `app/config.py`, `env.example` | `NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER` |
| ???????? ?????/???? ? ????? | `app/templates/dashboard.html` | ???? GPU, ?????? `nvidia_custom_driver_installer_typ...

---

## 6. ????? ?????? ??? ???????? VM ? ????????? ?????????

1. ??????????? ???????? GPU ?? `NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER` ? ??? ???????????? ??? ??????????.
2. ? `vm_routes.py` ??? ???????? VM ?????????? `use_custom_nvidia_driver`, ??????????? `nvidia_custom_driver_callba...
3. ? `generate_vm_manifest` ????? ??????????? ? ????????? VM.
4. ? `generate_cloud_init_userdata` ?????????? `nvidia_custom_driver_callback_url` ? `nvidia_custom_driver_installer_...
5. ????? ????????? ???????? (?? cloud-init ??? ?? ?????????? ????) ?????????? ?????? ?? callback URL ? `token` ? `na...
6. ?????????? `nvidia_custom_driver_installed` ????????? ?????; ??? ???????????? ? ??? ?????????? ????????????? ??...

?? ???? ??????? ????? ??????? ?????????? ? Docker ??? ????? ?????? ?????? ????????? ?????????? ????????.
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/DEVELOPMENT.md (6.4 KB)

```
# ??????????? ?? ?????????????

## ????????? ?????????? ????????? ??? Kubernetes

??? ??????? UI ??? ??????? ? Kubernetes ???????? ??????????? ????? `FRONTEND_STUB_MODE`.

### ??????? ????? ? Docker Compose

```bash
# 1. ????????? ?????????? ? stub ??????
docker-compose up -d

# 2. ???????? ???????
# UI: http://localhost:8083
# ?????: example / your-secret-password (?? ?????????)
```

?????????? ?????????? ? `FRONTEND_STUB_MODE=true` ? ????? ?????????? ????-??????:
- **demo-gpu-vm** - ?????????? VM ? GPU (8 CPU, 32GB RAM, 1x A100-80GB)
- **data-import-vm** - VM ? ???????? ???????? (4 CPU, 16GB RAM)
- **analytics-vm** - ???????????? VM (6 CPU, 24GB RAM)

????????? ?????? ??? Docker

```bash
# 1. ?????????? ??????????
pip install -r requirements.txt

# 2. ????????? ? stub ?????? (?????? ??? ????????? ?????????)
./run-stub-mode.sh

# ??? ? ???????? PostgreSQL (???????, ???????????, ????)
./run-stub-mode.sh --with-db

# 3. ???????? ???????
# UI: http://localhost:8970
# ? --with-db: ????? admin / admin (?????? ?????? ??????? bootstrap-??????)
```

??? ???????? ? FRONTEND_STUB_MODE

? **Dashboard**
- ?????? VM ? ??????? ?????????
- ?????????? ?????? VM
- GPU ??????? ?? ??????

? **?????? VM**
- ?????????? ? ???????? (CPU, RAM, Storage, GPU)
- ?????? VM
- ?????????? VM (Start/Stop/Delete) - ??????????

? **????????**
- ??????? CPU ? ?????? ? ???????????? ?????????? ???????
- ?????????? ?????? ? ???????? ???????
- ?????? ????????? ????????????

? **???????? VM**
- ????? ???????? VM
- ????????? ?????
- ???????? ???????? ????????

? **VNC Console**
- ?? ???????? ??? ????????? Kubernetes

Stub ????? ? ???????? ??
```

??? ??????? `./run-stub-mode.sh --with-db`:

- PostgreSQL ??????????? ? Docker (`docker-compose.stub.yml`, ???? 5433)
- ?????????? ??????????? ? ??, ??????? ??????? ??? ?????? ???????
- Bootstrap: ???? ???????????? ??? ? ????????? ????? `admin` / `admin` (??? `ADMIN_USERNAME` / `ADMIN_PASSWORD`)
- ????????: ?????? «???????????», ?????????? ??????, ??????????? ????????????
- VM ??-???????? ????? stub (??? K8s)

????????? PostgreSQL: `docker compose -f docker-compose.stub.yml down`

?????????? ????????? ??? ??????????

??????????	????????	????????
`FRONTEND_STUB_MODE`	`true`	???????? ????? ?????? (??? Kubernetes)
`DEBUG`	`true`	????? ???????
`DEMO_USERNAME`	`example`	????? ??? ??????? ??? ??
`DEMO_PASSWORD`	`your-secret-password`	?????? ??? ??????? ??? ??

????????? ????-??????

?????? ??? stub ?????? ????????? ? `app/utils/k8s_utils.py`:

```python
STUB_VM_TEMPLATES = [
    _build_stub_vm(
        name="my-custom-vm",
        printable_status="Running",
        cpu_cores=16,
        memory_gi=64,
        storage_gi=500,
        gpu_resource="nvidia.com/a100-80gb",
        gpu_count=2
    ),
    # ... ???????? ???? VM
]
```

???????

???? ? Docker

```bash
# ???????? ???? ? ???????? ???????
docker-compose logs -f

# ?????????? ???? ??????????
docker-compose logs kubevirt-api
```

???????? stub ??????

```bash
# ????????? /ready endpoint
curl http://localhost:8083/ready

# ????? ?????? ????????? "mode": "stub"
# {"status": "ready", "mode": "stub"}
```

??????????? ? ????????? ????????

??? ?????? ? ????????? Kubernetes:

```bash
# 1. ????????? ??? kubeconfig ????????
kubectl get nodes

# 2. ????????? stub ????? ? docker-compose.yml
# FRONTEND_STUB_MODE: "false"
```

```

# 3. ????????????
docker-compose down
docker-compose up -d
```

?????????? ????? ???????

1. **???????? ???? ? k8s_utils.py**
```python
if FRONTEND_STUB_MODE:
    # ???? ?????? ??? stub ??????
    return stub_data
```

2. **???????? ???????? ??????**
```python
# ???????? ?????? ? Kubernetes API
api = get_custom_api()
return api.get_namespaced_custom_object(...)
```

3. **????????????? ??? ??????**
- Stub: `FRONTEND_STUB_MODE=true python app/main.py`
- Real: `FRONTEND_STUB_MODE=false python app/main.py`

Troubleshooting

????? ??????

```bash
# ???????? ????? ? docker-compose.yml
ports:
  - "8084:8080"  # UI (?????? 8083)
  - "8005:8001"  # kubectl proxy (?????? 8004)
```

????????? ?? ??????????

```bash
# ??????????? ?????
docker-compose build --no-cache
docker-compose up -d
```

??? ?????? ? ????????

??????? ? stub ?????? ??????????? ????????? ???????.
????????? ??????? ???????? (F12) ?? ?????? JavaScript.

???????? ???????

```bash
# ?????????? ??????????
docker-compose down

# ?????????? ? ?????????
docker-compose up -d --build

# ?????????? ??????
docker-compose ps

# ????? ? ?????????
docker-compose exec kubevirt-api bash
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/DUAL-DISPLAY-VNC-GPU.md (8.7 KB)

```
# ?? Dual Display: VNC + GPU Passthrough

**??????**: 1.0-rc.0.0.1

## ????????

??? ????????????? GPU passthrough ? ?????????? ??????? ????????? ????????:

- **?? ????????? GPU ????????**: VNC ???????? ????????? (????? ???????????? bochs/vga ?????)
- **????? ????????? GPU ????????**: ?? ?????????? ????? ?? ?????????? GPU ? VNC ?????????? ?????? ?????
- **??????????**: ?????????? ????????? VM ???????? ????? VNC

## ???????: Dual Display

VM ? GPU ????????????? ???????? **??? ???????????????** (QXL + ????????? GPU):

### ??????????

```
??????????????????????????????????????????????????
?              VM with GPU                       ?
?                                                ?
?  ???????????????      ?????????????????????  ?
?  ? QXL Display ?      ?  Physical GPU   ?  ?
?  ? (Secondary) ?      ?   (Primary)     ?  ?
?  ?             ?      ?                 ?  ?
?  ?  VNC Access ?      ? Monitor Output  ?  ?
?  ?  1920x1080  ?      ?    4K/144Hz     ?  ?
?  ???????????????      ?????????????????????  ?
?         ?                      ?            ?
??????????????????????????????????????????????????
          ?                      ?
          ?                      ?
    ???????????          ???????????
    ?   VNC    ?          ? Physical ?
    ?  Client  ?          ? Monitor  ?
    ???????????          ???????????
```

??? ?????????? ????????????

??? ???????? VM ? GPU (????? API ??? ???-?????????) ? manifest ??????????:

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: gpu-vm
spec:
  template:
    metadata:
      annotations:
        # ? ???????????? ??????????
        kubevirt.io/domain: |
          devices:
            video:
              - model:
                  type: 'qxl'
                  vram: '65536'
                  heads: '1'
    spec:
      domain:
        devices:
          # GPU passthrough (?????????? ????????????)
          gpus:
```
```

```
            - name: gpu0
              deviceName: nvidia.com/1080ti

          # Audio passthrough (??????????? ????????????)
          hostDevices:
            - name: audio0
              deviceName: nvidia.com/1080ti-audio
```

????????? ? ??

????? ???????? ?? ????????? **??? ???????**:

???????	???	?????????	????????????
Display 1	QXL (???????????)	1920x1080	VNC Remote Desktop
Display 2	NVIDIA GPU (??????????)	3840x2160	Primary Monitor

????????????

??????? 1: VNC ?????? (??????????)

```bash
# ??????????? ????? VNC (?????? ????????)
kubectl virt vnc <vm-name> -n <namespace>

# ??? ????? ???-?????????
# VM Details ? VNC Console
```

??? ?????: Secondary display (QXL) - ????? ???? ?????? ??????? ??????, ?? ?????? ????? ? ????????? ???? ????????.

??? ????: ?????????? ????????, ?????????, ????????? ??.

??????? 2: SSH + X11 forwarding (?????? ? GPU)

```bash
# ??????????? ?? SSH ? X11 forwarding
ssh -X user@<vm-ip>

# ??? ???????? NodePort ??? SSH ????? ???-?????????
# VM Details ? Services ? Add Service ? SSH (22)
```

??? ?????: Primary display (GPU) - ?????? ?????????????????.

??? ????: 3D ?????????, CUDA ??????????, ML.

??????? 3: ?????????? ???????

?????????? ??????? ? HDMI/DisplayPort ?????? GPU ?? ???????.

??? ????: ??????????? ????????????????? ??? ??????? ????????.

????????? ? Linux

??????????? Primary Display ?? QXL (??? VNC)

???? ?????? ??????????? VNC ??? ???????? ?????:

1. ??????????? ????? SSH ??? ?????????? ???????
2. ????????? X11/Wayland: ???????? QXL ??? primary display
3. VNC ????? ?????????? ??????? ????

????????????? ??

??	Dual Display
Linux (Ubuntu/Debian/Fedora)	? ??

??????????? ??????

??? ???????? annotation kubevirt.io/domain

KubeVirt ????????? ????????? ??????????? libvirt XML ????????? ????? annotations:

```yaml
metadata:
  annotations:
    kubevirt.io/domain: |
      devices:
        video:
          - model:
              type: 'qxl'
              vram: '65536'  # 64 MB ??????????
              heads: '1'      # ???? ???????
```

??? ???????? VM KubeVirt **??????????** ???? XML ? ???????????? ??????????????? ??????????:

```xml
<!-- ???????????? (default) -->
<devices>
  <video>
    <model type="bochs" vram="16384"/>
  </video>
  <graphics type="vnc" autoport="yes"/>
</devices>

<!-- + ????????? ????? annotation -->
<devices>
  <video>
    <model type="qxl" vram="65536" heads="1"/>
  </video>
</devices>

<!-- = ????????? ? libvirt domain -->
<devices>
  <video>
    <model type="bochs" vram="16384"/>  <!-- VNC default -->
  </video>
  <video>
    <model type="qxl" vram="65536"/>     <!-- ??? dual display -->
  </video>
  <graphics type="vnc"/>
  <hostdev mode="subsystem" type="pci">  <!-- GPU passthrough -->
    <source>
      <address domain="0x0000" bus="0x06" slot="0x00"/>
    </source>
  </hostdev>
</devices>
```

?????? QXL, ? ?? bochs/vga?

???	??????????????????	????????? ?????????	????????????
bochs	??????	1024x768	????????????
vga	??????	1280x1024	?????? ??
qxl	**???????**	**?? 4K**	**Modern OS**

QXL ????????????:
- ? 2D ????????? ??? VNC
- ? ??????? ?????????? (1920x1080+)
- ? ??????? ???????? ??? ????????? ???????
- ? ????????? copy-paste ????? SPICE/VNC

?????????? ? ????

Linux VMs (vm_utils.py)

```python
```

```
# app/utils/vm_utils.py, ?????? ~690

if gpu_count_int > 0:
    # ... GPU devices ...

    # Add secondary display for VNC
    manifest["spec"]["template"]["metadata"]["annotations"]["kubevirt.io/domain"] = """
devices:
  video:
    - model:
        type: 'qxl'
        vram: '65536'
        heads: '1'
"""
```

Troubleshooting

????????: VNC ??????????? ?????? ?????

????????: ?? ??????????? GPU ??? primary display.

????????: ??????????? ????? SSH ? ??????????? primary display ?? QXL.

????????: QXL device ??????????? ? VM

????????: VM ??????? ??? GPU (dual display ??????????? ?????? ??? GPU VMs).

????????: ??? ????????? ?????????. Dual display ????? ?????? ??? GPU passthrough.

??. ?????

- GPU-SETUP.md - ????????? GPU ? KubeVirt
- GPU-PASSTHROUGH-FIX.md - ??????????? ??????? ? GPU

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/GPU-PASSTHROUGH-FIX.md (4.8 KB)

```
# ?????????? ???????? "GPU fallen off the bus"

## ????????
```
NVRM: The NVIDIA GPU 0000:0e:00.0 (PCI ID: 10de:1b06) installed in this system has fallen off the bus and is not resp...
```

??? ?????? ????????, ??? GPU ???????? ????? ? VM ????? PCI passthrough.

## ???????
1. **?????????? IOMMU ????????** - GPU ?? ?????????? ??????? ???????
2. **???????????? machine type** - i440fx ?? ????????? ??? PCI passthrough
3. **?????????? UEFI** - ?????????? GPU ??????? UEFI ?????? BIOS
4. **????????? ??????????** - ???????????? ???????? CPU

## ?????????? ? ????

### 1. ???????? ???????? VM (vm_manifest_common.py)
??????? `add_gpu_to_manifest()` ? `app/utils/vm_manifest_common.py` ????????? ????????? ??? GPU VM:

```python
# Machine type Q35 (??????????? PCIe)
domain["machine"] = {"type": "q35"}

# UEFI firmware (?????????? ??? GPU)
domain["firmware"] = {"bootloader": {"efi": {"secureBoot": False}}}

# CPU host-passthrough ??? IOMMU
domain["cpu"]["model"] = "host-passthrough"

# IO threads ??? ??????????????????
domain["ioThreadsPolicy"] = "auto"
```

??????????? ? `vm_utils.py` ??? ???????? VM ? GPU.

???????? ?? ????? Kubernetes

1. ????????? IOMMU ???????
```bash
# ?? ???? ? GPU
dmesg | grep -i iommu
# ?????? ????: DMAR: IOMMU enabled

cat /proc/cmdline
# ?????? ?????????: intel_iommu=on iommu=pt
```

2. ????????? GPU ? IOMMU ??????
```bash
# ????? PCI ????? GPU
lspci | grep -i nvidia
# ??????: 0e:00.0 VGA compatible controller: NVIDIA Corporation GP102 [GeForce GTX 1080 Ti]

# ????????? IOMMU ??????
find /sys/kernel/iommu_groups/ -type l | grep 0e:00
```

3. ????????? ??? vfio-pci ????????
```bash
lsmod | grep vfio
# ?????? ????: vfio_pci, vfio_iommu_type1, vfio

# ????????? ??? GPU ?????????? vfio-pci ???????
lspci -k -s 0e:00.0
```
```

Kernel driver in use: vfio-pci
```

## ??????????? ???? (???? ?????)

### /etc/default/grub
```bash
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash intel_iommu=on iommu=pt vfio-pci.ids=10de:1b06,10de:10ef"
```

???:
- `10de:1b06` - PCI ID GPU (GTX 1080 Ti)
- `10de:10ef` - PCI ID HDMI Audio

????? ?????????:
```bash
sudo update-grub
sudo reboot
```

??????????? VM

????? ?????????? ??????????? ? ????:

1. **??????? ?????? VM**:
```bash
kubectl delete vm <vm-name> -n <namespace>
kubectl delete dv <vm-name>-rootdisk -n <namespace>
```

2. **??????? ????? VM** ????? UI ? GPU

3. **????????? ???????** ? ?????????:
```bash
kubectl logs -n <namespace> virt-launcher-<vm>-xxxxx -c compute
```

???????? ?????? VM

????? ???????? VM:

```bash
# ????????? GPU ?????
lspci | grep -i nvidia

# ?????????? ??????? (???????????? ????? cloud-init)
# ??? ???????:
sudo ubuntu-drivers autoinstall

# ????????? ??????? ????????
nvidia-smi

# ?? ?????? ???? ?????? "fallen off the bus"
dmesg | grep -i nvrm
```

????????????? ???????????

1. ??????????? dedicated CPU cores (??????????)
```yaml
spec:
  domain:
    cpu:
      dedicatedCpuPlacement: true
```

2. Huge Pages ??? ?????? (??????????)
```yaml
spec:
  domain:
    memory:
```
```

```
      hugepages:
        pageSize: "1Gi"
```

3. ????????? ??? GPU ?? ??????????? ??????
```bash
# ?? ????
nvidia-smi
# ?? ?????? ?????????? ????????
```

Troubleshooting

?????? ??? ???????? VM
```bash
kubectl describe vm <vm-name> -n <namespace>
kubectl describe vmi <vm-name> -n <namespace>
```

???? virt-launcher
```bash
kubectl logs -n <namespace> virt-launcher-<vm>-xxxxx -c compute
```

???????? KubeVirt
```bash
kubectl get kubevirt -n kubevirt kubevirt -o yaml
```

????????? ??? ????????:
```yaml
spec:
  configuration:
    developerConfiguration:
      featureGates:
        - GPU
        - HostDevices
```

?????? ?????????
- **?????**: 2025-11-27
- **?????? API**: 1.1.3
- **??????**: app/utils/vm_utils.py

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/GPU-SETUP.md (5.8 KB)

```
# ????????? GPU Passthrough ??? KubeVirt

## 1. ????????????? ????-??? ? GPU

?? ?????? ???? Kubernetes, ??? ??????????? GPU (NVIDIA):

```bash
# ????????? ????????? NVIDIA
sudo apt update
sudo apt install -y nvidia-driver-535 nvidia-utils-535

# ????????? NVIDIA Container Toolkit
distribution=$(. /etc/os-release;echo $ID$VERSION_ID)
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey | sudo gpg --dearmor -o /usr/share/keyrings/nvidia-con...
curl -s -L https://nvidia.github.io/libnvidia-container/$distribution/libnvidia-container.list | \
  sed 's#deb https://#deb [signed-by=/usr/share/keyrings/nvidia-container-toolkit-keyring.gpg] https://#g' | \
  sudo tee /etc/apt/sources.list.d/nvidia-container-toolkit.list

sudo apt update
sudo apt install -y nvidia-container-toolkit

# ????????? containerd ??? ?????? ? GPU
sudo nvidia-ctk runtime configure --runtime=containerd
sudo systemctl restart containerd

# ???????? ?????????
nvidia-smi
```

2. ????????? NVIDIA Device Plugin ? Kubernetes

```bash
# ????????? device plugin
kubectl create -f https://raw.githubusercontent.com/NVIDIA/k8s-device-plugin/v0.14.0/nvidia-device-plugin.yml

# ???????? ?????????
kubectl get pods -n kube-system | grep nvidia

# ???????? ??????? GPU ???????? ?? ?????
kubectl get nodes -o json | jq '.items[].status.capacity'
```

?? ?????? ??????? ?????? `nvidia.com/gpu` ?? ????? ? GPU.

3. ????????? KubeVirt ??? GPU Passthrough

```bash
# ???????? ?????? PCI ????????? NVIDIA ?? ????
lspci -nn | grep NVIDIA

# ????????? ?????:
# 01:00.0 VGA compatible controller [0300]: NVIDIA Corporation GA102 [GeForce RTX 3090] [10de:2204] (rev a1)
# 01:00.1 Audio device [0403]: NVIDIA Corporation GA102 High Definition Audio Controller [10de:1aef] (rev a1)

# ????????? KubeVirt ??? ?????????? GPU ????????? ? HDMI Audio ???????????
# ????????: ??? ?????? GPU Passthrough ??????????? ?????? ???? ??????? FeatureGate "HostDevices"
# ? ??????????? KubeVirt (kubevirt-cr.yaml).

kubectl patch kubevirt kubevirt -n kubevirt --type=merge -p '{
  "spec": {
    "configuration": {
      "developerConfiguration": {
        "featureGates": [
          "HostDevices"
        ]
```
```

```
      },
      "permittedHostDevices": {
        "pciHostDevices": [
          {
            "pciVendorSelector": "10DE:1B06",
            "resourceName": "nvidia.com/1080ti",
            "externalResourceProvider": false
          },
          {
            "pciVendorSelector": "10DE:10EF",
            "resourceName": "nvidia.com/1080ti-audio",
            "externalResourceProvider": false
          }
        ]
      }
    }
  }
}'

# ???? ?????? ???????? ????????? ??? NVIDIA GeForce GTX 1080 Ti.
# ID ?????????: 10DE:1B06 (?????) ? 10DE:10EF (?????).

# ???????? ???????????
kubectl get kubevirt kubevirt -n kubevirt -o yaml | grep -A 10 permittedHostDevices
```

4. ???????????? ? Web UI

????? ????????? ??????????????:

1. ???????? ???-????????? KubeVirt Manager
2. ??? ???????? VM ???????? **"????????? ???????????"**
3. ??????? ????????? GPU (?? 0 ?? 4)
4. ???????? VM

5. ???????? GPU ?????? VM

????? ???????? VM ? GPU ??????????? ? ??? ? ?????????:

```bash
# ????? ? ??????? VM
virtctl console <vm-name> -n <namespace>

# ?????? VM ????????? ????????
sudo apt update
sudo apt install -y nvidia-driver-535

# ???????????? VM
sudo reboot

# ????? ??????????? ????????? GPU
nvidia-smi
```

6. Troubleshooting

GPU ?? ????? ? VM

```bash
# ????????? ?? ????-????
lspci | grep NVIDIA

# ???????? pod VM
kubectl get vmi <vm-name> -n <namespace> -o yaml | grep -A 20 spec

# ????????? ???????
kubectl describe vmi <vm-name> -n <namespace>
```

Device Plugin ?? ????????

```bash
# ????????? ???? device plugin
kubectl logs -n kube-system -l name=nvidia-device-plugin-ds

# ???????????? device plugin
kubectl delete pods -n kube-system -l name=nvidia-device-plugin-ds
```

KubeVirt ?? ????? GPU ???????

```bash
# ????????? ??????????? KubeVirt
kubectl get kubevirt kubevirt -n kubevirt -o yaml

# ???????????? virt-handler
kubectl delete pods -n kubevirt -l kubevirt.io=virt-handler
```

7. ?????? ?????????

?? **GPU Passthrough ???????:**

- ????????? IOMMU ?? ????-???? (Intel VT-d ??? AMD-Vi)
- ????????? ???????????? ? BIOS
- ????????? ????????? GPU ?? ????
- ???? GPU ????? ????????????? ?????? ????? VM ???????????

?? **???????????:**

- GPU ?????? "????????" ????? ?????????? VM
- ??? vGPU ????? Enterprise-???????? NVIDIA
- ????????? GPU ?? ??????????? passthrough

?? **????????????????????:**

- GPU Passthrough ???? ~95-98% ???????? ?????????????????
- ????????? ??????? ?? ???????????? ?????????

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/KUBERNETES-DEPLOYMENT.md (5.8 KB)

```
# Kubernetes Deployment Guide

This guide describes how to deploy KubeVirt API Manager to a Kubernetes cluster, using either the bundled manifest or...

## Prerequisites

- **Kubernetes** 1.24+
- **KubeVirt** 1.0+
- **CDI** (Containerized Data Importer) for VM disk management
- **Storage classes**: `your-SC` (VM disks), `your-SC` (application config volume)
- **Optional**: Cert-manager, External-DNS, Prometheus Operator (for ServiceMonitor)

## Deployment Options

### Option 1: Bundled Manifest (Recommended)

Single-file deployment including all resources:

```bash
# 1. Create namespace
kubectl create namespace kvm

# 2. Create secrets (required before applying bundled.yaml)
kubectl create secret generic kubevirt-api-secrets \
  --from-literal=secret-key='your-random-secret-key-at-least-32-characters' \
  --from-literal=demo-username='admin' \
  --from-literal=demo-password='your-secure-password' \
  -n kvm

# 3. Create Juniper SSH key secret (required if using Juniper NAT)
kubectl create secret generic juniper-ssh-key \
  --from-file=id_rsa=./your_juniper_private_key \
  -n kvm

# 4. Apply bundled manifest
kubectl apply -f kubernetes_example/bundled.yaml

# 5. Verify deployment
kubectl get pods -n kvm
kubectl get ingress -n kvm
```

Option 2: Individual Manifests

For finer control or CI/CD pipelines:

```bash
# Apply in order (namespace first, then RBAC, then workload)
kubectl apply -f kubernetes_example/ns.yaml
kubectl apply -f kubernetes_example/rbac.yaml
kubectl apply -f kubernetes_example/kvm-clone-rbac.yaml   # Required for Windows
kubectl apply -f kubernetes_example/juniper-sidecar-cm.yaml
kubectl apply -f kubernetes_example/secrets-example.yaml          # Edit placeholders first!
kubectl apply -f kubernetes_example/service.yaml
kubectl apply -f kubernetes_example/statefulset.yaml
kubectl apply -f kubernetes_example/ingress.yaml
kubectl apply -f kubernetes_example/servicemonitor.yaml          # If Prometheus Operator installed
kubectl apply -f kubernetes_example/global-network-policy.yaml   # If Calico installed
```

What the Bundled Manifest Contains

Resource	Description
Namespace	`kvm`
```

PriorityClass	`kubevirt-manager-priority`
ServiceAccount	`kubevirt-api-manager-multus`
ClusterRoles	Cluster-wide + user namespace permissions
ClusterRoleBindings	Binds SA to ClusterRoles
Role + RoleBinding	`windows-iso-cloner` ? required for Windows DataVolume cloning
ConfigMap	`juniper-sidecar-code` ? Juniper sidecar Python script
Service	`kubevirt-api-manager` (NodePort 30180)
Ingress (login)	`/login` ? no basic auth (for session redirect)
Ingress (main)	`/` ? with basic auth and external-DNS annotations
ServiceMonitor	For Prometheus (requires Prometheus Operator)
Deployment	Main application + Juniper sidecar + dnat-logs
GlobalNetworkPolicy	VM egress restrictions (Calico)

Required Secrets

Secret	Keys	Purpose
`kubevirt-api-secrets`	`secret-key`, `demo-username`, `demo-password`	Flask session, demo login
`juniper-ssh-key`	`id_rsa`	SSH key for Juniper router (if using NAT)

Create from example:

```bash
cp kubernetes_example/secrets-example.yaml kubernetes_example/secrets.yaml
# Edit secrets.yaml with real values
kubectl apply -f kubernetes_example/secrets.yaml
```

Customization Before Apply

Edit `kubernetes_example/bundled.yaml` (or individual manifests) and replace:

Placeholder	Description
`CHANGE-ME.example.com`	Your Ingress hostname
`CHANGE-ME` (Ingress target)	External IP or LB address
`CHANGE-ME` (hostAliases)	IP for LLM hosts
`your-registry.example.com`	Container registry
`JUNIPER_HOST`, `JUNIPER_USER`, `JUNIPER_EXTERNAL_IP`	Juniper router settings
`your-SC`	Storage class for config volume
`your-SC`	Storage class for VM disks

Storage Classes

- **Config volume**: `your-SC` (or change `storageClassName` in `volumeClaimTemplates`)
- **VM disks**: `your-SC` (configurable via `STORAGE_CLASS_NAME` env)

Windows Support

For Windows VM creation (installer or golden image):

1. **RBAC**: The bundled manifest includes `windows-iso-cloner` Role. No extra steps.
2. **DataVolumes**: Create Windows ISO and golden image per [WINDOWS-GOLDEN-IMAGE-SETUP.md](WINDOWS-GOLDEN-IMAGE-SETU...
3. **Config**: Defaults use `kvm` namespace. Override via env:
   - `WINDOWS_ISO_NAMESPACE`
   - `WINDOWS_GOLDEN_IMAGE_NAMESPACE`

Health Checks

```bash
# In-cluster
kubectl exec -it deployment/kubevirt-api-manager -n kvm -- curl -s http://localhost:8080/health
kubectl exec -it deployment/kubevirt-api-manager -n kvm -- curl -s http://localhost:8080/ready

# Port-forward
kubectl port-forward svc/kubevirt-api-manager 8080:80 -n kvm
curl http://localhost:8080/health
curl http://localhost:8080/ready
```

Verifying Deployment

```bash
# Pods
kubectl get pods -n kvm
# Expect: kubevirt-api-manager-* with 3 containers (manager, juniper-sidecar, dnat-logs)

# Service
kubectl get svc -n kvm
# Expect: kubevirt-api-manager NodePort 80:30180

# Ingress
kubectl get ingress -n kvm
```

Troubleshooting

Pod not starting

- Check secrets exist: `kubectl get secrets -n kvm`
- Check PVC: `kubectl get pvc -n kvm`
- Check events: `kubectl describe pod -n kvm -l app=kubevirt-api-manager`

ServiceMonitor not selecting pods

- Ensure Service port has `name: http` (ServiceMonitor references `port: http`)
- Check Prometheus Operator is installed and watching the namespace

Windows cloning fails

- Ensure `windows-iso-cloner` Role and RoleBinding are applied
- DataVolumes `windows-10-iso` and `windows-10-golden-image` must exist in `kvm`

Juniper sidecar errors

- Verify `juniper-ssh-key` Secret contains valid `id_rsa`
- Check `JUNIPER_HOST`, `JUNIPER_USER`, `JUNIPER_EXTERNAL_IP` env vars in Deployment

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/KUBEVIRT-GPU-UNIFIED-EXAMPLE.md (3.7 KB)

```
# ??????????? ?????? KubeVirt: consumer GPU (1080 Ti + audio) ? enterprise vGPU (RTX Pro 6000)

? ????? ???????? ????? ???? **consumer GPU** ? ????????? PCI-?????????? HDMI-????? (1080 Ti, 3080 Ti, 3090) ? **ente...

## ????? ??? ?????

- **Consumer (1080 Ti ? ?.?.):** ? `permittedHostDevices` ???????? ? GPU (`nvidia.com/1080ti`), ? ????? (`nvidia.com/...
- **Enterprise (RTX Pro 6000 mdev):** ?????????? ?????-?????????? ???. ? ????????? VM ?????????? ?????? GPU; ?????? ...

?????????? ???????? ? ???? **?? ??????** ???????: ??? ???? ??? ????? ?????-?????????? ?? ????????????.

## ?????? ???????????? KubeVirt

```yaml
apiVersion: kubevirt.io/v1
kind: KubeVirt
metadata:
  name: kubevirt
  namespace: kubevirt
spec:
  certificateRotateStrategy: {}
  configuration:
    developerConfiguration:
      featureGates:
        - HostDevices          # ??? PCI passthrough (1080 Ti + audio)
        - GPU                  # ??? mdev/vGPU (???? ????????? ? ????? ??????)
        - DisableMDEVConfiguration  # ??? externalResourceProvider mdev (RTX Pro 6000)
    imagePullPolicy: IfNotPresent
    permittedHostDevices:
      # Consumer: PCI GPU + HDMI Audio (1080 Ti)
      pciHostDevices:
        - pciVendorSelector: "10DE:1B06"
          resourceName: nvidia.com/1080ti
        - pciVendorSelector: "10DE:10EF"
          resourceName: nvidia.com/1080ti-audio
      # Enterprise: vGPU (RTX Pro 6000 Blackwell, ??? ?????)
      mediatedDevices:
        - externalResourceProvider: true
          mdevNameSelector: NVIDIA RTX 6000 1-24Q
          resourceName: nvidia.com/NVIDIA_RTX_Pro_6000_Blackwell_DC-1-24Q
        - externalResourceProvider: true
          mdevNameSelector: NVIDIA RTX 6000 2-48Q
          resourceName: nvidia.com/NVIDIA_RTX_Pro_6000_Blackwell_DC-2-48Q
        - externalResourceProvider: true
          mdevNameSelector: NVIDIA RTX 6000 4-96Q
          resourceName: nvidia.com/NVIDIA_RTX_Pro_6000_Blackwell_DC-4-96Q
  customizeComponents: {}
  imagePullPolicy: IfNotPresent
  workloadUpdateStrategy: {}
```

?????????? ??????????

?????????, ??? ? ?????????? ??????:

- **GPU_RESOURCES_WITH_AUDIO** ? ?????? ???????? GPU, ??? ??????? ? ???????? ???? ????????? ?????-?????????? (`<resou...
- **NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER** ? ??? ? ??????, ?????? ???????? ?????????? ???????? (RTX Pro 6000 ? ?.?.).

????? ?????????? ?????? KubeVirt ? ??????????? ?????????? VM ? 1080 Ti ???????? GPU ? ?????, VM ? RTX Pro 6000 ? ???...
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/KUBEVIRT-UPGRADE.md (9.3 KB)

```
# ???? ????????????? KubeVirt ? v1.6.2 ?? v1.7.0

## ??????? ?????????

- **??????? ??????**: v1.6.2 (??????????? 22 ??????? 2025)
- **??????? ??????**: v1.7.0 (???????? 27 ?????? 2025)
- **????? ??????????**: ?????? ????????? ????? ????????? (??? Helm)
- **Namespace**: `kubevirt`

## ???????? ????????? ? v1.7.0

### ??????????? ???????????

- ? ?????????????? ????????? ???????? ? Multus/L3 ??????
- ? ??????????? ???????? ??????????? ????? (hotplug)
- ? ?????????????? ??????????? ???????? ??? ??????? ?????????
- ? ??????????? ??????? ? attachment pods ? Multus
- ? ????????? Kubernetes v1.33
- ? ??????????? ??????? ? ??????????

### ????? ???????????

- VMPool auto-healing ? scale-in ?????????
- ?????????????? ????????? GPU ? live migration ??? ImageVolume
- ????????? Intel TDX ? AMD SEV-SNP (????????????????)
- ?????????????? ??????????? Preferences
- ??????????? instancetypes v1.5.1

### ?????? ?????????

- ?? ?????? ?????????????? API `instancetype.kubevirt.io/v1alpha{1,2}`
- ?? ??????? ????????? ??????????? ppc64le
- ?? `virtctl ssh/scp` ?????? ?????? ?????????????? ????????? SSH ??????
- ?? Deprecated: `foregroundDeleteVirtualMachine` ? `kubevirt.io/foregroundDeleteVirtualMachine`

## ???????? ????????????

### 1. ????????? ??????? ?? ? ?????? API

```bash
# ????????? ??????????????? ?????? API ? VMs
kubectl get vm -A -o jsonpath='{range .items[*]}{.apiVersion}{"\t"}{.metadata.namespace}/{.metadata.name}{"\n"}{end}'...

# ????????? instancetypes (???? ???????????????)
kubectl get virtualmachineinstancetype,virtualmachineclusterinstancetype -A

# ????????? VirtualMachineSnapshots
kubectl get vmsnapshot,vmsnapshotcontent -A
```

2. ????????? ??????????????? GPU ? HostDevices

```bash
kubectl get kubevirt kubevirt -n kubevirt -o yaml | grep -A 20 "permittedHostDevices"
```

3. ??????? ????????? ????? ???????????????

```bash
# Backup KubeVirt CR
kubectl get kubevirt kubevirt -n kubevirt -o yaml > kubevirt-cr-backup-$(date +%Y%m%d).yaml

# Backup ???? VM
kubectl get vm -A -o yaml > all-vms-backup-$(date +%Y%m%d).yaml

```
```

Backup ???? VMI
kubectl get vmi -A -o yaml > all-vmis-backup-$(date +%Y%m%d).yaml
```

## ????????? ??????????

### ????? 1: ?????????? ????????? (????????????)

#### ??? 1: ??????????

```bash
# ?????????? ?????????? ??????
export KUBEVIRT_VERSION=v1.7.0

# ????????? ??????? ?????????? VM
kubectl get vmi -A
```

??? 2: ?????????? ????????

```bash
# ????????? ????? ???????? ????????
kubectl apply -f https://github.com/kubevirt/kubevirt/releases/download/${KUBEVIRT_VERSION}/kubevirt-operator.yaml
```

??? 3: ?????????? CR

```bash
# ????????? ????? ???????? CR
kubectl apply -f https://github.com/kubevirt/kubevirt/releases/download/${KUBEVIRT_VERSION}/kubevirt-cr.yaml
```

??? 4: ?????????? ??????????

```bash
# ??????? ?? ???????? ??????????
kubectl -n kubevirt get kubevirt kubevirt -w

# ????????? ?????? ?????
kubectl get pods -n kubevirt -w

# ????????? ???? ????????
kubectl logs -n kubevirt -l kubevirt.io=virt-operator -f
```

??? 5: ????????????? GPU ??????????

????? ?????????? ????? ??????????? ??????????? GPU ? HostDevices:

```bash
kubectl patch kubevirt kubevirt -n kubevirt --type=merge -p '
{
  "spec": {
    "configuration": {
      "developerConfiguration": {
        "featureGates": ["GPU", "HostDevices"]
      },
      "permittedHostDevices": {
        "pciHostDevices": [
          {
            "externalResourceProvider": false,
            "pciVendorSelector": "10de:1b06",
            "resourceName": "nvidia.com/1080ti"
          },
          {
            "externalResourceProvider": false,
            "pciVendorSelector": "10de:10ef",
            "resourceName": "nvidia.com/1080ti-audio"
          }
        ]
      }

```
    }
  }
}'
```

????? 2: ????? ?????????? ??????? ? KubeVirt CR

```bash
# ???????? ?????? ?????? ???????
kubectl patch kubevirt kubevirt -n kubevirt --type=merge -p '
{
  "spec": {
    "imageTag": "v1.7.0",
    "imageRegistry": "quay.io/kubevirt"
  }
}'
```

???????? ????? ??????????

1. ???????? ?????? ??????????

```bash
# ????????? ?????? KubeVirt
kubectl get kubevirt kubevirt -n kubevirt -o jsonpath='{.status.observedKubeVirtVersion}'

# ????????? ??? ??????????
kubectl get pods -n kubevirt -o custom-columns=NAME:.metadata.name,IMAGE:.spec.containers[0].image
```

2. ???????? ?????????????????

```bash
# ????????? ?????? KubeVirt
kubectl get kubevirt -n kubevirt

# ????????? ?????????? VM
kubectl get vmi -A

# ??????? ???????? VM
kubectl apply -f - <<EOF
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: test-upgrade-vm
  namespace: default
spec:
  running: false
  template:
    metadata:
      labels:
        kubevirt.io/vm: test-upgrade-vm
    spec:
      domain:
        devices:
          disks:
          - disk:
              bus: virtio
            name: containerdisk
        resources:
          requests:
            memory: 64Mi
      volumes:
      - containerDisk:
          image: quay.io/kubevirt/cirros-container-disk-demo
        name: containerdisk
EOF

# ????????? ???????? VM
kubectl patch vm test-upgrade-vm -n default --type merge -p '{"spec":{"running":true}}'
```

```
# ????????? ??????
kubectl get vmi test-upgrade-vm -n default

# ??????? ???????? VM
kubectl delete vm test-upgrade-vm -n default
```

3. ???????? GPU ? HostDevices

```bash
# ????????? ??????????? GPU
kubectl get kubevirt kubevirt -n kubevirt -o jsonpath='{.spec.configuration.permittedHostDevices}' | jq .

# ????????? ????????? GPU ??????? ?? ?????
kubectl get nodes -o json | jq '.items[] | {name: .metadata.name, gpu: .status.allocatable}'
```

????? ??? ?????????

???? ????????? ????????, ????? ?????????? ? ?????????? ??????:

```bash
export KUBEVIRT_VERSION=v1.6.2

# ???????? ????????
kubectl apply -f https://github.com/kubevirt/kubevirt/releases/download/${KUBEVIRT_VERSION}/kubevirt-operator.yaml

# ???????? CR
kubectl apply -f https://github.com/kubevirt/kubevirt/releases/download/${KUBEVIRT_VERSION}/kubevirt-cr.yaml

# ??? ??????????? ?? ??????
kubectl apply -f kubevirt-cr-backup-*.yaml
```

???????????

1. **????? ??????????**: ????????? ? ?????? ??????????? ????????
2. **VM**: ????????? VM ?? ????? ???????????, ?? ????? ??????? snapshot ????????? VM
3. **???????????**: ??????? ?? ????????? ? ?????? ? ???????? ??????????
4. **????????????**: ????? ????????? ????????????? ????????/???????? VM, ????????, hotplug
5. **GPU VM**: ????????? ?????? VM ? GPU ????? ?????????

????????????? ??????????

- ? [KubeVirt v1.7.0 Release Notes](https://github.com/kubevirt/kubevirt/releases/tag/v1.7.0)
- ? [KubeVirt Upgrade Documentation](https://kubevirt.io/user-guide/operations/installation/#updating-kubevirt)
- ? [Breaking Changes in v1.7.0](https://github.com/kubevirt/kubevirt/releases/tag/v1.7.0)

??????? ???????? ? Multus

?????????? ?? v1.7.0 ?????? ?????? ??????? ???????? ? "pod link is missing" ????????? ??????????:

- [PR #15630] Allow decentralized live migration on L3 networks
- ????????? ????????? Multus attachment pods
- ?????????? ? ?????? ? ???????? ???????????

????? ?????????? ???????????? ?????????? ?????????? VM ??? ?????????? ????? ??????????.

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/NVIDIA-CUSTOM-DRIVER.md (18.5 KB)

```
# ????????? ??????? NVIDIA ??? enterprise GPU (RTX Pro 6000 Blackwell)

??? GPU ? ??????????? ????????? (**RTX Pro 6000 Blackwell**, mdev/vGPU) ??????? ??????????? ????? ????????? PVC: ??...

??????? ? KubeVirt (?????? ?? `permittedHostDevices.mediatedDevices`):

- `nvidia.com/NVIDIA_RTX_Pro_6000_Blackwell_DC-1-24Q`
- `nvidia.com/NVIDIA_RTX_Pro_6000_Blackwell_DC-2-48Q`
- `nvidia.com/NVIDIA_RTX_Pro_6000_Blackwell_DC-4-96Q`

????????? ? ???????? ??: ?????????????? ?????? `linux-headers-$(uname -r)`, `build-essential`, `gcc-multilib`, `dkms...

## ??? ??? ????????

1. ? PVC ???????? **?????? ISO-?????** ? ????????? (???? ???? .iso). ??????: ???-????????? ? ???????????? PVC ? `/d...
2. ??? ???????? **Linux VM** ? ????????? ??????? GPU ?? ?????? «????????? ???????» ??????????? ???????? **??? ??????...
3. ????????? **?????** PVC ? namespace ???????????, ???? ??????????? ? VM ??? **CD-ROM (?????? ??????)**.
4. Cloud-init ????????? CD-ROM (???? ?? ???? ???? ???? .iso ? ????????? ??? ????? loop), ??????? ????????? ????????...

## ?????????

### 1. ISO ? PVC ? ?????????? ??? CD-ROM (?????? ??????)

????: ? PVC ????? **??????? ISO-?????** (????? ?????). ???? ????? PVC ??????????? ? VM ??? **CD-ROM** ? ? ???????? ?...

#### ???????? ISO ? ?????????

```bash
mkdir -p driver
cp NVIDIA-Linux-x86_64-*.run nvidia-*.deb nvidia-*.rpm driver/   # ?/??? .exe ??? Windows
chmod +x driver/*.run driver/*.sh 2>/dev/null || true

xorriso -as mkisofs -r -J -V "RTX-6000-Pro-GD" -o nvidia-driver.iso driver
# ????????: xorriso -indev nvidia-driver.iso -ls /
```

??????? A: DataVolume ? ?????? ISO ?? HTTP (????????????)

????????? ISO ?? HTTP (????????? ??????, MinIO, ????????? URL). ???????? DataVolume ? ????????? `http` ? URL ?? ??...

?????? ?????????: `kubernetes_example/nvidia-custom-driver-datavolume-iso.yaml`. ????????? ???? `url` ? `storage` (?...

```bash
kubectl apply -f kubernetes_example/nvidia-custom-driver-datavolume-iso.yaml
kubectl get datavolumes -n kvm    # ????????? phase: Succeeded
```

????? ????????? ??????? PVC ? ?????? ?? DataVolume (???????? `rtx-pro-6000`) ????? ??????????? ??? ????????; ???? ??...

??????? B: ???????? ??????? ? PVC (???-????????? + kubectl cp)

???? ?? ??????????? ?????? ?? HTTP: ???????? ??????? PVC ? ???????? ???????? ? ???-?????????, ????????? ? ??? **????...

???? ??? ???????? B (????? ??? ???? .iso ?? PVC):

1. ???????? PVC ? ???-????????? (??. `kubernetes/nvidia-pvc-dr.yaml` ? `kubernetes/nvidia-pod-dr.yaml` ??? ??????? ? ...
2. ????????? **? ??????? `/data/`** (??? ??????????? ? ???? ?? ???? `/data`; ???? ????????? ? `:/`, ???? ??????? ? ?...
   - ???? ISO: `kubectl cp ./NVidia-trx-6000-guest-driver.iso <namespace>/nvidia-driver-upload:/data/NVidia-trx-6000-...
   - ??? ????????? ?????: `kubectl cp ./NVIDIA-Linux-x86_64-580.126.09-grid.run <namespace>/nvidia-driver-upload:/data/`
3. ??????? ??? (PVC ????????).
4. **??? PVC ? namespace** ????????? ????? ?? ????????? ????????? (??? ???????? ? `app/config.py`): `NVIDIA_CUSTOM_...

?????: ???????? **PVC** ???????? ? ???????????? ?????? ??????. ????????? ????? `delete` ?? ???? PVC (RBAC, ????...

????????? ?????????
```

??????? ?????????? ????????? ?????????? (Deployment/StatefulSet ??? `.env`):
   - `NVIDIA_CUSTOM_DRIVER_PVC_NAME` ? ??? PVC (???????? `rtx-pro-6000` ??? `nvidia-580-grid-driver`);
   - `NVIDIA_CUSTOM_DRIVER_PVC_NAMESPACE` ? namespace ????? PVC (???????? `kvm`);
   - `APP_INTERNAL_BASE_URL` ? ?????????? URL API (???????? `http://kubevirt-api-manager.kvm.svc.cluster.local`), ???...

?????? ***:** ? `NVIDIA_CUSTOM_DRIVER_CLONE_SIZE` ??????? ?????? ?? ?????? ??????? ????????? PVC (? `app/config.p...

2. ??? ????????? ? ? `app/config.py`

??? ????? ? ???? ???????? ? **`app/config.py`** (????? ?????????? ????????? ??? ???????). ??? ????????? ???? ? ??????...

??????????	????????	??????
`NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER`	??????? GPU (KubeVirt), ????? ???????	`nvidia.com/NVIDIA_RTX_Pro_6000_Blac...
`NVIDIA_CUSTOM_DRIVER_PVC_NAME`	??? PVC ? ISO	`rtx-pro-6000`
`NVIDIA_CUSTOM_DRIVER_PVC_NAMESPACE`	Namespace PVC	`kvm`
`NVIDIA_CUSTOM_DRIVER_UPLOAD_POD_NAME`	??? ???? ??? ???????? ISO (? ??????????)	`nvidia-driver-upload`
`NVIDIA_CUSTOM_DRIVER_MOUNT_PATH`	???? ??????????? ? ???????? ??	`/mnt/nvidia-custom-driver`
`NVIDIA_CUSTOM_DRIVER_CLONE_SIZE`	?????? ????? DataVolume (? ??????? PVC ? ISO)	`5Gi`
`NVIDIA_CUSTOM_DRIVER_VOLUME_NAME`	??? ????/????? ? spec VM (CD-ROM)	`nvidia-custom-driver`
`NVIDIA_CUSTOM_DRIVER_DATAVOLUME_SUFFIX`	??????? ????? ????? DV: `vm_name` + ???????	`-nvidia-custom-driver`
`NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH`	??????? URL callback ????? `.../vm/<vm_name>/`	`nvidia-custom-driver-insta...
`NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY`	???? ????????? VM ??? ?????? callback	`vm.kubevirt.io/nvidia-custom...
`APP_INTERNAL_BASE_URL`	??????? URL ?????????? (callback)	`kubevirt-api-manager.kvm.svc.cluster.local`

????????? ??????? **?? ????????????**, ???? ?????: `NVIDIA_CUSTOM_DRIVER_PVC_NAME`, `NVIDIA_CUSTOM_DRIVER_PVC_NAMESPA...

3. ??????????? callback ?? ???? VM

????????? VM ?????? ????? ??????? ?????? ? API ????????? ?? `APP_INTERNAL_BASE_URL`. ?????? ??? ????????? Service, ...

- `http://<service-name>.<namespace>.svc.cluster.local`

????????, ??? ?? namespace ??????????? (??? ??????????? virt-launcher) ???? ??????? ?? ????? ??????.

??????????? GPU ? ????????

??? RTX Pro 6000 (? ?????? ????????? GPU) ? ???????? ?????? ???? ??????????????? ?????????????? ??????? (KubeVirt /...

????????? ??? ???????? VM

- ???? ?????? GPU **??** ?? ?????? ????????? (1080 Ti, 3080 Ti, 3090 ? ?.?.) ? ??????????? ????????? ?? ??????????? ...
- ???? ?????? GPU ?? ?????? ????????? ? ?????? PVC ? `APP_INTERNAL_BASE_URL` ? ????????? ???? PVC, ??????????? ? VM,...

???????????

- Callback ??????? ??????????? ???????, ?????????? ? ????????? VM ??? ????????; ??? ??????????? ?????? ? namespace ??...
- ????? ????????? callback ??? ? ????????? ????????? ?? VM ? ???? DataVolume ????????, ??????? ???????? ???????????...

CD-ROM: ?????? ??????, ??? ?????? ? ?????

??? ? ????????? ??????????? ? VM ??? **CD-ROM** (???? SATA), ? ???????? ?? ? `/dev/sr0` ??? `/dev/cdrom`. ?????????...

???????????? ???????: ISO-????? ??????? ?????????? ? PVC (DataVolume ? ???????? ?? HTTP, ??. ????). ?????????? P...

????????

- **??? PVC** ?????? ????????? ? `NVIDIA_CUSTOM_DRIVER_PVC_NAME` (?? ????????? `rtx-pro-6000`); **namespace** ? ? `NV...
- ??? ???????? VM ????????? ???? ????? PVC ? ??????????? ? VM ??? **CD-ROM (?????? ??????)**. Cloud-init ????????? `...
- ??? ???????? ? ???-??????????? (???????? ??????? ? PVC): ????? ????????? ? `/data/`; ????? ??????????? ??? ????? ??...

?????? ??? ?????? ? ????????? ????????? (????????)

???? ? VM ? enterprise GPU (RTX Pro 6000 ? ?.?.) **??? ?????????????? ?????** ? ????????, ????????? ?? ?????:

1. **??? ??? ?????????? ?? ??????:** ?????????? ????? ?? ?? ????????? ??? ?? ???????? ? `app/config.py` (`rtx-pro-600...

2. **??????????? ????? ??????? GPU:** ???????? ?? ????? «?????? GPU» ?????? **?????** ????????? ? ????? ?? ????? ? `NV...

3. **PVC ?????????? ? ????????:** ???? ????????? ?? PVC ? ????????? namespace. ?????????: `kubectl get pvc -n <NVIDIA...

4. **???? ??????????:** ??? ???????? Linux VM ? ????????? ??????? ?? ?????? ????????? ?????????? ???????? ??????????...

???????? ?????????? ?????? ???????? (PVC ? DataVolume) ????? kubectl

????? ????? ????? ???? **`<???_vm>-nvidia-custom-driver`** (? DataVolume, ? PVC ? ???? ? ?? ?? ???).

?????: ???? ??????? ?????? PVC, ? DataVolume ???????? ? ?????????? CDI ????? ?????????? PVC. ??????? **??????? ?...

??? 1: ??????? DataVolume

```bash
kubectl delete datavolume -n kv-admin-001 \
  test-nvidia-custom-driver \
  test2-nvidia-custom-driver \
  gpu-nvidia-custom-driver \
  testgpu-nvidia-custom-driver
```

??? 2: ?????????, ??? DV ?????????

```bash
kubectl get datavolumes -n kv-admin-001
```

???? ?????-?? DV ????? ? **Terminating**, ????? finalizers (?????????? ??? ??????? ?????? DV):

```bash
# ??? ??????? ????????? DV (???????? NAME ?? ??? ?? get datavolumes):
kubectl patch datavolume -n kv-admin-001 NAME -p '{"metadata":{"finalizers":null}}' --type=merge
```

??? ????? edit: `kubectl edit datavolume -n kv-admin-001 NAME` ? ??????? ???? `finalizers:` ? `metadata` ? ?????????.

??? 3: ????????? ??????????? DV ? ????????? PVC

????? ???? ??? DV ???????? ?? ?????? `kubectl get datavolumes`, PVC ?????? ????????? ????????. ????????:

```bash
kubectl get pvc -n kv-admin-001
```

???? PVC ???????? ? ??????? ??????? (?????? ????? ????, ??? ??????????????? DV ??? ???????):

```bash
kubectl delete pvc -n kv-admin-001 \
  test-nvidia-custom-driver \
  test2-nvidia-custom-driver \
  gpu-nvidia-custom-driver \
  testgpu-nvidia-custom-driver
```

????? ????????: ??????? DV ? ????? ????? finalizers (???? CDI ?????? ?? ? Terminating)

??????? ???????, ????? ??? ??????? ????? ????? finalizers, ????? DV ?? ?????? ? Terminating ? ?? ?????? ???????? PVC:

```bash
NS=kv-admin-001
for name in test-nvidia-custom-driver test2-nvidia-custom-driver gpu-nvidia-custom-driver testgpu-nvidia-custom-drive...
  kubectl delete datavolume -n "$NS" "$name" --ignore-not-found --wait=false
  sleep 1
  kubectl patch datavolume -n "$NS" "$name" -p '{"metadata":{"finalizers":null}}' --type=merge 2>/dev/null || true
done
kubectl get datavolumes -n "$NS"
kubectl get pvc -n "$NS"
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/RDP_QUICK_REFERENCE.md (5.4 KB)

```
# RDP Quick Reference Card

## ? Enable RDP in Windows VM

```powershell
# 1. Enable RDP
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name "fDenyTSConnections" -Value 0

# 2. Enable Firewall
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"

# 3. Restart Service
Restart-Service TermService -Force

# 4. Verify
Get-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' | Select-Object fDenyTSConnections
```

?? Create RDP Port Forwarding

Via Dashboard
1. Open **Dashboard**
2. Find Windows VM row
3. Click **RDP** button (green, next to VNC)
4. Note the external port in toast notification

Via VM Details
1. Open VM ? **Overview** tab
2. Section: **??????? ??????**
3. Click **RDP (3389)** button
4. Service appears in active services list

? Connect to Windows VM

Windows (Built-in)
```
Win + R ? mstsc
Computer: <EXTERNAL_IP>:<PORT>
Username: Administrator
Password: <your_password>
```

Command Line:
```cmd
mstsc /v:IP:PORT /f
```

Linux (Remmina)
```bash
# Install
sudo apt install remmina remmina-plugin-rdp

# Run
remmina
# Add New Connection ? RDP
# Server: <EXTERNAL_IP>:<PORT>
```

Linux (xfreerdp)
```bash
# Basic
```
```

```
xfreerdp /v:IP:PORT /u:Administrator /p:YourPass /cert-ignore

# With clipboard
xfreerdp /v:IP:PORT /u:Administrator /p:YourPass /cert-ignore /clipboard

# With file sharing
xfreerdp /v:IP:PORT /u:Administrator /cert-ignore /drive:share,/home/user/shared

# Fullscreen
xfreerdp /v:IP:PORT /u:Administrator /cert-ignore /f
```

macOS
```
1. Download "Microsoft Remote Desktop" from App Store
2. + Add PC
3. PC Name: <EXTERNAL_IP>:<PORT>
4. User Account: Administrator
5. Connect
```

? Troubleshooting

Problem: Can't connect to RDP

```powershell
# Check RDP is enabled
Get-ItemProperty 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name fDenyTSConnections
# Should return: fDenyTSConnections = 0

# Check Firewall
Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Where-Object {$_.Enabled -eq $false}
# Should return nothing

# Restart RDP service
Restart-Service TermService -Force

# Check service is running
Get-Service TermService
# Should return: Status = Running
```

Problem: Port not responding

```bash
# Check service exists
kubectl get svc -n <namespace> | grep rdp

# Check VM is running
kubectl get vmi -n <namespace>

# Test port from node
nc -zv <node-ip> <nodeport>
```

Problem: Authentication failed

```powershell
# Reset Administrator password
net user Administrator NewPassword123!

# Or create new user
New-LocalUser -Name "rdpuser" -Password (ConvertTo-SecureString "Pass123!" -AsPlainText -Force)
Add-LocalGroupMember -Group "Remote Desktop Users" -Member "rdpuser"
Add-LocalGroupMember -Group "Administrators" -Member "rdpuser"
```

? Quick Commands

Check RDP Status
```powershell
# RDP enabled?
(Get-ItemProperty 'HKLM:\System\CurrentControlSet\Control\Terminal Server').fDenyTSConnections -eq 0

# Firewall rules OK?
(Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Where-Object {$_.Enabled -eq $false}).Count -eq 0

# Service running?
(Get-Service TermService).Status -eq "Running"
```

View Active RDP Sessions
```powershell
# Current sessions
quser

# Recent logins (last 10)
Get-EventLog -LogName Security -InstanceId 4624 -Newest 10 | Select TimeGenerated, @{n='User';e={$_.ReplacementString...

# RDP connection history
Get-WinEvent -LogName 'Microsoft-Windows-TerminalServices-LocalSessionManager/Operational' -MaxEvents 20
```

Security Hardening
```powershell
# Enable Network Level Authentication
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server\WinStations\RDP-Tcp' -Name "UserAuthen...

# Account lockout policy (5 attempts, 30 min lockout)
net accounts /lockoutthreshold:5 /lockoutduration:30

# Disable Administrator account (use separate user)
Disable-LocalUser -Name "Administrator"
```

? Service Management

Check RDP Service via API
```bash
# List all services for VM
curl -X GET http://kvm.example.com/<username>/vm/<vm_name>/services \
  -H "Cookie: session=<your_session>"

# Response will include:
# {
#   "name": "rdp",
#   "port": 3389,
#   "nodePort": 30456,
#   "public_ip": "IP"
# }
```

Delete RDP Service
```bash
# Via Dashboard: Click RDP button again ? Confirm

# Via kubectl
kubectl delete svc <vm-name>-rdp -n <namespace>
```

? Feature Comparison

Feature	RDP	VNC (KubeVirt)

```
| Performance | ????? | ??? |
| File Transfer | ? | ? |
| Audio | ? | ? |
| Clipboard | ? | ?? Limited |
| Browser Access | ? | ? |
| Setup Required | ?? Yes | ? No |

**Recommendation**: Use VNC for initial setup, RDP for daily work.

---

## ? Full Documentation

? [WINDOWS-RDP-USAGE.md](docs/WINDOWS-RDP-USAGE.md) - Complete guide
? [WINDOWS-GOLDEN-IMAGE-SETUP.md](docs/WINDOWS-GOLDEN-IMAGE-SETUP.md) - Template creation
? [WINDOWS-GPU-SETUP.md](docs/WINDOWS-GPU-SETUP.md) - GPU passthrough

---

**Quick Help**: For detailed troubleshooting, see [WINDOWS-RDP-USAGE.md](docs/WINDOWS-RDP-USAGE.md#troubleshooting)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/SECURITY_IMPROVEMENTS.md (8.3 KB)

```
# Security and Configuration Improvements

## ??????????? ????????

### 1. ? ???????????

- **??????? ???????? ? ?????????? ?????????**: `SECRET_KEY`, `DEMO_USERNAME`, `DEMO_PASSWORD` ?????? ?????????? ?? e...
- **Session timeout**: ???????? ???????????? ??????? ?????? (?? ????????? 3600 ??????)
- **?????? ????????**: ????????? ??????????? ???????? ??? CPU, Memory, Storage

### 2. ? RBAC ?????

- ????????? ????? ?? `metrics.k8s.io/v1beta1` ??? ????????? ?????? VM
- ????????? ????? ?? ???????? `persistentvolumeclaims`

### 3. ? ?????????? ?????????????? ?????????

- ? `vm_routes.py` ???????????? ????????? ????????? `username` ? `vm_username`
- ???????? ????????? ? ????????? URL

### 4. ? ?????????? ????????? ??????

- ??? ??????? ? `k8s_utils.py` ?????? ????????? `(success: bool, error_message: str)`
- ???????? rollback ??? ?????? ???????? VM (???????? DataVolume)
- ????????? ?????????? ?????? ? ????????? ??????

### 5. ? ??????????? Nginx

- ??????? ?????? ?? ????????????? `/etc/nginx/auth.d/*.conf`
- ??????? ??????? ?????????? ? `debug` ?? `warn`
- ????????? ????????? ???????????

### 6. ? ????????? ??????? ??????

#### ????????? ??????? ?????????:

- **`validate_vm_name()`**: ???????? ????? VM ?? DNS-1123 ????????
- **`validate_ssh_key()`**: ???????? ??????? SSH ?????
- **????????? ????????**: CPU (1-16), Memory (1-64 GB), Storage (10-500 GB)
- **????????? URL**: ???????? ????????? http/https
- **???????????? ???????????** ? hostname, username, password

### 7. ? Debug mode ???????? ? production

- ??????? ?????????? ?????? ??????? ?? ????????? `DEBUG`
- Gunicorn ????????? ????????? `LOG_LEVEL` (?? ????????? `info`)
- Nginx ???????? ?? ?????? `warn` ?????? `debug`

### 8. ? Healthcheck endpoints

- **`/health`**: Liveness probe (???????? ?????? ?????????)
- **`/ready`**: Readiness probe (???????? ????????? ? ????????? ????????)
- ? `deployment.yaml` ????????? liveness ? readiness probes

### 9. ? ????????? ? ????????? ??????

- ???????? ????????? ? ?????? ??????? ??? cloud-init
- ????????? ????????? cloud-init ? ????????????? ?????????? ????????
- ??????? `delete_data_volume()` ??? ??????? orphan resources

### 10. ? UI/UX Improvements (v1.1.0)

- **???????? inline ????? ? CSS**: ??? `style="..."` ????????? ? CSS ?????
- **?????? visibility**: ???????????? `.hidden` ? `.visible` ?????? ??????? ?????????????? `style.display`
- **Wizard ?????????**: ???????? ?????? ??????????? ?????
```

- **GPU ?????????????**: ????? ??? GPU ?????? ? ????????? ???????
- **??????? ??????????????**: CPU ? %, Memory ? GB ? ??????????

11. ? Configuration Management (v1.1.0)

- **??????? ?????? ? config.py**: `DEFAULT_PACKAGES` ???????? ? ???????????
- **???????????????? ??????????**: ??? ????????? ???????? ? ????? ?????
- **?????? ????????????**: ????????? ???????? ??? ?????? ????

????????????

?????????? ?????????

????? ????????? ????? ?????????? ????????? ??? Kubernetes Secrets:

```bash
# ???????????
SECRET_KEY=your-random-secret-key-here
DEMO_USERNAME=admin
DEMO_PASSWORD=secure-password
SESSION_TIMEOUT=3600

# ?????? ????????
MAX_CPU_CORES=16
MAX_MEMORY_GB=64
MAX_STORAGE_GB=500

# ????? ???????
DEBUG=false
LOG_LEVEL=info

# Kubernetes
K8S_NAMESPACE_PREFIX=kv
STORAGE_CLASS_NAME=your-SC
```

???????? ???????? ? Kubernetes

```bash
kubectl create secret generic kubevirt-api-secrets \
  --from-literal=secret-key='your-random-secret-key-at-least-32-characters' \
  --from-literal=demo-username='admin' \
  --from-literal=demo-password='your-secure-password' \
  -n kvm
```

??? ?????????? ???? `kubernetes_example/secrets-example.yaml` (?? ???????? ???????? ????????):

```bash
# ????????????????? secrets-example.yaml
kubectl apply -f kubernetes_example/secrets-example.yaml
```

?????? ??????????? ?????????

```bash
# ????????? ?????????? RBAC ????? ? StatefulSet
kubectl apply -f kubernetes_example/bundled.yaml

# ??? ?? ??????????
kubectl apply -f kubernetes_example/rbac.yaml
kubectl apply -f kubernetes_example/statefulset.yaml
```

???????? health endpoints

```bash
# ?????? ????????
kubectl exec -it <pod-name> -n kvm -- curl http://localhost:8080/health
kubectl exec -it <pod-name> -n kvm -- curl http://localhost:8080/ready
```

```
# ????? port-forward
kubectl port-forward svc/kubevirt-api-manager 8080:80 -n kvm
curl http://localhost:8080/health
curl http://localhost:8080/ready
```

??? ?? ???? ??????????

- **CSRF ?????? (Flask-WTF)**: ??????? ????????????? ?????????? ?????????? Flask-WTF

??????????? ??? production

1. **??????????? ??????? ???????** ? `secrets-example.yaml` ????? ???????
2. **??????????? ??????? ??????? ?????????? ?????????** (Vault, Sealed Secrets)
3. **????????? ??????????** ?? ?????? `/health` ? `/ready` endpoints
4. **???????? TLS** ?? ?????? Ingress
5. **?????????? ?????? ????????** ????? ?????????? ????????? ??? ???? ?????
6. **????????? ???????????** ? ???????????????? ??????? (ELK, Loki)

???????? ? ??????

????????? ?????:

- ?? `app/config.py` - ???????? env vars, ??????
- ?? `app/main.py` - ???????? healthcheck endpoints, ????????? ??????????
- ?? `app/routes/__init__.py` - ???????????? config ??? credentials, session timeout
- ?? `app/routes/vm_routes.py` - ????????, ?????????? ????????? ??????, ????????????? ?????????
- ?? `app/utils/vm_utils.py` - ??????? ????????, ????????????, ???????? ?????????
- ?? `app/utils/k8s_utils.py` - ?????????? ????????? ??????, ??????? delete_data_volume
- ?? `conf/default.conf` - ???????? ????????????? include, ????????? log level
- ?? `entrypoint.sh` - ???????????? ????????? LOG_LEVEL
- ?? `kubernetes_example/rbac.yaml` - ???????? ????? ?? metrics API ? PVC delete
- ?? `kubernetes_example/deployment.yaml` - ???????? env vars, probes

????? ?????:

- ? `kubernetes_example/secrets-example.yaml` - ?????? ??????????? ????????
- ? `SECURITY_IMPROVEMENTS.md` - ???? ????????

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/SERVICE_MANAGEMENT_ARCHITECTURE.md (14.2 KB)

```
# ??????????? ?????????? ????????? VM

> **??????**: 2.0 (?????? 2026)
> **???????????**: ??????? ? ?????? "???? ?????? = ???? ????" ?? "???? ?????? ?? VM ? ???????????? ???????"

---

## ? ?????????

### ?????? ??????????? (?? v0.0.5)

```
VM: my-ubuntu-vm
??? my-ubuntu-vm-ssh       (Service, NodePort 30123 ? VM port 22)
??? my-ubuntu-vm-http      (Service, NodePort 30080 ? VM port 80)
??? my-ubuntu-vm-https     (Service, NodePort 30443 ? VM port 443)
??? my-ubuntu-vm-custom-1  (Service, NodePort 30888 ? VM port 8080)
```

????????:
- ? ????????? Kubernetes Services ?? ???? VM
- ? ????????? ?????????? (????????/???????? ????????? ????????)
- ? ??????????? metadata ? annotations

????? ??????????? (v0.0.5+)

```
VM: my-ubuntu-vm
??? my-ubuntu-vm-service   (Service ? ????????????? ???????)
    ??? port "ssh":      22 ? NodePort 30123
    ??? port "http":     80 ? NodePort 30080
    ??? port "https":   443 ? NodePort 30443
    ??? port "custom-1": 8080 ? NodePort 30888
```

???????????:
- ? ???? ?????? ?? VM
- ? ??????????? ??????????/???????? ?????? ????? `kubectl patch`
- ? ????????????? ???????? ??????? ??? ???????? ?????????? ?????
- ? ???????????????? ??????????? (labels, annotations)

? ??????????

Backend: `service_utils.py`

???????: `add_port_to_vm_service()`

```python
def add_port_to_vm_service(namespace, vm_name, target_port, service_type, node_port=None):
    """
    ????????? ???? ? ?????? VM. ??????? ?????? ???? ??? ???.

    ??????:
    1. ???????? ????????? ??????????? ?????? {vm_name}-service
    2. ???? ?????? ??????????:
       - ?????????, ?? ???????? ?? ??? ???? ????
       - ???????? ????????? NodePort
       - ????????? ????? ???? ????? api.patch_namespaced_service()
       - ??????????? Juniper NAT ??? ?????? ?????
    3. ???? ?????? ?? ?????????? (404):
       - ???????? _create_vm_service() ??? ???????? ?????? ???????
```
```

```

    Returns: (success, message, assigned_node_port)
    """
```

?????? ???????? ??????? ?????:
```python
success, msg, port = add_port_to_vm_service(
    namespace="kv-admin-default",
    vm_name="my-ubuntu-vm",
    target_port=22,
    service_type="ssh"
)
# ?????????: ?????? ?????? my-ubuntu-vm-service ? ?????? ssh:22?30123
```

?????? ?????????? ??????? ?????:
```python
success, msg, port = add_port_to_vm_service(
    namespace="kv-admin-default",
    vm_name="my-ubuntu-vm",
    target_port=80,
    service_type="http"
)
# ?????????: ?????? ?????????, ???????? ???? http:80?30080
```

???????: `remove_port_from_vm_service()`

```python
def remove_port_from_vm_service(namespace, vm_name, target_port):
    """
    ??????? ???? ?? ??????? VM. ??????? ???? ?????? ???? ??? ????????? ????.

    ??????:
    1. ?????? ?????? {vm_name}-service
    2. ??????? ???? ?? target_port (????????, 22, 80, 443)
    3. ??????? Juniper NAT ??? ????? ?????
    4. ???? ???????? ?????? ?????:
       - ?????? ??????, ?????? ???? ???? ?? spec.ports[]
    5. ???? ??? ??? ????????? ????:
       - ??????? ???? ?????? ????? api.delete_namespaced_service()

    Returns: (success, message)
    """
```

?????? ???????? ????? (?????? ????????):
```python
success, msg = remove_port_from_vm_service(
    namespace="kv-admin-default",
    vm_name="my-ubuntu-vm",
    target_port=80
)
# ?????????: ???? http:80 ??????, ?????? my-ubuntu-vm-service ??????? (???? ssh:22)
```

?????? ???????? ?????????? ?????:
```python
success, msg = remove_port_from_vm_service(
    namespace="kv-admin-default",
    vm_name="my-ubuntu-vm",
    target_port=22
)
# ?????????: ???? ssh:22 ??????, ?????? my-ubuntu-vm-service ?????? (?????? ?????? ???)
```

API Endpoints

POST `/<username>/vm/<vm_name>/services`

????????? ???? ? ?????? VM (??????? ?????? ???? ?????).

Request:
```json
{
  "port": 8080,
  "type": "custom-1"
}
```

Response (success):
```json
{
  "message": "Port 8080 added successfully",
  "port": 30888,
  "target_port": 8080,
  "type": "custom-1"
}
```

DELETE `/<username>/vm/<vm_name>/services/<service_type>`

??????? ???? ?? ??????? VM ?? ???? (????????, "ssh", "http", "custom-1").

Request:
```http
DELETE /admin/vm/my-ubuntu-vm/services/ssh
```

Response (success):
```json
{
  "message": "Port 22 removed successfully"
}
```

Response (????????? ????):
```json
{
  "message": "Port 22 removed, service deleted (no ports left)"
}
```

GET `/<username>/vm/<vm_name>/services`

?????????? ?????? ???? ???????? ?????? VM.

Response:
```json
{
  "services": [
    {
      "name": "ssh",
      "port": 22,
      "nodePort": 30123,
      "protocol": "TCP"
    },
    {
      "name": "http",
      "port": 80,
      "nodePort": 30080,
      "protocol": "TCP"
    }
```

```
  ]
}
```

Kubernetes Manifest

?????? ??????? ? ?????????????? ???????:

```yaml
apiVersion: v1
kind: Service
metadata:
  name: my-ubuntu-vm-service
  namespace: kv-admin-default
  labels:
    kubevirt-manager.io/vm-name: my-ubuntu-vm
    kubevirt-manager.io/managed: "true"
  annotations:
    juniper.kubevirt.io/dest-ip: "10.0.1.50"
spec:
  type: NodePort
  selector:
    vmi.kubevirt.io/id: my-ubuntu-vm
  ports:
  - name: ssh
    protocol: TCP
    port: 22
    targetPort: 22
    nodePort: 30123
  - name: http
    protocol: TCP
    port: 80
    targetPort: 80
    nodePort: 30080
  - name: custom-1
    protocol: TCP
    port: 8080
    targetPort: 8080
    nodePort: 30888
```

? ????????? ???? ???????

???????? 1: ???????? ??????

```
1. VM ???????, ?????? ???
   ?? Kubernetes: ?????? my-ubuntu-vm-service ?? ??????????

2. ??????????? ????????? SSH (???? 22)
   ?? Backend: add_port_to_vm_service(...)
      ?? api.read_namespaced_service() ? 404
      ?? ????? _create_vm_service()
      ?? ??????? NodePort: 30123
      ?? ?????? ?????? my-ubuntu-vm-service ? ?????? ssh:22?30123

3. ??????????? ????????? HTTP (???? 80)
   ?? Backend: add_port_to_vm_service(...)
      ?? api.read_namespaced_service() ? ?????? ??????????
      ?? ??????? NodePort: 30080
      ?? service.spec.ports.append({http:80?30080})
      ?? api.patch_namespaced_service() ?

4. ????: 1 ??????, 2 ?????
   ?? my-ubuntu-vm-service
      ?? ssh:22 ? 30123
      ?? http:80 ? 30080
```

```
```

???????? 2: ???????? ??????

```
????????? ?????????: 2 ????? (ssh, http)

1. ??????????? ??????? HTTP (???? 80)
   ?? Backend: remove_port_from_vm_service(..., target_port=80)
      ?? api.read_namespaced_service() ? ?????? ??????
      ?? ?????? ???? http:80 ? NodePort 30080
      ?? delete_port_forward(vm_name, node_ip, 30080) ? Juniper NAT ??????
      ?? remaining_ports = [ssh:22]
      ?? service.spec.ports = remaining_ports
      ?? api.patch_namespaced_service() ?

2. ????: 1 ??????, 1 ????
   ?? my-ubuntu-vm-service
      ?? ssh:22 ? 30123

3. ??????????? ??????? SSH (???? 22)
   ?? Backend: remove_port_from_vm_service(..., target_port=22)
      ?? api.read_namespaced_service() ? ?????? ??????
      ?? ?????? ???? ssh:22 ? NodePort 30123
      ?? delete_port_forward(vm_name, node_ip, 30123) ? Juniper NAT ??????
      ?? remaining_ports = [] ? ?????? ??????
      ?? api.delete_namespaced_service() ? ?????? ??????

4. ????: ??????? ?? ??????????
```

? ???????????

?????? ??????????? ????? kubectl

```bash
# 1. ??????? VM
kubectl apply -f my-vm.yaml

# 2. ???????? SSH ???? ????? API
curl -X POST http://kvm.example.com/admin/vm/my-ubuntu-vm/services \
  -H "Content-Type: application/json" \
  -d '{"port": 22, "type": "ssh"}'

# 3. ????????? ??????
kubectl get svc my-ubuntu-vm-service -n kv-admin-default -o yaml
# ?????? ???? 1 ????: ssh:22

# 4. ???????? HTTP ????
curl -X POST http://kvm.example.com/admin/vm/my-ubuntu-vm/services \
  -H "Content-Type: application/json" \
  -d '{"port": 80, "type": "http"}'

# 5. ????????? ??????
kubectl get svc my-ubuntu-vm-service -n kv-admin-default -o yaml
# ?????? ???? 2 ?????: ssh:22, http:80

# 6. ??????? HTTP
curl -X DELETE http://kvm.example.com/admin/vm/my-ubuntu-vm/services/http

# 7. ????????? ??????
kubectl get svc my-ubuntu-vm-service -n kv-admin-default -o yaml
# ?????? ???????? 1 ????: ssh:22

# 8. ??????? SSH
curl -X DELETE http://kvm.example.com/admin/vm/my-ubuntu-vm/services/ssh
```

```
# 9. ????????? ??????
kubectl get svc my-ubuntu-vm-service -n kv-admin-default
# Error from server (NotFound): services "my-ubuntu-vm-service" not found ?
```

? ????????? ??????????????????

????????	?????? ???????????	????? ???????????
???????? 1 ????	1 Service CREATE	1 Service CREATE
???????? 5 ??????	5 Service CREATE	1 CREATE + 4 PATCH
??????? 1 ????	1 Service DELETE	1 Service PATCH
??????? ????????? ????	1 Service DELETE	1 Service DELETE
?????? ??????	`list_namespaced_service()` ? ????????	`read_namespaced_service()`
Kubernetes ???????? ?? VM	5 Services	1 Service

?????: ????? ??????????? **? 5 ??? ???????????** ?? ????????? Kubernetes ???????? ? API calls ??? ???????????? ...

? ???????????

???????? namespace

??? ???????? ????????? ???????? namespace:

```python
user_ns = session.get('namespace')
if not user_ns or not user_ns.startswith(f'{K8S_NAMESPACE_PREFIX}-{username}-'):
    return jsonify({'error': 'Access denied'}), 403
```

Label selector

?????? ??????? label:

```yaml
labels:
  kubevirt-manager.io/vm-name: my-ubuntu-vm
  kubevirt-manager.io/managed: "true"
```

??? ?????????:
- ???????? ??????????? ??????? ?? ??????
- ??????? orphaned ??????? ??? ???????? VM

? ???????? ?? ?????? ???????????

????????????? ???????? (TODO)

```python
def migrate_legacy_services(namespace, vm_name):
    """
    ????????? ?????? ??????? (my-vm-ssh, my-vm-http) ? ????? ??????.

    1. ???? ??? ??????? ? ????????? {vm_name}-
    2. ??? ??????? ??????? ???????:
       - ?????? NodePort ? ????
       - ????????? ? ????? ?????? ????? add_port_to_vm_service()
       - ??????? ?????? ??????
    """
    api = get_core_api()
    label_selector = f"kubevirt-manager.io/vm-name={vm_name}"

    old_services = api.list_namespaced_service(
        namespace=namespace,
        label_selector=label_selector
```

```
    )

    for svc in old_services.items:
        # Skip new-format service
        if svc.metadata.name == f"{vm_name}-service":
            continue

        # Extract port info from old service
        for port_spec in svc.spec.ports:
            add_port_to_vm_service(
                namespace, vm_name,
                target_port=port_spec.port,
                service_type=port_spec.name,
                node_port=port_spec.node_port  # Preserve NodePort
            )

        # Delete old service
        api.delete_namespaced_service(svc.metadata.name, namespace)
```

? ????????????? ?????????

- [Kubernetes Services Documentation](https://kubernetes.io/docs/concepts/services-networking/service/)
- [Kubernetes API: PATCH vs UPDATE](https://kubernetes.io/docs/reference/using-api/api-concepts/#patch)
- [KubeVirt Networking](https://kubevirt.io/user-guide/virtual_machines/interfaces_and_networks/)

???? ??????????: 29 ?????? 2026
?????: KubeVirt API Manager Team

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/VM-SCHEDULING-CSI.md (2.2 KB)

```
# ?????????? ????????? CDI ?? ????? ? CSI (Ceph RBD)

?????? `driver name rbd.csi.ceph.com not found in the list of registered CSI drivers` ?????????, ????? **??? ????????...

????? ???? ????????? ?????????? ?????? ?? ????? ? CSI (???????? Ceph RBD), ??????? **nodeSelector** ????? ??????????...

## 1. ???????? ???? ? CSI

???????? ????, ?? ??????? ????????? Ceph RBD CSI (??? ???? ???? `ceph-rbd-ceph-csi-rbd-nodeplugin-*`):

```bash
kubectl label node f9.example.com f3.example.com c16.example.com csi-rbd=ceph --overwrite
# ???????? ????????? ???? ??: kubectl get pod -A -o wide | grep csi-ceph-rbd
```

2. ??????? VM_STORAGE_NODE_SELECTOR

Docker / .env:

```bash
VM_STORAGE_NODE_SELECTOR='{"csi-rbd":"ceph"}'
```

Kubernetes (Deployment/StatefulSet):

```yaml
env:
  - name: VM_STORAGE_NODE_SELECTOR
    value: '{"csi-rbd":"ceph"}'
```

????? ????? ???? CDI importer/transfer (importer-prime-*) ????? ???????????? ?????? ?? ???? ? ???? ??????. VM ??-???...

??? ????????????

- **???:** `app/scheduling_config.py` ? `get_storage_node_selector()` ??? ?????????; affinity/tolerations ??? VM.
- **DataVolume:** ??? ???????? DataVolume (HTTP-?????? ? dataVolumeTemplates ? Windows) ? spec ???????????? `nodePla...

??. ????? `env.example` ? `app/utils/k8s_utils.py` (create_data_volume), `app/utils/windows_utils.py` (dataVolumeTemp...

```

## [FILE] viirtuoz-vps-dev-rewrite-back-2/docs/VM_RUN_STRATEGY.md (18.4 KB)

```
? VM Run Strategy - ????????? ????????? VM

????????

?? ????????? KubeVirt ????????? ????????? `runStrategy: "Always"`, ??? ????????:
- ? VM ???????????? ??????????? ??? ????????
- ? **????? ????????? ?? ???????? ?? = ????????? VM**
- ? KubeVirt ?? ????????? graceful shutdown ? reboot

????????:
```bash
# ? Windows VM:
Start ? Shutdown ? VM ??????????????! ?

# ? Linux VM:
sudo shutdown -h now ? VM ??????????????! ?
```

? ???????: runStrategy ?????????

??????? ? ?????? ???????, ????????????? ??????????????? ????????? ??????? VM ????? ????????? ????????? `VM_RUN_STRA...

????????? ?????????

?????????	Shutdown ?? ??	Reboot ?? ??	?????????	???????????
Manual ?	????????? ?	????????? ??	???	**?? ?????????** (?????? ???????? ???????????)
RerunOnFailure	????????? ?	????????? ?	??	Production (????????? + ????-???????)
Always	????????? ?	????????? ?	??	Legacy (?????? ????????)
Halted	-	-	???	VM ?????????

? ??? ????????

Manual (???????????? ?? ?????????) ?

?????????:
- VM ?? ??????????? ???????????? ??? ???????? (????????? ? ????????? `Stopped`)
- ??????????? ??????? ????????? VM ????? Dashboard (?????? "?????????") ??? `virtctl start`
- **Shutdown ?? ??** ? VM ??????????????? (??????: `Stopped`)
- **Reboot ?? ??** ? VM ??????????????? (?? ???????????????!) ??

???????:

```bash
# ???????? VM
kubectl apply -f vm.yaml   # VM ???????, ?? ?? ???????? ??

# ?????? ??????? ????? virtctl
virtctl start my-vm        # ?????? VM ???????? ?

# ??? ????? Dashboard
# ?????? ?????? "?????????" ?? ???????? VM

# ????????? ?? ??
$ sudo shutdown -h now     # VM ??????????????? ?
$ sudo reboot              # VM ??????????????? (?? ???????????????!) ??
```

??????? ? KubeVirt:
```yaml
# ????? ???????? (?? ??????? ???????)
```
```

```
status:
  ready: false
  printableStatus: Stopped

# ????? shutdown ??? reboot
status:
  ready: false
  printableStatus: Stopped
```

????? ????????????:
- ? **?? ????????? ??? ???? VM** (????????????)
- ? ?????? ???????? ??? ????????/????????????? VM
- ? ???????? ???????? (VM ?? ???????? ??? ?????? ???????? ????????????)
- ? Development/Testing VM
- ? VM ??? ?????????? ????????????
- ?? ??????? ??????? ??????? ????? reboot ?? ??

RerunOnFailure (????????? + ????-???????)

?????????:
- VM ???????????????? ?????????????? ??? ????????
- **Graceful shutdown** (exit code 0) ? VM ??????????????????? (??????: `Stopped`)
- **Reboot** ?? ?? ? VM ??????????????????? ????????????????
- **Crash/Kernel Panic** (exit code != 0) ? VM ??????????????????? ????????????????

???????:

```bash
# Windows VM
C:\ > shutdown /s   # ?????????? ? VM ??????????????????? ?
C:\ > shutdown /r   # ??????????????? ? VM ??????????????????? ?

# Linux VM
$ sudo shutdown -h now   # ?????????? ? VM ??????????????????? ?
$ sudo reboot            # ??????????????? ? VM ??????????????????? ?
$ sudo systemctl poweroff # ?????????? ? VM ??????????????????? ?
```

??????? ? KubeVirt:
```yaml
# ????? shutdown
status:
  ready: false
  printableStatus: Stopped

# ????? reboot (VM ???????????????????)
status:
  ready: true
  printableStatus: Running
```

????? ????????????:
- Production VM, ??????? ?????? ???????? 24/7
- VM ? ?????????????? ??? ????????
- VM, ????????? ?????????????????? ??????????????? ?? ??
- ????????? ??????? ? ?????????????????? ??????????????????? ????? ?????

Always (?????? ?????????)

?????????:
- VM ?????? ?????? ????????
- **????? ??????????** ? VM ??????????????????? (???? graceful shutdown)

???????:

```bash
```

```
# Windows VM
C:\ > shutdown /s   # ??????????? ? VM ??????????????? ?

# Linux VM
$ sudo shutdown -h now   # ??????????? ? VM ??????????????? ?
```

????? ???????????:
- Legacy compatibility (?????? VM ?? KubeVirt v0.x)
- ????????? ???????, ??????? ?????? ?????? ???????? (???? ??? ????????? shutdown)
- ?? ?? ???????????? ??? ????? deployments

Manual (?????? ????????)

?????????:
- VM ?? ?????????? ???????????? ??? ????????
- ????? ??????? ????????? ????? Dashboard ??? `virtctl start`
- **Shutdown/Reboot** ? VM ??????????????? (?? ???????????????)

???????:

```bash
# ???????? VM
kubectl apply -f vm.yaml   # VM ???????, ?? ?? ???????? ??

# ?????? ???????
virtctl start my-vm        # ?????? VM ???????? ?

# ??????????? ?? ??
$ sudo shutdown -h now     # VM ??????????????? ?
$ sudo reboot              # VM ??????????????? (?? ???????????????!) ??
```

????? ???????????:
- ? **?? ?????????** (???????????? ? ?????? 2.0)
- Development/Testing VM
- VM ??? ??????????? ????????????
- ?????? ?????? ???????? ??? ????????? ??????

Halted (??????????)

?????????:
- VM ????????? ? ?? ???????????
- ??????????? ??? ????????? ????????? VM

???????:

```bash
# ??????????? VM
kubectl patch vm my-vm --type=merge -p '{"spec":{"runStrategy":"Halted"}}'

# VM ?????????? ? ?? ??????????? ?? ????? ????????
```

?? ???????????

??????????? ??????????? ????????? `VM_RUN_STRATEGY`:

```bash
# ? .env ????? (??? Docker Compose)
VM_RUN_STRATEGY=Manual   # ???????????? (?? ?????????)

# ? Kubernetes Secret/ConfigMap
apiVersion: v1
kind: Secret
```

```
metadata:
  name: kubevirt-api-secrets
stringData:
  VM_RUN_STRATEGY: "RerunOnFailure"
```

???????? ?? ?????????:
- ???? ?? ??????: `RerunOnFailure` ?
- ?????????? ????????: `Always`, `RerunOnFailure`, `Manual`, `Halted`

2. ????????? ?????????

Docker Compose:
```bash
# ???????? .env
echo "VM_RUN_STRATEGY=RerunOnFailure" >> .env

# ?????????? ?????????
docker-compose down
docker-compose up -d
```

Kubernetes:
```bash
# ???????? Secret
kubectl edit secret kubevirt-api-secrets -n kvm

# ???????????? deployment
kubectl rollout restart deployment/kubevirt-api-manager -n kvm
```

3. ????????

```bash
# ????????? ???? ?????????
docker-compose logs web | grep "VM Run Strategy"
# ??? ? Kubernetes:
kubectl logs deployment/kubevirt-api-manager -n kvm | grep "VM Run Strategy"

# ?????? ????:
? VM Run Strategy: RerunOnFailure
```

? ???????????

Windows VM

```powershell
# 1. ???????? Windows VM ????? Dashboard
# 2. ??????????? ????? RDP
# 3. ???????? PowerShell

# ???? 1: Graceful Shutdown
shutdown /s /t 0

# ????????? ????????? (RerunOnFailure):
# - VM ??????????
# - ?????? ? Dashboard: "Stopped"
# - VM ?? ?????????????? ???????????? ?

# ????????? VM ??????? ????? Dashboard
# ??????????? ?????

# ???? 2: Reboot
shutdown /r /t 0

# ????????? ????????? (RerunOnFailure):
# - VM ??????????????
# - ?????? ? Dashboard: "Running" (????? ???????????)
```

```
# - VM ???????? ????? ?
```

Linux VM

```bash
# 1. ???????? Linux VM ????? Dashboard
# 2. ??????? ????? SSH
# 3. ????????? ?????

# ???? 1: Graceful Shutdown
sudo shutdown -h now

# ????????? ????????? (RerunOnFailure):
# - VM ??????????
# - ?????? ? Dashboard: "Stopped"
# - VM ?? ?????????????? ???????????? ?

# ????????? VM ??????? ????? Dashboard
# ??????????? ?????

# ???? 2: Reboot
sudo reboot

# ????????? ????????? (RerunOnFailure):
# - VM ??????????????
# - ?????? ? Dashboard: "Running" (????? ???????????)
# - VM ???????? ????? ?

# ???? 3: Systemctl poweroff
sudo systemctl poweroff

# ????????? ????????? (RerunOnFailure):
# - VM ??????????
# - ?????? ? Dashboard: "Stopped"
# - VM ?? ?????????????? ?
```

? Troubleshooting

VM ??? ????? ?????????????? ??? shutdown

???????? 1: ????????, ??? VM_RUN_STRATEGY ????????

```bash
# ????????? ??????????? ? ?????
docker-compose logs web | grep "VM Run Strategy"

# ?????? ????:
? VM Run Strategy: RerunOnFailure
```

???????? 2: ????????? ???????? VM

```bash
kubectl get vm <vm-name> -n <namespace> -o yaml | grep runStrategy

# ?????? ????:
  runStrategy: RerunOnFailure
```

???????? 3: ??????????? VM

???? VM ???? ??????? ?? ????????? ???????????, ????? ??????????:

```bash
# ??????? VM
kubectl delete vm <vm-name> -n <namespace>
```

???????? ?????? ????? Dashboard
????? VM ????? ??????????? RerunOnFailure
```

---

### ??????????? VM ?? ?????????

**???????:** ????????? `VM_RUN_STRATEGY` ??????????? ?????? ? **????? VM**.

**??????? 1: ???????? ????????? (??????)**

```bash
#!/bin/bash
# ???????? runStrategy ??? ???? VM ? namespace

NAMESPACE="kv-user-default"

for VM_NAME in $(kubectl get vm -n $NAMESPACE -o name | cut -d'/' -f2); do
    echo "Updating $VM_NAME..."
    kubectl patch vm $VM_NAME -n $NAMESPACE --type=merge \
        -p '{"spec":{"runStrategy":"RerunOnFailure"}}'
done
```

??????? 2: ?????? ????????? ????? kubectl

```bash
kubectl patch vm <vm-name> -n <namespace> --type=merge \
    -p '{"spec":{"runStrategy":"RerunOnFailure"}}'
```

??????? 3: ??????????? VM (??? ????????? ?????????)

```bash
# 1. ????????? PVC (???? VM)
kubectl get pvc -n <namespace> | grep <vm-name>

# 2. ??????? VM (PVC ??????????!)
kubectl delete vm <vm-name> -n <namespace>

# 3. ???????? VM ?????? ????? Dashboard
# ??????????? ??? ?? PVC (???? ????? ?????? ????)
```

VM ?? ??????????? ????????????

???????: ??????????? ????????? `Manual` ??? `Halted`.

???????:

```bash
# ????????? ??????? ?????????
kubectl get vm <vm-name> -n <namespace> -o yaml | grep runStrategy

# ???????? ?? RerunOnFailure
kubectl patch vm <vm-name> -n <namespace> --type=merge \
    -p '{"spec":{"runStrategy":"RerunOnFailure"}}'

# ????????? VM
virtctl start <vm-name> -n <namespace>
# ??? ????? Dashboard: ?????? "?????????"
```

? ?????????

????? Dashboard
```

```

Dashboard ? VM Details ? ??????
```

????????? ???????:

????????	RerunOnFailure	Always
???????? VM	`Provisioning` ? `Running`	`Provisioning` ? `Running`
Shutdown ?? ??	`Running` ? `Stopped` ?	`Running` ? `Running` (??????????) ?
Reboot ?? ??	`Running` ? `Running` (??????????) ?	`Running` ? `Running` (??????????) ?
Crash	`Running` ? `Running` (??????????) ?	`Running` ? `Running` (??????????) ?

????? kubectl

```bash
# ??????? ?? ???????? VM
kubectl get vm <vm-name> -n <namespace> -w

# ??????? ?? VMI (instance)
kubectl get vmi <vm-name> -n <namespace> -w

# ????????? runStrategy
kubectl get vm <vm-name> -n <namespace> -o jsonpath='{.spec.runStrategy}'
```

???? ???????

```bash
# ??????? VM
kubectl describe vm <vm-name> -n <namespace>

# ???? KubeVirt (virt-controller)
kubectl logs -n kubevirt -l kubevirt.io=virt-controller --tail=100
```

? Best Practices

Default (Recommended for Most Users)

```bash
# ? ???????????? ?? ????????? (? ?????? 2.0)
VM_RUN_STRATEGY=Manual

# ???????????:
# - ?????? ?????? ???????? ??? ????????/?????????? VM
# - VM ?? ?????????? ??? ????? ??????? ??????????? (???????? ????????)
# - Graceful shutdown ? VM ??????????????
# - Reboot ? VM ??????????????? (??????? ??????? ???????)
# - ????????????? ?????????????????? ???????????? ????????

# ???????? ???:
# - ???????????????? VM (desktop, development environments)
# - VM ??? ????????? ????????????
# - Testing/QA environments
```

Production Services (Auto-start Required)

```bash
# ? ??? production ???????? ? ??????????
VM_RUN_STRATEGY=RerunOnFailure

# ???????????:
# - VM ???????????? ?????????? ??? ????????
# - Graceful shutdown ? VM ??????????????? (???????? ????????)
# - Crash ? VM ??????????????? (??????? ??????????)
# - Reboot ? VM ??????????????? (????????? ???????????)
```

```

# ???????? ???:
# - Production web servers, databases, API services
# - VM, ??????? ?????? ???????? 24/7
# - ????????? ??????? ? ????????????? ???????????????
```

Legacy/Critical Services

... [48 more lines truncated]

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/VM_RUN_STRATEGY_QUICKSTART.md (5.2 KB)

```md
# ? VM Run Strategy - ??????? ???????

## ??? ????

**VM_RUN_STRATEGY** - ?????????, ??????????????:
- ? **?????????? VM ??? ????????** (??/???)
- ? **????????? ??? shutdown ?? ??** (?????????/??????????)
- ? **????????? ??? reboot ?? ??** (??????????/?????????)

---

## ? ????? ????????? (Quick Start)

### Manual (?? ?????????) ?

**??????????? ?????:**
- ???????????????? VM (desktop, development)
- ????? ?????? ???????? ???????/????????
- VM ??? ?????????? ????????????

```bash
# ??? ?????????? ?? ?????????!
VM_RUN_STRATEGY=Manual

? VM ????????? ? ????????? Stopped
? ??????? ??????? ??????? ????? Dashboard
? Shutdown/Reboot ? VM ??????????????
? ???????? ???????? (VM ?? ???????? ??? ???????)
```

RerunOnFailure (Production)

??????????? ?????:
- Production ??????? (web servers, databases, API)
- VM ?????? ???????? 24/7 ? ??????????
- ????????? ????????????? ??????????? ?? ??

```bash
# ???????? ? .env (Docker Compose) ??? Secret (Kubernetes)
VM_RUN_STRATEGY=RerunOnFailure

? VM ???????????? ?????????? ??? ????????
? Shutdown ? VM ??????????????
? Reboot ? VM ??????????????
? Crash ? VM ??????????????? (??????? ??????????)
```

?? ????????? (Manual ? RerunOnFailure)

???? ????? ????????? ??? production VM:

Docker Compose

```bash
# 1. ???????? ? .env
echo "VM_RUN_STRATEGY=RerunOnFailure" >> .env

# 2. ????????????
docker-compose down && docker-compose up -d

# 3. ?????????
docker-compose logs web | grep "VM Run Strategy"
# ?????? ????: ? VM Run Strategy: RerunOnFailure
```
```

```
```

Kubernetes

```bash
# 1. ???????? Secret
kubectl edit secret kubevirt-api-secrets -n kvm
# ????????: VM_RUN_STRATEGY: "RerunOnFailure"

# 2. ????????????? deployment
kubectl rollout restart deployment/kubevirt-api-manager -n kvm

# 3. ?????????
kubectl logs deployment/kubevirt-api-manager -n kvm | grep "VM Run Strategy"
# ?????? ????: ? VM Run Strategy: RerunOnFailure
```

? ?????????

Manual (?? ?????????)

????????	?????????
???????? VM	?????????: Stopped (??????? ??????? ???????) ?
Shutdown ?? ??	VM ??????????????? ?
Reboot ?? ??	VM ??????????????? (??????? ??????? ???????) ??
Crash/Panic	VM ??????????????? ??

RerunOnFailure (production)

????????	?????????
???????? VM	VM ???????????? ?????????? ?
Shutdown ?? ??	VM ??????????????? ?
Reboot ?? ??	VM ??????????????? ?
Crash/Panic	VM ??????????????? (????????????????????) ?

?? ?????

????? VM ?????????? ??????? ????????? VM_RUN_STRATEGY!

- ?? ?????????: `Manual` (VM ????????? ? Stopped ?????????)
- ??? production: ?????????? `RerunOnFailure` (??. ????)

?????????? ??????????? VM

???? ????? ???????? ????????? ??? ??? ????????? VM:

```bash
# ???????? ?? Manual (?????? ????????)
kubectl patch vm <vm-name> -n <namespace> --type=merge \
    -p '{"spec":{"runStrategy":"Manual"}}'

# ???????? ?? RerunOnFailure (????????? + ????-???????)
kubectl patch vm <vm-name> -n <namespace> --type=merge \
    -p '{"spec":{"runStrategy":"RerunOnFailure"}}'

# ???????? ?????????? ???? VM ?? Manual
for VM in $(kubectl get vm -n <namespace> -o name | cut -d'/' -f2); do
    kubectl patch vm $VM -n <namespace> --type=merge \
        -p '{"spec":{"runStrategy":"Manual"}}'
done

# ???????? ?????????? ???? VM ?? RerunOnFailure
for VM in $(kubectl get vm -n <namespace> -o name | cut -d'/' -f2); do
    kubectl patch vm $VM -n <namespace> --type=merge \
        -p '{"spec":{"runStrategy":"RerunOnFailure"}}'
```

```
done
```

? ?????? ???????????

[docs/VM_RUN_STRATEGY.md](VM_RUN_STRATEGY.md)

????:
- ????????? ???????? ???? ????????? (Always, RerunOnFailure, Manual, Halted)
- ??????????? shutdown/reboot
- Troubleshooting
- Best practices

? **??????!** ?????? shutdown ?? Windows/Linux ???????????? VM ?????? ???????????.

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/WINDOWS-GOLDEN-IMAGE-EXPORT.md (24.4 KB)

```
# ? Windows Golden Image - ????????-?????????? (????? Sysprep)

## ? ????

????????? ???? VM ????? sysprep ??? **golden image** ??? ???????? ???????? ????? Windows VM.

**????? ??????????:** 5 ????? + 3-5 ????? ???????????

---

## ? ??????? ????? (5 ?????)

### ?????????????? ???????

- ? Windows ????????? ? ????????
- ? Cloudbase-Init ????????? ? ????????
- ? **Sysprep ????????** (`/generalize /oobe /shutdown`)
- ? **VM ?????????**

---

### 1?? ??????? ???? VM ? PVC

```bash
# ???????? ?? ???? ????????
VM_NAME="goimage"        # ? ??? ????? VM ????? sysprep
NAMESPACE="kv-example-001"   # ? ??? namespace
PVC_NAME="${VM_NAME}-rootdisk"

# ????????? ??? VM ??????????
kubectl get vm $VM_NAME -n $NAMESPACE
# ?????? ???? ? ???????: Stopped
```

2?? ? ????????: ??????? ownerReferences

?? ??? ????? ???? PVC ???????? ??? ???????? VM!

```bash
# ??????? ownerReferences (?????? PVC ?????????? ?? VM)
kubectl patch pvc $PVC_NAME -n $NAMESPACE --type=json \
  -p='[{"op": "remove", "path": "/metadata/ownerReferences"}]'

# ???????? (?????? ???? ?????)
kubectl get pvc $PVC_NAME -n $NAMESPACE -o jsonpath='{.metadata.ownerReferences}'
# ????????? ?????: [] (?????)
```

? **?????? PVC ?? ???????? ??? ???????? VM!**

3?? ????????? ? kvm namespace

```bash
kubectl apply -f - <<EOF
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-golden-image
  namespace: kvm
spec:
  source:
    pvc:
      name: $PVC_NAME
      namespace: $NAMESPACE
  pvc:
```
```

```
    accessModes: [ReadWriteOnce]
    resources:
      requests:
        storage: 80Gi  # ? ?????? ?????? ????? (??? ??????)
    storageClassName: your-SC
EOF
```

4?? ????????? ?????????? ???????????

```bash
# ??????? ?? ????????? (?????? 3-5 ?????)
kubectl get dv windows-10-golden-image -n kvm -w

# ????? ??????? Succeeded:
# NAME                      PHASE       PROGRESS
# windows-10-golden-image   Succeeded   100.0%
```

5?? ??????? ????????? VM (?????? ?????????!)

```bash
# ?????? ????? ??????? VM
kubectl delete vm $VM_NAME -n $NAMESPACE

# ??????????: ??????? ?????? PVC ?? user namespace
kubectl delete pvc $PVC_NAME -n $NAMESPACE
```

? ???????? ??????????

```bash
# 1. Golden image ???????
kubectl get pvc windows-10-golden-image -n kvm

# ?????? ????:
# NAME                        STATUS   VOLUME        CAPACITY   ACCESS MODES
# windows-10-golden-image     Bound    pvc-xxx...    80Gi       RWO

# 2. ??? ownerReferences?
kubectl get pvc windows-10-golden-image -n kvm -o jsonpath='{.metadata.ownerReferences}'
# ?????? ???? ?????!

# 3. ?????????? ????? ? golden image?
kubectl get deployment kubevirt-api-manager -n kvm -o yaml | grep WINDOWS_GOLDEN

# ?????? ????:
# - name: WINDOWS_GOLDEN_IMAGE_NAME
#   value: windows-10-golden-image
# - name: WINDOWS_GOLDEN_IMAGE_NAMESPACE
#   value: kvm
```

? ??????! ?????????? ????? VM

????? Web UI (Dashboard) - ???????????? ?

```
1. ????????: https://kvm.example.com/<username>/dashboard
2. ???????: "??????? ??????????? ??????"
3. ????????: Windows 10 ? Golden Image (??????? ????????)
4. ????????? ???????:
   - CPU: 4-8 cores
   - RAM: 8-16 GB
   - Storage: 80-120 GB (??????? 80 GB!)
5. ??????? ?????? ????????????? (??????????!)
6. ??????? ? ?????? ?? 3-5 ?????! ?
```

```
```

????????????:
- ? **3-5 ?????** ???????? ?????? 30-60 ????? ?????????
- ? ????????????????? ????????? ????? Cloudbase-Init
- ?? RDP ?????? ?? ???????
- ? ?????? ????????? ????????? (Manual ?? ?????????, ??????????? RerunOnFailure)

? ????????????????? ??????????

?????? ???????????????

- WINDOWS-GOLDEN-IMAGE-SETUP.md - ?????? ??????????? ?? ????????
- WINDOWS-GOLDEN-IMAGE-QUICK-START.md - ???????? VM ?? ??????
- VM_RUN_STRATEGY.md - ??????????? shutdown/reboot
- WINDOWS-RDP-USAGE.md - ????????? RDP ???????

? Troubleshooting

????????: PVC ???????? ??? ???????? VM

????????: ?? ??????? `ownerReferences` ????? ????????? VM.

????????: ??????????? golden image ??????:
1. ????????? ? VM ?? sysprep (???? ???? ???????)
2. ??? ??????????? VM ? ???? ?? WINDOWS-GOLDEN-IMAGE-SETUP.md

????????: DataVolume ??????? ?? ImportInProgress

```bash
# ????????? ??????
kubectl describe dv windows-10-golden-image -n kvm

# ????????? CDI pods
kubectl get pods -n cdi

# ???? CDI ?? ???????? - ????????????
kubectl rollout restart deployment/cdi-deployment -n cdi
```

????????: Golden image ?? ????? ? Dashboard

```bash
# ????????? ?????????? ?????????
kubectl get deployment kubevirt-api-manager -n kvm -o yaml | grep WINDOWS_GOLDEN

# ???? ??? ??? ???????? - ???????? Secret ? ????????????
kubectl edit secret kubevirt-api-secrets -n kvm
# ????????:
# WINDOWS_GOLDEN_IMAGE_NAME: windows-10-golden-image
# WINDOWS_GOLDEN_IMAGE_NAMESPACE: kvm

kubectl rollout restart deployment/kubevirt-api-manager -n kvm
```

????????: VM ?????????, ?? Cloudbase-Init ?? ????????

```bash
# ????????? ???? serial console
POD=$(kubectl get pods -n <namespace> -l kubevirt.io/domain=<vm-name> -o name)
kubectl logs $POD -c guest-console-log -n <namespace>

# ?????? ???? ??????:
# [Cloudbase-Init] Starting initialization...
# [Cloudbase-Init] Cloudbase-Init complete

# ???? ??? - ????????? ????????? Cloudbase-Init ? golden image
```

```
```

? ??????? ?????????

```bash
# ????? sysprep ? ?????????? VM ?????????:

# 1. ??????? ownerReferences (??????????!)
kubectl patch pvc <vm>-rootdisk -n <ns> --type=json \
  -p='[{"op": "remove", "path": "/metadata/ownerReferences"}]'

# 2. ??????????? ? kvm (DataVolume yaml ????)

# 3. ????????? Succeeded
kubectl get dv windows-10-golden-image -n kvm -w

# 4. ??????? ????????? VM
kubectl delete vm <vm-name> -n <namespace>

# 5. ????????? ????? VM ????? Dashboard! ?
```

? ?????????? Golden Image

????? ????? ???????? ??????? ????? (Windows Updates, ????? ????????):

```bash
# 1. ???????? VM ?? ???????? golden image (????? Dashboard)
# 2. ?????????? ????????? ????? Windows Update
# 3. ?????????? ????????????? ????????/?????????
# 4. ????? sysprep:
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown

# 5. ????????? ????????? ???????? (???? 1-5 ????)
# 6. ???????? ?????? golden image ????? (??? ???????? ? ?????? ??????)
```

? ????????????? ????????

???????? 1: ????????? ?????? Golden Image

```bash
# ???????? ????????? ??????? ??? ?????? ?????
windows-10-golden-image          # ??????? ??????
windows-10-golden-image-dev      # ?????? ??? ?????????? (Visual Studio, etc.)
windows-10-golden-image-gpu      # ?????? ? ?????????? GPU
```

???????? 2: ??????? Golden Image ?? ??????? storage

```bash
# ???????????? PVC ? ???? ??? ?????????? ??????????
virtctl image-download pvc windows-10-golden-image -n kvm \
  --output=windows-10-golden.img

# ??????????? ??????? (????????, ? ?????? ???????)
virtctl image-upload dv windows-10-golden-image-backup -n kvm \
  --image-path=windows-10-golden.img --size=80Gi
```

??????? ? ?? ??? ???????? ???????????? Windows ? KubeVirt

```


# ??????????? ????? DataVolume/PVC ????????? ? VM
kubectl get vm $VM_NAME -n $VM_NAMESPACE -o yaml | grep -A10 "dataVolume:"

# ??? ???????? ?????? PVC
kubectl get pvc -n $VM_NAMESPACE | grep $VM_NAME

# ?????? ??????:
# my-windows-setup-rootdisk   Bound    pvc-abc123...   80Gi       RWO            your-SC   2h

# ????????? ??? PVC
export SOURCE_PVC="my-windows-setup-rootdisk"
```

??? 3: ??????? namespace ??? golden images

```bash
# ???????? namespace kvm ???? ??? ???
kubectl create namespace kvm --dry-run=client -o yaml | kubectl apply -f -

# ?????????
kubectl get namespace kvm
```

??? 4: ??????????? ???? ? golden image

???? **??? ???????**: Clone ??? Snapshot.

?????? ?: Clone PVC (????????????)

```bash
# ???????? DataVolume-???? ?????? ?????
kubectl apply -f - <<EOF
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-golden-image
  namespace: kvm
  labels:
    app: kubevirt-golden-image
    os: windows
    version: "10"
spec:
  source:
    pvc:
      name: $SOURCE_PVC
      namespace: $VM_NAMESPACE
  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 80Gi  # ??? ?? ?????? ??? ? ? ?????????
    storageClassName: your-SC
EOF

# ??????? ?? ??????????? ???????????
kubectl get dv windows-10-golden-image -n kvm -w

# ????????? ?????:
# NAME                        PHASE       PROGRESS   AGE
# windows-10-golden-image     CloneInProgress   35.2%      1m
# windows-10-golden-image     CloneInProgress   67.8%      2m
# windows-10-golden-image     Succeeded         100.0%     3m
```

????? ????????????: 3-10 ????? ? ??????????? ?? ??????? ?????.

?????? ?: Snapshot (???????, ?? ??????? CSI driver ? ??????????? snapshots)

```
```

```bash
# ???????? VolumeSnapshot
kubectl apply -f - <<EOF
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
  name: windows-10-snapshot
  namespace: $VM_NAMESPACE
spec:
  volumeSnapshotClassName: csi-rbd-snapclass  # ????????? ???????? ? ????? ????????
  source:
    persistentVolumeClaimName: $SOURCE_PVC
EOF

# ????????? ????????? ????????
kubectl get volumesnapshot windows-10-snapshot -n $VM_NAMESPACE -w

# ???????? DataVolume ?? ????????
kubectl apply -f - <<EOF
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-golden-image
  namespace: kvm
spec:
  source:
    snapshot:
      name: windows-10-snapshot
      namespace: $VM_NAMESPACE
  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 80Gi
    storageClassName: your-SC
EOF
```

??? 5: ???????? golden image

```bash
# ????????? ??? DataVolume ??????
kubectl get dv windows-10-golden-image -n kvm

# ?????? ????:
# NAME                        PHASE       PROGRESS   AGE
# windows-10-golden-image     Succeeded   100.0%     5m

# ????????? PVC
kubectl get pvc windows-10-golden-image -n kvm

# ?????? ???? Bound
# NAME                        STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECL...
# windows-10-golden-image     Bound    pvc-def456...                              80Gi       RWO            your-SC  ...

# ????????? ?????????
kubectl describe dv windows-10-golden-image -n kvm | grep -A5 "Status:"
```

??? 6: ??????? ???????? VM (??????????)

```bash
# ???? ??? ?????? ?? ????? ???????? VM ??? ?????????
kubectl delete vm $VM_NAME -n $VM_NAMESPACE

# DataVolume/PVC ???? ????? ??????? (?????????? ?????)
kubectl delete dv ${VM_NAME}-rootdisk -n $VM_NAMESPACE
```

? ????????? ? Web UI

?????? ????? ????????? ?????????, ????? ??? ??????????? ??? golden image.

??????? 1: ??????? URL-based DataVolume (????????????)

???????? HTTP endpoint ??? ?????? ??????:

```bash
# 1. ???????????? ????? ? ????
kubectl exec -n kvm deployment/cdi-deployment -- \
  qemu-img convert -f raw -O qcow2 \
  /dev/rbd/pool/pvc-abc123... \
  /tmp/windows-10-golden.qcow2

# 2. ????????? ???? ?? HTTP ??????
# (Minio, Nginx, Apache ? ?.?.)
# URL ?????? ???? ???????? ?? ????????

# 3. ??????????? URL ? ?????????
# ????????: http://images.example.com/windows-10-golden.qcow2
```

??????? 2: Clone from PVC (??????? ??????)

???????? ??? ????????? ??? ??????????? ?? golden image:

? `app/utils/windows_utils.py`:

```python
def generate_windows_vm_manifest_from_golden(
    vm_name: str,
    namespace: str,
    cpu: int,
    memory: int,
    storage: int,
    golden_image_name: str = "windows-10-golden-image",
    golden_image_namespace: str = "kvm",
    gpu_model: Optional[str] = None,
    gpu_count: int = 0,
    gpu_node_selector: Optional[Dict[str, str]] = None,
) -> Dict[str, Any]:
    """
    Generates Windows VM manifest that clones from golden image
    """

    # DataVolume template ??? ???????????
    datavolume_template = {
        "metadata": {
            "name": f"{vm_name}-rootdisk"
        },
        "spec": {
            "source": {
                "pvc": {
                    "name": golden_image_name,
                    "namespace": golden_image_namespace
                }
            },
            "pvc": {
                "accessModes": ["ReadWriteOnce"],
                "resources": {
                    "requests": {
                        "storage": f"{storage}Gi"
                    }
                },
                "storageClassName": "your-SC"
            }
        }
    }

```

```
    # ... ????????? ??? VM manifest
```

??????? 3: ??????????? ??? ???? (????? ???????)

???????? create-vm.js ????? ??? ?????? Windows ???????????? ?????????? ????????:

```javascript
// ? app/static/js/create-vm.js

... [278 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/WINDOWS-GOLDEN-IMAGE-QUICK-START.md (9.8 KB)

```
# ? Windows Golden Image - ??????? ???????

## ???????? VM ?? ???????? ??????

????? ???? ??? ?? ??????? golden image (????? sysprep), ???????? ????? Windows VM ???????? **3-5 ?????** ?????? 30-60...

---

## ? ????? Web UI

### ??? 1: ???????? Dashboard
```
https://kvm.example.com/<username>/dashboard
```

### ??? 2: ??????? ??????????? ??????
1. ??????? **"??????? ??????????? ??????"**
2. ? ??????? **"??????????? ???????"** ????????:
   - **Windows 10 ? Golden Image** ? (??????? ????????)
   - ~~Windows 10 ? ??????????~~ (??? ???????? ????? golden images)

### ??? 3: ????????? ???????
```
CPU: 4-8 cores (????????????? 4+)
Memory: 8-16 GB (????????????? 8+)
Storage: 80-120 GB (??????? 80 GB)
GPU: ?? ??????? (??????????)
```

**?????**: Storage ?????? ???? **>= 80 GB** (?????? golden image).

### ??? 4: ???????????
```
??? VM: my-windows-vm
Username: Administrator (?? ????????)
Password: YourStrongPassword123! ?? ??????????!
```

**?????????? ? ??????**:
- ??????? 8 ????????
- ?????????????: ????? + ????? + ??????????

### ??? 5: ???????
??????? **"???????"** ? ????????? 3-5 ?????

---

## ? ??? ?????????? ??? ???????

```
1. CDI ????????? golden image (kvm/windows-10-golden-image)
   ? ????? PVC ? ????? namespace (3-5 ???)

2. ????????? Secret ? Cloudbase-Init userdata:
   - hostname: ??? VM
   - username: Administrator (??? ???)
   - password: ????????? ??????

3. ??????????? VM ?:
   - ???????????????? ?????? (Windows ??? ??????????!)
   - CloudInit volume (Cloudbase-Init ????????? ?????????)
   - Serial console (??? ????? Cloudbase-Init)

4. ??? ?????? ????????:
   - Windows ???????? OOBE (Out-of-Box Experience)
```
```

```
   - Cloudbase-Init ????????????? hostname
   - ????????? ??????????? ? ???????
   - Cloudbase-Init ????????? ?????? (???? ? serial console)

5. ??????! ?
   - RDP ???????? (???? 3389)
   - ????? ???????????
```

? ???? ? ???????

????? VNC (??????????)
```
Dashboard ? VM details ? Console ? VNC
Username: Administrator (??? ???)
Password: (????????? ??? ????????)
```

????? RDP (????????????)
```
Dashboard ? ?????? "RDP" ? ??????? ???????
? ???????? External IP:Port
? mstsc /v:IP:PORT
Username: Administrator
Password: (??? ??????)
```

? ??????????? Golden Image

????????	????????? ? ????	Golden Image
????? ????????	30-60 ?????	3-5 ????? ?
?????? ??????	?? (VNC, ????????)	??? ?
Cloudbase-Init	????? *************	????? ?
Serial Console	????? *************	????? ?
RDP	????? *************	????? ?
????????	???????????? ???????	?????? ?
?????????? Windows	???	???? (? ??????) ?

? Troubleshooting

VM ???????? ?? "Provisioning"
```bash
# ????????? ?????? ???????????
kubectl get dv -n <namespace>

# ???? ImportInProgress > 10 ?????:
kubectl describe dv <vm-name>-rootdisk -n <namespace>
```

Cloudbase-Init ?? ????????? ????????
```bash
# ????????? ???? serial console
kubectl get pods -n <namespace> -l kubevirt.io/domain=<vm-name>
kubectl logs <pod-name> -c guest-console-log -n <namespace>

# ?????? ????: "Cloudbase-Init complete"
```

?? ???? ????? ? ???????
```bash
# ????????? secret ? Cloudbase-Init
kubectl get secret <vm-name>-cloudinit -n <namespace> -o yaml

# userdata ?????? ????????? ??? ?????? (base64)
```

```
kubectl get secret <vm-name>-cloudinit -n <namespace> \
  -o jsonpath='{.data.userdata}' | base64 -d
```

Golden image ?? ??????
```bash
# ????????? ??????? golden image
kubectl get pvc windows-10-golden-image -n kvm

# ???? ??? - ???????? ?? ??????????:
# docs/WINDOWS-GOLDEN-IMAGE-EXPORT.md
```

? ?????????? ???????? VM

????? Web UI
```
1. Dashboard ? VM ?????????? ?? ???????? "Provisioning"
2. ????? 3-5 ????? ? "Starting"
3. ??? 2-3 ?????? ? "Running"
4. VM details ? Serial console ? ???? Cloudbase-Init
```

????? kubectl
```bash
# ??????? ?? DataVolume
kubectl get dv <vm-name>-rootdisk -n <namespace> -w

# ??????? ?? VM
kubectl get vm <vm-name> -n <namespace> -w

# ??????? ?? VMI (instance)
kubectl get vmi <vm-name> -n <namespace> -w

# ???? Cloudbase-Init
POD=$(kubectl get pods -n <namespace> -l kubevirt.io/domain=<vm-name> -o name)
kubectl logs -f $POD -c guest-console-log -n <namespace>
```

? ?????????? Golden Image

????? ????? ???????? ??????? ????? (????? ????????, Windows Updates):

```bash
# 1. ???????? ????????? VM ?? ???????? golden image
# (????? Web UI ??? kubectl)

# 2. ??????????? ????? VNC/RDP

# 3. ?????????? ??????????:
#    - Windows Updates
#    - ????????
#    - ??????????

# 4. ????? sysprep:
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown

# 5. ????? ?????????? - ???????? golden image:
kubectl delete dv windows-10-golden-image-old -n kvm
kubectl patch dv windows-10-golden-image -n kvm --type=json \
  -p '[{"op": "replace", "path": "/metadata/name", "value": "windows-10-golden-image-old"}]'

# ??????????? ????? ?????
kubectl patch dv <temp-vm>-rootdisk -n kvm --type=json \
  -p '[{"op": "replace", "path": "/metadata/name", "value": "windows-10-golden-image"}]'
```

? Best Practices

1. ?????? ?????
```
Golden image: 80 GB (???????)
????? VM:     80-120 GB (?? ??????????)

?? ?????? ??????????? ? ???? ?????? golden image!
? ????? ??????????? ? ???? ??????
```

2. ????????? ??????????? VM (VM Run Strategy)

????????: ??? ????????? Windows ????? Start ? Shutdown, VM ??????????????? ?????? ????????.

???????: ??????? ? ?????? 1.0-rc+ ?????? ??????????? **RerunOnFailure** ????????:

```bash
# ? .env (Docker Compose) ??? Secret (Kubernetes)
VM_RUN_STRATEGY=RerunOnFailure
```

????????:
- ? **Shutdown** ?? Windows ? VM ??????????????? (??????: Stopped)
- ? **Reboot** ?? Windows ? VM ???????????????
- ? **Crash** ? VM ???????????? ???????????????

????????: [VM Run Strategy Documentation](VM_RUN_STRATEGY.md)

3. ??????
```
? ??????: admin, password123, 12345678
? ???????: MyV1rtua!Passw0rd, Win$ecur3!2026
```

4. Hostname
```
? ????????????: ????????? ? ?????? VM
   VM name:  my-windows-app
   Hostname: my-windows-app

? ?? ????????????: ?????? ????? (????????)
```

5. ???????
```
??????????? (??? ??????):
  CPU: 2, RAM: 4GB, Storage: 80GB

???????????? (??? ??????):
  CPU: 4, RAM: 8GB, Storage: 100GB

? GPU (??? ML/Gaming):
  CPU: 8, RAM: 16GB, Storage: 120GB, GPU: 1x
```

? ??????? ??????? ????????

- [ ] Golden image ?????? ? `kvm/windows-10-golden-image`
- [ ] Golden image ????? ?????? `Succeeded`
- [ ] ? Web UI ???????? "Windows 10 ? Golden Image" ???????
- [ ] ?????? Windows 10 Golden Image
- [ ] ??????? ??????????? (CPU >= 4, RAM >= 8, Storage >= 80)
- [ ] ?????? ??????? ?????? (8+ ????????)
- [ ] ?????? ?????? "???????"
- [ ] VM ????????? (?????? "Provisioning")
- [ ] ????? 5 ????? VM ??????????? (?????? "Running")

```
- [ ] VNC ?????????? Windows (OOBE ??? ??????? ????)
- [ ] ????? ????? ? ????????? ???????
- [ ] RDP ???????? (?????? "RDP" ??????? ???????)

---

? **??????!** ?????? ?? ?????? ????????? Windows VM ?? ?????? ?????? ?????!

**????????????**:
- [??????? Golden Image](WINDOWS-GOLDEN-IMAGE-EXPORT.md)
- [????????? RDP](WINDOWS-RDP-USAGE.md)
- [Serial Console](WINDOWS-SERIAL-CONSOLE-SETUP.md)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/WINDOWS-GOLDEN-IMAGE-SETUP.md (26.6 KB)

```
# ? ????????? Windows 10 Golden Image

> ?? **?????????? ?????**: ????? ???????? golden image ?? ???????? VM ?? ??? ???, ???? ?? ????????? PVC ??? ?????????...

## ??? 1: ??????????

### 1.1 ???????? namespace ??? ??????? ???????

```bash
kubectl create namespace kvm
```

1.2 ???????? Windows 10 ISO

???????? Windows 10 LTSC 2021 ISO ? ??????????? ????? Microsoft:
- https://www.microsoft.com/en-us/evalcenter/evaluate-windows-10-enterprise
- ??? ?????????? Volume License ??????

1.3 ????????? ISO ? ???????

```bash
# ???????? DataVolume ??? ISO
kubectl apply -f - <<EOF
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-iso
  namespace: kvm
spec:
  source:
    upload: {}
  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 7Gi
    storageClassName: your-SC
EOF

# ????????? ????????? PVC
kubectl wait --for=condition=Ready dv/windows-10-iso -n kvm --timeout=300s

# ????????? ISO (????????? virtctl upload)
virtctl image-upload dv windows-10-iso -n kvm \
  --image-path=/path/to/windows-10.iso \
  --insecure \
  --uploadproxy-url=https://cdi-uploadproxy.cdi:443
```

????????????: ???? ISO ???????? ?? HTTP/HTTPS, ?????? ??????? URL ? ?????????.

??? 2: ???????? Golden Image DataVolume

```bash
# ????????? ????????
kubectl apply -f kubernetes_example/golden-Images/windows10/windows-golden-image-dundled.yaml

# ????????? ???????? DataVolume
kubectl get dv -n kvm

# ????????? ??????? Succeeded
kubectl wait --for=condition=Ready dv/windows-10-golden-image -n kvm --timeout=600s
```
```

```
```

??? 3: ????????? Windows

3.1 ????????? installer VM

```bash
# ????????? VM
kubectl patch vm windows-10-installer -n kvm \
  --type merge -p '{"spec":{"running":true}}'

# ????????? ??????
kubectl get vmi -n kvm
```

3.2 ??????????? ????? VNC

> ? **??????????**: ???? VM ??????? ? GPU, ? ??? ????? ??? ??????? (??. WINDOWS-GPU-SETUP.md)...

```bash
# ???????? VNC URL
virtctl vnc windows-10-installer -n kvm

# ??? ????? port-forward
kubectl port-forward -n kvm \
  svc/virt-vnc-windows-10-installer 5900:5900
```

???????? VNC ??????: `vnc://localhost:5900`

3.3 ????????? Windows

1. **???????? ????** ? ??????? / English
2. **??????? "Install now"**
3. **??????? Product Key** (??? ?????????? ??? ??????? ??????)
4. **???????? ??????**: Windows 10 Enterprise LTSC
5. **??? ?????????**: Custom (advanced)
6. **?? ????? - ???????? VirtIO ?????????**:
   - ??????? "Load driver"
   - ???????? CD-ROM ? VirtIO ?????????
   - ??????? ????? `viostor\w10\amd64`
   - ?????????? ??????? Red Hat VirtIO SCSI
   - ?????? ???? ????? ?????!
7. **???????? ????** ? Next
8. **????????? ?????????** (15-30 ?????)
9. **????????? ???????????** ? ???????? ?????????????

3.4 ????????? Windows ????? ????????

????? ?????? ????????:

```powershell
# 1. ?????????? ??? VirtIO ????????
# ???????? Device Manager ? ???????? ??? ?????????? ??????????
# ??????????? CD-ROM ? virtio-drivers

# 2. ?????????? QEMU Guest Agent
# ?? CD-ROM virtio-drivers ???????: guest-agent\qemu-ga-x86_64.msi
# ????????? ????????

# 3. ???????? RDP (??????????)
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' `
  -Name "fDenyTSConnections" -Value 0
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"

# 3a. ????????? ????? ????? ? Serial Console (COM1)
# ??? ???????? ?????? ???? Cloudbase-Init ????? kubectl logs
Set-ItemProperty -Path 'HKLM:\SYSTEM\CurrentControlSet\Services\EventLog\Application' `
  -Name "CustomSD" -Value "O:BAG:SYD:(A;;0xf0007;;;SY)(A;;0x7;;;BA)(A;;0x3;;;IU)(A;;0x3;;;SU)"
```

```
bcdedit /ems ON
bcdedit /emssettings EMSPORT:1 EMSBAUDRATE:115200

# 4. ????????? Windows Defender (??????????, ??? ??????????????????)
Set-MpPreference -DisableRealtimeMonitoring $true

# 5. ?????????? ?????????? ??
# - Google Chrome / Firefox
# - 7-Zip
# - Notepad++
# - ? ?.?.

# 6. ????????? Windows Updates
# Settings ? Update & Security ? Check for updates
```

3.5 (??????????) ?????????? Cloudbase-Init

Cloudbase-Init - ??? ?????? cloud-init ??? Windows, ????????? ??????????????? ?????????:

```powershell
# ???????? cloudbase-init
Invoke-WebRequest -Uri "https://cloudbase.it/downloads/CloudbaseInitSetup_Stable_x64.msi" `
  -OutFile "C:\CloudbaseInit.msi"

# ??????????
msiexec /i C:\CloudbaseInit.msi /qn /l*v C:\cloudbase_install.log

# ????????? cloudbase-init.conf
# C:\Program Files\Cloudbase Solutions\Cloudbase-Init\conf\cloudbase-init.conf
```

??????????? cloudbase-init:
```ini
[DEFAULT]
username=Administrator
groups=Administrators
inject_user_password=true
config_drive_raw_hhd=true
config_drive_cdrom=true
config_drive_vfat=true
bsdtar_path=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\bin\bsdtar.exe
mtools_path=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\bin\
verbose=true
debug=true
logdir=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\log\
logfile=cloudbase-init.log
default_log_levels=comtypes=INFO,suds=INFO,iso8601=WARN,requests=WARN
local_scripts_path=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\LocalScripts\
metadata_services=cloudbaseinit.metadata.services.configdrive.ConfigDriveService,
    cloudbaseinit.metadata.services.httpservice.HttpService,
    cloudbaseinit.metadata.services.ec2service.EC2Service

plugins=cloudbaseinit.plugins.common.mtu.MTUPlugin,
    cloudbaseinit.plugins.common.sethostname.SetHostNamePlugin,
    cloudbaseinit.plugins.windows.createuser.CreateUserPlugin,
    cloudbaseinit.plugins.windows.extendvolumes.ExtendVolumesPlugin,
    cloudbaseinit.plugins.common.userdata.UserDataPlugin,
    cloudbaseinit.plugins.common.localscripts.LocalScriptsPlugin

# ???????? serial logging ??? ?????? ? COM1
serial_log_port=COM1
log_serial_console_port=1
```

?????: ????? ????????? Cloudbase-Init ????????? ????????????? ??????????:

```powershell
# ???????? ?????? ??? ?????? ??????? ? serial console
$InitScript = @'
# Cloudbase-Init startup marker
```

```
Write-Host "[Cloudbase-Init] Starting initialization..."
Get-Date | Out-String | Write-Host
'@

$InitScript | Out-File -FilePath "C:\Program Files\Cloudbase Solutions\Cloudbase-Init\LocalScripts\cloudbase-init-sta...

# ????????? ????? ? COM1
$ComPort = New-Object System.IO.Ports.SerialPort "COM1", 115200
$ComPort.Open()
$ComPort.WriteLine("[Test] Serial console configured")
$ComPort.Close()
```

??? 4: ?????????? ? ??????????? (Sysprep)

?? **??????????? ?????** - ??? sysprep ????? ????? ????? ?????????? SID!

4.1 ????????? Sysprep

```powershell
# ???????? PowerShell ?? ????? ?????????????

# ??????? 1: Sysprep ? cloudbase-init (???? ??????????)
cd "C:\Program Files\Cloudbase Solutions\Cloudbase-Init\conf"
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown /unattend:Unattend.xml

# ??????? 2: ??????? sysprep (??? cloudbase-init)
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown
```

??? ?????? sysprep:
- ??????? ????????? ????????? ?????? (SID, ??? ??????????)
- ????????? ????????? Windows
- ????????????? ????? ? ???????????
- ????????? ???????

4.2 ????????? ????????? VM

```bash
# ????????? ??? VM ??????????
kubectl get vmi -n kvm

# ???? VMI ????? - ?????? ?????????? ???????
```

4.3 ????????? installer VM

```bash
# ????????? VM (???? ??? ????????)
kubectl patch vm windows-10-installer -n kvm \
  --type merge -p '{"spec":{"running":false}}'

# ??????????: ??????? installer VM (?? ?? DataVolume!)
kubectl delete vm windows-10-installer -n kvm
```

??? 5: ????????? RBAC ??? ???????????

????? ????? ????????? ?????????? DataVolume ?? `kvm`:

```bash
kubectl apply -f - <<EOF
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: datavolume-cloner
rules:
```

```
  - apiGroups: ["cdi.kubevirt.io"]
    resources: ["datavolumes"]
    verbs: ["get", "list"]
  - apiGroups: [""]
    resources: ["persistentvolumeclaims"]
    verbs: ["get", "list"]
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: kubevirt-manager-clone-binding
subjects:
  - kind: ServiceAccount
    name: kubevirt-manager-sa
    namespace: kvm
roleRef:
  kind: ClusterRole
  name: datavolume-cloner
  apiGroup: rbac.authorization.k8s.io
EOF
```

??? 6: ?????????? Golden Image

> ? **?????????? ?????**: VM ?????? ? `dataVolumeTemplates`, ??????? PVC ????? `ownerReferences` ?? VM. ??? ???????? ...

6.1 ??????? ??? ????? VM ? PVC

```bash
# ???????? ?? ???? ????????
VM_NAME="w10-gpu-gi"        # ? ??? ????? VM
NAMESPACE="kv-example-001"   # ? ??? namespace
PVC_NAME="${VM_NAME}-rootdisk"

# ????????? ??? VM ??????????
kubectl get vm $VM_NAME -n $NAMESPACE
# ?????? ???? ? ??????? Stopped

# ????????? PVC
kubectl get pvc $PVC_NAME -n $NAMESPACE
# PVC ?????? ??????????? ? ???? Bound
```

6.2 ? ??????? ownerReferences (??????????!)

??? ????? ???? PVC ???????? ??? ???????? VM!

```bash
# ??????? ownerReferences (?????? PVC ??????????? ?? VM)
kubectl patch pvc $PVC_NAME -n $NAMESPACE --type=json \
  -p='[{"op": "remove", "path": "/metadata/ownerReferences"}]'

# ????????? ??? ownerReferences ???????
kubectl get pvc $PVC_NAME -n $NAMESPACE -o jsonpath='{.metadata.ownerReferences}'
# ????? ?????? ???? ??????: []
```

? ?????? PVC **?? ????????** ??? ???????? VM!

6.3 ?????????? PVC ? namespace kvm

```bash
# ???????? DataVolume ??????? ????????? PVC
kubectl apply -f - <<EOF
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-golden-image
  namespace: kvm
```

```
spec:
  source:
    pvc:
      name: $PVC_NAME
      namespace: $NAMESPACE
  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 80Gi  # ? ?????? ?????? ????? (??? ??????)
    storageClassName: your-SC
EOF

# ??????? ?? ???????????? (?????? 3-5 ?????)
kubectl get dv windows-10-golden-image -n kvm -w

# ????? ??????? Succeeded:
# NAME                      PHASE       PROGRESS
# windows-10-golden-image   Succeeded   100.0%
```

6.4 ????????? golden image

```bash
# ????????? ??? PVC ??????
kubectl get pvc windows-10-golden-image -n kvm

# ?????? ????:
# NAME                        STATUS   VOLUME        CAPACITY   ACCESS MODES
# windows-10-golden-image     Bound    pvc-xxx...    80Gi       RWO

# ????????? ??? ??? ownerReferences
kubectl get pvc windows-10-golden-image -n kvm -o jsonpath='{.metadata.ownerReferences}'
# ????? ?????? ???? ??????!
```

?? ?????: ???? ????? ?? ?????? (???????? DataVolume ownerReference):

```bash
# ?????? ?????? ? ownerReferences:
# [{"apiVersion":"cdi.kubevirt.io/v1beta1","kind":"DataVolume",...}]

# ??? ?????????! DataVolume ??????? PVC ? ownerReference ?? ????
# ????? ??????? ownerReferences ????? ???????????:

kubectl patch pvc windows-10-golden-image -n kvm --type=json \
  -p='[{"op": "remove", "path": "/metadata/ownerReferences"}]'

# ???????? (?????? ???? ?????)
kubectl get pvc windows-10-golden-image -n kvm -o jsonpath='{.metadata.ownerReferences}'

# ?????? ????? ????????? ??????? DataVolume (PVC ?????????)
kubectl delete dv windows-10-golden-image -n kvm
```

6.5 ??????? ????????? VM (?????? ?????????!)

```bash
# ?????? ????? ??????? VM
kubectl delete vm $VM_NAME -n $NAMESPACE

# ??????????: ??????? ?????? PVC ?? user namespace
kubectl delete pvc $PVC_NAME -n $NAMESPACE
```

??? 8: ???????? Windows VM ?? Golden Image

8.1 ????? Web UI (Dashboard) - ???????????? ?

1. **???????? Dashboard**:
   ```
   https://kvm.example.com/<username>/dashboard
   ```

2. **??????? ??????????? ??????**:
   - ??????? ?????? **"??????? ??????????? ??????"**

3. **???????? ??????????? ???????**:
   - ? ??????? **"??????????? ???????"** ??????? **Windows 10**
   - ???????? **"Golden Image"** (??????? ????????) ?
   - **??** ????????? "?????????" (??? ??? ???????? ????? golden images)

4. **????????? ???????**:
   ```
   CPU:     4-8 cores (???????????? 4+)
   Memory:  8-16 GB (???????????? 8+)
   Storage: 80-120 GB (??????? 80 GB - ?????? golden image!)
   GPU:     ?? ??????? (??????????)
   ```

   ?? **??????**: Storage ?????? ???? **>= 80 GB** (?????? ?????? golden image)

5. **????????????**:
   ```
   ??? VM:   my-windows-vm
   Username: Administrator (?? ?????????)
   Password: YourStrongPassword123!  ?? ??????????!
   ```

   ?????????? ? ??????:
   - ??????? 8 ????????
   - ?????????????: ????? + ????? + ??????????

6. **????????**:
   - ??????? **"???????"**
   - ????????? 3-5 ?????

??? ????????? ??? ???????:

```
1. CDI ???????? golden image (kvm/windows-10-golden-image)
   ? ????? PVC ? ????? namespace (3-5 ???)

2. ????????? Secret ? Cloudbase-Init userdata:
   - hostname: ??? VM
   - username: Administrator (??? ???)
   - password: ????????? ??????

3. ??????????? VM ?:
   - ???????????? ?????? (Windows ??? ??????????!)
   - CloudInit volume (Cloudbase-Init ????????? ?????????)
   - Serial console (??? ????? Cloudbase-Init)

4. ??? ?????? ????????:
   - Windows ???????? OOBE (Out-of-Box Experience)
   - Cloudbase-Init ???????????? hostname
   - ????????? ??????????? ? ???????
   - Cloudbase-Init ????????? ?????? (???? ? serial console)

5. ??????! ?
   - RDP ???????? (???? 3389)
   - ????? ???????????
```

8.2 ????? kubectl (??? Advanced ????????????)

```bash
kubectl apply -f - <<EOF
apiVersion: kubevirt.io/v1
```

```
kind: VirtualMachine
metadata:
  name: my-windows-vm
  namespace: kv-user-default
spec:
  running: true
  dataVolumeTemplates:
    - metadata:
        name: my-windows-vm-rootdisk

... [311 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/WINDOWS-GPU-SETUP.md (18.5 KB)

```
# Windows VM ? GPU Passthrough - ??????????

**????**: 4 ??????? 2026
**??????**: 1.1

## ?????

?????? ?????????? ????????? ??????? ???????? Windows 10 VM ? GPU passthrough ???:
- ? Gaming
- ? Machine Learning (CUDA, TensorFlow, PyTorch)
- ? 3D Rendering
- ? Video Encoding

### ?? Dual Display (VNC + GPU)

**??????? ? ?????? 0.0.5-windows**, VM ? GPU ???????????? ???????? **??? ???????????????**:

1. **QXL Display** (??? VNC) - ???????? ????????? ??????
2. **Physical GPU** (??????????? ???????????) - ??? ??????????? ?????????????????

#### ??? ??? ???????

```yaml
# ???????????? ?????????? ? VM ? GPU:
spec:
  template:
    metadata:
      annotations:
        kubevirt.io/domain: |
          devices:
            video:
              - model:
                  type: 'qxl'
                  vram: '65536'
                  heads: '1'
    spec:
      domain:
        devices:
          gpus:
            - name: gpu0
              deviceName: nvidia.com/1080ti
```

?????? ? VM

????? ????????? ???????? NVIDIA ? ??? ????? **??? ???????** ??????????:

??????	????? ???????????	?????????????????? GPU
VNC Console	????????? Windows, ?????????	? ??? (QXL display)
Physical Monitor	????????? ?????? ?? ???????	? ?????? (GPU output)
RDP	????????? ?????? ? GPU	? ?????? (GPU rendering)

> ?? **?????**: Windows ???????????? ?????????? **primary display** ?? GPU. VNC ????????? ?????????, ?? ??????? **s...

??????????????? ???????????

1. ????????? ?????
- ? GPU NVIDIA (GTX 1080 Ti, RTX 3090, A100 ? ?.?.)
- ? CPU ? ????????? IOMMU (Intel VT-d / AMD-Vi)
- ? BIOS/UEFI: ???????? VT-x, VT-d, IOMMU

2. Kubernetes ???????
```

```bash
# ???????? ???? ? GPU
kubectl get nodes -l node-role.kubernetes.io/kubevirt=
kubectl describe node <gpu-node> | grep -A5 "nvidia.com"
```

????? ?????? ?????????:
```
nvidia.com/1080ti:     1
nvidia.com/1080ti-audio: 1
```

3. KubeVirt ???????????

????????? ??? ? KubeVirt CR ??????? feature gate `HostDevices`:

```bash
kubectl get kubevirt kubevirt -n kubevirt -o yaml
```

?????? ????:
```yaml
spec:
  configuration:
    developerConfiguration:
      featureGates:
        - HostDevices
    permittedHostDevices:
      pciHostDevices:
        - pciVendorSelector: "10DE:1B06"  # GPU
          resourceName: "nvidia.com/1080ti"
        - pciVendorSelector: "10DE:10EF"  # HDMI Audio
          resourceName: "nvidia.com/1080ti-audio"
```

??? 1: ???????? Windows VM ? GPU

????? Web UI

1. ???????? **Web UI** ? **?????????? ??????**
2. ??????? **"??????? VM"**

??? 1: ????? ??
- ???????? **Windows 10 Installer**

??? 2: ???????

Preset (???????????? ??? GPU):
- ???????? **"GPU ? Single"** preset:
  - 8 vCPU
  - 32 GB RAM
  - 200 GB Storage
  - 1x GPU (????????????)

??? Manual:
- CPU: ??????? **4 ????** (???????????? 8)
- Memory: ??????? **8 GB** (???????????? 16-32 GB)
- Storage: ??????? **60 GB** (???????????? 100+ GB)

GPU ????????????:
- ??? ???????: **GPU**
- ?????? GPU: ???????? ???? ?????? (????????, **NVIDIA GTX 1080 Ti**)
- ????????? GPU: **1** ??? **2**

?? **??????**: ????? ?????? Windows + GPU ???????? ????????????? ???? ? ??????????? ?? ????????? ?????????.

??? 3: ????????
- ??????? **"???????"**
- VM ????? ??????? ? ??????? **"Stopped"**

??? 2: ????????? Windows 10

2.1 ?????? VM

```bash
# ????? UI: ??????? "Start" ?? VM
# ??? ????? kubectl:
kubectl patch vm <vm-name> -n <namespace> --type merge -p '{"spec":{"running":true}}'
```

2.2 ?????????? ????? VNC

1. ? Web UI ???????? ?????? VM
2. ??????? **"VNC Console"**
3. ????????? ???????? Windows Setup

2.3 ????????? Windows

??? 1: ???????? ????????? VirtIO

Windows Installer **?? ??????** ???? ??? VirtIO ?????????:

1. ?? ?????? ?????? ????? ??????? **"Load Driver"**
2. ???????? ???? **E:\\ (VirtIO drivers CD)**
3. ????????? ?:
   ```
   E:\viostor\w10\amd64\
   ```
4. ??????? **"OK"** ? ??????? VirtIO SCSI ????? ????????
5. ?????? ?? ??????? **???? 200 GB** (??? ??? ??????, ??????? ???????)

??? 2: ????????? Windows

1. ???????? ???? ? ??????? **"Next"**
2. ????????? ?????????? ??????
3. ????? ??????????? ????????? ?????????? ????????? Windows:
   - ??????: Russia
   - Keyboard: Russian
   - ???????? ???????????: **Administrator** (??? ??????)
   - ??????: **?????????? ???????? ??????**

??? 3: ????????? ????????? VirtIO (? Windows)

????? ????????? Windows:

1. ???????? **????????? ?????????** (Win+X ? Device Manager)
2. ?? ??????? **?????????? ??????????** (Ethernet, Balloon, Serial Port)
3. ???????? ???? **E:\\ (VirtIO drivers)**
4. ????????? **virtio-win-gt-x64.msi** ??? **virtio-win-guest-tools.exe**
5. ?????????? ??? ????????
6. ???????????? VM

??? 3: ????????? NVIDIA GPU ?????????

3.1 ???????? GPU ? Windows

???????? **????????? ?????????**:
- ?????????? **"Display adapters"** ??? **"Other devices"**
- ?? ?????? ??????? **NVIDIA GPU** (????? ???? ??? "Unknown device")

3.2 ?????????? ????????? NVIDIA

? VM:

1. ???????? ??????? (Edge)
2. ????????? ??: https://www.nvidia.com/Download/index.aspx

3. ????????:
   - Product Type: **GeForce** (??? **Tesla** ??? A100)
   - Product Series: **GeForce GTX 10 Series** (??? GTX 1080 Ti)
   - Product: **GeForce GTX 1080 Ti**
   - Operating System: **Windows 10 64-bit**
   - Download Type: **Game Ready Driver** ??? **Studio Driver**
4. ??????? **"Search"** ? **"Download"**

3.3 ????????? ?????????

1. ????????? ????????? ???? (????????, `551.23-desktop-win10-win11-64bit-international-dch-whql.exe`)
2. ???????? **"NVIDIA Graphics Driver"**
3. ??? ?????????: **"Express"** (????????????)
4. ????????? ????????? ?????????
5. **????????????? VM**

3.4 ???????? GPU

????? ???????????:

1. ???????? **PowerShell** ??? **CMD**
2. ?????????:
   ```powershell
   nvidia-smi
   ```

?????:
```
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 551.23                 Driver Version: 551.23         CUDA Version: 12.4     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                  TCC/WDDM      | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA GeForce GTX 1080 Ti  WDDM   |   00000000:06:00.0 Off |                  N/A |
| 23%   35C    P8              9W /  250W |       0MiB /  11264MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
```

? **GPU ????????!**

??? 4: ????????? ???????? (Dual Display)

????? ????????? ???????? NVIDIA Windows ????????? **??? ???????**:

4.1 ???????? ????????

1. ?????? ???? ?? ??????? ????? ? **Display Settings**
2. ?? ???????:
   - **Display 1** - QXL Display (VNC, secondary)
   - **Display 2** - NVIDIA GPU (primary)

4.2 ????????? ??? VNC ??????? (??????????)

???? ?????? ??????????? VNC ??? ???????? ???????:

```powershell
# ????????? PowerShell ?? ????? ?????????????

# 1. ??????? QXL primary display
Set-DisplayResolution -Width 1920 -Height 1080 -DeviceID "\\.\DISPLAY1"

# 2. ??? ????????? GPU display (???? ?? ????? ?????????? ???????)
# Settings ? Display ? Multiple displays ? Show only on 1
```

4.3 ????????? ??? RDP ??????? (????????????)

RDP ???????????? ??????????? GPU ??? ???????????:

1. ???????? RDP ? Windows:
   ```powershell
   Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name "fDenyTSConnections" -Value 0
   Enable-NetFirewallRule -DisplayGroup "Remote Desktop"
   ```

2. ???????? RDP ???? ????? API:
   ```bash
   # ????? ???-?????????: VM Details ? Services ? Add Service ? RDP (3389)

   # ??? ????? curl:
   curl -X POST "https://dev-kubevirt.k8s.example.com/example/vm/<vm-name>/services" \
     -H "Content-Type: application/json" \
     -d '{"port": 3389, "type": "rdp"}'
   ```

3. ??????????? ????? RDP ??????:
   ```bash
   # Windows: Win+R ? mstsc.exe
   # Linux: remmina ??? freerdp
   # macOS: Microsoft Remote Desktop

   # ?????: <external-ip>:<allocated-port>
   ```

> ? **RDP + GPU = ?????? ?????????????????? ????????!**

??? 5: ??????????? GPU

5.1 CUDA Test

??????????? **CUDA Toolkit**:

1. ???????? ? https://developer.nvidia.com/cuda-downloads
2. ???????? **Windows 10**, **x86_64**, **exe (local)**
3. ??????????? CUDA Toolkit
4. ???????? CMD:
   ```cmd
   cd "C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v12.4\extras\demo_suite"
   deviceQuery.exe
   ```

?????:
```
CUDA Device Query (Runtime API) version (CUDART static linking)

Detected 1 CUDA Capable device(s)

Device 0: "NVIDIA GeForce GTX 1080 Ti"
  CUDA Driver Version / Runtime Version          12.4 / 12.4
  CUDA Capability Major/Minor version number:    6.1
  Total amount of global memory:                 11264 MBytes
  ...
  Result = PASS
```

5.2 Gaming Test

??????????? **GPU-Z**:
- ???????? ? https://www.techpowerup.com/gpuz/
- ????????? ? ??????? **"Graphics Card"**
- ?????????: GPU Name, Memory Size, Driver Version

??????????? ???? (Steam, Epic Games) ? ????????? FPS.

5.3 Machine Learning Test

Python + PyTorch:

```powershell
# ????????? Python 3.11
# ????????? PyTorch ? CUDA:
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121

# ????:
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}'); print(f'GPU: {torch.cuda.get_device_n...
```

?????:
```
CUDA available: True
GPU: NVIDIA GeForce GTX 1080 Ti
```

????????? ???????

???????? 1: GPU ?? ??????????? ? Device Manager

????????:
```bash
# ????????? ??? VM ???????????? ???????? GPU
kubectl describe vm <vm-name> -n <namespace> | grep -A5 gpus

# ????? ?????? ?????????:
# gpus:
# - deviceName: nvidia.com/1080ti
#   name: gpu0
```

???? GPU ??? ? ?????????:
1. ??????? VM
2. ??????????? ? GPU checkbox ?????????

???????? 2: "Code 43" ? Device Manager

???????: NVIDIA ??????? ???????? ????????????.

????????:

????????? ??? ? ????????? VM ????:
```yaml
spec:
  template:
    spec:
      domain:
        cpu:
          model: host-passthrough  # ? ?????!
        firmware:
          bootloader:
            efi:
              secureBoot: false
```

??? ???????????? ???????????? ??? ???????? Windows + GPU VM.

???????? 3: VM ?? ?????????? (Pending)

???????:
```bash
# ????????? ???? ? GPU
kubectl get nodes -l node-role.kubernetes.io/kubevirt=

# ????????? Pod VM
kubectl get pods -n <namespace> | grep virt-launcher
```

```
# ????????? ???????
kubectl describe vmi <vm-name> -n <namespace>
```

????????? ???????:
- ??? ??? ? label `node-role.kubernetes.io/kubevirt=`
- GPU ??? ??????????? ?????? VM
- ??????????? ???????? (CPU/RAM)

???????? 4: VNC ????????? ?????? ????? ????? ????????? GPU ????????

????????: Windows ??????????? primary display ?? ????????? GPU.

????????:

```bash
# ??????? 1: ??????????? RDP (????????????)
# ???????? RDP service ????? ???-?????????: VM Details ? Services ? Add Service ? RDP

# ??????? 2: ????????? ????????? ??????? ? GPU ?? ???????

# ??????? 3: ??????????? primary display ??????? ?? QXL
# ????? RDP ??? ????????? ???????:
# Settings ? Display ? Multiple displays ? Make this my main display (??????? Display 1)
```

???????? 5: ?????? ???????????????? GPU

????????:

1. ????????? ??? ??????????? **host-passthrough** CPU:
   ```bash
   kubectl get vmi <vm-name> -n <namespace> -o yaml | grep -A2 "cpu:"
   ```

2. ????????? **MSI mode** ??? GPU ? Windows:
   ```powershell
   # ???????? MSI Mode Utility
   # https://github.com/CHEF-KOCH/MSI-utility
   ```

3. ????????? **Windows power throttling**:
   ```powershell
   powercfg -setactive 8c5e7fda-e8bf-4a96-9a85-a6e23a8c635c  # High performance
   ```

??????????? ?? ???????????

1. ????????? Windows

```powershell
# ????????????? ?????????
Settings ? Update & Security ? Windows Update
```

2. Firewall

?? ????????? VM **???????????** ? namespace. ??? ??????? ?? RDP:

```bash
# ???????? NodePort service ??? RDP
kubectl expose vmi <vm-name> --type=NodePort --port=3389 --target-port=3389 -n <namespace>
```

3. ?????????

????????? **Windows Defender** ??? **?????? ??????????**.

```
492
493 ## ??????????? ?????????????????
494
495 ### 1. CPU Pinning (???????????)
496
497 ??? production workloads ????????? CPU pinning:
498
499 ```yaml
500 spec:
501
502 ... [77 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/WINDOWS-RDP-USAGE.md (14.1 KB)

```
# ? Windows Remote Desktop (RDP) - ??????????? ?? ????????????

## ?????

Remote Desktop Protocol (RDP) ????????? ??????????? ? Windows VM ???????? ? ?????? ??????????? ???????????. ? ??????...

- ? **?????? ??????????????????** - ???????????? ??? Windows
- ? **???????? ??????** - ??????????? ?????? ????? ????????? ?? ? VM
- ? **??????????????? ******* - ??????????????? ????? ? VM ?? ????????? ??
- ? **??????????? ?????????** - ?????? ? VM ?? ????????? ????????
- ? **???????????????? ?????????** - ?????? ?? ?????????? ?????????
- ? **????? ??????** - ??????????? ?????? ????? ?????????

---

## ? ??????? *****

### ??? 1: ???????? Windows VM

1. ???????? **Dashboard** ? **???????? ??????????? ??????**
2. ???????? **Windows Installer** ? ?????? ??????????? ??????
3. ????????? ??????? (???????????? ??????? 4 CPU, 8GB RAM)
4. ??????? **????????**

### ??? 2: ????????? Windows

1. ????????? ???????? VM
2. ???????? **VNC ????????** ??? ????????? Windows
3. ???????? ??????? ????????? Windows
4. **?????**: ????????? ?????? ?????????????!

### ??? 3: ????????? RDP

#### ? Windows VM (????? VNC):

```powershell
# ???????? Remote Desktop
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name "fDenyTSConnections" -Value 0

# ????????? RDP ????? Firewall
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"

# ????????? ??????
Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Select-Object DisplayName, Enabled
```

??? 4: ???????? RDP ??????? *****

????? Dashboard:

1. ???????? ?????? VM ? **Dashboard**
2. ??????? ???? Windows VM
3. ??????? ?????? **RDP** (??????????? ?????? ??? Windows VM)
4. ??????? ???????????? ??????? ??????? ????? 3389 ? ????????? ??????? ????

????? VM Details:

1. ???????? ???????? VM ? ??????? **Overview**
2. ? ?????? **???????? ??????** ??????? ?????? **RDP (3389)**
3. ????????? ???????? ??????? ????

? ??????????? ? Windows VM

```

### Windows (mstsc)

1. ??????? `Win + R`
2. ???????: `mstsc`
3. ? ???? **Computer**: `<EXTERNAL_IP>:<PORT>`
   - ??????: `IP:PORT`
4. ??????? **Connect**
5. ???????:
   - **Username**: `Administrator`
   - **Password**: ??????, ???????????? ??? ???????? Windows

**??????????? ?????** (?? ???????):
```
mstsc /v:IP:PORT /f
```
- `/f` - ???????????? ?????
- `/admin` - ????????? ??????
- `/multimon` - ??????????? ??? ????????

### Linux (Remmina)

```bash
# ????????? Remmina (???? ?? ??????????)
sudo apt update
sudo apt install remmina remmina-plugin-rdp

# ??????
remmina
```

? Remmina:
1. ???????? ????? ??????????: **RDP**
2. **Server**: `<EXTERNAL_IP>:<PORT>`
3. **Username**: `Administrator`
4. **Password**: ??? ??????
5. **Resolution**: ???????? ?????? ?????????
6. ????????? ? ???????????

macOS (Microsoft Remote Desktop)

1. ????????? [Microsoft Remote Desktop](https://apps.apple.com/app/microsoft-remote-desktop/id1295203466) ?? App Store
2. ??????? **+ Add PC**
3. **PC Name**: `<EXTERNAL_IP>:<PORT>`
4. **User Account**: Add User Account
   - **Username**: `Administrator`
   - **Password**: ??? ??????
5. ??????? **Add** ? **Connect**

Linux (xfreerdp - ????????? ??????)

```bash
# ?????????
sudo apt install freerdp2-x11

# ??????????
xfreerdp /v:IP:PORT /u:Administrator /p:YourPassword /cert-ignore

# ? ????????? ??????
xfreerdp /v:IP:PORT /u:Administrator /p:YourPassword /cert-ignore /drive:share,/home/user/shared

# ???????????? ?????
xfreerdp /v:IP:PORT /u:Administrator /p:YourPassword /cert-ignore /f

# ? ??????????? ?????? ??????
xfreerdp /v:IP:PORT /u:Administrator /p:YourPassword /cert-ignore /clipboard
```

? ???????????

????????: ?????????? ?? ?????????

??? ?????? ?????????? ???????? ????????????? ? ??????????:

Windows: ??????? **Yes** / **Connect anyway**

Linux/Remmina: ???????? **Accept certificate** ? **OK**

xfreerdp: ?????????? ???????? `/cert-ignore`

??????????? ?? ???????????

1. **??????????? ??????? ??????**:
   ```powershell
   # ? Windows VM
   net user Administrator NewStrongPassword123!
   ```

2. **???????? ?????????? ???????????** (?? ?????????? Administrator):
   ```powershell
   New-LocalUser -Name "rdpuser" -Password (ConvertTo-SecureString "StrongPass123!" -AsPlainText -Force)
   Add-LocalGroupMember -Group "Remote Desktop Users" -Member "rdpuser"
   ```

3. **????????? NLA (Network Level Authentication)**:
   ```powershell
   Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server\WinStations\RDP-Tcp' -Name "UserAut...
   ```

4. **?????????? ?????????? ????????? ??????? ?????**:
   ```powershell
   # Account Lockout Policy
   net accounts /lockoutthreshold:5 /lockoutduration:30 /lockoutwindow:30
   ```

? ????????????? ??????????

???????? ??????

Windows ? Windows VM:
1. ??????????? ????? RDP
2. ???????? **Local Resources** ? **More...**
3. ???????? **Drives** ? ???????? ????? ??? ???????????
4. ??????????? ? ? VM ???????? **This PC** ? **Redirected Drives**

Linux ? Windows VM (Remmina):
1. ? ?????????? ??????????: **Share folder**
2. ??????? ???? ? ????????? ?????: `/home/user/shared`
3. ? Windows VM ???????? **\\tsclient\share**

?????????????? ?????

Windows mstsc:
1. ? mstsc: **Show Options** ? **Local Resources**
2. **Remote audio**: Play on this computer
3. **Remote audio recording**: Record from this computer

Remmina:
1. ? ??????????: **Audio output mode** ? **Local**

????? ??????

????? ?????? ???????? ???????????? ??? ?????????? RDP ????????:
- ????????? ????? ?? ????????? ?? ? ????????? ? Windows VM
- ????????? ???? ?? ????????? ?? ? ????????? ? Windows VM (????????? ?????)

???????????????? ?????????

Windows mstsc:

```
mstsc /v:IP:PORT /multimon
```

Remmina:
1. **Resolution**: Use client resolution
2. **Multi monitor**: ? Enabled

? Troubleshooting

???????? 1: "Remote Desktop can't connect"

??????? 1: ????????? ??? RDP ??????? ? Windows:
```powershell
Get-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name "fDenyTSConnections"
# ?????? ????: fDenyTSConnections = 0
```

??????? 2: ????????? firewall:
```powershell
Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Select-Object DisplayName, Enabled
# ??? ??????? ?????? ???? Enabled = True
```

??????? 3: ???????????? ?????? RDP:
```powershell
Restart-Service TermService -Force
```

???????? 2: ???? ?? ????????

???????? ? Kubernetes:
```bash
# ????????? ??????
kubectl get svc -n <namespace> | grep rdp

# ????????? ??? ???? ?????? ?? ????
nc -zv <node-ip> <nodeport>

# ????????? ??? VM ????????
kubectl get vmi -n <namespace>
```

???????? 3: ????????? ??????????

????????: ???????????? ????????? RDP:

Windows mstsc:
1. **Show Options** ? **Experience**
2. **Connection speed**: LAN (10 Mbps or higher)
3. ?????????:
   - Desktop background
   - Menu and window animation
   - Themes

xfreerdp:
```bash
xfreerdp /v:IP:PORT /u:User /p:Pass /cert-ignore /compression /network:lan /gfx:rfx
```

???????? 4: ?????? ????? ????? ??????????

??????? 1: ????????? ???????? Windows (????? ?????? ?? 30 ??????)

??????? 2: ?????????? ? ??????????? ?????

??????? 3: ????????? ??? GPU ???????? ??????????:
```powershell
# ? Windows VM (????? VNC)
```

```
Get-WmiObject Win32_VideoController | Select-Object Name, DriverVersion
```

???????? 5: "Your credentials did not work"

????????: ????????? ??????:
```powershell
# ???????? ?????? Administrator
net user Administrator NewPassword123!

# ??? ??????? ?????? ???????????
New-LocalUser -Name "testuser" -Password (ConvertTo-SecureString "Test123!" -AsPlainText -Force)
Add-LocalGroupMember -Group "Administrators" -Member "testuser"
```

? ?????????: RDP vs VNC

???????	RDP	VNC (KubeVirt)
??????????????????	????? ???????	??? ??????
???????? ????????	????? ???????	??? ??????
???????? ??????	? ??	? ???
????? ??????	? ??	?? ???????????
????	? ??	? ???
????????	? ??	? ???
???????????????	? ??	? ???
??? ????????????? ?????????	?? ??????? ?????????	? ???????? ?? ???????
??????????? ??????	? ???	? ??
????????? Linux VM	? ???	? ??

?????????????:
- **??????????? VNC** ??? ????????? ????????? Windows ? ??????????
- **??????????? RDP** ??? ??????????? ?????? ? Windows VM

? Best Practices

1. ??????????? ????????? ?????? ??? ?????? ?????

```
- RDP (3389)  ? ????????? ??????? ????
- SSH (22)    ? PowerShell remoting ????? OpenSSH
- HTTPS (443) ? ???-??????????
```

2. ???????????? ????????? RDP

???????? PowerShell ?????? `setup-rdp.ps1`:

```powershell
# Enable RDP
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name "fDenyTSConnections" -Value 0

# Enable firewall rules
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"

# Enable NLA
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server\WinStations\RDP-Tcp' -Name "UserAuthen...

# Restart RDP service
Restart-Service TermService -Force

Write-Host "RDP configured successfully!" -ForegroundColor Green
```

????????? ????? VNC ??? ???????? ? Cloudbase-Init.

3. ?????????? RDP ???????????

```powershell
# ??????? RDP ??????
quser

# ??????? ??????
Get-EventLog -LogName Security -InstanceId 4624 -Newest 10 | Select-Object TimeGenerated, Message

# RDP ????
Get-WinEvent -LogName 'Microsoft-Windows-TerminalServices-LocalSessionManager/Operational' -MaxEvents 50
```

? ????????????? ???????

- [Microsoft RDP Documentation](https://docs.microsoft.com/en-us/windows-server/remote/remote-desktop-services/client...
- [Remmina Documentation](https://remmina.org/how-to-install-remmina/)
- [FreeRDP Documentation](https://github.com/FreeRDP/FreeRDP/wiki)
- [Windows Firewall ??? RDP](https://docs.microsoft.com/en-us/troubleshoot/windows-server/remote/enable-remote-desktop)

? ??????? ??????? ??????????

- [ ] Windows VM ??????? ? ????????
- [ ] Windows ?????????? ????? VNC
- [ ] RDP ??????? ? Windows (`Set-ItemProperty`)
- [ ] Firewall ???????? (`Enable-NetFirewallRule`)
- [ ] RDP ???? ?????? ? Dashboard ??? VM Details
- [ ] ??????? IP ? ???? ?????????
- [ ] RDP ?????? ?????????? ?? ????????? ??
- [ ] ?????? Administrator ????????
- [ ] ???????? ?????????? ????? RDP!

??????! ? ?????? ?? ?????? ?????????? ???????? ? Windows VM ????? Remote Desktop.

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/WINDOWS-SERIAL-CONSOLE-SETUP.md (12.6 KB)

```
# ? Windows Serial Console Setup

## ?????

Serial Console (COM1) ????????? ???????? ???? Cloudbase-Init ?? Windows VM ? KubeVirt pod, ??? ???? ??????????:

- ? ?????? ?????? ???????????? Windows VM ? ???????? ???????
- ? ?????? ???? ????? `kubectl logs` ??? ??? Linux VM
- ? ?????????? ?????? "????????..." ? "????????????..." ? "?????? ? ??????" ? Web UI
- ? ??????? ??????? ? Cloudbase-Init ??? VNC

---

## ??????????

```
Windows VM
    ?
Cloudbase-Init ??????? ???? ? COM1 (Serial Port)
    ?
KubeVirt Serial Device (virtio-serial)
    ?
guest-console-log container (? virt-launcher pod)
    ?
kubectl logs -c guest-console-log
    ?
Web UI ????????? ?????? ????????????
```

??? 1: ?????????? Serial Device ? VM Manifest

Serial device ??? ???????????? ?????????? ??? ???????? Windows VM ????? `generate_windows_installer_vm_manifest()`:

```python
# app/utils/windows_utils.py
"devices": {
    "disks": [...],
    "interfaces": [...],
    # Serial console for Cloudbase-Init logs
    "serials": [
        {
            "type": "serial",
            "name": "serial0"
        }
    ],
}
```

????????:

```bash
kubectl get vmi <vm-name> -n <namespace> -o yaml | grep -A3 "devices:" | grep -A2 serials

# ?????? ????:
# serials:
# - name: serial0
#   type: serial
```

??? 2: ????????? Windows ??? Serial Output

```

### 2.1 ????????? Emergency Management Services (EMS)

EMS ????????? Windows ???????? ????????? ????????? ? serial port.

```powershell
# ? Windows VM (????? VNC ??? RDP):

# ???????? EMS
bcdedit /ems ON

# ??????????? ???? COM1, ???????? 115200
bcdedit /emssettings EMSPORT:1 EMSBAUDRATE:115200

# ????????? ?????????
bcdedit /enum

# ????? ?????? ?????????:
# ems                 Yes
# emssettings         EMSPORT:1 EMSBAUDRATE:115200
```

???????????????? VM ??? ?????????? ????????.

2.2 ????????? ???? ??????? ? Event Log (???????????)

??? ?????? ??????? ?????????? ? serial console:

```powershell
Set-ItemProperty -Path 'HKLM:\SYSTEM\CurrentControlSet\Services\EventLog\Application' `
  -Name "CustomSD" -Value "O:BAG:SYD:(A;;0xf0007;;;SY)(A;;0x7;;;BA)(A;;0x3;;;IU)(A;;0x3;;;SU)"
```

??? 3: ????????? Cloudbase-Init

3.1 ?????????????? cloudbase-init.conf

????: `C:\Program Files\Cloudbase Solutions\Cloudbase-Init\conf\cloudbase-init.conf`

???????? ??????:

```ini
[DEFAULT]
username=Administrator
groups=Administrators
inject_user_password=true
config_drive_raw_hhd=true
config_drive_cdrom=true
config_drive_vfat=true
bsdtar_path=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\bin\bsdtar.exe
mtools_path=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\bin\
verbose=true
debug=true
logdir=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\log\
logfile=cloudbase-init.log
default_log_levels=comtypes=INFO,suds=INFO,iso8601=WARN,requests=WARN
local_scripts_path=C:\Program Files\Cloudbase Solutions\Cloudbase-Init\LocalScripts\

# ? ?????: Serial console logging
serial_log_port=COM1
log_serial_console_port=1

metadata_services=cloudbaseinit.metadata.services.configdrive.ConfigDriveService,
    cloudbaseinit.metadata.services.httpservice.HttpService,
    cloudbaseinit.metadata.services.ec2service.EC2Service

plugins=cloudbaseinit.plugins.common.mtu.MTUPlugin,
    cloudbaseinit.plugins.common.sethostname.SetHostNamePlugin,
    cloudbaseinit.plugins.windows.createuser.CreateUserPlugin,
    cloudbaseinit.plugins.windows.extendvolumes.ExtendVolumesPlugin,
```

```
    cloudbaseinit.plugins.common.userdata.UserDataPlugin,
    cloudbaseinit.plugins.common.localscripts.LocalScriptsPlugin
```

3.2 ???????? ?????????? ???????

???????? ???? ??? ?????? ????????? ? serial console:

`C:\Program Files\Cloudbase Solutions\Cloudbase-Init\LocalScripts\serial-logger.ps1`

```powershell
# Serial Console Logger for Cloudbase-Init
# ??????? ?????? ???????????? ? COM1

try {
    # ????????? COM1
    $ComPort = New-Object System.IO.Ports.SerialPort "COM1", 115200
    $ComPort.Open()

    # ??????? ????????? ????????
    $timestamp = Get-Date -Format "yyyy-MM-dd HH:mm:ss"
    $ComPort.WriteLine("[$timestamp] [Cloudbase-Init] Starting initialization...")
    $ComPort.WriteLine("[$timestamp] [Cloudbase-Init] Hostname: $env:COMPUTERNAME")
    $ComPort.WriteLine("[$timestamp] [Cloudbase-Init] User: $env:USERNAME")

    # ????????? ????
    $ComPort.Close()

    Write-Host "Serial console logger: OK"
} catch {
    Write-Host "Serial console logger failed: $_"
}
```

3.3 ???????? ?????????? ???????

`C:\Program Files\Cloudbase Solutions\Cloudbase-Init\LocalScripts\serial-complete.ps1`

```powershell
# Cloudbase-Init completion marker for serial console

try {
    $ComPort = New-Object System.IO.Ports.SerialPort "COM1", 115200
    $ComPort.Open()

    $timestamp = Get-Date -Format "yyyy-MM-dd HH:mm:ss"
    $ComPort.WriteLine("[$timestamp] [Cloudbase-Init] Execution of Cloudbase-Init is done")
    $ComPort.WriteLine("[$timestamp] [Cloudbase-Init] VM is ready for use")

    $ComPort.Close()
} catch {
    # ????????? ??????
}
```

?????: ????????? ??? `serial-complete.ps1` ?????????? **?????????** ? ?????? LocalScripts (????? ??????????? ?...

??? 4: ??????????? Serial Console

4.1 ???????? ???????? Windows VM

```bash
# ???????? Windows VM ????? Web UI ??? kubectl
# ????????? ??? ? ????????? ???? serial device (??. ??? 1)
```

4.2 ?????? VM

```bash
```

```
kubectl patch vm <vm-name> -n <namespace> --type merge -p '{"spec":{"running":true}}'
```

4.3 ???????? ????? serial console

```bash
# ??????? pod virt-launcher
kubectl get pods -n <namespace> -l kubevirt.io/domain=<vm-name>

# ??????? ???? ?? guest-console-log ??????????
kubectl logs -f -n <namespace> <virt-launcher-pod> -c guest-console-log
```

????????? ?????:

```
...
[2026-02-03 12:34:56] [Cloudbase-Init] Starting initialization...
[2026-02-03 12:34:56] [Cloudbase-Init] Hostname: WIN-ABC123
[2026-02-03 12:34:56] [Cloudbase-Init] User: Administrator
...
[2026-02-03 12:35:12] [Cloudbase-Init] Execution of Cloudbase-Init is done
[2026-02-03 12:35:12] [Cloudbase-Init] VM is ready for use
```

??? 5: ?????????? ? Web UI

Backend ??? ???????????? ?????????? Windows VM ? ???? ?????????????? ??????? ? ?????:

```python
# app/routes/vm_details_routes.py

# ?????????? ??? ??
vm_os_type = vm_obj.get("metadata", {}).get("labels", {}).get("vm.kubevirt.io/os", "linux")
is_windows = "windows" in vm_os_type.lower()

if is_windows:
    # ??? Windows ???? ???????? Cloudbase-Init
    if (
        "Cloudbase-Init complete" in console_log
        or "Execution of Cloudbase-Init is done" in console_log
        or "Cloudbase-Init finished" in console_log
    ):
        cloudinit_finished = True
        cloudinit_status = "completed"
        cloudinit_message = "?????? ? ??????"
```

? Web UI ????? ???????????:

- **????????...** - VM ????????, Cloudbase-Init ??? ?? ?????
- **????????????...** - Cloudbase-Init ????????
- **?????? ? ??????** ? - Cloudbase-Init ???????? ??????

Troubleshooting

???????? 1: ???? ?? ?????????? ? serial console

???????:

1. ???????? ??? serial device ???????? ? VM:
   ```bash
   kubectl get vmi <vm-name> -n <namespace> -o yaml | grep -A2 serials
   ```

2. ???????? ??? EMS ??????? ? Windows:
   ```powershell
   ```

```
   bcdedit /enum | findstr /i "ems"
   ```

3. ????????? ????????? Cloudbase-Init:
   ```powershell
   Get-Content "C:\Program Files\Cloudbase Solutions\Cloudbase-Init\conf\cloudbase-init.conf" | Select-String -Patter...
   ```

4. ??????? ???????????????? COM1:
   ```powershell
   $ComPort = New-Object System.IO.Ports.SerialPort "COM1", 115200
   $ComPort.Open()
   $ComPort.WriteLine("Test message")
   $ComPort.Close()

   # ????????? ? kubectl logs
   ```

???????? 2: "Access denied" ??? ???????? COM1

???????:

????????? ??? ??????? ??????????? ?? ????? SYSTEM (Cloudbase-Init ???????? ??? ??????):

```powershell
# ????????? ??????
Get-Service cloudbase-init

# ?????? ???? Running
```

???????? 3: Cloudbase-Init ?? ??????????? ????? Sysprep

???????:

????? sysprep ????????? ??? ?????? cloudbase-init ????????? ?? ??????????:

```powershell
Set-Service -Name cloudbase-init -StartupType Automatic
```

Golden Image Setup

??? ???????? Golden Image ??? ???????????:

1. ? ?????????? Cloudbase-Init
2. ? ????????? `cloudbase-init.conf` ? serial logging
3. ? ???????? LocalScripts (`serial-logger.ps1`, `serial-complete.ps1`)
4. ? ???????? EMS ????? `bcdedit`
5. ? ????????? Sysprep

????? ??????????? ?????? ????? VM ????? ???????????????? ???????? ???? ? serial console!

???????????

?? Serial Console:
- ? ?????????? ?????? ?????? ???????????????? Windows ??? VNC
- ? ??? ????? ? kubectl
- ? Web UI ?? ????? ???????? ???????? ????????

????? Serial Console:
- ? ???? Cloudbase-Init ???????? ????? `kubectl logs`
- ? Web UI ?????????? ??????: "????????..." ? "?????????????..." ? "?????? ? ??????"
- ? ??????? ??????? ??? VNC
- ? ??????????????? ?????? ? Linux VM (?????????? UX)

```
```

????????????? ??????????

????? ????????? ???????

????? ????????? ????? Windows Event Log ? serial console:

```powershell
# ???????? ?????? ??? ?????? ??????????? ???????
$Action = New-ScheduledTaskAction -Execute 'PowerShell.exe' -Argument '-File "C:\Scripts\event-logger.ps1"'
$Trigger = New-ScheduledTaskTrigger -AtLogon
Register-ScheduledTask -Action $Action -Trigger $Trigger -TaskName "SerialEventLogger" -User "SYSTEM"
```

`C:\Scripts\event-logger.ps1`:

```powershell
$ComPort = New-Object System.IO.Ports.SerialPort "COM1", 115200
$ComPort.Open()

Get-EventLog -LogName System -EntryType Error -Newest 5 | ForEach-Object {
    $ComPort.WriteLine("[System Error] $($_.Message)")
}

$ComPort.Close()
```

Boot Progress Monitoring

```powershell
# ? startup script
$ComPort = New-Object System.IO.Ports.SerialPort "COM1", 115200
$ComPort.Open()

$ComPort.WriteLine("[Boot] Windows is starting...")
$ComPort.WriteLine("[Boot] Loading services...")
# ... ? ?.?.

$ComPort.Close()
```

?????????

Serial Console ??? Windows VM ?????????:
- ? ???????????? ????????? Linux ? Windows VM
- ? ?????? ?????? ???????????? ? ???????? ???????
- ? ????????? ???????? ??? VNC
- ? ???????? UX ? Web UI

???????????? ??????????? ??? ???? production Windows VM!

[FILE] viirtuoz-vps-dev-rewrite-back-2/docs/WINDOWS-USAGE.md (5.5 KB)

```
# ? ???????????? Windows ? KubeVirt API Manager

## ??????? ?????

### 1. ????????? (???? ???)

????????? ??? Windows ISO DataVolume ?????? ? namespace `kvm`:

```bash
kubectl get dv windows-10-iso -n kvm

# ?????? ???? Succeeded
# NAME              PHASE       PROGRESS
# windows-10-iso    Succeeded   100.0%
```

?????: ISO ????????? ? ????? namespace `kvm`, ?? ??? ???????? VM ?? ???????????? ?????????? ? namespace ???????...

2. ???????? VM ??? ????????? Windows

1. ???????? ???????
2. ??????? **"??????? ?????????? ??????"**
3. ???????? **"Windows 10 (?????????)"** ????????
4. ????????? ???????:
   - CPU: 4 cores (????????????)
   - Memory: 8 GB (???????)
   - Storage: 60 GB (??????? ??? Windows 10)
   - GPU: ?????????? (???? ????? GPU ? golden image)
5. ??????? **"???????"**

3. ????????? Windows

????? ???????? VM:

```bash
# ????????? VM
kubectl patch vm <vm-name> -n <namespace> --type merge -p '{"spec":{"running":true}}'

# ??????????? ????? VNC
virtctl vnc <vm-name> -n <namespace>
```

? VNC:

1. ???????? ???? ? ?????
2. **?????**: ??????? **"Load driver"**
   - ???????? CD-ROM **E:** (VirtIO drivers)
   - ??????? `viostor\w10\amd64`
   - ????????? **Red Hat VirtIO SCSI**
3. ?????? ???? ????? ?????!
4. ???????? ???? ? ?????
5. ????????? ????????? (15-30 ?????)
6. ????????? ???????????

4. ????? ????????

1. **?????????? VirtIO ????????**:
   - Device Manager ? Update all unknown devices
   - ?????????? CD-ROM E: ? VirtIO drivers

2. **?????????? QEMU Guest Agent**:
   - ?? CD E: ??????? `guest-agent\qemu-ga-x86_64.msi`
   - ????????? ?????????

3. **????????? ???????**:
```

```
   - ?????????? ?????? ??
   - ????????? Windows Update
   - ????????? ?????????

4. **?????????? ? ??????????? (Sysprep)**:
   ```powershell
   # PowerShell ?? ?????????????
   C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown
   ```

5. **???????????? DataVolume ? golden image**:
   ```bash
   # ????? ?????????? VM ??????????? PVC
   kubectl get pvc -n <namespace>

   # ???????? snapshot ??? ?????????? PVC ? kvm ??? windows-10-golden-image
   ```

5. ???????????? ???????? ?????? (TODO)

????? ???????? golden image ?????? ?????? **"Windows 10"** ?????? ???????? ? ????? ?????????????? ??????? ???????.

??????? ??????

? **???????????:**
- ???????? VM ? ISO ??? ????????? Windows
- ????????????? ?????????????? VirtIO drivers
- Windows-?????????????? ?????????????? (Hyper-V enlightenments)
- ????????? GPU
- ??????? cloud-init ?????? ??? Windows

? **TODO (????????? ????):**
- ??????????? ?? golden image
- ???????????? ????? cloudbase-init
- ????????? Windows ????? KMS
- RDP ?????? ????? NodePort

?????????????? ??????

??? ????????? ??? Windows VM

- **Blank DataVolume**: ?????? ???? 60GB ??? ????????? Windows
- **Windows ISO DataVolume (????)**: ISO ?????????????? ?? `kvm` ? namespace ??????????? (7GB)
- **VirtIO Drivers**: ?????????????? ??? containerDisk (?? ???????? storage)
- **Machine Type**: q35 (?????????????? ??? Windows)
- **Hyper-V Features**: ??? ?????????????? ????????
- **Boot Order**: 1) ISO, 2) HDD

????? storage: ~67GB (60GB ???? + 7GB ISO) ??? ????? Windows VM ? ???????? ?????????

????? ???????????

- `app/config.py` - Windows settings (WINDOWS_* env vars)
- `app/utils/windows_utils.py` - `generate_windows_installer_vm_manifest()`, `generate_windows_golden_vm_manifest()`
- `app/routes/vm_routes.py` - ????????? `image_url == "windows-installer"` ? golden image
- `app/templates/partials/os_cards.html` - Windows ???????? ? UI

??????? ???????

???? ?? ????? ??? ?????????

? ????????? VirtIO ??????? viostor\w10\amd64

VM ?? ??????????????
```

```
? ????????? ??? Windows ISO DataVolume ??????????:
```bash
kubectl get dv windows-10-iso -n kvm
```

VNC ?? ???????????

? ????????? ???? VMI ??????????:
```bash
kubectl get vmi -n <namespace>
```

?????? ?????????????????

? ????????? ??? Hyper-V enlightenments ???????? (???????????? ? ????)

```

## [FILE] viirtuoz-vps-dev-rewrite-back-2/entrypoint.sh (851.0 B)

```
#!/bin/bash
set -euo pipefail

if [-d "/docker-entrypoint.d"]; then
  for f in /docker-entrypoint.d/*.sh; do
    [-f "$f"] && . "$f"
  done
fi

if [[-f "/etc/juniper-secret/id_rsa"]]; then
  mkdir -p /root/.ssh
  cp /etc/juniper-secret/id_rsa /root/.ssh/id_rsa
  cp /etc/juniper-secret/id_rsa.pub /root/.ssh/id_rsa.pub
  chmod 700 /root/.ssh
  chmod 600 /root/.ssh/id_rsa
  chmod 644 /root/.ssh/id_rsa.pub
fi

uv run uvicorn app.main:app \
  --host 127.0.0.1 \
  --port 8000 \
  --workers 1 \
  --log-level "${LOG_LEVEL:-info}" \
  --proxy-headers \
  --forwarded-allow-ips="*" &
APP_PID=$!

nginx -g "daemon off;" &
NGINX_PID=$!

cleanup() {
  kill -TERM "$APP_PID" "$NGINX_PID" 2>/dev/null || true
}
trap cleanup INT TERM

wait -n "$APP_PID" "$NGINX_PID"
EXIT_CODE=$?

cleanup
wait "$APP_PID" "$NGINX_PID" 2>/dev/null || true
exit "$EXIT_CODE"
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/entrypoint/90-oauth-proxy.sh (733.0 B)

```
#!/bin/sh
# Copyright 2025 Marcelo Parisi (github.com/feitnomore)
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

CONFIG_FILE="/etc/nginx/oauth.d/oauth2.conf"
if [ -f $CONFIG_FILE ]; then
    /bin/oauth2-proxy --config $CONFIG_FILE &
fi

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/entrypoint/91-startkubectl.sh (1.3 KB)

```
#!/bin/sh
# Copyright 2025 Marcelo Parisi (github.com/feitnomore)
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

# Point to the internal API server hostname
export APISERVER=https://kubernetes.default.svc

# Path to ServiceAccount token
export SERVICEACCOUNT=/var/run/secrets/kubernetes.io/serviceaccount

# Read this Pod's namespace
export NAMESPACE=$(cat ${SERVICEACCOUNT}/namespace)

# Read the ServiceAccount bearer token
export TOKEN=$(cat ${SERVICEACCOUNT}/token)

# Reference the internal certificate authority (CA)
export CACERT=${SERVICEACCOUNT}/ca.crt

# Start kubectl proxy in the background with API prefix
echo "Starting kubectl proxy..."
kubectl proxy --address='0.0.0.0' --accept-hosts='^.*$' --api-prefix=/k8s/ --port=8001 &

KUBECTL_PID=$!
echo "kubectl proxy started on port 8001 with /k8s/ prefix (PID: $KUBECTL_PID)"

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/env.example (3.9 KB)

```
# Example environment variables for kubevirt-api-manager
# Copy to .env and fill in real values. DO NOT commit .env!
# For docker-compose: cp env.example .env
# For Kubernetes: use Secrets or ConfigMap, see kubernetes_example/secrets-example.yaml

# Flask
SECRET_KEY=
DEBUG=

# Auth (demo mode, when DATABASE_URL is not set)
DEMO_USERNAME=
DEMO_PASSWORD=
DEMO_USER_ID=
SESSION_TIMEOUT=

# Database (PostgreSQL). Required for multi-user mode with admin/user roles
DATABASE_URL=

# Bootstrap: create first admin on startup when DB has no users
ADMIN_USERNAME=
ADMIN_PASSWORD=

# Kubernetes
K8S_NAMESPACE_PREFIX=
STORAGE_CLASS_NAME=

# Allowed image domains for VM creation (comma-separated hostnames)
ALLOWED_IMAGE_DOMAINS=cloud-images.ubuntu.com,download.fedoraproject.org,cloud.centos.org,download.rockylinux.org,cdi...

# VM scheduling: CDI importer pods (importer-prime-*) only on nodes with CSI
# JSON nodeSelector so the importer that creates/copies disks runs on nodes with the CSI driver (e.g. Ceph RBD).
# VMs themselves stay on KubeVirt nodes only. Example: {"csi-rbd":"ceph"}
VM_STORAGE_NODE_SELECTOR=
# Optional affinity for CDI importer pods (importer-prime-*).
# Use "kubevirt" to reuse kubevirt-node affinity, or provide full JSON affinity object.
VM_STORAGE_NODE_AFFINITY=

# GPU (Linux VM with GPU)
DEFAULT_GPU_RESOURCE=nvidia.com/1080ti
# Comma-separated list of GPU resources that have a separate HDMI audio device in KubeVirt (e.g. 1080 Ti). Do NOT inc...
GPU_RESOURCES_WITH_AUDIO=nvidia.com/1080ti,nvidia.com/3080ti,nvidia.com/3090
# Driver versions for cloud-init install: comma-separated; "auto" = ubuntu-drivers autoinstall, numeric = nvidia-driv...
NVIDIA_DRIVER_VERSIONS=auto,535,545,550,555
DEFAULT_NVIDIA_DRIVER=535

# VM resource limits (validation when creating VMs)
MAX_CPU_CORES=16
MAX_MEMORY_GB=64
MAX_STORAGE_GB=500
MAX_GPU_COUNT=2

# Custom NVIDIA driver: GPU resources that use custom install (your logic, e.g. Docker). Callback URL and token are s...
NVIDIA_GPU_RESOURCES_CUSTOM_DRIVER=nvidia.com/<allocated-name-1>,nvidia.com/<allocated-name-2>
NVIDIA_CUSTOM_DRIVER_CALLBACK_PATH=nvidia-custom-driver-installed
NVIDIA_CUSTOM_DRIVER_ANNOTATION_TOKEN_KEY=vm.kubevirt.io/nvidia-custom-driver-token
APP_INTERNAL_BASE_URL=

# VM Configuration
# VM_RUN_STRATEGY: Controls VM behavior on shutdown/reboot AND auto-start on creation
# Options: Manual (recommended - VM created stopped, requires manual start),
#          RerunOnFailure (VM auto-starts, restart on crash only),
#          Always (old - always restart), Halted (stopped)
VM_RUN_STRATEGY=Manual

# Windows VM (installer + golden image)
```

```
WINDOWS_ISO_DATAVOLUME=
WINDOWS_ISO_NAMESPACE=
WINDOWS_GOLDEN_IMAGE_NAME=
WINDOWS_GOLDEN_IMAGE_NAMESPACE=
WINDOWS_ISO_STORAGE_GB=
WINDOWS_DEFAULT_CPU=4
WINDOWS_DEFAULT_MEMORY=8
WINDOWS_DEFAULT_STORAGE=60
WINDOWS_MACHINE_TYPE=q35
WINDOWS_DEFAULT_USERNAME=Administrator
VIRTIO_DRIVERS_IMAGE=

# LLM API (legacy: ?????? ?????????? ? ??????)
LLM_API_URL=
# DEPRECATED: ???? ?? ?????, ???????????? ??????? ?? LLM_CHAT_SHARED_TOKEN
LLM_API_KEY=
LLM_MODEL=

# LLM per-token: llm-service URL. ?????? ?? SK-?????? ?? ?????? (? UI).
LLM_SERVICE_URL=
# ????? ????? ??? ??????? AI-??? (???????? ??????????? ?? ??????????).
LLM_CHAT_SHARED_TOKEN=
# ????? ?????? ?? ??????????? ?? ??????? «?????».
LLM_USER_KEYS_MAX=10
# ???? ????????? ??? ???????? API-?????? ? ?? (??????? ????????? ??????).
LLM_KEYS_ENCRYPTION_KEY=

# Juniper router (NAT / port forwarding)
JUNIPER_HOST=
JUNIPER_USER=
JUNIPER_SIDECAR_URL=
JUNIPER_EXTERNAL_IP_POOL=

# Network (for policies)
POD_NETWORK_CIDR=
SERVICE_NETWORK_CIDR=
SERVER_NETWORK_CIDRS=

# Stub / dev
FRONTEND_STUB_MODE=
STUB_NODE_IP=

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/bundled.yaml (19.8 KB)

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: kvm
  labels:
    name: kvm
---
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
  name: kubevirt-manager-priority
value: 1000000
globalDefault: false
description: Priority class for kubevirt-manager
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: kubevirt-api-manager-multus
  namespace: kvm
---
# ClusterRole for minimal cluster-wide permissions (only nodes and GPU discovery)
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: kubevirt-api-manager-cluster-role-multus
rules:
  - apiGroups: [""]
    resources: ["nodes"]
    verbs: ["get", "list"]
  - apiGroups: [""]
    resources: ["namespaces"]
    verbs: ["get", "list", "watch"]
  - apiGroups: ["metrics.k8s.io"]
    resources: ["pods"]
    verbs: ["get", "list"]
---
# ClusterRoleBinding for minimal cluster-wide access
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: kubevirt-api-manager-cluster-binding-multus
subjects:
  - kind: ServiceAccount
    name: kubevirt-api-manager-multus
    namespace: kvm
roleRef:
  kind: ClusterRole
  name: kubevirt-api-manager-cluster-role-multus
  apiGroup: rbac.authorization.k8s.io
---
# Role for namespace-specific permissions (kv-* namespaces)
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: kubevirt-api-manager-user-ns-role-multus
rules:
  - apiGroups: ["kubevirt.io"]
    resources:
      - virtualmachines
      - virtualmachineinstances
      - virtualmachineinstancepresets
      - virtualmachineinstancereplicasets
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["subresources.kubevirt.io"]
```

```
    resources:
      ["virtualmachineinstances/vnc", "virtualmachineinstances/console"]
    verbs: ["get"]
  - apiGroups: ["subresources.kubevirt.io"]
    resources:
      ["virtualmachines/start", "virtualmachines/stop", "virtualmachines/restart"]
    verbs: ["update"]
  - apiGroups: ["cdi.kubevirt.io"]
    resources: ["datavolumes"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: [""]
    resources:
      ["pods", "endpoints", "persistentvolumeclaims", "events", "configmaps"]
    verbs: ["get", "list", "watch"]
  - apiGroups: [""]
    resources: ["services"]
    verbs: ["get", "list", "watch", "create", "delete", "update", "patch"]
  - apiGroups: [""]
    resources: ["secrets"]
    verbs: ["get", "list", "watch", "create", "delete"]
  - apiGroups: [""]
    resources: ["persistentvolumeclaims"]
    verbs: ["delete"]
  - apiGroups: [""]
    resources: ["pods/log"]
    verbs: ["get"]
  - apiGroups: [""]
    resources: ["namespaces"]
    verbs: ["create"]
  - apiGroups: ["k8s.cni.cncf.io"]
    resources: ["network-attachment-definitions"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["crd.projectcalico.org"]
    resources: ["ippools"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["crd.projectcalico.org"]
    resources: ["globalnetworkpolicies"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["networking.k8s.io"]
    resources: ["networkpolicies"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
---
# ClusterRoleBinding for user namespace access (kv-* pattern)
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: kubevirt-api-manager-user-ns-binding-multus
subjects:
  - kind: ServiceAccount
    name: kubevirt-api-manager-multus
    namespace: kvm
roleRef:
  kind: ClusterRole
  name: kubevirt-api-manager-user-ns-role-multus
  apiGroup: rbac.authorization.k8s.io
---
# Role for Windows DataVolume cloning (required for Windows Golden Image / ISO)
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: windows-iso-cloner
  namespace: kvm
rules:
  - apiGroups: ["cdi.kubevirt.io"]
    resources: ["datavolumes/source"]
    verbs: ["create"]
  - apiGroups: ["cdi.kubevirt.io"]
    resources: ["datavolumes"]
    verbs: ["get"]
    resourceNames: ["windows-10-iso", "windows-10-golden-image"]
  - apiGroups: [""]
```

```
    resources: ["persistentvolumeclaims"]
    verbs: ["get"]
    resourceNames: ["windows-10-iso", "windows-10-golden-image"]
---
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: windows-iso-cloner-binding
  namespace: kvm
subjects:
  - kind: Group
    name: system:serviceaccounts
    apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: windows-iso-cloner
  apiGroup: rbac.authorization.k8s.io
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: juniper-sidecar-code
  namespace: kvm
data:
  juniper_sidecar.py: |
    import logging
    import subprocess
    import os
    from flask import Flask, request, jsonify

    MAX_LEN = 31
    SSH_CONTROL = "/tmp/juniper-ssh-%r@%h:%p"

    def _juniper_name(vm_name, namespace, service_name, port):
        ns = (namespace or "").replace("kv-", "").replace("-", "")[:6] or "x"
        vm = (vm_name or "x")[:8]
        svc = (service_name or "x")[:4]
        name = f"kv_{vm}_{ns}_{svc}_{port}"
        return name[:MAX_LEN] if len(name) > MAX_LEN else name

    # Configure logging
    logging.basicConfig(
        level=logging.DEBUG,
        format='%(asctime)s [%(levelname)s] %(name)s: %(message)s'
    )
    logger = logging.getLogger("juniper-sidecar")

    app = Flask(__name__)

    # Configuration (set via env in StatefulSet; defaults are placeholders)
    JUNIPER_HOST = os.getenv('JUNIPER_HOST', 'CHANGE-ME')
    JUNIPER_USER = os.getenv('JUNIPER_USER', 'CHANGE-ME')
    JUNIPER_EXTERNAL_IP = os.getenv('JUNIPER_EXTERNAL_IP', 'CHANGE-ME')
    SSH_KEY_PATH = "/root/.ssh/id_rsa"

    def execute_ssh_command(commands):
        """Execute commands on Juniper via SSH"""
        cmd_string = "; ".join(commands)
        full_cmd_string = f"edit; {cmd_string}; show | compare; commit; quit"

        ssh_command = [
            "ssh",
            "-i", SSH_KEY_PATH,
            "-o", "StrictHostKeyChecking=no",
            "-o", "UserKnownHostsFile=/dev/null",
            "-o", "PubkeyAcceptedAlgorithms=+ssh-rsa",
            "-o", "HostKeyAlgorithms=+ssh-rsa",
            "-o", "ControlMaster=auto",
            "-o", f"ControlPath={SSH_CONTROL}",
            "-o", "ControlPersist=60",
            f"{JUNIPER_USER}@{JUNIPER_HOST}",
```

```
            full_cmd_string
        ]

        logger.info(f"Connecting to {JUNIPER_USER}@{JUNIPER_HOST}...")
        logger.info(f"Commands to execute:\n{full_cmd_string}")

        try:
            # Debug: Check key permissions before running
            subprocess.run(["ls", "-l", SSH_KEY_PATH], check=False)

            result = subprocess.run(
                ssh_command,
                capture_output=True,
                text=True,
                timeout=45
            )

            if result.returncode != 0:
                logger.error("---------------- JUNIPER SSH ERROR ----------------")
                logger.error(f"Return Code: {result.returncode}")
                logger.error(f"STDERR:\n{result.stderr}")
                logger.error(f"STDOUT:\n{result.stdout}")
                logger.error("--------------------------------------------------")
                return False, result.stderr

            logger.info("---------------- JUNIPER SUCCESS ----------------")
            logger.info(f"STDOUT:\n{result.stdout}")
            logger.info("-------------------------------------------------")
            return True, result.stdout

        except Exception as e:
            logger.exception("Exception during SSH execution")
            return False, str(e)

    @app.route('/health', methods=['GET'])
    def health():
        return jsonify({"status": "ok"}), 200

    @app.route('/add_rule', methods=['POST'])
    def add_rule():
        data = request.json
        vm_name = data.get('vm_name')
        dest_ip = data.get('dest_ip')
        port = data.get('port')
        service_name = data.get('service_name', 'unknown')
        namespace = data.get('namespace', 'unknown')
        external_ip = data.get('external_ip') or JUNIPER_EXTERNAL_IP

        if not all([vm_name, dest_ip, port]):
            return jsonify({"error": "Missing required fields"}), 400

        common_name = _juniper_name(vm_name, namespace, service_name, port)

        dst_nat_pool_name = common_name
        rule_name = common_name
        policy_name = common_name
        app_name = common_name

        commands = [
            f"set security nat destination pool {dst_nat_pool_name} address {dest_ip}/32 port {port}",
            f"set security nat destination rule-set dst-nat rule {rule_name} match destination-address {external_ip}/...
            f"set security nat destination rule-set dst-nat rule {rule_name} match destination-port {port}",
            f"set security nat destination rule-set dst-nat rule {rule_name} then destination-nat pool {dst_nat_pool_...
            f"set security zones security-zone trust address-book address {common_name} {dest_ip}/32",
            f"set applications application {app_name} protocol tcp destination-port {port}",
            f"set security policies from-zone untrust to-zone trust policy {policy_name} match source-address any",
            f"set security policies from-zone untrust to-zone trust policy {policy_name} match destination-address {c...
            f"set security policies from-zone untrust to-zone trust policy {policy_name} match application {app_name}",
            f"set security policies from-zone untrust to-zone trust policy {policy_name} then permit"
        ]

```

```
        success, msg = execute_ssh_command(commands)
        if success:
            return jsonify({"status": "success", "output": msg}), 200
        else:
            return jsonify({"status": "error", "error": msg}), 500

    @app.route('/delete_rule', methods=['POST'])
    def delete_rule():
        data = request.json
        vm_name = data.get('vm_name')
        port = data.get('port')
        service_name = data.get('service_name', 'unknown')
        namespace = data.get('namespace', 'unknown')

        if not all([vm_name, port]):
            return jsonify({"error": "Missing required fields"}), 400

        common_name = _juniper_name(vm_name, namespace, service_name, port)

        commands = [
            f"delete security nat destination pool {common_name}",
            f"delete security nat destination rule-set dst-nat rule {common_name}",
            f"delete security zones security-zone trust address-book address {common_name}",
            f"delete applications application {common_name}",
            f"delete security policies from-zone untrust to-zone trust policy {common_name}"
        ]

        success, msg = execute_ssh_command(commands)
        # Success even if it fails because it might not exist
        return jsonify({"status": "success", "output": msg}), 200

    if __name__ == '__main__':
        print("Starting Juniper Sidecar on 5001...")
        app.run(host='0.0.0.0', port=5001)
---
apiVersion: v1
kind: Service
metadata:
  name: kubevirt-api-manager
  namespace: kvm
  labels:
    app: kubevirt-api-manager
spec:
  type: NodePort
  ports:
    - name: http
      port: 80
      targetPort: 8080
      nodePort: 30180
  selector:
    app: kubevirt-api-manager
---
# Login path - NO Basic Auth (so session-expired redirect works)
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: kubevirt-manager-ingress-login
  namespace: kvm
  annotations:
    cert-manager.io/cluster-issuer: letsencrypt-prod
    nginx.ingress.kubernetes.io/ssl-redirect: "false"
spec:
  ingressClassName: nginx
  tls:
    - hosts:
        - CHANGE-ME.example.com
      secretName: kubevirt-manager-tls
  rules:
    - host: CHANGE-ME.example.com
      http:
        paths:
```

```yaml
          - path: /login
            pathType: Prefix
            backend:
              service:
                name: kubevirt-api-manager
                port:
                  number: 80
---
# All other paths - WITH Basic Auth
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: kubevirt-manager-ingress
  namespace: kvm
  annotations:
    cert-manager.io/cluster-issuer: letsencrypt-prod
    external-dns.alpha.kubernetes.io/hostname: "CHANGE-ME.example.com"
    external-dns.alpha.kubernetes.io/target: "CHANGE-ME"
    external-dns.alpha.kubernetes.io/ttl: "3600"
    nginx.ingress.kubernetes.io/ssl-redirect: "false"
    nginx.ingress.kubernetes.io/auth-realm: "Enter your credentials"
    nginx.ingress.kubernetes.io/auth-secret: "htpasswd"
    nginx.ingress.kubernetes.io/auth-type: "basic"
spec:
  ingressClassName: nginx
  tls:
    - hosts:
        - CHANGE-ME.example.com
      secretName: kubevirt-manager-tls
  rules:
    - host: CHANGE-ME.example.com
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: kubevirt-api-manager
                port:
                  number: 80
---
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  name: kubevirt-manager-monitor
  namespace: kvm
  labels:
    app: kubevirt-api-manager
spec:
  selector:
    matchLabels:
      app: kubevirt-api-manager
  endpoints:
    - port: http
      interval: 30s
  namespaceSelector:
    matchNames:
      - kvm
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kubevirt-api-manager
  namespace: kvm
  labels:
    app: kubevirt-api-manager
spec:
  replicas: 1
  selector:
    matchLabels:
      app: kubevirt-api-manager
```

```yaml
  template:
    metadata:
      labels:
        app: kubevirt-api-manager
    spec:
      affinity:
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
              - matchExpressions:
                  - key: node-role.kubernetes.io/control-plane
                    operator: Exists
                  - key: node-role.kubernetes.io/gpu-worker
                    operator: DoesNotExist
              - matchExpressions:
                  - key: node-role.kubernetes.io/master
                    operator: Exists
                  - key: node-role.kubernetes.io/gpu-worker
                    operator: DoesNotExist
              - matchExpressions:
                  - key: node-role.kubernetes.io/worker
                    operator: Exists
                  - key: node-role.kubernetes.io/gpu-worker
                    operator: DoesNotExist
      tolerations:
        - key: node-role.kubernetes.io/master
          operator: Exists
          effect: NoSchedule
        - key: node-role.kubernetes.io/control-plane
          operator: Exists
          effect: NoSchedule
        - key: cephfs
          operator: Exists
          effect: NoSchedule
        - key: data
          operator: Exists
          effect: NoSchedule
      serviceAccountName: kubevirt-api-manager-multus
      hostAliases:
        - ip: "CHANGE-ME"
          hostnames:
            - "llm.CHANGE-ME"
      containers:
        - name: manager
          image: your-registry.example.com/kubevirt/kv_api_manager:1.0-rc.0.0.1
          imagePullPolicy: Always
          ports:
            - containerPort: 8080
              name: http
          env:
            - name: FLASK_ENV
              value: "production"
            - name: DEBUG
              value: "false"
            - name: LOG_LEVEL
              value: "info"
            - name: JUNIPER_USER
              value: "CHANGE-ME"
            - name: JUNIPER_HOST
              value: "CHANGE-ME"
            - name: JUNIPER_EXTERNAL_IP
              value: "CHANGE-ME"
            - name: STORAGE_CLASS_NAME
              value: "your-SC"
            # VM_STORAGE_NODE_SELECTOR: JSON nodeSelector so VMs run only on nodes with CSI (e.g. Ceph RBD). Example:...
            - name: VM_STORAGE_NODE_SELECTOR
              value: ""
            - name: SECRET_KEY
              valueFrom:
                secretKeyRef:
                  name: kubevirt-api-secrets
```

```
492                   key: secret-key
493                   optional: true
494             - name: DEMO_USERNAME
495               valueFrom:
496                 secretKeyRef:
497                   name: kubevirt-api-secrets
498                   key: demo-username
499                   optional: true
500             - name: DEMO_PASSWORD
501
502 ... [128 more lines truncated]
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/deployment.yaml (5.0 KB)

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kubevirt-api-manager
  namespace: kvm
  labels:
    app: kubevirt-api-manager
spec:
  replicas: 1
  selector:
    matchLabels:
      app: kubevirt-api-manager
  template:
    metadata:
      labels:
        app: kubevirt-api-manager
    spec:
      affinity:
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
            - matchExpressions:
              - key: node-role.kubernetes.io/control-plane
                operator: Exists
              - key: node-role.kubernetes.io/gpu-worker
                operator: DoesNotExist
            - matchExpressions:
              - key: node-role.kubernetes.io/master
                operator: Exists
              - key: node-role.kubernetes.io/gpu-worker
                operator: DoesNotExist
            - matchExpressions:
              - key: node-role.kubernetes.io/worker
                operator: Exists
              - key: node-role.kubernetes.io/gpu-worker
                operator: DoesNotExist
      tolerations:
      - key: node-role.kubernetes.io/master
        operator: Exists
        effect: NoSchedule
      - key: node-role.kubernetes.io/control-plane
        operator: Exists
        effect: NoSchedule
      - key: cephfs
        operator: Exists
        effect: NoSchedule
      - key: data
        operator: Exists
        effect: NoSchedule
      serviceAccountName: kubevirt-api-manager-multus
      hostAliases:
        - ip: "CHANGE-ME"
          hostnames:
            - "llm.CHANGE-ME"
      containers:
        - name: manager
          image: your-registry.example.com/kubevirt/kv_api_manager:1.0-rc.0.0.1
          imagePullPolicy: Always
          ports:
            - containerPort: 8080
              name: http
          env:
            - name: FLASK_ENV
              value: "production"
            - name: DEBUG
```

```
              value: "false"
            - name: LOG_LEVEL
              value: "info"
            - name: JUNIPER_USER
              value: "CHANGE-ME"
            - name: JUNIPER_HOST
              value: "CHANGE-ME"
            - name: STORAGE_CLASS_NAME
              value: "your-SC"
            - name: VM_STORAGE_NODE_SELECTOR
              value: ""
            - name: SECRET_KEY
              valueFrom:
                secretKeyRef:
                  name: kubevirt-api-secrets
                  key: secret-key
                  optional: true
            - name: DEMO_USERNAME
              valueFrom:
                secretKeyRef:
                  name: kubevirt-api-secrets
                  key: demo-username
                  optional: true
            - name: DEMO_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: kubevirt-api-secrets
                  key: demo-password
                  optional: true
          livenessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 30
            periodSeconds: 10
            timeoutSeconds: 5
            failureThreshold: 3
          readinessProbe:
            httpGet:
              path: /ready
              port: 8080
            initialDelaySeconds: 10
            periodSeconds: 5
            timeoutSeconds: 3
            failureThreshold: 3
          resources:
            requests:
              cpu: "1000m"
              memory: "1024Mi"
            limits:
              cpu: "2000m"
              memory: "2Gi"
          volumeMounts:
            - name: juniper-ssh-key
              mountPath: /etc/juniper-secret
              readOnly: true
            - name: log-volume
              mountPath: /app/logs
        - name: juniper-sidecar
          image: your-registry.example.com/kubevirt/kv_api_manager:1.0-rc.0.0.1
          command: ["/bin/bash", "-c"]
          args:
            - |
              mkdir -p /root/.ssh
              cp /etc/juniper-secret/id_rsa /root/.ssh/id_rsa
              chmod 600 /root/.ssh/id_rsa
              echo "Starting Sidecar..."
              python /app/sidecar/juniper_sidecar.py
          env:
            - name: JUNIPER_USER
              value: "CHANGE-ME"
```

```yaml
            - name: JUNIPER_HOST
              value: "CHANGE-ME"
            - name: JUNIPER_EXTERNAL_IP
              value: "CHANGE-ME"
          volumeMounts:
            - name: juniper-sidecar-code
              mountPath: /app/sidecar
            - name: juniper-ssh-key
              mountPath: /etc/juniper-secret
              readOnly: true
            - name: log-volume
              mountPath: /app/logs
        - name: dnat-logs
          image: busybox
          args: [/bin/sh, -c, 'mkdir -p /app/logs && touch /app/logs/dnat.log && tail -f /app/logs/dnat.log']
          volumeMounts:
            - name: log-volume
              mountPath: /app/logs
      volumes:
        - name: juniper-sidecar-code
          configMap:
            name: juniper-sidecar-code
        - name: log-volume
          emptyDir: {}
        - name: juniper-ssh-key
          secret:
            secretName: juniper-ssh-key
            defaultMode: 0400

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/global-network-policy.yaml (1.4 KB)

```yaml
---
# GlobalNetworkPolicy to deny VM pods access to internal networks
# This policy applies to all VM pods (pods with kubevirt.io/domain label)
# across all namespaces

apiVersion: crd.projectcalico.org/v1
kind: GlobalNetworkPolicy
metadata:
  name: vm-deny-internal-networks
  labels:
    app: kubevirt-api-manager
    policy-type: vm-network-restriction
spec:
  # Select VM pods only (pods with kubevirt.io/domain label)
  selector: has(kubevirt.io/domain)

  # Lower order = higher priority
  order: 1000

  types:
    - Ingress
    - Egress

  ingress:
    # Allow all incoming traffic to VMs
    - action: Allow

  egress:
    # Deny egress to Pod network (default: 10.244.0.0/16)
    # Update POD_NETWORK_CIDR in config.py if your pod network is different
    - action: Deny
      destination:
        nets:
          - "192.168.192.0/18"  # Pod network CIDR

    # Deny egress to Service network (default: 10.96.0.0/12)
    # Update SERVICE_NETWORK_CIDR in config.py if your service network is different
    - action: Deny
      destination:
        nets:
          - "10.96.0.0/18"  # Service network CIDR

    # Deny egress to server networks
    # Update SERVER_NETWORK_CIDRS in config.py with your server network CIDRs
    - action: Deny
      destination:
        nets:
          - "CHANGE-ME"  # Example server network CIDR

    # Allow all other egress traffic
    - action: Allow

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/golden-Images/windows10/datavolume.yaml (505.0 B)

```yaml
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-iso
  namespace: kvm  # Match WINDOWS_ISO_NAMESPACE from config
spec:
  source:
    # SECURITY: Do NOT commit presigned URLs with credentials. Use one of:
    # - upload: {} then virtctl image-upload
    # - http: { url: "YOUR_ISO_URL" } (set via env/kustomize)
    upload: {}
  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 7Gi
    storageClassName: your-SC

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/golden-Images/windows10/windows-golden-image-dur

```yaml
---
# Windows 10 Golden Image DataVolume
# ???? DataVolume ????????? ???? ??? ? ???????? ???????????? Windows 10
# ??? ???????? ????? VM ?? ????? ???????????? ? ????? PVC

apiVersion: v1
kind: Namespace
metadata:
  name: kvm
  labels:
    name: kvm
    purpose: golden-images

---
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-golden-image
  namespace: kvm
  labels:
    app: kubevirt
    os: windows-10
    type: golden-image
  annotations:
    cdi.kubevirt.io/storage.bind.immediate.requested: "true"
    description: "Windows 10 LTSC 2021 - Golden Image for cloning"
spec:
  source:
    # ??????? 1: ????????? ??????? qcow2 ????? (???????????? ???? ????)
    # http:
    #   url: "https://your-storage.example.com/windows-10-ltsc-2021.qcow2"

    # ??????? 2: ??????? ?????? ???? ??? ????????? ? ISO
    blank: {}

  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 60Gi  # Windows 10 ??????? ??????? 40-60GB
    storageClassName: your-SC
    volumeMode: Filesystem

---
# ????????? VM ??? ????????? Windows (?????????? ???? source: blank)
# ????? ????????? Windows ??????? ??? VM, ?? ???????? DataVolume
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: windows-10-installer
  namespace: kvm
  labels:
    app: windows-installer
    temporary: "true"
spec:
  running: false  # ????????? ???????: kubectl patch vm windows-10-installer -n kvm --type merge -p '{"spec":{"runnin...
  template:
    metadata:
      labels:
        kubevirt.io/vm: windows-10-installer
    spec:
      domain:
        cpu:
          cores: 4
```

```
          sockets: 1
          threads: 1
        devices:
          disks:
            # ???????? ???? ??? ????????? Windows
            - name: rootdisk
              disk:
                bus: virtio
              bootOrder: 2

            # Windows ISO (????? ??????? ????????? DataVolume ? ISO)
            - name: windows-iso
              cdrom:
                bus: sata
              bootOrder: 1

            # VirtIO ???????? ??? Windows
            - name: virtio-drivers
              cdrom:
                bus: sata

          interfaces:
            - name: default
              masquerade: {}

        machine:
          type: q35  # ??????????? ??? Windows

        resources:
          requests:
            memory: 8Gi
            cpu: 4

        # Windows ??????????? (Hyper-V enlightenments)
        features:
          acpi:
            enabled: true
          apic: {}
          hyperv:
            relaxed:
              enabled: true
            vapic:
              enabled: true
            spinlocks:
              enabled: true
              spinlocks: 8191
            vpindex:
              enabled: true
            runtime:
              enabled: true
            synic:
              enabled: true
            stimer:
              enabled: true
            reset:
              enabled: true
            frequencies:
              enabled: true
            reenlightenment:
              enabled: true
            tlbflush:
              enabled: true
            ipi:
              enabled: true

      networks:
        - name: default
          pod: {}

      volumes:
        # ???????? ???? - ?????? ?? golden image DataVolume
```

```yaml
        - name: rootdisk
          dataVolume:
            name: windows-10-golden-image

        # Windows ISO - ????? ??????? ???????? (??. ????)
        - name: windows-iso
          persistentVolumeClaim:
            claimName: windows-10-iso

        # VirtIO ???????? ?? containerDisk
        - name: virtio-drivers
          containerDisk:
            image: quay.io/kubevirt/virtio-container-disk:latest

---
# ?????: ???????? ???? DataVolume ????? ??????????? Windows
# ????????? Windows 10 ISO ? ??????????? ????? Microsoft
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: windows-10-iso
  namespace: kvm
  labels:
    app: kubevirt
    type: installation-media
spec:
  source:
    # ??????? URL ? Windows 10 ISO (6-7GB)
    http:
      url: "https://software-download.microsoft.com/download/pr/..."  # ???????? ?? ???????? URL

    # ??? ??????????? upload ??? ???????? ? ????????? ??????:
    # upload: {}

  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 7Gi
    storageClassName: your-SC

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/ingress.yaml (1.7 KB)

```yaml
# Login path - NO Basic Auth (so session-expired redirect works)
# Must be more specific than / so it matches first
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: kubevirt-manager-ingress-login
  namespace: kvm
  annotations:
    cert-manager.io/cluster-issuer: letsencrypt-prod
    nginx.ingress.kubernetes.io/ssl-redirect: "false"
spec:
  ingressClassName: nginx
  tls:
    - hosts:
        - CHANGE-ME.example.com
      secretName: kubevirt-manager-tls
  rules:
    - host: CHANGE-ME.example.com
      http:
        paths:
          - path: /login
            pathType: Prefix
            backend:
              service:
                name: kubevirt-api-manager
                port:
                  number: 80
---
# All other paths - WITH Basic Auth
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: kubevirt-manager-ingress
  namespace: kvm
  annotations:
    cert-manager.io/cluster-issuer: letsencrypt-prod
    external-dns.alpha.kubernetes.io/hostname: 'CHANGE-ME.example.com'
    external-dns.alpha.kubernetes.io/target: 'CHANGE-ME'
    external-dns.alpha.kubernetes.io/ttl: '3600'
    nginx.ingress.kubernetes.io/ssl-redirect: "false"
    nginx.ingress.kubernetes.io/auth-realm: 'Enter your credentials'
    nginx.ingress.kubernetes.io/auth-secret: 'htpasswd'
    nginx.ingress.kubernetes.io/auth-type: 'basic'
spec:
  ingressClassName: nginx
  tls:
    - hosts:
        - CHANGE-ME.example.com
      secretName: kubevirt-manager-tls
  rules:
    - host: CHANGE-ME.example.com
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: kubevirt-api-manager
                port:
                  number: 80

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/juniper-sidecar-cm.yaml (6.7 KB)

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: juniper-sidecar-code
  namespace: kvm
data:
  juniper_sidecar.py: |
    import logging
    import subprocess
    import os
    from flask import Flask, request, jsonify

    MAX_LEN = 31  # Juniper identifier limit
    SSH_CONTROL = "/tmp/juniper-ssh-%r@%h:%p"  # Connection multiplexing

    def _juniper_name(vm_name, namespace, service_name, port):
        """Short readable name: kv_{vm8}_{ns6}_{svc}_{port}."""
        ns = (namespace or "").replace("kv-", "").replace("-", "")[:6] or "x"
        vm = (vm_name or "x")[:8]
        svc = (service_name or "x")[:4]
        name = f"kv_{vm}_{ns}_{svc}_{port}"
        return name[:MAX_LEN] if len(name) > MAX_LEN else name

    # Configure logging
    logging.basicConfig(
        level=logging.DEBUG,
        format='%(asctime)s [%(levelname)s] %(name)s: %(message)s'
    )
    logger = logging.getLogger("juniper-sidecar")

    app = Flask(__name__)

    # Configuration (set via env in StatefulSet; defaults are placeholders)
    JUNIPER_HOST = os.getenv('JUNIPER_HOST', 'CHANGE-ME')
    JUNIPER_USER = os.getenv('JUNIPER_USER', 'CHANGE-ME')
    JUNIPER_EXTERNAL_IP = os.getenv('JUNIPER_EXTERNAL_IP', 'CHANGE-ME')
    SSH_KEY_PATH = "/root/.ssh/id_rsa"

    def execute_ssh_command(commands):
        """Execute commands on Juniper via SSH"""
        cmd_string = "; ".join(commands)
        full_cmd_string = f"edit; {cmd_string}; show | compare; commit; quit"

        ssh_command = [
            "ssh",
            "-i", SSH_KEY_PATH,
            "-o", "StrictHostKeyChecking=no",
            "-o", "UserKnownHostsFile=/dev/null",
            "-o", "PubkeyAcceptedAlgorithms=+ssh-rsa",
            "-o", "HostKeyAlgorithms=+ssh-rsa",
            "-o", "ControlMaster=auto",
            "-o", f"ControlPath={SSH_CONTROL}",
            "-o", "ControlPersist=60",
            f"{JUNIPER_USER}@{JUNIPER_HOST}",
            full_cmd_string
        ]

        logger.info(f"Connecting to {JUNIPER_USER}@{JUNIPER_HOST}...")
        logger.info(f"Commands to execute:\n{full_cmd_string}")

        try:
            # Debug: Check key permissions before running
            subprocess.run(["ls", "-l", SSH_KEY_PATH], check=False)

            result = subprocess.run(
```

```
                ssh_command,
                capture_output=True,
                text=True,
                timeout=45
            )

            if result.returncode != 0:
                logger.error("---------------- JUNIPER SSH ERROR ----------------")
                logger.error(f"Return Code: {result.returncode}")
                logger.error(f"STDERR:\n{result.stderr}")
                logger.error(f"STDOUT:\n{result.stdout}")
                logger.error("--------------------------------------------------")
                return False, result.stderr

            logger.info("---------------- JUNIPER SUCCESS ----------------")
            logger.info(f"STDOUT:\n{result.stdout}")
            logger.info("--------------------------------------------------")
            return True, result.stdout

        except Exception as e:
            logger.exception("Exception during SSH execution")
            return False, str(e)

    @app.route('/health', methods=['GET'])
    def health():
        return jsonify({"status": "ok"}), 200

    @app.route('/add_rule', methods=['POST'])
    def add_rule():
        data = request.json
        vm_name = data.get('vm_name')
        dest_ip = data.get('dest_ip')
        port = data.get('port')
        service_name = data.get('service_name', 'unknown')
        namespace = data.get('namespace', 'unknown')
        external_ip = data.get('external_ip') or JUNIPER_EXTERNAL_IP

        if not all([vm_name, dest_ip, port]):
            return jsonify({"error": "Missing required fields"}), 400

        # Juniper limit 31 chars; use readable name or hash
        common_name = _juniper_name(vm_name, namespace, service_name, port)

        dst_nat_pool_name = common_name
        rule_name = common_name
        policy_name = common_name
        app_name = common_name

        commands = [
            f"set security nat destination pool {dst_nat_pool_name} address {dest_ip}/32 port {port}",
            f"set security nat destination rule-set dst-nat rule {rule_name} match destination-address {external_ip}/...
            f"set security nat destination rule-set dst-nat rule {rule_name} match destination-port {port}",
            f"set security nat destination rule-set dst-nat rule {rule_name} then destination-nat pool {dst_nat_pool_...
            f"set security zones security-zone trust address-book address {common_name} {dest_ip}/32",
            f"set applications application {app_name} protocol tcp destination-port {port}",
            f"set security policies from-zone untrust to-zone trust policy {policy_name} match source-address any",
            f"set security policies from-zone untrust to-zone trust policy {policy_name} match destination-address {c...
            f"set security policies from-zone untrust to-zone trust policy {policy_name} match application {app_name}",
            f"set security policies from-zone untrust to-zone trust policy {policy_name} then permit"
        ]

        success, msg = execute_ssh_command(commands)
        if success:
            return jsonify({"status": "success", "output": msg}), 200
        else:
            return jsonify({"status": "error", "error": msg}), 500

    @app.route('/delete_rule', methods=['POST'])
    def delete_rule():
        data = request.json
        vm_name = data.get('vm_name')
```

```
        port = data.get('port')
        service_name = data.get('service_name', 'unknown')
        namespace = data.get('namespace', 'unknown')

        if not all([vm_name, port]):
            return jsonify({"error": "Missing required fields"}), 400

        # Same hash as add_rule
        common_name = _juniper_name(vm_name, namespace, service_name, port)

        commands = [
            f"delete security nat destination pool {common_name}",
            f"delete security nat destination rule-set dst-nat rule {common_name}",
            f"delete security zones security-zone trust address-book address {common_name}",
            f"delete applications application {common_name}",
            f"delete security policies from-zone untrust to-zone trust policy {common_name}"
        ]

        success, msg = execute_ssh_command(commands)
        # Success even if it fails because it might not exist
        return jsonify({"status": "success", "output": msg}), 200

    if __name__ == '__main__':
        print("Starting Juniper Sidecar on 5001...")
        app.run(host='0.0.0.0', port=5001)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/kvm-clone-rbac.yaml (1.3 KB)

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: windows-iso-cloner
  namespace: kvm
rules:
  # CDI ??????? ????? 'create' ?? ?????? 'datavolumes/source' ? ???????? namespace ??? ???????????
  - apiGroups: ["cdi.kubevirt.io"]
    resources: ["datavolumes/source"]
    verbs: ["create"]
  # ????? ?? ?????? ?????????? DataVolumes
  - apiGroups: ["cdi.kubevirt.io"]
    resources: ["datavolumes"]
    verbs: ["get"]
    resourceNames: ["windows-10-iso", "windows-10-golden-image"]
  # ????? ?? ?????? ?????????????? PVC
  - apiGroups: [""]
    resources: ["persistentvolumeclaims"]
    verbs: ["get"]
    resourceNames: ["windows-10-iso", "windows-10-golden-image"]
---
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: windows-iso-cloner-binding
  namespace: kvm
subjects:
  # ???? ?????? ???? ServiceAccounts ???????? (????? ???????? ??? ???? namesapces ????????????: kv-user1, kv-user2 ?...
  - kind: Group
    name: system:serviceaccounts
    apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: windows-iso-cloner
  apiGroup: rbac.authorization.k8s.io

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/ns.yaml (83.0 B)

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: kvm
  labels:
    name: kvm

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/nvidia-custom-driver-datavolume-iso.yaml (1.8 KB)

```yaml
# DataVolume: ?????? ISO-?????? ? ?????????? ?? HTTP ? PVC.
# ????????? ? PVC ? ????? ?????????? ISO; ???? ????? PVC ??????????? ? VM ??? CD-ROM (?????? ??????).
# ??? PVC = ??? DataVolume (???????? rtx-pro-6000) ? ??? ??????? ? NVIDIA_CUSTOM_DRIVER_PVC_NAME.
#
# 1) ???????? ISO (xorriso -as mkisofs -r -J -V "LABEL" -o nvidia-driver.iso driver/)
# 2) ????????? ISO ?? HTTP (python -m http.server, nginx, MinIO ? ?.?.)
# 3) ????????? url ? ?????? storage (?? ?????? ??????? ISO), ????????? ????????
# 4) ?????????: kubectl get datavolumes -n kvm  -> phase: Succeeded
#
# ???? CDI ?????? ?????? (????????, ??????????? ISO ??? ????? ?????), ??????????
# ??????? B ? ???????????: PVC ? ???????? ???????? ? ???-?????????, ????????? ? ???? ????? ?????????.
# ??. docs/NVIDIA-CUSTOM-DRIVER.md
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: rtx-pro-6000
  namespace: <namespace>
  labels:
    app.kubernetes.io/component: nvidia-custom-driver-source
  annotations:
    description: "NVIDIA driver ISO; do not delete unless replacing"
spec:
  source:
    http:
      url: "http://YOUR_SERVER/nvidia-driver.iso"   # ???????? ?? ???????? URL ISO
  pvc:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 5Gi   # ?? ?????? ??????? ISO (???????? 3Gi ??? ~2.2 GB)
    storageClassName: <storage class>  # ???????? ?? ???? (???????? your-SC)

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/nvidia-custom-driver-pvc.yaml (962.0 B)

```yaml
# PVC for custom NVIDIA driver installer (.run or install.sh).
# Used as source for CDI clone when creating VMs with RTX Pro 6000 / custom driver GPU.
# Fill the PVC using the upload pod (nvidia-custom-driver-upload-pod.yaml) and kubectl cp.
# See docs/NVIDIA-CUSTOM-DRIVER.md.
#
# IMPORTANT: Deleting this PVC will permanently remove the driver image. Deleting the
# upload pod does NOT delete this PVC ? data remains. To prevent accidental PVC deletion,
# use RBAC or avoid granting delete on this PVC to non-admins.
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: rtx-pro-6000
  namespace: <namespace>
  labels:
    app.kubernetes.io/component: nvidia-custom-driver-source
  annotations:
    description: "Source for NVIDIA custom driver; do not delete unless replacing driver"
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 5Gi
  storageClassName: <storege class>   # change to your cluster's storage class

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/nvidia-custom-driver-upload-pod.yaml (1.2 KB)

```yaml
# One-off pod to copy the NVIDIA driver .run file into the PVC.
# 1. Apply the PVC first: kubectl apply -f nvidia-custom-driver-pvc.yaml
# 2. Apply this pod: kubectl apply -f nvidia-custom-driver-upload-pod.yaml
# 3. Wait: kubectl wait --for=condition=Ready pod/nvidia-driver-upload -n kvm --timeout=120s
# 4. Copy file to /data/ (PVC is mounted there; :/ would copy into container root, not the volume):
#    kubectl cp ./NVIDIA-Linux-x86_64-580.126.09-grid.run <namespace>/nvidia-driver-upload:/data/
#    or one .iso: kubectl cp ./NVidia-trx-6000-guest-driver.iso <namespace>/nvidia-driver-upload:/data/driver.iso
# 5. Delete pod: kubectl delete pod nvidia-driver-upload -n <namespace>
#
# claimName must match NVIDIA_CUSTOM_DRIVER_PVC_NAME in app config (default rtx-pro-6000).
# See docs/NVIDIA-CUSTOM-DRIVER.md.
apiVersion: v1
kind: Pod
metadata:
  name: nvidia-driver-upload
  namespace: <namespace>
spec:
  restartPolicy: Never
  containers:
    - name: upload
      image: busybox:1.36
      command: ["sleep", "3600"]
      volumeMounts:
        - name: driver-data
          mountPath: /data
  volumes:
    - name: driver-data
      persistentVolumeClaim:
        claimName: rtx-pro-6000   # must match NVIDIA_CUSTOM_DRIVER_PVC_NAME

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/pc.yaml (191.0 B)

```yaml
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
  name: kubevirt-manager-priority
value: 1000000
globalDefault: false
description: "Priority class for kubevirt-manager"

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/pvc.yaml (379.0 B)

```yaml
# Superseded by volumeClaimTemplates in StatefulSet (config-volume, your-SC).
# Kept for reference only. Do not apply unless not using StatefulSet.

apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: kubevirt-api-config
  namespace: kvm
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 100Mi
  storageClassName: your-SC

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/rbac.yaml (3.4 KB)

```yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: kubevirt-api-manager-multus
  namespace: kvm
---
# ClusterRole for minimal cluster-wide permissions (only nodes and GPU discovery)
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: kubevirt-api-manager-cluster-role-multus
rules:
  - apiGroups: [""]
    resources: ["nodes"]
    verbs: ["get", "list"]
  - apiGroups: [""]
    resources: ["namespaces"]
    verbs: ["get", "list", "watch"]
  - apiGroups: ["metrics.k8s.io"]
    resources: ["pods"]
    verbs: ["get", "list"]
---
# ClusterRoleBinding for minimal cluster-wide access
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: kubevirt-api-manager-cluster-binding-multus
subjects:
  - kind: ServiceAccount
    name: kubevirt-api-manager-multus
    namespace: kvm
roleRef:
  kind: ClusterRole
  name: kubevirt-api-manager-cluster-role-multus
  apiGroup: rbac.authorization.k8s.io
---
# Role for namespace-specific permissions (kv-* namespaces)
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: kubevirt-api-manager-user-ns-role-multus
rules:
  - apiGroups: ["kubevirt.io"]
    resources: ["virtualmachines", "virtualmachineinstances", "virtualmachineinstancepresets", "virtualmachineinstanc...
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["subresources.kubevirt.io"]
    resources: ["virtualmachineinstances/vnc", "virtualmachineinstances/console"]
    verbs: ["get"]
  - apiGroups: ["subresources.kubevirt.io"]
    resources: ["virtualmachines/start", "virtualmachines/stop", "virtualmachines/restart"]
    verbs: ["update"]
  - apiGroups: ["cdi.kubevirt.io"]
    resources: ["datavolumes"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: [""]
    resources: ["pods", "endpoints", "persistentvolumeclaims", "events", "configmaps"]
    verbs: ["get", "list", "watch"]
  - apiGroups: [""]
    resources: ["services"]
    verbs: ["get", "list", "watch", "create", "delete", "update", "patch"]
  - apiGroups: [""]
    resources: ["secrets"]
    verbs: ["get", "list", "watch", "create", "delete"]
  - apiGroups: [""]
    resources: ["persistentvolumeclaims"]
```

```
    verbs: ["delete"]
  - apiGroups: [""]
    resources: ["pods/log"]
    verbs: ["get"]
  - apiGroups: [""]
    resources: ["namespaces"]
    verbs: ["create"]
  - apiGroups: ["k8s.cni.cncf.io"]
    resources: ["network-attachment-definitions"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["crd.projectcalico.org"]
    resources: ["ippools"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["crd.projectcalico.org"]
    resources: ["globalnetworkpolicies"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
  - apiGroups: ["networking.k8s.io"]
    resources: ["networkpolicies"]
    verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]
---
# ClusterRoleBinding for user namespace access (kv-* pattern)
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: kubevirt-api-manager-user-ns-binding-multus
subjects:
  - kind: ServiceAccount
    name: kubevirt-api-manager-multus
    namespace: kvm
roleRef:
  kind: ClusterRole
  name: kubevirt-api-manager-user-ns-role-multus
  apiGroup: rbac.authorization.k8s.io

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/secrets-example.yaml (757.0 B)

```yaml
# Example secrets file for kubevirt-api-manager
# DO NOT commit this file with real credentials!
# All sensitive data: set via env vars or Kubernetes Secrets.
# See app/config.py for full list of configurable variables.
#
# kubectl create secret generic kubevirt-api-secrets \
#   --from-literal=secret-key='your-random-secret-key' \
#   --from-literal=demo-username='admin' \
#   --from-literal=demo-password='secure-password' \
#   --from-literal=llm-api-key='sk-...' \
#   -n kvm

apiVersion: v1
kind: Secret
metadata:
  name: kubevirt-api-secrets
  namespace: kvm
type: Opaque
stringData:
  secret-key: "CHANGE-ME-random-secret-key-at-least-32-characters-long"
  demo-username: "admin"
  demo-password: "CHANGE-ME-secure-password"

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/service.yaml (292.0 B)

```yaml
apiVersion: v1
kind: Service
metadata:
  name: kubevirt-api-manager
  namespace: kvm
  labels:
    app: kubevirt-api-manager
spec:
  type: NodePort
  ports:
    - name: http
      port: 80
      targetPort: 8080
      nodePort: 30180
  selector:
    app: kubevirt-api-manager

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/servicemonitor.yaml (335.0 B)

```yaml
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  name: kubevirt-manager-monitor
  namespace: kvm
  labels:
    app: kubevirt-api-manager
spec:
  selector:
    matchLabels:
      app: kubevirt-api-manager
  endpoints:
  - port: http
    interval: 30s
  namespaceSelector:
    matchNames:
    - kvm

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/statefulset.yaml (5.1 KB)

```yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: kubevirt-api-manager
  namespace: kvm
  labels:
    app: kubevirt-api-manager
spec:
  serviceName: kubevirt-api-manager
  replicas: 1
  selector:
    matchLabels:
      app: kubevirt-api-manager
  template:
    metadata:
      labels:
        app: kubevirt-api-manager
    spec:
      affinity:
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
            - matchExpressions:
              - key: node-role.kubernetes.io/control-plane
                operator: Exists
              - key: node-role.kubernetes.io/gpu-worker
                operator: DoesNotExist
            - matchExpressions:
              - key: node-role.kubernetes.io/master
                operator: Exists
              - key: node-role.kubernetes.io/gpu-worker
                operator: DoesNotExist
            - matchExpressions:
              - key: node-role.kubernetes.io/worker
                operator: Exists
              - key: node-role.kubernetes.io/gpu-worker
                operator: DoesNotExist
      tolerations:
      - key: node-role.kubernetes.io/master
        operator: Exists
        effect: NoSchedule
      - key: node-role.kubernetes.io/control-plane
        operator: Exists
        effect: NoSchedule
      - key: cephfs
        operator: Exists
        effect: NoSchedule
      - key: data
        operator: Exists
        effect: NoSchedule
      serviceAccountName: kubevirt-api-manager-multus
      hostAliases:
        - ip: "CHANGE-ME"
          hostnames:
            - "llm.CHANGE-ME"
      containers:
        - name: manager
          image: your-registry.example.com/kubevirt/kv_api_manager:1.0-rc.0.0.1
          imagePullPolicy: Always
          ports:
            - containerPort: 8080
              name: http
          env:
            - name: FLASK_ENV
              value: "production"
```

```
            - name: DEBUG
              value: "false"
            - name: LOG_LEVEL
              value: "info"
            - name: JUNIPER_USER
              value: "CHANGE-ME"
            - name: JUNIPER_HOST
              value: "CHANGE-ME"
            - name: STORAGE_CLASS_NAME
              value: "your-SC"
            - name: VM_STORAGE_NODE_SELECTOR
              value: ""
            - name: SECRET_KEY
              valueFrom:
                secretKeyRef:
                  name: kubevirt-api-secrets
                  key: secret-key
                  optional: true
            - name: DEMO_USERNAME
              valueFrom:
                secretKeyRef:
                  name: kubevirt-api-secrets
                  key: demo-username
                  optional: true
            - name: DEMO_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: kubevirt-api-secrets
                  key: demo-password
                  optional: true
          livenessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 30
            periodSeconds: 10
            timeoutSeconds: 5
            failureThreshold: 3
          readinessProbe:
            httpGet:
              path: /ready
              port: 8080
            initialDelaySeconds: 10
            periodSeconds: 5
            timeoutSeconds: 3
            failureThreshold: 3
          resources:
            requests:
              cpu: "1000m"
              memory: "1024Mi"
            limits:
              cpu: "2000m"
              memory: "2Gi"
          volumeMounts:
            - name: juniper-ssh-key
              mountPath: /etc/juniper-secret
              readOnly: true
            - name: log-volume
              mountPath: /app/logs
        - name: juniper-sidecar
          image: your-registry.example.com/kubevirt/kv_api_manager:1.0-rc.0.0.1
          command: ["/bin/bash", "-c"]
          args:
            - |
              mkdir -p /root/.ssh
              cp /etc/juniper-secret/id_rsa /root/.ssh/id_rsa
              chmod 600 /root/.ssh/id_rsa
              echo "Starting Sidecar..."
              python /app/sidecar/juniper_sidecar.py
          env:
            - name: JUNIPER_USER
```

```
              value: "CHANGE-ME"
            - name: JUNIPER_HOST
              value: "CHANGE-ME"
            - name: JUNIPER_EXTERNAL_IP
              value: "CHANGE-ME"
          volumeMounts:
            - name: juniper-sidecar-code
              mountPath: /app/sidecar
            - name: juniper-ssh-key
              mountPath: /etc/juniper-secret
              readOnly: true
            - name: log-volume
              mountPath: /app/logs
        - name: dnat-logs
          image: busybox
          args: [/bin/sh, -c, 'mkdir -p /app/logs && touch /app/logs/dnat.log && tail -f /app/logs/dnat.log']
          volumeMounts:
            - name: log-volume
              mountPath: /app/logs
      volumes:
        - name: juniper-sidecar-code
          configMap:
            name: juniper-sidecar-code
        - name: log-volume
          emptyDir: {}
        - name: juniper-ssh-key
          secret:
            secretName: juniper-ssh-key
            defaultMode: 0400

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/kubernetes_example/templates/custom-network-policy-template.yaml (4.4 K

```yaml
---
# Custom NetworkPolicy Template for VM Namespaces
# This template allows users to define custom network policies for their VMs
#
# Usage:
# 1. Copy this template to your namespace
# 2. Customize the rules according to your needs
# 3. Apply with: kubectl apply -f custom-network-policy.yaml -n <namespace>
#
# This policy will be applied in addition to the default vm-namespace-policy

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: custom-vm-network-policy
  namespace: <NAMESPACE>  # Replace with your namespace
  labels:
    app: kubevirt-api-manager
    policy-type: custom-vm-policy
spec:
  # Select VM pods (pods with kubevirt.io/domain label)
  podSelector:
    matchExpressions:
      - key: kubevirt.io/domain
        operator: Exists

  policyTypes:
    - Ingress
    - Egress

  ingress:
    # Example: Allow specific ports from specific sources
    # Uncomment and customize as needed

    # - from:
    #     - namespaceSelector:
    #         matchLabels:
    #           name: allowed-namespace
    #     - podSelector:
    #         matchLabels:
    #           app: allowed-app
    #     - ipBlock:
    #         cidr: 192.168.1.0/24  # Allow from specific IP range
    #   ports:
    #     - protocol: TCP
    #       port: 22    # SSH
    #     - protocol: TCP
    #       port: 80    # HTTP
    #     - protocol: TCP
    #       port: 443   # HTTPS

    # Default: Allow all ingress (can be restricted above)
    - {}

  egress:
    # Example: Allow egress to specific destinations
    # Uncomment and customize as needed

    # - to:
    #     - namespaceSelector:
    #         matchLabels:
    #           name: allowed-namespace
    #     - podSelector:
    #         matchLabels:
    #           app: allowed-app
```

```yaml
    #     - ipBlock:
    #         cidr: 8.8.8.8/32  # Allow to specific IP
    #   ports:
    #     - protocol: TCP
    #       port: 53    # DNS
    #     - protocol: UDP
    #       port: 53    # DNS

    # Example: Deny egress to specific networks
    # - to:
    #     - ipBlock:
    #         cidr: 10.0.0.0/8  # Deny access to private networks
    #   ports:
    #     - protocol: TCP
    #   action: Deny  # Note: This requires a CNI that supports deny actions

    # Default: Allow all egress (can be restricted above)
    - {}

---
# Example: Custom NetworkPolicy with port restrictions
# This example shows how to allow only specific ports

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: custom-vm-restricted-ports
  namespace: <NAMESPACE>  # Replace with your namespace
spec:
  podSelector:
    matchExpressions:
      - key: kubevirt.io/domain
        operator: Exists

  policyTypes:
    - Ingress

  ingress:
    # Allow SSH from anywhere
    - from:
        - {}
      ports:
        - protocol: TCP
          port: 22

    # Allow HTTP/HTTPS from anywhere
    - from:
        - {}
      ports:
        - protocol: TCP
          port: 80
        - protocol: TCP
          port: 443

    # Allow custom port from specific namespace
    - from:
        - namespaceSelector:
            matchLabels:
              name: trusted-namespace
      ports:
        - protocol: TCP
          port: 8080

---
# Example: Custom NetworkPolicy with network restrictions
# This example shows how to restrict egress to specific networks

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: custom-vm-restricted-egress
```

```yaml
  namespace: <NAMESPACE>  # Replace with your namespace
spec:
  podSelector:
    matchExpressions:
      - key: kubevirt.io/domain
        operator: Exists

  policyTypes:
    - Egress

  egress:
    # Allow DNS
    - to:
        - namespaceSelector:
            matchLabels:
              name: kube-system
        - podSelector:
            matchLabels:
              k8s-app: kube-dns
      ports:
        - protocol: UDP
          port: 53
        - protocol: TCP
          port: 53

    # Allow HTTPS to external
    - to:
        - ipBlock:
            cidr: 0.0.0.0/0
            except:
              - 10.0.0.0/8      # Deny private networks
              - 172.16.0.0/12   # Deny private networks
              - 192.168.0.0/16  # Deny private networks
      ports:
        - protocol: TCP
          port: 443

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/pyproject.toml (444.0 B)

```toml
[project]
name = "viirtuoz-vps"
version = "0.1.0"
description = "Add your description here"
readme = "README.md"
requires-python = ">=3.11"
dependencies = [
    "asyncpg>=0.30.0",
    "bcrypt>=4.0.0",
    "fastapi>=0.129.0",
    "httpx>=0.27.0",
    "itsdangerous>=2.2.0",
    "jinja2>=3.1.0",
    "kubernetes>=35.0.0",
    "pydantic-settings>=2.13.0",
    "python-multipart>=0.0.22",
    "requests>=2.31.0",
    "uvicorn[standard]>=0.30.0",
]

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/requirements.txt (406.0 B)

```
fastapi>=0.115.0
uvicorn[standard]>=0.30.0
starlette>=0.40.0
pydantic>=2.0
pydantic-settings>=2.0

# Database
asyncpg>=0.30.0

# Auth
bcrypt>=4.0,<5

# Async HTTP client (replaces requests for LLM proxy)
httpx>=0.27.0

# Kubernetes
kubernetes>=29.0.0

# Templates
jinja2>=3.0.0

# YAML (cloud-init templates)
PyYAML>=6.0

# File locking (scheduling_config)
filelock>=3.12.0

# ASGI server
gunicorn>=22.0.0

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/run-stub-mode.sh (4.4 KB)

```
#!/bin/bash
# ?????? ??? ???????? ??????? ?????????? ? stub ?????? (??? Kubernetes)
#
# --with-db   ??????? PostgreSQL ? Docker ? ?????????? ? ?????????
#             (???????, ???????????, ???? ????? ???????? ? ???????? ??)

WITH_DB=false
for arg in "$@"; do
    case "$arg" in
        --with-db) WITH_DB=true ;;
    esac
done

echo "? ?????? KubeVirt API Manager ? STUB ?????? (??? Kubernetes)"
echo "==================================================="

# ???? ??? stub-?????? (8080 ?? ?????????, 8970 ???? 8080 ?????)
STUB_PORT="${PORT:-8970}"
if command -v fuser >/dev/null 2>&1; then
    fuser -k 8970/tcp 2>/dev/null; fuser -k 8080/tcp 2>/dev/null
    sleep 2
fi
export PORT="$STUB_PORT"

# ????????? ????????? ????????
export FRONTEND_STUB_MODE=true
export DEBUG=true
export DEMO_USERNAME=admin
export DEMO_PASSWORD=your-secret-password
export SECRET_KEY=dev-secret-key-for-stub-mode
# LLM: ??????? ? .env
export LLM_SERVICE_URL="${LLM_SERVICE_URL}"
export LLM_CHAT_SHARED_TOKEN="${LLM_CHAT_SHARED_TOKEN}"
export LLM_MODEL="${LLM_MODEL}"
export LLM_AUTH_SERVICE_URL="${LLM_AUTH_SERVICE_URL}"
export LLM_AUTH_ADMIN_TOKEN="${LLM_AUTH_ADMIN_TOKEN}"
export LLM_USER_KEYS_MAX="${LLM_USER_KEYS_MAX:-10}"
export LLM_KEYS_ENCRYPTION_KEY="${LLM_KEYS_ENCRYPTION_KEY}"
if [ "$WITH_DB" = true ]; then
    COMPOSE_FILE="$(dirname "$0")/docker-compose.stub.yml"
    if [ ! -f "$COMPOSE_FILE" ]; then
        echo "? ???? $COMPOSE_FILE ?? ??????"
        exit 1
    fi
    echo "? ?????? PostgreSQL..."
    if command -v docker >/dev/null 2>&1; then
        if docker compose version >/dev/null 2>&1; then
            DC="docker compose"
        else
            DC="docker-compose"
        fi
    else
        echo "? Docker ?? ??????. ?????????? Docker ??? --with-db"
        exit 1
    fi
    $DC -f "$COMPOSE_FILE" up -d postgres
    echo "   ???????? ????????? PostgreSQL..."
    for i in {1..30}; do
        if $DC -f "$COMPOSE_FILE" exec -T postgres pg_isready -U viirtuoz 2>/dev/null; then
            break
        fi
        sleep 1
    done
    export DATABASE_URL="postgresql://viirtuoz:stub@127.0.0.1:5434/viirtuoz"
    export ADMIN_USERNAME="${ADMIN_USERNAME:-admin}"
```

```
    export ADMIN_PASSWORD="${ADMIN_PASSWORD:-your-secret-password}"
    echo "? PostgreSQL ??????? (???? 5434)"
    echo "   DATABASE_URL=postgresql://viirtuoz:***@127.0.0.1:5434/viirtuoz"
    echo "   ?????? ?????? ??????? ??????: $ADMIN_USERNAME / $ADMIN_PASSWORD"
    echo ""
else
    # ??? ??: ????-????? ? ????? ???????????? ?? env (?????????? ?????? ? ????????? ??? .env)
    export DATABASE_URL=""
fi

echo "? ?????????? ????????? ??????????:"
echo "   FRONTEND_STUB_MODE=true"
echo "   DEBUG=true"
echo "   DEMO_USERNAME=admin"
echo "   DEMO_PASSWORD=your-secret-password"
[ -n "${DATABASE_URL+x}" ] && echo "   DATABASE_URL=*** (???????? ??)"
[ -n "${LLM_AUTH_ADMIN_TOKEN}" ] && echo "   LLM_AUTH_ADMIN_TOKEN=*** (?????)"
[ -n "${LLM_CHAT_SHARED_TOKEN}" ] && echo "   LLM_CHAT_SHARED_TOKEN=*** (?????)"
echo ""

# ???????? ???????????
if ! python3 -c "import fastapi" 2>/dev/null; then
    echo "? FastAPI ?? ??????????"
    echo "   ?????????? ??????????: pip install -r requirements.txt"
    exit 1
fi

# Config: ???????? example, ???? config.py ??????????
if [ ! -f app/config.py ]; then
    if [ -f app/config.py.example ]; then
        cp app/config.py.example app/config.py
        echo "   app/config.py ?????? ?? example"
    fi
fi

echo "? ?????????? ???????"
echo ""

# ?????? ??????????
echo "? ?????? ??????????..."
echo "   UI: http://localhost:$STUB_PORT"
echo "   ?????: admin / ??????: your-secret-password"
echo "   LLM: ??? ?????????? ????? ????? ?? LLM_CHAT_SHARED_TOKEN"
echo ""
echo "? ???? VM: demo-gpu-vm, data-import-vm, analytics-vm"
[ "$WITH_DB" = true ] && echo "??  ? ??: docker compose -f docker-compose.stub.yml down ? ????????? PostgreSQL"
echo ""
echo "??????? Ctrl+C ??? ?????????"
echo "==================================================="
echo ""

cd "$(dirname "$0")"
python3 -m app.main

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/scripts/__init__.py (0.0 B)

```
1
```

[FILE] viirtuoz-vps-dev-rewrite-back-2/scripts/check_llm_balance.py (1022.0 B)

```python
#!/usr/bin/env python3
"""Check LLM balance for a user in auth-service. Usage: python -m scripts.check_llm_balance <username>"""

import asyncio
import sys
from pathlib import Path

# Ensure project root in path (for direct run: python scripts/check_llm_balance.py)
_root = Path(__file__).resolve().parent.parent
if str(_root) not in sys.path:
    sys.path.insert(0, str(_root))

from app.services.llm_auth_client import LLMAuthClient


async def main():
    username = sys.argv[1] if len(sys.argv) > 1 else "ops"
    client = LLMAuthClient()
    if not client.available():
        print("LLM auth-service not configured (LLM_AUTH_SERVICE_URL, LLM_AUTH_ADMIN_TOKEN)")
        sys.exit(1)
    uid = await client.get_user_id(username)
    if not uid:
        print(f"User '{username}' not found in auth-service")
        sys.exit(1)
    balance = await client.get_balance(uid)
    print(f"{username}: user_id={uid}, llm_balance={balance}")


if __name__ == "__main__":
    asyncio.run(main())

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/scripts/check_main_balance.py (987.0 B)

```python
#!/usr/bin/env python3
"""Check main (PostgreSQL) balance for a user. Usage: python -m scripts.check_main_balance <username>"""

import asyncio
import sys
from pathlib import Path

_root = Path(__file__).resolve().parent.parent
if str(_root) not in sys.path:
    sys.path.insert(0, str(_root))

from app.config import settings
from app.database import create_db


async def main():
    username = sys.argv[1] if len(sys.argv) > 1 else "ops"
    if not settings.DATABASE_URL:
        print("DATABASE_URL not configured")
        sys.exit(1)
    async with create_db(settings.DATABASE_URL) as db:
        user = await db.users.get_by_username(username)
        if not user:
            print(f"User '{username}' not found")
            sys.exit(1)
        bal, ver = await db.users.get_balance(user.id) or (None, None)
        print(f"{username}: id={user.id}, main_balance={bal}, balance_version={ver}")


if __name__ == "__main__":
    asyncio.run(main())

```

[FILE] viirtuoz-vps-dev-rewrite-back-2/uv.lock (203.2 KB)

```
version = 1
revision = 3
requires-python = ">=3.11"

[[package]]
name = "annotated-doc"
version = "0.0.4"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/57/ba/046ceea27344560984e26a590f90bc7f4a75b06701f65322245892...
wheels = [
    { url = "https://files.pythonhosted.org/packages/1e/d3/26bf1008eb3d2daa8ef4cacc7f3bfdc11818d111f7e2d0201bc6e3b49d...
]

[[package]]
name = "annotated-types"
version = "0.7.0"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/ee/67/531ea369ba64dcff5ec9c3402f9f51bf748cec26dde048a2f973a4...
wheels = [
    { url = "https://files.pythonhosted.org/packages/78/b6/6307fbef88d9b5ee7421e68d78a9f162e0da4900bc5f5793f6d3d0e34f...
]

[[package]]
name = "anyio"
version = "4.12.1"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "idna" },
    { name = "typing-extensions", marker = "python_full_version < '3.13'" },
]
sdist = { url = "https://files.pythonhosted.org/packages/96/f0/5eb65b2bb0d09ac6776f2eb54adee6abe8228ea05b20a5ad0e4945...
wheels = [
    { url = "https://files.pythonhosted.org/packages/38/0e/27be9fdef66e72d64c0cdc3cc2823101b80585f8119b5c112c2e8f5f7d...
]

[[package]]
name = "asyncpg"
version = "0.31.0"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/fe/cc/d18065ce2380d80b1bcce927c24a2642efd38918e33fd724bc4bca...
wheels = [
    { url = "https://files.pythonhosted.org/packages/08/17/cc02bc49bc350623d050fa139e34ea512cd6e020562f2a7312a7bcae4b...
    { url = "https://files.pythonhosted.org/packages/a4/62/4ded7d400a7b651adf06f49ea8f73100cca07c6df012119594d1e3447a...
    { url = "https://files.pythonhosted.org/packages/d6/5b/4179538a9a72166a0bf60ad783b1ef16efb7960e4d7b9afe9f77a55516...
    { url = "https://files.pythonhosted.org/packages/e6/35/c27719ae0536c5b6e61e4701391ffe435ef59539e9360959240d6e47c8...
    { url = "https://files.pythonhosted.org/packages/43/f4/01ebb9207f29e645a64699b9ce0eefeff8e7a33494e1d29bb53736f776...
    { url = "https://files.pythonhosted.org/packages/3e/f4/03ff1426acc87be0f4e8d40fa2bff5c3952bef0080062af9efc2212e3b...
    { url = "https://files.pythonhosted.org/packages/c7/39/cc788dfca3d4060f9d93e67be396ceec458dfc429e26139059e58c2c24...
    { url = "https://files.pythonhosted.org/packages/28/fc/735af5384c029eb7f1ca60ccb8fa95521dbdaeef788edf4cecfc604c3c...
    { url = "https://files.pythonhosted.org/packages/2a/a6/59d0a146e61d20e18db7396583242e32e0f120693b67a8de43f1557033...
    { url = "https://files.pythonhosted.org/packages/36/01/ffaa189dcb63a2471720615e60185c3f6327716fdc0fc04334436fbb7c...
    { url = "https://files.pythonhosted.org/packages/9f/62/3f699ba45d8bd24c5d65392190d19656d74ff0185f42e19d0bbd973bb3...
    { url = "https://files.pythonhosted.org/packages/8c/d1/a867c2150f9c6e7af6462637f613ba67f78a314b00db220cd26ff559d5...
    { url = "https://files.pythonhosted.org/packages/7a/1a/cce4c3f246805ecd285a3591222a2611141f1669d002163abef999b60f...
    { url = "https://files.pythonhosted.org/packages/40/ae/0fc961179e78cc579e138fad6eb580448ecae64908f95b8cb8ee2f241f...
    { url = "https://files.pythonhosted.org/packages/52/b2/b20e09670be031afa4cbfabd645caece7f85ec62d69c312239de568e05...
    { url = "https://files.pythonhosted.org/packages/b5/f0/f2ed1de154e15b107dc692262395b3c17fc34eafe2a78fc21159315617...
    { url = "https://files.pythonhosted.org/packages/95/11/97b5c2af72a5d0b9bc3fa30cd4b9ce22284a9a943a150fdc768763caf0...
    { url = "https://files.pythonhosted.org/packages/1b/71/157d611c791a5e2d0423f09f027bd499935f0906e0c2a416ce712ba51e...
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    { url = "https://files.pythonhosted.org/packages/7a/10/f882167cd207fbdd743e55534d5d9620e095089d176d55cb22d5322f2a...
    { url = "https://files.pythonhosted.org/packages/89/66/c7a9e1b7429be72123441bfdbaf2bc13faab3f90b933f664db506dea59...
    { url = "https://files.pythonhosted.org/packages/c4/26/b9924fa27db384bdcd97ab83b4f0a8058d96ad9626ead570674d5e737d...
    { url = "https://files.pythonhosted.org/packages/af/8f/3ed4bfa0c0c72a7ca17f0380cd9e4dd842b09f664e780c13cff1dcf2ef...
    { url = "https://files.pythonhosted.org/packages/2a/35/7051599bd493e62411d6ede36fd5af83a38f37c4767b92884df7301db2...
    { url = "https://files.pythonhosted.org/packages/10/9a/97c8d48ef10d6cd4fcead2415523221624bf58bcf68a802721a6bc807c...
    { url = "https://files.pythonhosted.org/packages/10/bf/979224a919a1b606c82bd2c5fa49b5c6d5727aa47b4312bb27b1734f53...
    { url = "https://files.pythonhosted.org/packages/ba/33/0ad65587441fc730dc7bd90e9716b30b4702dc7b617e6ba4997dc86514...
    { url = "https://files.pythonhosted.org/packages/67/ed/331d6b249259ee71ddea93f6f2f0a56cfebd46938bde6fcc6f7b9a3d0e...
    { url = "https://files.pythonhosted.org/packages/67/ff/f6b948ca32e4f2a4576aa129d8bed61f2e0543bf9f5f2b7fc3758ed005...
    { url = "https://files.pythonhosted.org/packages/16/85/276033dcbcc369eb176594de22728541a925b2632f9716428c851b149e...
    { url = "https://files.pythonhosted.org/packages/9e/f2/6a2a1f722b6aba37050e626530a46a68f74e63683947a8acff92569f97...
    { url = "https://files.pythonhosted.org/packages/60/bb/2186cb2f2bbaea6338cad15ce23a67f9b0672929744381e28b05926768...
    { url = "https://files.pythonhosted.org/packages/7d/a5/bf6f13b772fbb2a90360eb620d52ed8f796f3c5caee8398c3b2eb7b1c6...
    { url = "https://files.pythonhosted.org/packages/df/c5/d1be898bf0dc3ef9030c3825e5d3b83f2c528d207d246cbabe24596680...
    { url = "https://files.pythonhosted.org/packages/a5/42/90c1f7b9341eef50c8a1cb3f098ac43b0508413f33affd762855f67a41...
    { url = "https://files.pythonhosted.org/packages/76/be/4d3ee471e8145d12795ab655ece37baed0929462a86e72372fd2585904...
    { url = "https://files.pythonhosted.org/packages/b0/6f/8f7af07237c34a1defe7defc565a9bc1807762f672c0fde711a4b22bf9...
    { url = "https://files.pythonhosted.org/packages/4b/51/8ade005e5ca5b0d80fb4aff72a3775b325bdc3d27408c8113811a7cbe6...
    { url = "https://files.pythonhosted.org/packages/da/5f/6b8f83a55bb8278772c5ae54a577f3099025f9ade59d0136ac24a0df4b...
    { url = "https://files.pythonhosted.org/packages/0a/4c/925909008ed5a988ccbb72dcc897407e5d6d3bd72410d69e051fc0c146...
]

[[package]]
name = "click"
version = "8.3.1"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "colorama", marker = "sys_platform == 'win32'" },
]
sdist = { url = "https://files.pythonhosted.org/packages/3d/fa/656b739db8587d7b5dfa22e22ed02566950fbfbcdc203119934836...
wheels = [
    { url = "https://files.pythonhosted.org/packages/98/78/01c019cdb5d6498122777c1a43056ebb3ebfeef2076d9d026bfe15583b...
]

[[package]]
name = "colorama"
version = "0.4.6"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/d8/53/6f443c9a4a8358a93a6792e2acffb9d9d5cb0a5cfd8802644b7b1c...
wheels = [
    { url = "https://files.pythonhosted.org/packages/d1/d6/3965ed04c63042e047cb6a3e6ed1a63a35087b6a609aa3a15ed8ac56c2...
]

[[package]]
name = "durationpy"
version = "0.10"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/9d/a4/e44218c2b394e31a6dd0d6b095c4e1f32d0be54c2a4b250032d717...
wheels = [
    { url = "https://files.pythonhosted.org/packages/b0/0d/9feae160378a3553fa9a339b0e9c1a048e147a4127210e286ef18b730f...
]

[[package]]
name = "fastapi"
version = "0.129.0"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "annotated-doc" },
    { name = "pydantic" },
    { name = "starlette" },
    { name = "typing-extensions" },
    { name = "typing-inspection" },
]
sdist = { url = "https://files.pythonhosted.org/packages/48/47/75f6bea02e797abff1bca968d5997793898032d9923c1935ae2efd...
wheels = [
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    { url = "https://files.pythonhosted.org/packages/9e/dd/d0ee25348ac58245ee9f90b6f3cbb666bf01f69be7e0911f9851bddbda...
]

[[package]]
name = "h11"
version = "0.16.0"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/01/ee/02a2c011bdab74c6fb3c75474d40b3052059d95df7e73351460c85...
wheels = [
    { url = "https://files.pythonhosted.org/packages/04/4b/29cac41a4d98d144bf5f6d33995617b185d14b22401f75ca86f384e87f...
]

[[package]]
name = "httpcore"
version = "1.0.9"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "certifi" },
    { name = "h11" },
]
sdist = { url = "https://files.pythonhosted.org/packages/06/94/82699a10bca87a5556c9c59b5963f2d039dbd239f25bc2a63907a0...
wheels = [
    { url = "https://files.pythonhosted.org/packages/7e/f5/f66802a942d491edb555dd61e3a9961140fd64c90bce1eafd741609d33...
]

[[package]]
name = "httptools"
version = "0.7.1"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/b5/46/120a669232c7bdedb9d52d4aeae7e6c7dfe151e99dc70802e2fc7a...
wheels = [
    { url = "https://files.pythonhosted.org/packages/9c/08/17e07e8d89ab8f343c134616d72eebfe03798835058e2ab579dcc8353c...
    { url = "https://files.pythonhosted.org/packages/aa/06/c9c1b41ff52f16aee526fd10fbda99fa4787938aa776858ddc4a1ea825...
    { url = "https://files.pythonhosted.org/packages/cc/cc/10935db22fda0ee34c76f047590ca0a8bd9de531406a3ccb10a90e12ea...
    { url = "https://files.pythonhosted.org/packages/0e/84/875382b10d271b0c11aa5d414b44f92f8dd53e9b658aec338a79164fa5...
    { url = "https://files.pythonhosted.org/packages/30/e1/44f89b280f7e46c0b1b2ccee5737d46b3bb13136383958f20b580a821c...
    { url = "https://files.pythonhosted.org/packages/6f/7e/b9287763159e700e335028bc1824359dc736fa9b829dacedace91a39b3...
    { url = "https://files.pythonhosted.org/packages/b3/07/5b614f592868e07f5c94b1f301b5e14a21df4e8076215a3bccb830a687...
    { url = "https://files.pythonhosted.org/packages/53/7f/403e5d787dc4942316e515e949b0c8a013d84078a915910e9f391ba9b3...
    { url = "https://files.pythonhosted.org/packages/2a/0d/7f3fd28e2ce311ccc998c388dd1c53b18120fda3b70ebb022b135dc983...
    { url = "https://files.pythonhosted.org/packages/84/a6/b3965e1e146ef5762870bbe76117876ceba51a201e18cc31f5703e4545...
    { url = "https://files.pythonhosted.org/packages/11/7d/71fee6f1844e6fa378f2eddde6c3e41ce3a1fb4b2d81118dd544e3441e...
    { url = "https://files.pythonhosted.org/packages/22/a5/079d216712a4f3ffa24af4a0381b108aa9c45b7a5cc6eb141f81726b18...
    { url = "https://files.pythonhosted.org/packages/e9/9e/025ad7b65278745dee3bd0ebf9314934c4592560878308a6121f7f8120...
    { url = "https://files.pythonhosted.org/packages/6d/de/40a8f202b987d43afc4d54689600ff03ce65680ede2f31df348d7f368b...
    { url = "https://files.pythonhosted.org/packages/09/8f/c77b1fcbfd262d422f12da02feb0d218fa228d52485b77b953832105bb...
    { url = "https://files.pythonhosted.org/packages/0a/1a/22887f53602feaa066354867bc49a68fc295c2293433177ee90870a7d5...
    { url = "https://files.pythonhosted.org/packages/32/6a/6aaa91937f0010d288d3d124ca2946d48d60c3a5ee7ca62afe870e3ea0...
    { url = "https://files.pythonhosted.org/packages/6d/70/023d7ce117993107be88d2cbca566a7c1323ccbaf0af7eabf2064fe356...
    { url = "https://files.pythonhosted.org/packages/32/4d/9dd616c38da088e3f436e9a616e1d0cc66544b8cdac405cc4e81c8679f...
    { url = "https://files.pythonhosted.org/packages/1d/3a/a6c595c310b7df958e739aae88724e24f9246a514d909547778d776799...
    { url = "https://files.pythonhosted.org/packages/fd/82/88e8d6d2c51edc1cc391b6e044c6c435b6aebe97b1abc33db1b0b24cd5...
    { url = "https://files.pythonhosted.org/packages/34/50/9d095fcbb6de2d523e027a2f304d4551855c2f46e0b82befd718b8b200...
    { url = "https://files.pythonhosted.org/packages/07/f0/89720dc5139ae54b03f861b5e2c55a37dba9a5da7d51e1e824a1f34362...
    { url = "https://files.pythonhosted.org/packages/b3/cb/eea88506f191fb552c11787c23f9a405f4c7b0c5799bf73f2249cd4f52...
    { url = "https://files.pythonhosted.org/packages/e0/4a/a548bdfae6369c0d078bab5769f7b66f17f1bfaa6fa28f81d6be695906...
    { url = "https://files.pythonhosted.org/packages/4d/31/14df99e1c43bd132eec921c2e7e11cda7852f65619bc0fc5bdc2d0cb12...
    { url = "https://files.pythonhosted.org/packages/22/d2/b7e131f7be8d854d48cb6d048113c30f9a46dca0c9a8b08fcb3fcd588c...
    { url = "https://files.pythonhosted.org/packages/53/cf/878f3b91e4e6e011eff6d1fa9ca39f7eb17d19c9d7971b04873734112f...
]

[[package]]
name = "httpx"
version = "0.28.1"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "anyio" },
    { name = "certifi" },
    { name = "httpcore" },
    { name = "idna" },
]
```

```
sdist = { url = "https://files.pythonhosted.org/packages/b1/df/48c586a5fe32a0f01324ee087459e112ebb7224f646c0b5023f5e7...
wheels = [
    { url = "https://files.pythonhosted.org/packages/2a/39/e50c7c3a983047577ee07d2a9e53faf5a69493943ec3f6a384bdc792de...
]

[[package]]
name = "idna"
version = "3.11"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/6f/6d/0703ccc57f3a7233505399edb88de3cbd678da106337b9fcde432b...
wheels = [
    { url = "https://files.pythonhosted.org/packages/0e/61/66938bbb5fc52dbdf84594873d5b51fb1f7c7794e9c0f5bd885f30bc50...
]

[[package]]
name = "itsdangerous"
version = "2.2.0"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/9c/cb/8ac0172223afbccb63986cc25049b154ecfb5e85932587206f4231...
wheels = [
    { url = "https://files.pythonhosted.org/packages/04/96/92447566d16df59b2a776c0fb82dbc4d9e07cd95062562af01e408583f...
]

[[package]]
name = "jinja2"
version = "3.1.6"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "markupsafe" },
]
sdist = { url = "https://files.pythonhosted.org/packages/df/bf/f7da0350254c0ed7c72f3e33cef02e048281fec7ecec5f032d4aac...
wheels = [
    { url = "https://files.pythonhosted.org/packages/62/a1/3d680cbfd5f4b8f15abc1d571870c5fc3e594bb582bc3b64ea099db13e...
]

[[package]]
name = "kubernetes"
version = "35.0.0"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "certifi" },
    { name = "durationpy" },
    { name = "python-dateutil" },
    { name = "pyyaml" },
    { name = "requests" },
    { name = "requests-oauthlib" },
    { name = "six" },
    { name = "urllib3" },
    { name = "websocket-client" },
]
sdist = { url = "https://files.pythonhosted.org/packages/2c/8f/85bf51ad4150f64e8c665daf0d9dfe9787ae92005efb9a4d1cba59...
wheels = [
    { url = "https://files.pythonhosted.org/packages/0c/70/05b685ea2dffcb2adbf3cdcea5d8865b7bc66f67249084cf845012a0ff...
]

[[package]]
name = "markupsafe"
version = "3.0.3"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/7e/99/7690b6d4034fffd95959cbe0c02de8deb3098cc577c67bb6a24fe5...
wheels = [
    { url = "https://files.pythonhosted.org/packages/08/db/fefacb2136439fc8dd20e797950e749aa1f4997ed584c62cfb8ef7c2be...
    { url = "https://files.pythonhosted.org/packages/e1/2e/5898933336b61975ce9dc04decbc0a7f2fee78c30353c5efba7f2d6ff2...
    { url = "https://files.pythonhosted.org/packages/1d/09/adf2df3699d87d1d8184038df46a9c80d78c0148492323f4693df54e17...
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    { url = "https://files.pythonhosted.org/packages/19/ae/31c1be199ef767124c042c6c3e904da327a2f7f0cd63a0337e1eca2967...
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    { url = "https://files.pythonhosted.org/packages/a4/28/6e74cdd26d7514849143d69f0bf2399f929c37dc2b31e6829fd2045b27...
    { url = "https://files.pythonhosted.org/packages/62/7e/a145f36a5c2945673e590850a6f8014318d5577ed7e5920a4b3448e086...
    { url = "https://files.pythonhosted.org/packages/0f/62/d9c46a7f5c9adbeeeda52f5b8d802e1094e9717705a645efc71b0913a0...
    { url = "https://files.pythonhosted.org/packages/83/8a/4414c03d3f891739326e1783338e48fb49781cc915b2e0ee052aa490d5...
```

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    { url = "https://files.pythonhosted.org/packages/35/73/893072b42e6862f319b5207adc9ae06070f095b358655f077f69a35601...
    { url = "https://files.pythonhosted.org/packages/5a/72/147da192e38635ada20e0a2e1a51cf8823d2119ce8883f7053879c2199...
    { url = "https://files.pythonhosted.org/packages/9a/81/7e4e08678a1f98521201c3079f77db69fb552acd56067661f8c2f534a7...
    { url = "https://files.pythonhosted.org/packages/1e/2c/799f4742efc39633a1b54a92eec4082e4f815314869865d876824c257c...
    { url = "https://files.pythonhosted.org/packages/3c/2e/8d0c2ab90a8c1d9a24f0399058ab8519a3279d1bd4289511d74e909f06...
    { url = "https://files.pythonhosted.org/packages/2c/54/887f3092a85238093a0b2154bd629c89444f395618842e8b0c41783898...
    { url = "https://files.pythonhosted.org/packages/c9/2f/336b8c7b6f4a4d95e91119dc8521402461b74a485558d8f238a68312f1...
    { url = "https://files.pythonhosted.org/packages/32/43/67935f2b7e4982ffb50a4d169b724d74b62a3964bc1a9a527f5ac4f1ee...
    { url = "https://files.pythonhosted.org/packages/89/e0/4486f11e51bbba8b0c041098859e869e304d1c261e59244baa3d295d47...
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    { url = "https://files.pythonhosted.org/packages/aa/5b/bec5aa9bbbb2c946ca2733ef9c4ca91c91b6a24580193e891b5f7dbe8e...
    { url = "https://files.pythonhosted.org/packages/e5/f1/216fc1bbfd74011693a4fd837e7026152e89c4bcf3e77b6692fba99231...
    { url = "https://files.pythonhosted.org/packages/38/2f/907b9c7bbba283e68f20259574b13d005c121a0fa4c175f9bed27c4597...
    { url = "https://files.pythonhosted.org/packages/9c/d9/5f7756922cdd676869eca1c4e3c0cd0df60ed30199ffd775e319089cb3...
    { url = "https://files.pythonhosted.org/packages/00/07/575a68c754943058c78f30db02ee03a64b3c638586fba6a6dd56830b30...
    { url = "https://files.pythonhosted.org/packages/a9/21/9b05698b46f218fc0e118e1f8168395c65c8a2c750ae2bab54fc4bd4e0...
    { url = "https://files.pythonhosted.org/packages/7f/71/544260864f893f18b6827315b988c146b559391e6e7e8f7252839b1b84...
    { url = "https://files.pythonhosted.org/packages/c2/28/b50fc2f74d1ad761af2f5dcce7492648b983d00a65b8c0e0cb457c82eb...
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    { url = "https://files.pythonhosted.org/packages/bc/e6/fa0ffcda717ef64a5108eaa7b4f5ed28d56122c9a6d70ab8b72f9f715c...
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    { url = "https://files.pythonhosted.org/packages/80/d6/2d1b89f6ca4bff1036499b1e29a1d02d282259f3681540e16563f27ebc...
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    { url = "https://files.pythonhosted.org/packages/14/c7/ca723101509b518797fedc2fdf79ba57f886b4aca8a7d31857ba3ee828...
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[[package]]
name = "oauthlib"
version = "3.3.1"
source = { registry = "https://pypi.org/simple" }
sdist = { url = "https://files.pythonhosted.org/packages/0b/5f/19930f824ffeb0ad4372da4812c50edbd1434f678c90c2733e1188...
wheels = [
    { url = "https://files.pythonhosted.org/packages/be/9c/92789c596b8df838baa98fa71844d84283302f7604ed565dafe5a6b504...
]

[[package]]
name = "pydantic"
version = "2.12.5"
source = { registry = "https://pypi.org/simple" }
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492 dependencies = [
493     { name = "annotated-types" },
494     { name = "pydantic-core" },
495     { name = "typing-extensions" },
496     { name = "typing-inspection" },
497 ]
498 sdist = { url = "https://files.pythonhosted.org/packages/69/44/36f1a6e523abc58ae5f928898e4aca2e0ea509b5aa6f6f392a5d88...
499 wheels = [
500     { url = "https://files.pythonhosted.org/packages/5a/87/b70ad306ebb6f9b585f114d0ac2137d792b48be34d732d60e597c2f846...
501
502 ... [528 more lines truncated]
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